CHAPTER

38

WATER/WASTE

CHAPTER 38
WATER/WASTE

Subject/Page	Date	COC
38-EFFECTIVE PAGES		
1 thru 7	Sep 15/2026	
8	BLANK	
38-CONTENTS		
1	May 15/2024	
R 2	Sep 15/2026	
3	Jan 15/2025	
4	Jan 15/2025	
5	Jan 15/2026	
6	Jan 15/2026	
O 7	Sep 15/2026	
O 8	Sep 15/2026	
9	May 15/2024	
10	May 15/2024	
11	May 15/2024	
12	May 15/2025	
13	May 15/2025	
14	BLANK	
38-00-00		
901	May 15/2024	
902	May 15/2024	
903	May 15/2024	
904	May 15/2024	
905	May 15/2024	
906	May 15/2024	
907	May 15/2024	
908	May 15/2024	
38-10-00		
201	Sep 15/2025	
R 202	Sep 15/2026	
R 203	Sep 15/2026	
R 204	Sep 15/2026	
R 205	Sep 15/2026	
R 206	Sep 15/2026	
O 207	Sep 15/2026	
O 208	Sep 15/2026	
O 209	Sep 15/2026	
210	May 15/2026	
38-10-00 (cont)		
211	May 15/2026	
212	May 15/2026	
213	May 15/2026	
214	BLANK	
38-10-00		
701	Jan 15/2025	
702	Jan 15/2025	
703	Jan 15/2025	
704	Jan 15/2025	
705	Jan 15/2025	
706	Jan 15/2025	
707	Jan 15/2025	
708	Jan 15/2025	
709	Jan 15/2025	
710	Jan 15/2025	
711	Jan 15/2025	
712	BLANK	
38-11-01		
401	May 15/2024	
402	Jan 15/2025	
403	May 15/2025	
404	May 15/2025	
405	Jan 15/2025	
406	May 15/2024	
407	May 15/2025	
408	Sep 15/2024	
409	Sep 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2025	
413	May 15/2025	
414	May 15/2025	
415	May 15/2025	
416	May 15/2025	
38-11-03		
401	May 15/2024	
402	Jan 15/2025	
38-11-03 (cont)		
403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2026	
408	May 15/2026	
409	May 15/2026	
410	May 15/2026	
411	May 15/2026	
412	May 15/2026	
413	May 15/2026	
414	BLANK	
38-11-05		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
R 206	Sep 15/2026	
O 207	Sep 15/2026	
O 208	Sep 15/2026	
209	May 15/2024	
210	May 15/2024	
211	May 15/2024	
212	May 15/2024	
213	May 15/2024	
214	May 15/2024	
215	May 15/2024	
R 216	Sep 15/2026	
217	May 15/2024	
218	May 15/2024	
219	May 15/2024	
220	May 15/2024	
221	May 15/2024	
222	May 15/2024	
223	May 15/2024	

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Subject/Page	Date	COC
38-11-05 (cont)		
224	May 15/2024	
38-11-06		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	BLANK	
38-11-06		
401	May 15/2024	
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403	May 15/2024	
404	May 15/2024	
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406	BLANK	
38-11-07		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	
38-11-08		
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403	Jan 15/2025	
404	Sep 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
407	Sep 15/2025	
408	Sep 15/2025	
409	Sep 15/2025	
410	Sep 15/2025	
38-11-09		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Jan 15/2025	
405	May 15/2024	
38-11-09 (cont)		
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
38-13-01		
R 401	Sep 15/2026	
402	May 15/2024	
R 403	Sep 15/2026	
404	May 15/2024	
R 405	Sep 15/2026	
406	BLANK	
38-13-01		
R 501	Sep 15/2026	
502	May 15/2024	
38-14-01		
401	May 15/2025	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	BLANK	
38-14-01		
501	Sep 15/2025	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	BLANK	
38-31-00		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
38-31-00 (cont)		
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	BLANK	
38-31-00		
701	May 15/2024	
702	Sep 15/2024	
703	May 15/2024	
704	May 15/2024	
705	May 15/2024	
706	BLANK	
38-31-01		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	Jan 15/2025	
405	Jan 15/2025	
406	Jan 15/2025	
407	May 15/2026	
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413	Jan 15/2025	
414	Jan 15/2025	
415	Jan 15/2025	
416	Jan 15/2025	
417	Jan 15/2025	
R 418	Sep 15/2026	
O 419	Sep 15/2026	
O 420	Sep 15/2026	
O 421	Sep 15/2026	
422	BLANK	

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38-31-06		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Jan 15/2026	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
38-32-00		
201	Sep 15/2024	
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225	Jan 15/2026	
226	Jan 15/2026	
227	Jan 15/2026	
38-32-00 (cont)		
228	Jan 15/2026	
229	Jan 15/2026	
230	Jan 15/2026	
231	Jan 15/2026	
232	Jan 15/2026	
233	Jan 15/2026	
234	Jan 15/2026	
235	Jan 15/2026	
236	Jan 15/2026	
237	Jan 15/2026	
238	Jan 15/2026	
239	Jan 15/2026	
240	BLANK	
38-32-00		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Sep 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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38-32-00		
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503	May 15/2024	
504	May 15/2024	
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508	May 15/2025	
38-32-00		
701	May 15/2024	
702	May 15/2024	
703	May 15/2024	
704	May 15/2025	
705	May 15/2024	
38-32-00 (cont)		
706	May 15/2024	
707	May 15/2024	
708	May 15/2024	
709	May 15/2024	
710	May 15/2024	
711	May 15/2024	
712	Sep 15/2024	
713	Sep 15/2024	
714	May 15/2024	
715	May 15/2024	
716	May 15/2024	
717	May 15/2024	
718	May 15/2024	
719	May 15/2024	
720	May 15/2024	
721	May 15/2024	
722	May 15/2024	
723	May 15/2024	
724	May 15/2024	
725	Sep 15/2024	
726	May 15/2025	
727	May 15/2024	
728	May 15/2025	
729	May 15/2025	
730	May 15/2024	
731	May 15/2025	
732	May 15/2025	
733	May 15/2024	
734	May 15/2025	
735	May 15/2024	
736	May 15/2024	
737	May 15/2024	
738	May 15/2024	
739	May 15/2024	
740	May 15/2024	
741	May 15/2024	

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38-32-00 (cont)		
742	May 15/2024	
743	May 15/2024	
744	May 15/2024	
745	May 15/2024	
746	Sep 15/2024	
747	May 15/2025	
748	May 15/2024	
749	May 15/2024	
750	May 15/2024	
751	May 15/2024	
752	May 15/2024	
753	May 15/2024	
754	May 15/2024	
755	May 15/2024	
756	BLANK	
38-32-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
R 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
O 414	Sep 15/2026	
O 415	Sep 15/2026	
O 416	Sep 15/2026	
417	May 15/2024	
R 418	Sep 15/2026	
O 419	Sep 15/2026	
420	May 15/2024	

Subject/Page	Date	COC
38-32-01 (cont)		
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	
R 424	Sep 15/2026	
R 425	Sep 15/2026	
R 426	Sep 15/2026	
O 427	Sep 15/2026	
428	May 15/2024	
429	May 15/2024	
430	May 15/2024	
431	May 15/2024	
432	BLANK	
38-32-02		
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202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	BLANK	
38-32-03		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	Sep 15/2025	
413	May 15/2024	
414	May 15/2024	
415	May 15/2024	

Subject/Page	Date	COC
38-32-03 (cont)		
416	May 15/2024	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	
420	May 15/2024	
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	
424	May 15/2024	
425	May 15/2024	
426	May 15/2024	
427	May 15/2024	
428	May 15/2024	
38-32-03		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
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38-32-04		
A 401	Sep 15/2026	
A 402	Sep 15/2026	
A 403	Sep 15/2026	
A 404	Sep 15/2026	
38-32-04 Config 1		
D 401	Sep 15/2026	
D 402	Sep 15/2026	
D 403	Sep 15/2026	
D 404	Sep 15/2026	
38-32-05		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	

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406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
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414	May 15/2024	
38-32-06		
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38-32-07		
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215	May 15/2024	
216	May 15/2024	
217	May 15/2024	
218	May 15/2024	
219	May 15/2024	
38-32-07 (cont)		
220	May 15/2024	
221	May 15/2024	
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38-32-08		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	Jan 15/2026	
406	Jan 15/2026	
407	Jan 15/2026	
408	BLANK	
38-32-09		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
38-32-10		
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202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	May 15/2024	
38-32-11		
201	May 15/2024	
R 202	Sep 15/2026	
203	May 15/2024	
R 204	Sep 15/2026	
205	May 15/2024	
R 206	Sep 15/2026	
207	Sep 15/2025	
208	May 15/2024	
38-32-11 (cont)		
209	May 15/2024	
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38-32-12		
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402	May 15/2024	
403	May 15/2024	
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38-33-00		
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507	May 15/2024	
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38-33-01		
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408	May 15/2026	
38-33-01		
701	Sep 15/2024	
702	May 15/2024	
703	May 15/2024	
704	May 15/2024	
38-33-02		
401	May 15/2024	
402	May 15/2024	

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403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
R 407	Sep 15/2026	
408	May 15/2024	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
38-33-02		
701	May 15/2024	
702	May 15/2024	
703	May 15/2024	
704	May 15/2024	
38-33-03		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
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210	May 15/2024	
38-42-00		
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203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	BLANK	
38-42-00		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	

Subject/Page	Date	COC
38-42-00 (cont)		
504	May 15/2024	
38-42-01		
401	May 15/2024	
402	Sep 15/2025	
403	Sep 15/2025	
404	May 15/2024	
405	May 15/2024	
406	Sep 15/2025	
407	Sep 15/2025	
408	Sep 15/2025	
409	Sep 15/2025	
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38-42-02		
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403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
38-42-06		
201	Jan 15/2025	
202	Jan 15/2025	
203	Jan 15/2025	
204	May 15/2024	
205	Jan 15/2025	
206	Jan 15/2025	
207	Jan 15/2025	
208	May 15/2024	
38-42-07		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	Jan 15/2025	
205	Jan 15/2025	
206	May 15/2024	

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207	May 15/2024	
208	May 15/2024	
209	Jan 15/2025	
210	May 15/2024	
211	May 15/2024	
212	May 15/2024	
38-42-08		
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403	Jan 15/2025	
404	May 15/2024	
405	Jan 15/2025	
406	May 15/2024	
407	Jan 15/2025	
408	BLANK	
38-42-09		
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403	May 15/2025	
404	May 15/2025	
405	May 15/2025	
406	May 15/2025	
407	May 15/2025	
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412	May 15/2025	
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414	May 15/2025	
38-42-10		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
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38-42-10 (cont)		
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R 408	Sep 15/2026	
409	Jan 15/2025	
410	Sep 15/2025	
411	Jan 15/2025	
412	BLANK	

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
WATER/WASTE - DDG MAINTENANCE PROCEDURES	38-00-00		901	PTF ALL
DDG 38-10-01A Potable Water System Individual Components Inoperative - Preparation TASK 38-00-00-040-805			901	PTF ALL
DDG 38-10-01A Restoration - Potable Water System Individual Components Deactivated/Isolated TASK 38-00-00-440-805			902	PTF ALL
DDG 38-10-01B Potable Water System Drained and Not Serviced - Preparation TASK 38-00-00-040-806			902	PTF ALL
DDG 38-10-01B Potable Water System Drained and Not Serviced - Restoration TASK 38-00-00-440-806			902	PTF ALL
DDG 38-30-01A Lavatory Waste System Individual Components Inoperative - Preparation TASK 38-00-00-040-807			903	PTF ALL
DDG 38-30-01A Lavatory Waste System Individual Components Inoperative - Restoration TASK 38-00-00-440-807			903	PTF ALL
DDG 38-30-01B Lavatory Waste System Not Used - Preparation TASK 38-00-00-040-808			904	PTF ALL
DDG 38-30-01B Lavatory Waste System Not Used - Restoration TASK 38-00-00-440-808			904	PTF ALL
DDG 38-30-01-01 Lavatory Waste System Vacuum Blower Inoperative - Preparation TASK 38-00-00-040-809			905	PTF ALL
DDG 38-30-01-01 Lavatory Waste System Vacuum Blower Inoperative - Restoration TASK 38-00-00-440-809			908	PTF ALL
POTABLE WATER SYSTEM - MAINTENANCE PRACTICES	38-10-00		201	PTF ALL
Potable Water System Disinfectant TASK 38-10-00-600-801			201	PTF ALL

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Potable Water System - Leak Test TASK 38-10-00-790-801			206	PTF ALL
POTABLE WATER SYSTEM - CLEANING	38-10-00		701	PTF ALL
Manually Clean the Potable Water Tank TASK 38-10-00-100-801			701	PTF ALL
Descale Cleaning of the Potable Water System TASK 38-10-00-100-802			708	PTF ALL
WATER TANK - REMOVAL/INSTALLATION	38-11-01		401	PTF ALL
Water Tank Removal TASK 38-11-01-000-801			401	PTF ALL
Water Tank Installation TASK 38-11-01-400-801			411	PTF ALL
FILL/OVERFLOW VALVE AND CONTROL CABLE - REMOVAL/INSTALLATION	38-11-03		401	PTF ALL
Fill/Overflow Valve Removal TASK 38-11-03-000-801			401	PTF ALL
Fill/Overflow Valve Installation TASK 38-11-03-400-801			407	PTF ALL
Fill/Overflow Valve Control Cable Removal TASK 38-11-03-000-802			409	PTF ALL
Fill/Overflow Valve Control Cable Installation TASK 38-11-03-400-802			410	PTF ALL
LAVATORY FAUCET - MAINTENANCE PRACTICES	38-11-05		201	PTF ALL
Lavatory Faucet - Removal TASK 38-11-05-000-801			201	PTF ALL
Lavatory Faucet - Installation TASK 38-11-05-400-801			206	PTF ALL
Cartridge Assembly - Replacement TASK 38-11-05-960-801			207	PTF ALL
I Lavatory Faucet - Adjustment TASK 38-11-05-820-801			216	PTF ALL

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CHAPTER 38
WATER/WASTE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FORWARD LAVATORY DRAIN VALVE - MAINTENANCE PRACTICES	38-11-06		201	PTF ALL
Forward Lavatory Drain Valve Operational Check TASK 38-11-06-000-801			201	PTF ALL
FORWARD LAVATORY DRAIN VALVE - REMOVAL/INSTALLATION	38-11-06		401	PTF ALL
Drain Valve Removal TASK 38-11-06-000-802			401	PTF ALL
Drain Valve Installation TASK 38-11-06-400-801			404	PTF ALL
LAVATORY WATER SUPPLY SHUTOFF VALVE - REMOVAL/INSTALLATION	38-11-07		401	PTF ALL
Shutoff Valve Removal TASK 38-11-07-000-801			401	PTF ALL
Shutoff Valve Installation TASK 38-11-07-400-801			404	PTF ALL
WATER TANK DRAIN VALVE AND CONTROL CABLE - REMOVAL/INSTALLATION	38-11-08		401	PTF ALL
Water Tank Drain Valve Removal TASK 38-11-08-000-801			401	PTF ALL
Water Tank Drain Valve Installation TASK 38-11-08-400-801			405	PTF ALL
Water Tank Drain Valve Control Cable Removal TASK 38-11-08-020-801			406	PTF ALL
Water Tank Drain Valve Control Cable Installation TASK 38-11-08-420-801			408	PTF ALL
POTABLE WATER FILL FITTING - REMOVAL/INSTALLATION	38-11-09		401	PTF ALL
Potable Water Fill Fitting Removal TASK 38-11-09-000-801			401	PTF ALL
Potable Water Fill Fitting Installation TASK 38-11-09-400-801			406	PTF ALL
WATER HEATER - REMOVAL/INSTALLATION	38-13-01		401	PTF ALL
Water Heater Removal TASK 38-13-01-000-801			401	PTF ALL

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WATER/WASTE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Water Heater Installation TASK 38-13-01-400-801			404	PTF ALL
WATER HEATER - ADJUSTMENT/TEST	38-13-01		501	PTF ALL
Water Heater Adjustment TASK 38-13-01-730-801			501	PTF ALL
WATER QUANTITY TRANSMITTER - REMOVAL/INSTALLATION	38-14-01		401	PTF ALL
Water Quantity Transmitter Removal TASK 38-14-01-000-801			401	PTF ALL
Water Quantity Transmitter Installation TASK 38-14-01-400-801			406	PTF ALL
WATER QUANTITY TRANSMITTER - ADJUSTMENT/TEST	38-14-01		501	PTF ALL
Water Quantity Transmitter Adjustment TASK 38-14-01-800-801			501	PTF ALL
GRAY WATER/DRAIN SYSTEM - MAINTENANCE PRACTICES	38-31-00		201	PTF ALL
Gray Water/Drain System - Maintenance Practice TASK 38-31-00-910-801			201	PTF ALL
GRAY WATER/DRAIN SYSTEM - CLEANING/PAINTING	38-31-00		701	PTF ALL
Gray Water System - Cleaning TASK 38-31-00-100-801			701	PTF ALL
DRAIN MAST - REMOVAL/INSTALLATION	38-31-01		401	PTF ALL
Forward Drain Mast Removal TASK 38-31-01-000-801			401	PTF ALL
Forward Drain Mast Installation TASK 38-31-01-400-801			407	PTF ALL
Aft Drain Mast Removal TASK 38-31-01-000-802			412	PTF ALL
Aft Drain Mast Installation TASK 38-31-01-400-802			418	PTF ALL

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WATER/WASTE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
SILL DRAIN FLOAT VALVE - REMOVAL/INSTALLATION	38-31-06		401	PTF ALL
Forward Sill Drain Float Valve Removal TASK 38-31-06-000-801			401	PTF ALL
Forward Sill Drain Float Valve Installation TASK 38-31-06-400-801			404	PTF ALL
Aft Sill Drain Float Valve Removal TASK 38-31-06-000-802			404	PTF ALL
Aft Sill Drain Float Valve Installation TASK 38-31-06-400-802			408	PTF ALL
VACUUM WASTE SYSTEM - MAINTENANCE PRACTICES	38-32-00		201	PTF ALL
Standard Practices for Work with the Toilet Waste and Equipment TASK 38-32-00-910-801			201	PTF ALL
Toilet Deactivation TASK 38-32-00-040-801			204	PTF ALL
Toilet Activation TASK 38-32-00-420-801			208	PTF ALL
Waste System - Vacuum Pressure Test TASK 38-32-00-780-801			210	PTF ALL
Toilet Blockage Removal TASK 38-32-00-160-801			213	PTF ALL
Vacuum Line Blockage Inspection TASK 38-32-00-280-801			216	PTF ALL
Waste Tube - Maintenance Practice TASK 38-32-00-910-802			220	PTF ALL
Vacuum Waste Line Blockage Removal TASK 38-32-00-160-802			220	PTF ALL
Waste Tank Drain Line Blockage Removal TASK 38-32-00-160-803			226	PTF ALL
Waste Drain Line Blockage Removal TASK 38-32-00-160-804			231	PTF ALL
Vacuum Waste Blower - Deactivation TASK 38-32-00-040-802			235	PTF ALL

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CHAPTER 38
WATER/WASTE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Vacuum Waste Blower - Activation TASK 38-32-00-440-801			237	PTF ALL
Power Cycle of the Vacuum Toilet Assembly Integrated System TASK 38-32-00-820-801			237	PTF ALL
VACUUM WASTE SYSTEM - REMOVAL/INSTALLATION	38-32-00		401	PTF ALL
Vacuum Waste Tubes Removal TASK 38-32-00-020-801			401	PTF ALL
Vacuum Waste Tubes Installation TASK 38-32-00-420-802			404	PTF ALL
VACUUM WASTE SYSTEM - ADJUSTMENT/TEST	38-32-00		501	PTF ALL
Vacuum Waste System - Leak Test TASK 38-32-00-700-801			501	PTF ALL
Toilet - Operational Test TASK 38-32-00-700-802			507	PTF ALL
VACUUM WASTE SYSTEM - CLEANING/PAINTING	38-32-00		701	PTF ALL
Periodic Flush - Vacuum Waste System Cleaning TASK 38-32-00-100-801			702	PTF ALL
Pressure Washer Cleaning of the Vacuum Waste Lines TASK 38-32-00-100-802			706	PTF ALL
Soak Method for Cleaning of the Vacuum Waste Lines TASK 38-32-00-100-803			712	PTF ALL
Recirculate Method of Cleaning the Vacuum Waste Lines TASK 38-32-00-110-801			725	PTF ALL
Recirculating Waste System Cleaning TASK 38-32-00-110-802			738	PTF ALL
Pressure Test Procedure TASK 38-32-00-780-802			746	PTF ALL
Vacuum Waste Tube Off Airplane Soak TASK 38-32-00-100-804			749	PTF ALL

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WATER/WASTE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Overboard Vent Tube Cleaning - Cleaning/Painting TASK 38-32-00-140-801			752	PTF ALL
TOILET ASSEMBLY - REMOVAL/INSTALLATION	38-32-01		401	PTF ALL
Toilet Shroud Removal TASK 38-32-01-000-801			401	PTF ALL
Toilet Shroud Installation TASK 38-32-01-400-801			404	PTF ALL
Vacuum Toilet Assembly Removal TASK 38-32-01-000-802			405	PTF ALL
Vacuum Toilet Assembly Installation TASK 38-32-01-400-802			411	PTF ALL
Rinse Water Supply Cleaning TASK 38-32-01-100-801			414	PTF ALL
Rinse Valve Removal TASK 38-32-01-000-804			415	PTF ALL
Rinse Valve Installation TASK 38-32-01-400-804			418	PTF ALL
Flush Valve Removal TASK 38-32-01-000-803			420	PTF ALL
Flush Valve Installation TASK 38-32-01-400-803			424	PTF ALL
Integrated System Controller (ISC) Removal TASK 38-32-01-000-805			426	PTF ALL
Integrated System Controller (ISC) Installation TASK 38-32-01-400-805			430	PTF ALL
LIQUID SEPARATOR - MAINTENANCE PRACTICES	38-32-02		201	PTF ALL
Liquid Separator Removal TASK 38-32-02-000-801			201	PTF ALL
Liquid Separator Installation TASK 38-32-02-400-801			204	PTF ALL
Liquid Separator and Liquid Separator Filter Cleaning TASK 38-32-02-100-801			206	PTF ALL

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CHAPTER 38
WATER/WASTE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
WASTE DRAIN BALL VALVE AND LINKAGE ASSEMBLY - REMOVAL/INSTALLATION	38-32-03		401	PTF ALL
Waste Drain Ball Valve Removal TASK 38-32-03-000-801			401	PTF ALL
Waste Drain Ball Valve Installation TASK 38-32-03-400-801			409	PTF ALL
Waste Drain Ball Valve Linkage Assembly Removal TASK 38-32-03-000-802			413	PTF ALL
Waste Drain Ball Valve Linkage Assembly Installation TASK 38-32-03-400-802			419	PTF ALL
Waste Drain Ball Valve Limit Switch Removal TASK 38-32-03-000-803			421	PTF ALL
Waste Drain Ball Valve Limit Switch Installation TASK 38-32-03-400-803			426	PTF ALL
WASTE DRAIN BALL-VALVE AND CONTROL ROD - ADJUSTMENT/TEST	38-32-03		501	PTF ALL
Waste Drain Ball Valve and Control Rod - Operational Test TASK 38-32-03-710-801			501	PTF ALL
Waste Drain Ball Valve and Control Rod - Adjustment TASK 38-32-03-820-801			503	PTF ALL
FLUSH SWITCH - REMOVAL/INSTALLATION	38-32-04		401	PTF ALL
Flush Switch Removal TASK 38-32-04-000-804			401	PTF ALL
Flush Switch Installation TASK 38-32-04-400-804			404	PTF ALL
VACUUM BLOWER - REMOVAL/INSTALLATION	38-32-05		401	PTF ALL
Vacuum Blower Removal TASK 38-32-05-000-801			401	PTF ALL
Vacuum Blower Installation TASK 38-32-05-400-801			406	PTF ALL

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CHAPTER 38
WATER/WASTE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Vacuum Blower Filter Restore (Clean) or Replace TASK 38-32-05-960-801			409	PTF ALL
VACUUM CHECK VALVE - REMOVAL/INSTALLATION	38-32-06		401	PTF ALL
Vacuum Check Valve Removal TASK 38-32-06-000-801			401	PTF ALL
Vacuum Check Valve Installation TASK 38-32-06-400-801			405	PTF ALL
WASTE TANK - MAINTENANCE PRACTICES	38-32-07		201	PTF ALL
Waste Tank Removal TASK 38-32-07-000-801			201	PTF ALL
Waste Tank Installation TASK 38-32-07-400-801			210	PTF ALL
Waste Tank Cleaning TASK 38-32-07-100-801			215	PTF ALL
WASTE DRAIN VALVE ASSEMBLY - REMOVAL/INSTALLATION	38-32-08		401	PTF ALL
Waste Drain Valve Assembly Removal TASK 38-32-08-000-801			401	PTF ALL
Waste Drain Valve Assembly Installation TASK 38-32-08-400-801			405	PTF ALL
WASTE TANK RINSE FITTING ASSEMBLY - REMOVAL/INSTALLATION	38-32-09		401	PTF ALL
Waste Tank Rinse Fitting Assembly Removal TASK 38-32-09-000-801			401	PTF ALL
Waste Tank Rinse Fitting Assembly Installation TASK 38-32-09-400-801			405	PTF ALL
WASTE TANK RINSE NOZZLE - MAINTENANCE PRACTICES	38-32-10		201	PTF ALL
Waste Tank Rinse Nozzle Removal TASK 38-32-10-000-801			201	PTF ALL
Waste Tank Rinse Nozzle Installation TASK 38-32-10-400-801			204	PTF ALL
Waste Tank Rinse Nozzle Cleaning TASK 38-32-10-100-801			206	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
WASTE TANK RINSE FILTER - MAINTENANCE PRACTICES	38-32-11		201	PTF ALL
Waste Tank Rinse Filter Removal TASK 38-32-11-000-801			201	PTF ALL
Waste Tank Rinse Filter Installation TASK 38-32-11-400-801			206	PTF ALL
Waste Tank Rinse Filter Element Cleaning TASK 38-32-11-100-801			207	PTF ALL
VACUUM BLOWER BAROMETRIC SWITCH - REMOVAL/INSTALLATION	38-32-12		401	PTF ALL
Vacuum Blower Barometric Switch Removal TASK 38-32-12-000-801			401	PTF ALL
Vacuum Blower Barometric Switch Installation TASK 38-32-12-400-801			404	PTF ALL
WASTE TANK QUANTITY INDICATION - ADJUSTMENT/TEST	38-33-00		501	PTF ALL
LAVS INOP BITE Test at the Attendant Panel TASK 38-33-00-740-801			501	PTF ALL
LCM BITE Test TASK 38-33-00-740-802			502	PTF ALL
Auto Zero Test TASK 38-33-00-710-801			503	PTF ALL
Attendant Panel Waste Tank Quantity Indication Operational Test TASK 38-33-00-710-802			505	PTF ALL
WASTE TANK POINT LEVEL SENSOR - REMOVAL/INSTALLATION	38-33-01		401	PTF ALL
Waste Tank Point Level Sensor Removal TASK 38-33-01-000-801			401	PTF ALL
Waste Tank Point Level Sensor Installation TASK 38-33-01-400-801			406	PTF ALL
WASTE TANK POINT LEVEL SENSOR - CLEANING/PAINTING	38-33-01		701	PTF ALL
Waste Tank Point Level Sensor Cleaning TASK 38-33-01-100-801			701	PTF ALL

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WATER/WASTE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
WASTE TANK CONTINUOUS LEVEL SENSOR - REMOVAL/INSTALLATION	38-33-02		401	PTF ALL
Waste Tank Continuous Level Sensor Removal TASK 38-33-02-000-801			401	PTF ALL
Waste Tank Continuous Level Sensor Installation TASK 38-33-02-400-801			407	PTF ALL
WASTE TANK CONTINUOUS LEVEL SENSOR - CLEANING/PAINTING	38-33-02		701	PTF ALL
Waste Tank Continuous Level Sensor Cleaning TASK 38-33-02-100-801			701	PTF ALL
LOGIC CONTROL MODULE - MAINTENANCE PRACTICES	38-33-03		201	PTF ALL
Logic Control Module Removal TASK 38-33-03-000-801			201	PTF ALL
Logic Control Module Installation TASK 38-33-03-400-801			206	PTF ALL
Logic Control Module Operational Test TASK 38-33-03-710-801			208	PTF ALL
WATER TANK PRESSURIZATION - MAINTENANCE PRACTICES	38-42-00		201	PTF ALL
Potable Water System - Deactivation TASK 38-42-00-800-801			201	PTF ALL
Potable Water System - Activation TASK 38-42-00-800-802			206	PTF ALL
WATER TANK PRESSURIZATION - ADJUSTMENT/TEST	38-42-00		501	PTF ALL
Water Tank Pressurization System - Operational Test TASK 38-42-00-800-803			501	PTF ALL
Pressure Regulator - Operational Test TASK 38-42-00-800-804			503	PTF ALL
AIR COMPRESSOR - REMOVAL/INSTALLATION	38-42-01		401	PTF ALL
Air Compressor Removal TASK 38-42-01-000-801			401	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Air Compressor Installation TASK 38-42-01-400-801			406	PTF ALL
COMPRESSOR INLET AIR FILTER - REMOVAL/INSTALLATION	38-42-02		401	PTF ALL
Air Filter Element Removal TASK 38-42-02-000-801			401	PTF ALL
Air Filter Element Installation TASK 38-42-02-400-801			405	PTF ALL
PRESSURE RELIEF VALVE - MAINTENANCE PRACTICES	38-42-06		201	PTF ALL
Pressure Relief Valve Removal TASK 38-42-06-000-801			201	PTF ALL
Pressure Relief Valve Installation TASK 38-42-06-400-801			205	PTF ALL
Pressure Relief Valve - Operational Test TASK 38-42-06-700-801			206	PTF ALL
WATER PRESSURE CUT-OUT SWITCH - MAINTENANCE PRACTICES	38-42-07		201	PTF ALL
Water Pressure Cut-Out Switch Removal TASK 38-42-07-000-801			201	PTF ALL
Water Pressure Cut-Out Switch Installation TASK 38-42-07-400-801			206	PTF ALL
Water Pressure Cut-Out Switch Operational Test TASK 38-42-07-700-801			208	PTF ALL
PRESSURE REGULATOR - REMOVAL/INSTALLATION	38-42-08		401	PTF ALL
Pressure Regulator Removal TASK 38-42-08-000-801			401	PTF ALL
Pressure Regulator Installation TASK 38-42-08-400-801			405	PTF ALL
CHECK VALVES - REMOVAL/INSTALLATION	38-42-09		401	PTF ALL
Compressor Supply Check Valve Removal TASK 38-42-09-020-801			401	PTF ALL
Compressor Check Valve Installation TASK 38-42-09-400-801			405	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Pneumatic Supply Check Valve Removal TASK 38-42-09-020-802			407	PTF ALL
Pneumatic Supply Check Valve Installation TASK 38-42-09-400-802			410	PTF ALL
BLEED AIR FILTER - REMOVAL/INSTALLATION	38-42-10		401	PTF ALL
Bleed Air Filter Assembly Removal TASK 38-42-10-000-801			401	PTF ALL
Bleed Air Filter Assembly Installation TASK 38-42-10-400-801			406	PTF ALL
Bleed Air Filter Element Replacement TASK 38-42-10-960-801			408	PTF ALL

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WATER/WASTE - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has the maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks prepare the airplane for flight with systems/components that are inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. These are the tasks for the components in the water systems.

(1) DDG 38-10-01A Potable Water Systems Individual Components Inoperative - Preparation

(2) DDG 38-10-01A Potable Water Systems Individual Components Inoperative - Restoration

(3) DDG 38-10-01B Potable Water Systems Drained and Not Serviced - Preparation

(4) DDG 38-10-01B Potable Water Systems Drained and Not Serviced - Restoration

(5) DDG 38-30-01A Lavatory Waste Systems Individual Components Inoperative - Preparation

(6) DDG 38-30-01A Lavatory Waste Systems Individual Components Inoperative - Restoration

(7) DDG 38-30-01B Lavatory Waste Systems Not Used - Preparation

(8) DDG 38-30-01B Lavatory Waste Systems Not Used - Restoration

(9) DDG 38-30-01-01 Lavatory Waste System Vacuum Blower Inoperative - Preparation

(10) DDG 38-30-01-01 Lavatory Waste System Vacuum Blower Inoperative - Restoration

TASK 38-00-00-040-805

2. DDG 38-10-01A Potable Water System Individual Components Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the individual components of the potable water system inoperative.

NOTE: Due to the wide variety of galley, lavatory, and drinking fountain installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

C. Potable Water System Preparation

SUBTASK 38-00-00-040-001

(1) Deactivate or isolate inoperative components.

SUBTASK 38-00-00-210-001

(2) Make sure that the system components do not have leaks.

NOTE: Parts of the system that operate correctly can be used.

END OF TASK

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TASK 38-00-00-440-805

3. DDG 38-10-01A Restoration - Potable Water System Individual Components Deactivated/Isolated

A. General

(1) This task puts the airplane back to its usual condition after operation with the individual components of the potable water system inoperative.

NOTE: Due to the wide variety of galley, lavatory, and drinking fountain installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

C. Potable Water System Preparation Restoration

SUBTASK 38-00-00-440-001

(1) Reactivate applicable water system components.

——— END OF TASK ———

TASK 38-00-00-040-806

4. DDG 38-10-01B Potable Water System Drained and Not Serviced - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the potable water system drained and not serviced.

B. References

Reference	Title
12-14-00-600-801	Potable Water System - Drain (P/B 301)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

D. Potable Water System Preparation

SUBTASK 38-00-00-680-001

(1) Drain the potable water system (TASK 12-14-00-600-801).

SUBTASK 38-00-00-800-001

(2) Make sure that the potable water system is not serviced.

——— END OF TASK ———

TASK 38-00-00-440-806

5. DDG 38-10-01B Potable Water System Drained and Not Serviced - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the potable water system drained and not serviced.

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B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

D. Potable Water System Restoration

SUBTASK 38-00-00-610-001

(1) Fill the potable water system (TASK 12-14-00-600-802).

——— END OF TASK ———

TASK 38-00-00-040-807

6. DDG 38-30-01A Lavatory Waste System Individual Components Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the individual components off the lavatory waste system inoperative.

NOTE: Due to the wide variety of lavatory installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

C. Lavatory Waste System Preparation

SUBTASK 38-00-00-040-002

(1) Deactivate or isolate inoperative components.

(2) Make sure that the system components do not have leaks.

——— END OF TASK ———

TASK 38-00-00-440-807

7. DDG 38-30-01A Lavatory Waste System Individual Components Inoperative - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the individual components of the lavatory waste system inoperative.

NOTE: Due to the wide variety of lavatory installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

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C. Lavatory Waste System Restoration

SUBTASK 38-00-00-440-002

(1) Reactivate applicable lavatory system components.

——— END OF TASK ———

TASK 38-00-00-040-808

8. DDG 38-30-01B Lavatory Waste System Not Used - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the lavatory waste system not used.

NOTE: Due to the wide variety of lavatory installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

D. Lavatory Waste System Preparation

SUBTASK 38-00-00-040-003

(1) Deactivate or isolate inoperative components.

SUBTASK 38-00-00-040-004

(2) If necessary, drain the leaking waste tank (TASK 12-17-01-610-801).

SUBTASK 38-00-00-040-005

(3) Close, lock, and placard the lavatory door.

——— END OF TASK ———

TASK 38-00-00-440-808

9. DDG 38-30-01B Lavatory Waste System Not Used - Restoration

A. General

(1) This task restores the airplane back to its usual condition after operation with the lavatory waste system not used.

NOTE: Due to the wide variety of lavatory installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

B. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

C. Lavatory Waste System Restoration

SUBTASK 38-00-00-440-003

(1) Reactivate the applicable lavatory system components.

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SUBTASK 38-00-00-440-004

(2) Remove the placard and unlock the lavatory door.

END OF TASK

TASK 38-00-00-040-809

10. DDG 38-30-01-01 Lavatory Waste System Vacuum Blower Inoperative - Preparation

(Figure 901)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the lavatory vacuum blower inoperative.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Lavatory Waste System Preparation

SUBTASK 38-00-00-860-001

(1) Deactivate the inoperative vacuum blower:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(b) Open power distribution panel, P91.

(c) Open this circuit breaker and install safety lock:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
F	2	C01389	VACUUM WASTE BLOWER

(d) Close power distribution panel, P91.

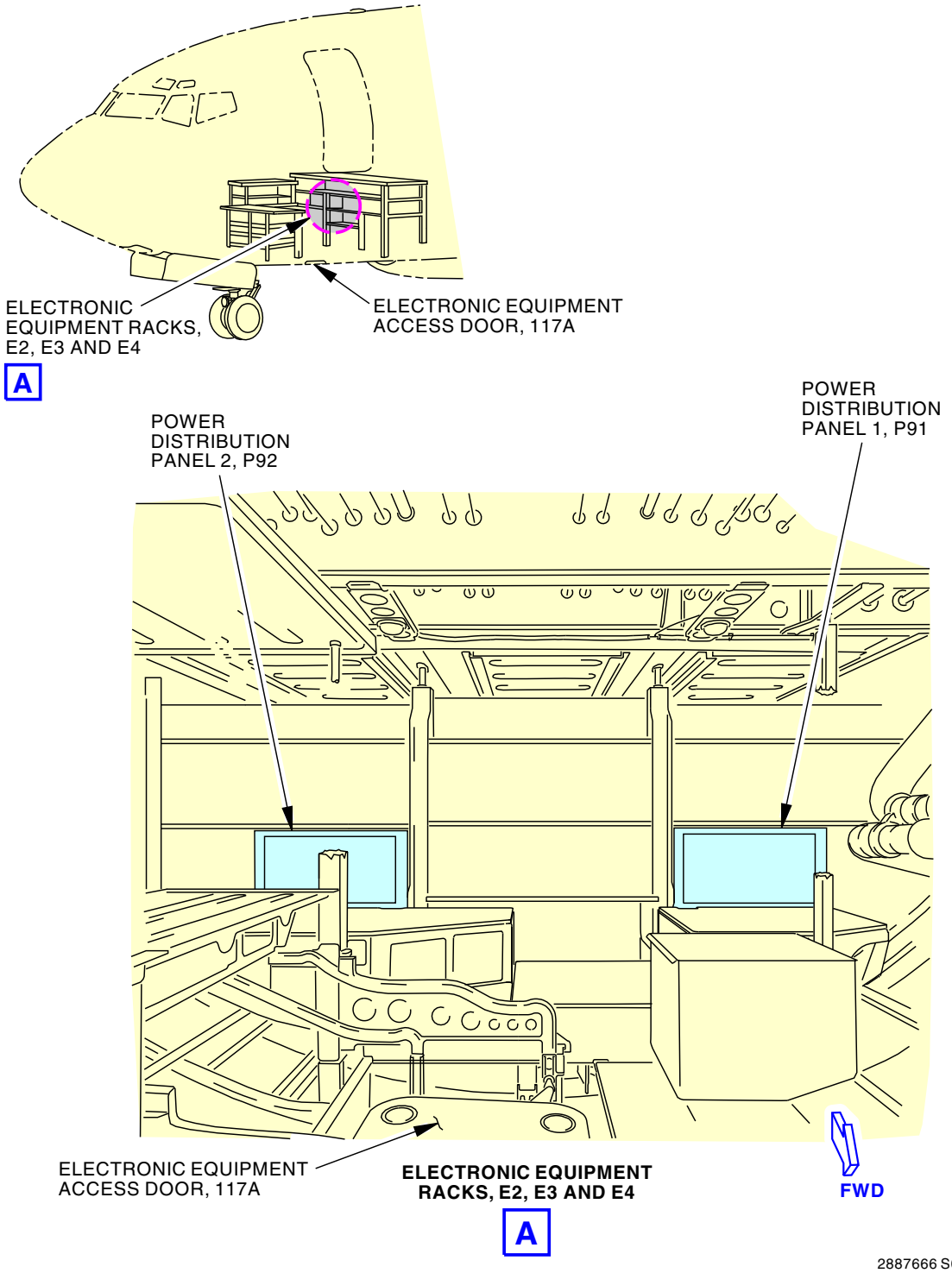
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

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Electrical Control Panels P91, P92
Figure 901/38-00-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-00-00-440-809

11. DDG 38-30-01-01 Lavatory Waste System Vacuum Blower Inoperative - Restoration

(Figure 901)

A. General

(1) This task restores the airplane back to its usual condition after operation with the lavatory vacuum blower inoperative.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Lavatory Waste System Restoration

SUBTASK 38-00-00-860-002

(1) Reactivate the inoperative vacuum blower:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(b) Open power distribution panel, P91.

(c) Remove the safety lock and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
F	2	C01389	VACUUM WASTE BLOWER

(d) Close power distribution panel, P91.

(e) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

POTABLE WATER SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Clean the potable water system with a disinfectant.

(2) A test for leakage of the potable water system.

B. It is necessary to flush and then use a disinfectant in the water system when you have these conditions:

(1) Contamination in the water system

(2) After repairs on the water system

TASK 38-10-00-600-801

2. Potable Water System Disinfectant

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The intent of this task is to supply a solution of the disinfectant to the potable water system that gives a concentration of 100 Parts Per Million (PPM). When you prepare the disinfectant solution, the stabilized chlorine dioxide (Purogene) and the citric acid are measured by volume (fluid ounces or liters).

(2) You can disinfect the potable water system by an ozone disinfectant procedure as an alternative to a chemical disinfectant. Make sure that the procedure supplies an ozone concentration of 3.5 PPM or less. Refer to the ozone equipment/cart supplier instructions when you do the ozone disinfectant procedure. Too much of the ozone can cause a degradation of the elastomeric components (O-rings) in the potable water system.

B. References

Reference	Title
12-14-00-600-801	Potable Water System - Drain (P/B 301)
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
25-31-00-960-801	Galley Water Filter Replacement (P/B 201)
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
COM-1141	Equipment - Potable Water Servicing Part #: 19-4301-0000 Supplier: 59603 Part #: 19-4303-0000 Supplier: 59603 Part #: 2198-310 Supplier: 6L481 Part #: CPW-100 Supplier: 32526 Part #: CPW-100-1 Supplier: 32526 Part #: CPW-100-2 Supplier: 32526 Part #: PW100 Supplier: 1CSJ2 Part #: SPW-350 Supplier: 32526 Part #: SPW-500PL Supplier: 32526 Part #: WC100E (ID 2026505) Supplier: 0T652 Part #: WSP-900 Supplier: 6L481 Part #: WTM-900 Supplier: 6L481 Opt Part #: 2303 Supplier: 00365 Opt Part #: SPW-250 Supplier: 32526
COM-25776	Socket - Quick Disconnect 1/4" Part #: 2HL25MS1 Supplier: 73992
COM-25777	Plug - Quick Disconnect 1/4" Part #: 2KLF16 Supplier: 73992
STD-478	Hose - Drain
STD-739	Respirator - Half-Mask, Air Purifying (with filters/cartridges)
STD-13465	Bucket - 2.5 Gallon Capacity

D. Consumable Materials

Reference	Description	Specification
B00637	Acid, Citric	A-A-59147
B01033	Cleaner - Disinfectant - Lysol	
B50246	Cleaner - Glyco-San Potable Water System Cleaner	AMS 1550B
G00022	Compound - Chlorine Dioxide For Water Purification - Purogene or Oxine	
G50140	Gloves - Protective, Latex or Nitrile	
G50318	Metallic Wool - Steel	

E. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

F. Access Panels

Number	Name/Location
146AR	Water Service Door

G. Prepare to Disinfect the Potable Water System

SUBTASK 38-10-00-680-001

WARNING DRAIN, OR USE THE POTABLE WATER SYSTEM A MINIMUM OF ONE TIME EACH THREE DAYS. IF YOU DO NOT DRAIN, OR USE THE WATER SYSTEM FREQUENTLY, IT CAN CAUSE BACTERIA IN THE WATER. IF YOU DRINK WATER WITH BACTERIA IN IT, ILLNESS CAN OCCUR.

(1) Do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

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SUBTASK 38-10-00-860-021

(2) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-10-00-020-001

(3) If installed, remove and discard the filter canisters from the water system in the galleys (TASK 25-31-00-960-801).

SUBTASK 38-10-00-210-001

(4) Make sure that the top of the housing is in its position without the filter canister.

SUBTASK 38-10-00-020-002

I (5) If installed, disconnect the coffee makers, steam ovens, and water boilers from the water system in the galleys and cap the water supply lines.

SUBTASK 38-10-00-860-006

(6) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

H. Potable Water System Disinfectant

SUBTASK 38-10-00-610-001

(1) Fill the potable water system approximately half full with water (TASK 12-14-00-600-802).

(a) Make sure that you remove the filter cartridges from the water service equipment and airplane's potable water system.

NOTE: If the cartridges were not replaced in the potable water system after the disinfecting, and the cartridges were not cleaned properly, contamination and bacteria can stay in the cartridge, and/or new bacteria can grow.

SUBTASK 38-10-00-670-001

(2) Make the disinfectant solution as follows:

NOTE: This disinfectant mixture will supply a solution that is 100 parts of chlorine dioxide for each one million parts of water. This mixture is calculated on the total water tank(s) capacity plus a quantity for waterlines.

(a) Purogene or Oxine compound, G00022 (stabilized 2%), and citric acid, B00637 (crystals or powder) (recommended disinfectant).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT BREATHE IN CHLORINE DIOXIDE GAS. MAKE SURE THAT YOU PUT ON A RESPIRATOR, AND GOGGLES. IF YOU BREATHE CHLORINE DIOXIDE GAS IT CAN CAUSE INJURIES TO PERSONNEL.

1) Mix the following amounts of Purogene or Oxine compound, G00022, with citric acid, B00637, in a clean plastic container:

NOTE: All measurements are volume measurements, and based on 22 gal (83 l) + 3 gal (11 l). To determine the actual amounts of each solution component for your tank capacity, use a scaling factor. The scaling factor can be calculated by dividing your total tank capacity by 25. The quantity of each solution component you require can be calculated by multiplying the individual amounts of each solution component with the scaling factor.

NOTE: Boeing uses the conversion factor of 29.574 to convert from fl-oz to ml. 1 ml equals 1 gram.

NOTE: Each gallon of tank capacity requires 0.64 fl-oz (19 ml) of Purogene or Oxine compound, G00022, activated by 0.08 fl-oz (2.37 ml) of citric acid, B00637 (crystals or powder).

NOTE: When you mix the Purogene solution, use sufficient airflow. You can use protective gloves, G50140, and a half-mask respirator, STD-739.

a) 16 fl-oz (473 ml) chlorine dioxide, Purogene or Oxine compound, G00022.

b) 2 fl-oz (59.15 ml) (59.15 grams) citric acid, B00637 (crystals or powder), or 3.25 fl-oz (96.11 ml) liquid citric acid, B00637 (50% solution).

(b) Use a clean instrument to mix the solution fully.

(c) Stop for 15 to 30 minutes.

(d) Add water to make approximately 5 gal (19 l) of solution.

SUBTASK 38-10-00-010-001

(3) Open this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-10-00-670-002

(4) Add the disinfectant solution to the potable water servicing equipment, COM-1141.

SUBTASK 38-10-00-670-003

(5) Put the disinfectant solution into the potable water system through the water fill connection at the water service panel.

SUBTASK 38-10-00-610-002

(6) Add water to the potable water system until the water tank is full (TASK 12-14-00-600-802).

SUBTASK 38-10-00-860-002

(7) Open each of the lavatory and galley faucets.

SUBTASK 38-10-00-160-002

(8) Purge each water supply line at the rail or quick-disconnect in the potable water system as follows:

(a) Use the bucket, STD-13465, and drain hose, STD-478, with the quick disconnect socket, COM-25776, or quick disconnect plug, COM-25777, to drain the water supply line for the galley equipment at the quick-disconnect port.

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(b) Let the water flow until you smell and see the disinfectant solution from each faucet.

NOTE: The disinfectant has a light yellow color.

(c) Close each faucet after you see the disinfectant flow.

(d) Remove the bucket, STD-13465, and drain hose, STD-478, with the quick disconnect socket, COM-25776, or quick disconnect plug, COM-25777, from the water supply line after you see the disinfectant flow.

SUBTASK 38-10-00-160-001

(9) Clean and disinfect the aerator assembly as follows:

NOTE: Optional to remove and install new aerator assembly.

(a) Put a plug in the sink.

(b) Remove the aerator assembly from the faucet assembly.

(c) Soak the aerator assembly in Lysol cleaner, B01033, or Glyco-San potable water system cleaner, B50246, for 30 minutes.

(d) If it is necessary, clean the aerator assembly with steel wool, G50318, or equivalent.

(e) Flush the aerator assembly with clean water.

(f) Install the aerator assembly to the faucet assembly.

(g) Remove the plug from the sink.

SUBTASK 38-10-00-170-001

(10) Flush each toilet a minimum of two times with a 15 second delay between flushes or until water with disinfectant appears.

SUBTASK 38-10-00-860-004

(11) Fill the potable water system with water until it is full (TASK 12-14-00-600-802).

SUBTASK 38-10-00-670-004

WARNING KEEP THE DISINFECTANT IN THE WATER FOR A MINIMUM OF ONE HOUR. LESS TIME WILL NOT KILL ALL OF THE BACTERIA. BACTERIA CAN CAUSE ILLNESS.

(12) Keep the disinfectant in the potable water between one and two hours.

SUBTASK 38-10-00-680-002

CAUTION DO NOT KEEP THE DISINFECTANT IN THE SYSTEM LONGER THAN TWO HOURS. CORROSION CAN OCCUR AFTER TWO HOURS AND CAUSE DAMAGE TO THE SYSTEM.

(13) Drain the potable water system (TASK 12-14-00-600-801).

SUBTASK 38-10-00-680-003

(14) Purge each lavatory and galley faucets as follows:

(a) Open each of the lavatory and galley faucets.

(b) Let the water drain from lavatory and galley supply lines.

(c) Close the faucets.

I. Flush the Potable Water System with Clean Water

SUBTASK 38-10-00-860-005

(1) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-10-00-860-007

(2) Purge each lavatory and galley faucets as follows:

(a) Open each of the lavatory and galley faucets.

(b) Let the water drain until clean water flows through each for one minute.

(c) Close the faucets.

SUBTASK 38-10-00-170-002

(3) Flush each toilet a minimum of two times with a 15 second delay between each flush or until clean water without the disinfectant appears.

SUBTASK 38-10-00-680-004

(4) If the water has a bad taste, a bad scent, or a color, do these steps:

(a) Do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

(b) Open each of the lavatory faucets, galley faucets, and water boiler drain plugs or valves (depends on water boiler configuration) until clean water flows through each for one minute.

(c) Flush each toilet a minimum of two times with a 15 second delay between each flush or until clean water without the disinfectant appears.

(d) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-10-00-860-008

(1) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-10-00-420-001

(2) If it is necessary, remove the water supply line caps and reconnect the coffee makers, steam ovens, and water boilers.

SUBTASK 38-10-00-420-002

(3) If removed, install new filter canisters in the water filters for the galleys (TASK 25-31-00-960-801).

SUBTASK 38-10-00-860-010

(4) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-10-00-860-009

(5) If it is necessary, do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-10-00-700-001

(6) Make sure that there is no leakage from applicable galley insert connections.

——— END OF TASK ———

TASK 38-10-00-790-801

3. Potable Water System - Leak Test

(Figure 201, Figure 202)

A. General

(1) Use this procedure to make sure the potable water system does not have a leak.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
38-31-00-910-801	Gray Water/Drain System - Maintenance Practice (P/B 201)
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1951	Drain Mast Plug - Waste Water Part #: C38001-29 Supplier: 81205 Opt Part #: C38001-23 Supplier: 81205
STD-77	Air Source, Regulated - Dry Filtered, 0-50 psig
STD-1091	Gauge - Air Pressure, 0-100 PSIG (0-690 KPa)

D. Consumable Materials

Reference	Description	Specification
D50011	Grease - Perfluoropolyether - Christo-lube MCG111	
D50063	Grease - Perfluoropolyether, fuel and oxygen resistant - Krytox 240AC	MIL-PRF- 27617 Type III

E. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right
200	Upper Half of Fuselage

F. Access Panels

Number	Name/Location
822	Aft Cargo Door

G. Potable Water System - Leakage Test

SUBTASK 38-10-00-860-012

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 38-10-00-010-002

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-10-00-010-007

(3) Remove the electrical equipment rack, E6, aft access panel.

SUBTASK 38-10-00-020-004

(4) Remove the cap from the pressure test equipment connection tee fitting in the potable water line (Figure 201).

SUBTASK 38-10-00-480-001

(5) Install the pressure gauge (0-100 PSIG) (0-690 KPa), STD-1091, in the pressure test equipment connection tee fitting of the potable water line (Figure 201).

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SUBTASK 38-10-00-610-003

(6) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-10-00-480-002

(7) Install the regulated air source, STD-77, in the pressure test equipment connection tee fitting of the potable water line to pressurize the potable water system to 50 psig (344.7 kPa) (Figure 201).

SUBTASK 38-10-00-860-013

(8) Pressurize the potable water system to 50 psig (344.7 kPa).

SUBTASK 38-10-00-860-014

(9) Open the faucets in each lavatory and galley and let the water flow for 2 minutes.

(a) Keep the pressure at 50 psig (344.7 kPa) (in the potable water system).

SUBTASK 38-10-00-860-015

(10) Close all the water faucets.

SUBTASK 38-10-00-860-016

(11) Keep the pressure at 50 psig (344.7 kPa) for at least 5 minutes.

SUBTASK 38-10-00-790-001

(12) Make sure that there are no air or water leaks in these locations:

(a) At the water tank.

(b) At the water fill connection, on the water service panel.

(c) At the water drain connections, on the airplane skin.

(d) In the water tubes and hoses that supply water to the lavatories and galleys.

SUBTASK 38-10-00-860-017

(13) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-10-00-080-001

(14) Remove the pressure gage from the potable water line.

SUBTASK 38-10-00-420-003

(15) Install the cap on the potable water line.

WARNING

KRYTOX 240AC IS AN AGENT THAT IS POISONOUS AND AN IRRITANT. MAKE SURE ALL PERSONS OBEY THE PRECAUTIONS WHEN KRYTOX 240AC IS USED.

- DO NOT USE IN AREAS WHERE THERE IS HIGH HEAT, SPARKS, OR FLAMES.
- CLOSE THE CONTAINER WHEN NOT USED.
- DO NOT GET KRYTOX 240AC IN THE EYES, ON THE SKIN, OR ON YOUR CLOTHES.
- DO NOT BREATHE THE GAS.
- DO NOT EAT KRYTOX 240AC.

(a) Apply a thin layer Krytox 240AC perfluoropolyether grease, D50063, or Christo-lube MCG111 grease, D50011, to the external threads of the potable water line.

(b) Install the cap.

SUBTASK 38-10-00-860-018

(16) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Forward Gray Water System - Leakage Test

SUBTASK 38-10-00-480-003

(1) Install the waste water drain mast plug, SPL-1951, to seal the forward drain mast.

SUBTASK 38-10-00-860-019

(2) Put water in the most forward galley or the lavatory sink until the drain line is full of water.

(a) Locate the most forward sink (TASK 38-31-00-910-801).

NOTE: The drain line is full of water when water stays in the sink for five minutes.

NOTE: When the airplane is nose down, it is important to use the most forward sink to prevent possible backflows or spills from a more forward sink.

SUBTASK 38-10-00-790-002

(3) Make sure that the drain lines connected to the forward drain mast have no leaks.

SUBTASK 38-10-00-080-002

(4) Remove the waste water drain mast plug, SPL-1951, from the forward drain mast to let the water drain.

I. Aft Gray Water System - Leakage Test

SUBTASK 38-10-00-480-004

(1) Install the waste water drain mast plug, SPL-1951, to seal the aft drain mast.

SUBTASK 38-10-00-860-020

(2) Put water in the most forward galley or lavatory sink in the aft waste water system until the drain line is full of water.

(a) Locate the most forward sink (TASK 38-31-00-910-801).

NOTE: The drain line is full of water when water stays in the sink for five minutes.

NOTE: When the airplane is nose down, it is important to use the most forward sink to prevent possible backflows or spills from a more forward sink.

SUBTASK 38-10-00-790-003

(3) Make sure that the drain lines connected to the aft drain mast have no leaks.

SUBTASK 38-10-00-080-003

(4) Remove the waste water drain mast plug, SPL-1951, from the aft drain mast to let the water drain.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-10-00-410-009

(1) Install the electrical equipment rack, E6, aft access panel.

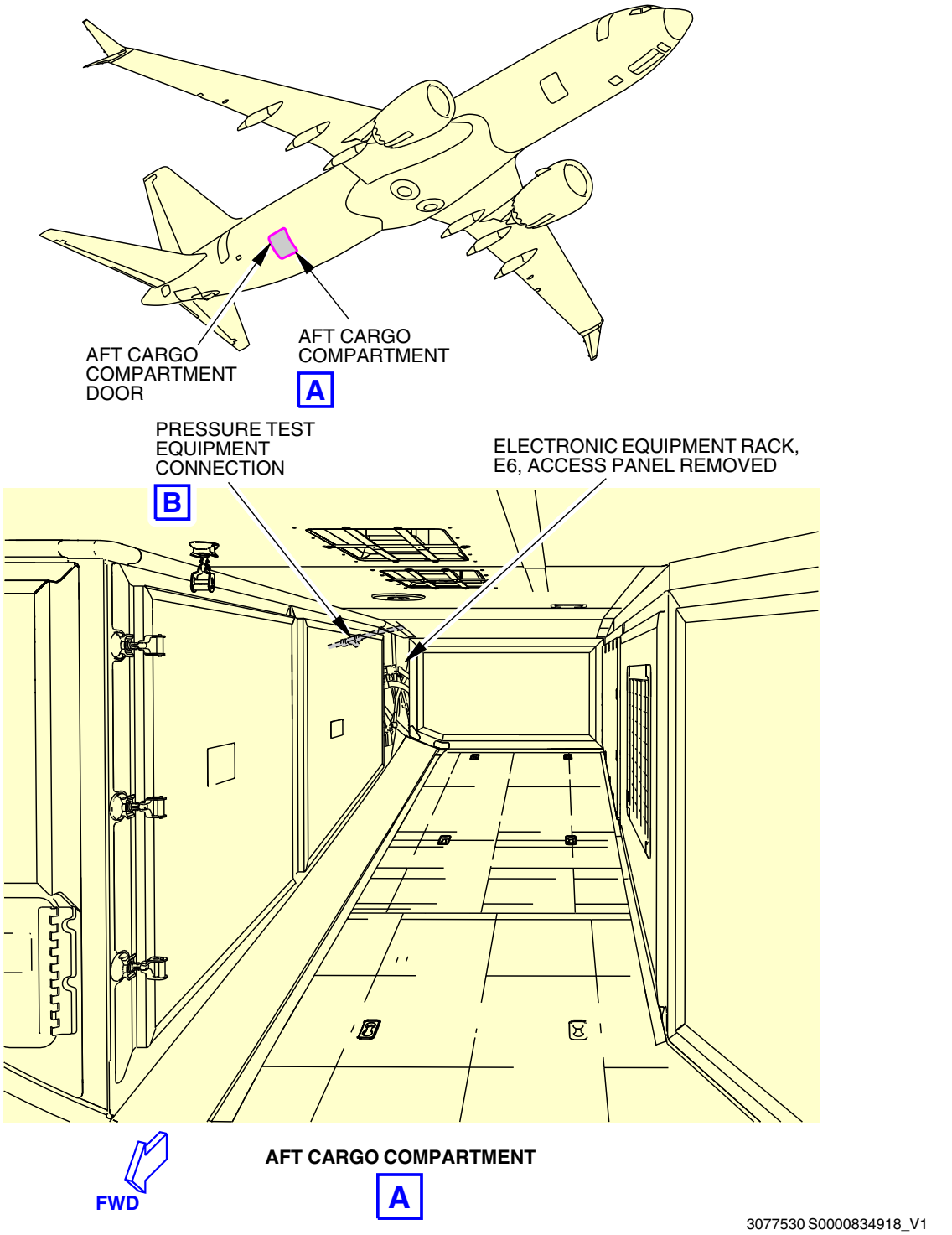
SUBTASK 38-10-00-410-002

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

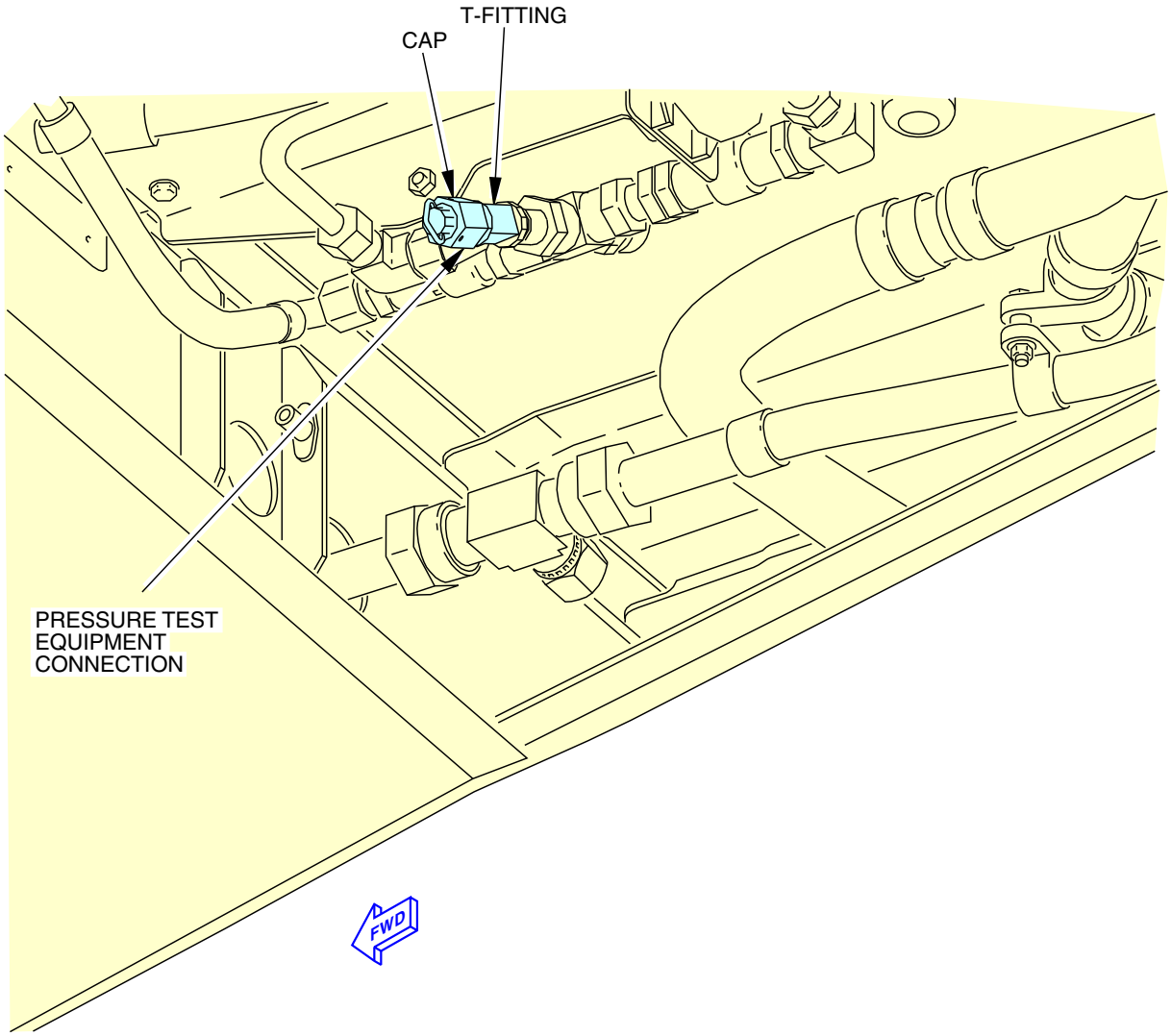
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Potable Water System Leak Test
Figure 201/38-10-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



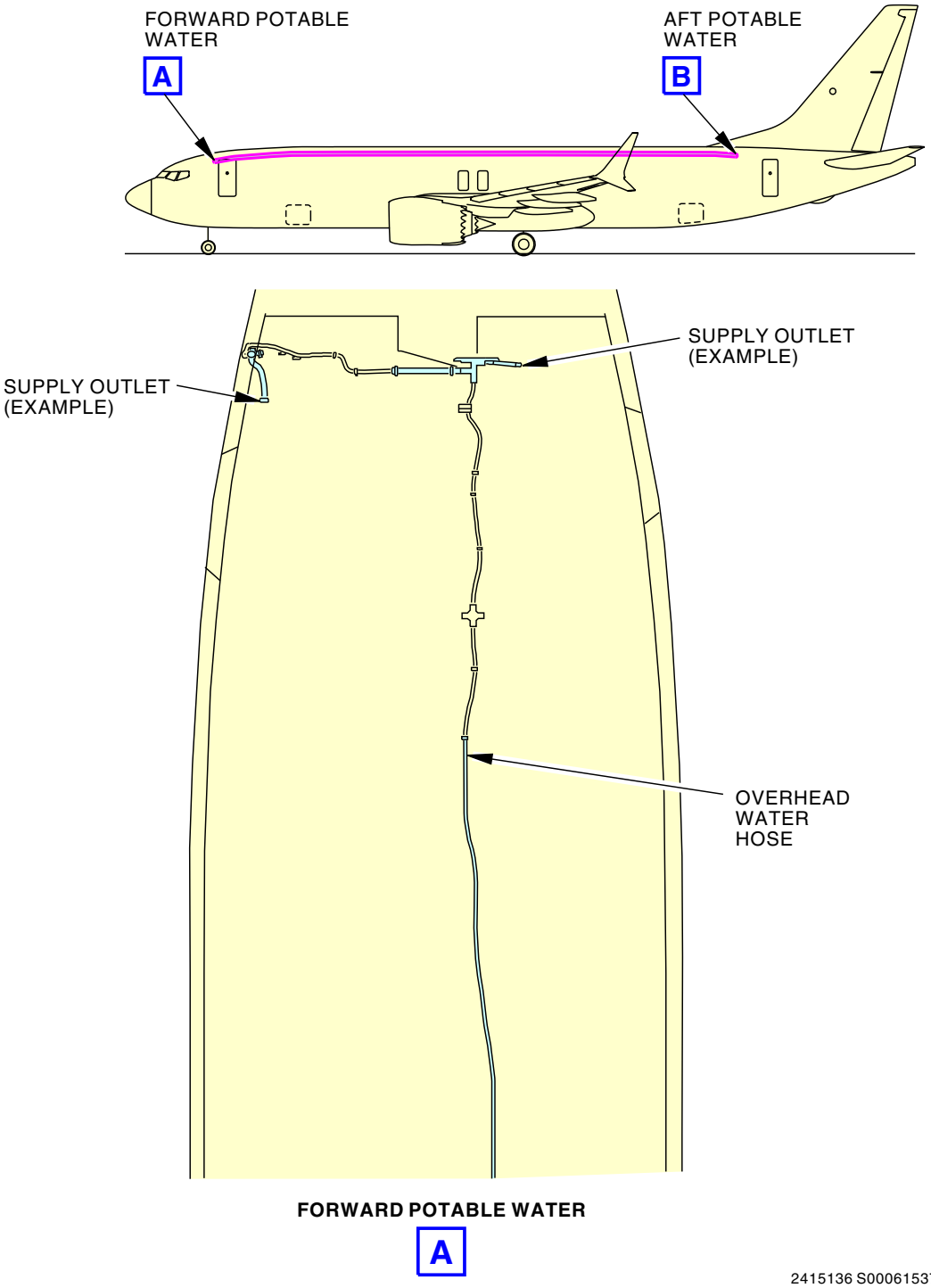
PRESSURE TEST EQUIPMENT CONNECTION

B

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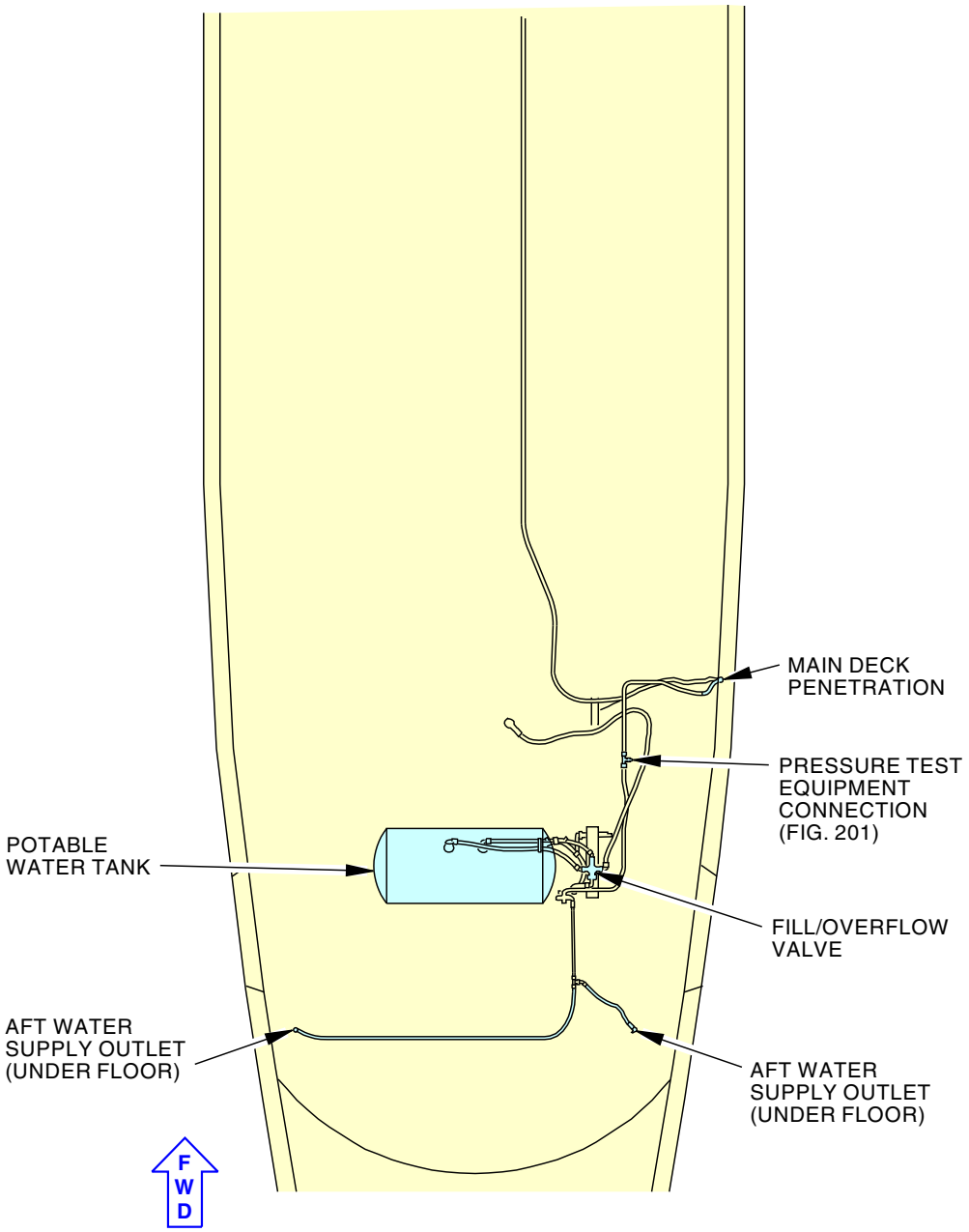
Potable Water System Leak Test
Figure 201/38-10-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Potable Water System - Primary Routing
Figure 202/38-10-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Potable Water System - Primary Routing
Figure 202/38-10-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

POTABLE WATER SYSTEM - CLEANING

1. General

A. This procedure has these tasks:

(1) Manually clean the potable water tank.

TASK 38-10-00-100-801

2. Manually Clean the Potable Water Tank

(Figure 701)

A. General

(1) This task gives the instructions to manual clean the potable water tank.

(2) This should only be used when the disinfecting procedure is not effective.

(3) For the normal process for maintenance of the water system is refer to portable water system disinfectant (TASK 38-10-00-600-801).

B. References

Reference	Title
12-14-00-600-801	Potable Water System - Drain (P/B 301)
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
38-10-00-600-801	Potable Water System Disinfectant (P/B 201)
38-11-01-000-801	Water Tank Removal (P/B 401)
38-11-01-400-801	Water Tank Installation (P/B 401)
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4911	Rotation Tool - Potable Water Tank End Cap Part #: 2MIT2A264-0812-1 Supplier: 1TGN3 Part #: D78T15348-1 Supplier: 1TGN3
STD-123	Brush - Soft Bristle

D. Consumable Materials

Reference	Description	Specification
B50026	Soap - Clear Liquid Dishwashing	

E. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
822	Aft Cargo Door

G. Prepare to Clean the Water Tank

SUBTASK 38-10-00-840-001

(1) Do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

SUBTASK 38-10-00-840-002

(2) To release the pressure from the potable water system, do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-10-00-020-005

(3) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-10-00-020-009

(4) To remove the end wall in the aft cargo compartment, do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

SUBTASK 38-10-00-020-010

(5) If it is necessary, remove the electrical equipment rack, E6, aft access panel.

SUBTASK 38-10-00-020-006

(6) If it is necessary remove the potable water tank, do this task: Water Tank Removal, TASK 38-11-01-000-801.

SUBTASK 38-10-00-010-003

(7) Do these steps to gain access to the inside of the water tank:

WARNING DEPRESSURIZE WATER TANK BEFORE OPENING TANK END CAP. IF THE TANK END CAP SNAP RING IS REMOVED PRIOR TO TANK DEPRESSURIZATION, THE TANK END CAP CAN INJURE MAINTENANCE PERSONEL.

(a) Remove the insulation on the end of the water tank.

(b) Remove lockwire (if installed).

(c) Put the rotation tool, COM-4911, in its position on the end cap(s) and then remove the end cap(s).

(d) Pull the end caps and o-rings from the tank.

H. Clean the Water Tank

SUBTASK 38-10-00-110-001

(1) Do these steps to clean the inside of the water tank.

(a) Mix 1 gal (4 l) of water and 3 fl-oz (89 ml) of soap, B50026, in a 2 gal (8 l) plastic container.

(b) Put the mixture into the water tank.

(c) Clean the inside of the tank and covers with a long handled, soft bristle brush, STD-123.

(d) If the water tank has been removed, manually drain the tank into a 5 gal (19 l) plastic container.

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(e) If the water tank has not been removed, do this task: Potable Water System - Drain, TASK 12-14-00-600-801

SUBTASK 38-10-00-680-005

(2) Do these steps to flush the inside of the water tank.

(a) Flush the inside of the tank with water.

(b) If the water tank has been removed, manually drain the tank into a 5 gal (19 l) plastic container.

(c) If the water tank has not been removed, do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

(d) Do the step above three times.

(e) Wash the tank cover(s) with clean water.

SUBTASK 38-10-00-420-004

(3) Do these steps to install the cover on the water tank.

(a) Make sure that the o-ring is not dislodged, pinched or damaged during assembly.

(b) Put the rotation tool, COM-4911, in its position on the end cap(s) and install the covers in the tank so that the spherical surface is inside the tank.

(c) Tighten the end cap(s) to 600 in-lb (68 N·m) - 800 in-lb (90 N·m).

(d) To install the lockwire, do this task: Lockwire, Cotter Pins, and Lockrings - Installation, TASK 20-10-44-400-801.

SUBTASK 38-10-00-420-005

(4) Make sure that the drain valve is closed.

SUBTASK 38-10-00-420-006

(5) If it is necessary, do this task: Water Tank Installation, TASK 38-11-01-400-801.

SUBTASK 38-10-00-110-002

(6) Do this task: Potable Water System Disinfectant, TASK 38-10-00-600-801.

SUBTASK 38-10-00-680-006

(7) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-10-00-840-003

(8) To restore pressure to the potable water system, do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-10-00-790-004

(9) Examine all the connections for leakage.

(a) Make sure that there is no leakage.

SUBTASK 38-10-00-420-007

(10) Install the insulation cover on the water tank.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-10-00-410-003

WARNING SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 38-10-00-410-010

(2) If removed, install the electrical equipment rack, E6, aft access panel.

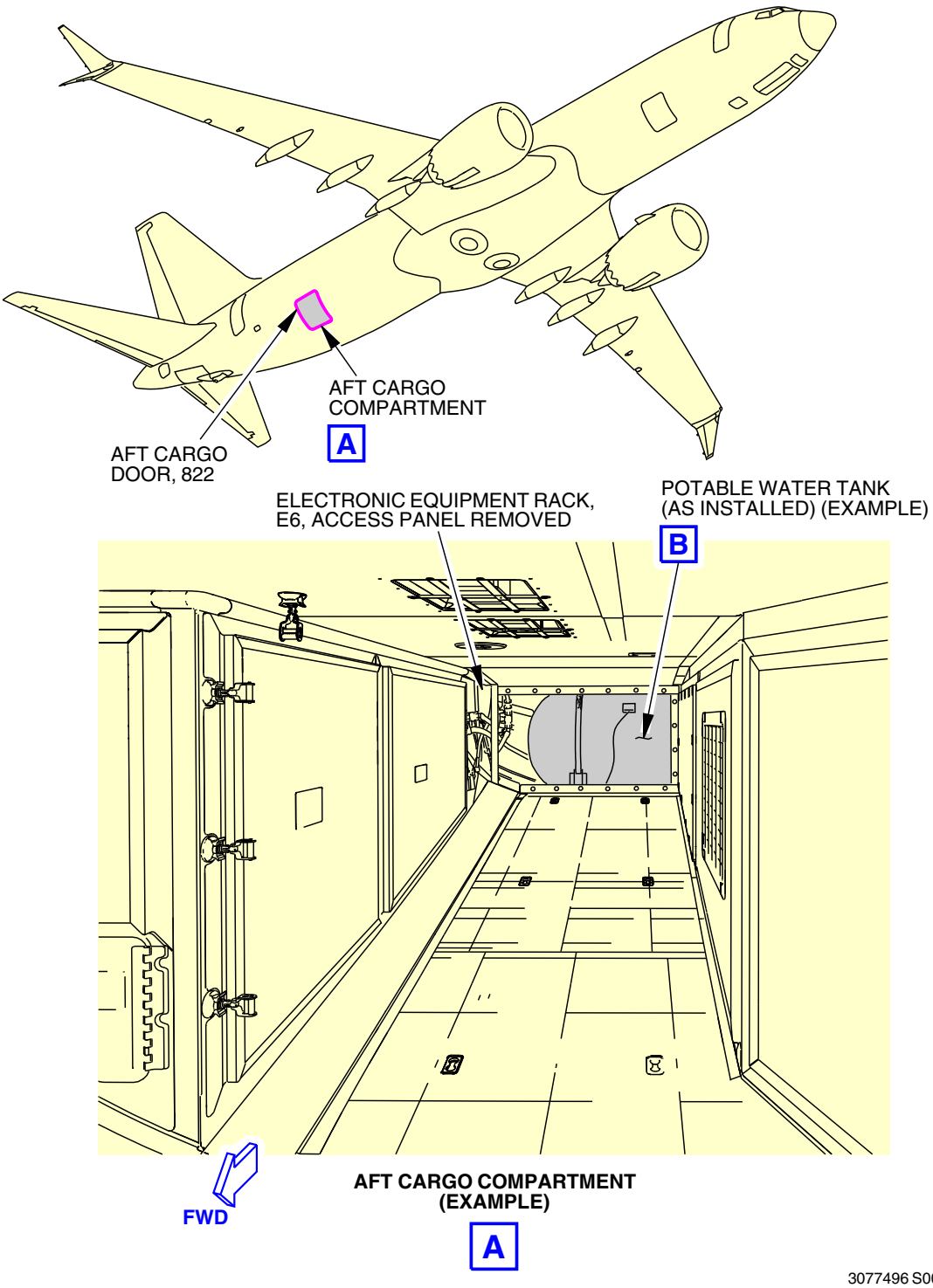
SUBTASK 38-10-00-410-004

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

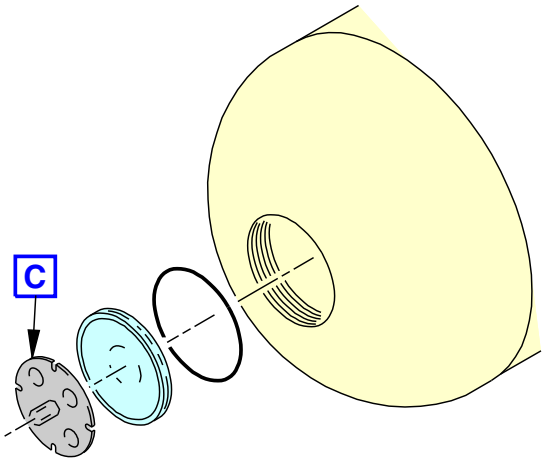
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



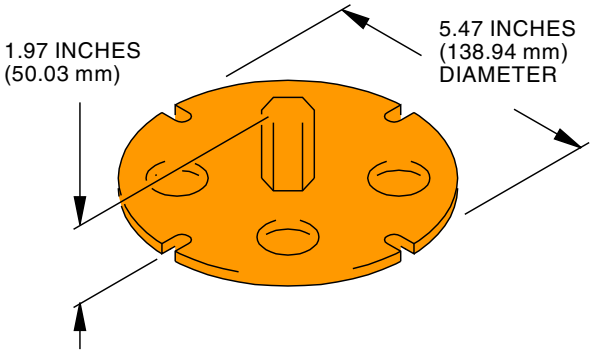
Water Tank Cleaning
Figure 701/38-10-00-990-803 (Sheet 1 of 3)

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WATER TANK END CAP

B



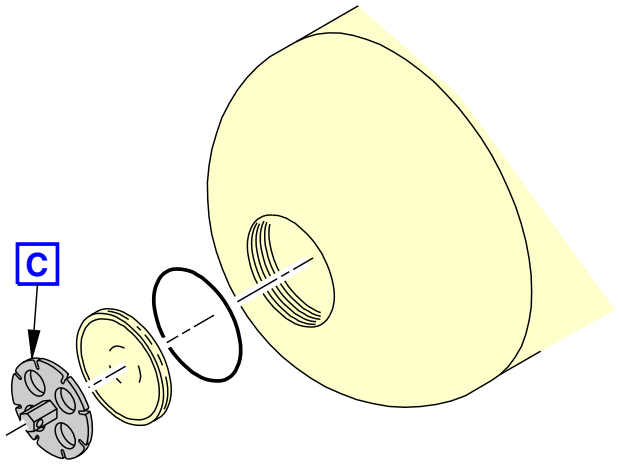
SPECIAL TOOL

C

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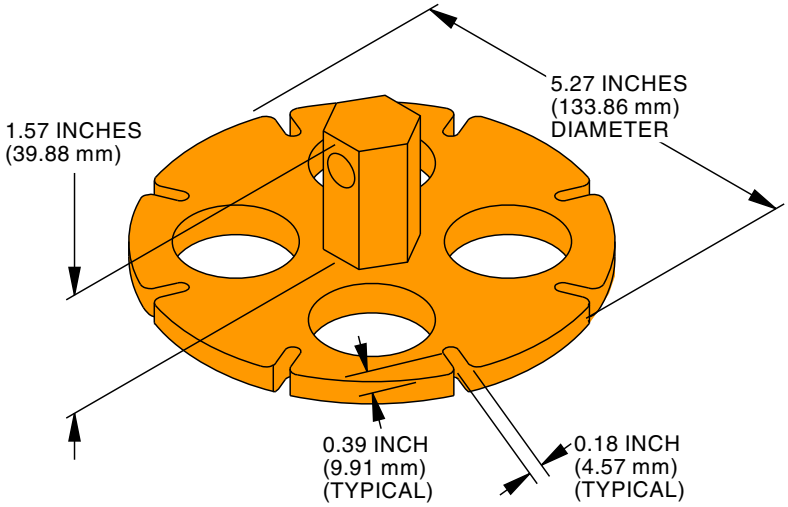
Water Tank Cleaning
Figure 701/38-10-00-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WATER TANK END CAP

B



SPECIAL TOOL

C

2767977 S0000629069_V1

Water Tank Cleaning
Figure 701/38-10-00-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-10-00-100-802

3. Descale Cleaning of the Potable Water System

A. General

(1) The intent of this task is to provide a descale cleaning solution to the potable water system. Glyco-San is an aggressive cleaner and disinfectant, for use when mineral scale is present. Boeing does not recommend general or frequent use as a disinfectant.

(2) Initial soak is recommended when descale cleaning has not been performed in over 360 days.

(3) Periodic soak is recommended when descale cleaning is performed every 180 to 360 days.

(4) It may take several flush steps to completely remove the cleaner from the system.

B. References

Reference	Title
12-14-00-600-801	Potable Water System - Drain (P/B 301)
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1141	Equipment - Potable Water Servicing Part #: 19-4301-0000 Supplier: 59603 Part #: 19-4303-0000 Supplier: 59603 Part #: 2198-310 Supplier: 6L481 Part #: CPW-100 Supplier: 32526 Part #: CPW-100-1 Supplier: 32526 Part #: CPW-100-2 Supplier: 32526 Part #: PW100 Supplier: 1CSJ2 Part #: SPW-350 Supplier: 32526 Part #: SPW-500PL Supplier: 32526 Part #: WC100E (ID 2026505) Supplier: 0T652 Part #: WSP-900 Supplier: 6L481 Part #: WTM-900 Supplier: 6L481 Opt Part #: 2303 Supplier: 00365 Opt Part #: SPW-250 Supplier: 32526

D. Consumable Materials

Reference	Description	Specification
B50246	Cleaner - Glyco-San Potable Water System Cleaner	AMS 1550B

E. Access Panels

Number	Name/Location
146AR	Water Service Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare to Descale the Potable Water System

SUBTASK 38-10-00-680-007

WARNING DRAIN, OR USE THE POTABLE WATER SYSTEM A MINIMUM OF ONE TIME EACH THREE DAYS. IF YOU DO NOT DRAIN, OR USE THE WATER SYSTEM FREQUENTLY, BACTERIA CAN GROW IN THE WATER. IF YOU DRINK WATER WITH BACTERIA IN IT, ILLNESS CAN OCCUR.

(1) Drain the water tanks. This is the task: Potable Water System - Drain, TASK 12-14-00-600-801.

SUBTASK 38-10-00-020-007

(2) To remove the filter canisters (if installed) from the water system in the galleys, refer to the galley manufacturers instructions.

NOTE: Make sure the top of the housing is in its position without the filter canister installed.

G. Descale the Potable Water System

SUBTASK 38-10-00-110-003

(1) Make the cleaning solution as follows:

(a) Mix 20 gal (76 l) of Glyco-San potable water system cleaner, B50246 with 60 gal (227 l) of water for the tank.

SUBTASK 38-10-00-010-004

(2) Open this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-10-00-110-004

(3) Add the cleaning solution to the potable water servicing equipment, COM-1141 for the potable water system.

NOTE: The filter cartridge must not be installed in the water service equipment.

SUBTASK 38-10-00-110-005

(4) Add the cleaning solution to the potable water system:

(a) To fill the tanks, do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-10-00-110-006

(5) Let the potable water flow for one (1) minute or until the cleaning solution is visible, do this step:

NOTE: The cleaning solution may be noticeable by its slight yellow and foamy appearance.

(a) Open all the lavatory faucets (hot and cold), galley faucets, water boiler valves, and coffee maker valves (as equipped) in the potable water system.

SUBTASK 38-10-00-110-007

(6) After one (1) minute of water flow or if the cleaning solution is visible, do this step:

(a) Close each faucet and valve.

SUBTASK 38-10-00-110-008

(7) Flush each toilet at least twice with a 15 second delay between flushes or until cleaning solution appears.

NOTE: The cleaning solution may be noticeable by its slight yellow and foamy appearance.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-10-00-110-009

(8) For initial soak, let the cleaning solution soak for a minimum of 6 hours

NOTE: Initial soak is recommended when descale cleaning has not been performed in over 360 days.

SUBTASK 38-10-00-110-010

(9) For periodic soak, let the cleaning solution soak for a minimum of 4 hours.

NOTE: Periodic soak is recommended when descale cleaning is performed every 180 to 360 days.

SUBTASK 38-10-00-170-003

(10) After the soak is complete, do these steps:

(a) Open each of the galley faucets, water boiler valves, and coffee maker valves (as equipped) to let the water in the galley supply lines, water boilers, and coffee makers drain.

(b) Do this task: Potable Water System - Drain, TASK 12-14-00-600-801

(c) Close each of the galley faucets, water boiler valves, and coffee maker valves (as equipped).

SUBTASK 38-10-00-170-004

(11) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802

SUBTASK 38-10-00-170-005

(12) Let the potable water flow for one (1) minute or until the cleaning solution is no longer visible, do this step:

NOTE: The cleaning solution may be noticeable by its slight yellow and foamy appearance.

(a) Open each lavatory faucet, galley faucet, water boiler valve, and coffee maker valve.

SUBTASK 38-10-00-170-006

(13) After one (1) minute of water flow or if the cleaning solution is no longer visible, do this step:

(a) Close the faucets, galley water boiler valves, and coffee maker valves.

SUBTASK 38-10-00-170-007

(14) Flush each toilet at least twice with a 15 second delay between flushes or until water without the cleaning solution appears.

NOTE: The cleaning solution may be noticeable by its slight yellow and foamy appearance.

SUBTASK 38-10-00-170-008

(15) Flush the potable water system again, do this task: Potable Water System - Drain, TASK 12-14-00-600-801

SUBTASK 38-10-00-170-009

(16) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802

SUBTASK 38-10-00-170-010

(17) Let the potable water flow for one (1) minute, do this step:

(a) Open each lavatory faucet, galley faucet, water boiler valve, and coffee maker valve (as equipped).

SUBTASK 38-10-00-170-011

(18) After one (1) minute of water flow, do this step :

(a) Close the faucets, galley water boiler valves, and coffee maker valves.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-10-00-170-012

(19) Flush each toilet at least twice with a 15 second delay between flushes.

H. Potable Water System Check

SUBTASK 38-10-00-700-002

(1) Check the potable water, do these steps:

(a) Open one of the galley or lavatory faucets. Half fill a glass container.

(b) Make sure that the water is clear.

1) If the water is clear, continue the procedure.

2) If the water is not clear, do the tasks to drain, fill and flush the system with clean water.

(c) With the cap on the container, shake the container.

(d) Examine the water for foam or bubbles.

1) If no foam or bubbles the system is flushed.

2) If foam or bubbles are visible, do the tasks to drain, fill and flush the system with clean water.

I. Put the Airplane Back to its Usual Condition

SUBTASK 38-10-00-420-008

(1) Install the new galley filter canisters (if equipped).

SUBTASK 38-10-00-790-005

(2) Make sure that there are no leaks in any filter canisters (if installed) or galley water units.

SUBTASK 38-10-00-410-005

(3) Close this access panel:

Number	Name/Location
146AR	Water Service Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER TANK - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the water tank

(2) An installation of the water tank.

B. In this procedure, the main water tank shall be referred to as the water tank.

TASK 38-11-01-000-801

2. Water Tank Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the water tank.

B. References

Reference	Title
12-14-00-600-801	Potable Water System - Drain (P/B 301)
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
38-42-07-000-801	Water Pressure Cut-Out Switch Removal (P/B 201)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-11-01-860-001

(1) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-11-01-680-001

(2) Do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

SUBTASK 38-11-01-860-006

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	13	C00104	LAVATORY WATER HEATER A

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(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	14	C01073	LAVATORY WATER HEATER D
F	15	C01096	LAVATORY WATER HEATER E

SUBTASK 38-11-01-010-003

(4) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-11-01-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-11-01-010-001

(6) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-01-010-002

(7) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

SUBTASK 38-11-01-010-004

(8) Remove the electrical equipment rack, E6, aft access panel.

SUBTASK 38-11-01-020-001

(9) Do these steps to get access to the water tank [1].

(a) Remove the screw [3], screw [4], and screw [5] to disconnect the support assembly [2].

(b) Remove the screw [4] to disconnect the stanchion assembly [6].

(c) Pull the pull pin [8] to disconnect the stanchion assembly [7].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Water Tank Removal

SUBTASK 38-11-01-020-002

(1) Disconnect the water quantity transmitter as follows:

(a) Remove the screws [16] and washers [17] that attach the cover [15] to the water tank [1].

(b) Remove the screw [18] and washer [19] that attach the connector [20] to the terminal.

SUBTASK 38-11-01-020-003

CAUTION MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(2) Use a teflon-jawed pliers, STD-664, to disconnect the electrical connector [12] from the pressure switch [11] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [12] from the pressure switch [11].

(b) Install the protective covers on the electrical receptacle of the pressure switch [11] and electrical connector [12].

SUBTASK 38-11-01-020-004

(3) To remove the pressure switch [11], do this task: Water Pressure Cut-Out Switch Removal, TASK 38-42-07-000-801.

SUBTASK 38-11-01-020-005

(4) Disconnect the pressurization connection [9] from the top of the water tank [1].

SUBTASK 38-11-01-020-016

(5) Remove the union [44] and packing [45] from the water tank [1] fitting.

(a) Discard the old packing [45].

SUBTASK 38-11-01-020-006

(6) Remove the clamshells [13] and packings [14] to disconnect the water hoses from the top of the water tank [1].

SUBTASK 38-11-01-020-014

(7) If necessary, remove the tank fitting [26] and packing [27].

SUBTASK 38-11-01-020-007

(8) Remove the clamshell [34], sleeve [35], and packings [36] to disconnect the drain fitting [37] from the tank fitting [33].

SUBTASK 38-11-01-020-015

(9) If necessary, remove the tank fitting [33], washer [42], and packing [43].

SUBTASK 38-11-01-020-008

(10) Remove the nut [38], washers [39], and screw [40] to disconnect the ground strap [41] from the water tank [1].

SUBTASK 38-11-01-860-003

(11) Remove the bolt [21], washer [22], washer [23], and nut [24] to disconnect the tie rod assembly [25].

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SUBTASK 38-11-01-860-004

(12) Disconnect the strap assemblies [10] that hold the water tank [1] in its position.

SUBTASK 38-11-01-020-009

(13) Remove the bolts [29], washers [30], and nuts [31] to disconnect the forward mounts [32].

(a) Remove the right side forward mount [32].

(b) Remove the left side forward mount [32].

SUBTASK 38-11-01-020-010

(14) If it is necessary to prevent damage, remove the insulation blankets from the water tank [1].

SUBTASK 38-11-01-020-011

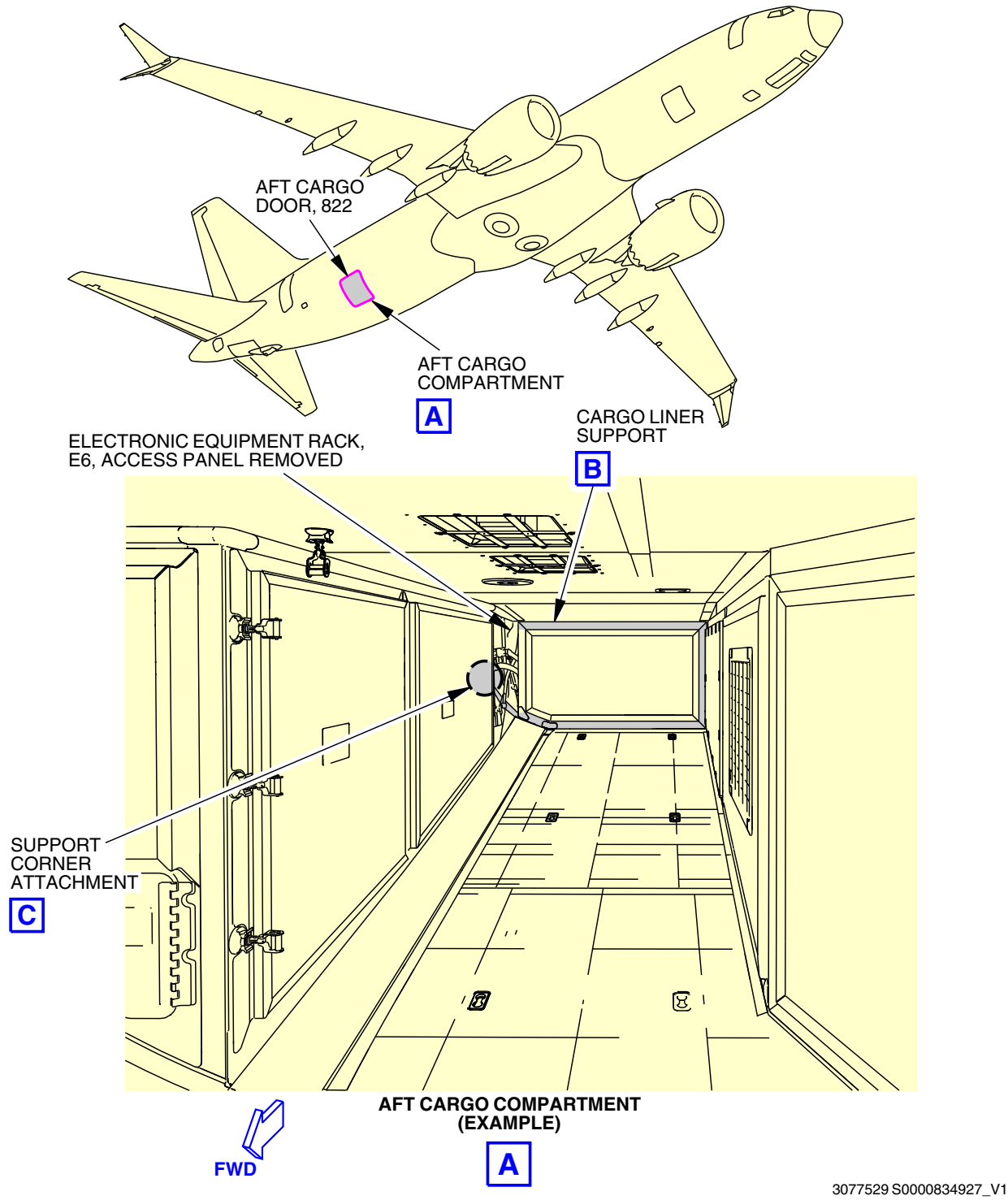
(15) Remove the water tank [1].

SUBTASK 38-11-01-020-013

(16) Install the protective covers on all open connections.

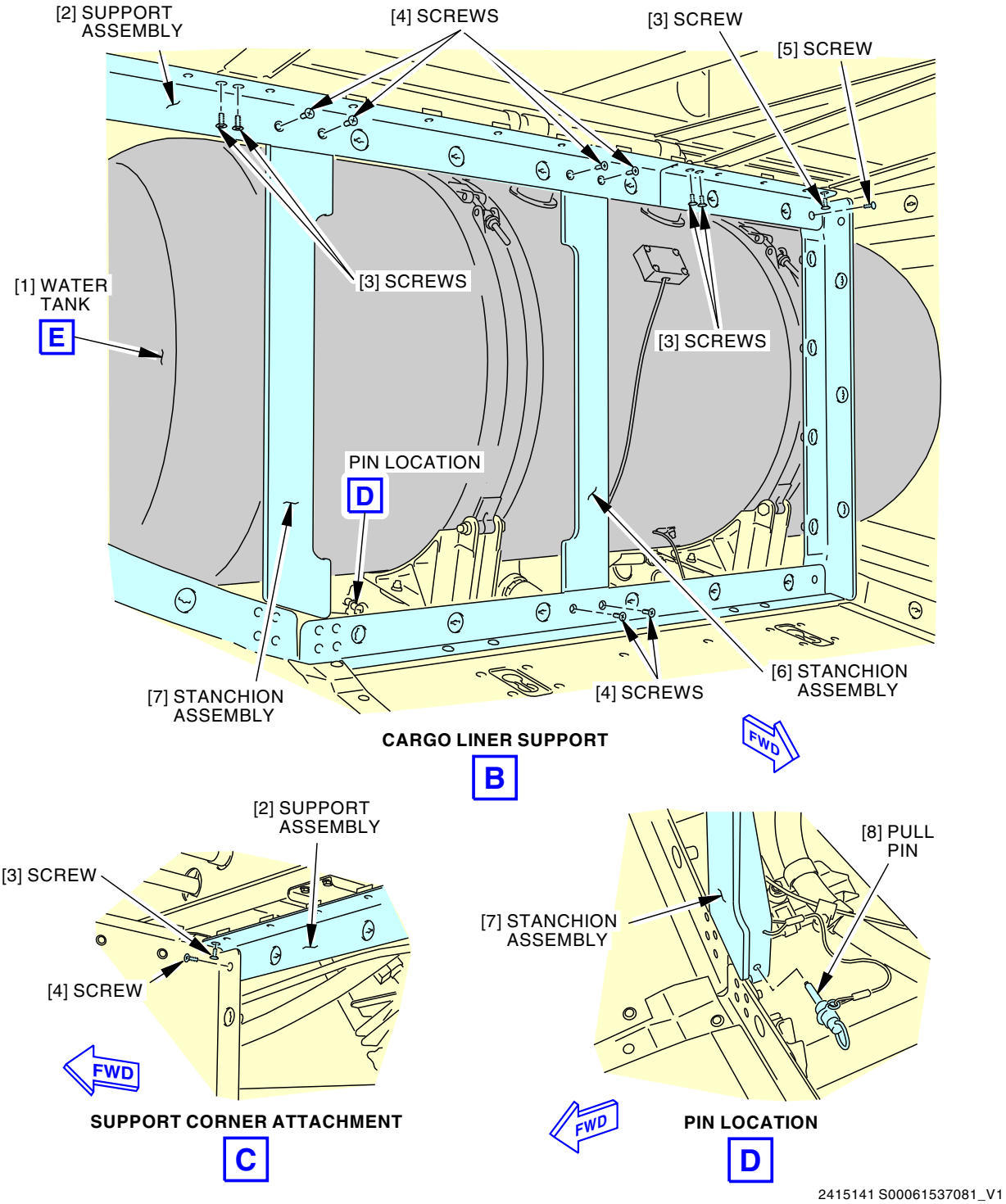
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



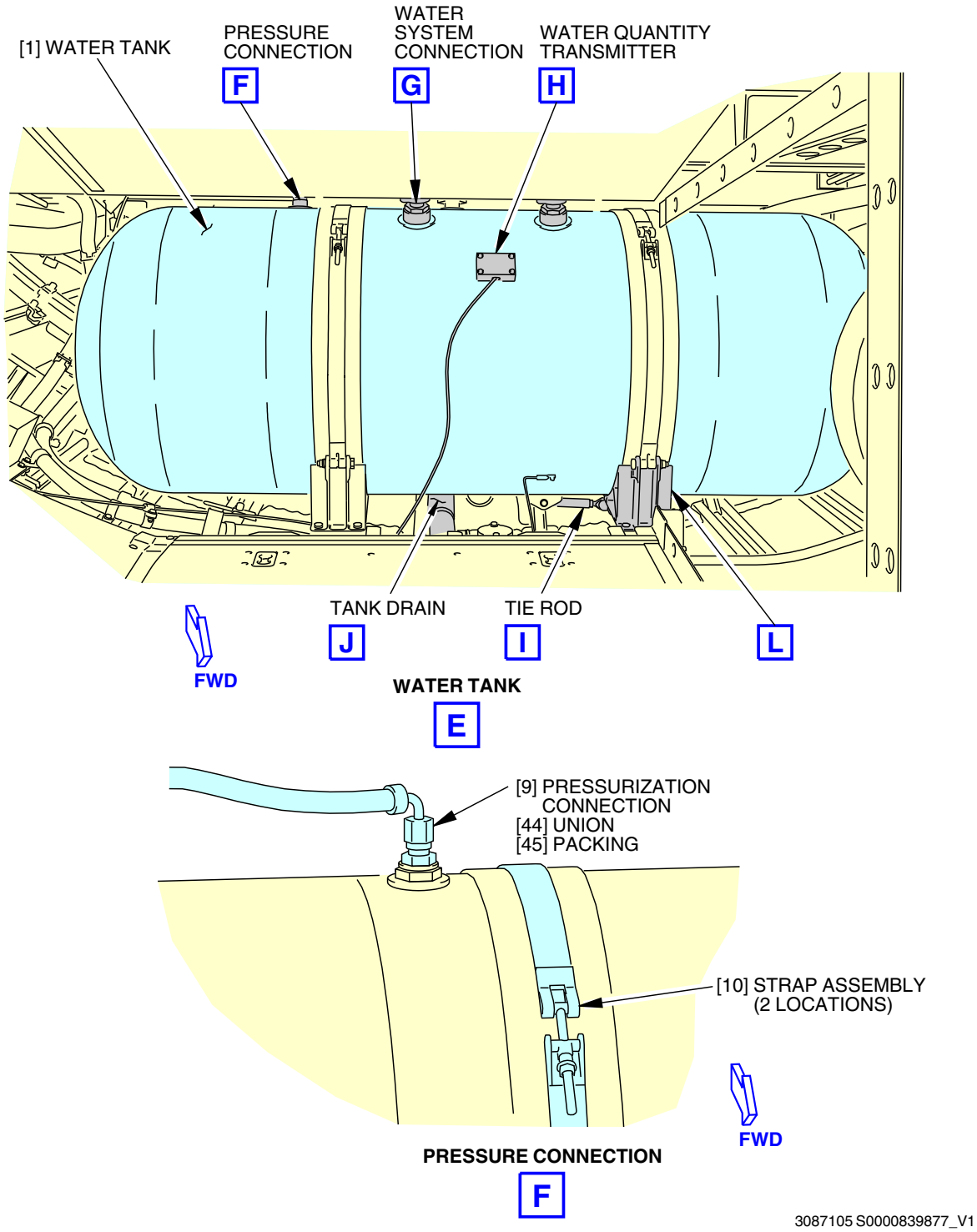
Potable Water Tank Installation
Figure 401/38-11-01-990-802 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



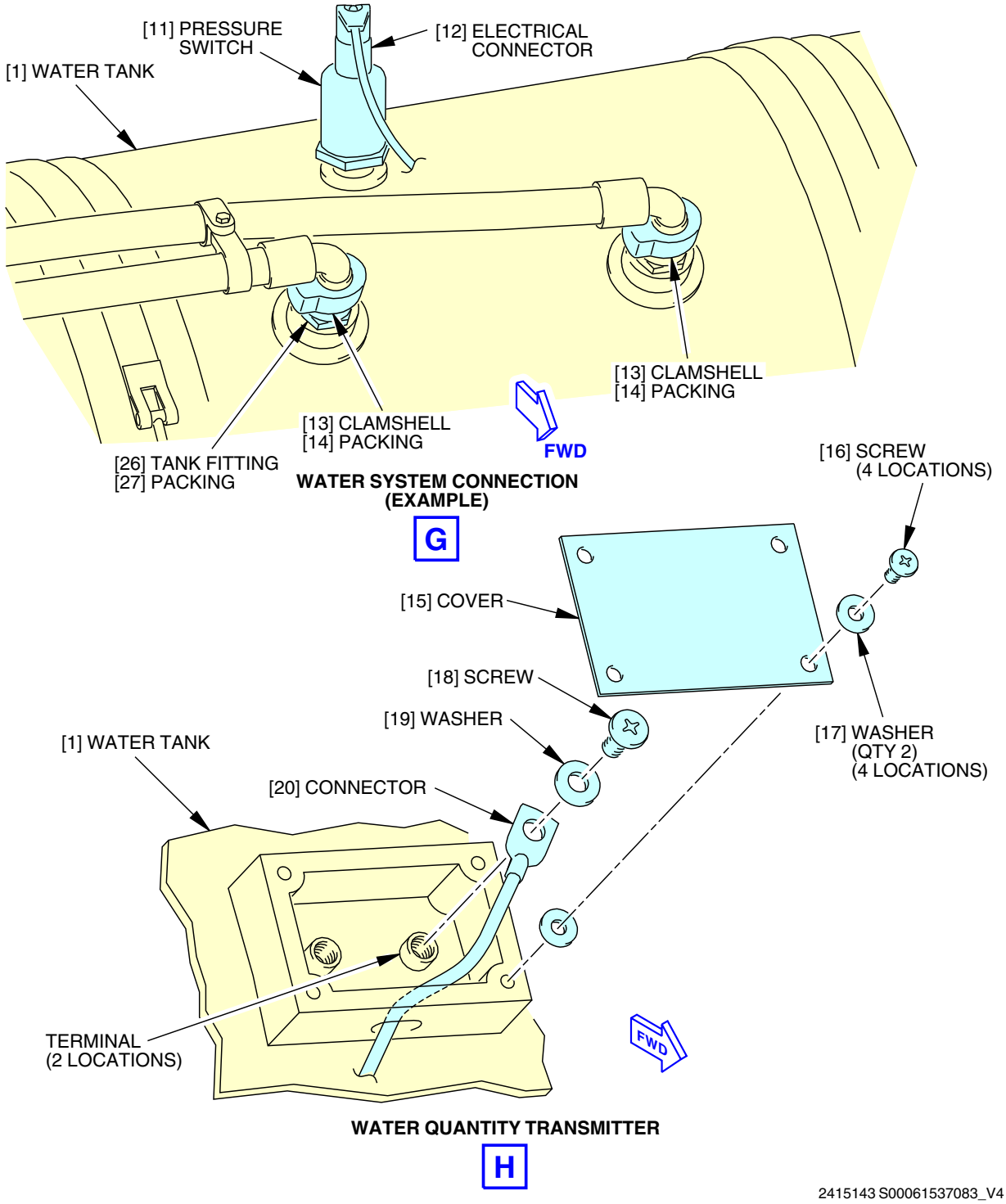
Potable Water Tank Installation
Figure 401/38-11-01-990-802 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



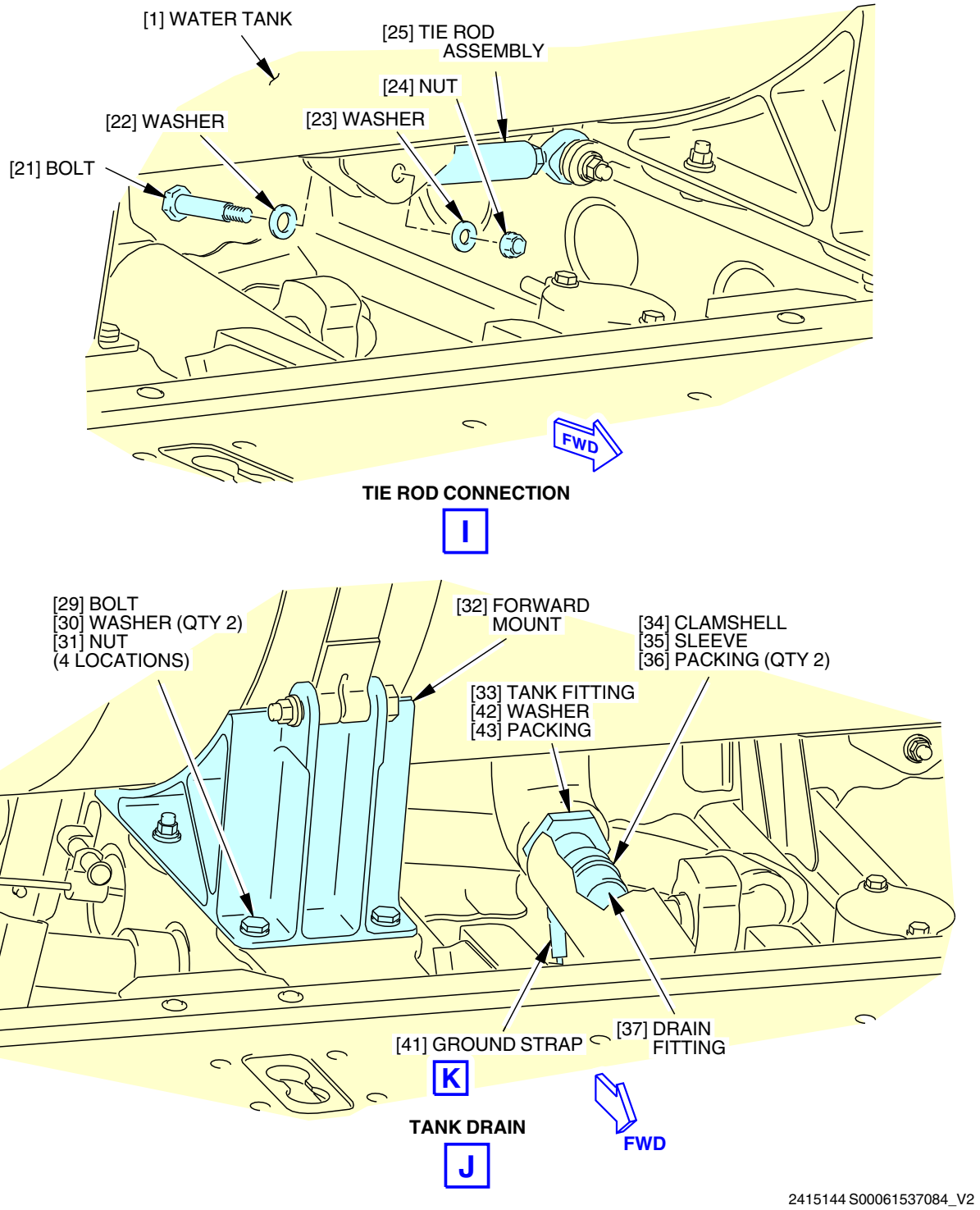
Potable Water Tank Installation
Figure 401/38-11-01-990-802 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



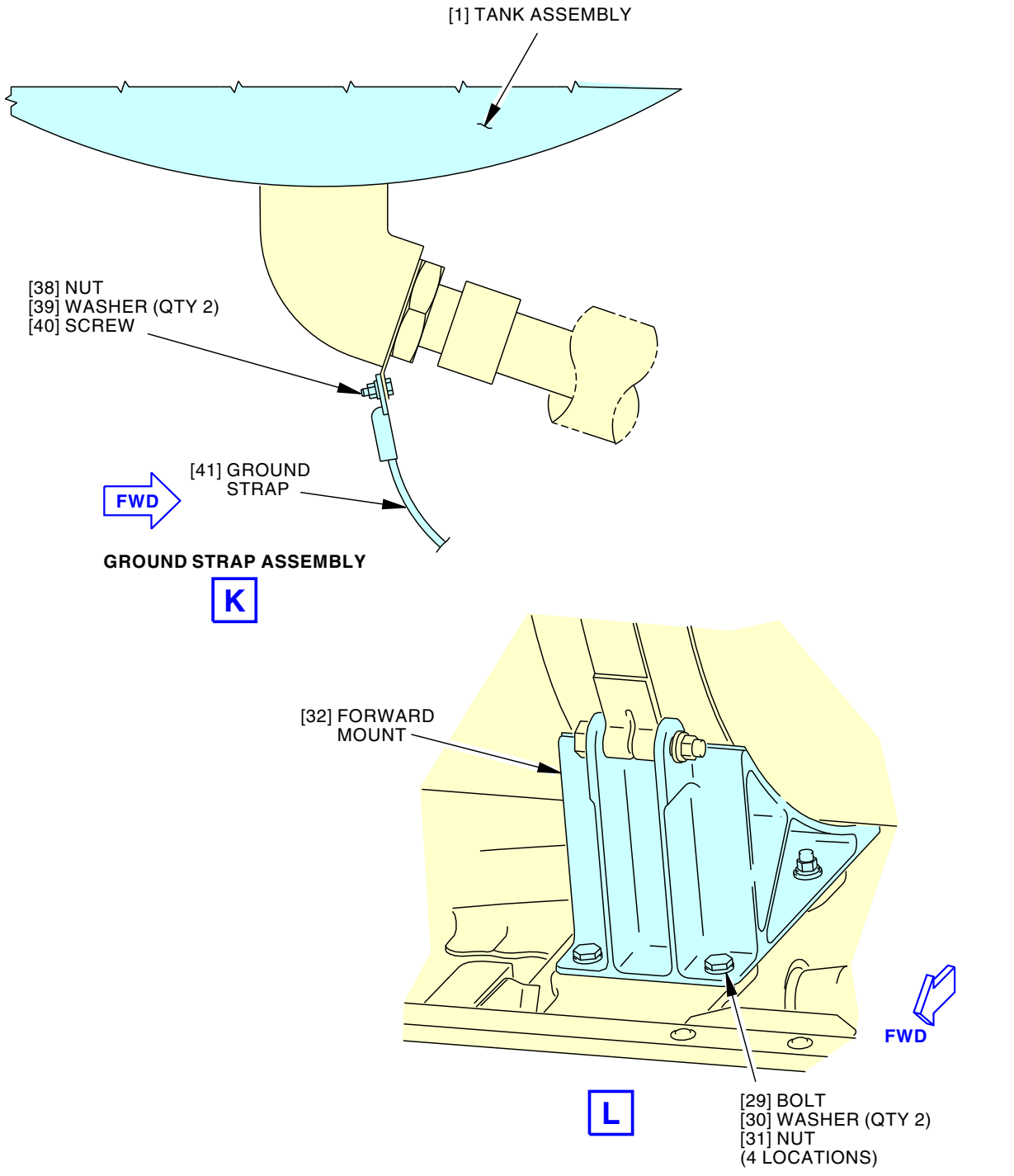
Potable Water Tank Installation
Figure 401/38-11-01-990-802 (Sheet 4 of 6)

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Potable Water Tank Installation
Figure 401/38-11-01-990-802 (Sheet 5 of 6)

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Potable Water Tank Installation
Figure 401/38-11-01-990-802 (Sheet 6 of 6)

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TASK 38-11-01-400-801

3. Water Tank Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the water tank.

B. References

Reference	Title
38-10-00-600-801	Potable Water System Disinfectant (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
38-42-07-400-801	Water Pressure Cut-Out Switch Installation (P/B 201)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	

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(Continued)

Reference	Description	Specification
D00463	Grease - Food Processing Equipment	DOD-G-24650
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Water tank	38-11-01-15-330	PTF ALL
14	Packing	38-11-01-15-085	PTF ALL
27	Packing	38-11-01-15-080	PTF ALL
36	Packing	38-11-01-15-085	PTF ALL
43	Packing	38-11-01-15-080	PTF ALL
45	Packing	38-11-01-15-075	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

H. Water Tank Installation

SUBTASK 38-11-01-420-014

(1) Remove the protective covers from all open connections.

SUBTASK 38-11-01-420-001

(2) Put the potable water tank [1] into the aft cargo compartment.

SUBTASK 38-11-01-420-002

(3) Put the water tank [1] in its position.

SUBTASK 38-11-01-420-003

(4) Put the right side and left side forward mounts [32] in position.

SUBTASK 38-11-01-420-004

(5) Install the bolts [29], washers [30], and nuts [31] to connect the forward mounts [32].

SUBTASK 38-11-01-420-005

(6) Install the bolt [21], washer [22], washer [23], and nut [24] to connect the tie rod assembly [25].

SUBTASK 38-11-01-420-006

(7) Install the strap assemblies [10].

SUBTASK 38-11-01-420-007

(8) Install the insulation blankets on the water tank [1].

SUBTASK 38-11-01-420-015

(9) Do these steps to install the tank fitting [33], if necessary:

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(a) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packing [43].

(b) Install the packing [43], washer [42], and tank fitting [33] on the water tank [1].

(c) Tighten the tank fitting [33] to 550 ±40 in-lb (62.1 ±4.5 N·m).

SUBTASK 38-11-01-640-001

(10) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packings [36].

SUBTASK 38-11-01-420-008

(11) Install the nut [38], washers [39], and screw [40] to connect the ground strap [41] to the water tank [1] as follows:

(a) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the nut [38], washers [39], screw [40], ground strap [41], and water tank [1].

(b) Apply sealant, A50110, or sealant, A50155, or sealant, A50231, or sealant, A50296, to the mating surfaces of the nut [38], washers [39], screw [40], and ground strap [41].

(c) Install the nut [38], washers [39], screw [40], and ground strap [41].

(d) Tighten the nut [38] to 8 in-lb (1 N·m).

1) Make sure that there is squeeze out around all the interfaces of each component.

(e) Measure the electrical resistance between the ground strap [41] and water tank [1] (SWPM 20-20-00).

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

SUBTASK 38-11-01-420-009

(12) Install the clamshell [34], sleeve [35], and packings [36] to connect the tank fitting [33] to the drain fitting [37].

SUBTASK 38-11-01-420-016

(13) Do these steps to install the tank fitting [26], if necessary:

(a) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packing [27].

(b) Install the packing [27] and tank fitting [26] on the water tank [1].

(c) Tighten the tank fitting [26] to 550 ±40 in-lb (62.1 ±4.5 N·m).

SUBTASK 38-11-01-640-002

(14) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packings [14].

SUBTASK 38-11-01-420-010

(15) Install the clamshell [13] and packings [14] to connect the water hoses to the top of the water tank [1].

SUBTASK 38-11-01-410-006

(16) Install the new packing [45]:

(a) Clean the packing [45] with cotton wiper, G00034.

(b) Apply grease, D00463, to the packing [45].

(c) Install the new packing [45] to the water tank [1] fitting.

SUBTASK 38-11-01-410-007

(17) Install the union [44] to the water tank [1] fitting:

(a) Apply Molykote 111 Compound, D00189, to the union [44] threads.

(b) Tighten the union [44] to 265 ±13 in-lb (29.9 ±1.5 N·m).

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SUBTASK 38-11-01-410-001

(18) Connect the pressurization connection [9] to the top of the water tank [1].

(a) Use a wrench to prevent movement of the union [44].

SUBTASK 38-11-01-420-011

(19) To install the pressure switch [11], do this task: Water Pressure Cut-Out Switch Installation, TASK 38-42-07-400-801.

SUBTASK 38-11-01-420-012

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(20) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [12] to the pressure switch [11] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the pressure switch [11] and electrical connector [12].

(b) Connect the electrical connector [12] to the pressure switch [11].

1) Tighten the connector with the teflon-jawed pliers, STD-664.

SUBTASK 38-11-01-420-013

(21) Connect the water quantity transmitter as follows:

NOTE: There are two transmitter cable connector terminals on the water tank. The transmitter cable connector can be connected to either terminal. Each terminal is attached to the sensor grid within the tank sidewall.

(a) Use cotton wiper, G00034, moist in alcohol, B00130, to clean the mating surfaces of the screw [18], washer [19], and connector [20].

(b) Install the screw [18] and washer [19] to attach the connector [20] to the terminal.

(c) Tighten the screw [18] to 35 ±2 in-lb (4.0 ±0.2 N·m).

(d) Measure the electrical bonding resistance between the washer [19] and second terminal in the housing (SWPM 20-20-00).

1) Use the non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.008 ohm (8 milliohm) or less.

(e) Apply fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, to the screw [18], washer [19], and connector [20].

(f) Install the screws [16] and washers [17] to attach the cover [15] to the water tank [1].

SUBTASK 38-11-01-020-012

(22) Do these steps to install the cargo liner support.

(a) Install the stanchion assembly [7] and insert the pull pin [8].

(b) Install the screw [4] to connect the stanchion assembly [6].

(c) Install the screw [3], screw [4], and screw [5] to connect the support assembly [2].

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SUBTASK 38-11-01-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(23) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-11-01-410-004

(24) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-11-01-860-007

(25) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	13	C00104	LAVATORY WATER HEATER A
F	14	C01073	LAVATORY WATER HEATER D
F	15	C01096	LAVATORY WATER HEATER E

SUBTASK 38-11-01-670-002

(26) Do this task: Potable Water System Disinfectant, TASK 38-10-00-600-801.

SUBTASK 38-11-01-790-001

(27) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-11-01-710-001

(28) Examine the connections of the water hoses and pressurization connection [9] for leakage.

SUBTASK 38-11-01-710-002

(29) Examine the fittings on the bottom of the water tanks for leakage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-11-01-410-002

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 38-11-01-410-005

(2) Install the electrical equipment rack, E6, aft access panel.

SUBTASK 38-11-01-410-003

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FILL/OVERFLOW VALVE AND CONTROL CABLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the fill/overflow valve

(2) An installation of the fill/overflow valve

(3) A removal of the fill/overflow valve control cable

(4) An installation of the fill/overflow valve control cable.

TASK 38-11-03-000-801

2. Fill/Overflow Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fill/overflow valve.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-11-03-800-001

(1) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-11-03-010-008

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-11-03-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND

SUBTASK 38-11-03-010-013

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-03-010-003

(5) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

SUBTASK 38-11-03-010-015

(6) Remove the electrical equipment rack, E6, aft access panel.

F. Fill/Overflow Valve Removal

SUBTASK 38-11-03-020-001

(1) Remove the bolts [5] and washers [6] that attach the control cable [13] to the fill/overflow valve [23].

SUBTASK 38-11-03-020-002

(2) Remove the screws [1] and washers [2] that attach the fill/overflow valve [23] to the structure.

SUBTASK 38-11-03-020-003

(3) Disconnect these water hoses from the fill/overflow valve [23]:

(a) The water hose to the water service panel.

(b) The water hose to fill the water tanks.

(c) The water hose for the water tank overflow.

(d) The water hose to attach the overflow to the skin fitting.

SUBTASK 38-11-03-020-013

(4) Install the protective caps on the water hoses.

SUBTASK 38-11-03-020-004

(5) Remove the screws [3] that attach the fill/overflow valve [23] to the bracket [4].

SUBTASK 38-11-03-020-005

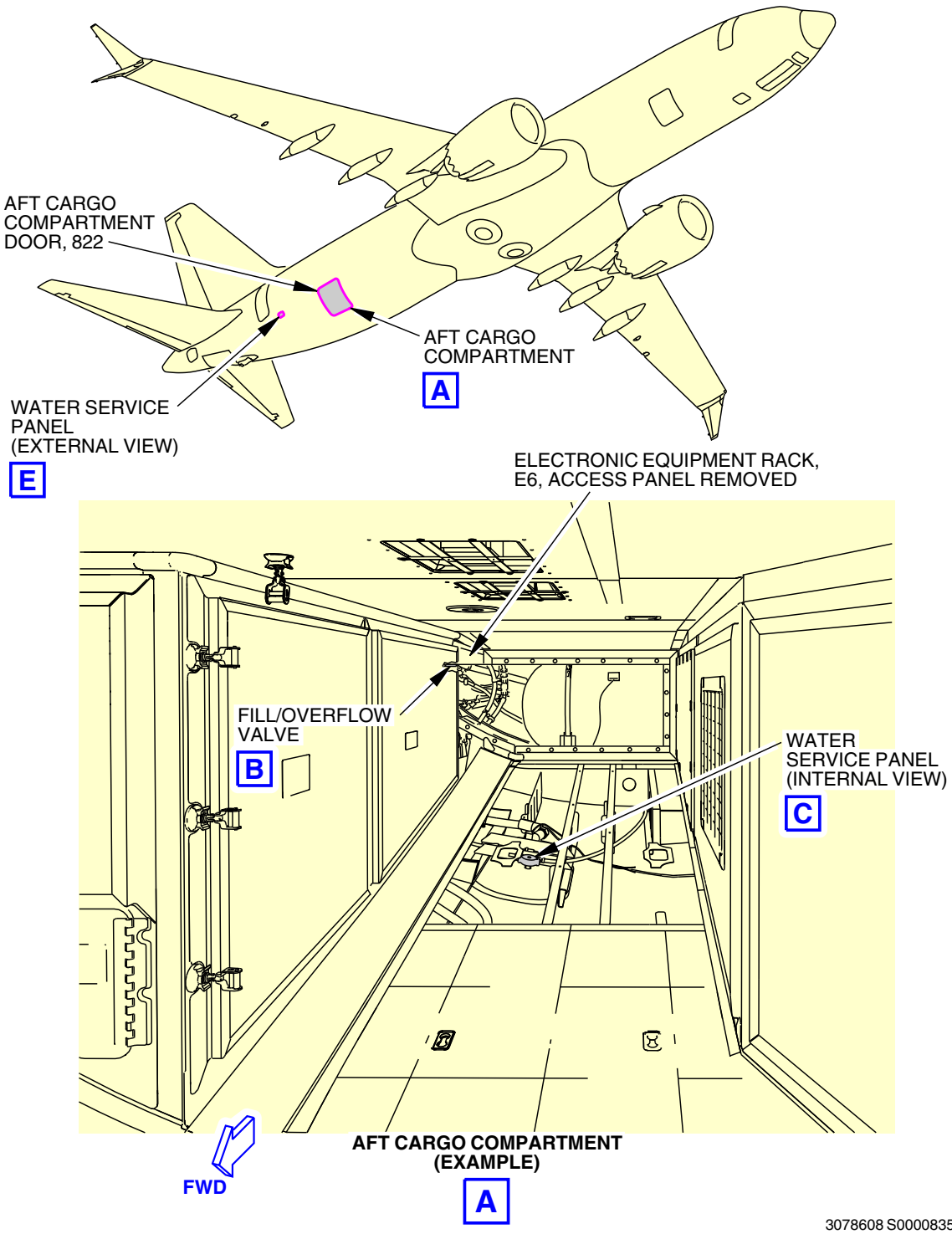
(6) Remove the fill/overflow valve [23].

SUBTASK 38-11-03-020-012

(7) Install the protective covers on the fill/overflow valve [23].

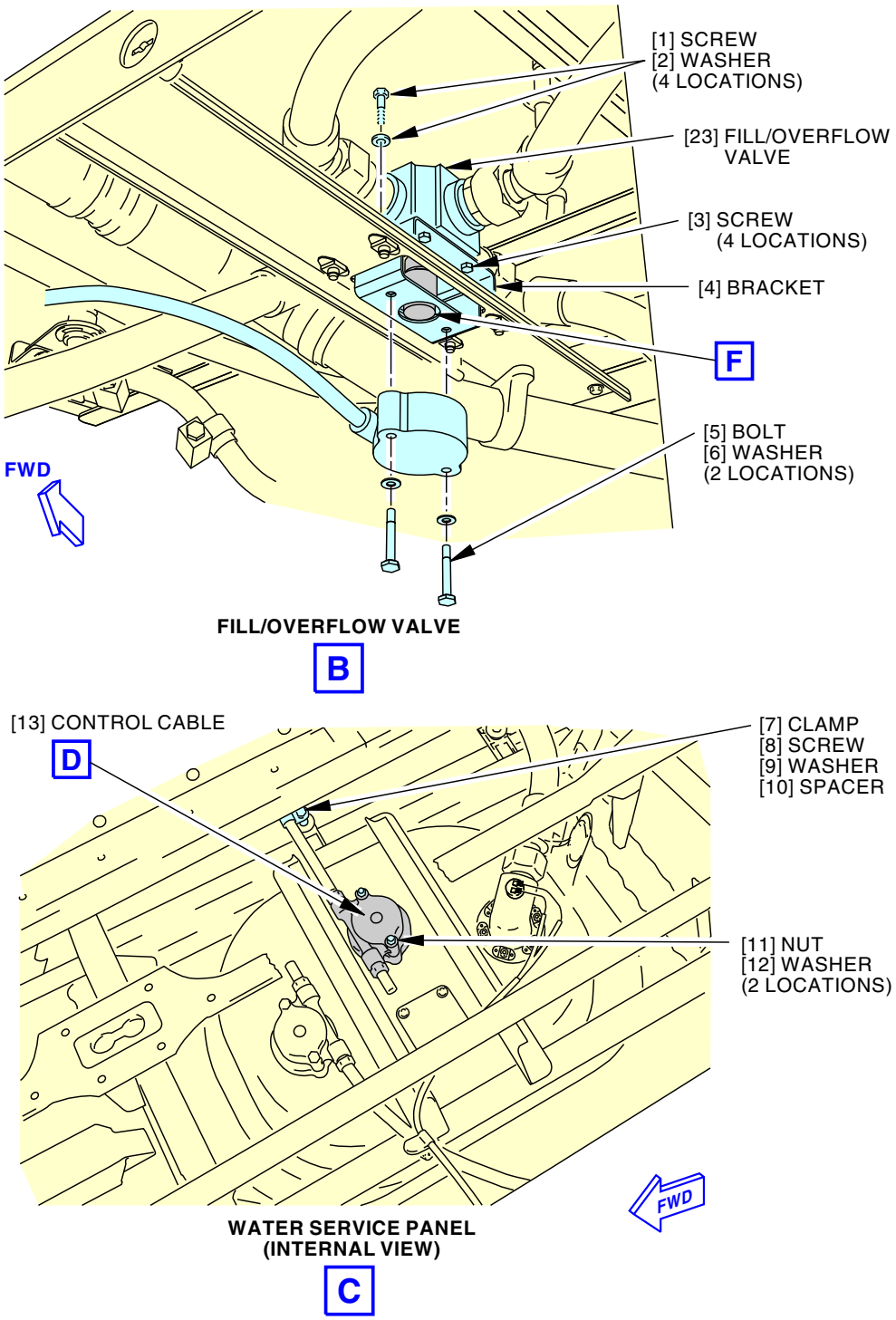
END OF TASK

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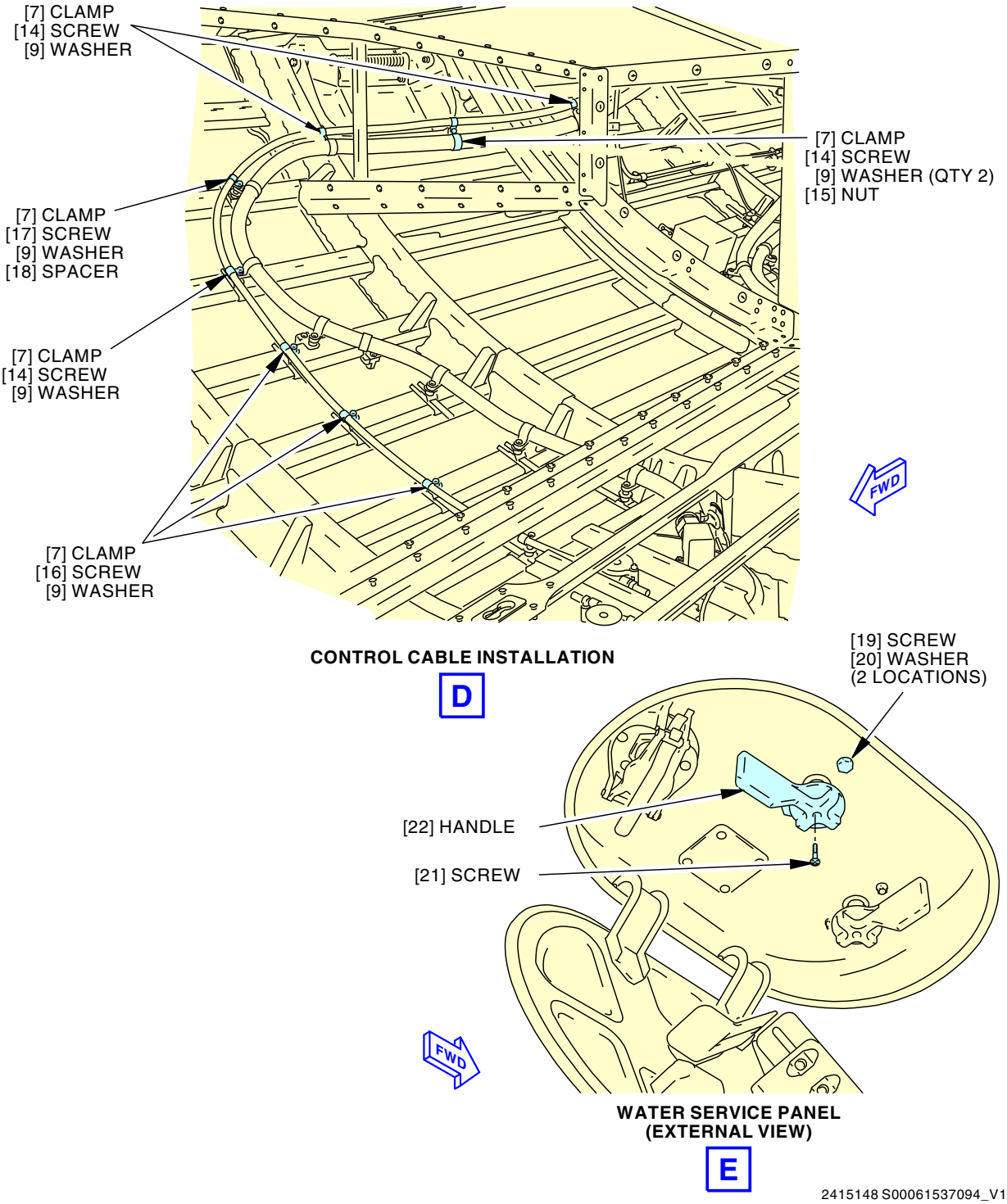
Fill/Overflow Valve and Control Cable Installation
Figure 401/38-11-03-990-802 (Sheet 1 of 4)

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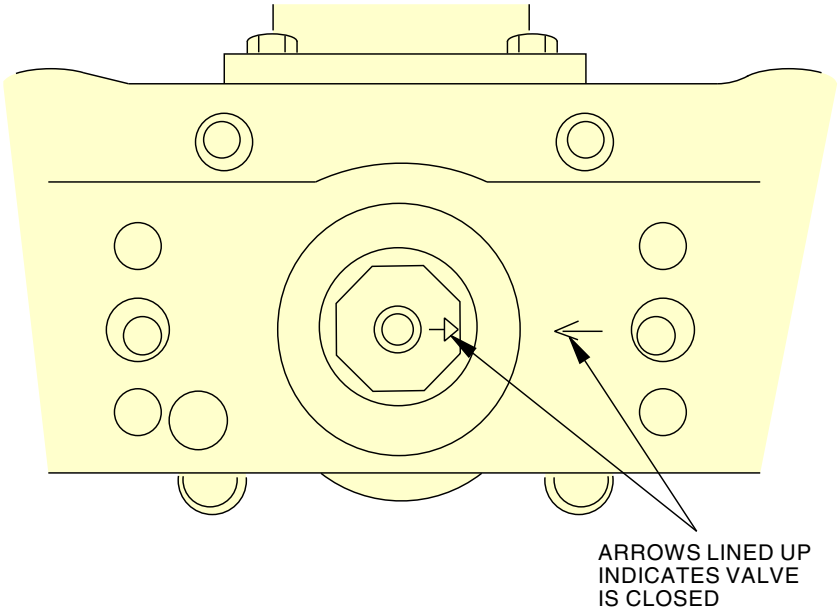
Fill/Overflow Valve and Control Cable Installation
Figure 401/38-11-03-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fill/Overflow Valve and Control Cable Installation
Figure 401/38-11-03-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FILL/OVERFLOW VALVE

F

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Fill/Overflow Valve and Control Cable Installation
Figure 401/38-11-03-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-11-03-400-801

3. Fill/Overflow Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fill/overflow valve.

B. References

Reference	Title
20-10-52-400-803	Flexible Hose Installation (P/B 401)
38-10-00-600-801	Potable Water System Disinfectant (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
23	Fill/overflow valve	38-11-03-03A-185	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
146AR	Water Service Door
822	Aft Cargo Door

F. Fill/Overflow Valve Installation

SUBTASK 38-11-03-420-012

(1) Remove the protective covers from the fill/overflow valve [23].

SUBTASK 38-11-03-420-001

(2) Install the screws [3] that attach the fill/overflow valve [23] to the bracket [4].

SUBTASK 38-11-03-420-002

(3) Put the fill/overflow valve [23] in its position.

SUBTASK 38-11-03-420-013

(4) Remove the protective caps from the water hoses.

SUBTASK 38-11-03-420-003

(5) Connect these water hoses to the fill/overflow valve [23] (TASK 20-10-52-400-803):

(a) The water hose to the water service panel.

1) Tighten the hose fitting nut to 40 ±5 in-lb (4.5 ±0.6 N·m).

(b) The water hose to fill the water tanks.

1) Tighten the hose fitting nut to 40 ±5 in-lb (4.5 ±0.6 N·m).

(c) The water hose for the water tank overflow.

1) Tighten the hose fitting nut to 40 ±5 in-lb (4.5 ±0.6 N·m).

(d) The water hose to drain the water at the skin fitting.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Tighten the hose fitting nut to 40 ±5 in-lb (4.5 ±0.6 N·m).

SUBTASK 38-11-03-420-004

(6) Install the screws [1] and washers [2] that attach the fill/overflow valve [23] to the structure.

SUBTASK 38-11-03-420-005

(7) Do these steps to attach the control cable [13] to the fill/overflow valve [23]:

(a) Make sure that the fill/overflow valve [23] is in the CLOSE position.

NOTE: Align the arrows to show that the valve is closed.

(b) Make sure that the control cable [13] is in the CLOSE position.

(c) Install the bolts [5] and washers [6] that attach the control cable [13] to the fill/overflow valve [23].

G. Fill/Overflow Valve Installation Test

SUBTASK 38-11-03-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND

SUBTASK 38-11-03-710-001

(2) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

(a) Make sure that the air compressor for the potable water system is on.

SUBTASK 38-11-03-010-011

(3) Open this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-11-03-710-002

(4) Turn the handle for the fill/overflow valve to the OPEN position.

(a) If the fill/overflow valve is not in the OPEN position, do a check for correct installation.

(b) Turn the handle for the fill/overflow valve to the CLOSED position.

H. Put Airplane Back to Its Usual Condition

SUBTASK 38-11-03-670-001

(1) Do this task: Potable Water System Disinfectant, TASK 38-10-00-600-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-03-010-014

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-03-410-001

WARNING SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(3) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 38-11-03-410-012

(4) Install the electrical equipment rack, E6, aft access panel.

SUBTASK 38-11-03-410-011

(5) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

TASK 38-11-03-000-802

4. Fill/Overflow Valve Control Cable Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fill/overflow valve control cable.

B. References

Reference	Title
50-10-04-000-801	Cargo Compartment Floor Panel Removal (P/B 401)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

D. Access Panels

Number	Name/Location
146AR	Water Service Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-11-03-020-006

(1) Open this access panel:

Number	Name/Location
146AR	Water Service Door

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SUBTASK 38-11-03-010-004

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-03-010-005

(3) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

SUBTASK 38-11-03-010-016

(4) Remove the electrical equipment rack, E6, aft access panel.

SUBTASK 38-11-03-010-006

(5) Do this task: Cargo Compartment Floor Panel Removal, TASK 50-10-04-000-801.

F. Fill/Overflow Valve Control Cable Removal

SUBTASK 38-11-03-020-007

(1) Remove the fill/overflow valve handle [22] and screw [21] for the control cable [13] at the water service panel.

SUBTASK 38-11-03-020-008

(2) Remove the hardware that follows to disconnect the control cable [13]:

(a) Remove the clamp [7], screw [8], washer [9], and spacer [10].

(b) Remove the clamps [7], screws [16], and washers [9].

(c) Remove the clamp [7], screw [14], and washer [9].

(d) Remove the clamp [7], screw [17], washer [9], and spacer [18]

(e) Remove the clamp [7], screw [14], and washer [9].

(f) Remove the clamp [7], screw [14], washers [9], and nut [15]

(g) Remove the clamp [7], screw [14], and washer [9].

SUBTASK 38-11-03-020-009

(3) Remove the screws [19], washers [20], nuts [11], and washers [12] that attach the control cable [13] to the water service panel.

SUBTASK 38-11-03-020-010

(4) Remove the bolts [5] and washers [6] that attach the control cable [13] to the fill/overflow valve [23].

SUBTASK 38-11-03-020-011

(5) Remove the control cable [13].

——— END OF TASK ———

TASK 38-11-03-400-802

5. Fill/Overflow Valve Control Cable Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fill/overflow valve control cable.

B. References

Reference	Title
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-10-04-400-801	Cargo Compartment Floor Panel Installation (P/B 401)

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(Continued)

Reference	Title
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
51-22-00-390-801	Non-Removable Faying (Mated) Surface Seal Application (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Control cable	38-11-03-03A-270	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
146AR	Water Service Door
822	Aft Cargo Door

G. Fill/Overflow Valve Control Cable Installation

SUBTASK 38-11-03-420-006

(1) Pull the control cable [13] from the water service panel to put the control cable assembly in its position.

SUBTASK 38-11-03-420-007

(2) Do these steps to attach the control cable [13] to the fill/overflow valve [23]:

(a) Make sure that the fill/overflow valve [23] is in the CLOSE position.

NOTE: Align the arrows to show that the valve is closed.

(b) Make sure that the control cable [13] is in the CLOSE position.

(c) Install the bolts [5] and washers [6] that attach the control cable [13] to the fill/overflow valve [23].

SUBTASK 38-11-03-420-008

(3) Install the hardware that follows to connect the control cable [13] to structure.

(a) Install the clamp [7], screw [8], washer [9], and spacer [10].

(b) Install the clamps [7], screws [16], and washers [9].

(c) Install the clamp [7], screw [14], and washer [9].

(d) Install the clamp [7], screw [17], washer [9], and spacer [18]

(e) Install the clamp [7], screw [14], and washer [9].

(f) Install the clamp [7], screw [14], washers [9], and nut [15].

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(g) Install the clamp [7], screw [14], and washer [9].

SUBTASK 38-11-03-420-009

(4) Get access to the water service panel.

SUBTASK 38-11-03-390-001

(5) To apply sealant, A00247, a fay surface seal to the inner area of the water service panel around the control cable [13], do this task: Non-Removable Faying (Mated) Surface Seal Application, TASK 51-22-00-390-801.

SUBTASK 38-11-03-420-010

(6) Loosely install the screws [19], washers [20], nuts [11], and washers [12] that attach the control cable [13] to the water service panel.

SUBTASK 38-11-03-420-011

(7) Install the handle [22] and screw [21].

SUBTASK 38-11-03-410-004

(8) Do this task: Cargo Compartment Floor Panel Installation, TASK 50-10-04-400-801.

H. Control Cable Installation Test

SUBTASK 38-11-03-710-003

(1) Make sure that the load you use to operate the control cable [13] is less than 50 in-lb (5.6 N·m).

SUBTASK 38-11-03-010-009

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-11-03-860-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND
D	11	C00873	POT WATER COMPRESSOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-03-410-010

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-11-03-710-004

(5) Make sure that the potable water system is activated (TASK 38-42-00-800-802).

(a) Make sure that the air compressor for the potable water system is on.

SUBTASK 38-11-03-610-001

(6) Turn the handle for the fill/overflow valve to the OPEN position.

(a) Make sure that the fill/overflow valve is completely open.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-11-03-410-005

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 38-11-03-410-013

(2) Install the electrical equipment rack, E6, aft access panel.

SUBTASK 38-11-03-410-006

(3) Do this task: Cargo Compartment Floor Panel Installation, TASK 50-10-04-400-801.

SUBTASK 38-11-03-410-007

(4) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-03-410-008

(5) Close this access panel:

Number	Name/Location
146AR	Water Service Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY FAUCET - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Lavatory faucet removal

(2) Lavatory faucet installation

(3) Cartridge valve assembly replacement

(4) Lavatory faucet adjustment.

TASK 38-11-05-000-801

2. Lavatory Faucet - Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove the lavatory faucet.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Prepare for the Removal

SUBTASK 38-11-05-000-001

(1) Get access to the lavatory module.

SUBTASK 38-11-05-010-001

(2) Open the shroud and access door on the lower vanity.

SUBTASK 38-11-05-860-001

(3) Put the water heat switch in the off position.

SUBTASK 38-11-05-860-002

(4) Turn the water shutoff valve to the OFF position.

NOTE: The water shutoff valve is below the sink in the lavatory.

SUBTASK 38-11-05-680-001

(5) On the forward lavatories, open the drain valve.

NOTE: Only the forward lavatories have the drain valve.

D. Lavatory Faucet Removal

SUBTASK 38-11-05-020-001

(1) Remove the set screws [2] from the faucet assembly [1].

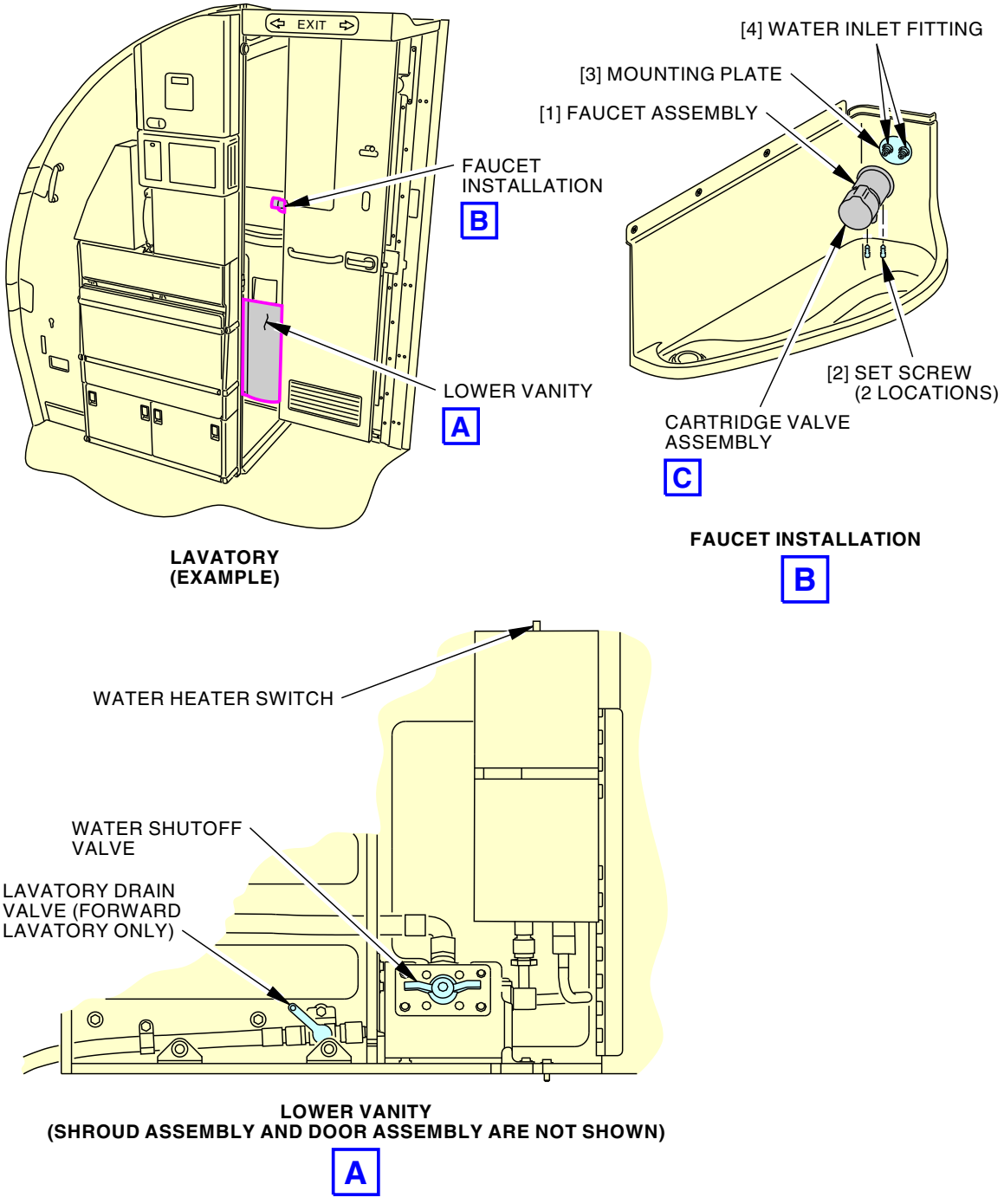
SUBTASK 38-11-05-020-002

(2) Remove the faucet assembly [1] from the mounting plate [3].

(a) Pull the faucet assembly [1] from the mounting plate [3] until the faucet disconnects from the water inlet fitting [4].

END OF TASK

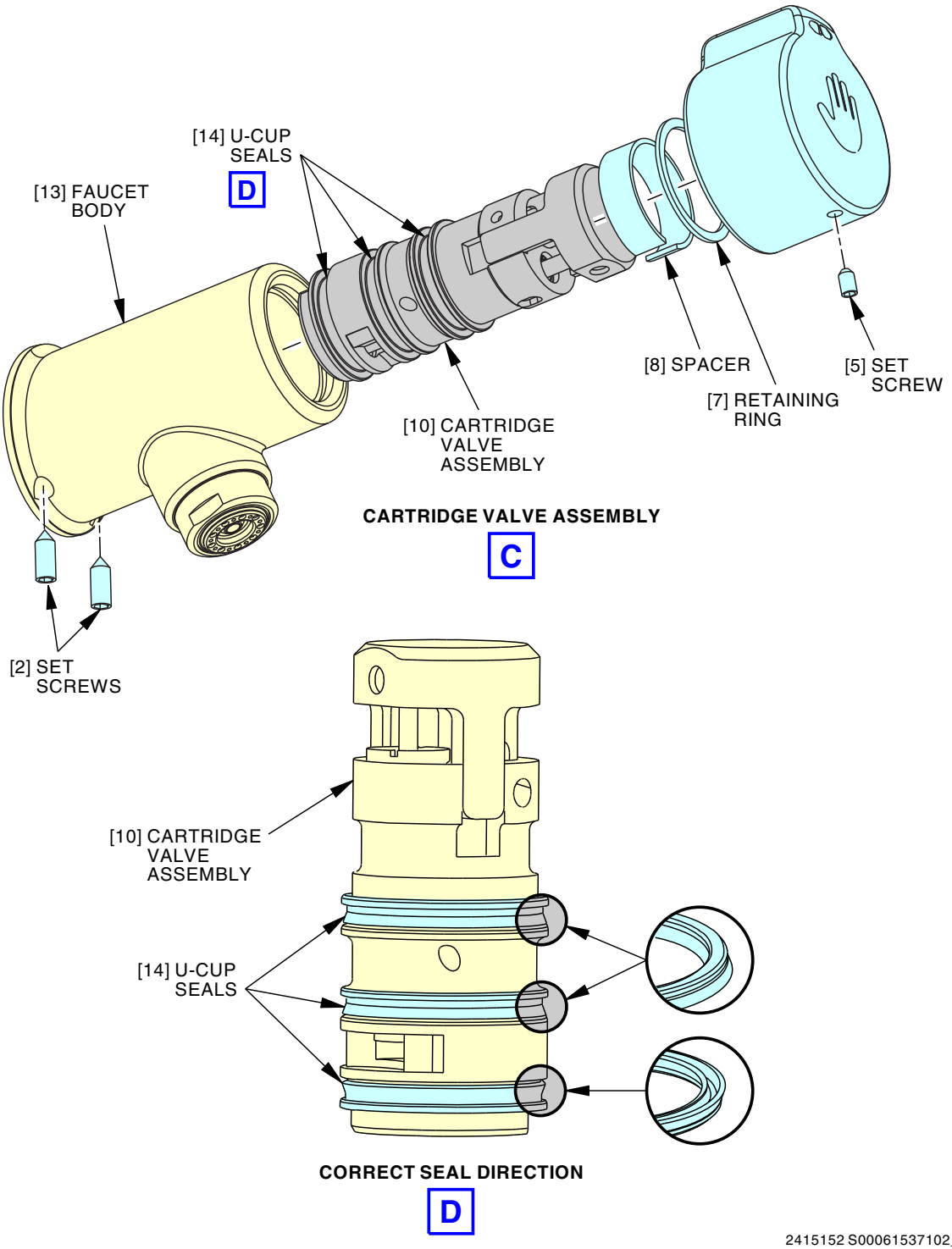
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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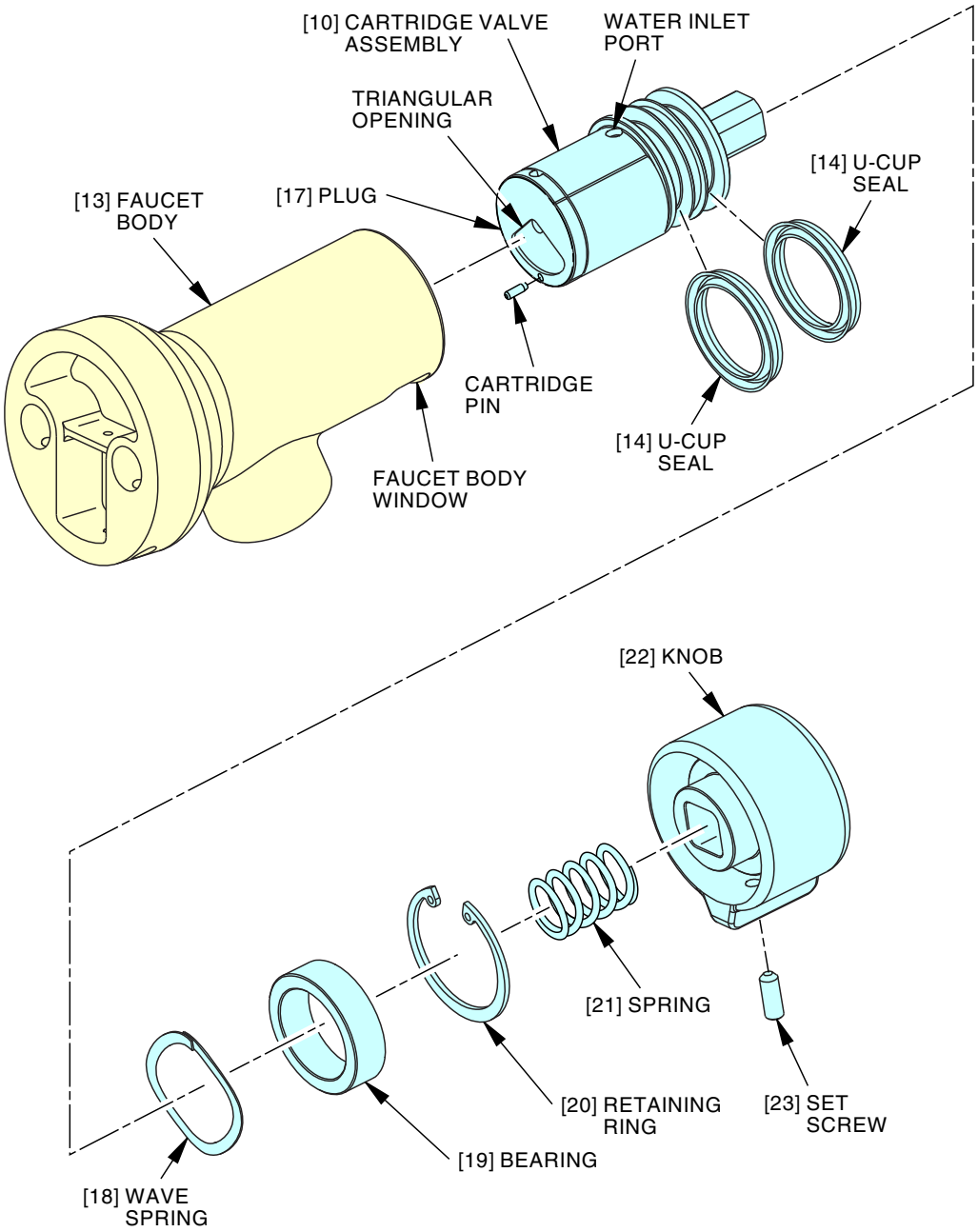
Lavatory Faucet Installation
Figure 201/38-11-05-990-802 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



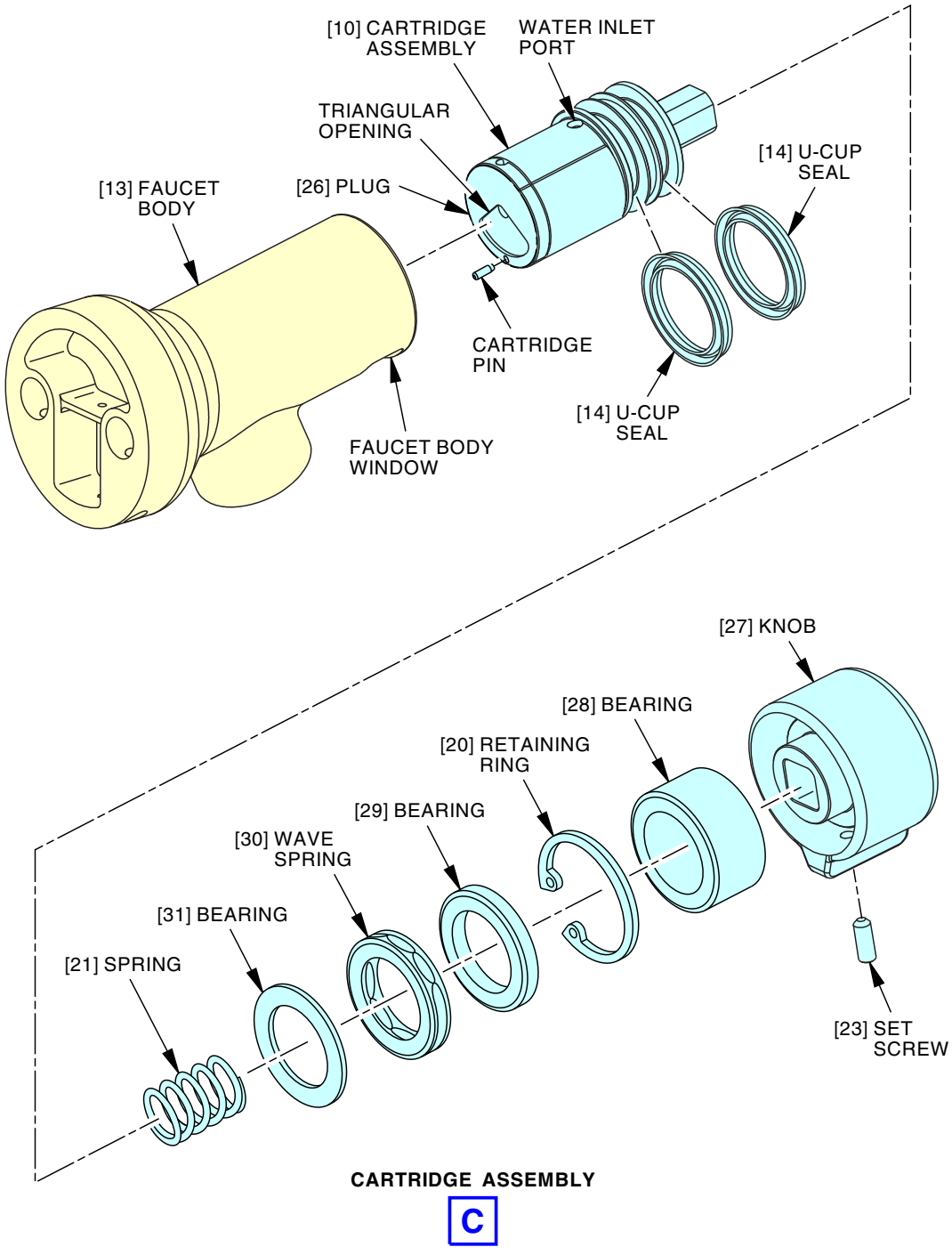
Lavatory Faucet Installation
Figure 201/38-11-05-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Faucet Installation
Figure 201/38-11-05-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Faucet Installation
Figure 201/38-11-05-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-11-05-400-801

3. Lavatory Faucet - Installation

(Figure 201)

A. General

(1) This task gives the instructions to install the lavatory faucet.

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Faucet assembly	25-41-11-05P-132	PTF ALL
		25-41-11-05P-133	PTF ALL
		25-41-11-06-127	PTF ALL
		25-41-11-06-128	PTF ALL
		25-41-60-06D-060	PTF ALL

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Lavatory Faucet Installation

SUBTASK 38-11-05-420-001

(1) Install the faucet assembly [1] on the mounting plate [3].

(a) Examine the O-rings on the water inlet fitting [4].

1) Replace the O-rings if damaged.

(b) Push the faucet assembly [1] onto the water inlet fitting [4].

NOTE: Lubrication is not required.

SUBTASK 38-11-05-420-052

(2) Install the set screws [2] in the faucet assembly [1].

(a) Tighten the set screws [2] to 26 in-lb (2.9 N·m) - 33.5 in-lb (3.8 N·m).

E. Lavatory Faucet Installation Test

SUBTASK 38-11-05-860-003

(1) On the forward lavatories, close the drain valve.

NOTE: Only the forward lavatories have the drain valve.

SUBTASK 38-11-05-860-004

(2) Turn the water shutoff valve to the ON position.

NOTE: The water shutoff valve is below the sink in the lavatory.

SUBTASK 38-11-05-790-001

(3) Make sure that the faucet assembly and water lines do not have a leak.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 38-11-05-860-005

(1) Put the water heater switch in the ON position.

SUBTASK 38-11-05-410-001

(2) Close the access door and shroud on the lower vanity.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-05-410-002

(3) Close the access to the lavatory module.

END OF TASK

TASK 38-11-05-960-801

4. Cartridge Assembly - Replacement

(Figure 201 and Figure 202)

A. General

(1) This task gives the instructions to replace the cartridge assembly.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-16537	Adapter-Installation Part #: T5776 Supplier: 80477

C. Consumable Materials

Reference	Description	Specification
A00648	Compound - Retaining - Loctite 242	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D50011	Grease - Perfluoropolyether - Christo-lube MCG111	
D50232	Compound - Release, Molykote 7 (Formerly Dow Corning 7)	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	Cartridge assembly	38-11-05-01-215	PTF ALL
		38-11-05-02-250	PTF ALL
		38-11-05-03-155	PTF ALL
14	U-cup seal	38-11-05-01-130	PTF ALL
		38-11-05-02-155	PTF ALL
		38-11-05-03-160	PTF ALL

E. Location Zones

Zone	Area
200	Upper Half of Fuselage

F. Prepare for the Replacement

SUBTASK 38-11-05-010-002

(1) Get access to the lavatory module.

SUBTASK 38-11-05-010-003

(2) Open the shroud and access door on the lower vanity.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-05-860-006

(3) Put the water heat switch in the off position.

SUBTASK 38-11-05-860-007

CAUTION

DEPRESSURIZE WATER SYSTEM BEFORE DISCONNECTING ANY WATER LINES OR WATER SYSTEM COMPONENTS TO PREVENT WATER SPILLAGE.

(4) Turn the water shutoff valve to the OFF position.

NOTE: The water shutoff valve is below the sink in the lavatory.

SUBTASK 38-11-05-860-008

(5) On the forward lavatories, open the drain valve.

NOTE: Only the forward lavatories have the drain valve.

PTF ALL; AIRPLANES WITH FAUCET P/N AR9472-3W1D492

G. Cartridge Assembly Removal

SUBTASK 38-11-05-020-003

WARNING

WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(1) Loosen the set screw [5] and remove the knob [6].

SUBTASK 38-11-05-010-004

(2) Remove the retaining ring [7] and spacer [8] from the groove in the faucet body [13].

SUBTASK 38-11-05-010-005

(3) Remove the cartridge assembly [10].

(a) Remove and discard u-cup seals [14] from the groove in the cartridge assembly [10].

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-1

H. Cartridge Assembly Removal

SUBTASK 38-11-05-020-008

(1) Loosen the set screw [23] and remove the knob [22].

SUBTASK 38-11-05-020-009

(2) Remove the spring [21] and the retaining ring [20].

SUBTASK 38-11-05-020-010

(3) Remove the cartridge assembly [10] with the bearing [19] and wave spring [18].

(a) Make sure that the plug [17] stays open on the cartridge assembly [10].

(b) Remove and discard the u-cup seals [14] from the groove in the cartridge assembly [10].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-4

I. Cartridge Assembly Removal

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-020-011

(1) Loosen the set screw [23] and remove the knob [27].

SUBTASK 38-11-05-020-012

(2) Remove the spring [21] and the retaining ring [20].

SUBTASK 38-11-05-020-013

(3) Remove the cartridge assembly [10] with the bearing [29], wave spring [30], and bearing [31].

(a) Make sure that the plug [26] stays open on the cartridge assembly [10].

(b) Remove and discard the u-cup seal [14] from the groove in the cartridge assembly [10].

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-5 OR 4101800-001

J. Cartridge Assembly Removal

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-020-014

(1) Loosen the set screw [23] and remove the knob [27].

SUBTASK 38-11-05-020-015

(2) Remove the spring [21] and the retaining ring [20].

SUBTASK 38-11-05-020-016

(3) Remove the cartridge assembly [10] with the bearing [29], wave spring [30], and bearing [31].

(a) Make sure that the plug [26] stays open on the cartridge assembly [10].

(b) Remove and discard the u-cup seal [14] from the groove in the cartridge assembly [10].

PTF ALL; AIRPLANES WITH FAUCET P/N AR9472-3W1D492

K. Cartridge Assembly Installation

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-420-012

(1) Install the new u-cup seals [14] onto the cartridge assembly [10].

(a) Lubricate the u-cup seals [14] with Molykote 7 release compound, D50232.

(b) Make sure that each of the u-cup seals [14] faces the correct direction.

SUBTASK 38-11-05-410-005

(2) Install the cartridge assembly [10] into the faucet body [13] (Figure 202).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N AR9472-3W1D492 (Continued)

SUBTASK 38-11-05-410-006

(3) Install the spacer [8], then install the retaining ring [7] in the groove of the faucet body [13].

SUBTASK 38-11-05-410-007

(4) Install the knob [6] and set screw [5] on the cartridge assembly [10].

NOTE: Use the installation adapter, COM-16537, if it is necessary.

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-1

L. Cartridge Assembly Installation

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-100-001

(1) Clean the faucet body [13] bore with a clean, lint free cloth to remove the contamination.

SUBTASK 38-11-05-420-020

(2) Make sure that the cartridge pin is in its position before installation of the cartridge assembly [10].

(a) If the pin is loose, use the Molykote 111 Compound, D00189, to hold the pin in its position during installation.

SUBTASK 38-11-05-390-001

(3) Apply thin layer of the Molykote 111 Compound, D00189, to u-cup seals [14].

SUBTASK 38-11-05-390-002

(4) Install two new u-cup seals [14] onto the cartridge assembly [10].

SUBTASK 38-11-05-420-021

(5) Point the cartridge assembly [10] with the water inlet port up, and the curved edge of triangular opening face down on the plug.

NOTE: The faucet temperature will not work correctly if the cartridge assembly [10] is not in the correct position.

SUBTASK 38-11-05-420-022

(6) Insert the cartridge assembly [10] into the faucet body [13] as follows:

(a) Carefully push the cartridge assembly [10] to let the u-cup seals [14] to move into the faucet body [13].

NOTE: Use a probe or other thin object to move the u-cup seals past the faucet body window. Use a wiping movement, and only sufficient pressure to push the u-cup seals in to the u-cups.

(b) Carefully move the cartridge assembly [10] to the bottom of the faucet body [13] bore.

(c) Make sure that the cartridge assembly [10] stays in the correct position.

SUBTASK 38-11-05-420-023

(7) Put the wave spring [18] and the bearing [19] into the faucet body [13].

SUBTASK 38-11-05-420-024

(8) Use the retaining ring tool to install the retaining ring [20] into the faucet body [13].

(a) Move the retaining ring [20] to the bearing [19].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-1 (Continued)

(b) Make sure that the retaining ring [20] engages in the faucet body groove.

SUBTASK 38-11-05-420-025

(9) Install the spring [21] in the faucet body [13].

SUBTASK 38-11-05-420-026

(10) Install the knob [22] on the faucet body [13].

SUBTASK 38-11-05-420-027

(11) Apply the Loctite 242 compound, A00648, to the set screw [23].

SUBTASK 38-11-05-420-028

(12) Install the set screw [23].

NOTE: Do not tighten the set screw [23] until you adjust the faucet timing.

SUBTASK 38-11-05-820-003

(13) Do this task: Lavatory Faucet - Adjustment, TASK 38-11-05-820-801.

NOTE: The faucet will not work correctly if the faucet timing is not correct.

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-4

M. Cartridge Assembly Installation

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-100-002

(1) Clean the faucet body [13] bore with a clean, lint free cloth to remove the contamination.

SUBTASK 38-11-05-420-029

(2) Make sure that the cartridge pin is in its position before installation of the cartridge assembly [10].

(a) If the pin is loose, use the Christo-lube MCG111 grease, D50011, to hold the pin in its position during installation.

SUBTASK 38-11-05-390-003

(3) Apply thin layer of the Christo-lube MCG111 grease, D50011, to two new u-cup seals [14].

SUBTASK 38-11-05-390-004

(4) Install two new u-cup seals [14] onto the cartridge assembly [10].

SUBTASK 38-11-05-420-030

(5) Point the cartridge assembly [10] with the water inlet port up, and the curved edge of triangular opening face down on the plug.

NOTE: The faucet temperature will not work correctly if the cartridge assembly is not in the correct position.

SUBTASK 38-11-05-420-031

(6) Insert the cartridge assembly [10] into the faucet body [13] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-4 (Continued)

(a) Carefully push the cartridge assembly [10] to let the u-cup seals [14] to move into the faucet body [13].

NOTE: Use a probe or other thin object to move the u-cup seals past the faucet body window. Use a wiping movement, and only sufficient pressure to push the u-cup seals in to the u-cups.

(b) Carefully move the cartridge assembly [10] to the bottom of the faucet body [13] bore.

(c) Make sure that the cartridge assembly [10] stays in the correct position.

SUBTASK 38-11-05-420-032

(7) Put the bearing [31], wave spring [30], and the bearing [29] into the faucet body [13].

SUBTASK 38-11-05-420-033

(8) Use the retaining ring tool to install the retaining ring [20] into the faucet body [13].

(a) Move the retaining ring [20] to the bearing [28].

(b) Make sure that the retaining ring [20] engages in the faucet body groove.

SUBTASK 38-11-05-420-034

(9) Install the spring [21] in the cartridge assembly [10].

SUBTASK 38-11-05-420-035

(10) Install the knob [27] on the faucet body [13].

SUBTASK 38-11-05-420-036

(11) Apply the Loctite 242 compound, A00648, to the set screw [23].

SUBTASK 38-11-05-420-037

(12) Install the set screw [23].

NOTE: Do not tighten the set screw until you adjust the faucet timing.

SUBTASK 38-11-05-820-004

(13) Do this task: Lavatory Faucet - Adjustment, TASK 38-11-05-820-801.

NOTE: The faucet will not work correctly if the faucet timing is not correct.

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-5 OR 4101800-001

N. Cartridge Assembly Installation

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-100-003

(1) Clean the faucet body [13] bore with a clean, lint free cloth to remove the contamination.

SUBTASK 38-11-05-420-043

(2) Make sure that the cartridge pin is in its position before installation of the cartridge assembly [10].

(a) If the pin is loose, use the Molykote 111 Compound, D00189, to hold the pin in its position during installation.

SUBTASK 38-11-05-390-005

(3) Apply thin layer of the Molykote 111 Compound, D00189, to two new u-cup seals [14].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-5 OR 4101800-001 (Continued)

SUBTASK 38-11-05-390-006

(4) Install two new u-cup seals [14] onto the cartridge assembly [10].

SUBTASK 38-11-05-420-044

(5) Point the cartridge assembly [10] with the water inlet port up, and the curved edge of triangular opening face down on the plug.

NOTE: The faucet temperature will not work correctly if the cartridge assembly is not in the correct position.

SUBTASK 38-11-05-420-045

(6) Insert the cartridge assembly [10] into the faucet body [13] as follows:

(a) Carefully push the cartridge assembly [10] to let the u-cup seals [14] to move into the faucet body [13].

NOTE: Use a probe or other thin object to move the u-cup seals past the faucet body window. Use a wiping movement, and only sufficient pressure to push the u-cup seals in to the u-cups.

(b) Carefully move the cartridge assembly [10] to the bottom of the faucet body [13] bore.

(c) Make sure that the cartridge assembly [10] stays in the correct position.

SUBTASK 38-11-05-420-046

(7) Put the bearing [31], wave spring [30], and the bearing [29] into the faucet body [13].

SUBTASK 38-11-05-420-047

(8) Use the retaining ring tool to install the retaining ring [20] into the faucet body [13].

(a) Move the retaining ring [20] to the bearing [28].

(b) Make sure that the retaining ring [20] engages in the faucet body groove.

SUBTASK 38-11-05-420-048

(9) Install the spring [21] in the cartridge assembly [10].

SUBTASK 38-11-05-420-049

(10) Install the knob [27] on the faucet body [13].

SUBTASK 38-11-05-420-050

(11) Apply the Loctite 242 compound, A00648, to the set screw [23].

SUBTASK 38-11-05-420-051

(12) Install the set screw [23].

NOTE: Do not tighten the set screw until you adjust the faucet timing.

SUBTASK 38-11-05-820-006

(13) Do this task: Lavatory Faucet - Adjustment, TASK 38-11-05-820-801.

NOTE: The faucet will not work correctly if the faucet timing is not correct.

PTF ALL

O. Cartridge Assembly Installation Test

SUBTASK 38-11-05-860-009

(1) On the forward lavatories, close the drain valve.

NOTE: Only the forward lavatories have the drain valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-05-860-010

(2) Turn the water shutoff valve to the ON position.

NOTE: The water shutoff valve is below the sink in the lavatory.

SUBTASK 38-11-05-790-002

(3) Make sure that the faucet assembly does not have a leak.

P. Put the Airplane Back to its Usual Condition

SUBTASK 38-11-05-860-011

(1) Put the water heater switch in the ON position.

SUBTASK 38-11-05-410-003

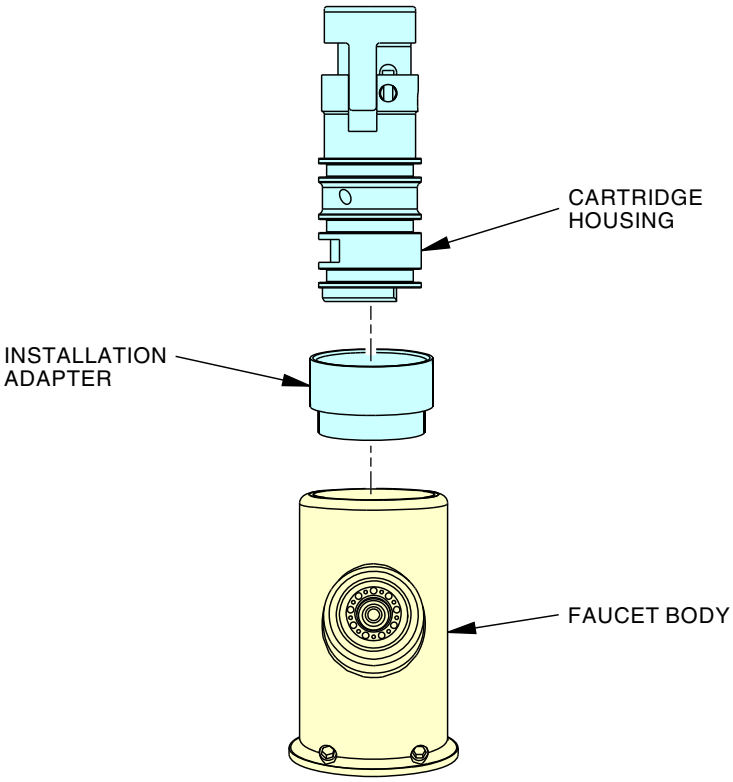
(2) Close the shroud and access door on the lower vanity.

SUBTASK 38-11-05-410-004

(3) Close the access to the lavatory module.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Positioning of Installation Adapter
Figure 202/38-11-05-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

TASK 38-11-05-820-801

5. Lavatory Faucet - Adjustment

(Figure 203)

A. General

(1) This task gives the instructions to adjust the lavatory faucet.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
25-40-00-440-801	Water Heater Power Activation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-836	Stopwatch
STD-1330	Wrench - Hexdrive, Allen Wrench

D. Consumable Materials

Reference	Description	Specification
A00648	Compound - Retaining - Loctite 242	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)
A50212	Compound - Threadlocking, Low-strength - Loctite 222MS	MIL-S-46163A, ASTM D5363

E. Location Zones

Zone	Area
200	Upper Half of Fuselage

F. Prepare for the Adjustment.

SUBTASK 38-11-05-860-012

(1) Before the hot water timing test, make sure that the water is hot (heated by the water heater), refer to Water Heater Power Activation, TASK 25-40-00-440-801.

SUBTASK 38-11-05-860-013

(2) Turn on the water supply to the faucet (TASK 12-14-00-600-802).

PTF ALL; AIRPLANES WITH FAUCET P/N AR9472-3W1D492

G. Water Faucet Adjustment

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-010-007

(1) Get access to the lavatory module.

SUBTASK 38-11-05-710-001

(2) To measure the length of time that the water flows, do these steps:

(a) To measure cold water flow, turn the faucet knob [6] to the cold side, push and hold the knob [6] down to let water flow for approximately ten seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N AR9472-3W1D492 (Continued)

(b) To measure hot water flow, turn the faucet knob [6] to the hot side, push and hold the knob [6] down to let water flow for approximately ten seconds.

(c) After ten seconds, let the knob [6] go back to the off position (no water flow).

(d) With the knob [6] in the same position, push it to its full on position and then let it go back to the off position.

1) Use a stopwatch, STD-836, to measure the time that it takes for the knob [6] to go back to the off position.

2) Repeat this step two more times.

3) Record the times and calculate the average flow time.

(e) The average flow time should be 4-8 seconds.

NOTE: Measure the flow time for the hot water and cold water before doing the adjustment steps.

SUBTASK 38-11-05-820-001

CAUTION

DEPRESSURIZE WATER SYSTEM BEFORE DISCONNECTING ANY WATER LINES OR WATER SYSTEM COMPONENTS TO PREVENT WATER SPILLAGE.

(3) If the length of time that the water flows is not satisfactory, do these steps:

(a) Remove the set screw [5] and remove the knob [6].

(b) Remove the retaining ring [7].

(c) Remove the spacer [8].

(d) Remove the lock pin [16].

(e) Remove the inner knob [9] from the cartridge assembly [10].

(f) Use a hexdrive allen wrench, STD-1330, to turn the hydraulic timer [15] to adjust the timing.

1) Turn the timer clockwise for slower timing.

2) Turn the timer counterclockwise for faster timing.

SUBTASK 38-11-05-420-013

(4) Install the inner knob [9] on the cartridge assembly [10].

SUBTASK 38-11-05-420-014

(5) Install the lock pin [16].

SUBTASK 38-11-05-420-015

(6) Install the spacer [8].

SUBTASK 38-11-05-420-016

(7) Install the retaining ring [7].

SUBTASK 38-11-05-420-017

(8) Install the knob [6] and tighten the set screw [5].

SUBTASK 38-11-05-420-038

(9) Push the faucet knob to start water flow.

(a) Repeat the test steps to measure the average flow time.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N AR9472-3W1D492 (Continued)

(b) If the average flow time is not in the required range, do the adjustment steps again.

SUBTASK 38-11-05-410-008

(10) Close the access to the lavatory module.

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-1

H. Water Faucet Adjustment

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-010-008

(1) Get access to the lavatory module.

SUBTASK 38-11-05-710-002

(2) To measure the length of time that the water flows, do these steps:

(a) Turn the faucet to the middle position (halfway between hot and cold).

(b) Push the knob [22] in fully and then release it.

1) Start the stopwatch, STD-836.

(c) When the faucet turns off, stop the stopwatch, STD-836, and record the time.

(d) Repeat these steps two more times (a total of three cycles).

(e) Turn the faucet to the full hot position and repeat the three cycles.

(f) Turn the faucet to the full cold position and repeat the three cycles.

(g) Calculate the average of the faucet times for each position.

NOTE: The average time for each position must be between 4 to 8 seconds.

1) If the average time is not between 4 to 8 seconds, continue with the steps that follow.

SUBTASK 38-11-05-820-002

CAUTION DEPRESSURIZE WATER SYSTEM BEFORE DISCONNECTING ANY WATER LINES OR WATER SYSTEM COMPONENTS TO PREVENT WATER SPILLAGE.

(3) If the length of time that the water flows is not satisfactory, do these steps:

(a) Loosen the set screw [23] to remove the knob [22].

(b) Loosen the set screw [25].

(c) Turn the plunger [24] to adjust timing.

1) Turn the plunger [24] 1/2 turn clockwise for slower timing.

2) Turn the plunger [24] 1/2 turn counterclockwise for faster timing.

SUBTASK 38-11-05-420-018

(4) Install the faucet knob [22].

(a) Do not tighten the set screw [23] until the faucet timing is correct.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-1 (Continued)

SUBTASK 38-11-05-420-039

(5) Repeat the test to time the faucet flow duration.

(a) If the faucet timing is not between 4 to 8 seconds then do the timing adjustment steps again.

(b) If the faucet timing is between 4 to 8 seconds, then continue with the steps that follow.

SUBTASK 38-11-05-420-019

(6) When the flow time is correct, do these steps:

(a) Remove the knob [22].

(b) Tighten the set screw [25] and torque to 1.5 ±0.5 in-lb (0.17 ±0.06 N·m).

(c) Install the knob [22].

(d) Apply Loctite 242 compound, A00648, to the set screw [23].

(e) Tighten the set screw [23] and torque to 8 ±1 in-lb (0.9 ±0.1 N·m).

SUBTASK 38-11-05-410-009

(7) Close the access to the lavatory module.

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-4 (4W200301-11) OR SC4L500301-5 (4101800-001).

I. Water Faucet Adjustment

WARNING WEAR SAFETY GOGGLES WHEN YOU REMOVE OR INSTALL COMPONENTS IN THE FAUCET. THE COMPONENTS ARE INSTALLED WITH COMPRESSION OR TENSION AND CAN RELEASE SUDDENLY. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 38-11-05-010-009

(1) Get access to the lavatory module.

SUBTASK 38-11-05-710-004

(2) To measure the length of time that the water flows, do these steps:

(a) Turn the faucet to the middle position (halfway between hot and cold).

(b) Push the knob [27] in fully and then release it.

1) Start the stopwatch, STD-836.

(c) When the faucet turns off, stop the stopwatch, STD-836, and record the time.

(d) Repeat these steps two more times (a total of three cycles).

(e) Turn the faucet to the full hot position and repeat the three cycles.

(f) Turn the faucet to the full cold position and repeat the three cycles.

(g) Calculate the average of the faucet times for each position.

NOTE: The average time for each position must be between 4 to 8 seconds.

1) If the average time is not between 4 to 8 seconds, continue with the steps that follow.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH FAUCET P/N SC4L500301-4 (4W200301-11) OR SC4L500301-5 (4101800-001). (Continued)

SUBTASK 38-11-05-820-005

CAUTION DEPRESSURIZE WATER SYSTEM BEFORE DISCONNECTING ANY WATER LINES OR WATER SYSTEM COMPONENTS TO PREVENT WATER SPILLAGE.

(3) If the length of time that the water flows is not satisfactory, do these steps:

(a) Loosen the set screw [23] to remove the knob [27].

(b) Loosen the plunger set screw [33].

(c) Turn the plunger [32] to adjust the flow duration as follows:

1) Turn the plunger [32] 1/2 turn clockwise to decrease flow duration.

2) Turn the plunger [32] 1/2 turn counterclockwise to increase flow duration.

SUBTASK 38-11-05-420-040

(4) Install the faucet knob [27].

(a) Do not tighten the set screw [23] until the faucet timing is correct.

SUBTASK 38-11-05-420-041

(5) Repeat the test to time the faucet flow duration.

(a) If the faucet timing is not between 4 to 8 seconds then do the timing adjustment steps again.

(b) If the faucet timing is between 4 to 8 seconds, then continue with the steps that follow.

SUBTASK 38-11-05-420-042

(6) When the flow time is correct, do these steps:

(a) Remove the knob [27].

(b) Remove the plunger set screw [33].

(c) Apply one drop (less than 0.04 in. (1 mm)) of Loctite 222MS compound, A50212, to the inside threads of the plunger [32] at the bottom corner of the plunger [32].

(d) Tighten the plunger set screw [33] and torque to 4.25 ±0.25 in-lb (0.48 ±0.03 N·m).

(e) Install the faucet knob [27].

(f) If required, apply Loctite 242 compound, A00648, to the set screw [23].

(g) Tighten the set screw [23] and torque to 8 ±1 in-lb (0.9 ±0.1 N·m).

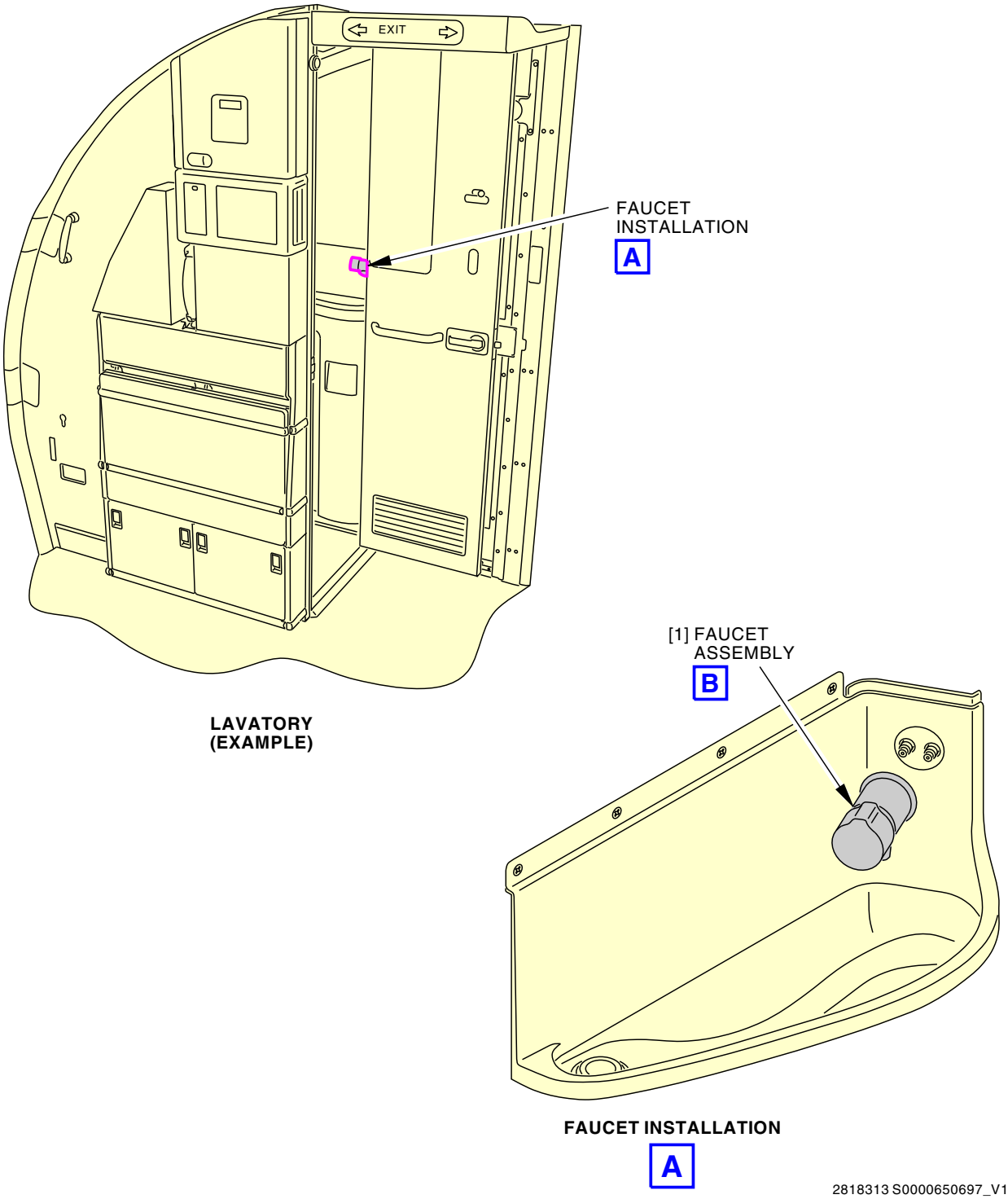
SUBTASK 38-11-05-410-010

(7) Close the access to the lavatory module.

PTF ALL

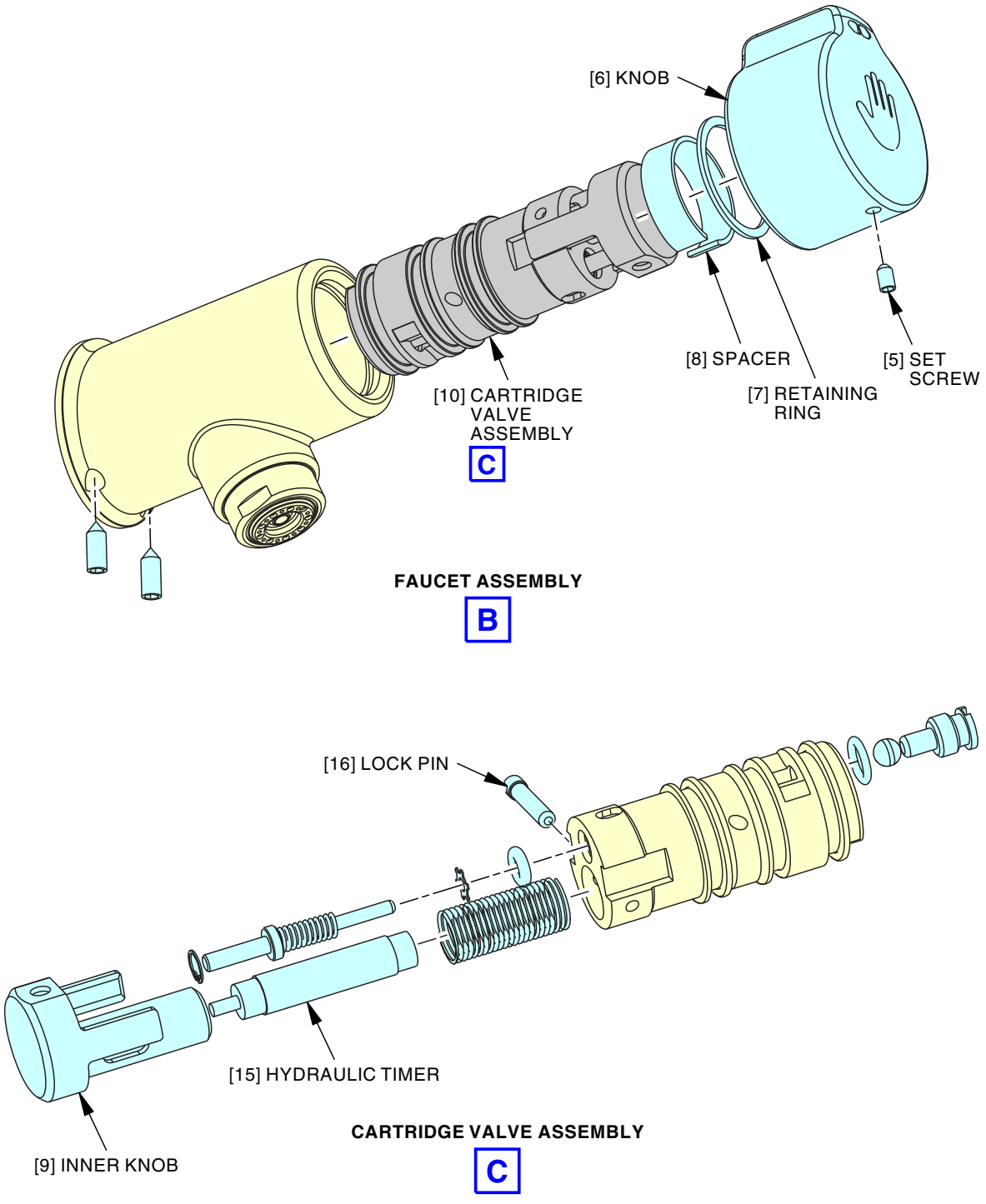
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



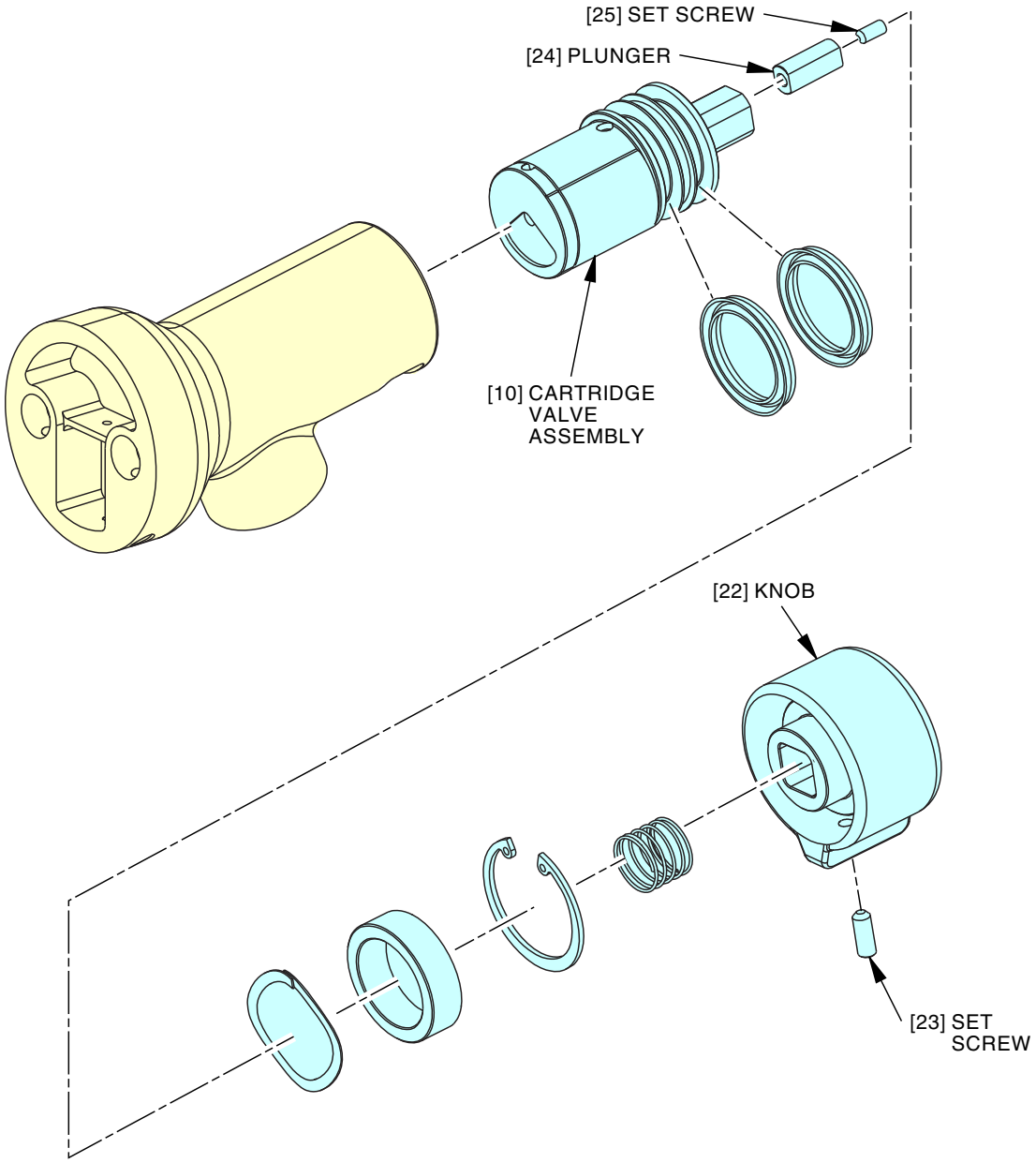
Lavatory Faucet Adjustment
Figure 203/38-11-05-990-804 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



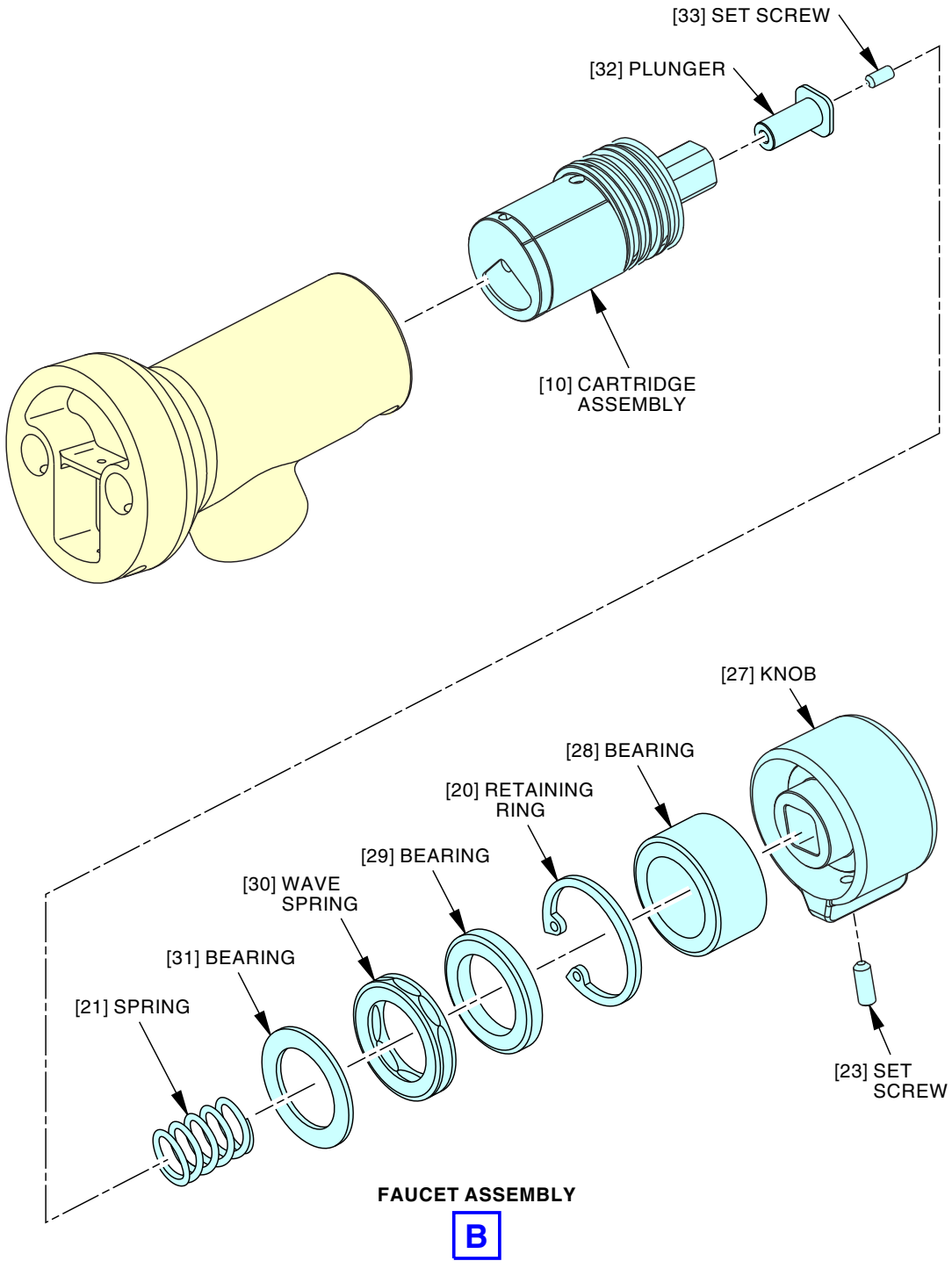
Lavatory Faucet Adjustment
Figure 203/38-11-05-990-804 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Faucet Adjustment
Figure 203/38-11-05-990-804 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Faucet Adjustment
Figure 203/38-11-05-990-804 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FORWARD LAVATORY DRAIN VALVE - MAINTENANCE PRACTICES

1. General

A. This procedure contains scheduled maintenance task data.

B. The Forward Lavatory Drain Valve is referred to as the Drain Valve in this procedure.

C. This procedure has this task:

(1) An operational check of the drain valve in the forward lavatory.

TASK 38-11-06-000-801

2. Forward Lavatory Drain Valve Operational Check

(Figure 201)

A. References

Reference	Title
38-42-00-800-802	Potable Water System - Activation (P/B 201)

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Prepare for the Operational Check

SUBTASK 38-11-06-860-001

(1) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-11-06-010-001

(2) Get access to the lavatory.

SUBTASK 38-11-06-010-002

(3) Open the shroud and access door on the lower vanity.

D. Drain Valve Operational Check

SUBTASK 38-11-06-860-002

(1) Make sure the water supply shutoff valve is in the SUPPLY ON position.

SUBTASK 38-11-06-790-001

(2) Make sure the handle for the drain valve is in the CLOSED position.

SUBTASK 38-11-06-860-003

(3) Open the water faucet, make sure water flows from the faucet..

NOTE: This step checks for water pressure. Water pressure is required for the drain valve leak check.

SUBTASK 38-11-06-860-004

(4) Close the water faucet.

SUBTASK 38-11-06-790-002

(5) Make sure that there is no water leakage from the drain port after approximately 5 minutes.

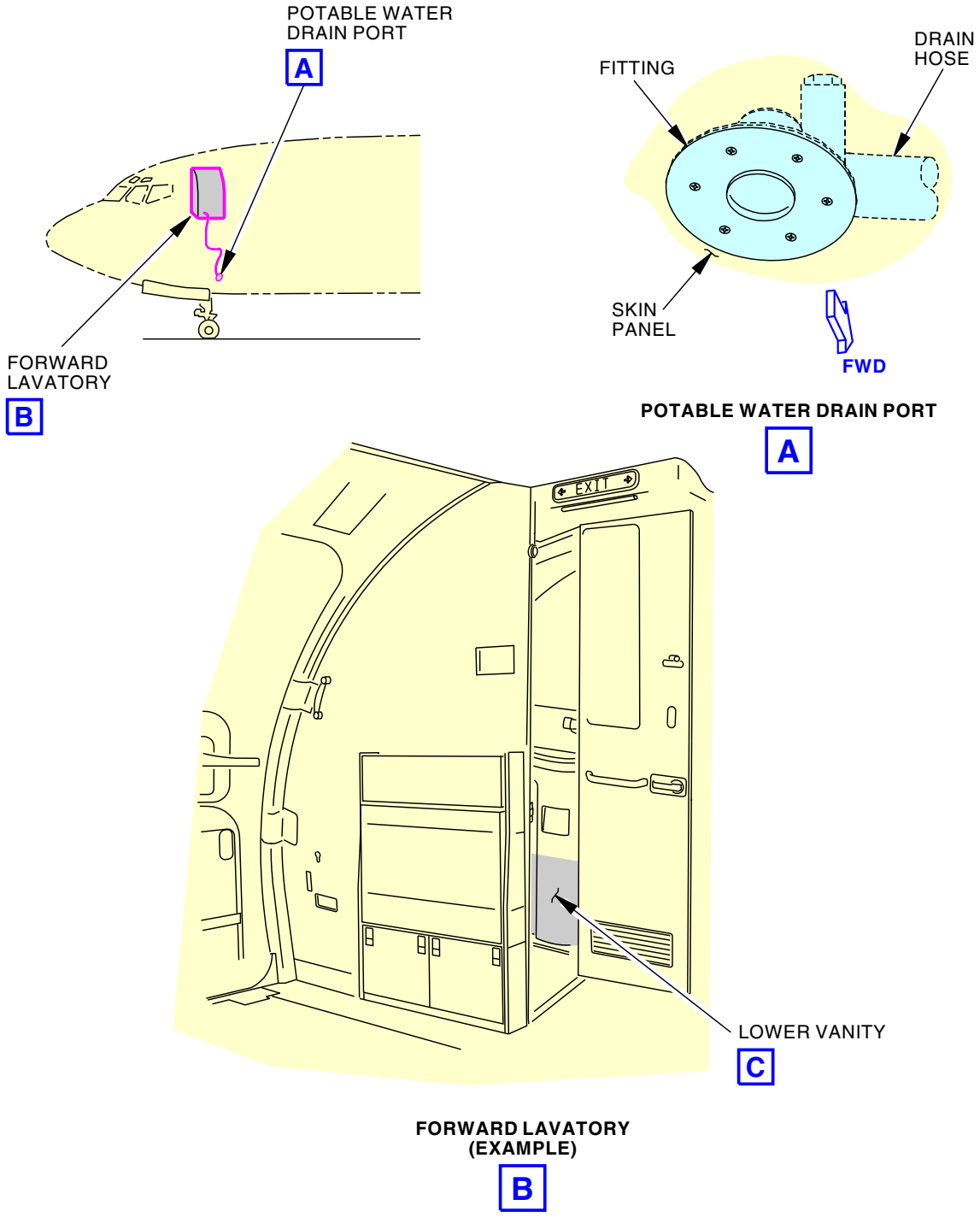
E. Put the Airplane Back in Its Usual Condition

SUBTASK 38-11-06-010-003

(1) Close the shroud and access door on the lower vanity.

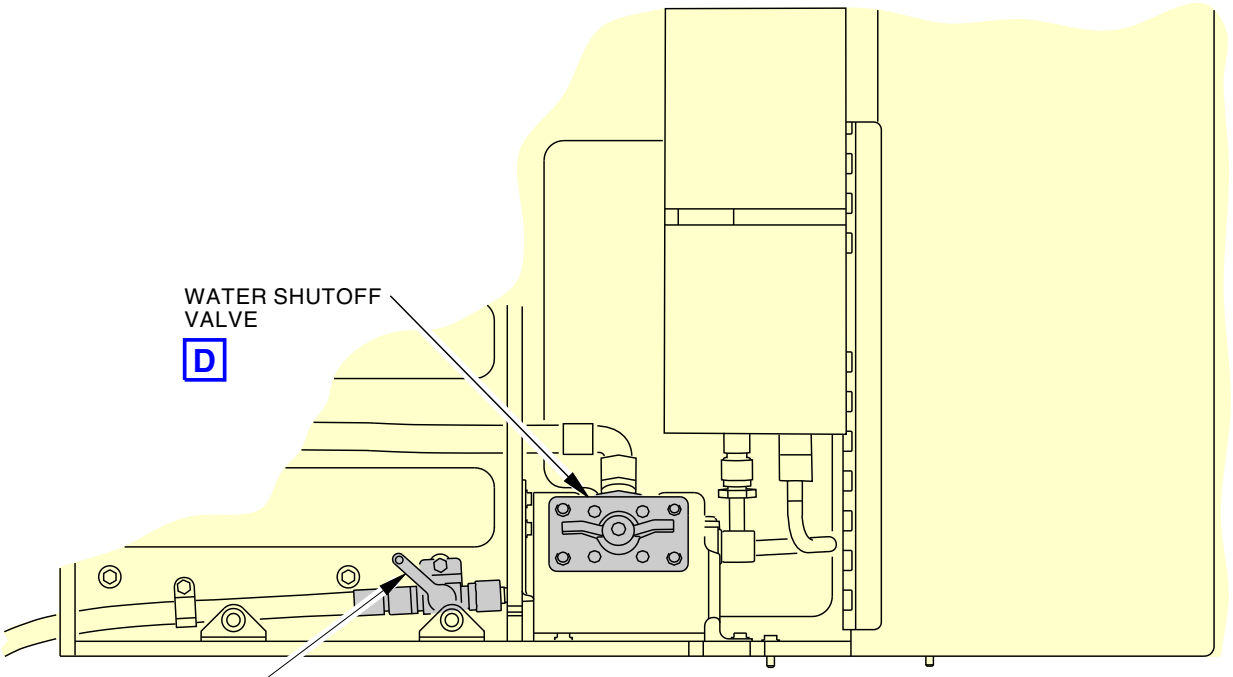
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



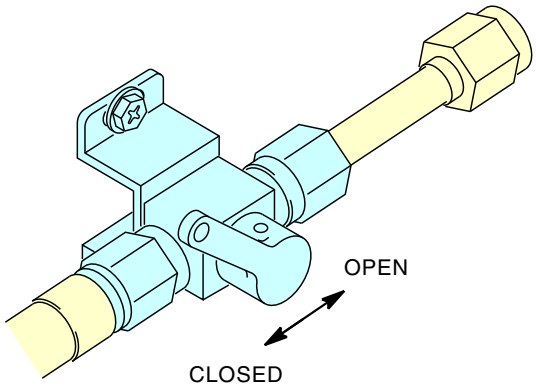
Forward Lavatory Water Drain - Operational Check
Figure 201/38-11-06-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



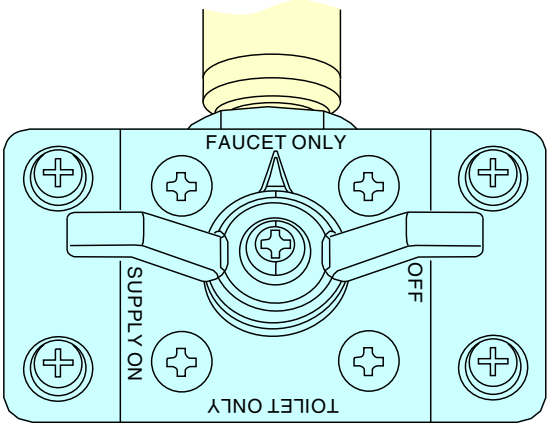
LOWER VANITY
(SHROUD ASSEMBLY AND DOOR ASSEMBLY ARE NOT SHOWN)

C



LAVATORY DRAIN VALVE

E



WATER SHUTOFF VALVE

D

2415154 S00061537112_V1

Forward Lavatory Water Drain - Operational Check
Figure 201/38-11-06-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FORWARD LAVATORY DRAIN VALVE - REMOVAL/INSTALLATION

1. General

A. The Forward Lavatory Drain Valve is referred to as the Drain Valve in this procedure.

B. This procedure has these tasks:

(1) A removal of the drain valve

(2) An installation of the drain valve.

TASK 38-11-06-000-802

2. Drain Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the drain valve.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for the Removal

SUBTASK 38-11-06-860-005

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	13	C00104	LAVATORY WATER HEATER A

SUBTASK 38-11-06-860-006

(2) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-11-06-010-004

(3) Get access to the lavatory.

SUBTASK 38-11-06-010-005

(4) Remove the shroud on the lower vanity to access the drain valve [1].

SUBTASK 38-11-06-010-006

(5) Open the door on the lower vanity to access the shutoff valve.

SUBTASK 38-11-06-860-007

(6) Turn the shutoff valve to the OFF position.

SUBTASK 38-11-06-860-008

(7) Move the drain valve handle to the OPEN position.

SUBTASK 38-11-06-860-012

(8) Open the faucet on the forward galley.

(a) Wait 1 minute to drain the remaining potable water or monitor the forward potable water drain port until water stops draining from it.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-06-860-013

(9) Close the faucet on the forward galley.

E. Drain Valve Removal

SUBTASK 38-11-06-020-001

(1) Loosen the B-nuts [4] for the hose fittings at the drain valve [1].

(a) Use a rag to catch the water in the hoses.

SUBTASK 38-11-06-020-002

(2) Remove the screw [3] from the drain valve [1].

SUBTASK 38-11-06-020-004

(3) If the washer [2] is installed on the drain valve [1], remove it.

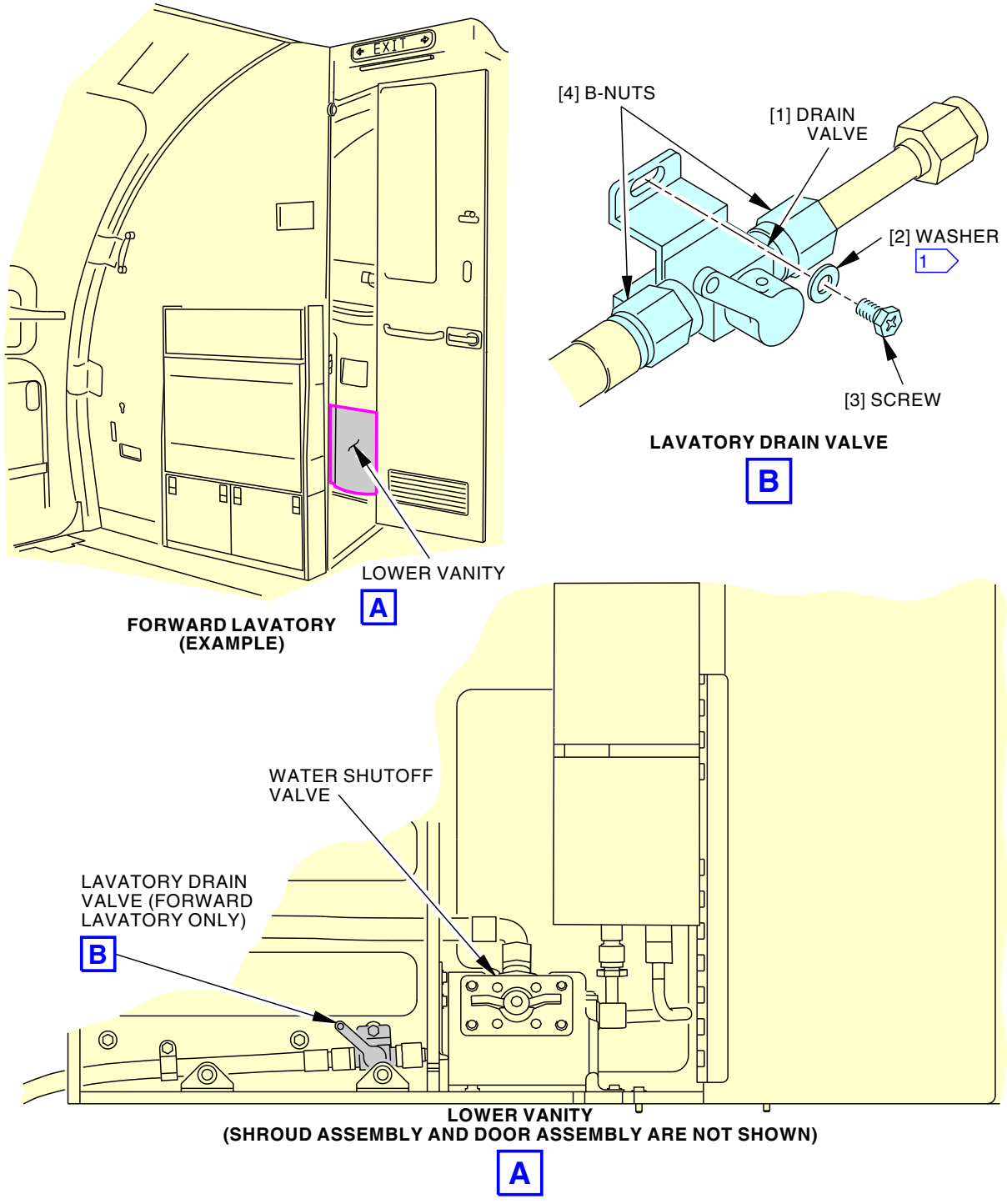
NOTE: The washer is installed not on all airplanes.

SUBTASK 38-11-06-020-003

(4) Remove the drain valve [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 THE WASHER SHOULD BE INSTALLED NOT ON ALL AIRPLANES

2415155 S00061537116_V2

Lavatory Water Drain Installation
Figure 401/38-11-06-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-11-06-400-801

3. Drain Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the drain valve.

B. References

Reference	Title
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Drain valve	38-11-06-01A-070	PTF ALL

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Drain Valve Installation

SUBTASK 38-11-06-410-001

(1) Put the drain valve [1] in position.

SUBTASK 38-11-06-420-003

(2) If the washer [2] was removed from the drain valve [1], install it.

NOTE: The washer is installed not on all airplanes.

SUBTASK 38-11-06-420-001

(3) Install the screw [3] to the drain valve [1].

SUBTASK 38-11-06-420-002

(4) Connect and tighten the hose fitting B-nuts [4] at the drain valve [1].

F. Drain Valve Installation Test

SUBTASK 38-11-06-440-001

(1) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-11-06-860-009

(2) Move the drain valve handle to the CLOSED position.

SUBTASK 38-11-06-860-010

(3) Turn the shutoff valve to the ON position.

SUBTASK 38-11-06-790-003

(4) Examine the drain valve [1] for leaks.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-11-06-860-011

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	13	C00104	LAVATORY WATER HEATER A

SUBTASK 38-11-06-410-002

(2) Install the shroud on the lower vanity.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-06-410-003

(3) Close the door on the lower vanity.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LAVATORY WATER SUPPLY SHUTOFF VALVE - REMOVAL/INSTALLATION

1. General

A. The Lavatory Water Supply Shutoff Valve is referred to as the Shutoff Valve in this procedure.

B. This procedure has two tasks:

(1) Shutoff valve removal

(2) Shutoff valve installation.

TASK 38-11-07-000-801

2. Shutoff Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the shutoff valve.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for the Removal

SUBTASK 38-11-07-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	13	C00104	LAVATORY WATER HEATER A
F	14	C01073	LAVATORY WATER HEATER D
F	15	C01096	LAVATORY WATER HEATER E

NOTE: You must open the applicable circuit breaker for the lavatory you work in.

SUBTASK 38-11-07-860-002

(2) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-11-07-010-001

(3) Get access to the lavatory.

SUBTASK 38-11-07-040-001

(4) Open the faucet to release the pressure and to drain some of the water.

SUBTASK 38-11-07-010-002

(5) Remove the shroud on the lower vanity to access the drain valve.

SUBTASK 38-11-07-010-003

(6) Open the door on the lower vanity to access the shutoff valve.

SUBTASK 38-11-07-860-003

(7) Move the drain valve handle to the OPEN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Shutoff Valve Removal

SUBTASK 38-11-07-020-001

(1) Loosen the hose fitting B-nut [14] and remove the water connections from the shutoff valve [11].

(a) Use a rag to catch the water in the water pipes.

SUBTASK 38-11-07-020-002

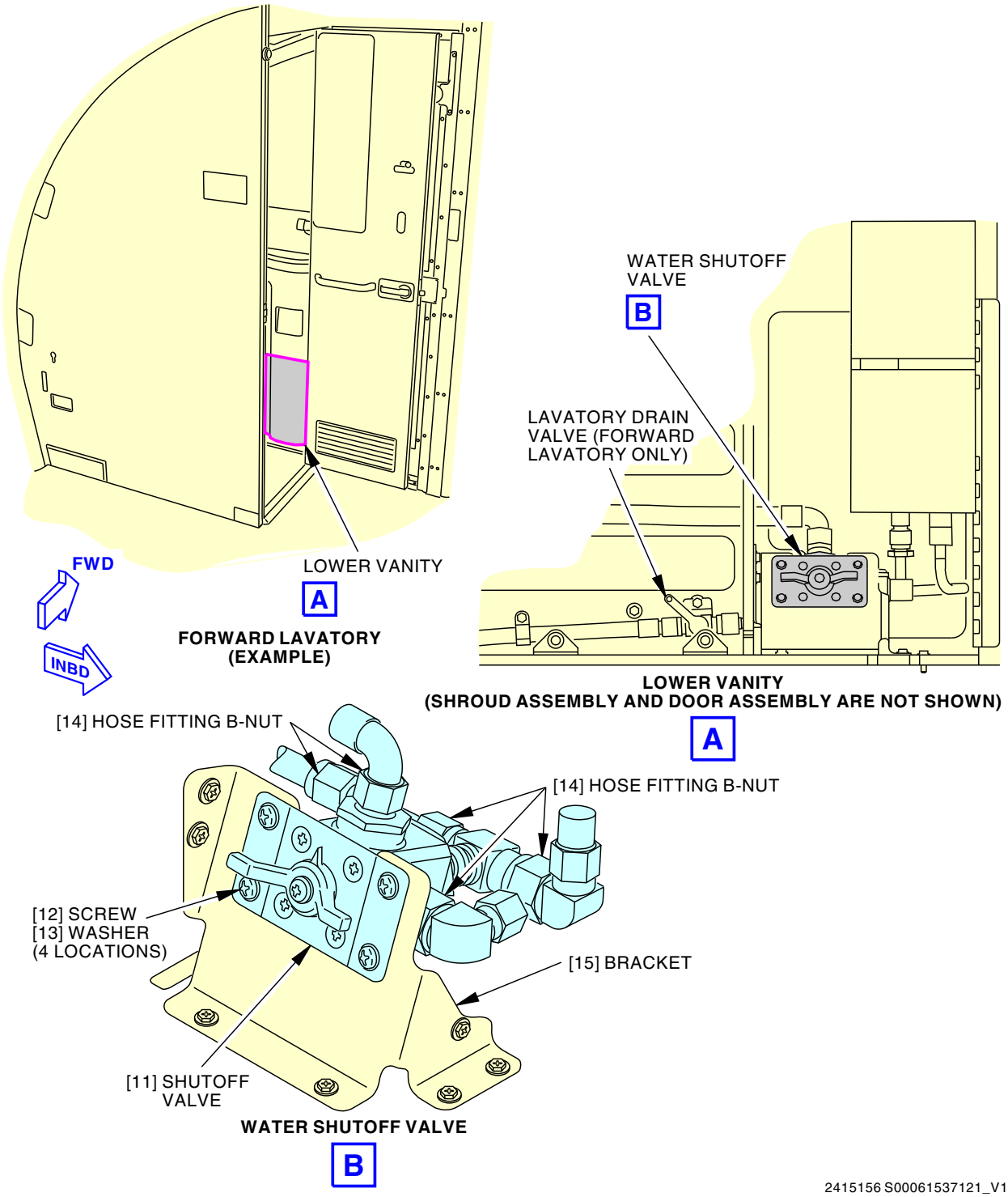
(2) Remove the screws [12] and washers [13].

SUBTASK 38-11-07-020-003

(3) Remove the shutoff valve [11] from the bracket [15].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Shutoff Valve Installation
Figure 401/38-11-07-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-11-07-400-801

3. Shutoff Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the shutoff valve.

B. References

Reference	Title
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Shutoff valve	38-11-51-11B-060	PTF ALL
		38-11-51-15B-285	PTF ALL

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Shutoff Valve Installation

SUBTASK 38-11-07-410-001

(1) Put the shutoff valve [11] in its position against the bracket [15].

SUBTASK 38-11-07-420-001

(2) Install the screws [12] and washers [13].

SUBTASK 38-11-07-420-002

(3) Connect the hose fitting B-nut [14].

F. Shutoff Valve Installation Test

SUBTASK 38-11-07-440-001

(1) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-11-07-860-004

(2) Move the drain valve handle to the CLOSED position.

SUBTASK 38-11-07-860-005

(3) Turn the shutoff valve to the ON position.

SUBTASK 38-11-07-790-001

(4) Check the shutoff valve connections for leaks.

G. Put the Airplane Back in Its Usual Condition

SUBTASK 38-11-07-860-006

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	13	C00104	LAVATORY WATER HEATER A
F	14	C01073	LAVATORY WATER HEATER D
F	15	C01096	LAVATORY WATER HEATER E

SUBTASK 38-11-07-410-002

(2) Install the shroud on the lower vanity.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-07-410-003

(3) Close the door on the lower vanity sink.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER TANK DRAIN VALVE AND CONTROL CABLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the water tank drain valve

(2) An installation of the water tank drain valve

(3) A removal of the water tank drain valve control cable

(4) An installation of the water tank drain valve control cable.

TASK 38-11-08-000-801

2. Water Tank Drain Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the water tank drain valve.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-11-08-860-001

(1) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-11-08-010-008

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-11-08-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-11-08-860-003

(4) To get access to the aft cargo compartment, open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-08-010-001

(5) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

F. Drain Valve Removal

SUBTASK 38-11-08-020-001

(1) Remove the clamshell [19] and packing [21] to disconnect the inlet from the drain valve [1].

SUBTASK 38-11-08-020-002

(2) Loosen the B-nut to disconnect the outlet from the drain valve [1].

SUBTASK 38-11-08-020-003

(3) Remove the screws [2] and washers [3].

SUBTASK 38-11-08-020-016

(4) Pull the control cable [12] and thermal isolator [20] to disconnect from the drain valve [1] bracket.

SUBTASK 38-11-08-020-004

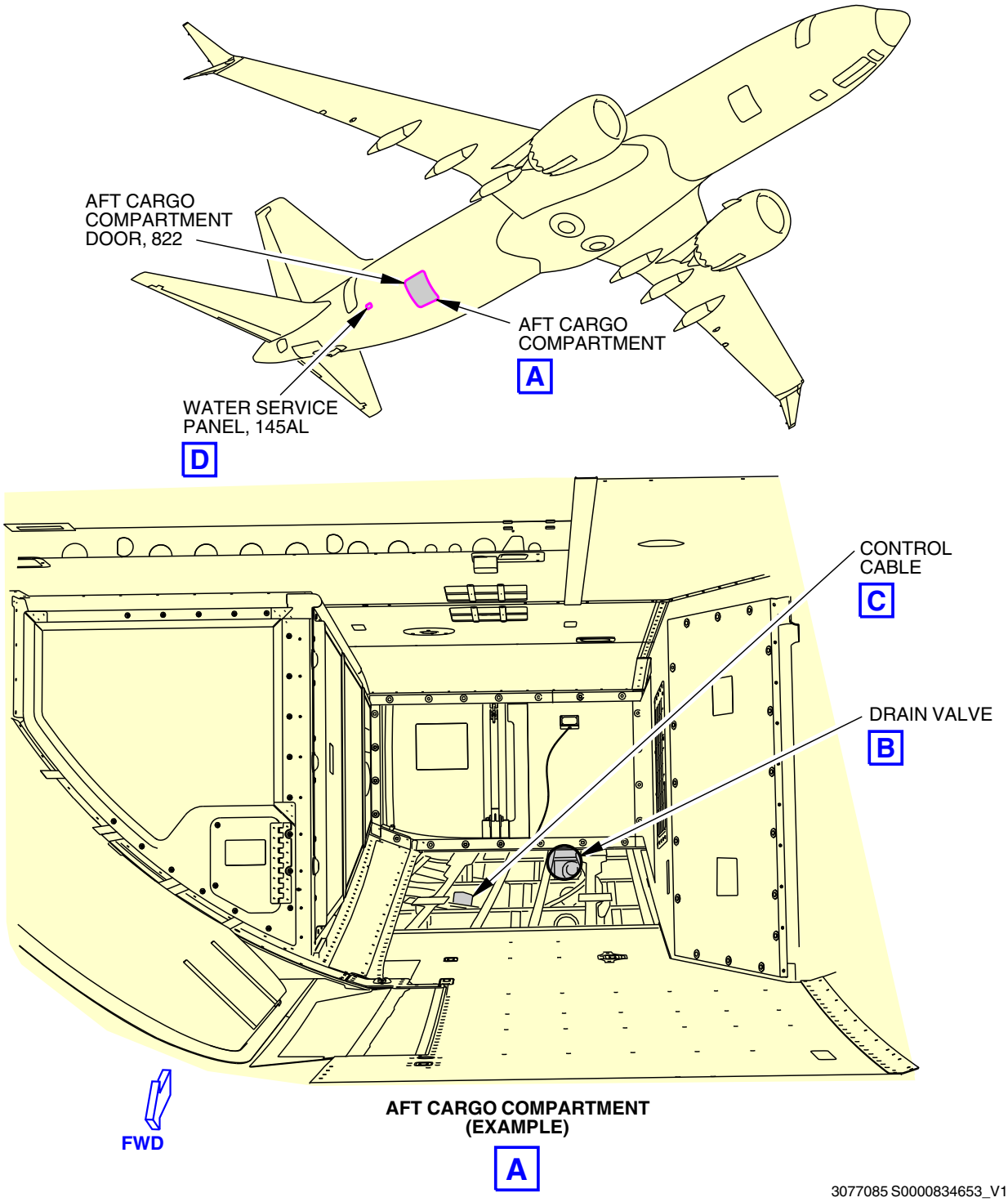
(5) Remove the bolts [4], nuts [6], and washers [5].

SUBTASK 38-11-08-020-005

(6) Remove the drain valve [1] and thermal isolators [23].

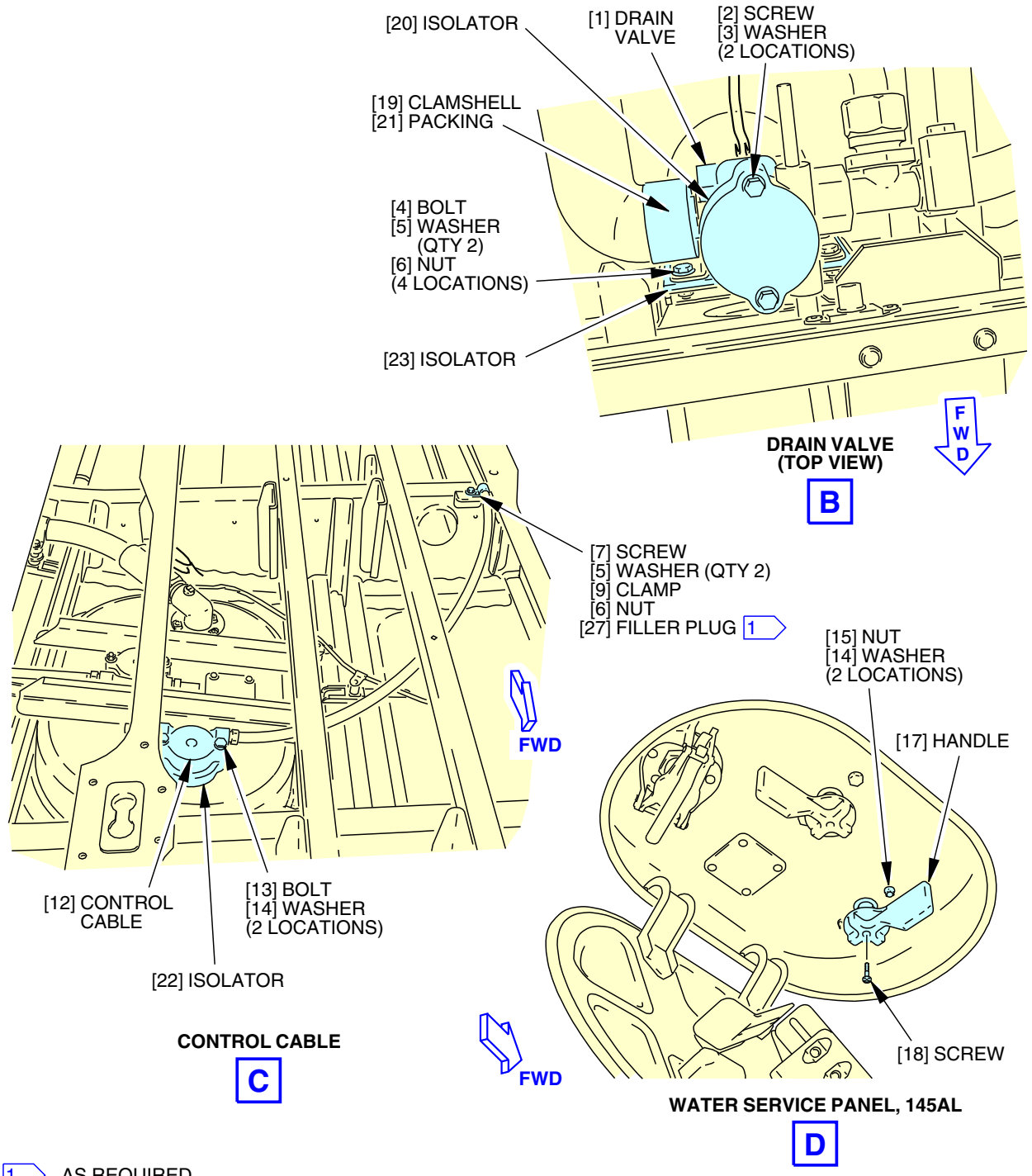
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Tank Drain Valve and Control Cable Installation
Figure 401/38-11-08-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Tank Drain Valve and Control Cable Installation
Figure 401/38-11-08-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-11-08-400-801

3. Water Tank Drain Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the water tank drain valve.

B. References

Reference	Title
38-10-00-600-801	Potable Water System Disinfectant (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Drain valve	38-11-08-03-090	PTF ALL
21	Packing	38-11-08-03-085	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Drain Valve Installation

SUBTASK 38-11-08-420-001

(1) Put the drain valve [1] and thermal isolators [23] in their position.

SUBTASK 38-11-08-420-002

(2) Install the bolts [4], nuts [6], and washers [5] for the drain valve [1].

SUBTASK 38-11-08-020-006

(3) Install the clamshell [19] and packing [21] to connect the drain valve [1].

SUBTASK 38-11-08-020-007

(4) Connect the outlet connection to the drain valve [1].

SUBTASK 38-11-08-420-003

(5) Align the arrows on the drain valve [1] and bracket to make sure that the drain valve [1] is in the closed position.

SUBTASK 38-11-08-420-004

(6) With the handle in the closed position at the service panel, align the drive shaft of the control cable [12] into the drain valve [1].

SUBTASK 38-11-08-420-005

(7) Install the screws [2] and washers [3] to connect the control cable [12] and thermal isolator [20] to the drain valve [1] bracket.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Drain Valve Installation Test

SUBTASK 38-11-08-860-004

> **WARNING** WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

> **WARNING** DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-11-08-410-007

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-11-08-860-006

(3) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-11-08-790-001

(4) Make sure that there are no leaks at the drain valve [1].

SUBTASK 38-11-08-670-001

(5) Do this task: Potable Water System Disinfectant, TASK 38-10-00-600-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-11-08-410-001

(1) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 38-11-08-410-002

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

TASK 38-11-08-020-801

4. Water Tank Drain Valve Control Cable Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the water tank drain valve control cable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
50-10-04-000-801	Cargo Compartment Floor Panel Removal (P/B 401)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

D. Access Panels

Number	Name/Location
146AR	Water Service Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-11-08-010-002

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-08-010-003

(2) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

SUBTASK 38-11-08-010-004

(3) To remove the floor panels for access to the control cable [12], do this task: Cargo Compartment Floor Panel Removal, TASK 50-10-04-000-801.

SUBTASK 38-11-08-010-005

(4) Open this access panel:

Number	Name/Location
146AR	Water Service Door

F. Water Tank Drain Valve Control Cable Removal

SUBTASK 38-11-08-020-008

(1) Remove the screws [2] and washers [3] that attach the control cable [12] to the drain valve [1].

SUBTASK 38-11-08-020-009

(2) Pull the control cable [12] and thermal isolator [20] from the support bracket at the drain valve [1].

SUBTASK 38-11-08-020-010

(3) Remove the screw [7], nut [6], and washers [5] to remove the clamp [9] from the control cable [12].

SUBTASK 38-11-08-020-011

(4) Remove the screw [18] to disconnect handle [17] for the control cable [12].

SUBTASK 38-11-08-020-012

(5) Remove the nuts [15] and washers [16] at the service panel.

SUBTASK 38-11-08-020-013

(6) Pull the control cable [12], bolts [13], and washers [14] from the thermal isolator [22] at the service panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-08-020-014

(7) Remove the control cable [12] and thermal isolator [22].

——— END OF TASK ———

TASK 38-11-08-420-801

5. Water Tank Drain Valve Control Cable Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the water tank drain valve control cable.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-10-04-400-801	Cargo Compartment Floor Panel Installation (P/B 401)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
12	Control cable	38-11-08-03-120	PTF ALL

D. Location Zones

Zone	Area
144	Area Below Aft Cargo Compartment - Right
146	Aft Cargo Compartment Equipment Bay - Right
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
146AR	Water Service Door
822	Aft Cargo Door

F. Water Tank Drain Valve Control Cable Installation

SUBTASK 38-11-08-420-006

(1) Put the bolts [13] and washers [14] into the control cable [12] at the service panel.

SUBTASK 38-11-08-420-007

(2) Put the control cable [12] and thermal isolator [22] in their position.

SUBTASK 38-11-08-420-008

(3) Install the nuts [15] and washers [16] for the control cable [12] at the service panel.

SUBTASK 38-11-08-420-009

(4) Install the handle [17] and screw [18] for the control cable [12].

SUBTASK 38-11-08-860-007

(5) Get access to the aft cargo compartment.

SUBTASK 38-11-08-020-015

(6) Install the screw [7], washers [5], clamp [9], and nut [6] on the control cable [12].

(a) If it is necessary, install the filler plug [27] in the 0.120 in. (3.0 mm) diameter hole of the clamp [9] to minimize the control cable [12] movement.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Trim the excess filler plug [27] 0.10 in. (2.5 mm) - 0.25 in. (6.4 mm) from the clamp [9].

SUBTASK 38-11-08-420-010

(7) Make sure that you align the arrows on the drain valve [1] and bracket to make sure that the drain valve [1] is in the closed position.

SUBTASK 38-11-08-420-011

(8) Put the valve end of the control cable [12] under the endwall support bracket.

SUBTASK 38-11-08-420-012

(9) With the handle in the closed position at the service panel, install loosely the screws [2], washers [3], and thermal isolator [20] for the control cable [12].

G. Water Tank Drain Valve Control Cable Installation Test

SUBTASK 38-11-08-010-006

(1) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-11-08-860-008

(2) Make sure that you have pressure in the potable water system.

(a) If you do not have pressure in the potable water system, do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-11-08-010-007

(3) Open this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-11-08-710-001

(4) Open the drain valve for the water tank to make sure that the water flows from the drain port.

SUBTASK 38-11-08-860-009

(5) Make sure that the drain valve for the water tank is fully open.

SUBTASK 38-11-08-710-002

(6) Close the drain valve for the aft system to make sure that the water flow stops.

SUBTASK 38-11-08-860-010

(7) Make sure that the drain valve for the water tank is fully closed.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-11-08-410-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Cargo Compartment Floor Panel Installation, TASK 50-10-04-400-801.

SUBTASK 38-11-08-410-004

(2) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-08-410-005

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-08-410-006

(4) Close this access panel:

Number	Name/Location
146AR	Water Service Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

POTABLE WATER FILL FITTING - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the potable water fill fitting

(2) An installation of the potable water fill fitting.

TASK 38-11-09-000-801

2. Potable Water Fill Fitting Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the potable water fill fitting.

B. References

Reference	Title
50-10-04-000-801	Cargo Compartment Floor Panel Removal (P/B 401)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
146AR	Water Service Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-11-09-860-004

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	18	C01463	WASTE/WTR LINE HEATERS
E	18	C01473	HOSE HEATERS

SUBTASK 38-11-09-010-005

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-09-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-11-09-010-001

(4) Open this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-11-09-040-001

(5) Pull the handle for the potable water fill valve to the OPEN position.

NOTE: This handle is on the water service panel.

SUBTASK 38-11-09-010-002

(6) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-09-010-003

(7) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

SUBTASK 38-11-09-010-004

(8) To remove the floor panel above the water service panel, do this task: Cargo Compartment Floor Panel Removal, TASK 50-10-04-000-801.

F. Potable Water Fill Fitting Removal

SUBTASK 38-11-09-020-006

(1) Remove the lockwire from the fill fitting heater [1] and cap assembly [4].

SUBTASK 38-11-09-020-001

(2) Disconnect the heated hose from the fill fitting heater [1].

SUBTASK 38-11-09-020-002

(3) Remove the bolts [5] that attach the cap assembly [4] to the structure.

SUBTASK 38-11-09-020-003

(4) Remove the thermal isolator [6] under the fill fitting heater [1].

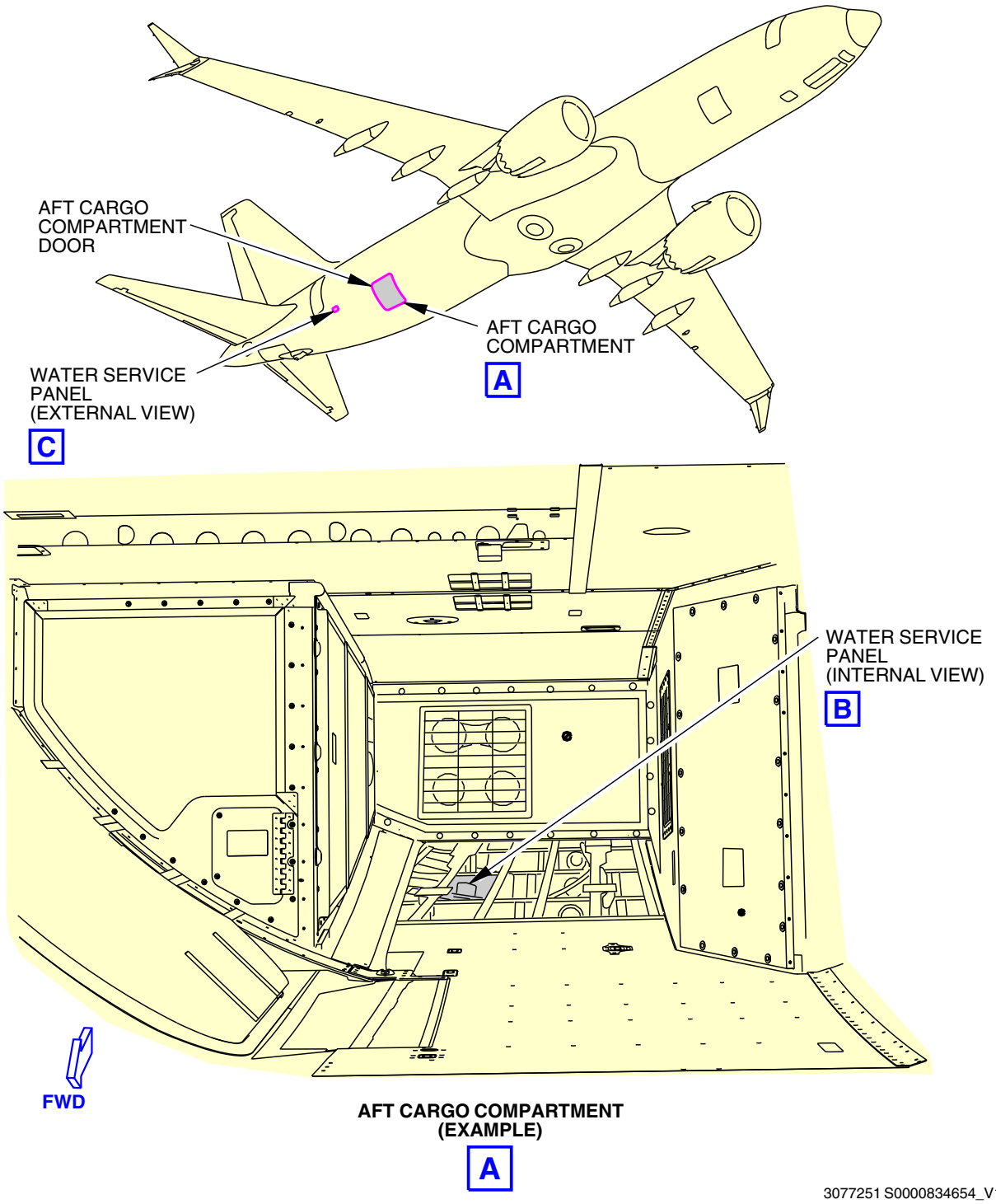
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-11-09-020-004

(5) Remove the thermal spacer [2] under the cap assembly [4].

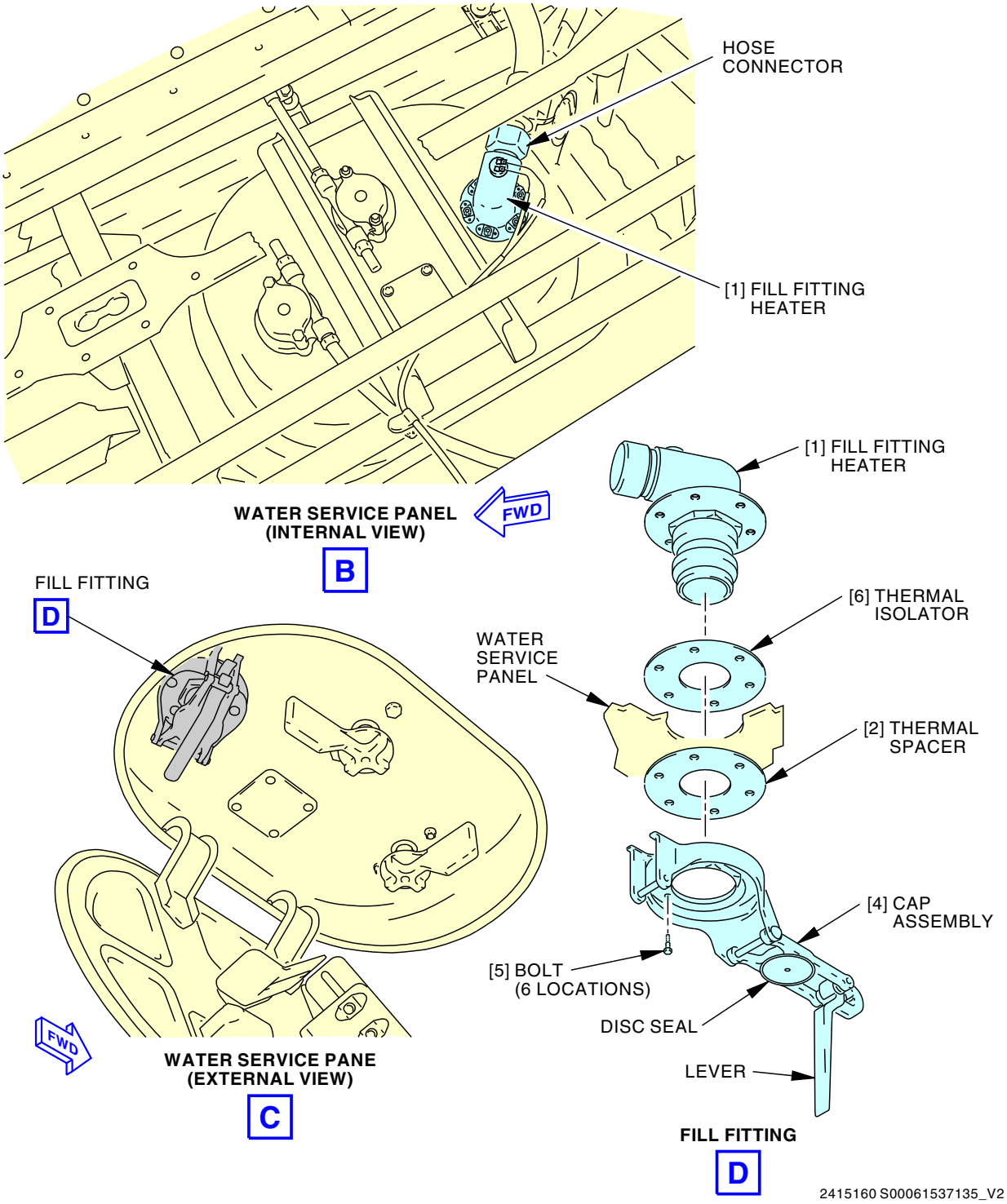
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Potable Water Fill Fitting Installation
Figure 401/38-11-09-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Potable Water Fill Fitting Installation
Figure 401/38-11-09-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-11-09-400-801

3. Potable Water Fill Fitting Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the potable water fill fitting.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
38-10-00-600-801	Potable Water System Disinfectant (P/B 201)
50-10-04-400-801	Cargo Compartment Floor Panel Installation (P/B 401)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fill fitting heater	30-71-01-01-050	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
146AR	Water Service Door
822	Aft Cargo Door

G. Potable Water Fill Fitting Installation

SUBTASK 38-11-09-420-003

(1) Apply sealant, A00247, to the thermal isolator [6], thermal spacer [2], fill fitting heater [1], and cap assembly [4] on the surface that is adjacent to the structure.

SUBTASK 38-11-09-420-004

(2) Put the fill fitting heater [1] and thermal spacer [2] in their positions.

SUBTASK 38-11-09-420-005

(3) Put the cap assembly [4] and thermal isolator [6] in their positions.

SUBTASK 38-11-09-420-006

(4) Install the bolts [5] wet with sealant, A00247, to connect the cap assembly [4] to the structure.

SUBTASK 38-11-09-420-007

(5) Connect the heated hose to the fill fitting heater [1].

(a) Tighten the heated hose to the fill fitting heater [1] to 620 ±80 in-lb (70 ±9 N·m).

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SUBTASK 38-11-09-420-008

(6) Use the double twist method to install the lockwire through the nipple and cap assembly [4] (TASK 20-10-44-400-801).

SUBTASK 38-11-09-670-001

(7) Do this task: Potable Water System Disinfectant, TASK 38-10-00-600-801.

H. Potable Water Fill Fitting Installation Test

SUBTASK 38-11-09-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-11-09-410-005

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-11-09-860-005

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	18	C01463	WASTE/WTR LINE HEATERS
E	18	C01473	HOSE HEATERS

SUBTASK 38-11-09-710-001

(4) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

(a) Make sure that there is no leakage at the water fill fitting.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put Airplane Back to Its Usual Condition

SUBTASK 38-11-09-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Cargo Compartment Floor Panel Installation, TASK 50-10-04-400-801.

SUBTASK 38-11-09-410-002

(2) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 38-11-09-410-003

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-11-09-410-004

(4) Close this access panel:

Number	Name/Location
146AR	Water Service Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER HEATER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Water heater removal

(2) Water heater installation.

B. The water heater is below the sink in the lavatory compartment.

TASK 38-13-01-000-801

2. Water Heater Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the water heater.

B. References

Reference	Title
12-14-00-600-801	Potable Water System - Drain (P/B 301)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Prepare for the Removal

SUBTASK 38-13-01-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	13	C00104	LAVATORY WATER HEATER A
F	14	C01073	LAVATORY WATER HEATER D
F	15	C01096	LAVATORY WATER HEATER E

NOTE: You must open the applicable circuit breaker.

SUBTASK 38-13-01-860-002

(2) Get access to the lavatory.

SUBTASK 38-13-01-040-003

(3) Put the toggle on/off switch to the OFF position.

SUBTASK 38-13-01-010-001

(4) Open the shroud and access door on the lower vanity.

SUBTASK 38-13-01-040-001

(5) Turn the water shutoff valve to the OFF position.

NOTE: The water shutoff valve is below the sink in the lavatory.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-13-01-680-001

(6) On the forward lavatories, open the drain valve.

NOTE: Only the forward lavatories have the drain valve.

SUBTASK 38-13-01-040-002

(7) Open the faucet to release the pressure and to drain some of the water.

SUBTASK 38-13-01-680-002

(8) If necessary, do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

F. Water Heater Removal

SUBTASK 38-13-01-020-001

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [3] from the water heater [1] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [3] from the water heater [1].

(b) Install the protective covers on the electrical receptacle of the water heater [1] and the electrical connector [3].

SUBTASK 38-13-01-020-002

(2) Disconnect the water outlet hose from the side of the water heater [1].

(a) Put a cloth below the water heater [1] to absorb the unwanted water.

SUBTASK 38-13-01-020-003

(3) Disconnect the water inlet from the bottom of the water heater [1].

(a) Put a cloth below the water heater [1] to absorb the unwanted water.

SUBTASK 38-13-01-020-004

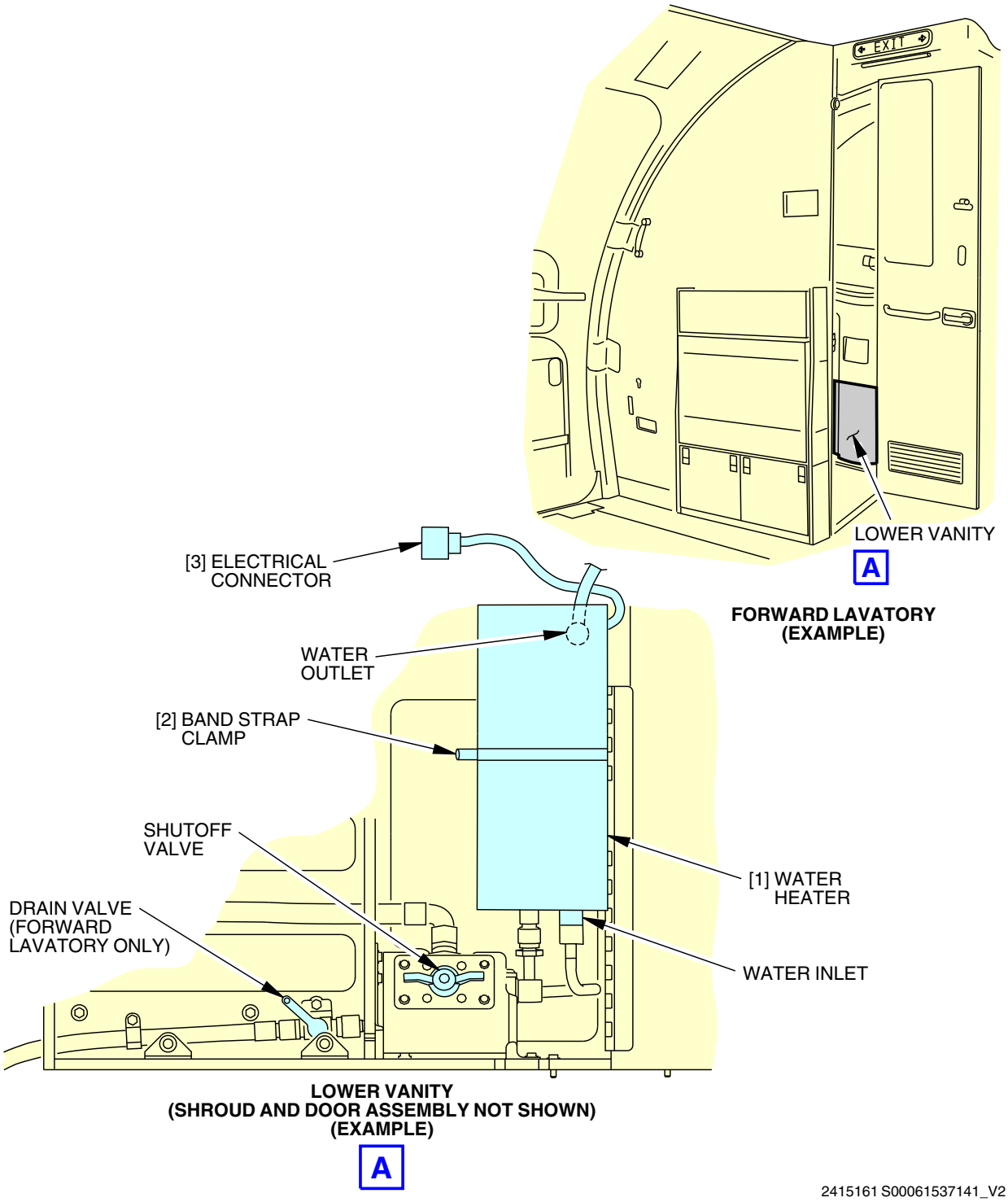
(4) Disconnect the band strap clamp [2].

SUBTASK 38-13-01-020-005

(5) Remove the water heater [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Heater Installation
Figure 401/38-13-01-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-13-01-400-801

3. Water Heater Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the water heater.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Water heater	38-13-01-01B-050	PTF ALL
		38-13-01-02A-050	PTF ALL
		38-13-01-03N-050	PTF ALL

E. Location Zones

Zone	Area
200	Upper Half of Fuselage

F. Water Heater Installation

SUBTASK 38-13-01-420-001

(1) Put the water heater [1] in its position.

SUBTASK 38-13-01-420-002

(2) Connect the band strap clamp [2] for the water heater [1].

NOTE: Do not tighten the clamp [2] until a step that follows.

SUBTASK 38-13-01-420-003

(3) Connect the water inlet to the bottom of the water heater [1].

SUBTASK 38-13-01-420-004

(4) Connect the water outlet hose to the side of the water heater [1].

SUBTASK 38-13-01-420-005

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(5) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [3] to the water heater [1] (TASK 70-00-01-910-803-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the protective covers from the electrical receptacle of the water heater [1] and the electrical connector [3].

(b) Connect the electrical connector [3] to the water heater [1].

1) Tighten the connector with the teflon-jawed pliers, STD-664.

SUBTASK 38-13-01-420-006

(6) Tighten the band strap clamp [2].

G. Put the Airplane Back in the Usual Condition.

SUBTASK 38-13-01-680-003

(1) If necessary, do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-13-01-860-003

(2) On the forward lavatories, close the drain valve.

NOTE: Only the forward lavatories have the drain valve.

SUBTASK 38-13-01-440-001

(3) Turn the water shutoff valve to the ON position.

SUBTASK 38-13-01-860-004

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
F	13	C00104	LAVATORY WATER HEATER A
F	14	C01073	LAVATORY WATER HEATER D
F	15	C01096	LAVATORY WATER HEATER E

SUBTASK 38-13-01-440-002

(5) Put the toggle on/off switch to the ON position.

SUBTASK 38-13-01-790-001

(6) Make sure that the water hose connections at the water heater [1] do not have a leak.

SUBTASK 38-13-01-710-001

(7) After 5 minutes, open the hot water faucet to make sure that the water is hot.

(a) If the water is not at the correct temperature, move the temperature select switch to a higher or lower temperature position.

SUBTASK 38-13-01-020-006

(8) Remove the cloth from below the water heater [1].

SUBTASK 38-13-01-410-001

(9) Close the door to the cabinet below the sink.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER HEATER - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) Water heater adjustment

B. The water heater is below the sink in the lavatory compartment.

TASK 38-13-01-730-801

2. Water Heater Adjustment

(Figure 501)

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-933	Thermometer - Digital

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Water Heater Adjustment

SUBTASK 38-13-01-800-001

(1) If the power ON indicator is not illuminated do the steps that follow:

(a) Make sure that the power on/off switch is in the ON position.

(b) Push the manual RESET button.

SUBTASK 38-13-01-730-001

(2) Select the temperature with a three-position switch mounted on the top cover.

(a) LOW 105 ±5°F (40.56 ±2.78°C):

(b) MED: 115 ±5°F (46.11 ±2.78°C)

(c) HIGH: 125 ±5°F (51.67 ±2.78°C)

SUBTASK 38-13-01-880-001

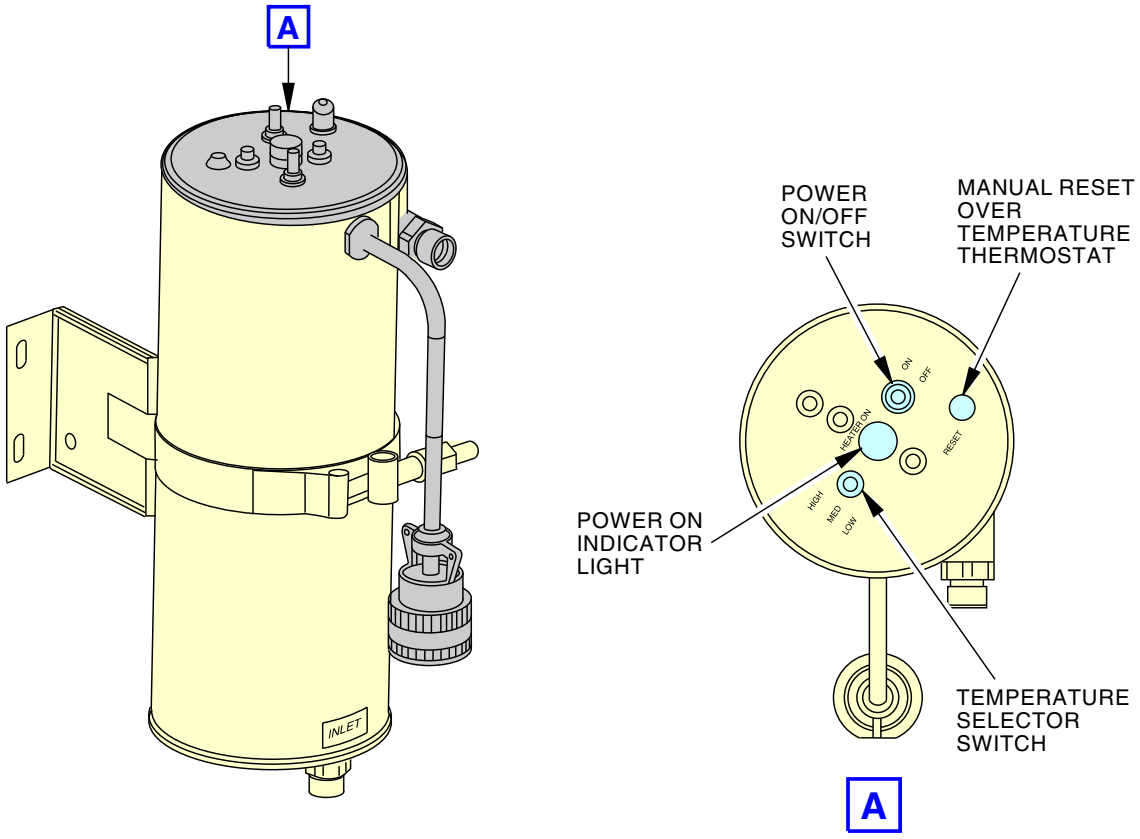
(3) Do a check of the water temperature as follows:

(a) Open the faucet and run the water in the sink for a minimum of 60 seconds then close the faucet.

(b) Wait a minimum of 5 minutes then open the faucet and with a digital thermometer, COM-933 check the water temperature to the applicable temperature setting above.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Water Heater Adjustment/Test
Figure 501/38-13-01-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER QUANTITY TRANSMITTER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the water quantity transmitter

(2) An installation of the water quantity transmitter.

TASK 38-14-01-000-801

2. Water Quantity Transmitter Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the water quantity transmitter.

B. References

Reference	Title
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2481	Sealant Removal Tool (Meets BSS7384 Requirements) Part #: 1-6390-A Supplier: 63318 Part #: 10810 Supplier: $0855 Part #: 10811 Supplier: $0855 Part #: 10812 Supplier: $0855 Part #: 234350 Supplier: 5HCF1 Part #: 235072 Supplier: 5HCF1 Part #: 235073 Supplier: 5HCF1 Part #: 235074 Supplier: 5HCF1 Part #: 235075 Supplier: 5HCF1 Part #: 235076 Supplier: 5HCF1 Part #: 311/03 Supplier: F6892 Part #: 311/14 Supplier: F6892 Part #: 311/25 Supplier: F6892 Part #: 311/37 Supplier: F6892 Part #: AS1 Supplier: $1351 Part #: AS2 Supplier: $1351 Part #: AS3 Supplier: $1351 Part #: DAD5013 Supplier: 7RKH2 Part #: DFD5019 Supplier: 7RKH2 Part #: JNT411B60 Supplier: 3DN12 Part #: JNT411B90 Supplier: 3DN12 Part #: PT6529-S Supplier: 81205 Part #: SCD5019 Supplier: 7RKH2 Part #: ST982LF-9 Supplier: 81205 Part #: TS1275-4 Supplier: 22975
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-14-01-010-005

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-14-01-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-14-01-010-001

(3) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-14-01-860-002

(4) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

SUBTASK 38-14-01-010-007

(5) Remove the electrical equipment rack, E6, aft access panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Water Quantity Transmitter Removal

SUBTASK 38-14-01-020-001

> **CAUTION** MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

> **CAUTION** USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [17] from the water quantity transmitter [1] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [17] from the water quantity transmitter [1].

(b) Install the protective covers on the electrical receptacle of the water quantity transmitter [1] and electrical connector [17].

SUBTASK 38-14-01-020-002

(2) Remove the screws [18] and washers [19] that attach the cover [20] to the water tank.

SUBTASK 38-14-01-020-003

(3) Remove the screw [21] and washer [22] that attach the connector [23] to the terminal.

(a) Use the sealant removal tool, COM-2481, or equivalent, to carefully remove sealant that was applied to the connector [23].

SUBTASK 38-14-01-020-004

(4) Remove the screws [2], washers [3], clamp [4], clamp [5], and nuts [6] to disconnect the coaxial wire from the structure.

SUBTASK 38-14-01-020-005

(5) Remove the screws [7], washers [8], clamp [9], and spacer [10] to disconnect the coaxial wire from the structure.

SUBTASK 38-14-01-020-006

(6) Remove the screws [15] and nuts [16] to disconnect the transmitter connector from the bracket [13].

SUBTASK 38-14-01-020-007

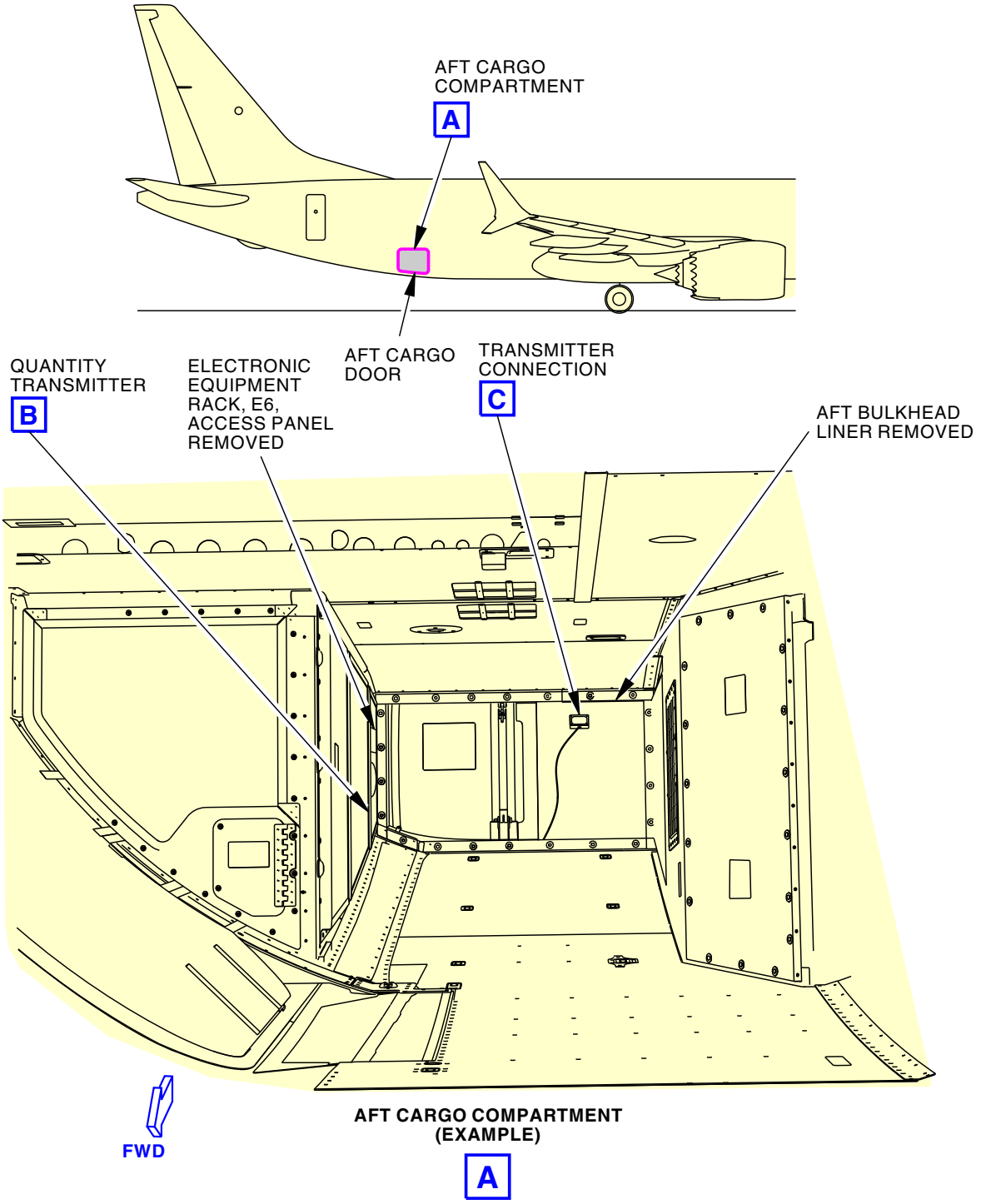
(7) Remove the screws [11] and washers [12].

SUBTASK 38-14-01-020-008

(8) Remove the water quantity transmitter [1].

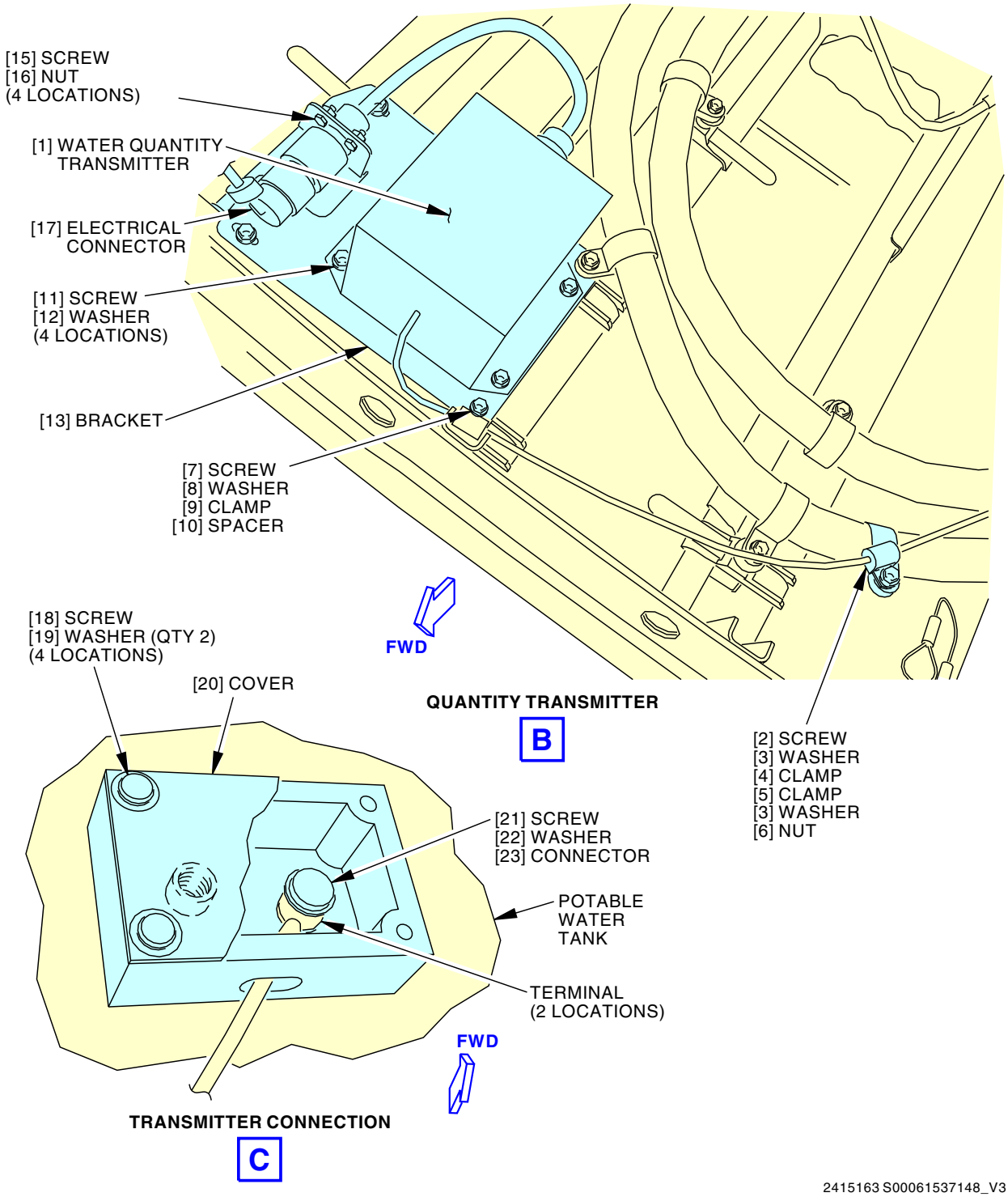
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Quantity Transmitter Installation
Figure 401/38-14-01-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Quantity Transmitter Installation
Figure 401/38-14-01-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-14-01-400-801

3. Water Quantity Transmitter Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the water quantity transmitter.

B. References

Reference	Title
38-14-01-800-801	Water Quantity Transmitter Adjustment (P/B 501)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00130	Alcohol - Isopropyl	TT-I-735
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Water quantity transmitter	38-14-01-04-075	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

H. Water Quantity Transmitter Installation

SUBTASK 38-14-01-420-001

(1) Put the water quantity transmitter [1] in its position on the bracket [13].

SUBTASK 38-14-01-420-002

(2) Install the screws [11] and washers [12] to connect the water quantity transmitter [1] to the bracket [13].

SUBTASK 38-14-01-420-003

(3) Install the screws [15] and nuts [16] to connect the transmitter connector to the bracket [13].

SUBTASK 38-14-01-420-004

(4) Install the screws [7], washers [8], clamp [9], and spacer [10] to connect the coaxial wire to the structure.

SUBTASK 38-14-01-420-005

(5) Install the screws [2], washers [3], clamp [4], clamp [5], and nuts [6] to connect the coaxial wire to the structure.

SUBTASK 38-14-01-420-006

(6) Install the connector [23] in its position on the water tank as follows:

NOTE: There are two transmitter cable connector terminals on the water tank. The transmitter cable connector may be connected to either terminal. Each terminal is attached to the sensor grid within the tank sidewall.

(a) Use cotton wiper, G00034, moist in alcohol, B00130, to clean the mating surfaces of the screw [21], washer [22], and connector [23].

(b) Install the screw [21] and washer [22] to attach the connector [23] to the terminal.

(c) Tighten the screw [21] to 35 ±2 in-lb (4.0 ±0.2 N·m).

(d) Measure the electrical bonding resistance between the washer [22] and second connector terminal in the housing (SWPM 20-20-00).

1) Use the non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.008 ohm (8 milliohm) or less.

(e) Apply fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, to the screw [21], washer [22], and connector [23].

SUBTASK 38-14-01-420-007

(7) Install the screws [18] and washers [19] to attach the cover [20] to the water tank.

SUBTASK 38-14-01-420-008

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(8) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [17] to the water quantity transmitter [1] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the water quantity transmitter [1] and electrical connector [17].

(b) Connect the electrical connector [17] to the water quantity transmitter [1].

1) Tighten the electrical connector [17] with the teflon-jawed pliers, STD-664.

I. Water Quantity Transmitter Installation Test

SUBTASK 38-14-01-860-003

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-14-01-410-006

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-14-01-820-001

(3) Do this task: Water Quantity Transmitter Adjustment, TASK 38-14-01-800-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-14-01-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER QUANTITY TRANSMITTER - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) An adjustment of the water quantity transmitter.

TASK 38-14-01-800-801

2. Water Quantity Transmitter Adjustment

(Figure 501)

A. References

Reference	Title
12-14-00-600-801	Potable Water System - Drain (P/B 301)
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1793	Multimeter - Digital/Analog (or equivalent meter meets task requirements) Part #: 117 Supplier: 89536 Part #: 260-8XPI Supplier: 55026 Part #: 287 Supplier: 89536 Part #: 289 Supplier: 89536 Part #: 87V Supplier: 89536 Part #: FLUKE 27 II Supplier: 89536 Part #: FLUKE-77-4 Supplier: 89536 Part #: WA2025 Supplier: 4MJL8 Opt Part #: 187 Supplier: 89536 Opt Part #: 189 Supplier: 89536 Opt Part #: 21 Supplier: 89536 Opt Part #: 27 Supplier: 89536 Opt Part #: 77 SERIES III Supplier: 89536 Opt Part #: 87 Supplier: 89536 Opt Part #: FLUKE 27 Supplier: 89536 Opt Part #: MODEL 27 Supplier: 89536
SPL-1949	Cable - Test, Potable Water Quantity Transmitter Part #: A38010-11 Supplier: 81205 Part #: A38010-45 Supplier: 81205

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Adjustment

SUBTASK 38-14-01-410-003

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-14-01-010-008

(2) Remove the electrical equipment rack, E6, aft access panel.

SUBTASK 38-14-01-610-002

(3) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

NOTE: Keep the water hose connected after the water tank is filled.

SUBTASK 38-14-01-680-001

(4) Do this task: Potable Water System - Drain, TASK 12-14-00-600-801.

NOTE: The inner area of the water tank must be wet to make an accurate calibration of the water quantity transmitter.

SUBTASK 38-14-01-010-004

(5) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-14-01-860-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND

SUBTASK 38-14-01-020-009

(7) Disconnect the D192 airplane electrical connector from the adapter cable.

SUBTASK 38-14-01-480-001

(8) Install the potable water quantity transmitter test cable, SPL-1949, between the D192 airplane electrical connector and adapter cable.

SUBTASK 38-14-01-480-002

(9) Connect the digital/analog multimeter, COM-1793, to the test points on the test cable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-14-01-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(10) Close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND

F. Water Quantity Transmitter Adjustment

SUBTASK 38-14-01-820-002

(1) Measure the voltage between the test points.

SUBTASK 38-14-01-820-003

(2) If the voltage is not 0 ±0.2 vdc, do these steps:

(a) On the potable water transmitter, find the adjustment screw labeled "Empty".

(b) Turn the adjustment screw until the voltage is 0 ±0.2 vdc on the multimeter.

SUBTASK 38-14-01-610-003

(3) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-14-01-820-004

(4) Measure the voltage between the test points on the cable.

SUBTASK 38-14-01-820-005

(5) If the voltage is not -10 ±0.2 vdc, do these steps:

(a) Find the adjustment screw with the FULL mark.

(b) Turn this FULL adjustment screw until the voltage is -10 ±0.2 vdc on the multimeter.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-14-01-860-007

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND

SUBTASK 38-14-01-080-001

(2) Disconnect the digital/analog multimeter, COM-1793, from the cable.

SUBTASK 38-14-01-080-002

(3) Remove the potable water quantity transmitter test cable, SPL-1949.

SUBTASK 38-14-01-420-009

(4) Connect the D192 airplane electrical connector to the adapter cable.

SUBTASK 38-14-01-860-009

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	9	C00138	WATER QTY IND

SUBTASK 38-14-01-410-008

(6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-14-01-860-010

(7) Make sure that the indicator on the Attendant Control Panel (ACP) shows FULL.

SUBTASK 38-14-01-410-009

(8) Install the electrical equipment rack, E6, aft access panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

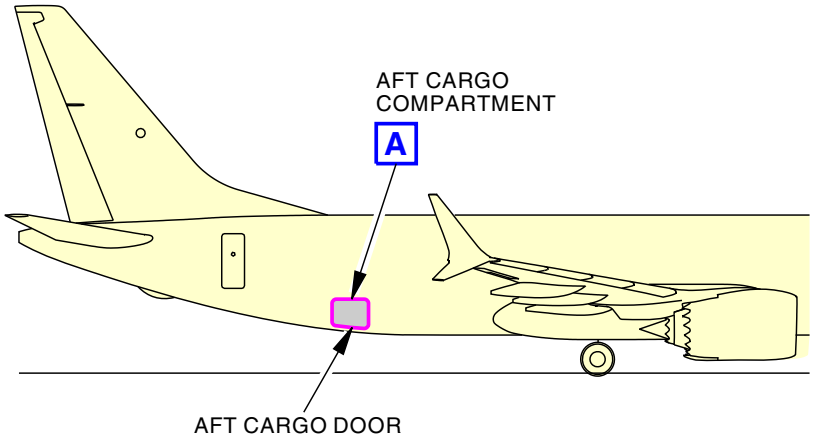
SUBTASK 38-14-01-410-005

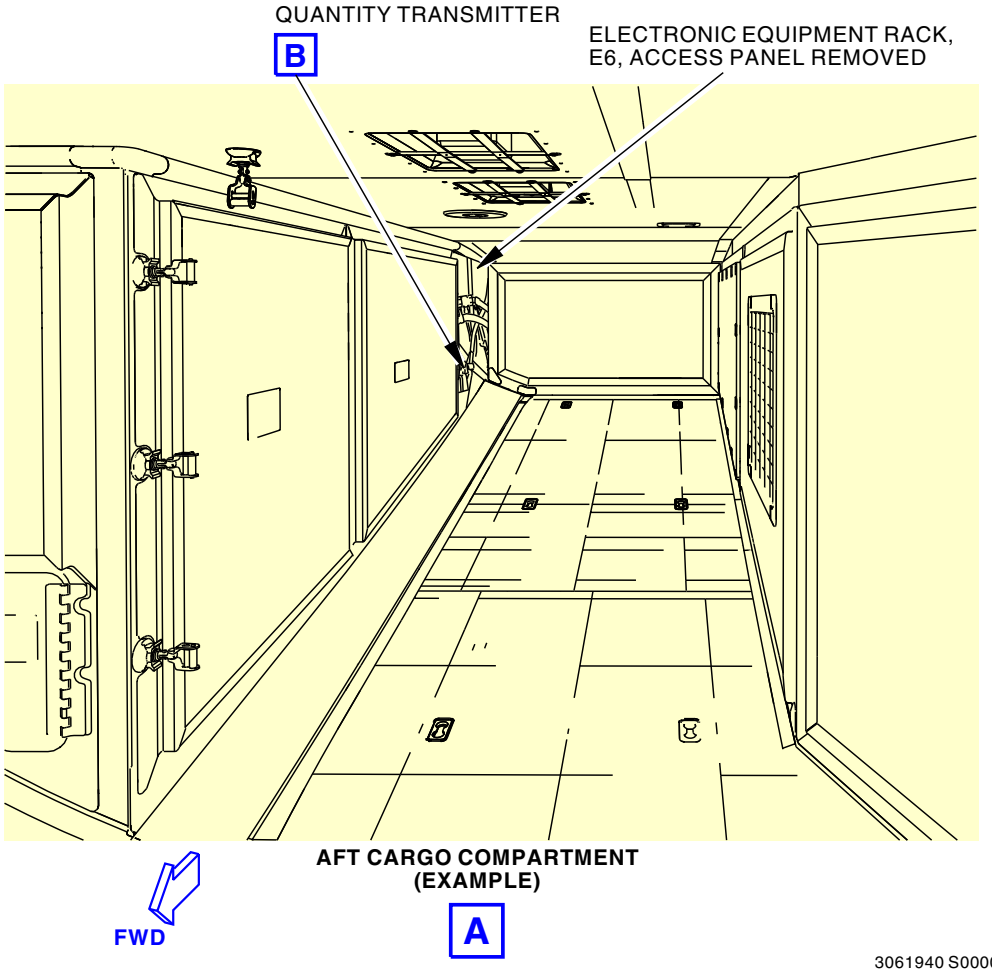
(9) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

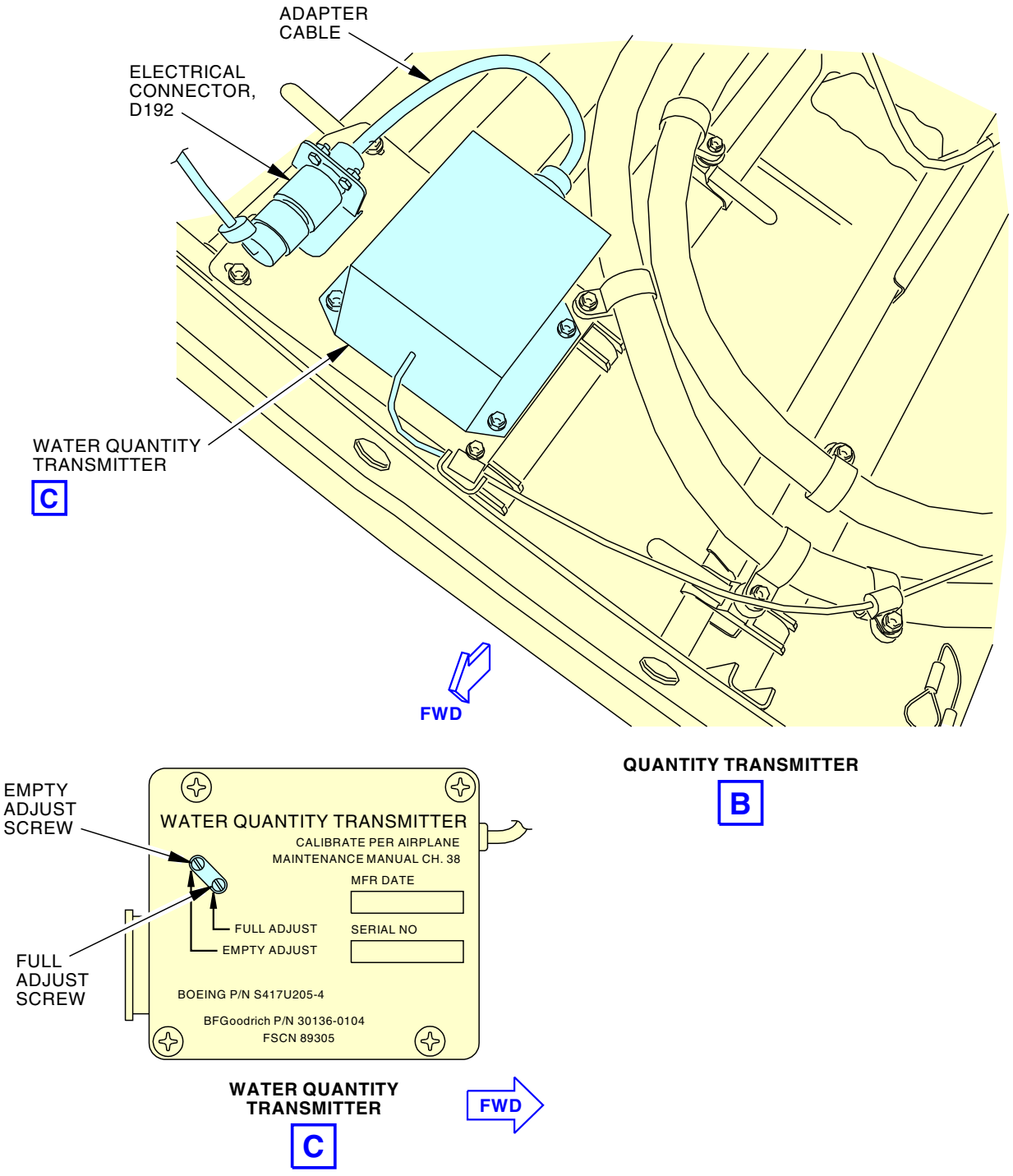




3061940 S0000824968_V1

Water Quantity Transmitter Adjustment
Figure 501/38-14-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Quantity Transmitter Adjustment
Figure 501/38-14-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GRAY WATER/DRAIN SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) The primary routing for the location of these systems.

(a) The gray water drain system

(b) The potable water drain system

(c) The drain system for the area at the doors.

B. This procedure gives the primary configuration items that are in the passenger compartment. This procedure is only a basic configuration of the gray water drains and potable water drains. For the accurate configuration of your airplane, it is necessary for you to refer to the approved configuration data.

C. This procedure has these illustrations:

(1) (Figure 201) shows the primary routing of the tubes for the gray water system.

(2) (Figure 202) shows the primary routing of the tubes for the drain system.

(3) (Figure 203) shows the primary routing of the tubes for the door drain system.

TASK 38-31-00-910-801

2. Gray Water/Drain System - Maintenance Practice

(Figure 201,Figure 202,Figure 203)

A. Location Zones

Zone	Area
100	Lower Half of Fuselage

B. Gray Water/Drain System - Maintenance Practice

SUBTASK 38-31-00-860-001

(1) To access the gray water system components use (Figure 201).

SUBTASK 38-31-00-860-002

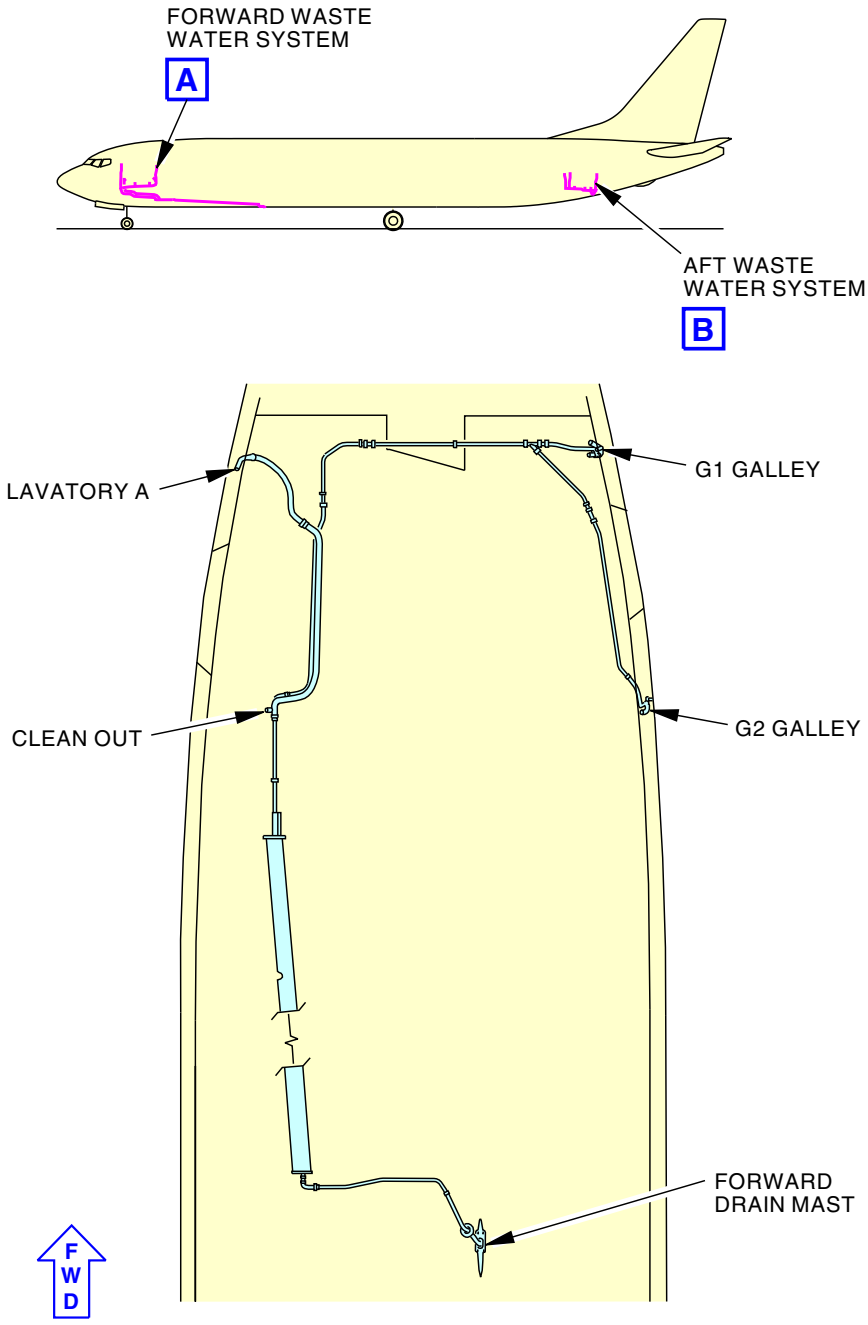
(2) To access the water drain system components use (Figure 202).

SUBTASK 38-31-00-860-003

(3) To access the door drain system components use (Figure 203).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



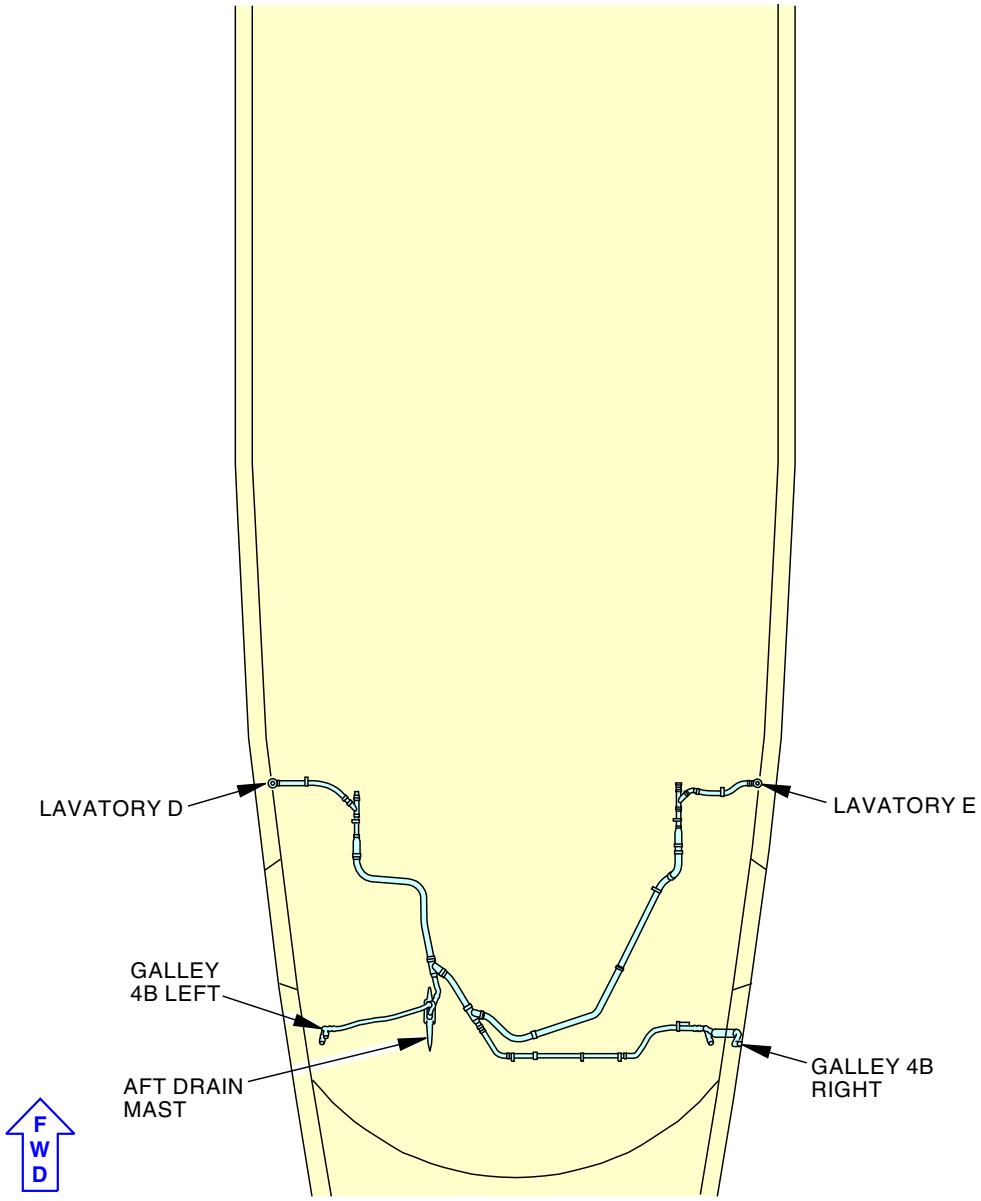
FORWARD WASTE WATER SYSTEM

A

2429941 S0000562411_V1

Waste Water System - Maintenance Practice
Figure 201/38-31-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



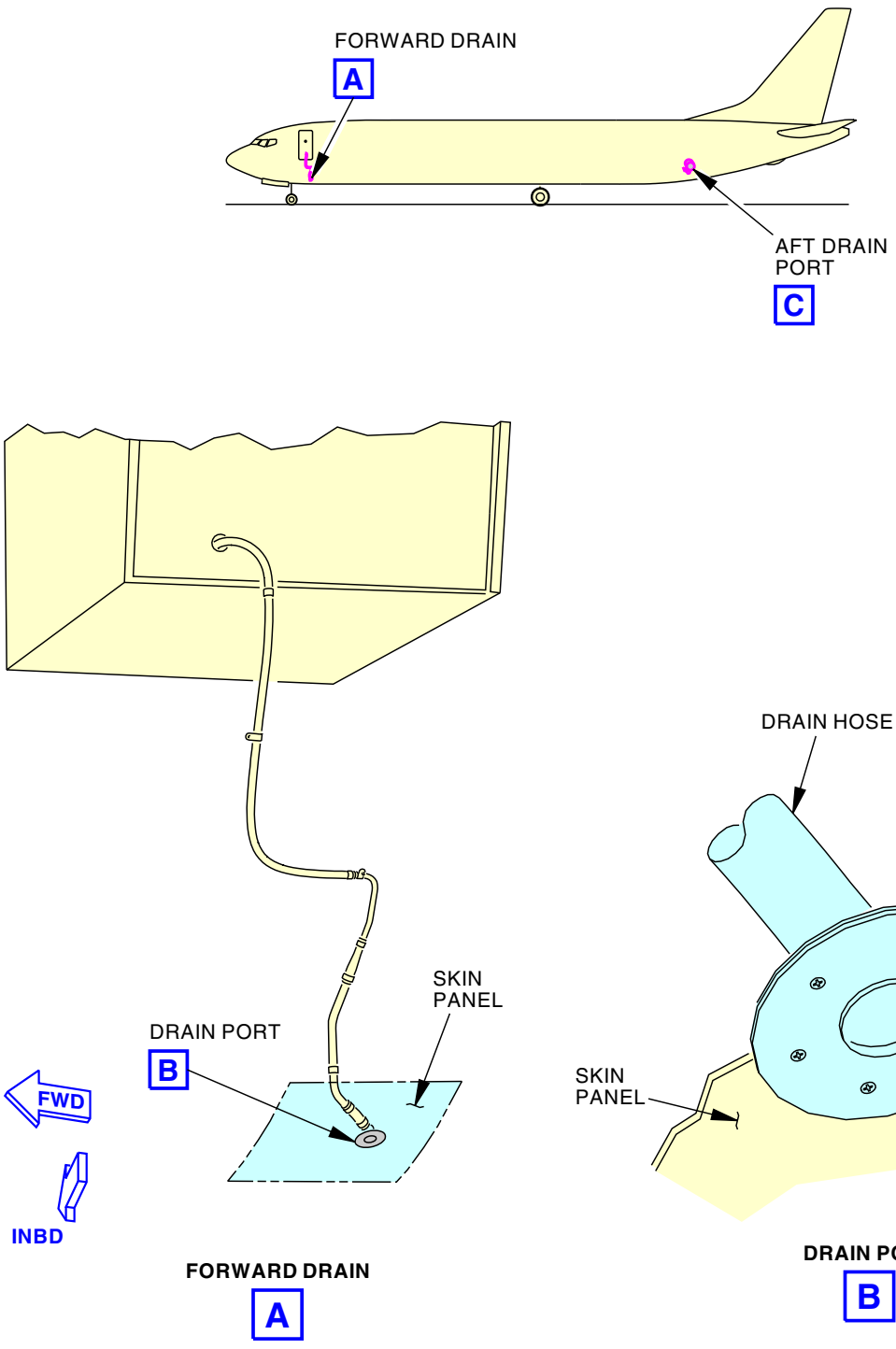
AFT WASTE WATER SYSTEM

B

2415166 S00061537157_V1

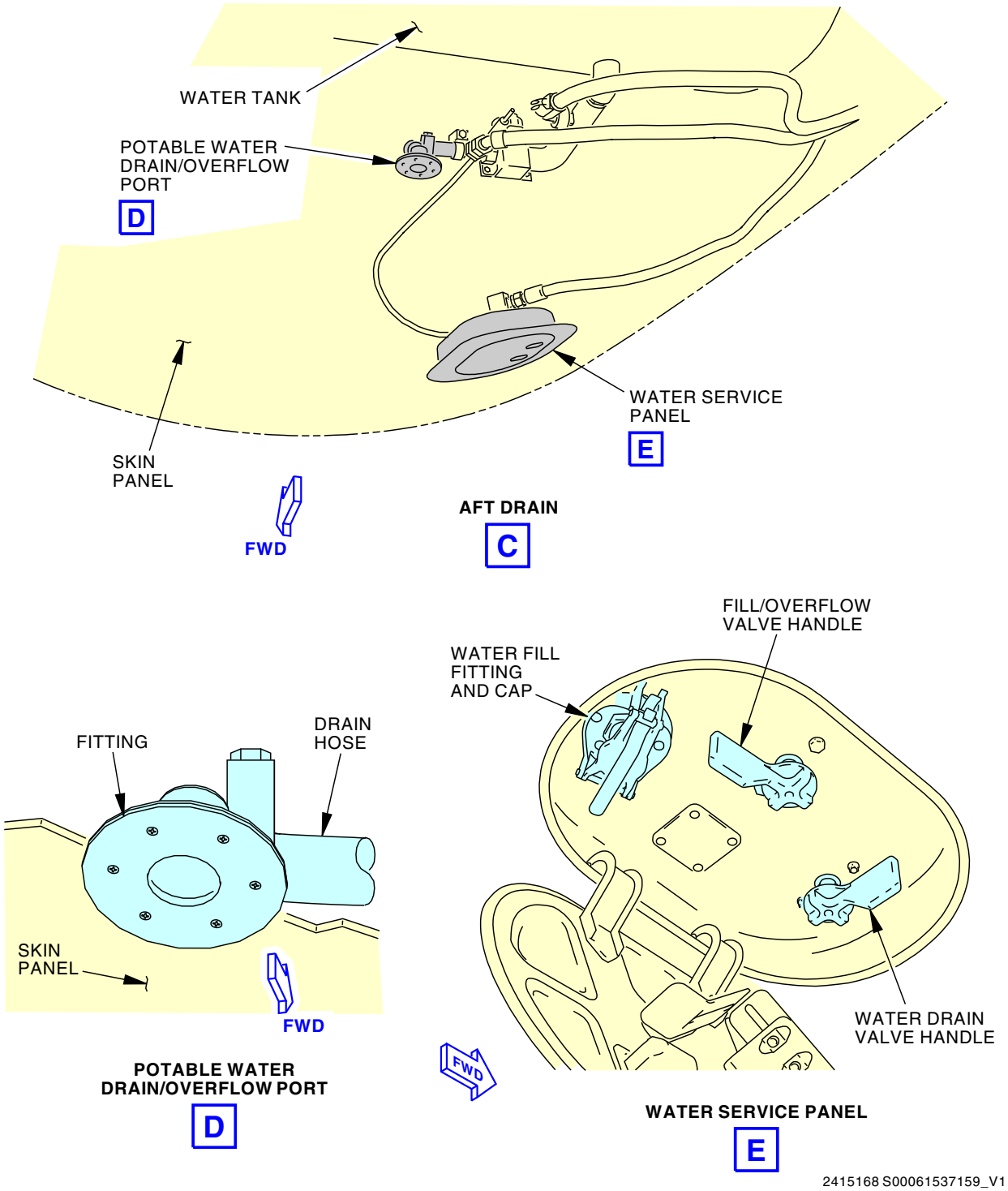
Waste Water System - Maintenance Practice
Figure 201/38-31-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



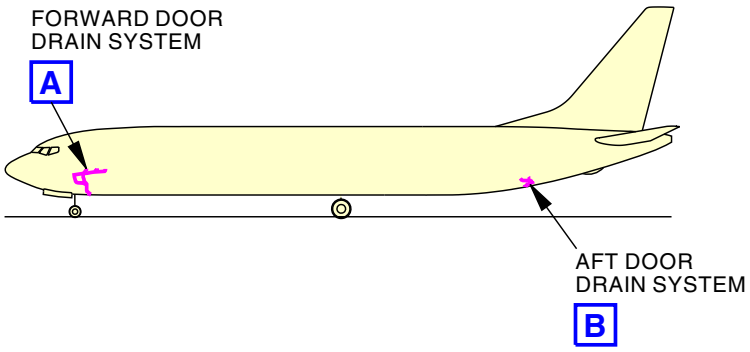
Water Drain System
Figure 202/38-31-00-990-802 (Sheet 1 of 2)

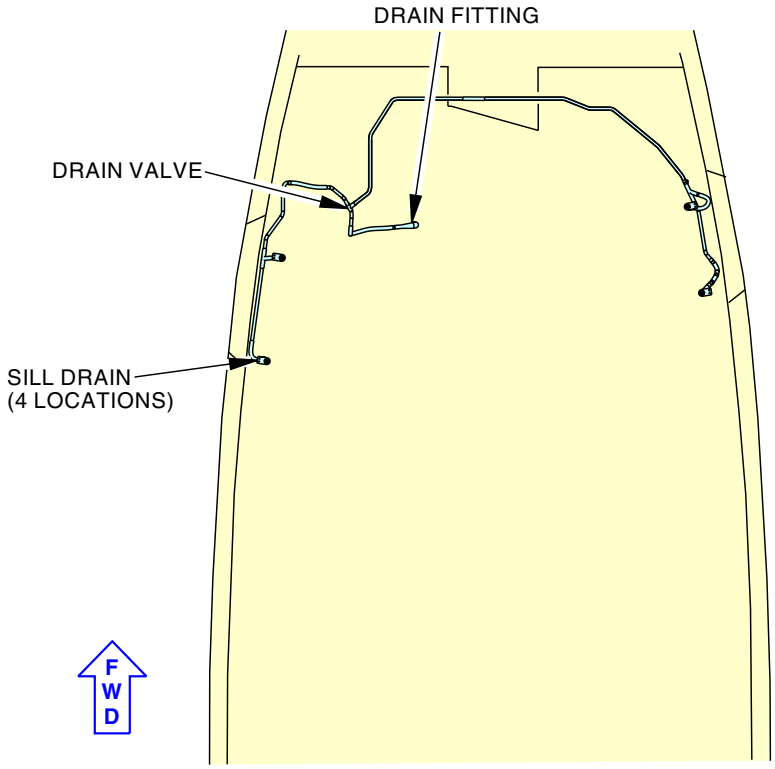
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Drain System
Figure 202/38-31-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





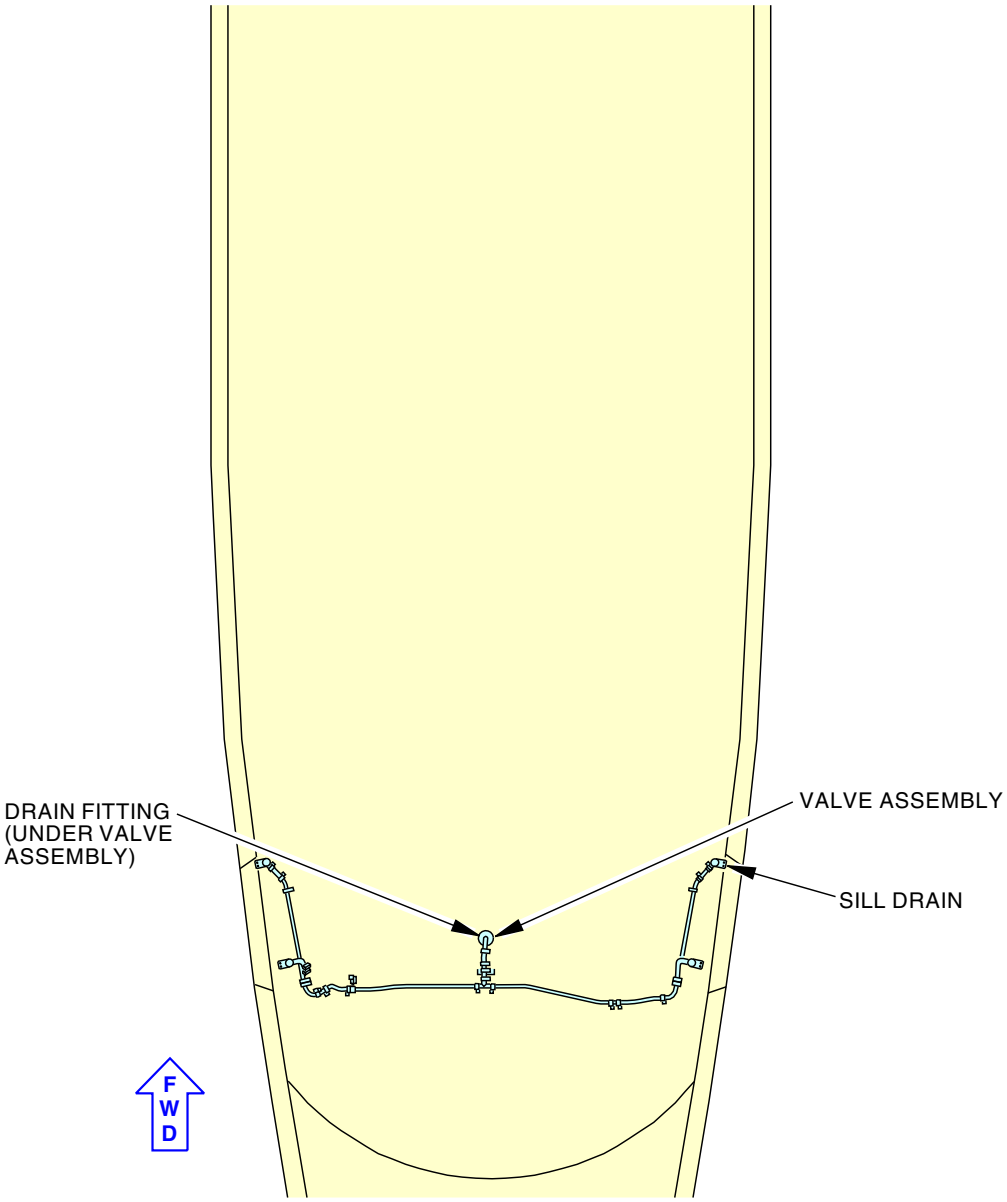
FORWARD DOOR DRAIN SYSTEM

A

2415169 S00061537160_V1

Door Sill Drain System - Maintenance Practice
Figure 203/38-31-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AFT DOOR DRAIN SYSTEM

B

2415170 S00061537161_V1

Door Sill Drain System - Maintenance Practice
Figure 203/38-31-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

GRAY WATER/DRAIN SYSTEM - CLEANING/PAINTING

1. General

A. This procedure describes the steps to remove blockages from clogged drain lines.

B. The lavatory and galley drain lines can become clogged when material other than water is allowed into the drain lines. Only use the drain lines for water.

TASK 38-31-00-100-801

2. Gray Water System - Cleaning

A. General

(1) Use one or more of the procedures that follow to remove obstructions from a clogged drain line.

NOTE: If the clogged drain line has a strainer, begin with procedure 1

(a) Procedure 1 removes the drain line strainer (if equipped) and checks the strainer for obstructions.

(b) Procedure 2 pressurizes the drain line with air or water to remove the obstruction.

(c) Procedure 3 uses tools (snake) to remove the obstruction.

(d) Procedure 4 uses chemical compounds to remove the obstruction.

(e) Procedure 5 removes the effected drain lines to clear a major obstruction.

(f) Procedure 1 is intended for clearing drain lines equipped with strainers:

1) Galley sink drains commonly have strainers mounted in the drain line just below the sink.

(g) Procedures 2, 3, and 4 are intended for small or local blockages.

(h) Procedure 5 is intended for large blockages.

B. References

Reference	Title
30-71-02-000-801	Gray Water Drain Line Heater Removal (P/B 401)
30-71-02-400-801	Gray Water Drain Line Heater Installation (P/B 401)
50-10-02-000-801	Cargo Compartment Sidewall Liners Removal (P/B 401)
50-10-02-400-801	Cargo Compartment Sidewall Liners Installation (P/B 401)
50-10-03-000-801	Cargo Compartment Ceiling Liner Removal (P/B 401)
50-10-03-400-801	Cargo Compartment Ceiling Liner Installation (P/B 401)
50-10-04-000-801	Cargo Compartment Floor Panel Removal (P/B 401)
50-10-04-400-801	Cargo Compartment Floor Panel Installation (P/B 401)
50-11-02-000-801	Forward Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-11-02-400-801	Forward Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1951	Drain Mast Plug - Waste Water Part #: C38001-29 Supplier: 81205 Opt Part #: C38001-23 Supplier: 81205
STD-1174	Drain Snake

D. Consumable Materials

Reference	Description	Specification
B00126	Cleaner - Non-Etch Alkaline, Turco Altrex 24 (Supersedes Altrex Cleaner)	BAC5749
B00636	Acid, Acetic (Vinegar) (5%-10% Acetic Acid)	Commercially available product

E. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

F. Procedure 1

SUBTASK 38-31-00-020-001

(1) If installed, remove the strainer from the drain line.

NOTE: The strainer for the galley sink drain line is found below the galley sink. The strainer for the door sill drain is found at the drain inlet.

SUBTASK 38-31-00-210-001

(2) Make sure the strainer is not clogged.

SUBTASK 38-31-00-210-002

(3) Make sure the drain tube upstream of the strainer is not clogged (as applicable).

SUBTASK 38-31-00-100-001

(4) If you did not find the blockage in the area near the strainer, use procedure 2 through 5.

NOTE: In sink drain lines, you must use the procedures 2, 3, 4, or 5 at the drain line located below the strainer, not at the sink.

G. Procedure 2

SUBTASK 38-31-00-020-002

(1) Install plugs for all the other drain lines connected to the drain tube to be pressurized.

SUBTASK 38-31-00-170-001

(2) If the drain tube has a strainer:

Remove the strainer and pressurize the drain tube downstream of the strainer with air or water (do not use more than 35 psi (241 kPa)).

NOTE: The strainer for the galley sink drain line is found below the galley sink. The strainer for the door sill drain is found at the drain inlet.

SUBTASK 38-31-00-170-002

(3) If the drain tube does not have a strainer:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Pressurize the drain tube at the sink with air or water (do not use more than 35 psi (241 kPa)).

SUBTASK 38-31-00-420-001

(4) After you remove the blockage, remove the plugs from the other drain lines.

H. Procedure 3

SUBTASK 38-31-00-140-001

(1) Use a drain snake, STD-1174 to mechanically clean the drain tube until the clogged drain line is clear.

NOTE: If the drain tube has a strainer, put the snake in the drain tube downstream of the strainer.

I. Procedure 4

SUBTASK 38-31-00-110-001

(1) If the sinks or gray water lines drain slowly do these steps:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	3	C00234	HEATERS DRAIN MAST GND
E	4	C00700	HEATERS DRAIN MAST AIR

(b) Install a plug on the floor drain lines.

WARNING DO NOT TOUCH THE DRAIN MASTS UNTIL THEY BECOME COOL. DRAIN MASTS BECOME VERY HOT. INJURIES TO PERSONNEL CAN OCCUR.

(c) Use one of these plugs in the drain mast outlet.

1) waste water drain mast plug, SPL-1951

WARNING OBEY THE VENDOR SAFETY PROCEDURES FOR THE ALKALINE CLEANER. IF YOU TOUCH THE ALKALINE CLEANER, YOU CAN IRRITATE OR BURN THE SKIN OR EYES.

CAUTION DO NOT GET THE SOLUTION ON THE AIRPLANE OR IN THE STRAINERS FOR THE DRAIN LINES. IT CAN CAUSE CORROSION.

(d) Use a suitable chemical cleaner Turco Altrex 24 Cleaner, B00126 or Acetic acid, B00636 to remove the blockage from the drain line.

NOTE: Do not use both solutions at the same time.

NOTE: Altrex is a stronger solution than the acetic acid and generally is most effective on organic clogs. The acetic acid is most effective on scale or lime obstructions.

1) If you use the Turco Altrex 24 Cleaner, B00126, mix 8 fl-oz (237 ml) of the Turco Altrex 24 Cleaner, B00126 per 1 gal (4 l) of hot water.

2) If you use the Acetic acid, B00636, use it full strength and do these steps:

a) Put 1-2 gallons of Acetic acid, B00636 in a container.

b) Heat the Acetic acid, B00636 to approximately 160°F (71°C).

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(e) Fill the drain tube with the cleaner solution or acetic acid. Add the cleaner through the most forward galley or lavatory sink, as applicable. This is to prevent possible backflows into, and spills from, a more forward sink, when the airplane is nose down.

NOTE: If the drain tube has a strainer, put the cleaner solution in the drain tube downstream of the strainer.

NOTE: Both cleaners should not be used together (at the same time).

NOTE: Altrex is a stronger solution than Acetic Acid and generally is most effective on organic clogs. The acetic acid works best on scale or lime clogs.

(f) Keep the drain tube full of the cleaner solution for approximately one hour.

NOTE: Let the cleaner solution soak for the applicable length of time. Use your experience to adjust the length of time.

(g) Drain the cleaner solution through the drain mast plug, waste water drain mast plug, SPL-1951, or remove the drain mast plug, waste water drain mast plug, SPL-1951.

(h) Remove the drain mast plug, waste water drain mast plug, SPL-1951, from the drain mast outlet.

(i) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	3	C00234	HEATERS DRAIN MAST GND
E	4	C00700	HEATERS DRAIN MAST AIR

J. Procedure 5

SUBTASK 38-31-00-160-001

(1) Do the steps that follow to remove the section of the drain tube that you think is clogged.

NOTE: You will usually find the blockage at one of these locations: the elbow fitting below the sink, the drain tubes in the area of the drain mast, or the "Y" fittings where the drain tubes connect.

(a) Do the applicable steps to get access to the drain lines:

1) Do this task: Cargo Compartment Sidewall Liners Removal, TASK 50-10-02-000-801.

2) Do this task: Cargo Compartment Floor Panel Removal, TASK 50-10-04-000-801.

3) Do this task: Cargo Compartment Ceiling Liner Removal, TASK 50-10-03-000-801.

4) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

5) Do this task: Forward Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-11-02-000-801.

(b) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	3	C00234	HEATERS DRAIN MAST GND
E	4	C00700	HEATERS DRAIN MAST AIR

(c) If the drain tube has heater, do this task: Gray Water Drain Line Heater Removal, TASK 30-71-02-000-801.

(d) Remove the clamps and fasteners for the drain tube.

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(e) Remove the drain tube.

(f) Blow the blockage out of the drain tube with air.

(g) If it is necessary, use a drain snake, STD-1174 to remove the blockage from the drain tube.

(h) Clean the drain tube with the alkaline cleaner.

(i) Install the drain tube.

(j) If the drain tube has a heater, do this task: Gray Water Drain Line Heater Installation, TASK 30-71-02-400-801.

K. Put the Airplane Back to Its Usual Condition.

SUBTASK 38-31-00-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do the applicable steps to complete this task:

(a) Do this task: Cargo Compartment Sidewall Liners Installation, TASK 50-10-02-400-801.

(b) Do this task: Cargo Compartment Floor Panel Installation, TASK 50-10-04-400-801.

(c) Do this task: Cargo Compartment Ceiling Liner Installation, TASK 50-10-03-400-801.

(d) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

(e) Do this task: Forward Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-11-02-400-801.

(f) If removed, install the strainer.

SUBTASK 38-31-00-860-004

(2) Close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	3	C00234	HEATERS DRAIN MAST GND
E	4	C00700	HEATERS DRAIN MAST AIR

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

DRAIN MAST - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Forward drain mast removal

(2) Forward drain mast installation

(3) Aft drain mast removal

(4) Aft drain mast installation.

B. Drain Mast electrical wiring can be identified as follows:

(1) Electrical power wires are identified with a one inch Black heat shrinkable sleeve.

(2) Ground wires are terminated with 0.25 inch terminal lugs, and:

(a) The heater ground wire is identified with a one inch White heat shrinkable sleeve.

(b) The static ground wires are identified by having no heat shrinkable sleeve.

TASK 38-31-01-000-801

2. Forward Drain Mast Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the forward drain mast.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box

E. Access Panels

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

F. Prepare for the Removal

SUBTASK 38-31-01-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	3	C00234	HEATERS DRAIN MAST GND
E	4	C00700	HEATERS DRAIN MAST AIR

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SUBTASK 38-31-01-010-004

CAUTION DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(2) To open the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

G. Forward Drain Mast Removal

SUBTASK 38-31-01-020-001

(1) Loosen the clamp [13] to disconnect the hose assembly [11] from the drain tube [10].

SUBTASK 38-31-01-020-002

(2) Remove the cable ties [14] and sleeves [41] to release the drain mast heater wires [12].

SUBTASK 38-31-01-020-003

CAUTION MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(3) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [5] from the drain mast heaters (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [5] from the drain mast heaters.

(b) Install the protective covers on the electrical receptacle of the drain mast heaters and the electrical connector [5].

SUBTASK 38-31-01-020-004

(4) Remove the nut [2], washer [3], bonding jumper [6], and ground wire [4] from the ground stud.

SUBTASK 38-31-01-020-020

(5) Remove the nut [7], washer [8], bonding jumper [9], and bonding jumper [6] from the ground stud.

SUBTASK 38-31-01-020-005

(6) Remove the bolts [17] and the washers [18] that attach the forward drain mast [1] to the airplane structure.

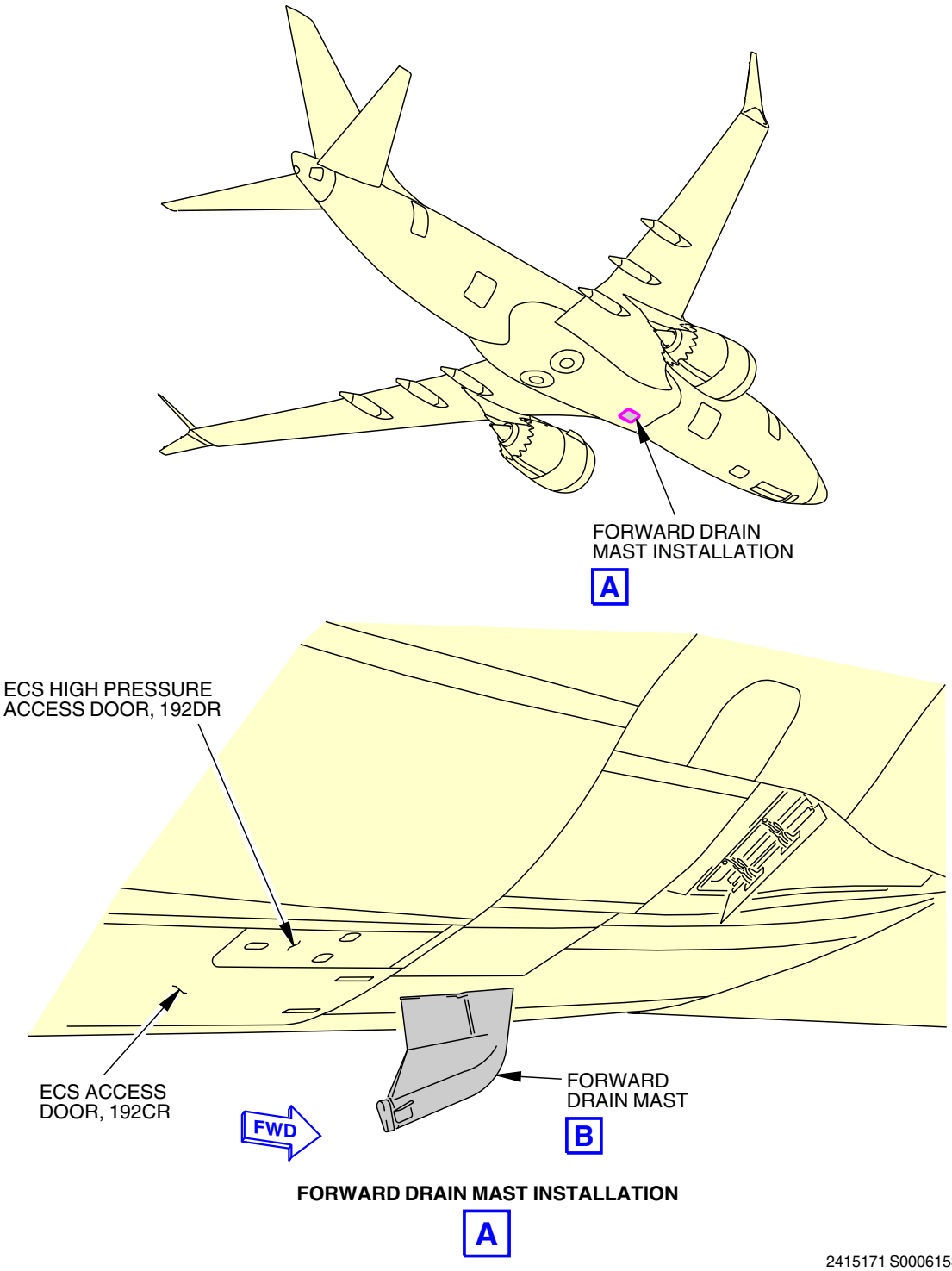
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-31-01-020-006

(7) Remove the forward drain mast [1].

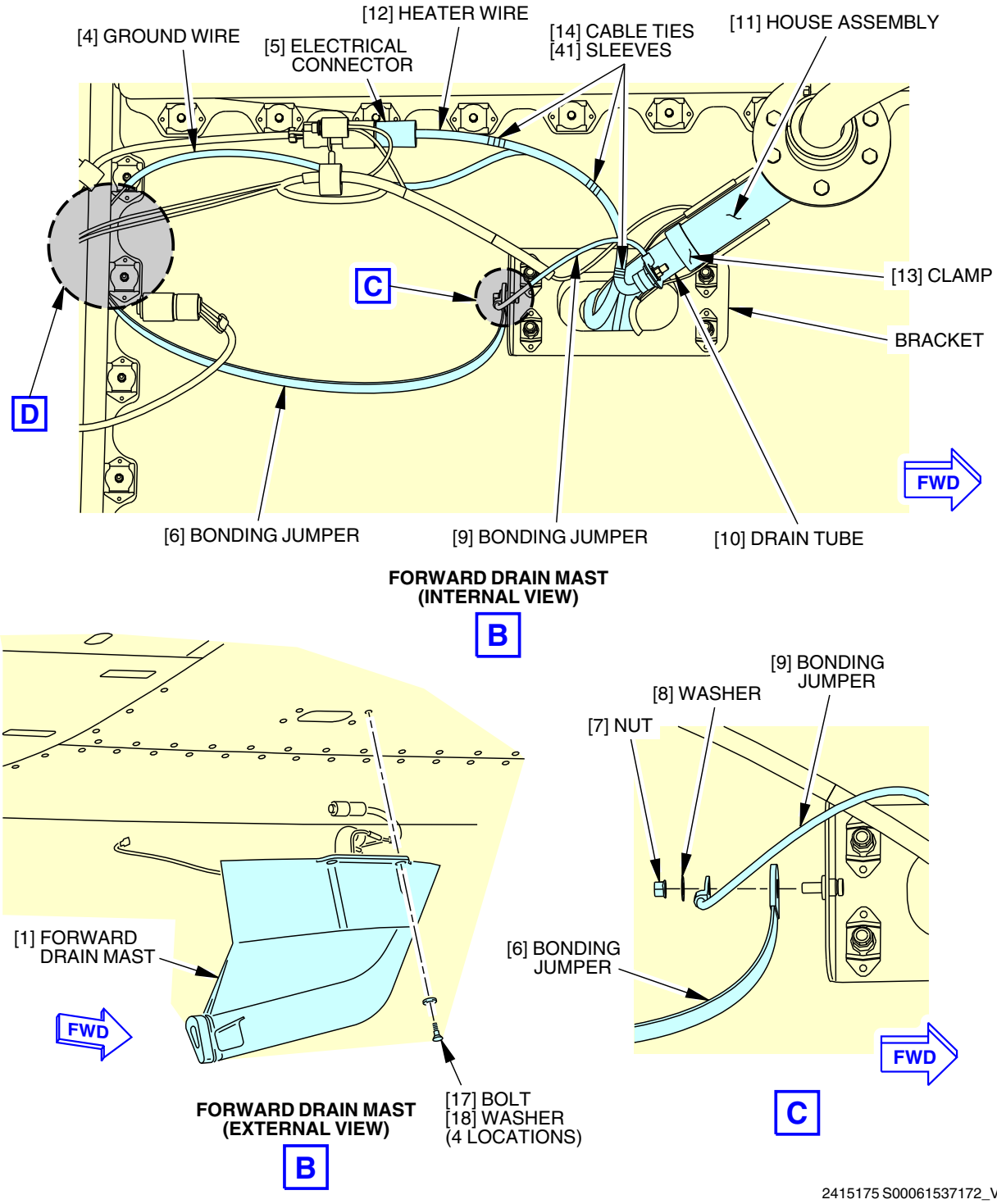
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



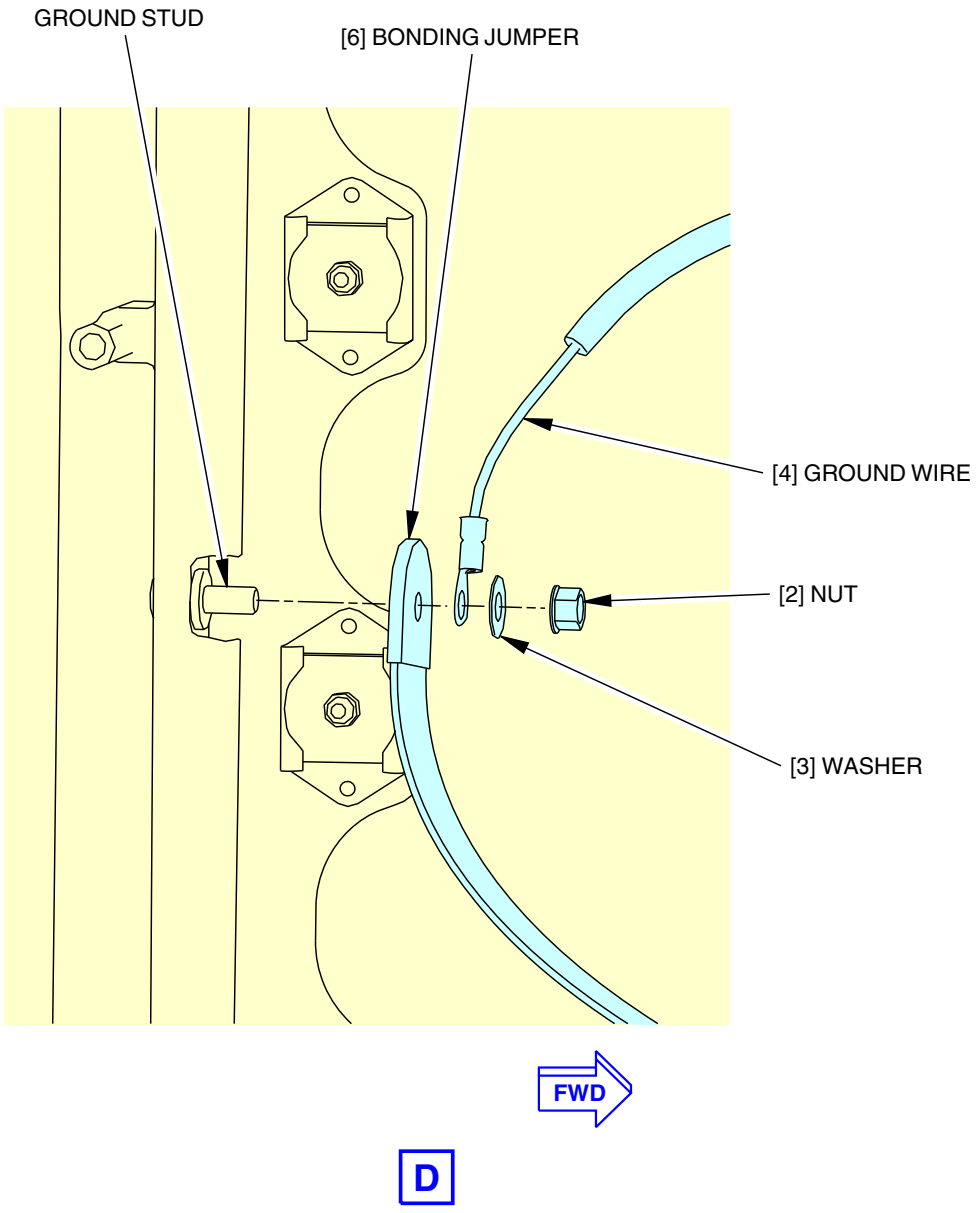
Forward Drain Mast Installation
Figure 401/38-31-01-990-803 (Sheet 1 of 3)

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Forward Drain Mast Installation
Figure 401/38-31-01-990-803 (Sheet 2 of 3)

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Forward Drain Mast Installation
Figure 401/38-31-01-990-803 (Sheet 3 of 3)

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TASK 38-31-01-400-801

3. Forward Drain Mast Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the forward drain mast.

(2) This task covers the installation of the aluminum drain mast and cannot be used to install composite drain masts.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
30-71-00-720-801	Drain Mast Heater Ground and Air Mode Test (P/B 501)
51-22-00-390-809	Aerodynamic Smoother Application (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B

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(Continued)

Reference	Description	Specification
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00068	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002, MIL-E-51454 Type II
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00175	Primer - Urethane Compatible, Corrosion Resistant	BMS10-79 Type III
C50075	Coating - Protective Enamel (BAC 707 Gray Color)	BMS10-60 Type II
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01148	Sleeve - Insulation, Electrical, Heat Shrinkable - RT-876	
G51477	Tubing - Heat Shrinkable, Teflon Sleeve - TFE-4X	AMS-DTL-23053/12 Classes 3 and 5

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward drain mast	38-31-01-08A-035	PTF ALL

F. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box
192	Lower Wing-To-Body Fairing - Under Wing Box

G. Access Panels

Number	Name/Location
192CR	ECS Access Door
192DR	ECS High Pressure Access Door

H. Forward Drain Mast Installation

SUBTASK 38-31-01-010-005

(1) Make sure that this access panel is open:

Number	Name/Location
192CR	ECS Access Door

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SUBTASK 38-31-01-390-001

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Clean the seal surface at the forward drain mast [1] and on the airplane skin with alcohol, B00068, or solvent, B00184 (SWPM 20-20-00, paragraph 2).

SUBTASK 38-31-01-390-007

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Do these steps to prepare the electrical bonding fay surface areas of the forward drain mast [1]:

(a) Clean the mating surface of the forward drain mast [1] and the area around the fastener holes (SWPM 20-20-00, paragraph 2).

(b) Clean the washers [18] and bolts [17] with cotton wiper, G00034, moist in alcohol, B00130 (SWPM 20-20-00, paragraph 2).

SUBTASK 38-31-01-420-001

(4) Install the forward drain mast [1], do these steps:

(a) Install the washers [18] and bolts [17] to attach the forward drain mast [1].

(b) Tighten the bolts [17] to 45 in-lb (5.1 N·m) - 55 in-lb (6.2 N·m).

(c) Measure the electrical bonding resistance between the forward drain mast [1] and bolt [17] head (SWPM 20-20-00, paragraph 2).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

(d) Apply fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, on the washers [18] and bolts [17].

(e) Apply finish with primer, C00175, and coating, C50075, on the forward drain mast [1].

SUBTASK 38-31-01-760-001

(5) Do a check of the electrical resistance of the forward drain mast [1] installation:

(a) With a non-intrinsically safe bonding meter, COM-614, make sure that the resistance between the drain hole on the forward drain mast [1] and the structure of the airplane is 0.01 ohm (10 milliohms) or less.

(b) With a non-intrinsically safe bonding meter, COM-614, make sure that the resistance between the forward drain mast [1] and airplane structure is 0.0005 ohm (0.5 milliohms) or less.

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SUBTASK 38-31-01-390-002

(6) Apply the aerodynamic seal with sealant, A02315, to the surfaces of the forward drain mast [1] (TASK 51-22-00-390-809).

NOTE: It is not necessary to apply the sealant immediately if the cure time will cause a flight delay. However, in order to prevent moisture damage to the airplane, sealant should be applied as soon as it is convenient for the operator.

SUBTASK 38-31-01-420-010

(7) Install the bonding jumper [6], ground wire [4], nut [2], and washer [3] to the ground stud.

(a) Clean the terminals of the bonding jumper [6], ground wire [4], ground stud, nut [2], and washers [3] with cotton wiper, G00034, moist in alcohol, B00130.

(b) Install the bonding jumper [6], ground wire [4], nut [2], and washer [3] to the ground stud.

(c) Tighten the nut [2] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

(d) Measure the electrical bonding resistance between the ground wire [4] and airplane structure (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

(e) Apply fillet seal on the bonding jumper [6], ground wire [4], ground stud, nut [2], and washer [3] with sealant, A50337, or sealant, A50231, or sealant, A50359.

SUBTASK 38-31-01-020-007

(8) Install the bonding jumper [6], bonding jumper [9], nut [7], and washer [8] to the ground stud.

(a) Clean the terminals of the bonding jumper [6], bonding jumper [9], ground stud, nut [7], and washer [8] with cotton wiper, G00034, moist in alcohol, B00130.

(b) Install the bonding jumper [6], bonding jumper [9], nut [7], and washer [8] to the ground stud.

(c) Tighten the nut [7] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

(d) Measure the electrical bonding resistance between the bonding jumper [9] and bracket (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

(e) Apply fillet seal on the bonding jumper [6], bonding jumper [9], nut [7], washer [8], and ground stud with sealant, A50337, or sealant, A50231, or sealant, A50359.

SUBTASK 38-31-01-420-002

(9) Install the clamp [13] and hose assembly [11] to the drain tube [10].

SUBTASK 38-31-01-420-007

(10) Tighten the clamp [13] to 30 in-lb (3.4 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 38-31-01-020-008

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

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(CAUTION PRECEDES)

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(11) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [5] to the drain mast heaters (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the drain mast heaters and the electrical connector [5].

(b) Connect the electrical connector [5] to the drain mast heaters.

1) Tighten the connector with the teflon-jawed pliers, STD-664.

SUBTASK 38-31-01-420-011

(12) Install the RT-876 sleeve, G01148, or tubing, G51477, on the heater wire [12].

SUBTASK 38-31-01-020-009

(13) Install the cable ties [14] approximately every 3 in. (76.2 mm) for the heater wires [12].

I. Forward Drain Mast Installation Test

SUBTASK 38-31-01-040-001

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	3	C00234	HEATERS DRAIN MAST GND
E	4	C00700	HEATERS DRAIN MAST AIR

SUBTASK 38-31-01-860-003

(2) Open the water faucet for a forward lavatory.

SUBTASK 38-31-01-710-001

(3) Make sure that the water flows from the forward drain mast without leakage.

SUBTASK 38-31-01-710-002

(4) Make sure that the drain mast heater operates (TASK 30-71-00-720-801).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-31-01-410-002

(1) To close the right ECS access door, do these steps:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(b) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-31-01-000-802

4. Aft Drain Mast Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the aft drain mast.

B. References

Reference	Title
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)
38-11-01-000-801	Water Tank Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

C. Location Zones

Zone	Area
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

D. Prepare for the Removal

SUBTASK 38-31-01-860-004

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	3	C00234	HEATERS DRAIN MAST GND
E	4	C00700	HEATERS DRAIN MAST AIR

SUBTASK 38-31-01-010-003

(2) To get access to the aft drain mast, do one of these steps:

(a) Do this task: Passenger Cabin Floor Panel Removal, TASK 53-21-00-000-801.

(b) Do this task: Aft Outflow Valve Assembly Removal, TASK 21-31-03-000-801.

(c) Do this task: Water Tank Removal, TASK 38-11-01-000-801.

E. Aft Drain Mast Removal

SUBTASK 38-31-01-020-010

(1) Loosen the hose clamp [22] to disconnect hose assembly [23] from the aft drain mast assembly [21].

SUBTASK 38-31-01-020-011

(2) Remove the wire ties to release the heater wires [25].

SUBTASK 38-31-01-020-012

(3) Disconnect the electrical splice connections to the heater wires [25].

SUBTASK 38-31-01-020-013

(4) Remove the nut [33], lock washer [34], washer [35], bonding jumper [26], and thermostat [39] from the drain tube clamp [40].

(5) Remove the nut [37], washers [36], bolt [38], bonding jumper [26], and ground wire [31] from the structure.

SUBTASK 38-31-01-020-014

(6) Remove the packings [29], bolts [27], and washers [28] that attach the aft drain mast assembly [21] to the structure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-31-01-030-001

(7) Remove the ground wire [30] and washers [28] on the bolt [27].

SUBTASK 38-31-01-020-015

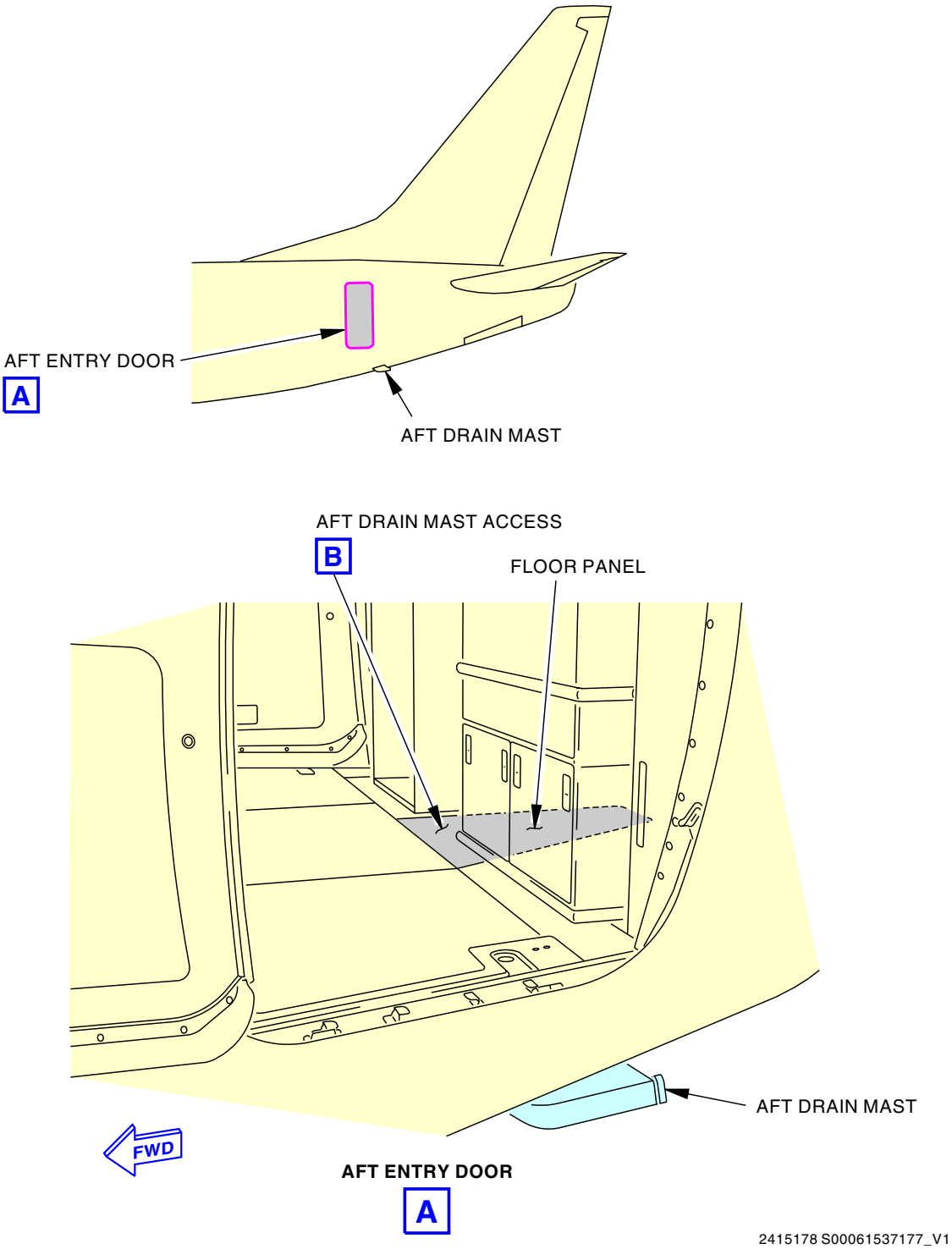
(8) Remove the ground wire [24] and washers [28] on the bolt [27].

SUBTASK 38-31-01-020-016

(9) Remove the aft drain mast assembly [21].

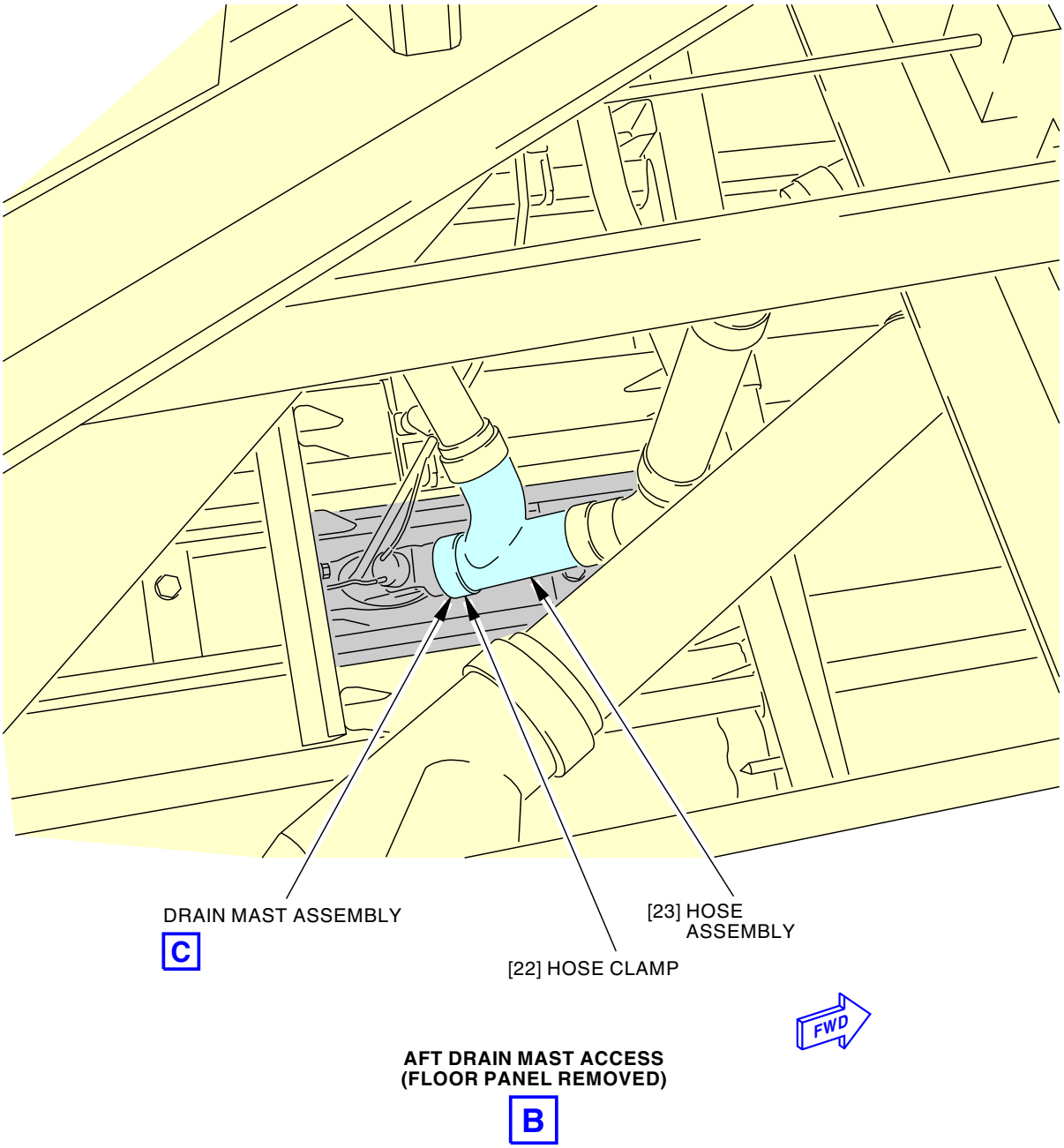
——— END OF TASK ———

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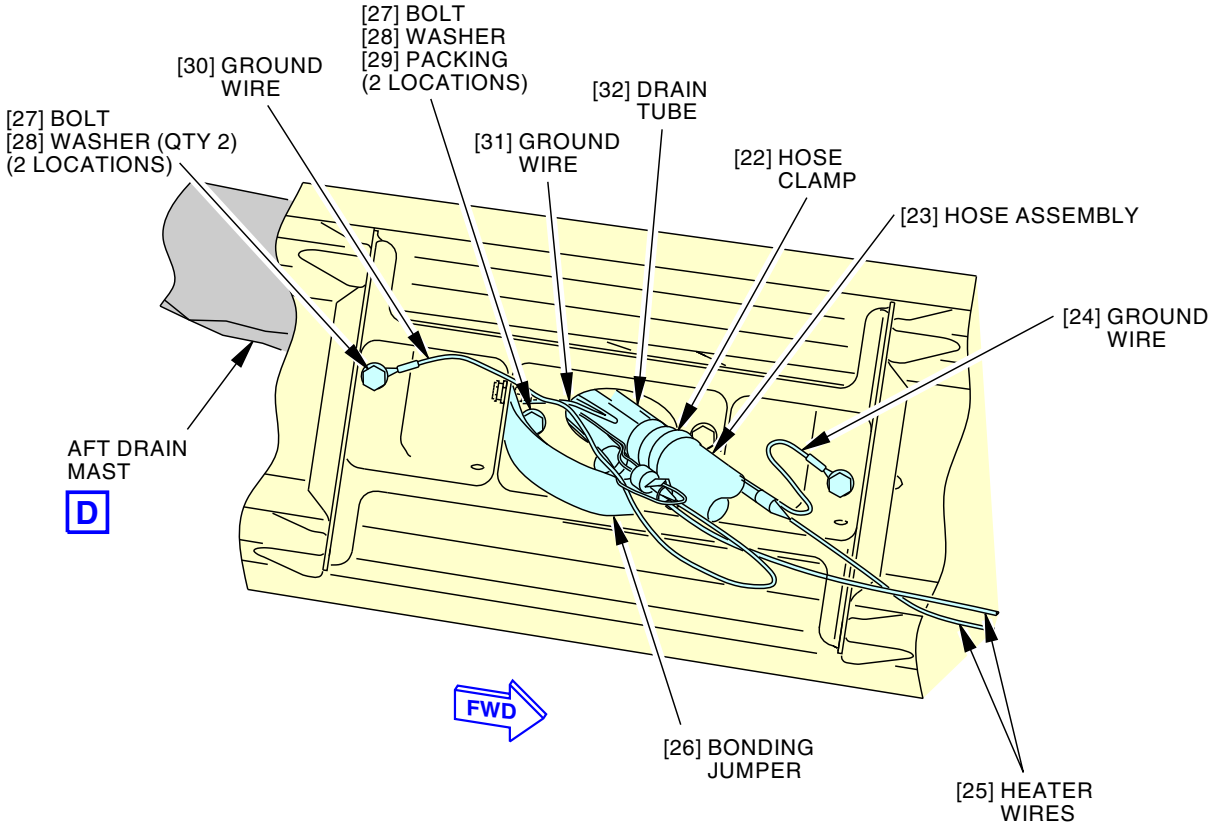
Aft Drain Mast Installation
Figure 402/38-31-01-990-804 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Drain Mast Installation
Figure 402/38-31-01-990-804 (Sheet 2 of 4)

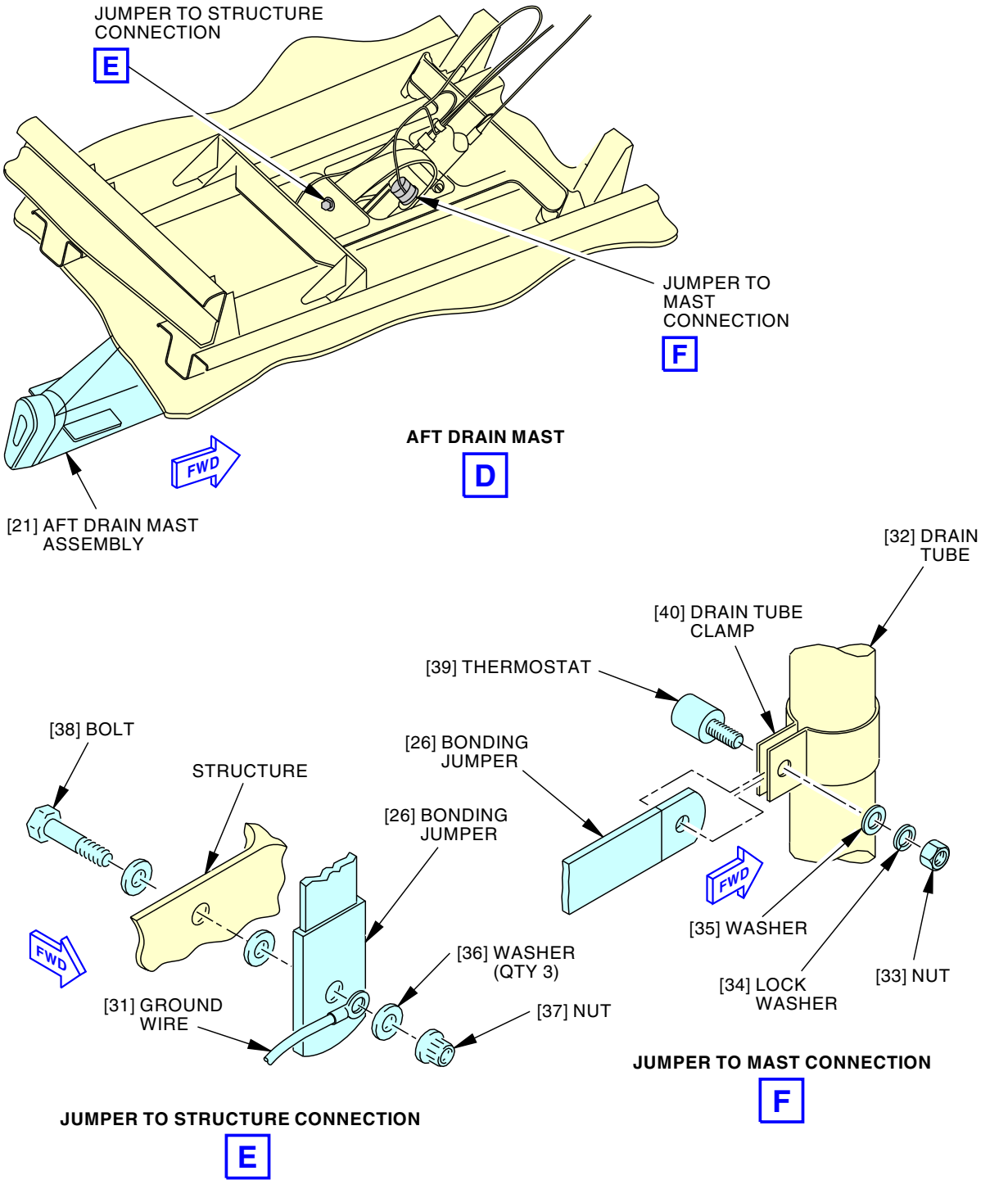
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aft Drain Mast Installation
Figure 402/38-31-01-990-804 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Drain Mast Installation
Figure 402/38-31-01-990-804 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-31-01-400-802

5. Aft Drain Mast Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the aft drain mast.

B. References

Reference	Title
21-31-03-400-801	Aft Outflow Valve Assembly Installation (P/B 401)
30-71-00-720-801	Drain Mast Heater Ground and Air Mode Test (P/B 501)
38-11-01-400-801	Water Tank Installation (P/B 401)
51-22-00-390-801	Non-Removable Faying (Mated) Surface Seal Application (P/B 201)
51-22-00-390-809	Aerodynamic Smoother Application (P/B 201)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A00767	Sealant - Fuel Tank	BMS5-45
A01076	Adhesive - Synthetic Rubber	BAC5010 Type 93 (BMS5-95 Class B)
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
A50009	Sealant - Low Density, Non-Chromate Type. (Formerly Chromate - Synthetic Rubber)	BMS5-142 Type II Class B-1 or B-2
B00068	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002, MIL-E-51454 Type II
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Aft drain mast assembly	38-31-01-05-100	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Location Zones

Zone	Area
240	Subzone - Passenger Compartment - Body Station 663.75 to Body Station 1016.00

G. Aft Drain Mast Installation

SUBTASK 38-31-01-390-003

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

(1) Clean the seal surface of the drain mast and airplane skin with solvent, B00184, or alcohol, B00068.

SUBTASK 38-31-01-110-001

(2) Clean the faying surfaces of the mast drain tube [32], the drain tube clamp [40], and the bonding jumper [26] terminals for electrical bonding (SWPM 20) with the solvent, B00184, or alcohol, B00068.

SUBTASK 38-31-01-390-004

(3) Apply a fay surface seal with sealant, A00247, between the airplane skin and the aft drain mast assembly [21] (TASK 51-22-00-390-801).

SUBTASK 38-31-01-420-003

(4) Put the aft drain mast assembly [21] in its position.

SUBTASK 38-31-01-430-001

(5) Put the ground wire [30] and washers [28] on the bolt [27].

(6) Put the ground wire [24] and washers [28] on the bolt [27].

SUBTASK 38-31-01-420-004

(7) Loosely install the packings [29], bolts [27], and washers [28] for the aft drain mast assembly [21].

SUBTASK 38-31-01-420-005

(8) Tighten the bolts [27], in an alternating pattern, to 50 ±5 in-lb (5.6 ±0.6 N·m).

SUBTASK 38-31-01-390-005

(9) Apply the aerodynamic seal with sealant, A02315, to the surfaces of the ground wire [31] (TASK 51-22-00-390-809).

NOTE: It is not necessary to apply the sealant immediately if the cure time will cause a flight delay. But to prevent moisture damage to the airplane, sealant should be applied as soon as it is convenient for the operator.

SUBTASK 38-31-01-430-002

(10) Install the ground connections.

(a) Put the drain tube clamp [40] in its position on the drain tube [32].

1) Put the thermostat [39] in position as near to the top of the tube as possible without interfering with the hose assembly [23] or hose clamp [22].

(b) Connect one end of the bonding jumper [26], thermostat [39], washer [35], lock washer [34], and nut [33] to the drain tube clamp [40].

(c) Tighten the nut to 25 in-lb (2.8 N·m) - 30 in-lb (3 N·m).

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(d) Connect the other end of the bonding jumper [26] to the structure.

1) Clean the faying surfaces of the structure, and the bonding jumper [26] terminal for electrical bonding (SWPM 20) with the solvent, B00184, or alcohol, B00068.

2) Install the bonding jumper [26], ground wire [31], washers [36], bolt [38], and nut [37] to the structure.

3) Tighten the nut [37] to 28 in-lb (3.2 N·m) - 35 in-lb (4 N·m).

(e) Make sure that the resistance between the bonding jumper [26] terminal and the drain tube [32] is 0.0015 ohm (1.5 milliohms) or less.

1) Use the intrinsically safe approved bonding meter, COM-1550, or equivalent.

(f) Make sure that the resistance between the bonding jumper [26] terminal and the structure is 0.0015 ohm (1.5 milliohms) or less.

1) Use the intrinsically safe approved bonding meter, COM-1550, or equivalent.

(g) Apply the sealant, A00767 (optional sealant, A02315), to cover the bonding jumper [26] terminal, the thermostat [39] stud, washer [35], lock washer [34] and nut [33].

(h) Apply the adhesive, A01076, or sealant, A50009, to cover the bonding jumper [26] terminal, ground wire [31], washers [36], bolt [38], and nut [37].

(i) Apply the sealant, A02315, to the faying surfaces of the drain tube [32] and drain tube clamp [40].

SUBTASK 38-31-01-420-006

(11) Install the hose clamp [22] and the hose assembly [23] to the drain tube [32].

SUBTASK 38-31-01-420-008

(12) Tighten the hose clamp [22] to 35 in-lb (4.0 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 38-31-01-020-017

(13) Connect the electrical splice connections to the heater wires [25].

SUBTASK 38-31-01-020-018

(14) Install the wire ties to the heater wires [25] as necessary.

H. Aft Drain Mast Installation Test

SUBTASK 38-31-01-040-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	3	C00234	HEATERS DRAIN MAST GND
E	4	C00700	HEATERS DRAIN MAST AIR

SUBTASK 38-31-01-860-006

(2) Open the water faucet for an aft lavatory.

SUBTASK 38-31-01-710-003

(3) Make sure that the water flows from the aft drain mast without leakage.

SUBTASK 38-31-01-710-004

(4) Make sure that the drain mast heater is operational (TASK 30-71-00-720-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-31-01-410-001

(1) To close the access for the aft drain mast, do one of these steps:

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WARNING

SEAL THE PASSENGER COMPARTMENT WITH THE FLOOR PANELS. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE FLOOR PANELS. IF YOU INSTALL THE FLOOR PANELS INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE AND CAUSE INJURY.

(a) Do this task: Passenger Cabin Floor Panel Installation, TASK 53-21-00-400-801.

(b) Do this task: Aft Outflow Valve Assembly Installation, TASK 21-31-03-400-801.

(c) Do this task: Water Tank Installation, TASK 38-11-01-400-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SILL DRAIN FLOAT VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Forward sill drain float valve removal

(2) Forward sill drain float valve installation

(3) Aft sill drain float valve removal

(4) Aft sill drain float valve installation.

TASK 38-31-06-000-801

2. Forward Sill Drain Float Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the forward sill drain float valve.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 38-31-06-010-001

(1) To get access to the forward sill drain float valve, do this step:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Forward Sill Drain Float Valve Removal

SUBTASK 38-31-06-020-001

(1) Loosen the clamps [6] and disconnect the drain lines from each end of the forward float valve [1], bladder assembly.

SUBTASK 38-31-06-020-002

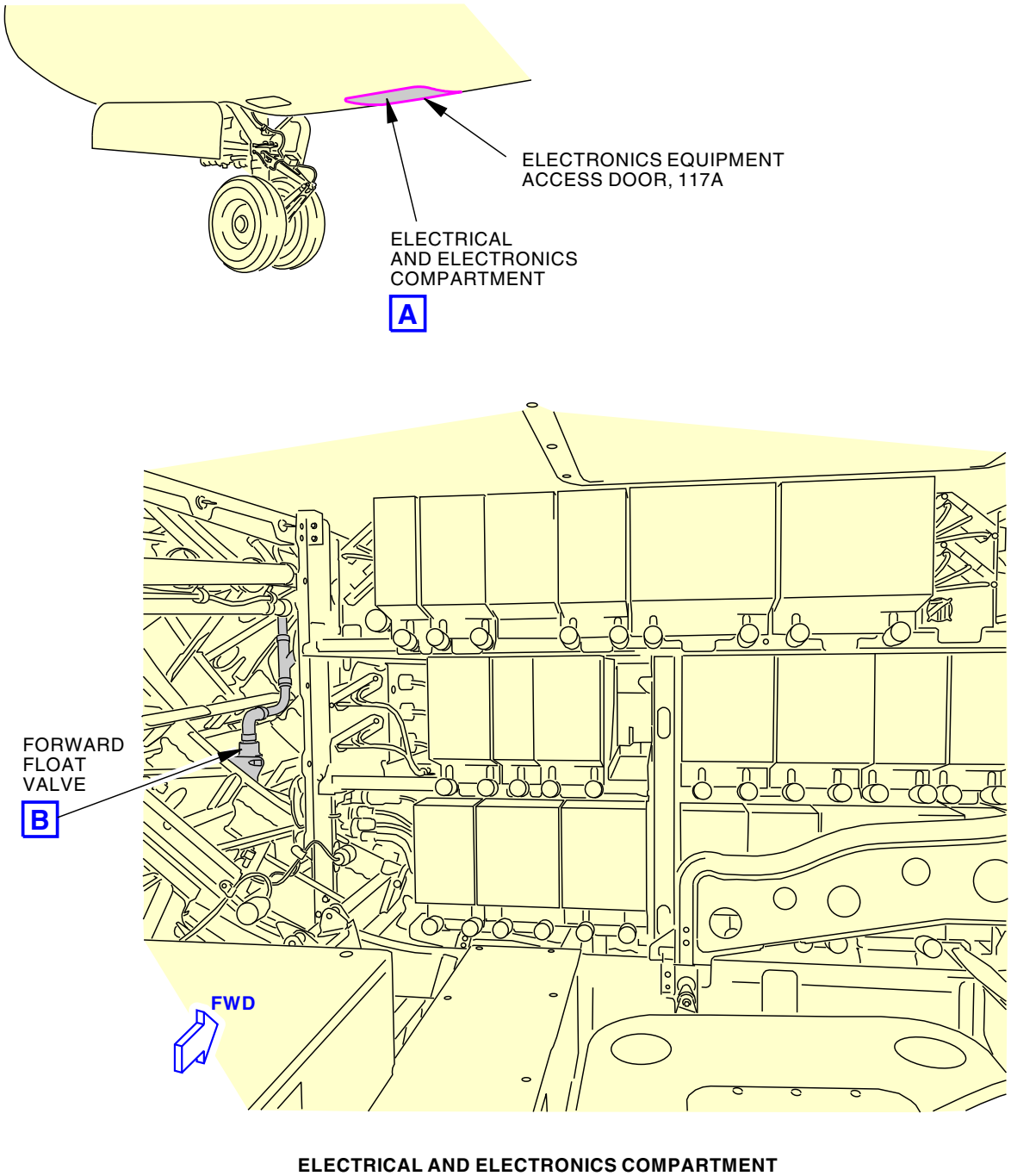
(2) Remove the screws [2] and washers [3] that attach the strap [4], saddle [5], and forward float valve [1] to the structure.

SUBTASK 38-31-06-020-003

(3) Remove the forward float valve [1].

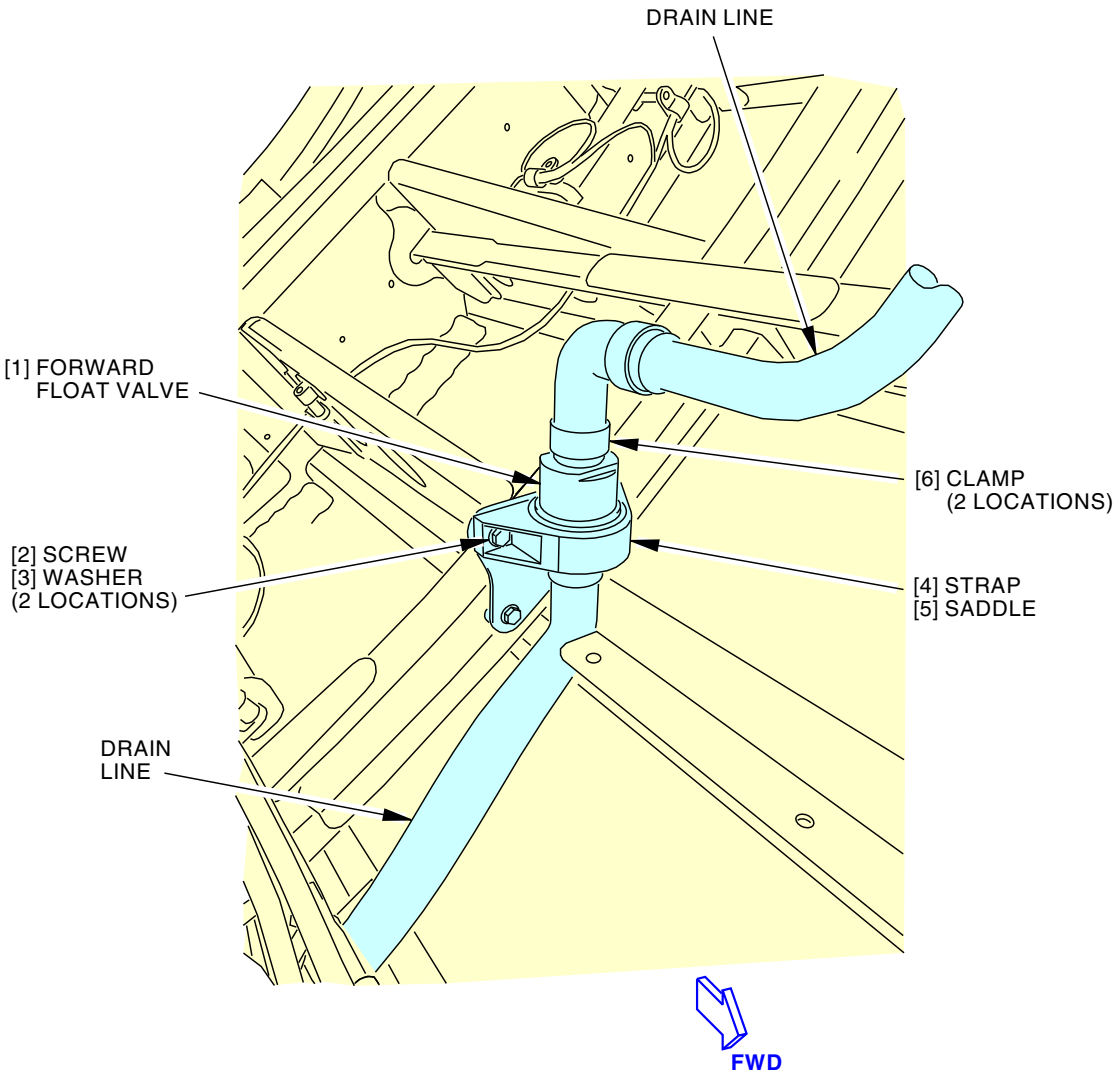
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Sill Drain Float Valve Installation
Figure 401/38-31-06-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD FLOAT VALVE

B

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Forward Sill Drain Float Valve Installation
Figure 401/38-31-06-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-31-06-400-801

3. Forward Sill Drain Float Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the forward sill drain float valve.

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward float valve	38-31-06-02-035	PTF ALL

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Forward Sill Drain Float Valve Installation

SUBTASK 38-31-06-420-001

(1) Install the screws [2] and washers [3] that attach the strap [4], saddle [5], and forward float valve [1] to the structure.

NOTE: The lip on the forward float valve [1] should be above the strap.

SUBTASK 38-31-06-390-001

(2) Install each end of the forward float valve [1] to the drain lines and tighten the clamps [6].

NOTE: The large end of the float valve should be up.

F. Forward Sill Drain Float Valve Assembly Installation Test

SUBTASK 38-31-06-860-001

(1) Pour water into the sill drain at the forward entry door.

SUBTASK 38-31-06-710-001

(2) Make sure that the water flows thru the forward sill drain valve assembly without leakage.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-31-06-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 38-31-06-000-802

4. Aft Sill Drain Float Valve Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the aft sill drain float valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
21-31-03-000-801	Aft Outflow Valve Assembly Removal (P/B 401)
38-11-01-000-801	Water Tank Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

C. Location Zones

Zone	Area
145	Aft Cargo Compartment Equipment Bay - Left

D. Prepare for the Removal

SUBTASK 38-31-06-010-002

(1) To get access to the Aft Sill Drain Valve Assembly, do one of these steps:

(a) Do this task: Passenger Cabin Floor Panel Removal, TASK 53-21-00-000-801.

(b) Do this task: Aft Outflow Valve Assembly Removal, TASK 21-31-03-000-801.

(c) Do this task: Water Tank Removal, TASK 38-11-01-000-801.

E. Aft Sill Drain Float Valve Removal

SUBTASK 38-31-06-020-004

(1) Loosen the clamps [26] and disconnect the drain line from each end of the aft float valve [21].

SUBTASK 38-31-06-020-005

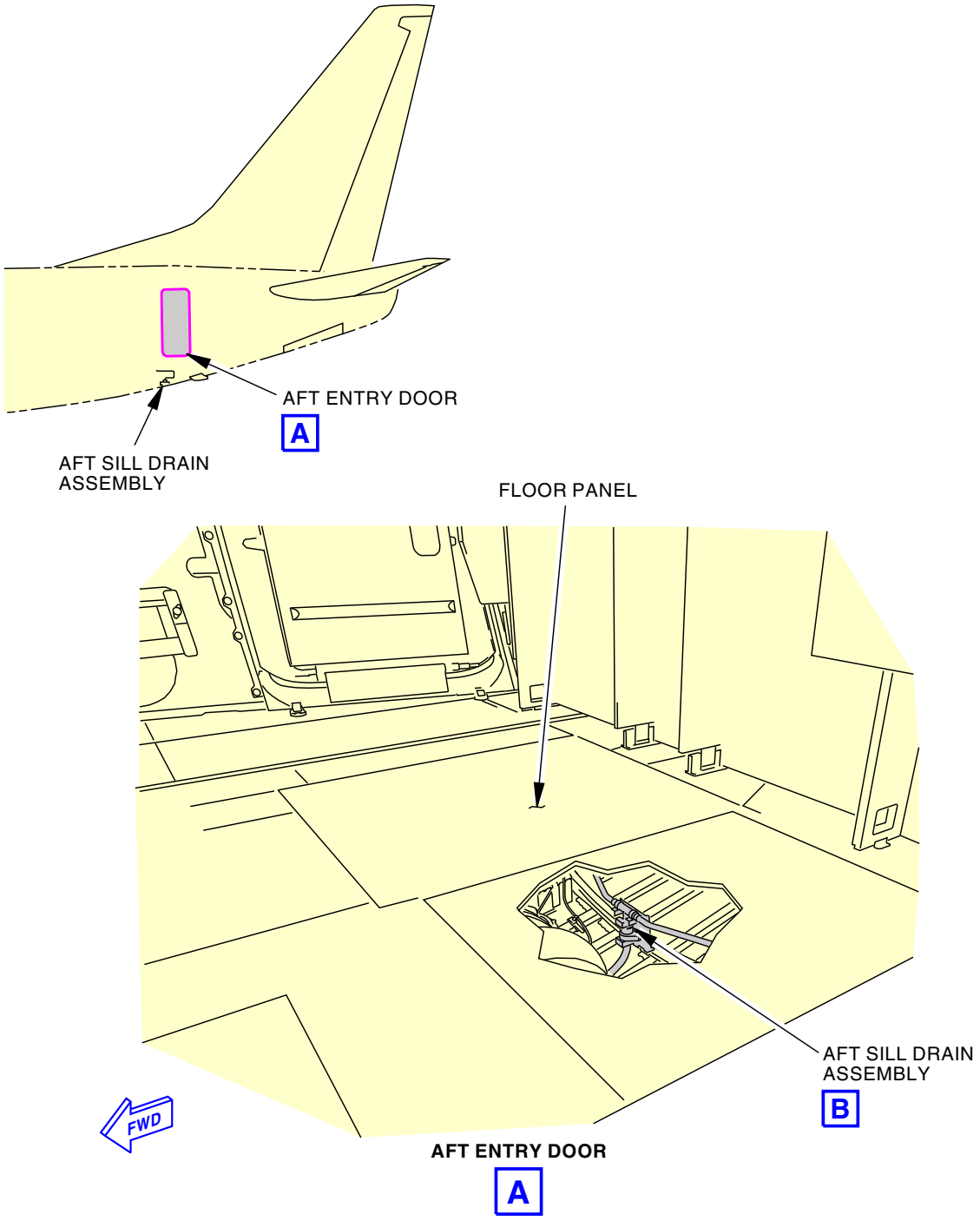
(2) Remove the screw [22] and washer [23] that attach the strap [24], saddle [25], and aft float valve [21] to the structure.

SUBTASK 38-31-06-020-006

(3) Remove the aft float valve [21].

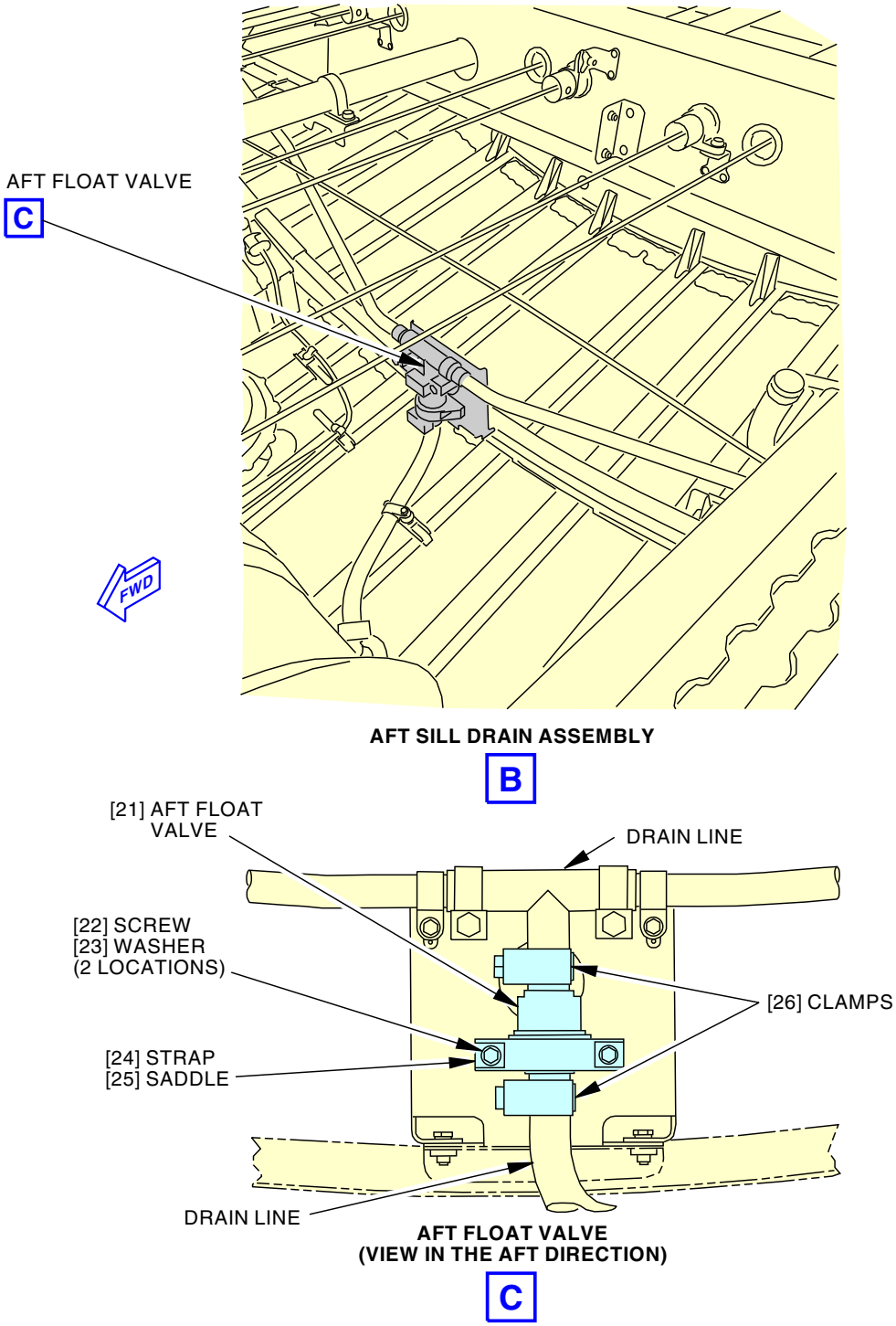
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Sill Drain Float Valve Installation
Figure 402/38-31-06-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Sill Drain Float Valve Installation
Figure 402/38-31-06-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-31-06-400-802

5. Aft Sill Drain Float Valve Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the aft sill drain float valve.

B. References

Reference	Title
21-31-03-400-801	Aft Outflow Valve Assembly Installation (P/B 401)
38-11-01-400-801	Water Tank Installation (P/B 401)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Aft float valve	38-31-06-01-200	PTF ALL

D. Location Zones

Zone	Area
191	Lower Wing-To-Body Fairing - Forward of Wing Box

E. Aft Sill Drain Float Valve Installation

SUBTASK 38-31-06-390-002

(1) Install each end of the aft float valve [21] to the drain lines and tighten the clamps [26].

NOTE: The large end of the float valve should be up.

SUBTASK 38-31-06-420-002

(2) Install the screw [22] and washer [23] that attach the strap [24], saddle [25], and aft float valve [21] to the structure.

NOTE: The lip on the float valve should be above the strap.

F. Aft Sill Drain Float Valve Installation Test

SUBTASK 38-31-06-860-002

(1) Pour water into the aft sill drain.

SUBTASK 38-31-06-710-002

(2) Make sure that the water flows to the aft sill drain float valve without leakage.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-31-06-410-002

(1) To close the access for the Aft Sill Drain Valve Assembly, do one of these steps:

(a) Do this task: Passenger Cabin Floor Panel Installation, TASK 53-21-00-400-801.

(b) Do this task: Aft Outflow Valve Assembly Installation, TASK 21-31-03-400-801.

(c) Do this task: Water Tank Installation, TASK 38-11-01-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VACUUM WASTE SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Standard Practices for Work with the Toilet Waste and Equipment

(2) Waste Tube Maintenance Practice

(3) A deactivation of the toilet.

(4) An activation of the toilet.

(5) A test of the vacuum pressure in the waste system.

NOTE: Use this task when you think there is leakage of the vacuum waste system.

(6) A removal of a blockage from the toilet.

NOTE: This task is for a blockage that is upstream of the toilet flush valve.

(7) An inspection of the vacuum lines to isolate a blockage.

(8) A removal of a blockage from the vacuum waste lines.

NOTE: This task is for a blockage that is between the toilet flush valve and the waste tank.

(9) A removal of a blockage from the drain line for the waste tank.

NOTE: This task is for a blockage that is between the waste tank and the service panel for the waste tank.

(10) A removal of a blockage from the drain line for the waste tank from outside the airplane.

NOTE: This task is for a blockage that is between the waste tank and the service panel for the waste tank.

(11) A deactivation of the vacuum waste blower.

(12) An activation of the vacuum waste blower.

(13) Power Cycle of the Vacuum Toilet Assembly Integrated System.

TASK 38-32-00-910-801

2. Standard Practices for Work with the Toilet Waste and Equipment

A. General

(1) You must obey local regulatory procedures and company policies when working with toilet waste and toilet waste equipment.

(2) Special procedures are given for work on the equipment that follows:

(a) The toilet assembly

(b) The vacuum waste lines

(c) The waste tanks

(d) The waste drain ball valves.

(3) If you operate the vacuum waste system in hangar areas, attach venting equipment to the vacuum system exhaust to remove the exhaust from personnel and hangar areas.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-419	Gloves - Rubber, Elbow Length
STD-1136	Facepiece (Mask) - Particulate Filtering, Disposable (NIOSH N95 approved)
STD-1137	Glasses - Safety
STD-1138	Tubing - Flexible, 1.5 Inch Outside Diameter

D. Consumable Materials

Reference	Description	Specification
G02315	Clothing - Disposable Gown, Gloves For Sewage Handling	

E. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

F. Standard Practices for Work with the Toilet Waste and Equipment

SUBTASK 38-32-00-910-001

(1) When servicing or maintaining the toilet system and its components, make sure that you wear the applicable personal protection equipment such as the following:

(a) elbow length rubber gloves, STD-419

(b) filtering facepiece, STD-1136

(c) safety glasses, STD-1137

(d) Disposable clothing, G02315.

SUBTASK 38-32-00-910-002

(2) You must thoroughly disinfect anything you come in contact with when working with toilet waste.

SUBTASK 38-32-00-910-003

WARNING MAKE SURE THAT YOU WEAR RUBBER GLOVES WHEN YOU DO WORK ON THE TOILET SYSTEM. CLEAN YOUR HANDS FULLY WITH SOAP AND WATER AFTER THE END OF THE PROCEDURE. OBEY THIS INSTRUCTION TO PREVENT THE ILLNESS TO PERSONNEL.

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(3) You must thoroughly disinfect yourself and your clothing before working on other aircraft system components.

NOTE: Never service the airplane potable water system, or maintain potable water system components after servicing or maintaining the toilet waste system and it's components without thoroughly disinfecting yourself.

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SUBTASK 38-32-00-910-004

WARNING ALL TOILET SYSTEM COMPONENTS MUST BE CLEAN OF WASTE AND DISINFECTED BEFORE USUAL HANDLING AND SHIPPING. OBEY THIS INSTRUCTION TO PREVENT THE ILLNESS TO PERSONNEL OR OTHERS.

(4) Thoroughly clean and disinfect any toilet waste system components removed from the airplane prior to normal handling and shipping operations.

SUBTASK 38-32-00-910-005

(5) Thoroughly disinfect the following components:

(a) The toilet bowl

(b) The flush valve

(c) Any other contaminated components.

SUBTASK 38-32-00-910-006

WARNING MAKE SURE THAT YOU OBEY THE MANUFACTURER'S INSTRUCTIONS WHEN YOU USE THE CHEMICAL PRECHARGE. THE CHEMICAL PRECHARGE CONTAINS CORROSIVE, AND POISONOUS MATERIALS. OBEY THESE INSTRUCTIONS TO PREVENT INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

CAUTION DO NOT ADD THE CHEMICAL PRECHARGE TO THE TANKS IF THE AIRPLANE WILL BE KEPT IN AN AREA WHERE IT CAN FREEZE. THE CHEMICAL PRECHARGE CAN CAUSE DAMAGE TO THE WASTE SYSTEM IF IT FREEZES.

(6) Add approximately 1 quart (1 liter) of chemical precharge to the toilet bowl before you remove the waste material from the bowl.

NOTE: Use a chlorine solution to clean the items you wear as an alternative to chemical precharge.

SUBTASK 38-32-00-160-001

(7) Do these steps to remove the contents of the toilet bowl, if a waste system is serviceable:

(a) Install a length of 1.5 inch outside diameter flexible tubing, STD-1138, from a serviceable toilet to the blocked toilet.

(b) Flush the serviceable toilet to drain the toilet bowl of the blocked toilet.

(c) Make sure the toilet bowl is empty.

(d) If the toilet bowl is not empty, flush the serviceable toilet again.

(e) Remove the 1.5 in. (3.8 cm) flexible tubing.

SUBTASK 38-32-00-160-002

(8) If no waste system is serviceable, manually remove the contents of the toilet bowl.

SUBTASK 38-32-00-910-007

(9) Do the steps that follow before you remove these parts:

(a) The waste tank

(b) The waste tank components

(c) The waste drain ball valve

(d) The drain lines downstream from the waste tank

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) The waste drain valve assembly.

1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

2) After the waste tank is serviced, drain the chemical precharge.

——— END OF TASK ———

TASK 38-32-00-040-801

3. Toilet Deactivation

(Figure 201)

A. General

(1) This task is for the deactivation of a toilet while the remaining toilets in the system are serviceable.

B. References

Reference	Title
38-32-01-000-801	Toilet Shroud Removal (P/B 401)
38-32-01-400-801	Toilet Shroud Installation (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Toilet Deactivation

SUBTASK 38-32-00-010-001

(1) Get access to the lavatory for the toilet removal.

SUBTASK 38-32-00-010-033

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-00-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-00-860-002

(4) Set the water shutoff valve for the lavatory to the FAUCET ONLY or OFF position.

SUBTASK 38-32-00-910-008

(5) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-020-001

(6) To remove the toilet shroud, do this task: Toilet Shroud Removal, TASK 38-32-01-000-801.

F. Toilet Deactivation

SUBTASK 38-32-00-020-002

(1) Disconnect the electrical connector.

SUBTASK 38-32-00-040-001

(2) Push the manual override lever down to close the toilet flush valve.

NOTE: The override lever is on the flush valve at the back of the toilet.

SUBTASK 38-32-00-930-001

(3) Identify the applicable lavatory that is not serviceable.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-410-024

(1) To install the toilet shroud, do this task: Toilet Shroud Installation, TASK 38-32-01-400-801.

SUBTASK 38-32-00-860-057

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

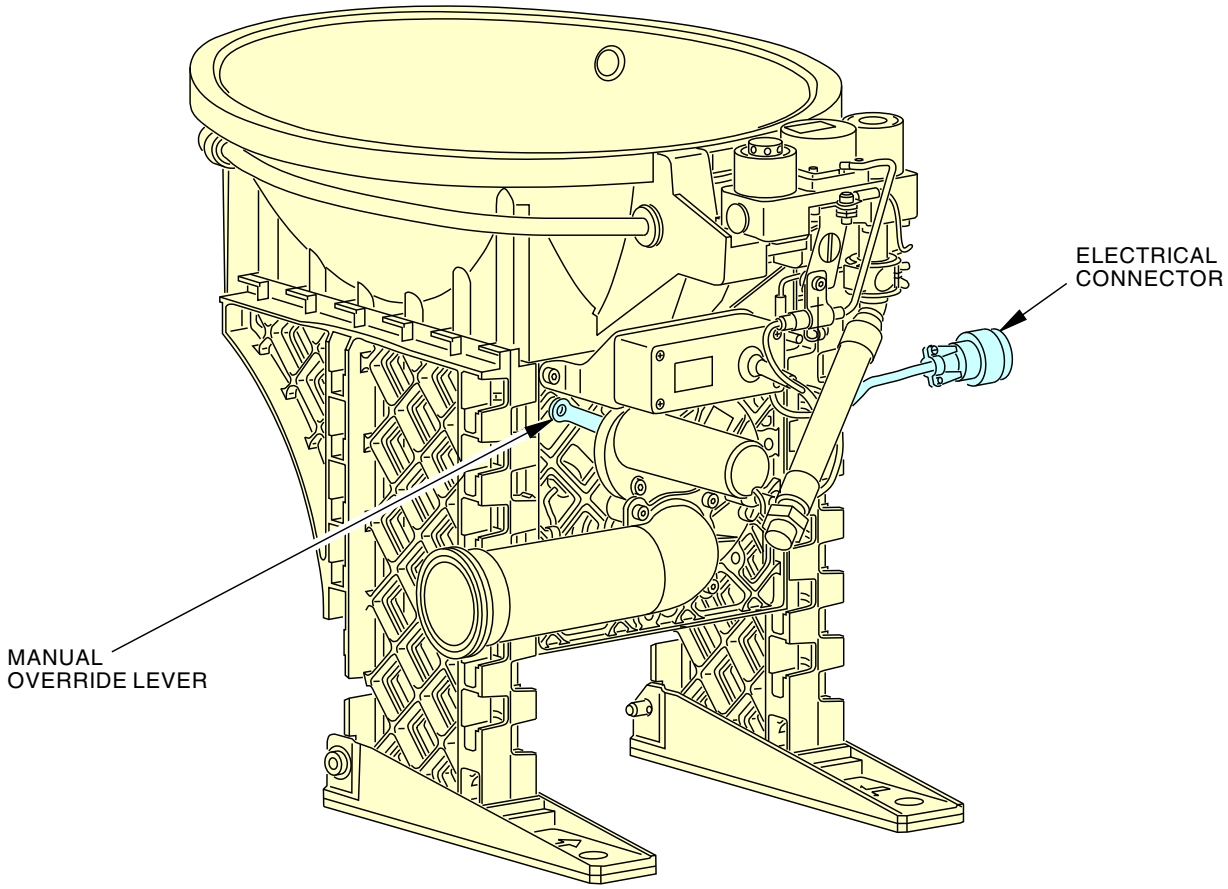
SUBTASK 38-32-00-410-025

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**TOILET ASSEMBLY
(REAR VIEW)**

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**Toilet Deactivation
Figure 201/38-32-00-990-801**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-420-801

4. Toilet Activation

(Figure 201)

A. References

Reference	Title
38-32-01-000-801	Toilet Shroud Removal (P/B 401)
38-32-01-400-801	Toilet Shroud Installation (P/B 401)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for Toilet Activation

SUBTASK 38-32-00-010-002

(1) Get access to the lavatory for the toilet removal.

SUBTASK 38-32-00-010-034

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-00-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-00-860-004

(4) Set the water shutoff valve for the lavatory to the SUPPLY ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-910-009

(5) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-020-003

(6) To remove the toilet shroud, do this task: Toilet Shroud Removal, TASK 38-32-01-000-801.

E. Toilet Activation

SUBTASK 38-32-00-440-001

(1) Push the manual override lever down to open the toilet flush valve.

NOTE: The override lever is on the flush valve at the back of the toilet.

SUBTASK 38-32-00-420-001

(2) Connect the electrical connector.

SUBTASK 38-32-00-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-00-410-026

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-00-710-001

(5) Do these steps to make sure the toilet operates correctly:

(a) Operate the flush switch several times.

(b) Make sure the toilet flushes correctly.

SUBTASK 38-32-00-930-002

(6) Remove the “not serviceable” identification from the lavatory.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-410-027

(1) To install the toilet shroud, do this task: Toilet Shroud Installation, TASK 38-32-01-400-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-780-801

5. Waste System - Vacuum Pressure Test

(Figure 202)

A. General

(1) This task uses the vacuum toilet tester to examine the vacuum during the operation of the toilet flush system.

(2) The Vacuum Toilet Tester is referred to as the Tester in this procedure.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1945	Tester - Toilet Vacuum Part #: A38006-19 Supplier: 81205

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Procedure

SUBTASK 38-32-00-910-010

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-860-006

(2) Put the tester, SPL-1945 in its position as follows:

(a) Make sure the valve on the tester, SPL-1945 is in the closed position.

(b) Put the end of the tester, SPL-1945 with the flexible seal into the toilet bowl on the exit hole.

(c) Push and then hold the tester tightly against the exit hole.

SUBTASK 38-32-00-780-001

(3) Do these steps to do a test of the vacuum at the toilet.

(a) Push the flush switch for the toilet.

(b) Make sure the water stays in the toilet bowl during the toilet flush operation.

(c) Write down the highest indication (inches of mercury) you see on the vacuum gage during the toilet flush operation.

(d) Do the steps above two more times.

SUBTASK 38-32-00-860-007

(4) Move the valve to the open position and then remove the tester, SPL-1945 from the toilet.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

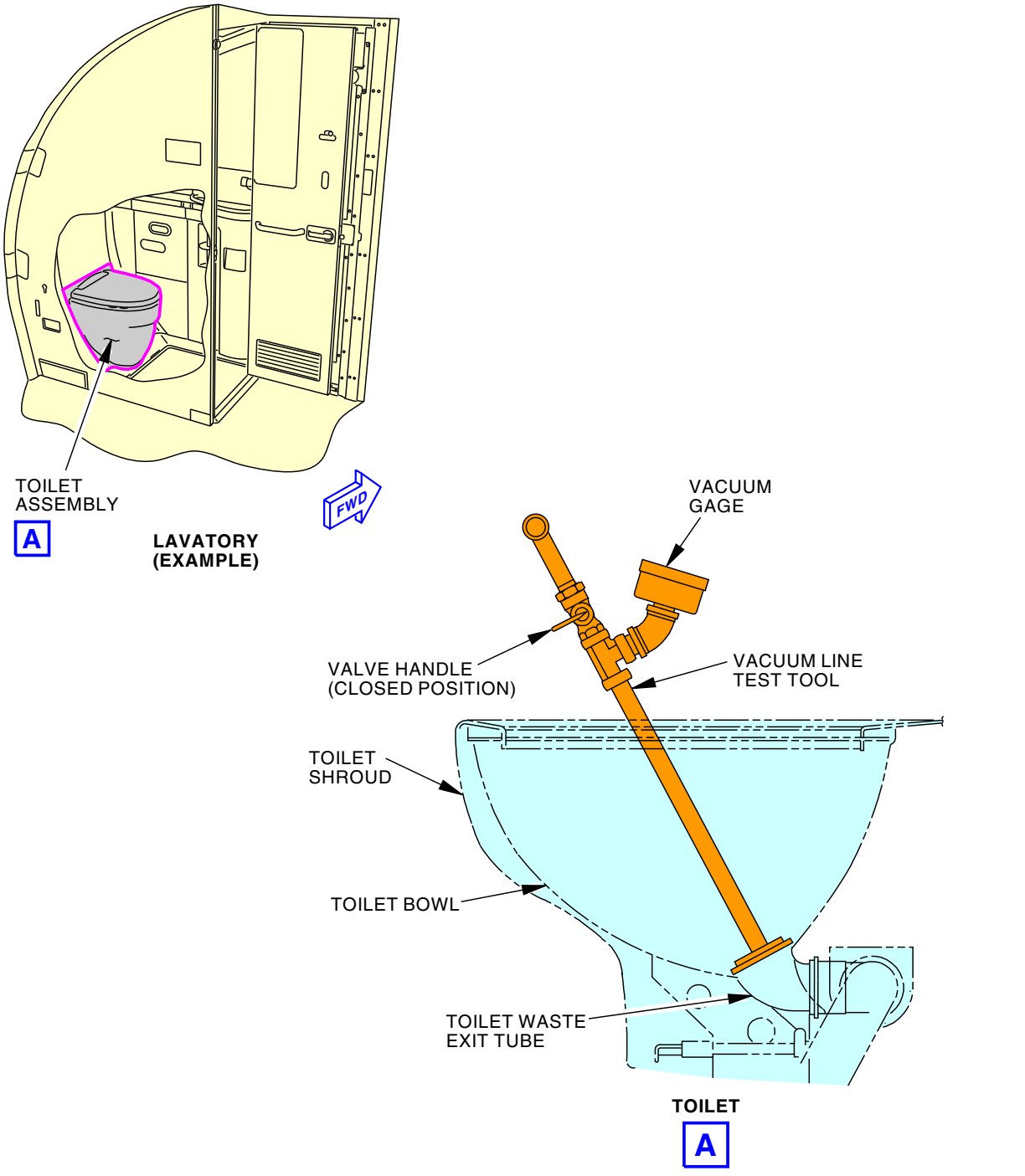
SUBTASK 38-32-00-220-001

(5) Make sure the average of the indications on the vacuum gage is from 5 to 10 inches of mercury.

NOTE: An average of less than 5 inches of mercury shows a blockage or leakage of the vacuum waste system. The blockage or leakage is between the toilet and the waste tank. An average of less than 5 inches of mercury for all of the vacuum waste system shows a problem with the vacuum blower.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Line Test
Figure 202/38-32-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-160-801

6. Toilet Blockage Removal

(Figure 203)

A. General

(1) This task is for a blockage that is upstream of the toilet flush valve.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1943	Remover - Blockage, Vacuum Toilet Part #: A38003-30 Supplier: 81205

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Toilet Blockage Removal

SUBTASK 38-32-00-910-011

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-860-008

(2) For the toilet that has a blockage, do the necessary step:

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ROW-F CIRCUIT-BREAKERS IN THE P91 AND P92 PANELS. IF POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER BEFORE YOU ACCESS THE CIRCUIT-BREAKERS ON THE INSIDE OF THE P91 AND P92 PANELS. THE P91 AND P92 PANELS CONTAIN HIGH VOLTAGES AND CURRENTS THAT CAN CAUSE INJURIES TO PERSONS.

(a) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-480-001

(3) Carefully put the vacuum toilet blockage remover, SPL-1943 into the toilet bowl.

SUBTASK 38-32-00-160-003

CAUTION

DO NOT PUSH THE BLOCKAGE REMOVER TOOL INTO THE TOILET BOWL WITH TOO MUCH FORCE. THE END OF THE BLOCKAGE REMOVER TOOL IS SHARP. IT CAN CAUSE DAMAGE TO THE TOILET, OR THE FLUSH VALVE.

(4) Move the blockage remover into the toilet bowl until it touches the blockage or until it engages in the toilet.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-160-004

(5) When the blockage remover touches the blockage, turn its handle clockwise to engage the blockage.

SUBTASK 38-32-00-160-005

(6) Pull the blockage remover carefully out of the toilet to pull the blockage out of the toilet.

SUBTASK 38-32-00-860-009

(7) After you remove the blockage, do the necessary step:

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ROW-F CIRCUIT-BREAKERS IN THE P91 AND P92 PANELS. IF POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER BEFORE YOU ACCESS THE CIRCUIT-BREAKERS ON THE INSIDE OF THE P91 AND P92 PANELS. THE P91 AND P92 PANELS CONTAIN HIGH VOLTAGES AND CURRENTS THAT CAN CAUSE INJURIES TO PERSONS.

(a) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-710-002

(8) Put approximately one gallon (4 liters) of water in the toilet bowl.

SUBTASK 38-32-00-710-003

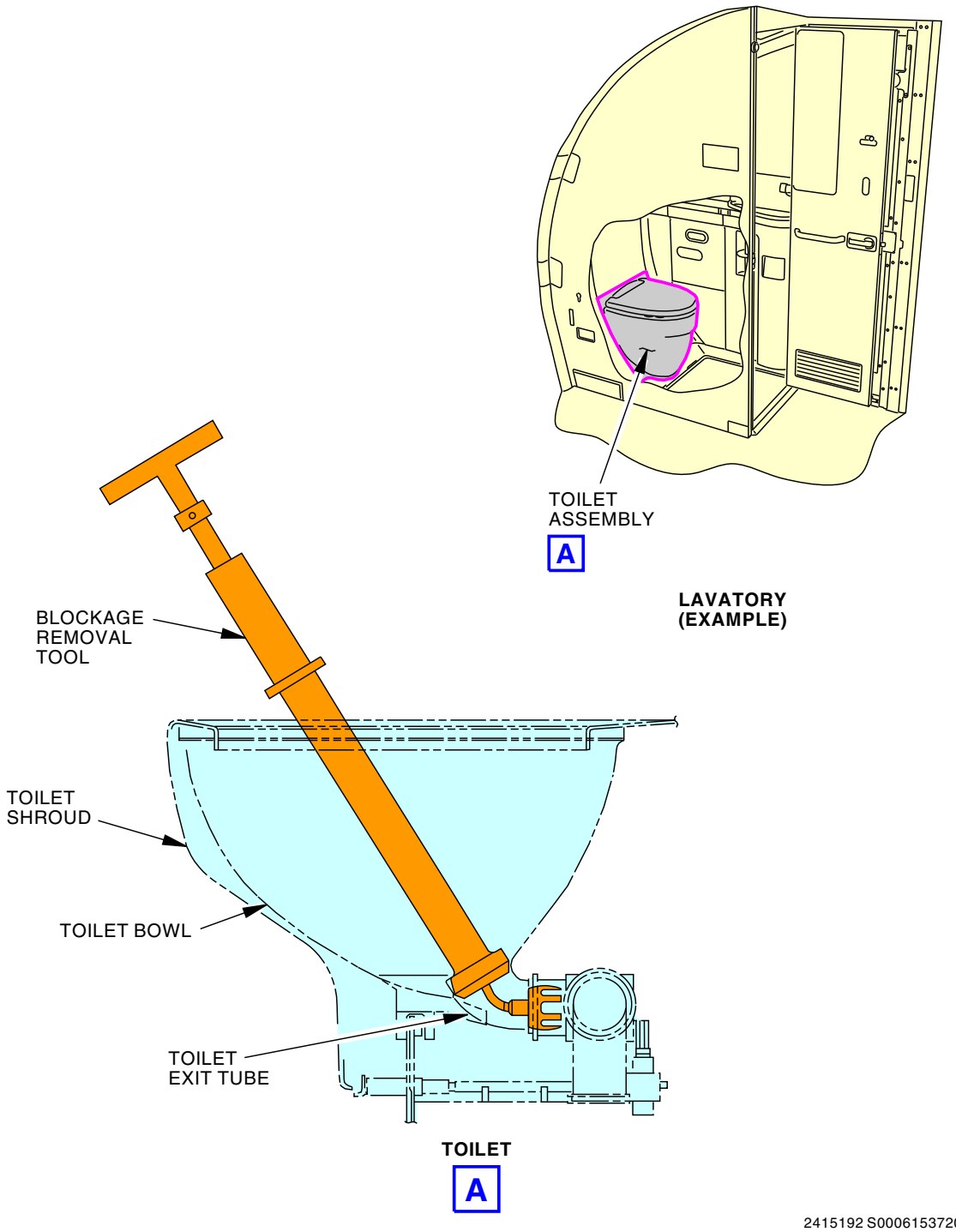
(9) Push the flush switch for the toilet.

SUBTASK 38-32-00-710-004

(10) Make sure the toilet operates correctly.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Toilet Blockage Removal
Figure 203/38-32-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-280-801

7. Vacuum Line Blockage Inspection

Figure 204

A. General

(1) This task uses the "coin tap" test to isolate blockages in the vacuum waste lines. Use a small metal object (a coin) when you do this test.

B. Consumable Materials

Reference	Description	Specification
B00636	Acid, Acetic (Vinegar) (5%-10% Acetic Acid)	Commercially available product
B00638	Cleaner - Acidic Liquid - Honey Bee 60 (McGean-Rohco)	

C. Location Zones

Zone	Area
100	Lower Half of Fuselage

D. Procedure

SUBTASK 38-32-00-750-001

(1) Do the "coin tap" test as follows:

(a) Hit the vacuum waste line lightly with the metal object for each inch of the tube.

1) A sharp metal ring shows that the section has no waste layer or blockages.

2) A dull muffled sound shows that the section has a waste layer or a blockage.

SUBTASK 38-32-00-020-004

(2) Remove from the airplane all sections of the vacuum waste line that show a waste layer or a blockage.

SUBTASK 38-32-00-110-001

(3) Do these steps to clean out the vacuum waste line with a waste layer or a blockage:

(a) Fill or soak the removed line with Honey Bee 60 cleaner, B00638, acid, B00636 or phosphoric acid, 15%.

(b) Keep the line in the solution until the layer of waste is loose or removed.

(c) If you soak the line in the solution, give the labels that are on the line necessary protection.

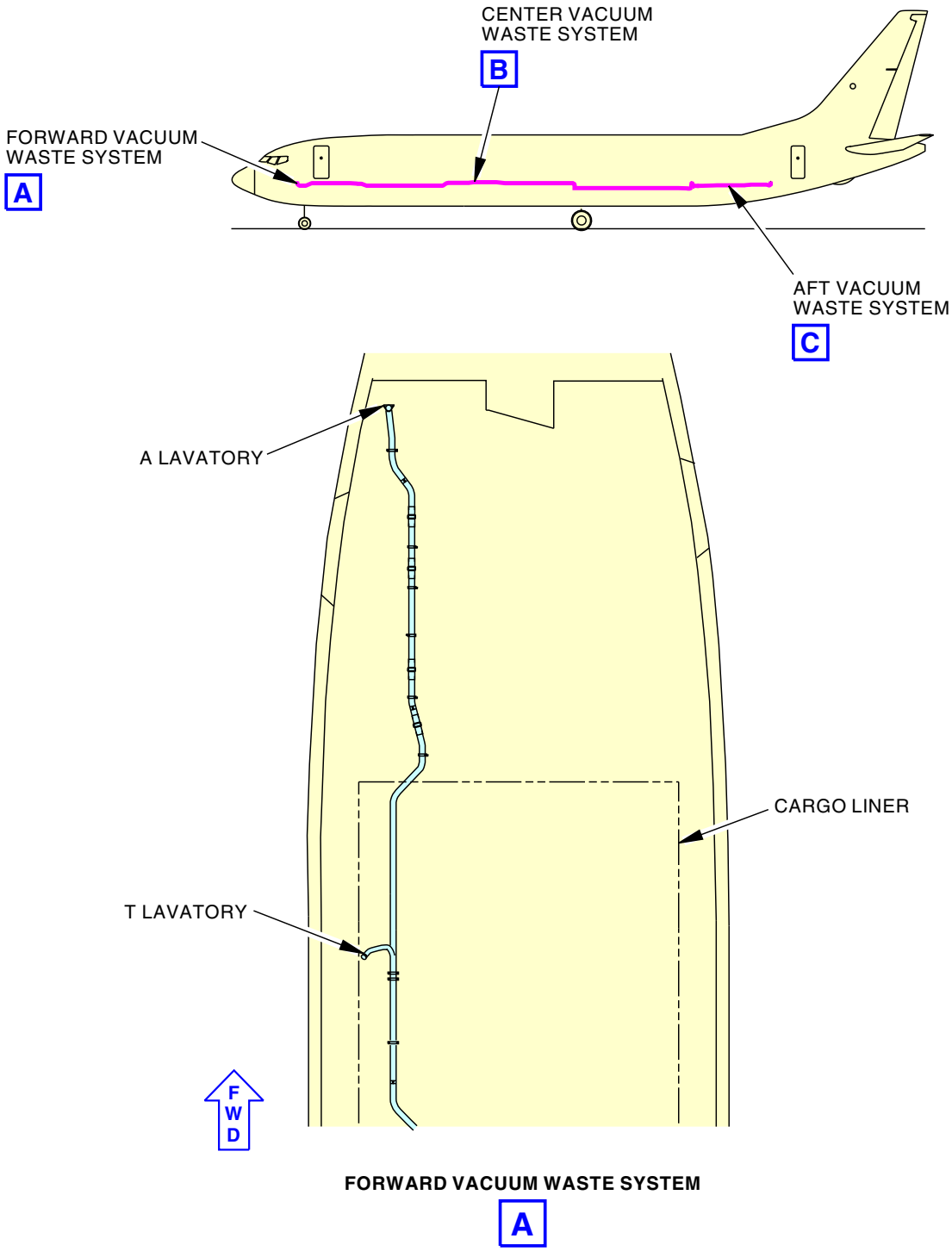
(d) Do the “coin tap” test to make sure that the line is clear.

SUBTASK 38-32-00-420-002

(4) Install the vacuum waste lines.

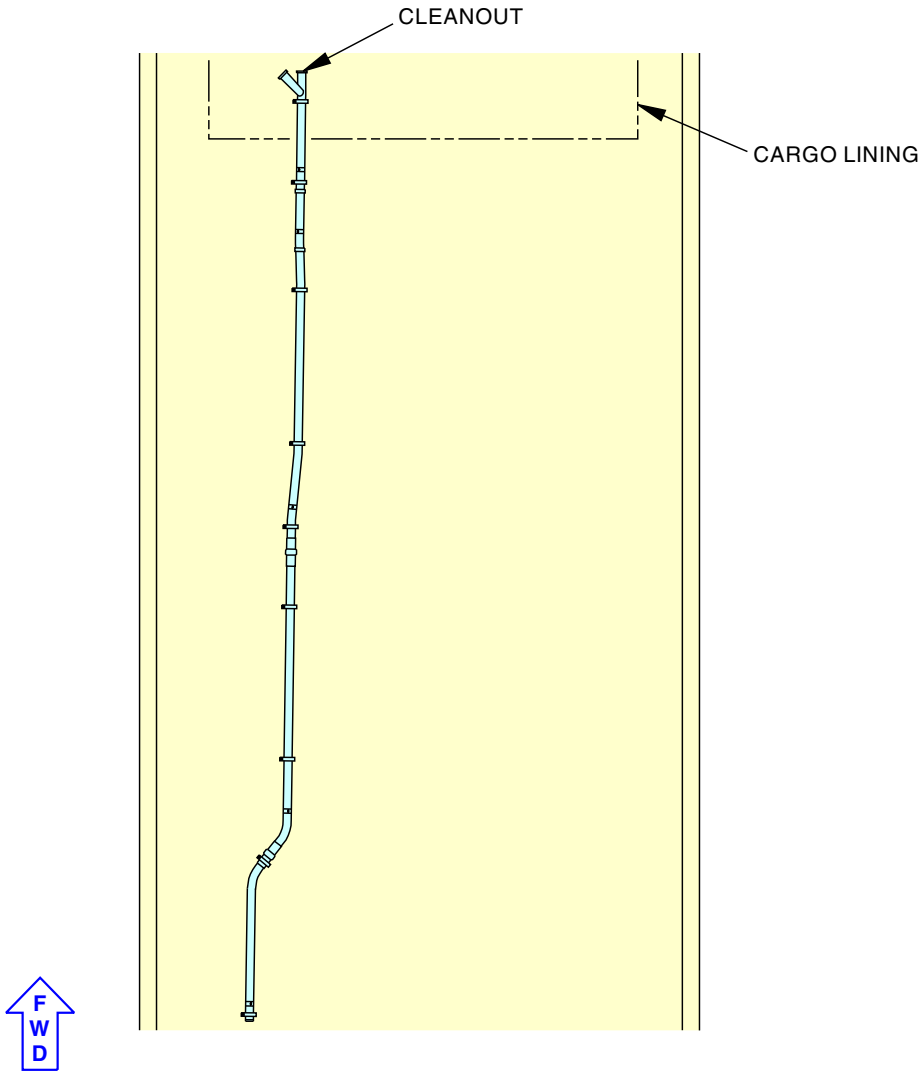
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Waste System - Primary Routing
Figure 204/38-32-00-990-805 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



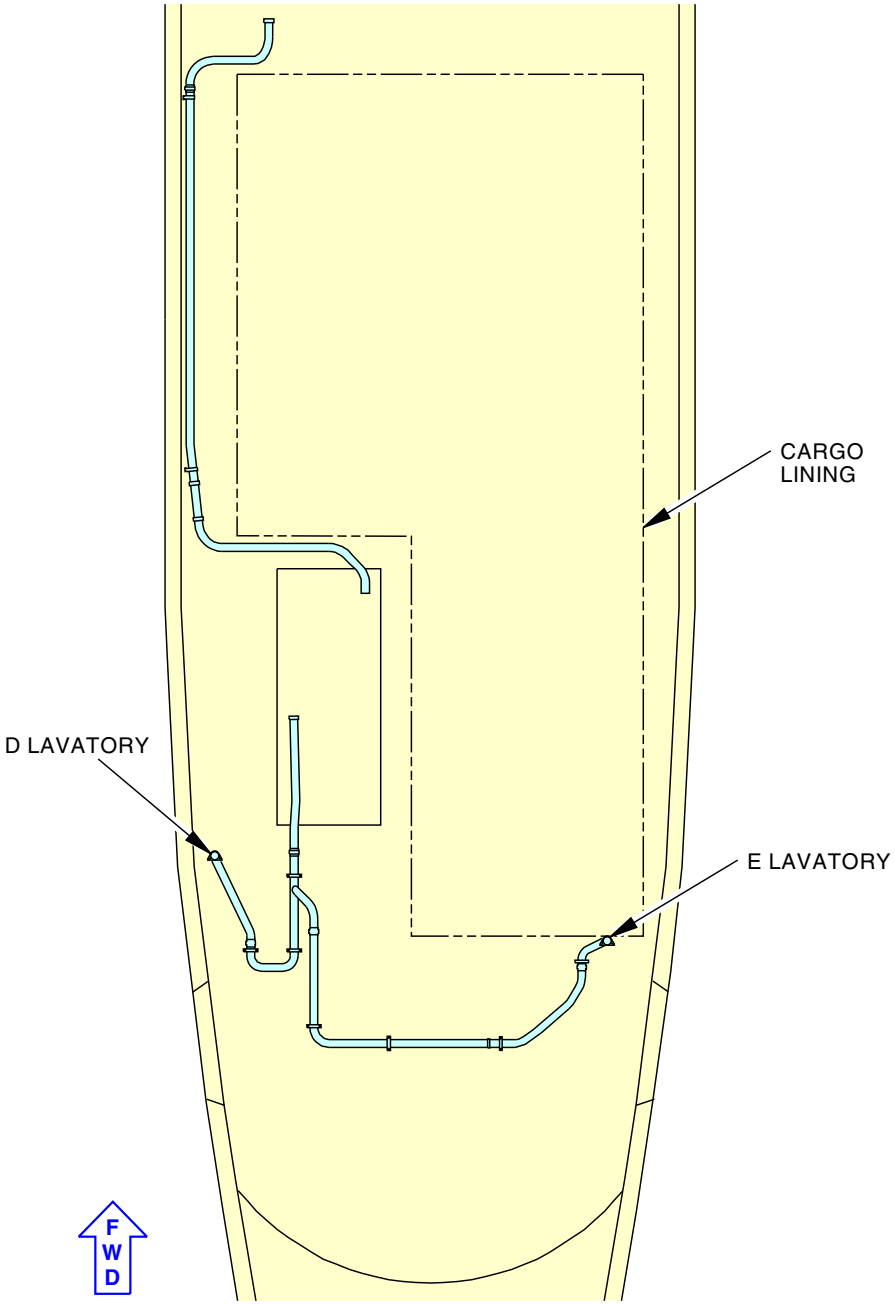
CENTER VACUUM WASTE SYSTEM

B

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Vacuum Waste System - Primary Routing
Figure 204/38-32-00-990-805 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AFT VACUUM WASTE SYSTEM

C

2415198 S00061537213_V1

Vacuum Waste System - Primary Routing
Figure 204/38-32-00-990-805 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-910-802

8. Waste Tube - Maintenance Practice

(Figure 204)

A. References

Reference	Title
50-10-03-000-801	Cargo Compartment Ceiling Liner Removal (P/B 401)
50-10-03-400-801	Cargo Compartment Ceiling Liner Installation (P/B 401)

B. Location Zones

Zone	Area
100	Lower Half of Fuselage

C. Waste Tube - Maintenance Practice

SUBTASK 38-32-00-860-011

(1) To access the waste tube and components, use (Figure 204).

SUBTASK 38-32-00-860-012

(2) To remove the plug from the cleanout port, do the steps that follow:

(a) Do this task: Cargo Compartment Ceiling Liner Removal, TASK 50-10-03-000-801.

(b) Remove the plug from the cleanout port.

SUBTASK 38-32-00-860-013

(3) To install the plug from the cleanout port, do the steps that follow:

(a) Make sure the cleanout port is clean.

(b) Install the plug from the cleanout port.

(c) Do this task: Cargo Compartment Ceiling Liner Installation, TASK 50-10-03-400-801.

——— END OF TASK ———

TASK 38-32-00-160-802

9. Vacuum Waste Line Blockage Removal

(Figure 205)

A. General

(1) This task is for a blockage that is between the toilet flush valve and the waste tank.

NOTE: The removal of the toilet assembly is necessary to do this procedure when you do it from the upper half of the fuselage.

(2) This procedure is also applicable to material in the lines that are sealed with a cap. The removal of the line cap and the removal of ceiling lining panels for access is necessary to do this procedure. For removal and installation procedures, do this task: Waste Tube - Maintenance Practice, TASK 38-32-00-910-802.

B. References

Reference	Title
38-32-01-000-802	Vacuum Toilet Assembly Removal (P/B 401)
38-32-01-400-802	Vacuum Toilet Assembly Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1947	Equipment - Removal, Vacuum Lavatory System Blockage Part #: G38004-17 Supplier: 81205 Opt Part #: G38004-10 Supplier: 81205

D. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

E. Procedure

SUBTASK 38-32-00-910-012

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-860-014

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ROW-F CIRCUIT-BREAKERS IN THE P91 AND P92 PANELS. IF POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER BEFORE YOU ACCESS THE CIRCUIT-BREAKERS ON THE INSIDE OF THE P91 AND P92 PANELS. THE P91 AND P92 PANELS CONTAIN HIGH VOLTAGES AND CURRENTS THAT CAN CAUSE INJURIES TO PERSONS.

(2) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-160-006

(3) If it is necessary to remove the contents of the toilet bowl with the toilet flush valve open, do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-010-003

(4) To get access to the vacuum waste line for the lavatory upstream of the blockage, do this task: Vacuum Toilet Assembly Removal, TASK 38-32-01-000-802.

SUBTASK 38-32-00-420-003

(5) Install the kinetic water ram vacuum lavatory system blockage removal equipment, SPL-1947 on the vacuum waste line.

NOTE: You can use a plumbers snake if the kinetic water ram equipment is not available.

SUBTASK 38-32-00-610-001

(6) Fill the waste line with water from a toilet upstream of the blockage or through the kinetic water ram equipment.

SUBTASK 38-32-00-610-002

(7) Do these steps while you fill the waste line.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Monitor the air bleed line while you fill the system.

(b) Stop the water flow when you see water at the bleed line exit.

(c) Disconnect the air bleed line.

(d) Put a cap on the air bleed connection.

(e) Do these steps for all toilets upstream of the blockage.

SUBTASK 38-32-00-860-016

(8) Make sure the air pressure of the kinetic water ram equipment is 70 to 80 PSIG (480-550 kPa).

NOTE: You can use a shop air source to get this pressure.

SUBTASK 38-32-00-040-002

WARNING ATTACH DO-NOT-OPERATE TAGS ON THE TOILET FLUSH HANDLES FOR ALL TOILETS IN THE SYSTEM. IF YOU FLUSH THE TOILETS THAT ARE UPSTREAM OF THE BLOCKAGE, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(9) Attach a DO-NOT-OPERATE tag to the toilet flush handles for all toilets upstream of the blockage.

SUBTASK 38-32-00-860-017

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ROW-F CIRCUIT-BREAKERS IN THE P91 AND P92 PANELS. IF POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER BEFORE YOU ACCESS THE CIRCUIT-BREAKERS ON THE INSIDE OF THE P91 AND P92 PANELS. THE P91 AND P92 PANELS CONTAIN HIGH VOLTAGES AND CURRENTS THAT CAN CAUSE INJURIES TO PERSONS.

(10) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-710-005

(11) Flush a toilet in the same waste system with the blockage and then pull the handle of the kinetic water ram equipment.

NOTE: Flush the toilet and pull the handle of the kinetic water ram equipment at the same time to have the most effect on the blockage.

SUBTASK 38-32-00-710-006

(12) Make sure the blockage is clear.

SUBTASK 38-32-00-160-007

(13) If the blockage is not clear, pressurize the kinetic water ram equipment and do the blockage removal procedure again until the line is clear.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-860-018

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ROW-F CIRCUIT-BREAKERS IN THE P91 AND P92 PANELS. IF POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER BEFORE YOU ACCESS THE CIRCUIT-BREAKERS ON THE INSIDE OF THE P91 AND P92 PANELS. THE P91 AND P92 PANELS CONTAIN HIGH VOLTAGES AND CURRENTS THAT CAN CAUSE INJURIES TO PERSONS.

(14) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-020-005

(15) Remove the kinetic water ram equipment on the vacuum waste line.

SUBTASK 38-32-00-080-001

(16) To put the system back in its usual condition, do these steps:

(a) Remove the cap on the air bleed connection for all toilets upstream of the blockage.

(b) Connect the air bleed line for all toilets upstream of the blockage.

SUBTASK 38-32-00-420-004

(17) If you removed a toilet, do this task: Vacuum Toilet Assembly Installation, TASK 38-32-01-400-802.

SUBTASK 38-32-00-440-002

(18) Remove the DO-NOT-OPERATE tag to the toilet flush handles for all toilets upstream of the blockage.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ROW-F CIRCUIT-BREAKERS IN THE P91 AND P92 PANELS. IF POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER BEFORE YOU ACCESS THE CIRCUIT-BREAKERS ON THE INSIDE OF THE P91 AND P92 PANELS. THE P91 AND P92 PANELS CONTAIN HIGH VOLTAGES AND CURRENTS THAT CAN CAUSE INJURIES TO PERSONS.

(a) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

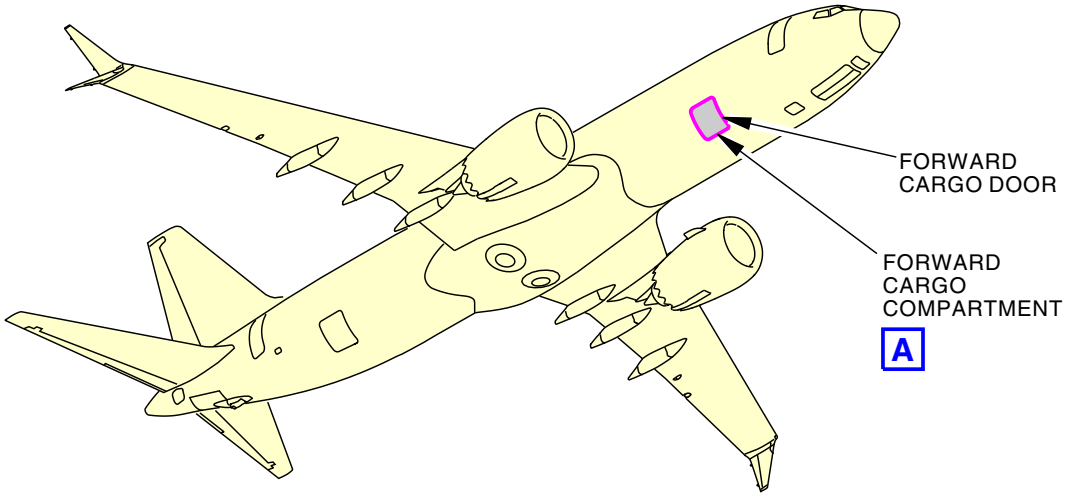
Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

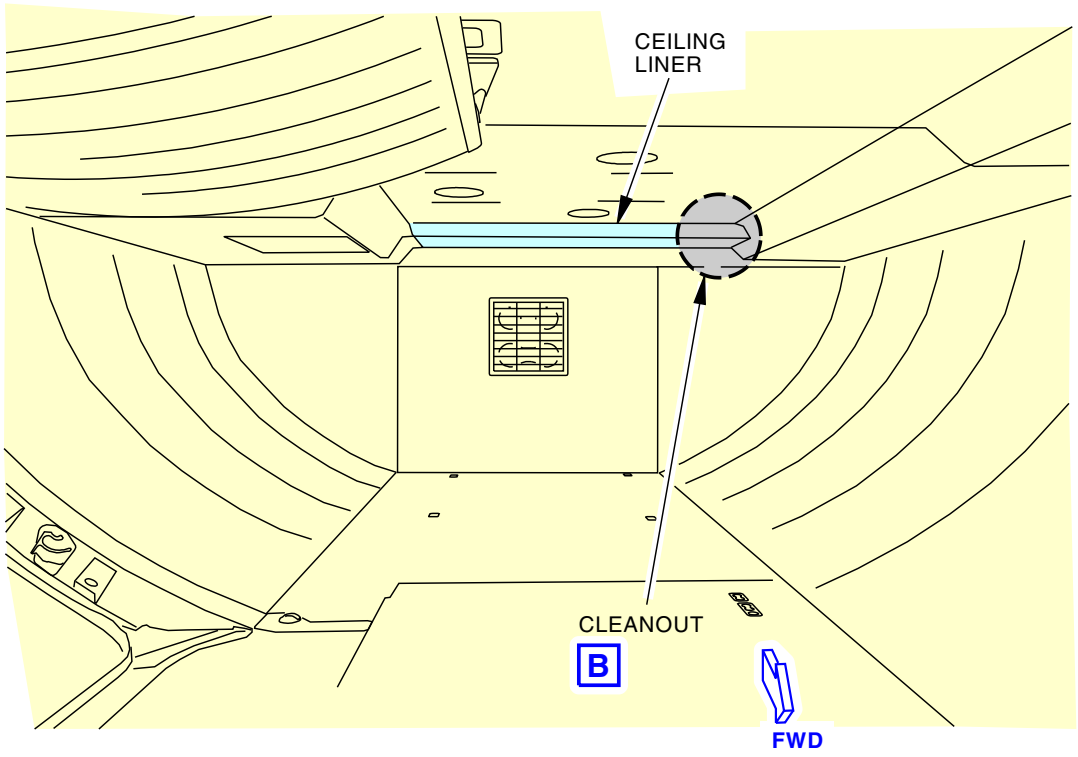
SUBTASK 38-32-00-710-007

(19) Flush the toilet to make sure the toilet operates correctly.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





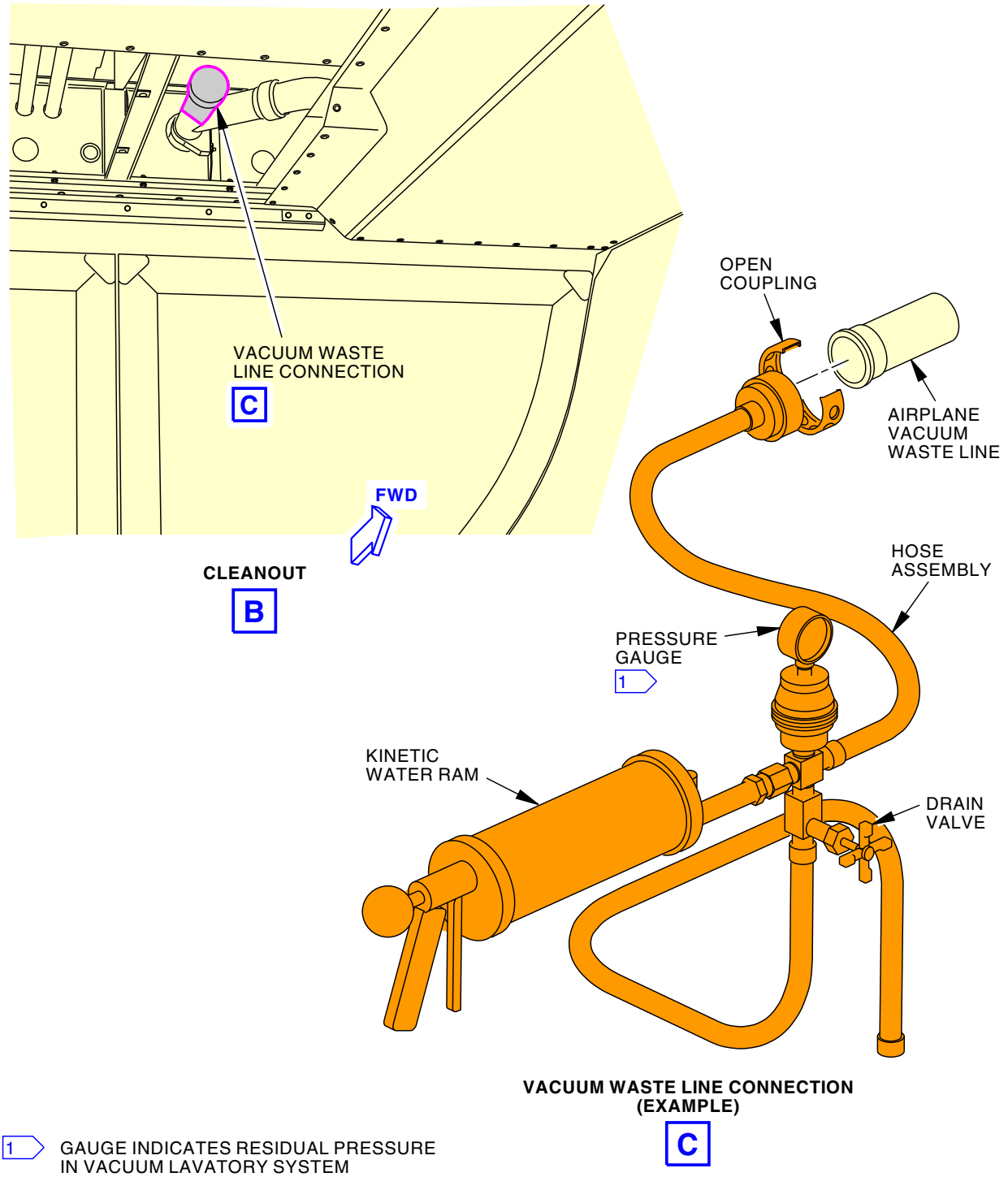
FORWARD CARGO COMPARTMENT

A

2415199 S00061537216_V1

Vacuum Line Blockage Removal
Figure 205/38-32-00-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Line Blockage Removal
Figure 205/38-32-00-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-160-803

10. Waste Tank Drain Line Blockage Removal

(Figure 206)

A. General

(1) This task is for a blockage that is between the waste tank and the waste drain valve assembly.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-33-00-740-802	LCM BITE Test (P/B 501)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1950	Equipment - Removal, Vacuum Waste System Blockage Part #: G38006-1 Supplier: 81205
STD-1142	Equipment - Waste System Servicing

D. Location Zones

Zone	Area
143	Area Below Aft Cargo Compartment - Left
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
145AL	Waste Service Door

F. Procedure

SUBTASK 38-32-00-910-013

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-010-004

(2) Get access to the aft cargo compartment.

SUBTASK 38-32-00-010-005

(3) To remove the waste tank enclosure, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-00-010-006

(4) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

SUBTASK 38-32-00-860-019

(5) Make sure the waste drain ball valve and the waste drain valve assembly are in the closed position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-020-006

(6) Remove the cap on the waste drain elbow.

SUBTASK 38-32-00-160-008

(7) Connect the pressurization vacuum waste system blockage removal equipment, SPL-1950 to the waste drain elbow.

SUBTASK 38-32-00-160-009

(8) Pressurize the waste drain elbow to 15 PSIG.

SUBTASK 38-32-00-010-007

(9) Open the waste drain ball valve.

SUBTASK 38-32-00-610-003

(10) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to service the waste tanks. Do not disconnect the servicing equipment.

SUBTASK 38-32-00-860-020

(11) Make sure the waste tank is empty.

SUBTASK 38-32-00-610-004

(12) If the waste tank is empty, add the chemical precharge.

(a) If the waste tank is not empty, do these steps:

1) Close the waste drain ball valve.

2) Do the steps again to pressurize the waste drain elbow and to open the waste drain ball valve.

(b) If the waste tank is not empty, then continue.

1) Close the waste drain ball valve.

CAUTION

DO NOT INCREASE THE PRESSURE TO MORE THAN 20 PSIG (135 KPA). A PRESSURE OF MORE THAN 20 PSIG CAN CAUSE DAMAGE TO THE WASTE DRAIN SYSTEM.

2) Increase the pressure to a maximum of 20 PSIG.

3) Do the steps again to pressurize the waste drain elbow and to open the waste drain ball valve.

(c) If the waste tank is not empty, remove the contents of the waste system manually.

SUBTASK 38-32-00-610-005

(13) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-00-710-008

(14) To make sure the system is operational, do this task: LCM BITE Test, TASK 38-33-00-740-802.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-080-002

(1) Disconnect the waste system servicing equipment, STD-1142 from the rinse fitting.

SUBTASK 38-32-00-080-013

(2) Disconnect the pressurization vacuum waste system blockage removal equipment, SPL-1950 from the waste drain elbow.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-410-021

(3) Install the cap on the waste drain elbow.

SUBTASK 38-32-00-410-001

(4) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-00-410-002

(5) Close the access to the aft cargo compartment.

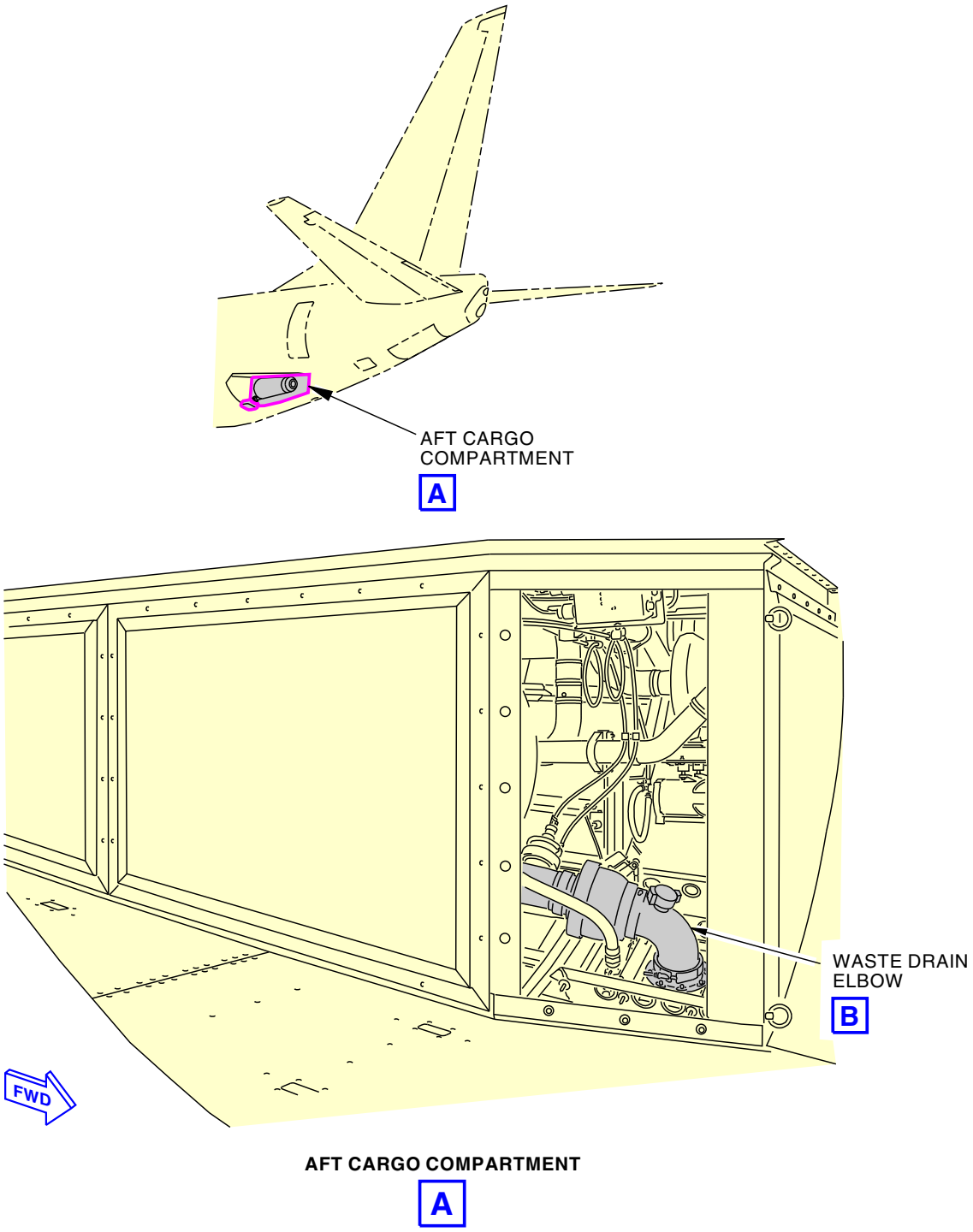
SUBTASK 38-32-00-410-003

(6) Close this access panel:

Number	Name/Location
145AL	Waste Service Door

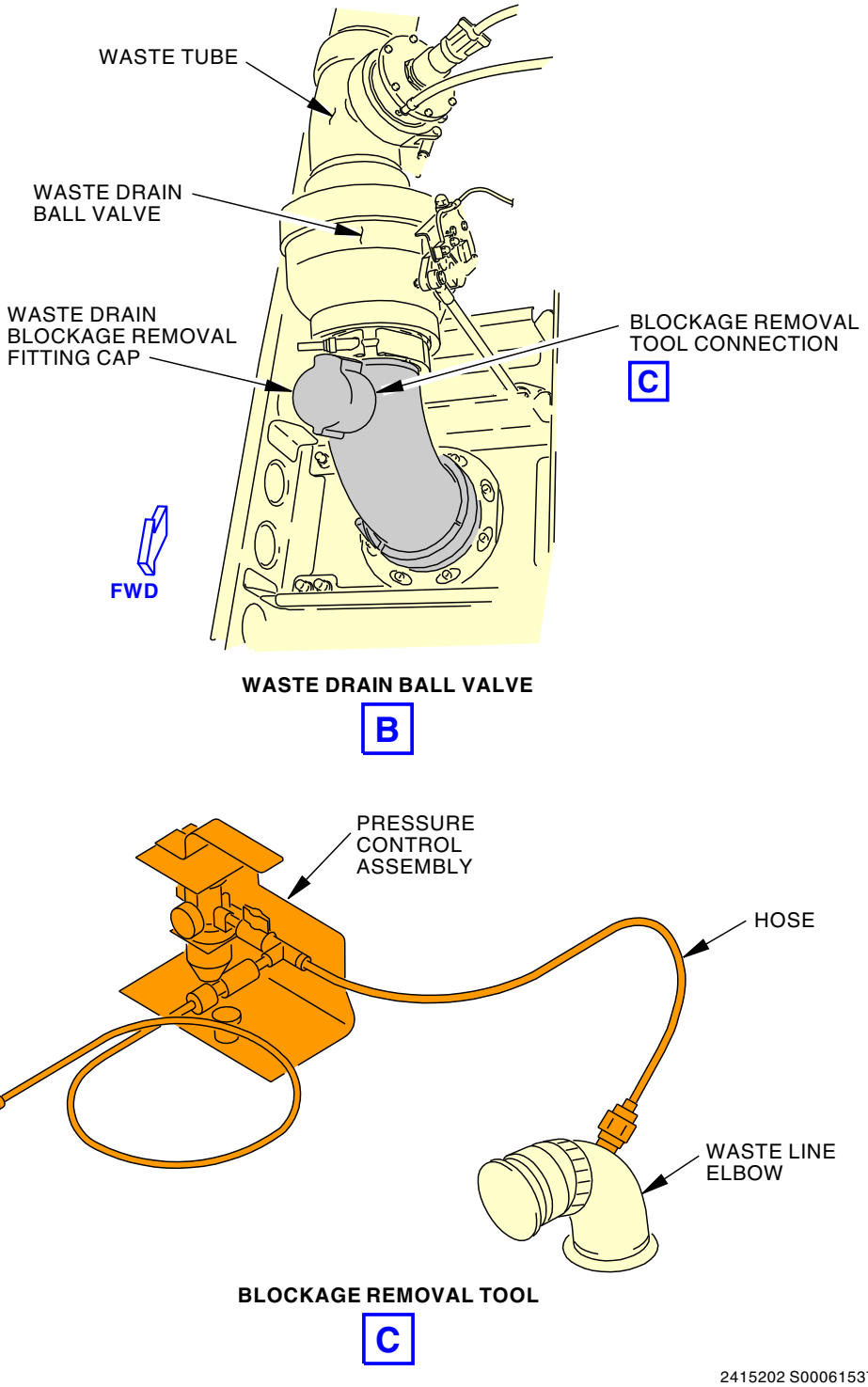
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Line Blockage Removal
Figure 206/38-32-00-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Line Blockage Removal
Figure 206/38-32-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-160-804

11. Waste Drain Line Blockage Removal

(Figure 207)

A. General

(1) This task is for a blockage that is between the waste tank and the waste drain valve assembly from outside the airplane. This task is an alternative to, do this task: Waste Tank Drain Line Blockage Removal, TASK 38-32-00-160-803.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-33-00-740-802	LCM BITE Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1950	Equipment - Removal, Vacuum Waste System Blockage Part #: G38006-1 Supplier: 81205
STD-77	Air Source, Regulated - Dry Filtered, 0-50 psig
STD-1142	Equipment - Waste System Servicing

D. Location Zones

Zone	Area
143	Area Below Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
145AL	Waste Service Door

F. Procedure

SUBTASK 38-32-00-910-014

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-010-008

(2) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

SUBTASK 38-32-00-010-009

(3) Open the cap on the waste drain valve assembly.

SUBTASK 38-32-00-480-002

(4) Connect the vacuum waste system blockage removal equipment, SPL-1950 to the waste drain valve assembly.

SUBTASK 38-32-00-480-003

(5) Attach the waste drain hose from the waste system servicing equipment, STD-1142 to the vacuum waste system blockage removal equipment, SPL-1950.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-860-021

(6) Make sure the hose clamp is in its position on the hose between the waste drain valve assembly and the waste drain hose.

SUBTASK 38-32-00-860-022

(7) Push the PRESS TO OPEN lever on the waste drain valve assembly.

SUBTASK 38-32-00-860-023

(8) Pull the handle at the waste service panel to open the waste drain ball valve.

SUBTASK 38-32-00-860-024

(9) Make sure the air shutoff valve on the vacuum waste system blockage removal equipment, SPL-1950 is in the closed position.

SUBTASK 38-32-00-860-025

(10) Attach an regulated air source, STD-77 to the vacuum waste system blockage removal equipment, SPL-1950.

SUBTASK 38-32-00-860-026

WARNING MAKE SURE THE PRESSURE TO THE WASTE BLOCKAGE REMOVAL EQUIPMENT IS NOT MORE THAN 25 PSIG (170 KPA). IF THE PRESSURE IS MORE THAN 25 PSIG (170 KPA), INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(11) Make sure the pressure to the vacuum waste system blockage removal equipment, SPL-1950 is 0 to 25 PSIG (170 kPa).

SUBTASK 38-32-00-160-010

(12) Open the air pressure valve of the vacuum waste system blockage removal equipment, SPL-1950 until the air flow is constant and the pressure does not increase or until the pressure drops, then close the air pressure valve.

SUBTASK 38-32-00-860-027

(13) Remove the hose clamp between the waste drain valve assembly and the waste drain hose.

SUBTASK 38-32-00-710-009

(14) Make sure the blockage is clear.

SUBTASK 38-32-00-610-006

(15) If the blockage is clear, then continue.

(a) If the blockage is not clear, do the blockage removal procedure again until the line is clear.

SUBTASK 38-32-00-080-003

(16) Disconnect the waste drain hose from the vacuum waste system blockage removal equipment, SPL-1950.

SUBTASK 38-32-00-080-004

(17) Disconnect the vacuum waste system blockage removal equipment, SPL-1950 from the waste drain valve assembly.

SUBTASK 38-32-00-610-007

(18) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-00-610-008

(19) To make sure the waste indication system is operational, do this task: LCM BITE Test, TASK 38-33-00-740-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

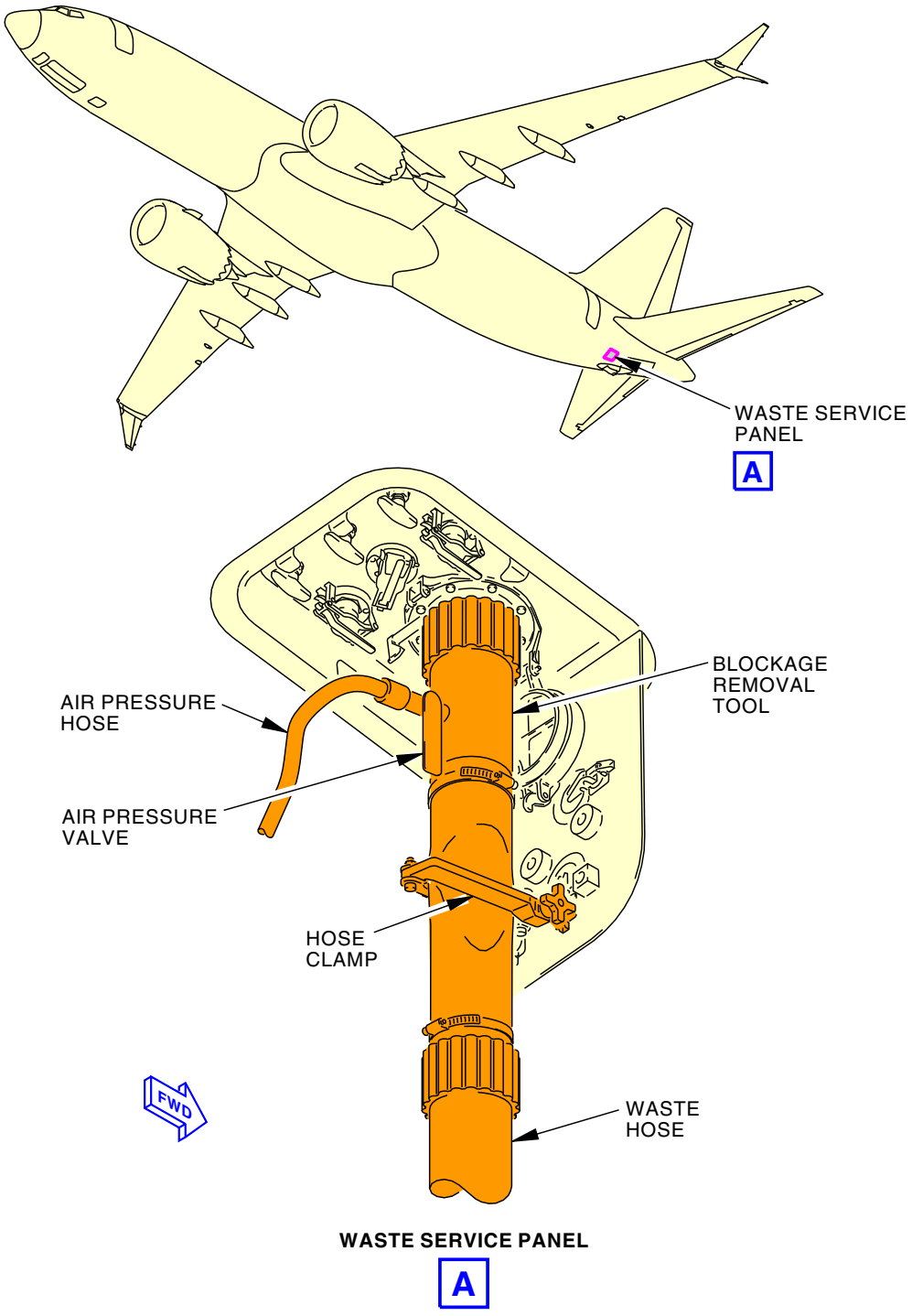
SUBTASK 38-32-00-010-010

(1) Close this access panel:

Number	Name/Location
145AL	Waste Service Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Waste Blockage Removal Tool
Figure 207/38-32-00-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-040-802

12. Vacuum Waste Blower - Deactivation

(Figure 208)

A. General

(1) This task will deactivate the Vacuum Waste Blower.

B. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

C. Procedure

SUBTASK 38-32-00-860-028

WARNING BE CAREFUL WHEN YOU OPEN OR CLOSE CIRCUIT BREAKERS IN THE P91 AND P92 PANELS WHILE THE PANELS HAVE POWER. ELECTRICAL SHOCK CAN CAUSE INJURIES TO PERSONNEL.

(1) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

D. Tryout

NOTE: This tryout is to make sure the Vacuum Waste Blower is in a zero energy state.

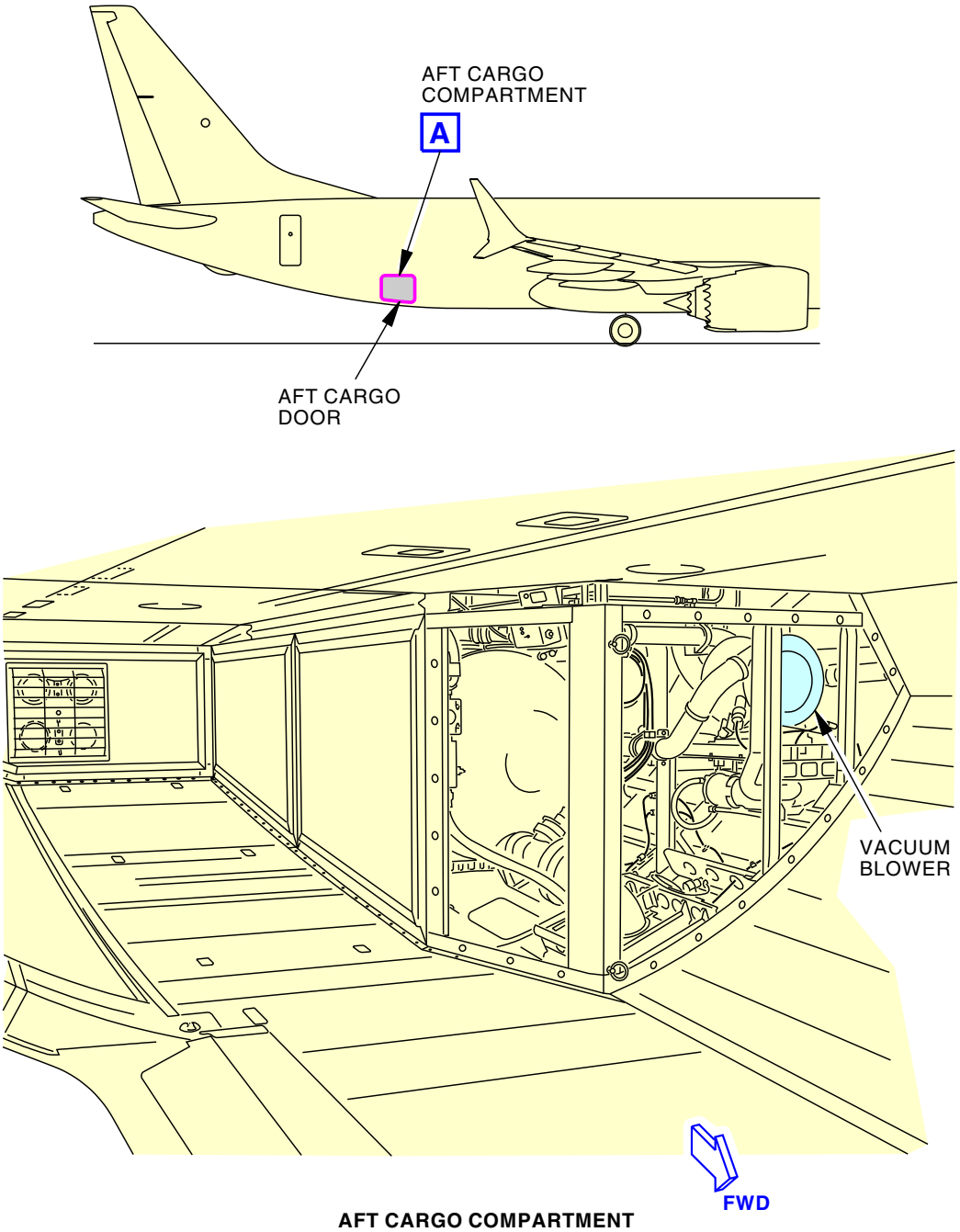
SUBTASK 38-32-00-710-010

(1) Flush a toilet in the waste system.

(a) Make sure that the toilet flush does not operate.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2415204 S00061537224_V1

Vacuum Waste Blower
Figure 208/38-32-00-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-440-801

13. Vacuum Waste Blower - Activation

(Figure 208)

A. General

(1) This task will activate the Vacuum Waste Blower.

B. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

C. Procedure

SUBTASK 38-32-00-860-029

WARNING BE CAREFUL WHEN YOU OPEN OR CLOSE CIRCUIT BREAKERS IN THE P91 AND P92 PANELS WHILE THE PANELS HAVE POWER. ELECTRICAL SHOCK CAN CAUSE INJURIES TO PERSONNEL.

(1) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

END OF TASK

TASK 38-32-00-820-801

14. Power Cycle of the Vacuum Toilet Assembly Integrated System

Figure 209

A. General

(1) This procedure are instructions to power cycle the Vacuum Toilet Assembly (VTA) Integrated System Controller (ISC).

B. References

Reference	Title
38-32-01-000-801	Toilet Shroud Removal (P/B 401)
38-32-01-400-801	Toilet Shroud Installation (P/B 401)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Procedure

SUBTASK 38-32-00-010-031

(1) Remove the toilet shroud: TASK 38-32-01-000-801.

SUBTASK 38-32-00-020-010

(2) Disconnect the electrical connector [1] from the toilet assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-860-053

(3) Move flush valve assembly override lever [2] to the mid point.

NOTE: The override lever [2] is on the flush valve at the back of the toilet. The flush valve is closed if the override lever [2] is in the up position and open when the override lever [2] is in the down position.

SUBTASK 38-32-00-420-009

(4) Wait 10-15 seconds, then reconnect the toilet assembly electrical connector [1].

NOTE: The ISC will go through a power cycle start up after the electrical connector [1] is connected.

(a) Make sure that the flush valve assembly override lever [2] will move down then up.

SUBTASK 38-32-00-710-017

(5) Push the flush switch for the toilet.

(a) If the toilet does not flush, repeat steps 2 thru 4 to power cycle the ISC.

(b) If the toilet flushes, then flush the toilet 4-10 times to verify continued operation.

NOTE: The flush switch is disabled for 15 seconds between flushes.

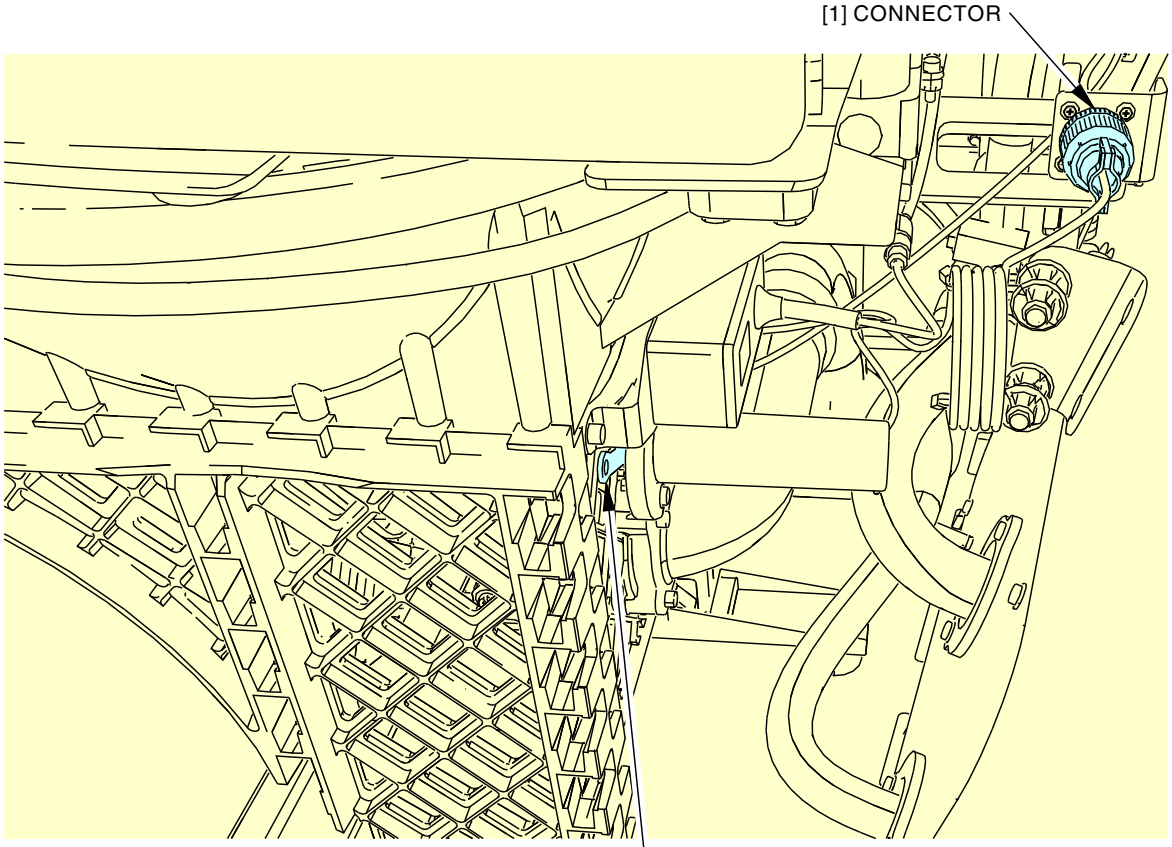
E. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-410-020

(1) Install the toilet shroud (TASK 38-32-01-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Power Cycle of the Vacuum Toilet Assembly Integrated System Controller
Figure 209/38-32-00-990-818

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VACUUM WASTE SYSTEM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Vacuum waste tubes removal

(2) Vacuum waste tubes installation.

TASK 38-32-00-020-801

2. Vacuum Waste Tubes Removal

A. General

(1) This task gives the instructions to remove a vacuum waste tube section for replacement or maintenance.

(2) You can do this procedure if you must remove and install a waste tube because of these conditions:

- If you can not remove a clog in a tube by an in-situ procedure (PAGEBLOCK 38-32-00/701).
- If it is necessary to remove a tube for access or for replacement.

B. References

Reference	Title
05-51-57-000-802	Corrosion Removal After Acid Spills (P/B 201)
38-32-00 P/B 701	VACUUM WASTE SYSTEM - CLEANING/PAINTING
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-32-00-990-805	Figure: Vacuum Waste System - Primary Routing (P/B 201)
53-11-00-990-803	Figure: Galley and Lavatory Areas (P/B 201)

C. Location Zones

Zone	Area
100	Lower Half of Fuselage

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 38-32-00-910-015

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-010-032

(2) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-865-001

WARNING BE CAREFUL WHEN YOU OPEN OR CLOSE CIRCUIT BREAKERS IN THE P91 AND P92 PANELS WHILE THE PANELS HAVE POWER. ELECTRICAL SHOCK CAN CAUSE INJURIES TO PERSONNEL.

(3) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-010-011

(4) Get access to the waste tube to remove. See the figures that follow:

- Vacuum Waste System - Primary Routing/Figure 38-32-00-990-805
- Galley and Lavatory Areas/Figure 53-11-00-990-803
- Vacuum Waste Tubes Installation/Figure 401

SUBTASK 38-32-00-030-001

(5) If necessary, remove the shroud [2] over the clampshell coupling [3] as required.

F. Vacuum Waste Tube Removal

SUBTASK 38-32-00-030-002

(1) To remove the clampshell coupling [3] do these steps:

NOTE: Be prepared to catch unwanted fluid from the waste line.

(a) Pull the latches up.

(b) Remove the clampshell coupling [3] from the flanges of the two ferrules and the sleeve [4].

(c) Move the sleeve [4] from the ferrules.

(d) Pull the tubes [1] apart.

NOTE: If necessary, loosen the support clamps on the adjacent waste lines.

(e) Remove and discard the O-ring [5] from each ferrule.

SUBTASK 38-32-00-020-007

(2) Remove the tube [1].

NOTE: Be prepared to catch unwanted fluid from the waste line.

(a) If necessary, do this task: Corrosion Removal After Acid Spills, TASK 05-51-57-000-802.

SUBTASK 38-32-00-620-001

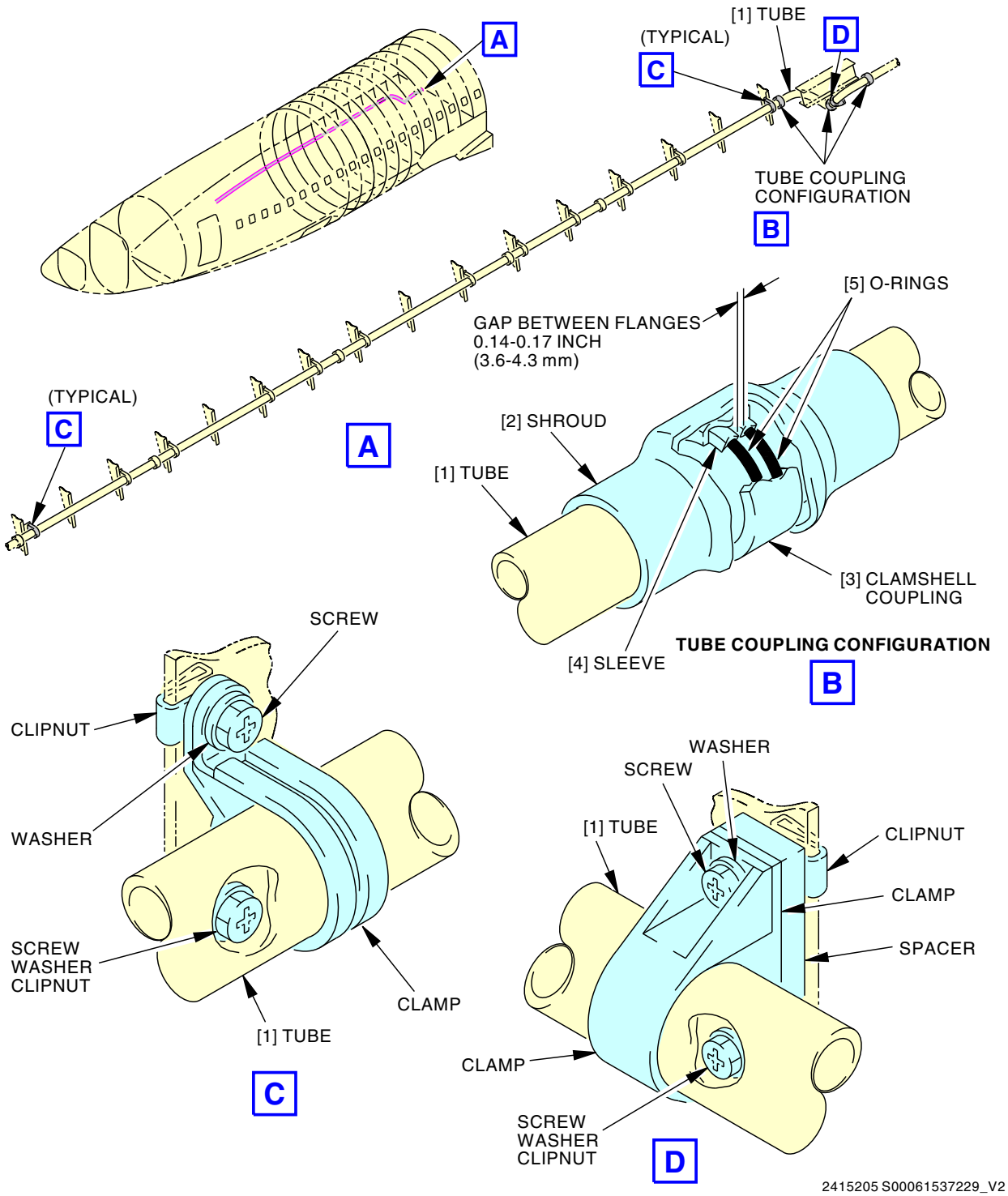
(3) Put a bag or cap on the ends of the tube [1] that you remove to contain unwanted material.

SUBTASK 38-32-00-620-002

(4) Put a bag or cap on the ends of the remaining tubing to contain unwanted material.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Waste Tubes Installation
Figure 401/38-32-00-990-817

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-420-802

3. Vacuum Waste Tubes Installation

A. General

(1) This task gives the instructions to install a vacuum waste tube section for replacement or after maintenance.

(2) You can do this procedure if you must remove and install a waste tube because of these conditions:

- If you can not remove a clog in a tube by an in-situ procedure (PAGEBLOCK 38-32-00/701).
- If it is necessary to remove a tube for access or for replacement.

B. References

Reference	Title
38-32-00 P/B 701	VACUUM WASTE SYSTEM - CLEANING/PAINTING

C. Consumable Materials

Reference	Description	Specification
B00316	Solvent - Aliphatic Naphtha (For Organic Coatings)	TT-N-95 Type I, ASTM D-3735 Type I
D00627 [P06-002]	Lubricant - White Petrolatum	PMC 9609
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Tube	38-32-00-02A-060	PTF ALL
		38-32-00-02A-065	PTF ALL
		38-32-00-02A-070	PTF ALL
		38-32-00-02A-075	PTF ALL
		38-32-00-02A-080	PTF ALL
		38-32-00-08-065	PTF ALL
		38-32-00-08-070	PTF ALL
		38-32-00-08-075	PTF ALL
		38-32-00-20A-055	PTF ALL
		38-32-51-05-150	PTF ALL
		38-32-51-05-155	PTF ALL
		38-32-51-05-160	PTF ALL
		38-32-51-05-165	PTF ALL
		38-32-51-05-185	PTF ALL
		38-32-51-05-190	PTF ALL
		38-32-51-09-030	PTF ALL
		38-32-51-09-035	PTF ALL
		38-32-51-09-040	PTF ALL
		38-32-51-11-275	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
1 (cont.)		38-32-51-11-295	PTF ALL
		38-32-51-11-300	PTF ALL
		38-32-51-11-325	PTF ALL
		38-32-51-11A-090	PTF ALL
		38-32-51-11A-095	PTF ALL
		38-32-51-11A-140	PTF ALL
		38-32-51-11B-130	PTF ALL
		38-32-51-11B-150	PTF ALL
		38-32-51-11B-175	PTF ALL
5	O-ring	38-32-00-08-060	PTF ALL
		38-32-00-20A-050	PTF ALL
		38-32-51-05-130	PTF ALL
		38-32-51-09-055	PTF ALL
		38-32-51-11-250	PTF ALL
		38-32-51-11A-080	PTF ALL
		38-32-51-11B-115	PTF ALL

E. Location Zones

Zone	Area
100	Lower Half of Fuselage

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Vacuum Waste Tubes Installation

SUBTASK 38-32-00-160-011

(1) Do these steps to clean the clampshell coupling [3]:

(a) Make sure that all the parts of the clampshell coupling [3] are free of:

- Cuts
- Scratches
- Dents
- Distortions
- Particles of unwanted material

(b) Clean the areas that seal with a lint-free gauze cotton wiper, G00034, moist with solvent, B00316

(c) Use a new, dry cotton wiper, G00034, to dry the sealing surfaces.

(d) Clean the new O-ring [5] with a clean, dry, lint-free gauze cotton wiper, G00034.

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SUBTASK 38-32-00-420-005

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(2) Install the clampshell coupling [3].

(a) Apply a thin layer of lubricant, D00627 [P06-002] or MIL-L-4343 on the inner surface of the sleeve [4].

(b) Install one new O-ring [5] on each ferrule.

(c) Put the tubes [1] into position for installation.

(d) Make sure that the ends between two adjacent tubing flanges that connect at the clampshell coupling [3] are 0.14 in. (3.56 mm) to 0.17 in. (4.32 mm) apart.

(e) Move the sleeve [4] to the center of the clampshell coupling [3].

(f) Make sure that the “S-tube” aligns with the adjacent tubing.

(g) Push each latch down.

(h) Make sure that each latch is on tight.

(i) Visually examine the latches to make sure that they are at the same level.

(j) Install the shroud [2] on the clampshell coupling [3] if necessary.

NOTE: The shroud [2] is used at joints to give more leakage protection.

SUBTASK 38-32-00-420-006

(3) Install the clamp

(a) Put the clamps onto the tubing.

(b) Install the screws, washers, and nuts on the clamps.

(c) Tighten the clamps.

SUBTASK 38-32-00-860-055

WARNING

MAKE SURE THAT YOU DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

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SUBTASK 38-32-00-410-023

(5) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VACUUM WASTE SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has these tasks:

(1) A leak test for the vacuum waste system.

(2) An operational test of the toilet.

TASK 38-32-00-700-801

2. Vacuum Waste System - Leak Test

(Figure 501)

A. General

(1) This procedure gives the instructions to make sure the vacuum waste system does not have a leak.

B. References

Reference	Title
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1942	Plugs - Vent, Vacuum Waste System Part #: A38001-22 Supplier: 81205
SPL-9132	Vacuum Pump Part #: A38001-68 Supplier: 81205
STD-1134	Vacuum Source, 24 Inches Hg Minimum

D. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
200	Upper Half of Fuselage

E. Access Panels

Number	Name/Location
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the test:

SUBTASK 38-32-00-860-030

WARNING

BE CAREFUL WHEN YOU ACCESS THE (ROW F) CIRCUIT BREAKERS ON THE INSIDE OF THE P91 AND P92 PANELS. IF POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER BEFORE YOU ACCESS THE INSIDE OF THE P91 AND P92 PANELS. THE P91 AND P92 PANELS CONTAIN HIGH VOLTAGES AND CURRENTS THAT MAY CAUSE INJURIES TO PERSONS.

(1) Make sure that these circuit breakers are open and have safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-010-012

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-00-010-013

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(3) To remove the sidewall panels to get access to the vacuum blowers, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-00-020-008

(4) Disconnect the inlet and the outlet connections of the vacuum blower.

SUBTASK 38-32-00-480-004

(5) Install the vent vacuum waste vent plug, SPL-1942 in the inlet and the outlet lines for the vacuum blower.

SUBTASK 38-32-00-480-005

(6) Install the vent plug assembly in the overboard vent outlet for the vacuum blower.

SUBTASK 38-32-00-480-006

(7) Connect the vacuum source (24 inches Hg minimum), STD-1134 source or vacuum test equipment vacuum pump, SPL-9132 to the vent plug assembly.

G. Vacuum Waste System - Leak Test

SUBTASK 38-32-00-790-001

(1) Supply a vacuum of 20 ±1 inches of mercury (500 mm of mercury) to the vacuum waste system.

SUBTASK 38-32-00-790-002

(2) Close the control valve at the vacuum source.

SUBTASK 38-32-00-790-003

(3) Write down the initial pressure after you close the control valve.

SUBTASK 38-32-00-790-004

(4) Write down the final pressure after 5 minutes.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-790-005

(5) Make sure the vacuum is between 10 and 20 inches of mercury (250 and 500 mm of mercury) after 5 minutes.

SUBTASK 38-32-00-790-006

(6) Open the control valve at the vacuum source to bleed the vacuum.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-080-005

(1) Disconnect the vacuum source from the vent plug assembly.

SUBTASK 38-32-00-080-006

(2) Remove the vent plug assembly from the overboard vent outlet for the vacuum blower.

SUBTASK 38-32-00-080-007

(3) Remove the vent plugs from the inlet and the outlet lines for the vacuum blower.

SUBTASK 38-32-00-420-007

(4) Connect the inlet and the outlet connections of the vacuum blower.

SUBTASK 38-32-00-410-004

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GET INTO THE PASSENGER COMPARTMENT DURING A FIRE.

(5) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-00-410-005

(6) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-00-860-031

WARNING

BE CAREFUL WHEN YOU ACCESS THE (ROW F) CIRCUIT BREAKERS ON THE INSIDE OF THE P91 AND P92 PANELS. IF POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER BEFORE YOU ACCESS THE INSIDE OF THE P91 AND P92 PANELS. THE P91 AND P92 PANELS CONTAIN HIGH VOLTAGES AND CURRENTS THAT MAY CAUSE INJURIES TO PERSONS.

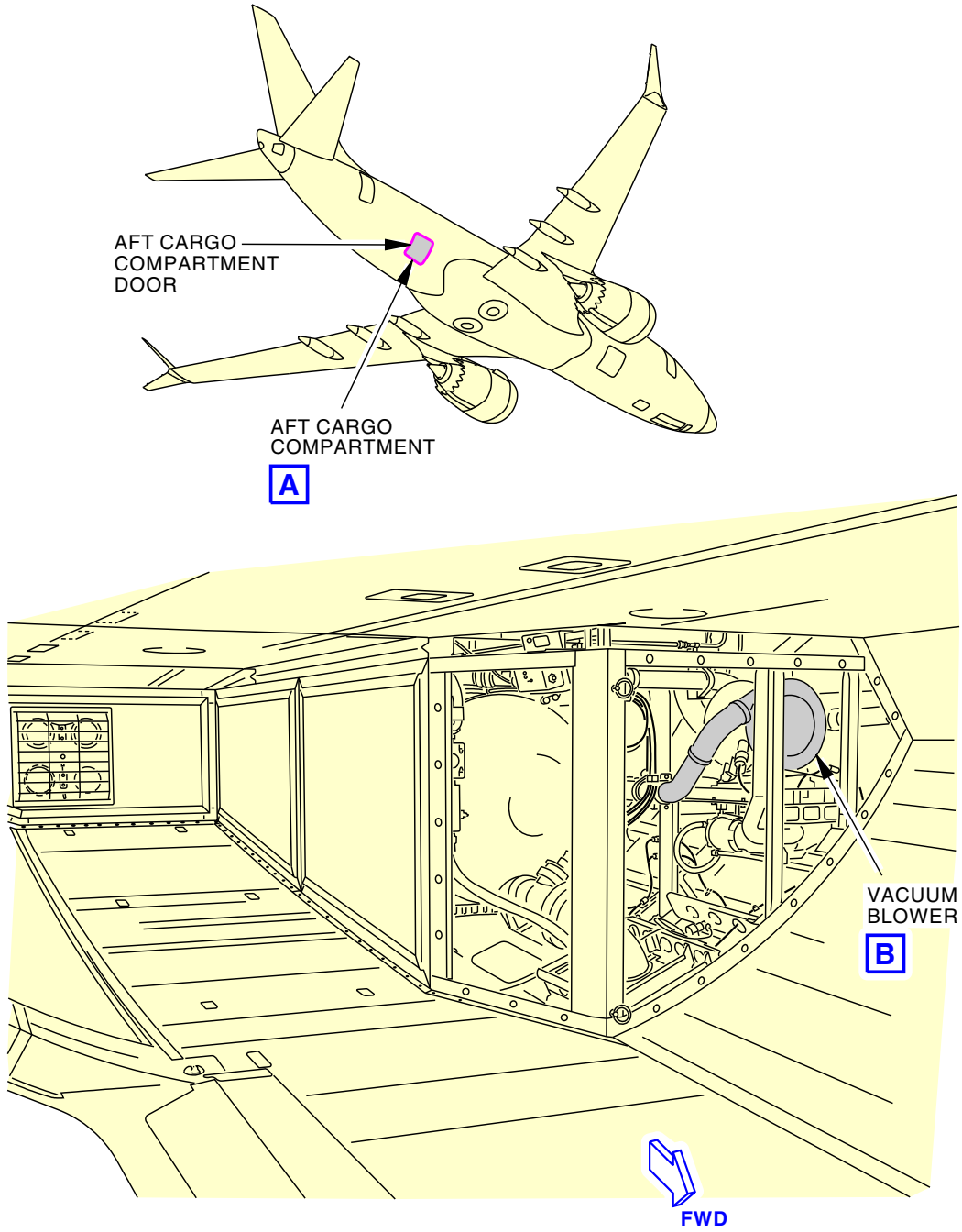
(7) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

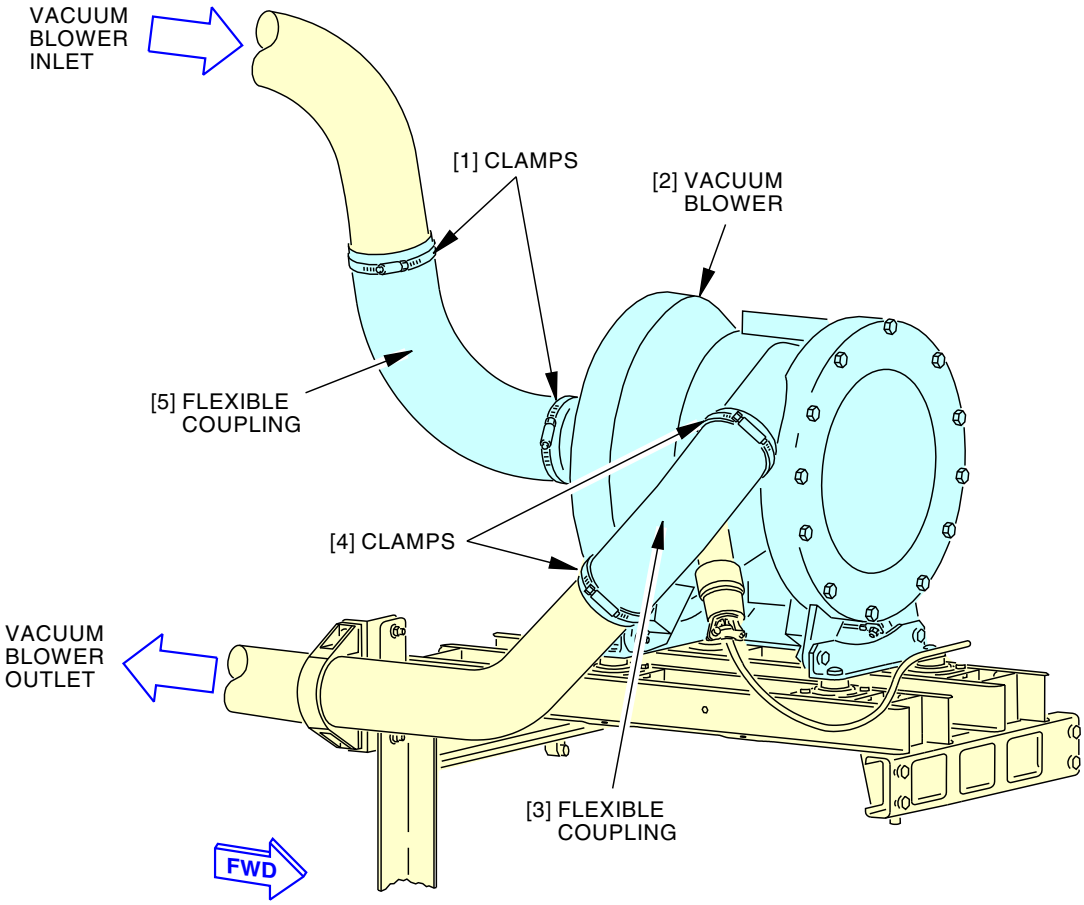
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum System - Leak Test
Figure 501/38-32-00-990-811 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



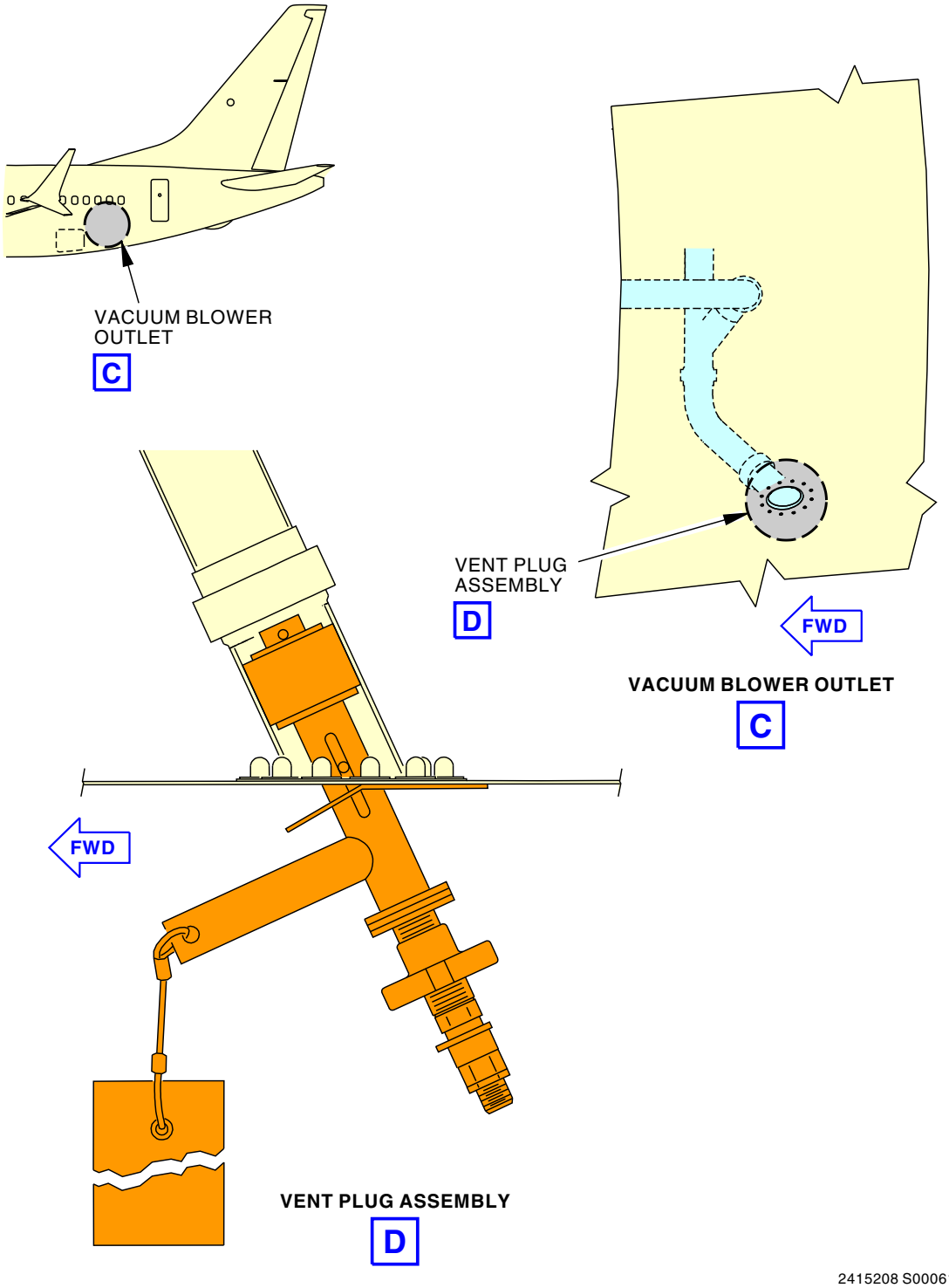
VACUUM BLOWER

B

2415207 S00061537233_V1

Vacuum System - Leak Test
Figure 501/38-32-00-990-811 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum System - Leak Test
Figure 501/38-32-00-990-811 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-700-802

3. Toilet - Operational Test

A. General

(1) This procedure gives the instructions to test the vacuum waste system from the toilet.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
38-32-01-000-801	Toilet Shroud Removal (P/B 401)
38-32-01-400-801	Toilet Shroud Installation (P/B 401)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for the Test

SUBTASK 38-32-00-010-014

(1) Get access to the lavatory for the toilet test.

SUBTASK 38-32-00-010-015

(2) To remove the toilet shroud, do this task: Toilet Shroud Removal, TASK 38-32-01-000-801.

E. Toilet - Operational Test

SUBTASK 38-32-00-860-034

(1) Make sure the potable water system is not empty.

(a) If the potable water system is empty, do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

(b) If the potable water system is not empty, then continue.

SUBTASK 38-32-00-860-056

(2) Make sure that the water shutoff valve for the lavatory is in the SUPPLY ON position.

SUBTASK 38-32-00-710-011

(3) Operate the toilet flush switch.

SUBTASK 38-32-00-790-007

(4) Make sure the vacuum blower operates immediately.

SUBTASK 38-32-00-710-012

(5) Make sure that water flows from all of the nozzles in the rinse ring of the toilet.

NOTE: The water must not come out of the toilet bowl as a mist.

SUBTASK 38-32-00-710-013

(6) Make sure the toilet flush valve opens to remove all of the water from the toilet bowl.

SUBTASK 38-32-00-710-014

(7) Make sure the toilet flush valve closes after all of the water is removed.

NOTE: A partially open valve will cause air noise at the toilet when any toilet on the system is activated.

SUBTASK 38-32-00-710-015

(8) Make sure the vacuum blower stops after approximately 15 seconds.

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SUBTASK 38-32-00-410-006

(9) To install the toilet shroud, do this task: Toilet Shroud Installation, TASK 38-32-01-400-801.

SUBTASK 38-32-00-710-016

(10) Do the Toilet - Operational Test again for the other toilets in the vacuum waste system.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-410-007

(1) Close the access to the lavatory.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VACUUM WASTE SYSTEM - CLEANING/PAINTING

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure gives the instructions to clean the vacuum waste tubes. Use this procedure to decrease the scale that collects in the vacuum waste tubes.

C. One or more of these tasks may be used as required to keep the waste lines clean. Which task(s) are used and their frequency of application should be based on local conditions and operator experience.

D. If there is excessive or quickly reoccurring scale buildup, check the operation of the toilet flush valves and make sure the valves are closing completely. A partially open valve can cause a drying effect in the tubes increasing the scale accumulation. A partially open valve will cause air noise at the toilet when any toilet on the system is activated.

E. This procedure has these tasks:

(1) A periodic flush of the vacuum waste system.

NOTE: You use this procedure to keep the vacuum waste lines clean of the waste build-up. To get the maximum effect, you must frequently do this task.

(2) A pressure washer cleaning of the vacuum waste lines.

NOTE: You use this procedure to clean the vacuum waste lines of the waste build-up. To get the maximum effect, you must use this procedure before you have frequent problems with the vacuum waste lines.

(3) A pressure test procedure.

NOTE: This procedure is used before the Soak or Recirculate procedure to make sure that the waste lines will not leak.

(4) A soak cleaning of the vacuum waste lines.

NOTE: You use this procedure to clean the vacuum waste lines of the waste build-up. To get the maximum effect, approximately 8 to 24 hours of soak time is needed to clean the vacuum waste lines.

(5) A recirculate method of cleaning the vacuum waste lines.

NOTE: You use this procedure to clean the vacuum waste lines of the heavy waste build-up. To get the maximum effect, you must use this procedure before the waste lines are fully blocked.

(6) A recirculate method of cleaning the vacuum waste lines and tank.

NOTE: You use this procedure to clean the vacuum waste lines of the heavy waste build-up. To get the maximum effect, you must use this procedure before the waste lines are fully blocked.

(7) A brush and vacuum cleaning of the overboard vent tubes.

NOTE: You use this procedure to clean the overboard vent lines when they are contaminated by waste material. Waste material can come from a soaked waste tank filter .

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TASK 38-32-00-100-801

2. Periodic Flush - Vacuum Waste System Cleaning

A. General

(1) This task contains the procedures necessary to clean by a periodic flush of a chemical through the vacuum waste system.

NOTE: You use these procedures to clean a waste build-up in the vacuum waste lines. To get the maximum effect, you must frequently do this task. You find the frequency and quantity of the cleaner by operator experience.

(2) The chemicals used in this task are qualified to the Boeing specified waste tank test in BSS7432. Alternate chemicals must be qualified to the Boeing specified waste tank test to prevent possible damage to the tank.

NOTE: Honey Bee 60 has higher viscosity than Honey Bee 60 T (thin) and 60 TNP (thin no perfume) and is the preferred Honey Bee 60 variant in applications where the surfaces to be cleaned will not be fully immersed in or continuously exposed to cleaning fluid, such as periodic flush cleaning procedures. The higher viscosity helps the cleaning fluid adhere to and clean these surfaces.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)

C. Consumable Materials

Reference	Description	Specification
B00636	Acid, Acetic (Vinegar) (5%-10% Acetic Acid)	Commercially available product
B00638	Cleaner - Acidic Liquid - Honey Bee 60 (McGean-Rohco)	
B50091	Cleaner - Gly-Vak Toilet Line Scale Remover	AMS 1550B
B50116	Toilet Soak Cleaner - Sani-Tank SP-8000N (Neutral pH and no phosphates)	AMS 1550B
B50329	Cleaner - Acidic Liquid - Honey Bee 60 T	
B50330	Cleaner - Acidic Liquid - Honey Bee 60 TNP	
G50461	Ice - Crushed, 0.50-inch (12.70 mm) to 0.75-inch (19.05 mm) maximum	

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Prepare for the Cleaning

SUBTASK 38-32-00-010-016

(1) Get access to the passenger compartment.

SUBTASK 38-32-00-610-009

(2) If the waste tanks are more than half full, do this task: Waste Tank Servicing, TASK 12-17-01-610-801 but do not do the steps add precharge.

SUBTASK 38-32-00-610-010

(3) For waste line cleaning with Procedure I or Procedure III, do this step:

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(a) Add approximately 20 gal (76 l) of water to each waste tank.

NOTE: The water will reduce the chance of damage to the tank from ice.

SUBTASK 38-32-00-860-035

(4) Make sure that this circuit breaker is closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

F. Waste Line Cleaning

NOTE: Use one of the following methods to periodically clean the waste lines.

NOTE: Ice is necessary for the best result of the procedure to clean, but this is optional.

SUBTASK 38-32-00-170-001

(1) Procedure I (Crushed Ice and Acid).

Do these steps, if you use the sulfamic acid Honey Bee 60 cleaner, B00638, Honey Bee 60 T cleaner, B50329, Honey Bee 60 TNP cleaner, B50330, or the alternative acetic acid, B00636, and crushed ice, G50461:

CAUTION DO NOT USE LARGE ICE CUBES AS AN ALTERNATIVE TO THE CRUSHED ICE. LARGE ICE CUBES CAN CAUSE DAMAGE TO THE POINT LEVEL SENSORS IN THE WASTE TANK.

(a) Starting with the forward most toilet, add about 0.5 gal (2 l) of crushed ice, G50461, to the toilet bowl.

NOTE: Keep the temperature of the ice at the coldest possible temperature. The ice should be about 0.5 in. (1 cm) to 0.75 in. (2 cm) in maximum dimension.

WARNING DO NOT GET THE TOILET SYSTEM CLEANER IN YOUR EYES, OR ON YOUR SKIN. FLUSH THE CLEANER FROM YOUR EYES AND SKIN WITH WATER. IF YOU DRINK THE TOILET SYSTEM CLEANER, GET MEDICAL AID. THE TOILET SYSTEM CLEANER IS AN ACID. IT CAN BURN YOU.

CAUTION DO NOT GET THE TOILET SYSTEM CLEANER ON THE STRUCTURE OF THE AIRPLANE. THE TOILET SYSTEM CLEANER IS AN ACID. IT CAN CAUSE DAMAGE TO THE AIRPLANE.

CAUTION USE A CLEANING SOLUTION FROM THE WASTE-TANK TEST SPECIFICATION. OTHER CLEANING SOLUTIONS CAN CAUSE DAMAGE TO THE TANK.

(b) Add about 2 qt (2 l) of the Honey Bee 60 cleaner, B00638, Honey Bee 60 T cleaner, B50329, Honey Bee 60 TNP cleaner, B50330, (recommended), or the 5 to 10% acetic acid, B00636 cleaner (optional), to the toilet bowl.

(c) Flush the toilet to put the toilet system cleaner into the waste line.

NOTE: The longer you leave the toilet system cleaner in the waste line, the better the cleaning results.

(d) Flush the toilet to remove the toilet system cleaner from the toilet bowl.

(e) Repeat steps (a), through (d) for each toilet, moving from forward to aft.

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(f) Let the toilet system cleaner stay in the waste lines as long as possible.

NOTE: The results are better if you let the cleaner stay in the waste line longer. No additional flushing of each toilet is required until the next passenger's normal usage of the lavatory. These cleaners are safe for long term exposure to the waste lines.

SUBTASK 38-32-00-170-002

(2) Procedure II (Chemical Without Crushed Ice).

Do these steps, if you use the glycolic acid Gly-Vak toilet line scale remover, B50091 (Gly-Vak®), or Honey Bee 60 cleaner, B00638, to prevent the formation of new scale:

WARNING DO NOT GET THE TOILET SYSTEM CLEANER IN YOUR EYES, OR ON YOUR SKIN. FLUSH THE CLEANER FROM YOUR EYES AND SKIN WITH WATER. IF YOU DRINK THE TOILET SYSTEM CLEANER, GET MEDICAL AID. THE TOILET SYSTEM CLEANER IS AN ACID. IT CAN BURN YOU.

CAUTION DO NOT GET THE TOILET SYSTEM CLEANER ON THE STRUCTURE OF THE AIRPLANE. THE TOILET SYSTEM CLEANER IS AN ACID. IT CAN CAUSE DAMAGE TO THE AIRPLANE.

CAUTION USE A CLEANING SOLUTION FROM THE WASTE-TANK TEST SPECIFICATION. OTHER CLEANING SOLUTIONS CAN CAUSE DAMAGE TO THE TANK.

(a) Starting with the forward most toilet, add approximately 1 qt (946 cc) of the Gly-Vak toilet line scale remover, B50091, or Honey Bee 60 cleaner, B00638, to the toilet bowl.

NOTE: The quantity of cleaner necessary to clean the vacuum toilet line changes with the length and condition of the line. The material manufacturer recommends 1 qt (946 cc). Operator experience shows that 1 qt (946 cc) is sufficient for short vacuum lines, but more than 1 qt (946 cc) is necessary for longer vacuum lines. The frequency of waste line cleaning and the quantity of Gly-Vak toilet line scale remover, B50091, or Honey Bee 60 cleaner, B00638, is established by the operator's experience.

(b) Flush the toilet to put the toilet system cleaner into the waste line.

(c) Repeat steps (a) and (b) for each toilet, moving from forward to aft.

(d) Let the toilet system cleaner stay in the waste lines as long as possible.

NOTE: The results are better if you let the cleaner stay in the waste line longer. No additional flushing of each toilet is required until the next passenger's normal usage of the lavatory. These cleaners are safe for long term exposure to the waste lines.

SUBTASK 38-32-00-170-003

(3) Procedure III (Alternate Chemical With Crushed Ice)

NOTE: Ice is necessary for the best result of the procedure to clean, but this is optional.

Do these steps, if you use the glycolic acid Sani-Tank SP-8000N toilet soak cleaner, B50116 (Sani-Tank® Neutral), and crushed ice, G50461, to clean up to 0.125 in. (3 mm) of scale:

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CAUTION DO NOT USE LARGE ICE CUBES AS AN ALTERNATIVE TO THE CRUSHED ICE. LARGE ICE CUBES CAN CAUSE DAMAGE TO THE POINT LEVEL SENSORS IN THE WASTE TANK.

(a) Starting with the forward most toilet, add about 1 qt (1 l) of crushed ice, G50461, to the toilet bowl.

NOTE: Keep the temperature of the ice at the coldest possible temperature. The ice should be about 0.5 in. (1 cm) to 0.75 in. (2 cm) in maximum dimension.

WARNING DO NOT GET THE TOILET SYSTEM CLEANER IN YOUR EYES, OR ON YOUR SKIN. FLUSH THE CLEANER FROM YOUR EYES AND SKIN WITH WATER. IF YOU DRINK THE TOILET SYSTEM CLEANER, GET MEDICAL AID. THE TOILET SYSTEM CLEANER IS AN ACID. IT CAN BURN YOU.

CAUTION DO NOT GET THE TOILET SYSTEM CLEANER ON THE STRUCTURE OF THE AIRPLANE. THE TOILET SYSTEM CLEANER IS AN ACID. IT CAN CAUSE DAMAGE TO THE AIRPLANE.

CAUTION USE A CLEANING SOLUTION FROM THE WASTE-TANK TEST SPECIFICATION. OTHER CLEANING SOLUTIONS CAN CAUSE DAMAGE TO THE TANK.

(b) Add about 1 qt (1 l) of Sani-Tank SP-8000N toilet soak cleaner, B50116, to the toilet bowl.

NOTE: The quantity of the cleaner necessary to clean a vacuum toilet line changes with the length and condition of the line. The material manufacturer recommends 1 qt (1 l). Operator experience shows that 1 qt (1 l) is sufficient for short vacuum toilet lines. For longer vacuum toilet lines, use more of the cleaner. The frequency you clean and the quantity of Sani-Tank SP-8000N toilet soak cleaner, B50116, is established by the operator experience.

(c) Flush the toilet to put the toilet system cleaner and crushed ice into the waste line.

(d) Repeat steps (a), (b) and (c) for each toilet, moving from forward to aft.

(e) Let the toilet system cleaner stay in the waste lines as long as possible.

NOTE: The results are better if you let the cleaner stay in the waste line longer. No additional flushing of each toilet is required until the next passenger's normal usage of the lavatory. These cleaners are safe for long term exposure to the waste lines.

(f) After you complete the full treatment with Sani-Tank SP-8000N toilet soak cleaner, B50116, and crushed ice, do this step:

1) Do the preventative maintenance using Gly-Vak toilet line scale remover, B50091, as shown in Procedure II.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-610-011

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-00-410-008

(2) Close the access to the passenger compartment.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-100-802

3. Pressure Washer Cleaning of the Vacuum Waste Lines

(Figure 701)

A. General

(1) To clean the vacuum waste lines with the pressure washer, you must get access at one of these locations:

(a) In the lavatory with the toilet assembly removed when you do not open the toilet flush valve.

(b) At one of the cleanout ports in the forward cargo compartment with which the removal of the ceiling panel is necessary.

(c) In the lavatory, with the toilet flush valve open. When you use this procedure, the bends of the pressure washer hose will be more than if you remove the toilet.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-700-801	Vacuum Waste System - Leak Test (P/B 501)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-32-00-910-802	Waste Tube - Maintenance Practice (P/B 201)
38-32-01-000-802	Vacuum Toilet Assembly Removal (P/B 401)
38-32-01-400-801	Toilet Shroud Installation (P/B 401)
50-10-03-000-801	Cargo Compartment Ceiling Liner Removal (P/B 401)
50-10-03-400-801	Cargo Compartment Ceiling Liner Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-419	Gloves - Rubber, Elbow Length
STD-1136	Facepiece (Mask) - Particulate Filtering, Disposable (NIOSH N95 approved)
STD-1137	Glasses - Safety
STD-1138	Tubing - Flexible, 1.5 Inch Outside Diameter
STD-1286	Equipment - Pressure Washer, Electric or Diesel Driven, 1500 PSIG to 2500 PSIG with Adjustable Output Pressure
STD-1287	Hose - Pressure Washer, 3/8 Inch I.D. 50 to 150 Foot Length
STD-1288	Nozzle - Rotating, for 3/8 Inch I.D. Hose (Self-Feeding Rotating Nozzle is Recommended)
STD-1289	Guide Spring - Nozzle, for 3/8 Inch I.D. Nozzle

D. Consumable Materials

Reference	Description	Specification
B00637	Acid, Citric	A-A-59147
G00022	Compound - Chlorine Dioxide For Water Purification - Purogene or Oxine	
G02315	Clothing - Disposable Gown, Gloves For Sewage Handling	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
200	Upper Half of Fuselage

F. Prepare for Cleaning

SUBTASK 38-32-00-010-017

(1) Get access to the passenger compartment.

SUBTASK 38-32-00-010-018

(2) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-910-016

(3) For the work on the waste equipment, make sure you wear the items for your protection.

NOTE: These are items that you can wear to give you protection when it is necessary.

(a) elbow length rubber gloves, STD-419

(b) filtering facepiece, STD-1136

(c) safety glasses, STD-1137

(d) Disposable clothing, G02315

SUBTASK 38-32-00-610-012

(4) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: After 5 minutes, drain the pre charge from the waste tank.

SUBTASK 38-32-00-860-037

(5) Start this procedure at a toilet which is upstream of the vacuum waste line to be cleaned.

NOTE: Areas upstream of the bends have the most waste material build-up on the waste tube walls. For further reference, (TASK 38-32-00-910-802).

SUBTASK 38-32-00-160-012

(6) If the toilet is full of waste, do these steps to remove the waste in the toilet bowl, if the waste system is serviceable:

(a) Install a length of 1.5 inch outside diameter flexible tubing, STD-1138 from a serviceable toilet to the blocked toilet.

(b) Flush the serviceable toilet to drain the toilet bowl of the blocked toilet.

(c) Make sure the toilet bowl is empty.

(d) If the toilet bowl is not empty, flush the serviceable toilet again.

(e) Remove the 1.5 inch outside diameter flexible tubing, STD-1138.

SUBTASK 38-32-00-160-013

(7) Use one to three gallons of the disinfectant that follows:

NOTE: The disinfectant mixture is a solution of chlorine dioxide, citric acid, and water.

(a) Mix the Chlorine dioxide (stabilized 2%) Purogene or Oxine compound, G00022 and citric acid, B00637 (crystals or powder) as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT BREATHE CHLORINE DIOXIDE GAS. PUT ON A RESPIRATOR, AND GOGGLES. WHEN THE TWO CHEMICALS MIX, THEY MAKE CHLORINE DIOXIDE GAS. CHLORINE DIOXIDE GAS CAN CAUSE INJURIES TO PERSONNEL IF THEY BREATHE IT.

1) Mix 20 fl-oz (0.6 l) of chlorine dioxide Purogene or Oxine compound, G00022 with 2 oz (57 g) of citric acid, B00637 (per gallon of solution needed) in a clean plastic container.
2) Stop for 5 minutes (activation period).
3) Use a clean instrument to mix the solution fully.
4) Add 1 gal (4 l) of water to the solution.

(b) Flush 1 gal (4 l) to 3 gal (11 l) of the disinfectant solution into the waste line.

NOTE: The quantity of solution necessary to disinfect the vacuum toilet line changes with the length and condition of the line. Operator experience shows that 1 gal (4 l) is sufficient for short vacuum lines, but more than 1 gal (4 l) is necessary for longer vacuum lines. The quantity of disinfectant necessary for a specific vacuum toilet line is established by the operator's experience.

SUBTASK 38-32-00-010-019

(8) Remove the toilet from the lavatory: Vacuum Toilet Assembly Removal, TASK 38-32-01-000-802.

NOTE: The toilet assembly removal is necessary to do this procedure from the passenger compartment when you do not open the toilet flush valve.

SUBTASK 38-32-00-010-020

(9) To access the main lines to the tanks, do the steps that follow:

NOTE: The removal of the ceiling lining panels is necessary to do this procedure.

(a) Do this task: Cargo Compartment Ceiling Liner Removal, TASK 50-10-03-000-801.

(b) Remove the cap on the cleanout port.

SUBTASK 38-32-00-860-038

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(10) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-930-003

(11) Make a mark on the pressure washer hose, STD-1287 at approximately 4 ft (1.2 m) from the rotating pressure washer nozzle, STD-1288 and nozzle guide spring, STD-1289.

NOTE: This location is for the removal of the hose from the waste tube to keep the pressurized water out of the work area.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Waste Line Cleaning

SUBTASK 38-32-00-170-004

(1) Put the pressure washer hose, STD-1287, rotating pressure washer nozzle, STD-1288 or nozzle guide spring, STD-1289 of the pressure washer (1500 PSIG to 2500 PSIG), STD-1286 approximately 4 ft (1.2 m) into the waste line.

NOTE: The distance necessary for the nozzle to be put into the waste tube can be different for different manufacturers nozzles. The distance given is to make sure that no pressurized water goes into the work area during this operation.

SUBTASK 38-32-00-170-005

CAUTION

DO NOT USE A PRESSURE OF MORE THAN 2500 PSIG (17,237 KPA) WHEN YOU OPERATE THE PRESSURE WASHER EQUIPMENT. TOO MUCH PRESSURE CAN CAUSE DAMAGE TO THE WASTE SYSTEM EQUIPMENT.

(2) Do the steps to operate the pressure washer (1500 PSIG to 2500 PSIG), STD-1286 as follows:

(a) One person must operate the pressure washer (1500 PSIG to 2500 PSIG), STD-1286 at the area you use to get access.

(b) A second person is necessary at a different toilet to flush that toilet one time between 45 to 75 seconds.

NOTE: It is necessary to flush a toilet to operate the vacuum blower to move the fluids to the waste tank. You can monitor the tank level from the waste system gage on the aft attendant's panel. When the waste tank is full, the waste system automatically stops and the LAVS INOP light comes on. To prevent a fluid spill, you must stop the pressure washer when the waste system automatically stops. You must do the waste system servicing before you start the pressure washer after a full tank shutdown.

(c) If the movement of the pressure washer hose, STD-1287 is not easy, put a small quantity of non-foaming dish soap on the hose.

NOTE: This is to lubricate the pressure washer hose, STD-1287 in the waste line.

(d) Turn on the water.

NOTE: If it is available, use hot water.

(e) Turn on the pressure washer (1500 PSIG to 2500 PSIG), STD-1286.

(f) Slowly put the pressure washer hose, STD-1287 of the pressure washer (1500 PSIG to 2500 PSIG), STD-1286 into the waste line in approximately 2 ft (61 cm) increments.

(g) Move the hose forward and then back, 3 or 4 times for the maximum effect in each of the 2 ft (61 cm) sections.

(h) When you put all the pressure washer hose, STD-1287 into the waste line or the pressure washer nozzle, STD-1288 is at the waste tank, remove the pressure washer hose, STD-1287.

NOTE: Clean the hose while you remove it to prevent contamination of the work area of the airplane.

(i) When the mark on the pressure washer hose, STD-1287 is in the work area, turn the water off.

(j) Remove the pressure washer hose, STD-1287, pressure washer nozzle, STD-1288 and nozzle guide spring, STD-1289.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-170-006

(3) Stop the operation of the vacuum blower.

SUBTASK 38-32-00-170-007

(4) Do this cleaning procedure again if it is necessary.

SUBTASK 38-32-00-170-008

(5) Clean the work area.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-410-009

(1) Do one of these steps to close the access:

(a) If you removed the toilet from the lavatory, do this task: Toilet Shroud Installation, TASK 38-32-01-400-801.

(b) If you removed the cap on the cleanout port, do the steps that follow:

1) Install the cleanout plug.

2) Do this task: Cargo Compartment Ceiling Liner Installation, TASK 50-10-03-400-801.

SUBTASK 38-32-00-610-013

(2) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-00-610-014

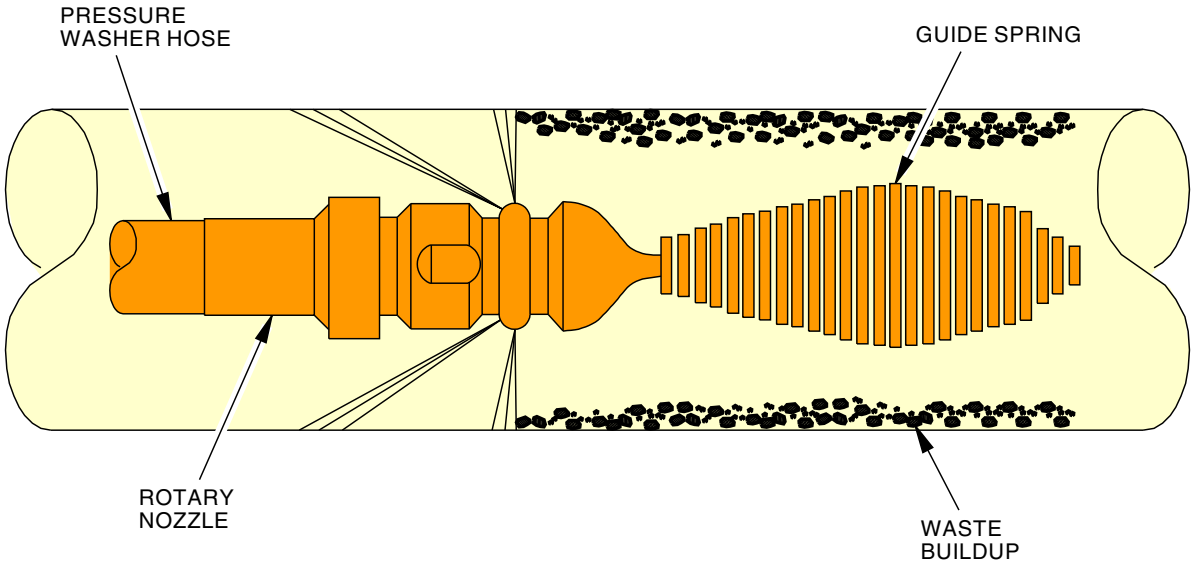
(3) Do this task: Vacuum Waste System - Leak Test, TASK 38-32-00-700-801.

SUBTASK 38-32-00-410-010

(4) Close the access to the passenger compartment.

——— END OF TASK ———

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Pressure Washer Cleaning
Figure 701/38-32-00-990-812

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-100-803

4. Soak Method for Cleaning of the Vacuum Waste Lines

(Figure 702)

A. General

(1) This procedure gives the instructions to clean the vacuum waste lines. Use this procedure to decrease the scale that collects in the vacuum waste lines.

(2) To clean the vacuum waste lines using the soak method, you access at one of these locations:

(a) In the most forward lavatory for each vacuum waste line with the toilet assembly removed.

(b) At the last tube prior to the entry of the waste tank for each vacuum waste line to be cleaned with the appropriate sidewall or ceiling lining panel removed.

B. References

Reference	Title
05-51-57-000-802	Corrosion Removal After Acid Spills (P/B 201)
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-700-801	Vacuum Waste System - Leak Test (P/B 501)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-32-00-910-802	Waste Tube - Maintenance Practice (P/B 201)
38-32-01-000-802	Vacuum Toilet Assembly Removal (P/B 401)
38-32-01-400-802	Vacuum Toilet Assembly Installation (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1953	Equipment - Pump, Static Soak, Vacuum Lav. System Part #: A38012-39 Supplier: 81205 Opt Part #: A38012-1 Supplier: 81205 Opt Part #: A38012-31 Supplier: 81205
STD-77	Air Source, Regulated - Dry Filtered, 0-50 psig
STD-419	Gloves - Rubber, Elbow Length
STD-1056	Container - Solvent Resistant, 5 Gallon (19 Liter)
STD-1136	Facepiece (Mask) - Particulate Filtering, Disposable (NIOSH N95 approved)
STD-1137	Glasses - Safety
STD-1439	Container - Drum, 55 Gallon (208 Liter)
STD-3926	Water Source - Cold, Regulated, 0 to 60 PSIG
STD-6928	Coupler Sleeve - AS1654T32
STD-6930	Coupling - AS1655A32

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D. Consumable Materials

Reference	Description	Specification
B00046	Acid - Corrosion Removing, Metal Conditioning, Phosphoric	MIL-C-10578
B00636	Acid, Acetic (Vinegar) (5%-10% Acetic Acid)	Commercially available product
B00637	Acid, Citric	A-A-59147
B00638	Cleaner - Acidic Liquid - Honey Bee 60 (McGean-Rohco)	
B01021	Cleaner - Drain (Enzyme) - Dispoz-All	
B01022	Cleaner - enzyme septic	
B50091	Cleaner - Gly-Vak Toilet Line Scale Remover	AMS 1550B
B50273	Cleaner - Acidic Liquid, Vacuum Toilet Line Cleaner - Sani-Vak G3	AMS 1640
B50329	Cleaner - Acidic Liquid - Honey Bee 60 T	
B50330	Cleaner - Acidic Liquid - Honey Bee 60 TNP	
D00627 [P06-002]	Lubricant - White Petrolatum	PMC 9609
G00022	Compound - Chlorine Dioxide For Water Purification - Purogene or Oxine	
G02315	Clothing - Disposable Gown, Gloves For Sewage Handling	

E. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
200	Upper Half of Fuselage

F. Prepare for Cleaning

SUBTASK 38-32-00-010-021

(1) Get access to the passenger compartment.

SUBTASK 38-32-00-010-022

(2) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-910-017

(3) For the work on the waste equipment, make sure you wear these items for your protection.

NOTE: These are items that you can wear to give you protection when it is necessary.

(a) elbow length rubber gloves, STD-419

(b) filtering facepiece, STD-1136

(c) safety glasses, STD-1137

(d) disposable clothing, G02315

SUBTASK 38-32-00-610-015

(4) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: After 5 minutes, drain the precharge from the waste tank.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-860-040

(5) Start this procedure at the most forward toilet of the vacuum waste line to be cleaned.

NOTE: Areas upstream of the bends have the most waste material build-up on the waste tube walls. For the primary layout of the vacuum waste lines see this reference, (TASK 38-32-00-910-802).

SUBTASK 38-32-00-160-014

(6) Use 1 gal (4 l) to 3 gal (11 l) of the disinfectant that follows:

NOTE: The disinfectant is a solution of chlorine, citric acid, and water.

(a) Flush the waste line with a chlorine mixture made of chlorine dioxide (stabilized 2%) Purogene or Oxine compound, G00022 and citric acid, B00637 (crystals or powder) as follows:

WARNING DO NOT BREATHE CHLORINE DIOXIDE GAS. PUT ON A RESPIRATOR, AND GOGGLES. WHEN THE TWO CHEMICALS MIX, THEY MAKE CHLORINE DIOXIDE GAS. CHLORINE DIOXIDE GAS CAN CAUSE INJURIES TO PERSONNEL IF THEY BREATHE IT.

1) For each 1 gal (4 l) of disinfectant solution desired, mix 20 fl-oz (0.6 l) of chlorine dioxide Purogene or Oxine compound, G00022 with2 fl-oz (0.06 l) of citric acid, B00637 in a plastic container.
2) Stop for 5 minutes (activation period).
3) Use a clean instrument to mix the solution fully.
4) Add water to make the desired quantity of solution.
5) Flush 1 gal (4 l) to 3 gal (11 l) of the disinfectant solution into the waste line.

NOTE: The quantity of solution necessary to disinfect the vacuum toilet line changes with the length and condition of the line. Operator experience shows that 1 gal (4 l) is sufficient for short vacuum lines, but more than 1 gal (4 l) is necessary for longer vacuum lines. The quantity of disinfectant necessary for a specific vacuum toilet line is established by the operator's experience.

SUBTASK 38-32-00-160-015

(7) Do this task: Periodic Flush - Vacuum Waste System Cleaning, TASK 38-32-00-100-801.

(a) Rinse the lines from the most forward toilet with approximately ½ gal (2 l) of water.

(b) Turn the water off to the toilet.

(c) Flush the toilet without water at least three times.

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WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(d) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-010-023

(8) To get access to the waste lines, do one of these steps:

(a) To remove the waste tank enclosure, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(b) Disconnect the last waste tube [1] section before the waste tank inlet.

NOTE: This is the waste line section that will be temporarily replaced by the pump static soak pump equipment, SPL-1953 during the procedure. Ensure that the tool will fit in its place. If not, a section of waste line upstream will need to be chosen.

1) Remove the waste tube [1] from its installed position, and connect the assembled drain valve [3] and flexible hose [16] to the airplane vacuum waste line, using the sleeve, STD-6928 and coupling, STD-6930, as necessary.

NOTE: Remaining liquid can stay in the waste lines. Be prepared to collect and then discard all of the liquid.

2) Make sure that the drain valve [3] is closed. If the drain valve is open, the pressure test that follows will be bad.

(c) For all lavatories on the waste line to be cleaned, do this task: Vacuum Toilet Assembly Removal, TASK 38-32-01-000-802.

NOTE: The toilet assembly removal is necessary to attach the tools for the pressure test and the drain/flush equipment.

G. Pressure Test Before Cleaning

SUBTASK 38-32-00-780-002

(1) Do this task: Pressure Test Procedure, TASK 38-32-00-780-802.

NOTE: This test is necessary to make sure the waste lines are water tight. If the waste line is not water tight, a leak of the cleaner solution on the airplane structure can occur.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Waste Line Cleaning

SUBTASK 38-32-00-110-002

(1) Prepare the chemical solution to soak the waste line with one of the chemical cleaners that follow:

NOTE: The quantity of the cleaner necessary for a waste line is 0.16 gallons for each linear foot (2.0 liters for each linear meter). Calculate the maximum quantity of the cleaner necessary for each line to prevent a spill when you fill the waste line.

(a) Honey Bee 60 cleaner, B00638, Honey Bee 60 T cleaner, B50329, or Honey Bee 60 TNP cleaner, B50330

(b) 5 to 10% Acetic acid, B00636

(c) 7% Phosphoric acid, B00046

(d) Dispoz-All cleaner, B01021

(e) enzyme septic cleaner, B01022

(f) Sani-Vak G3, B50273

(g) Gly-Vak toilet line scale remover, B50091.

SUBTASK 38-32-00-170-009

(2) Do these steps to fill the waste line with the cleaner solution:

(a) Assemble the static soak pump equipment, SPL-1953.

NOTE: This procedure is used to clean only one waste line at a time. Refer to Figure 702 (Sheet 5) for tool installation schematic.

1) At the forward most toilet waste connection [10], connect the flush plug assembly [18] and the air pressure assembly [8] (Figure 702 (Sheet 4)).

a) Do these steps if the coupling between the static soak pump equipment, SPL-1953 and the toilet waste connection [10] is male-male configuration Figure 702 (Sheet 5):

<1> Make sure that all the parts of the clamshell coupling are clean.

<2> Install the clamshell coupling

<3> Apply a thin layer of lubricant, D00627 [P06-002] or MIL-L-4343 on the inner surface of the sleeve.

<4> Install one new O-ring on each ferrule.

<5> Make sure that the ends between two adjacent tubing flanges that connect at the clamshell coupling are 0.11 in. (2.79 mm) to 0.17 in. (4.32 mm) apart.

<6> Move the sleeve to the center of the clamshell coupling.

<7> Push each latch down.

<8> Make sure that each latch is on tight.

<9> Visually examine the latches to make sure that they are at the same level.

2) Connect the overflow hose [13] and shutoff valve [14] at the quick-disconnect [15].

3) Put the end of the overflow hose [13] in a solvent resistant container (5 gal)(19 Liter), STD-1056 [12] to contain all possible flow of the cleaner solution into the work area in the airplane.

4) Connect an air hose [11] to the brass nipple on the air pressure assembly [8].

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5) In each lavatory between the most forward toilet and the waste tank, make sure a plug assembly [17] is installed in place of the toilet assembly.

6) Make sure that the flexible hose [16] and drain valve [3] is connected to the waste line, at the clamp where the segment was removed Figure 702 (Sheet 2).

7) Connect one end of the supply hose [4] to the drain valve [3] at the quick-disconnect [2].

8) Open the drain valve [3].

9) Connect the other end of the supply hose [4] with the shutoff valves [5] and the quick-disconnects and connect to the drum pump [6] (Figure 702 (Sheet 3)).

10) Connect an air hose [7] to the air motor on the drum pump [6].

11) Place the drum pump [6] in a 55 gallon (208 liter) drum container, STD-1439 full of the cleaner solution (Figure 702 (Sheet 3)).

CAUTION

MAKE SURE THAT THE PERSON WITH THE EQUIPMENT TO PREVENT LEAKAGE (RAGS, ETC), HAS ACCESS TO EACH LAVATORY IN THE WASTE SYSTEM THAT YOU FILL. MAKE SURE THAT THERE IS NO LEAKAGE OF THE CLEANER SOLUTION FROM THE WASTE SYSTEM INTO THE LAVATORY DURING THIS PROCEDURE. LEAKAGE OF CHEMICAL SOLUTIONS CAN CAUSE DAMAGE TO THE INTERIOR OF THE AIRCRAFT.

(b) Turn on the drum pump [6] to fill the waste lines with the cleaner solution slowly, to prevent leakage in the lavatories.

NOTE: When you smell some of the chemical solutions, they can cause persons to be offended while in the airplane. It can be necessary to supply temporary air removal equipment.

1) If you spill the cleaning solution, do this task: Corrosion Removal After Acid Spills, TASK 05-51-57-000-802.

(c) Fill the waste line until you see the chemical solution in the overflow hose [13] at the lavatory end of the line.

(d) Stop the drum pump [6].

NOTE: The level of the solution in the waste system can decrease and then adjust after a short time.

(e) Continue to fill the waste line until you see the chemical solution in the overflow hose [13] at the most forward lavatory.

(f) Close the drain valve [3] on the waste line at the cargo compartment.

(g) Close all of the valves on the adapters but open the shutoff valve [14] at the most forward lavatory.

NOTE: The most forward point of the waste line must stay open to let all gases come out of the line. The open valve will prevent pressurization and leakage of the system.

(h) Make sure you put the overflow hose [13] in the most forward lavatory in the solvent resistant container (5 gal)(19 Liter), STD-1056 to prevent leakage into the airplane.

NOTE: It is necessary to empty the container before the cleaner solution flows into the work area in the airplane.

(i) Close the drain valve [3] in the aft cargo compartment, and then disconnect the supply hose [4].

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(j) To soak one of the other waste lines, do these steps:

1) Connect the supply hose [4] to a different 2 inch adapter on the other waste line.

2) Connect the supply hose [4] to the other waste line.

NOTE: If you use more than one set of tools, other waste lines can be soaked at the same time.

3) Use these steps to fill the other waste line(s).

SUBTASK 38-32-00-170-010

(3) Let the cleaner solution stay in the waste line(s) for a minimum of 8 hours.

NOTE: To fully clean the waste line(s), more than 24 hours can be necessary to remove the heavy scale. The longer the cleaner is in the waste lines, the better the cleaning results. The length of time is established by the operator's experience.

NOTE: Gases are released during the chemical effect of the cleaner solution and the scale. Do a check of the container at regular times to prevent flow of the cleaner solution in the airplane.

I. Drain the Waste Line

SUBTASK 38-32-00-680-001

(1) Do these steps to drain the chemical solution from the system:

(a) Prepare to collect (for disposal) all of the chemical solution and water used to flush the system in the waste lines.

(b) Make sure you attach the supply hose [4] to the waste line to be drained.

(c) Connect the shutoff valve [19] to the drain end assembly [20] on the 55 gallon (208 liter) drum container, STD-1439 to hold used chemical solution.

(d) Open the drain valve [3] used to fill the waste line.

(e) Open the shutoff valve [19] in the hose at the container to drain the waste line.

(f) Open the pressure regulator [9] valve installed on the most forward toilet to let the air into the waste line.

(g) To flush the waste line, do the steps that follow:

1) Connect a 0 to 60 PSIG regulated cold water source, STD-3926 to the water supply assembly [41].

2) Connect the water supply assembly [41] to the flush plug assembly [18].

3) Flush the waste line with a minimum of 5 gal (19 l) of water for each toilet on the waste line.

4) Connect an regulated air source, STD-77 to the pressure regulator [9].

5) Dry the waste line with air flow (under 15 psi (103 kPa)) for approximately 5 minutes.

(h) Stop until the cleaner solution/water is fully drained from the waste line.

(i) If it is necessary to drain other waste lines, do these steps again to drain waste lines.

SUBTASK 38-32-00-860-041

(2) Do these steps to remove the tools:

(a) In the most forward lavatory, remove the air pressure assembly [8] and the flush plug assembly [18].

(b) In each lavatory between the most forward toilet and the waste tank, remove the plug assemblies [17].

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(c) Do this task: Vacuum Toilet Assembly Installation, TASK 38-32-01-400-802.

(d) Disconnect the drum pump [6] from the supply hose [4].

NOTE: Remaining liquid can stay in the waste lines. Be prepared to collect and then discard all of the liquid.

(e) In the aft cargo compartment, disconnect the flexible hose [16] and drain valve [3] from the waste line, at the clamp where the segment was removed, .

(f) Make sure all solid material is out of the waste line before you connect the waste line to the waste tank.

NOTE: Remaining solid material in the waste lines can cause blocked waste lines after you complete this procedure.

(g) Put the waste tube in its installed position.

(h) At the waste connection on the top of the waste tank, connect the first waste tube [1] to the waste tank inlet of the line that was cleaned.

NOTE: Repeat this procedure for each line to be cleaned.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-410-011

(1) Do these steps complete this task:

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(a) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-610-016

(2) To push other solid material out of the waste system, do this task: Periodic Flush - Vacuum Waste System Cleaning, TASK 38-32-00-100-801.

SUBTASK 38-32-00-610-017

(3) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-00-610-018

(4) Do this task: Vacuum Waste System - Leak Test, TASK 38-32-00-700-801.

SUBTASK 38-32-00-860-042

(5) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-00-410-012

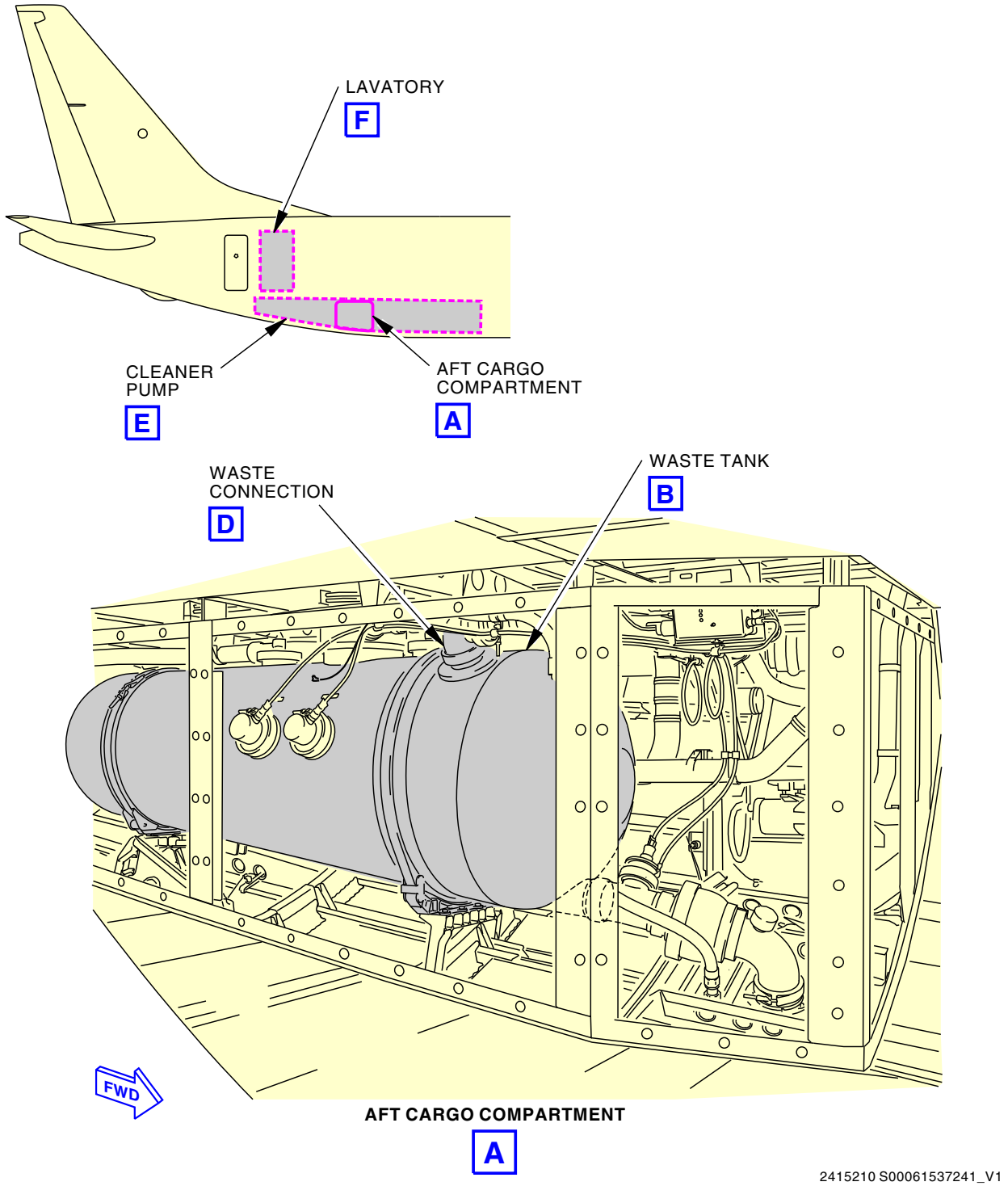
(6) Close the access to the passenger compartment.

SUBTASK 38-32-00-910-018

(7) Obey the local authorities to discard the chemical solution.

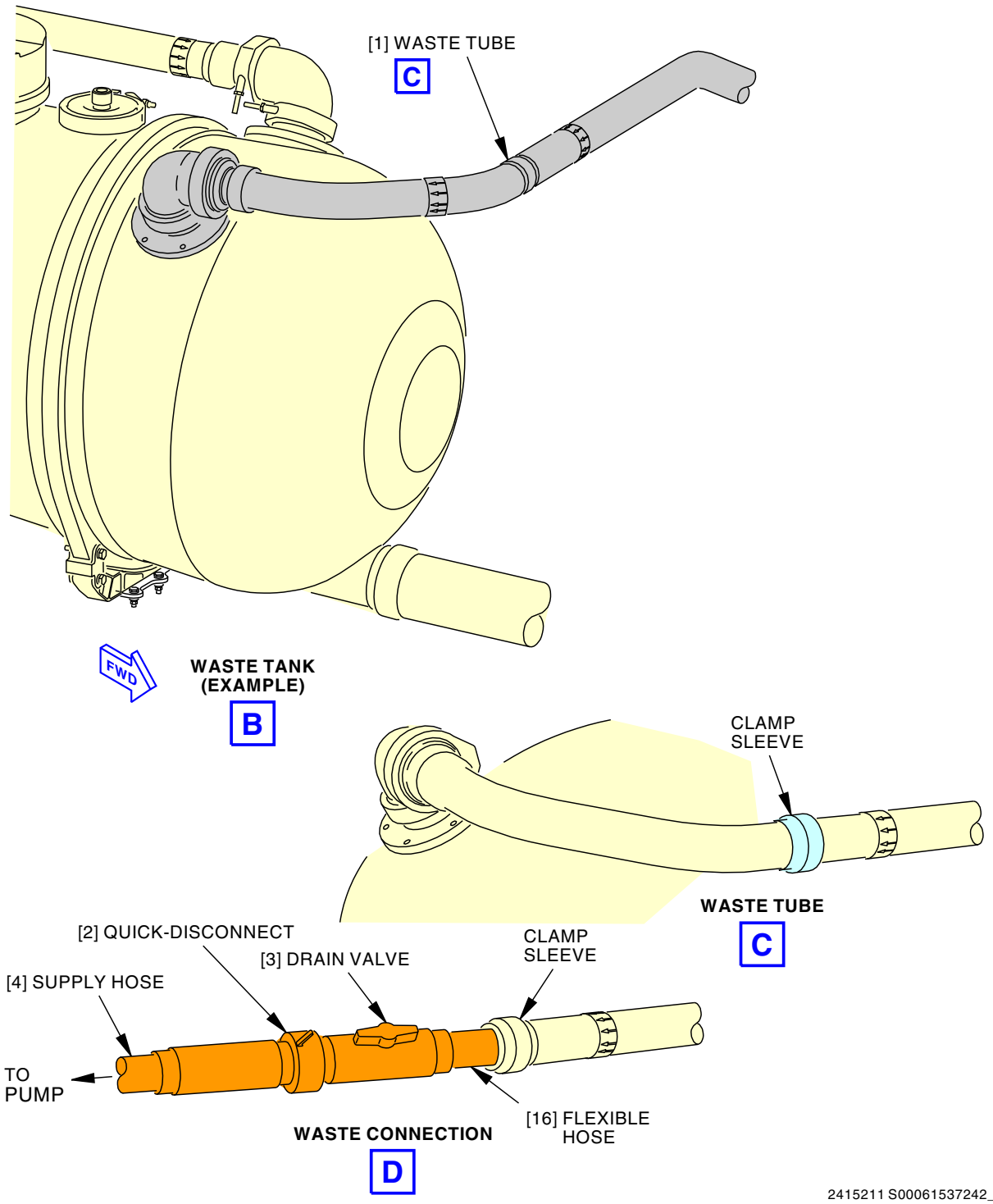
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



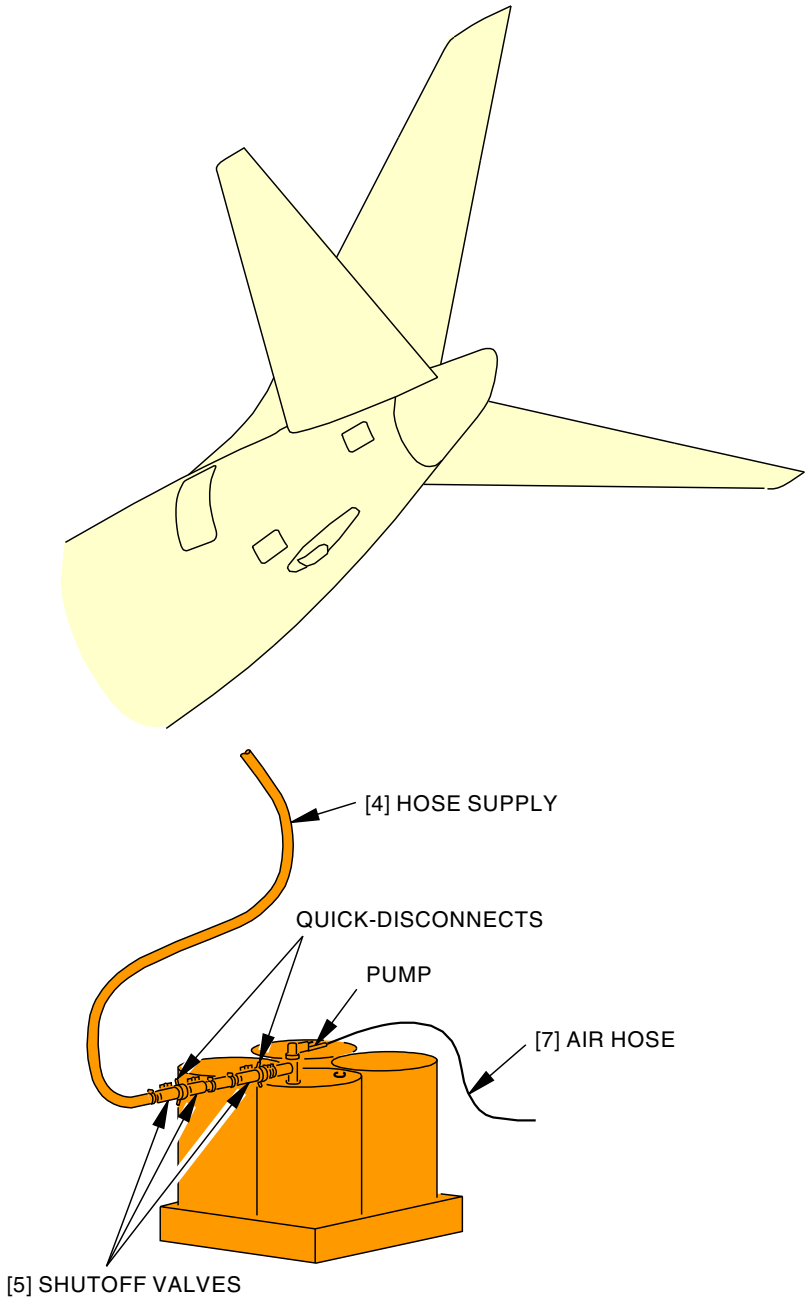
Soak Waste Line Cleaning
Figure 702/38-32-00-990-813 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Soak Waste Line Cleaning
Figure 702/38-32-00-990-813 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



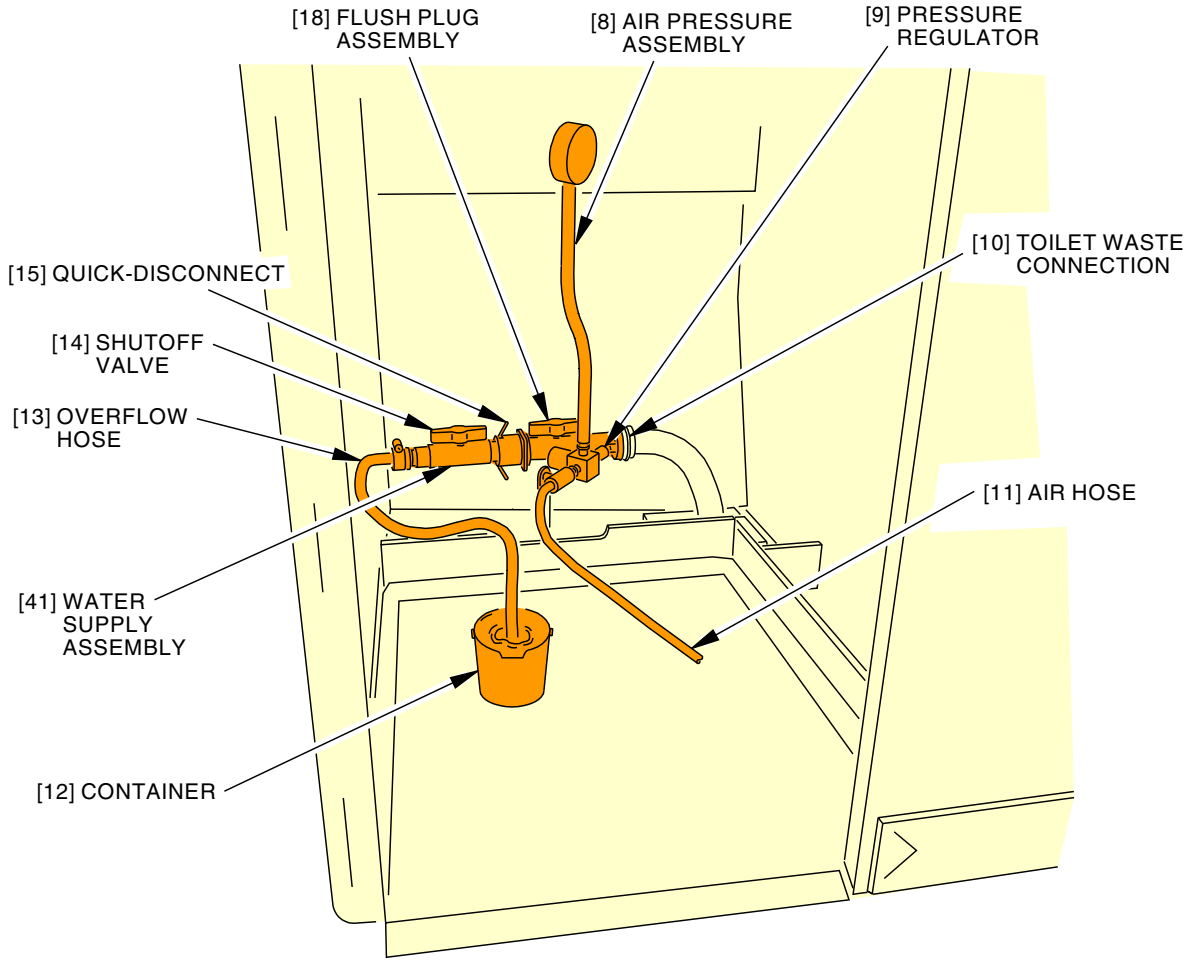
CLEANER PUMP

E

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Soak Waste Line Cleaning
Figure 702/38-32-00-990-813 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



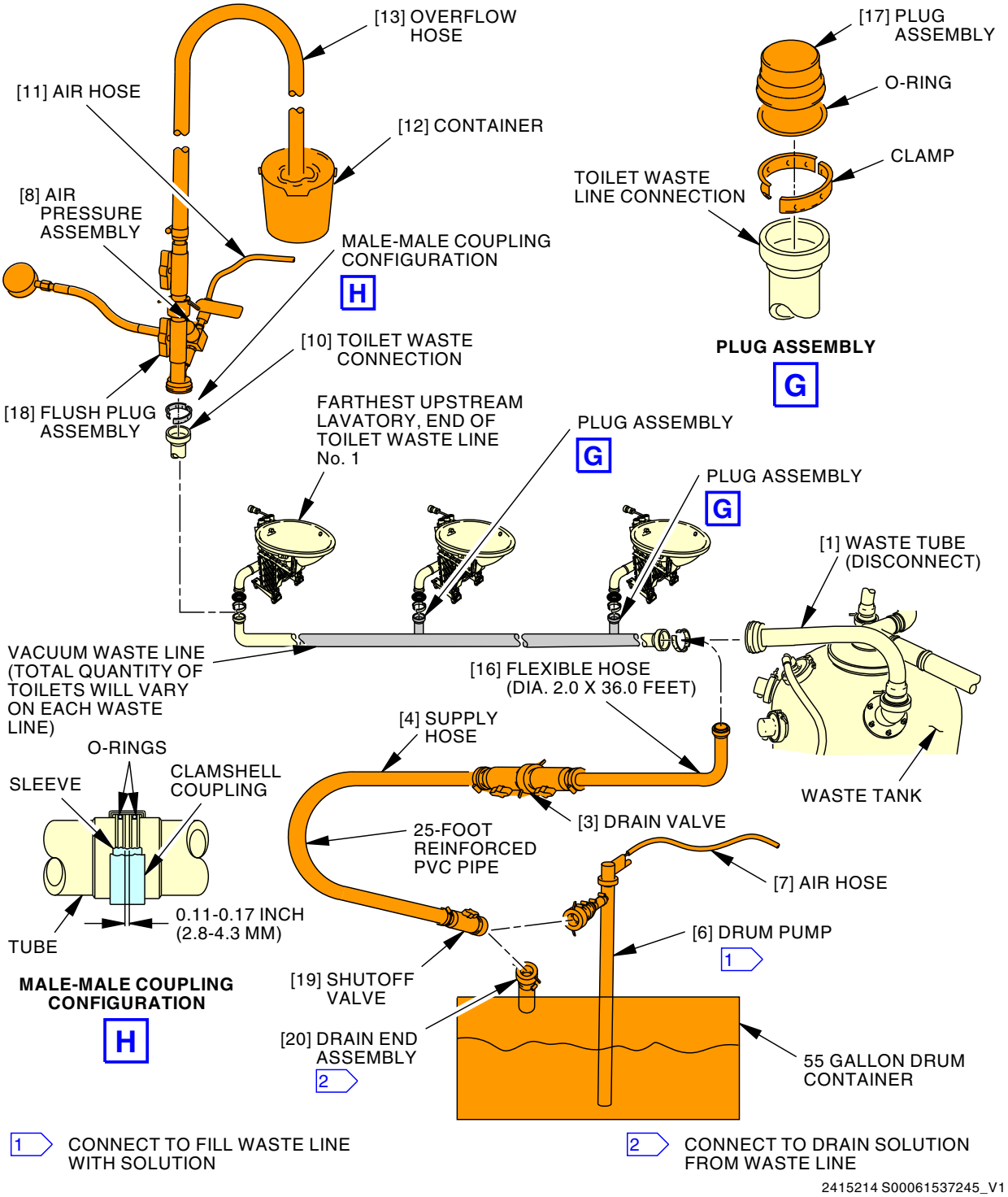
LAVATORY (EXAMPLE)

F

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Soak Waste Line Cleaning
Figure 702/38-32-00-990-813 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Soak Waste Line Cleaning
Figure 702/38-32-00-990-813 (Sheet 5 of 5)

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TASK 38-32-00-110-801

5. Recirculate Method of Cleaning the Vacuum Waste Lines

(Figure 703)

A. General

(1) This procedure gives the instructions to clean the vacuum waste tubes. Use this procedure to decrease the scale that collects in the vacuum waste tubes.

(2) One set of tools can connect two waste lines and will allow separate cleaning of the toilet systems. Additional tool sets will allow the different toilet systems to be cleaned at the same time.

(3) To clean the vacuum waste lines using the recirculate method, you must get access at these locations:

(a) In the lavatory with the toilet assembly removed.

(b) At the waste tube inlets to the waste tanks of the waste lines to be cleaned, where removal of the waste tank enclosure panel is necessary.

B. References

Reference	Title
05-51-57-000-802	Corrosion Removal After Acid Spills (P/B 201)
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-700-801	Vacuum Waste System - Leak Test (P/B 501)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-32-00-910-802	Waste Tube - Maintenance Practice (P/B 201)
38-32-01-000-802	Vacuum Toilet Assembly Removal (P/B 401)
38-32-01-400-802	Vacuum Toilet Assembly Installation (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-4780	Pump Equipment - Vacuum Lav System Part #: A38012-40 Supplier: 81205 Opt Part #: A38012-18 Supplier: 81205 Opt Part #: A38012-32 Supplier: 81205
STD-77	Air Source, Regulated - Dry Filtered, 0-50 psig
STD-419	Gloves - Rubber, Elbow Length
STD-1136	Facepiece (Mask) - Particulate Filtering, Disposable (NIOSH N95 approved)
STD-1137	Glasses - Safety
STD-3901	Container - Hydraulic Fluid Resistant, 50 Gallon (190 l)
STD-3910	Container - Plastic
STD-3926	Water Source - Cold, Regulated, 0 to 60 PSIG
STD-6928	Coupler Sleeve - AS1654T32
STD-6930	Coupling - AS1655A32

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D. Consumable Materials

Reference	Description	Specification
B00046	Acid - Corrosion Removing, Metal Conditioning, Phosphoric	MIL-C-10578
B00636	Acid, Acetic (Vinegar) (5%-10% Acetic Acid)	Commercially available product
B00637	Acid, Citric	A-A-59147
B00638	Cleaner - Acidic Liquid - Honey Bee 60 (McGean-Rohco)	
B01021	Cleaner - Drain (Enzyme) - Dispoz-All	
B01022	Cleaner - enzyme septic	
B50273	Cleaner - Acidic Liquid, Vacuum Toilet Line Cleaner - Sani-Vak G3	AMS 1640
B50329	Cleaner - Acidic Liquid - Honey Bee 60 T	
B50330	Cleaner - Acidic Liquid - Honey Bee 60 TNP	
D50007	Grease - Silicone-Based, Generic	
G00022	Compound - Chlorine Dioxide For Water Purification - Purogene or Oxine	
G02315	Clothing - Disposable Gown, Gloves For Sewage Handling	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
73	O-ring	38-32-07-50-060	PTF ALL

F. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
200	Upper Half of Fuselage

G. Prepare for Cleaning

SUBTASK 38-32-00-010-024

(1) Get access to the passenger compartment.

SUBTASK 38-32-00-010-025

(2) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-910-019

(3) For the work on the waste equipment, make sure that you wear these items for your protection.

NOTE: These are items that you can wear to give you protection when it is necessary.

(a) elbow length rubber gloves, STD-419

(b) filtering facepiece, STD-1136

(c) safety glasses, STD-1137

(d) Disposable clothing, G02315.

SUBTASK 38-32-00-610-019

(4) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: After 5 minutes, drain the precharge from the waste tank.

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SUBTASK 38-32-00-860-043

(5) Start this procedure at the toilet that is farthest upstream of the vacuum waste line to be cleaned.

NOTE: Areas near the bends have the most waste material build-up on the waste tube walls. For the primary layout of the vacuum waste lines, see this reference, (TASK 38-32-00-910-802).

SUBTASK 38-32-00-160-016

(6) Use 1 gal (4 l) - 3 gal (11 l) of the disinfectant that follows:

NOTE: The disinfectant is a solution of chlorine dioxide, citric acid, and water.

(a) Flush the waste line with a chlorine mixture made of chlorine dioxide (stabilized 2%) Purogene or Oxine compound, G00022, and citric acid, B00637(crystals or powder), from the toilet farthest from the waste tank on the waste line being cleaned, as follows:

WARNING DO NOT BREATHE CHLORINE DIOXIDE GAS. PUT ON A RESPIRATOR, AND GOGGLES. WHEN THE TWO CHEMICALS MIX, THEY MAKE CHLORINE DIOXIDE GAS. CHLORINE DIOXIDE GAS CAN CAUSE INJURIES TO PERSONNEL IF THEY BREATHE IT.

1) For each 1 gal (4 l) of final solution desired, mix 20 fl-oz (0.6 l) of chlorine dioxide Purogene or Oxine compound, G00022, with 2 oz (57 g) of citric acid, B00637, in a container, STD-3910.
2) Stop for 5 minutes (activation period).
3) Use a clean instrument to mix the solution fully.
4) Add water to make the desired quantity of solution.
5) Flush 1 gal (4 l) to 3 gal (11 l) of the disinfectant solution into the waste line.

NOTE: The quantity of solution necessary to disinfect the vacuum toilet line changes with the length and condition of the line. Operator experience shows that 1 gal (4 l) is sufficient for short vacuum lines, but more than 1 gal (4 l) is necessary for longer vacuum lines. The quantity of disinfectant necessary for a specific vacuum toilet line is established by the operator's experience.

SUBTASK 38-32-00-160-017

(7) Do this task: Periodic Flush - Vacuum Waste System Cleaning, TASK 38-32-00-100-801.

(a) Rinse the lines from the most forward toilet with approximately ½ gal (2 l) of water.

(b) Turn the water off to the toilet.

(c) Flush the toilet without water at least three times.

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WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(d) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-010-026

(8) To get access to the waste lines, do one of these steps:

(a) To remove the waste tank enclosure, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(b) For both lines to be cleaned, disconnect the last sections of the waste tubes [21] with the waste tank inlets [71] from the waste tank.

1) Remove the v-band tank couplings [72] to disconnect the waste tank inlets [71] from the waste tank.

2) Remove the o-rings [73] between the waste tank inlets [71] and waste tank.

a) If necessary, discard the o-rings [73].

(c) Replace the waste line section with the vacuum lav system pump equipment, SPL-4780.

1) Make sure that the vacuum lav system pump equipment, SPL-4780, holds in the position.

a) If the vacuum lav system pump equipment, SPL-4780, is not in its position, replace the upstream section of waste.

(d) Remove the waste tubes from their installed position, and connect the drain valve [23] to the airplane waste line, using the coupling, STD-6930, and sleeve, STD-6928, as necessary.

NOTE: Remaining liquid can stay in the waste lines. Be prepared to collect and then discard all of the liquid.

(e) Close the drain valves [23].

NOTE: If the drain valves [23] are open, the pressure test that follows will be bad.

(f) For all lavatories on the waste lines to be cleaned, do this task: Vacuum Toilet Assembly Removal, TASK 38-32-01-000-802.

NOTE: The toilet assembly removal is necessary to attach the tools for the pressure test and the drain/flush equipment.

SUBTASK 38-32-00-780-003

(9) Do this task: Pressure Test Procedure, TASK 38-32-00-780-802.

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H. Waste Line Cleaning

SUBTASK 38-32-00-840-001

(1) Prepare the chemical solution to clean the waste line with one of the chemical cleaners that follows:

NOTE: The quantity of the cleaner necessary for a waste line is 0.16 gallons for each linear foot (2.0 liters for each linear meter). Calculate the maximum quantity of the cleaner necessary for each line to prevent a spill when you fill the waste line.

(a) Honey Bee 60 cleaner, B00638, Honey Bee 60 T cleaner, B50329, or Honey Bee 60 TNP cleaner, B50330

(b) 5 to 10% Acetic acid, B00636

(c) 7% Phosphoric acid, B00046

(d) Dispoz-All cleaner, B01021

(e) enzyme septic cleaner, B01022

(f) Sani-Vak G3, B50273.

SUBTASK 38-32-00-110-003

CAUTION

DO NOT LET THE CLEANING SOLUTION LEAK FROM THE WASTE SYSTEM INTO THE AIRPLANE. MAKE SURE THAT A PERSON WITH SPILLAGE-PREVENTION EQUIPMENT HAS ACCESS TO EACH LAVATORY THAT IS CONNECTED TO THE WASTE LINE. THE CLEANING SOLUTION CAN CAUSE DAMAGE TO THE AIRPLANE STRUCTURE.

(2) Do these steps to connect the vacuum lav system pump equipment, SPL-4780 to fill the waste lines with the cleaner solution:

NOTE: This procedure is used to clean two waste lines simultaneously, designate a waste line #1 and a waste line #2 to help clarify tool usage. Refer to Figure 703 (Sheet 5) for a pump equipment installation schematic.

(a) Connect one end of the supply hose [24] to the quick-disconnect [22] on the drain valve [23] near the waste tank.

1) Install straps or blocking to hold the waste line connection.

NOTE: If the waste line is not held in its position, a possible leak below the floor can occur.

(b) Connect the other end of the supply hose [24] to the air pump [25] at the quick-disconnect [26].

(c) On the remaining line to be cleaned, connect one end of the drain hose [29] to the quick-disconnect [22] on the drain valve [23] near the waste tank.

(d) Connect the other end of the drain hose [29] to the drain end assembly [39] and put the hose end into the 50 Gallon (190 l) hydraulic fluid resistant container, STD-3901, of solution.

NOTE: Put a sufficient amount of the Drain End Hose into the container so that it will not come out when the air pump [25] is turned on.

(e) In each lavatory between the farthest upstream toilet and the waste tank make sure a flush plug assembly [38] is installed in place of the toilet assembly.

1) Make sure the bleed valve on each plug assembly [38] is closed.

(f) At the forward most lavatory of waste line #1 install the air pressure assembly [30] and the flush plug valve [32] on the toilet waste connection [31] (Figure 703 (Sheet 4)).

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(g) At the forward most lavatory of waste line #2, install the flush plug valve [32] at the toilet waste connection [36].

(h) Connect the jumper hose valves [34] and jumper hose [35] at the quick-disconnects [33] to join the two waste lines.

(i) Connect an air hose [37] to the brass nipple on the air pressure assembly [30].

(j) Open all the valves on the hoses and on the drain valves [23].

CAUTION

MAKE SURE THAT THE PERSON WITH THE EQUIPMENT TO PREVENT LEAKAGE (RAGS, ETC), HAS ACCESS TO EACH LAVATORY IN THE WASTE SYSTEM THAT YOU FILL. MAKE SURE THAT THERE IS NO LEAKAGE OF THE CLEANER SOLUTION FROM THE WASTE SYSTEM INTO THE LAVATORY DURING THIS PROCEDURE. LEAKAGE OF CHEMICAL SOLUTIONS CAN CAUSE DAMAGE TO THE INTERIOR OF THE AIRCRAFT.

(k) Put the air pump [25] intake into the container of solution and turn the pump on to fill the waste lines.

NOTE: It can take 3 to 4 minutes before the solution starts to come down the drain hose [29]. It can take more than an hour for all of the air to exit the lines.

(l) Slowly open the bleed valve in the waste line plug assembly [38] in each of the lavatories to bleed air from the toilet waste lines

(m) Close the bleed valve when you hear or see solution start to come through the valve.

SUBTASK 38-32-00-110-004

CAUTION

MAKE SURE THAT THE PERSON WITH THE EQUIPMENT TO PREVENT LEAKAGE (RAGS, ETC), HAS ACCESS TO EACH LAVATORY IN THE WASTE SYSTEM THAT YOU FILL. MAKE SURE THAT THERE IS NO LEAKAGE OF THE CLEANER SOLUTION FROM THE WASTE SYSTEM INTO THE LAVATORY DURING THIS PROCEDURE. LEAKAGE OF CHEMICAL SOLUTIONS CAN CAUSE DAMAGE TO THE INTERIOR OF THE AIRCRAFT.

(3) Let the cleaner solution circulate in the waste line(s) for as long as practical (a minimum of 8 hours is suggested).

NOTE: To fully clean the waste line(s), more than 24 hours can be necessary to remove the heavy scale. The longer you leave the cleaner in the waste lines, the better the results. The length of time is established by the operators experience.

(a) If you spill the cleaning solution, do this task: Corrosion Removal After Acid Spills, TASK 05-51-57-000-802.

SUBTASK 38-32-00-860-044

(4) To get cleaner lines and to prevent blockage caused by scale material you can reverse the flow of the solution during the procedure.

(a) Turn off the air pump [25] and close the drain valves [23] on the lines near the tanks.

(b) Close the valves on the drain hose [29] and the supply hose [24] where they connect to the drain valves [23] near the tanks.

(c) Disconnect the supply hose [24] and the drain hose [29]from the drain valves [23] and reconnect them to the opposite drain valve [23] on the opposite waste line.

(d) Restart the air pump [25].

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I. Drain the Waste Line

SUBTASK 38-32-00-840-002

(1) Do these steps to drain the chemical solution from the system:

(a) Stop the air pump [25].

(b) Close the supply hose valve [40] and disconnect it from the air pump [25] at the quick-disconnect [26].

(c) Put the end of the supply hose [24] into the container and open the supply hose valve [40]. The solution will start to drain from the hose.

(d) To flush the waste line, do the steps that follow:

1) Connect an regulated air source, STD-77, to the air pressure assembly [30] to force the solution out of the waste lines and into the container.

2) Connect a 0 to 60 PSIG regulated cold water source, STD-3926, to the air pressure assembly [30] adapter in the farthest upstream lavatory

3) Flush the waste line with a minimum of 5 gal (19 l) of water for each toilet on the waste line.

4) Connect an regulated air source, STD-77, to the pressure adapter.

5) Dry the waste line with air flow, under 15 psi (103 kPa) for approximately 5 minutes.

(e) Stop when the cleaning solution and flushing water is fully drained from the waste line.

(f) If it is necessary to drain other waste lines, do these steps again to drain waste lines.

SUBTASK 38-32-00-410-013

(2) Do these steps to remove the vacuum lav system pump equipment, SPL-4780:

(a) In the farthest upstream lavatory, remove the air pressure assembly [30] and jumper hose [35].

(b) In each lavatory between the farthest upstream toilet and the waste tank, remove the waste line plug assembly [38].

(c) Do this task: Vacuum Toilet Assembly Installation, TASK 38-32-01-400-802.

(d) Remove the drain valves [23] at the connection near the waste tanks for both lines that were cleaned.

NOTE: Remaining liquid can stay in the waste lines. Be prepared to collect and then discard all of the liquid.

(e) Make sure that all solid material is out of the waste line before you connect the waste line to the waste tank.

NOTE: Remaining solid material in the waste line can cause blocked waste lines after you complete this procedure.

(f) Connect the last section of the waste tubes [21], to the waste tank inlets [71] to the waste tank.

1) Apply silicone-based grease, D50007, to the o-rings [73].

2) Install the o-rings [73].

3) Install the v-band tank couplings [72] to connect the waste tank inlets [71].

4) Tighten the v-band tank couplings [72] to 40 in-lb (5 N·m)-50 in-lb (6 N·m).

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J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-860-045

(1) Do these steps complete this task:

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(a) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-170-011

(2) To push other solid material out of the waste system, do one of these procedures:

(a) Do this task: Periodic Flush - Vacuum Waste System Cleaning, TASK 38-32-00-100-801.

(b) Do this task: Pressure Washer Cleaning of the Vacuum Waste Lines, TASK 38-32-00-100-802.

SUBTASK 38-32-00-610-020

(3) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-00-610-021

(4) Do this task: Vacuum Waste System - Leak Test, TASK 38-32-00-700-801.

SUBTASK 38-32-00-860-046

(5) Do this task:Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801

SUBTASK 38-32-00-410-014

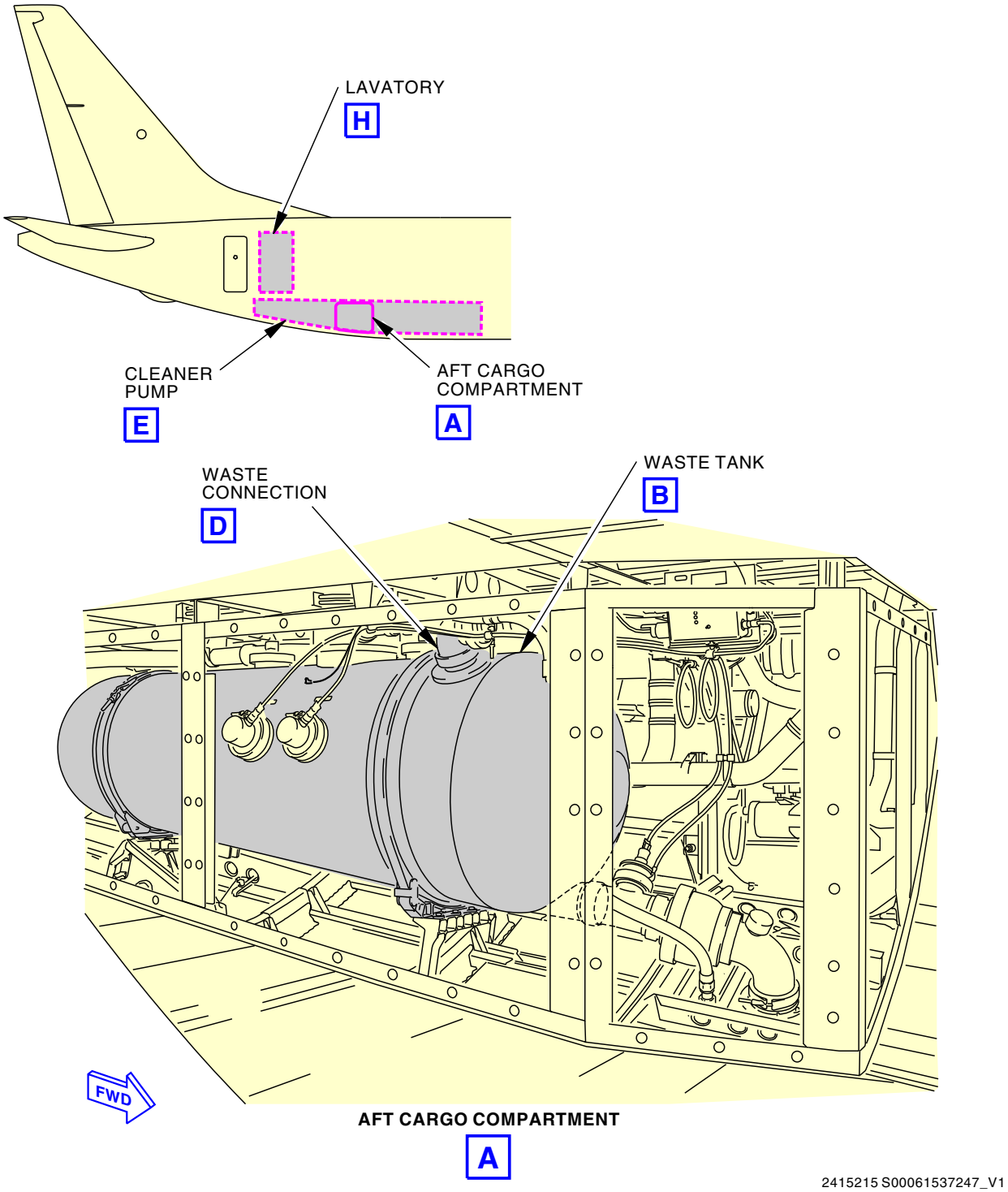
(6) Close the access to the passenger compartment.

SUBTASK 38-32-00-910-020

(7) Obey the local authorities to discard the chemical solution.

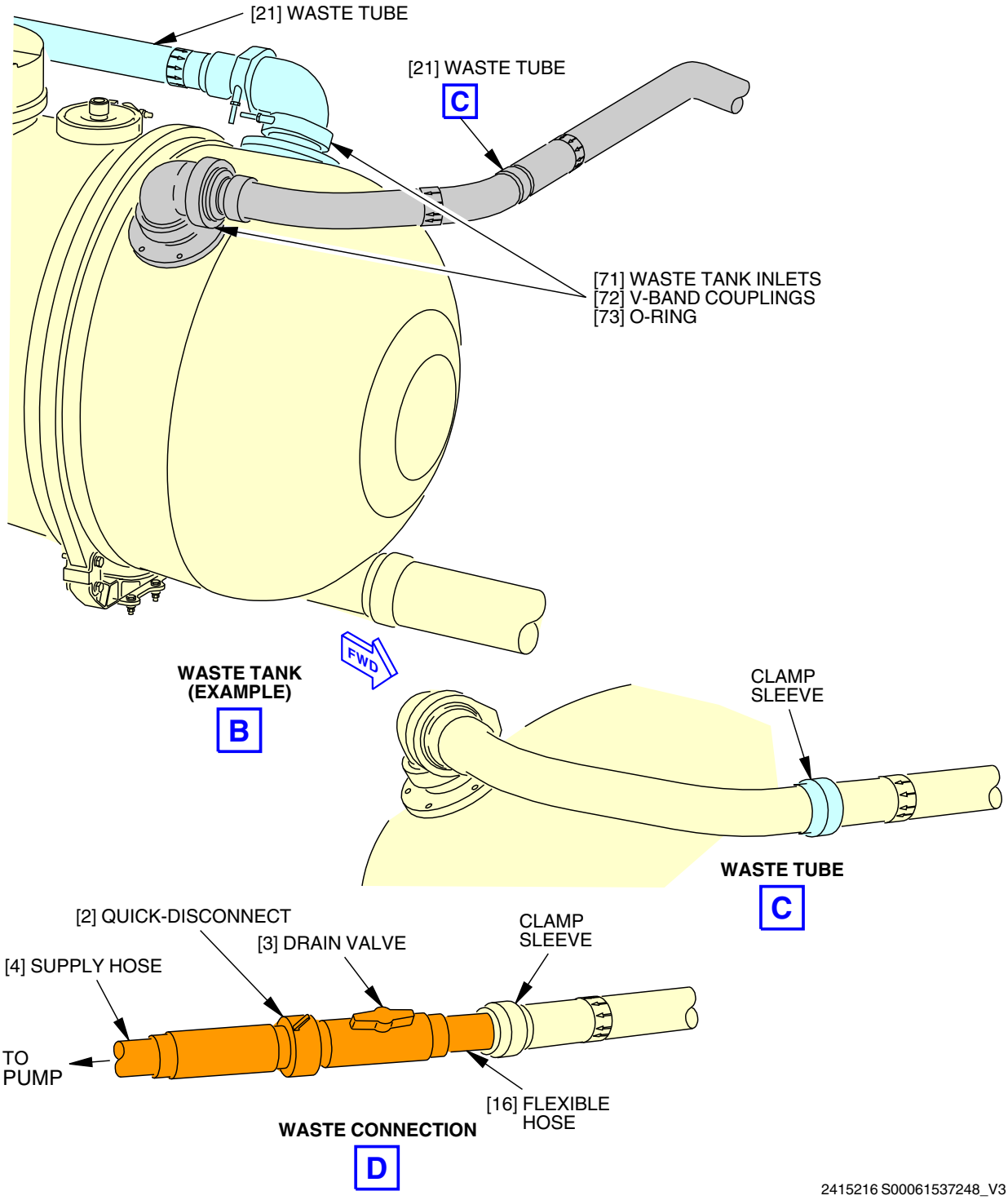
——— END OF TASK ———

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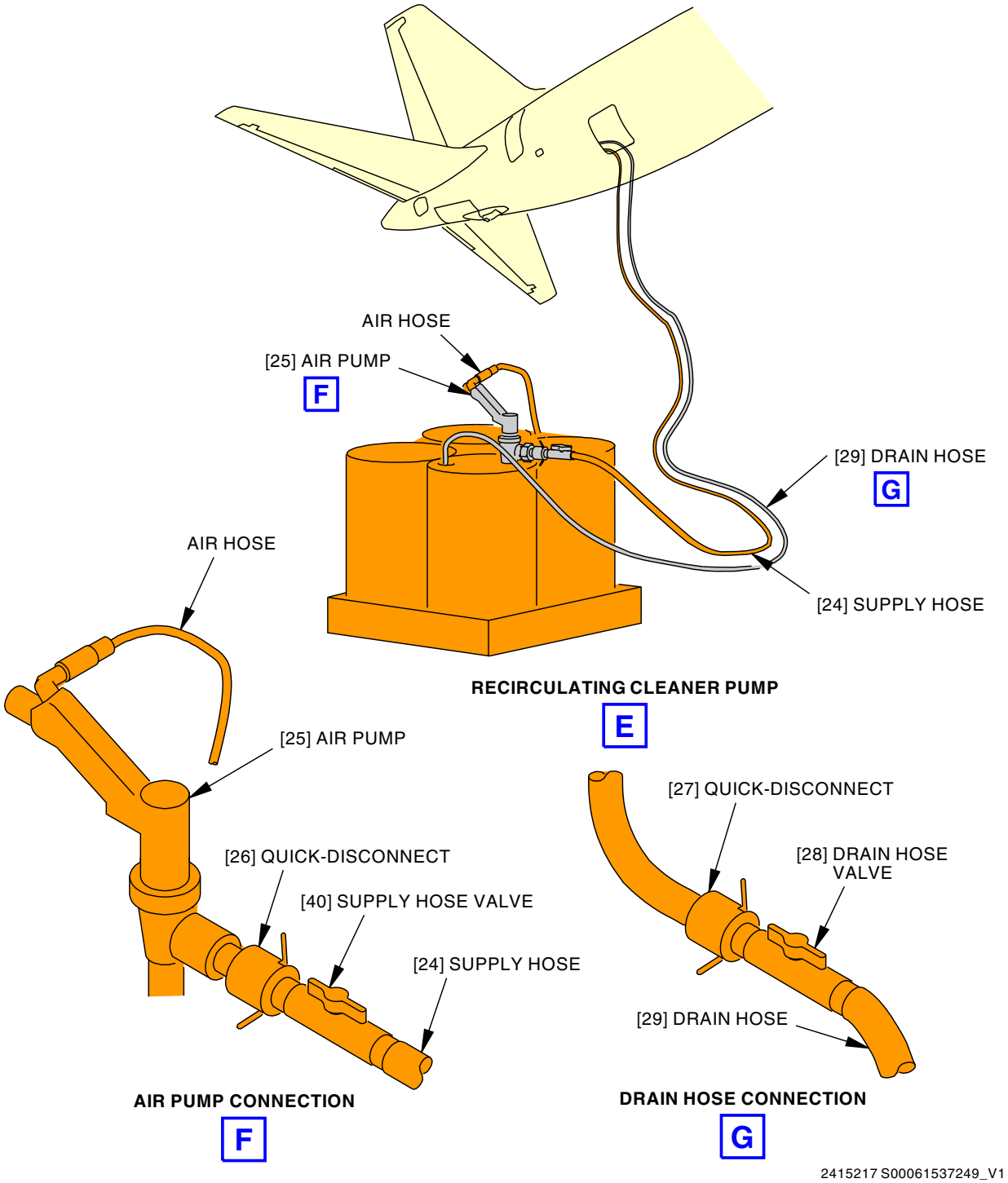
Recirculating Waste Line Cleaning
Figure 703/38-32-00-990-814 (Sheet 1 of 5)

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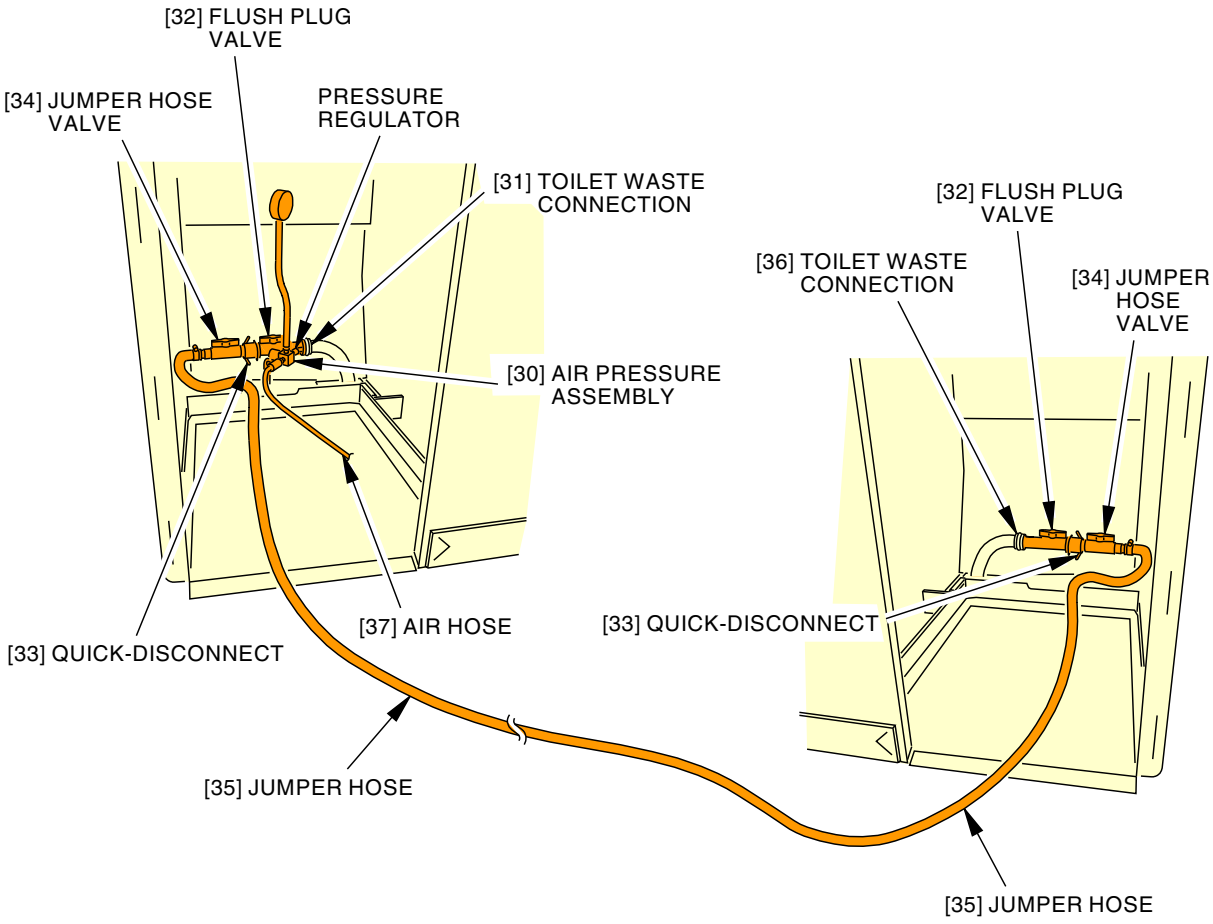
Recirculating Waste Line Cleaning
Figure 703/38-32-00-990-814 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Recirculating Waste Line Cleaning
Figure 703/38-32-00-990-814 (Sheet 3 of 5)

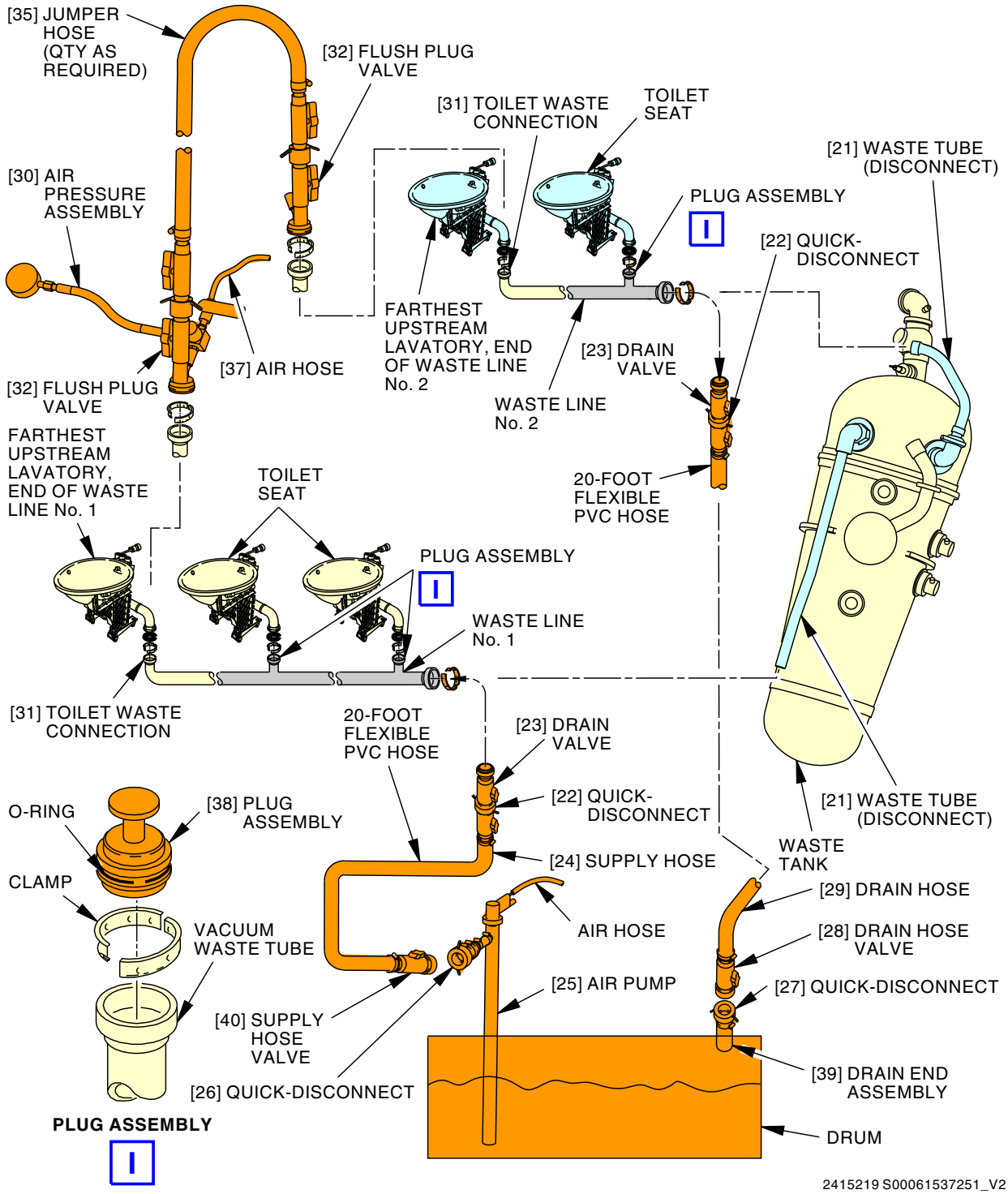
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Recirculating Waste Line Cleaning
Figure 703/38-32-00-990-814 (Sheet 4 of 5)

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Recirculating Waste Line Cleaning
Figure 703/38-32-00-990-814 (Sheet 5 of 5)

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TASK 38-32-00-110-802

6. Recirculating Waste System Cleaning

A. General

(1) This procedure gives the instructions to clean the vacuum waste tubes and tank. Use this procedure to decrease the scale that collects in the vacuum waste tubes.

(2) This procedure removes the toilets and fills the vacuum waste line with chemical cleaner at the toilet connection. The cleaner drains from the waste service panel and recirculates through the pump equipment, into the airplane. The time to the clean the vacuum waste system is 6-10 hours. The condition of the vacuum waste system changes the time necessary.

(3) This procedure uses negative (vacuum) pressure to recirculate the cleaner. A pressure check is not necessary.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-32-01-000-802	Vacuum Toilet Assembly Removal (P/B 401)
38-32-01-400-802	Vacuum Toilet Assembly Installation (P/B 401)
38-32-02-000-801	Liquid Separator Removal (P/B 201)
38-32-02-400-801	Liquid Separator Installation (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-13842	Equipment - Pump, Recirculating Heated Fluid, Vacuum Lav. System Part #: J38004-287 Supplier: 81205 Opt Part #: J38004-232 Supplier: 81205 Opt Part #: J38004-247 Supplier: 81205
STD-6842	Air Supply - Pneumatic, Hangar

D. Consumable Materials

Reference	Description	Specification
B50273	Cleaner - Acidic Liquid, Vacuum Toilet Line Cleaner - Sani-Vak G3	AMS 1640

E. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
200	Upper Half of Fuselage

F. Prepare for the Cleaning

SUBTASK 38-32-00-610-022

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

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SUBTASK 38-32-00-110-005

(2) Do this task: Periodic Flush - Vacuum Waste System Cleaning, TASK 38-32-00-100-801.

SUBTASK 38-32-00-610-023

(3) Do this task, but do not do the steps to add precharge: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-00-480-007

(4) Do these steps to isolate the overboard vent system.

NOTE: The plugs in the vent overboard tubes will prevent the chemical cleaner from spilling if the waste tank is filled above the liquid separator.

(a) Remove the waste tank enclosure, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

(b) Loosen the clamps and disconnect the flexible couplings from the intake and outflow of the vacuum blower.

(c) Install the plug [60] on the flexible coupling at the vacuum blower intake with the clamp.

(d) Install the plug outlet [61] on the flexible coupling at the vacuum blower outflow with the clamp.

(e) Install the vent plug assembly [62] in the vent port outlet, at the skin.

SUBTASK 38-32-00-480-008

(5) Assemble the recirculating pump equipment, SPL-13842 tool.

(a) Attach the clean supply hose [55] to the tool and tank 1.

(b) Attach the return hose [54] to the tool and tank 2.

(c) Attach the recirculation hose [57] to the tool and tank 2.

(d) Attach the drain hose [59] to the tool.

(e) Attach the supply hose [58] to the tool.

(f) Connect hangar pneumatic air supply, STD-6842 to the tool control panel

SUBTASK 38-32-00-860-047

(6) Connect the drain hose [59] to the service panel drain fitting.

SUBTASK 38-32-00-860-048

(7) Pull the drain valve handle open to the “DRAIN” position.

SUBTASK 38-32-00-010-027

(8) For all lavatories on the waste line to be cleaned, do this task: Vacuum Toilet Assembly Removal, TASK 38-32-01-000-802.

NOTE: The toilet assembly removal is necessary to attach the plug and flush plug equipment.

SUBTASK 38-32-00-480-009

CAUTION

DO NOT LET THE CLEANING SOLUTION LEAK FROM THE WASTE SYSTEM INTO THE AIRPLANE. MAKE SURE THAT A PERSON WITH SPILLAGE-PREVENTION EQUIPMENT HAS ACCESS TO EACH LAVATORY THAT IS CONNECTED TO THE WASTE LINE. THE CLEANING SOLUTION CAN CAUSE DAMAGE TO THE AIRPLANE STRUCTURE.

(9) Install the flush plug [51] and supply hose [58] to the most forward toilet.

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(a) Install straps or blocking to hold the waste line connection.

NOTE: If the waste line is not held in its position, a possible leak below the floor can occur.

SUBTASK 38-32-00-480-010

(10) Install a plug [52] in each lavatory between the most forward toilet and the waste tank

G. Waste Line Cleaning

SUBTASK 38-32-00-110-006

(1) Prepare the chemical solution to clean the waste line with one of the chemical cleaners that follows:

NOTE: The quantity of the cleaner necessary for the waste line and tool hoses is 0.16 gallons for each linear foot (2.0 liters for each linear meter).

(a) Sani-Vak G3, B50273

SUBTASK 38-32-00-110-007

(2) Fill tank 1 with cleaner.

SUBTASK 38-32-00-110-008

(3) Fill the waste lines with cleaner.

(a) Start the flow meter.

(b) Move the valve levers to the “CLEAN FLUID” AND “QUICK CLEAN” positions.

(c) Move the pump levers to “NEGATIVE PRESSURE PUMP OFF” AND “POSITIVE PRESSURE PUMP ON” positions.

(d) Make sure that the hose connections at the recirculating pump equipment, SPL-13842 tool and tanks do not have a leak.

(e) Make sure that the supply hose [58] to the toilet connection does not have a leak.

(f) Make sure that each lavatory plug [52] does not have a leak.

(g) Let the cleaning fluid fill the vacuum waste lines.

NOTE: The waste line will be full when you see the cleaner in the drain hose at the service panel.

SUBTASK 38-32-00-110-009

(4) Recirculate the cleaning fluid for a minimum of 6 hours.

NOTE: The time necessary to clean the waste lines is 6-10 hours. The condition of the vacuum waste system changes the time necessary

(a) Move the valve levers to the “RECIRCULATION” positions.

(b) Move the pump levers to “NEGATIVE PRESSURE PUMP ON” AND “POSITIVE PRESSURE PUMP OFF” positions.

(c) Record the flow rate.

(d) Record the differential pressure between the negative pump pressure and the heater pressure.

(e) If heat is necessary, start the heater.

1) Make sure that the cleaner you use is satisfactory for heat. Follow the manufacturers directions.

NOTE: The heater will not let the temperature be more than 120°F (49°C).

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SUBTASK 38-32-00-110-010

(5) Make sure that the lavatories and hose connections do not have a leak.

SUBTASK 38-32-00-110-011

(6) Make sure that the recirculation flow continues.

(a) Look at the drain hose [59] attached to the service panel.

(b) If the flow from the drain hose [59] falls or stops, do these steps to clean the waste tank:

NOTE: Scale from the waste lines or waste tank can collect and cause a blockage at the waste tank outlet.

1) Move the pump lever to “NEGATIVE PRESSURE PUMP OFF”.

2) Do this task: Liquid Separator Removal, TASK 38-32-02-000-801.

3) Remove the loose scale from the bottom of the waste tank.

4) Do this task: Liquid Separator Installation, TASK 38-32-02-400-801

5) Move the pump lever to “NEGATIVE PRESSURE PUMP ON”.

(c) Look at the flow meter and pressure gages at the control panel.

(d) If the flow rate decreases, or the differential pressure between the negative pump and the heater increases, do these steps to clean the filter:

NOTE: Scale from the waste lines or waste tank can collect and cause a blockage at the filter. If the flow rate or the differential pressure increases more than 30%, it is necessary to clean the filter. It is necessary to drain the system to clean the filter.

1) Move the valve lever to the “CLEAN FLUID” position.

2) When all of the cleaning fluid has drained from the airplane, move the pump lever to “NEGATIVE PRESSURE PUMP OFF”.

NOTE: This step will start to drain the waste line. If there is cleaning fluid in tank 1, it will pump into the waste line before the system drains.

3) Open the filter [53] housing cap.

4) Empty or replace the filter bags.

5) Close the filter [53] housing cap.

6) Move the valve levers to the “CLEAN FLUID” AND “QUICK CLEAN” positions.

7) Move the pump levers to “NEGATIVE PRESSURE PUMP OFF” AND “POSITIVE PRESSURE PUMP ON” positions.

8) Let the cleaning fluid fill the vacuum waste lines.

NOTE: The waste line will be full when you see the cleaner in the drain hose at the service panel.

9) Move the valve levers to the “RECIRCULATION” positions.

10) Move the pump levers to “NEGATIVE PRESSURE PUMP ON” AND “POSITIVE PRESSURE PUMP OFF” positions.

SUBTASK 38-32-00-110-012

(7) Drain the cleaning fluid from the waste lines.

(a) Move the valve lever to the “CLEAN FLUID” position.

(b) When all of the cleaning fluid has drained from the airplane, move the pump lever to “NEGATIVE PRESSURE PUMP OFF”.

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SUBTASK 38-32-00-170-012

(8) Flush the vacuum waste system and recirculating pump equipment, SPL-13842 with water.

(a) Put the 50 gallons of water in tank 1.

(b) Move the valve levers to the “CLEAN FLUID” AND “QUICK CLEAN” positions.

(c) Move the pump levers to “NEGATIVE PRESSURE PUMP OFF” AND “POSITIVE PRESSURE PUMP ON” positions.

(d) Let the water fill the vacuum waste lines.

NOTE: The waste line will be full when you see the water in the drain hose at the service panel.

(e) Move the pump levers to “NEGATIVE PRESSURE PUMP ON” AND “POSITIVE PRESSURE PUMP OFF” positions.

NOTE: This will let most of the water drain from the airplane.

(f) When the water has drained from the airplane, move the pump lever to “POSITIVE PRESSURE PUMP OFF”.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-080-008

(1) Remove the plug [52] from each lavatory.

SUBTASK 38-32-00-080-009

(2) Remove the flush plug [51] and supply hose [58] from the most forward toilet.

SUBTASK 38-32-00-410-015

(3) Install all of the toilets. Do this task: Vacuum Toilet Assembly Installation, TASK 38-32-01-400-802.

SUBTASK 38-32-00-080-010

(4) Do these steps to restore the overboard vent system.

(a) Remove the vent plug assembly [62] in the vent port outlet, at the skin.

(b) Remove the plug outlet [61] on the flexible coupling at the vacuum blower outflow.

(c) Remove the plug [60] on the flexible coupling at the vacuum blower intake.

(d) Connect the flexible couplings to the intake and outflow of the vacuum blower and tighten the clamps.

(e) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-00-160-018

(5) Clean the recirculating pump equipment, SPL-13842 filter.

(a) Open the filter housing cap.

(b) Clean or replace the filter bags.

(c) Close the filter housing cap.

SUBTASK 38-32-00-080-011

(6) Disassemble the recirculating pump equipment, SPL-13842 tool.

(a) Remove the clean supply hose [55] from the tool and tank 1.

(b) Remove the recirculation hose [57] and the return hose [54] from the tool and tank 2.

(c) Remove the drain hose [59] from the tool.

(d) Remove the supply hose [58] from the tool.

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SUBTASK 38-32-00-860-049

(7) Push the drain valve handle open to the “CLOSED” position.

SUBTASK 38-32-00-080-012

(8) Remove the drain hose [59] from the service panel drain fitting.

SUBTASK 38-32-00-610-024

(9) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-00-410-016

(10) Do this task:Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801

SUBTASK 38-32-00-410-017

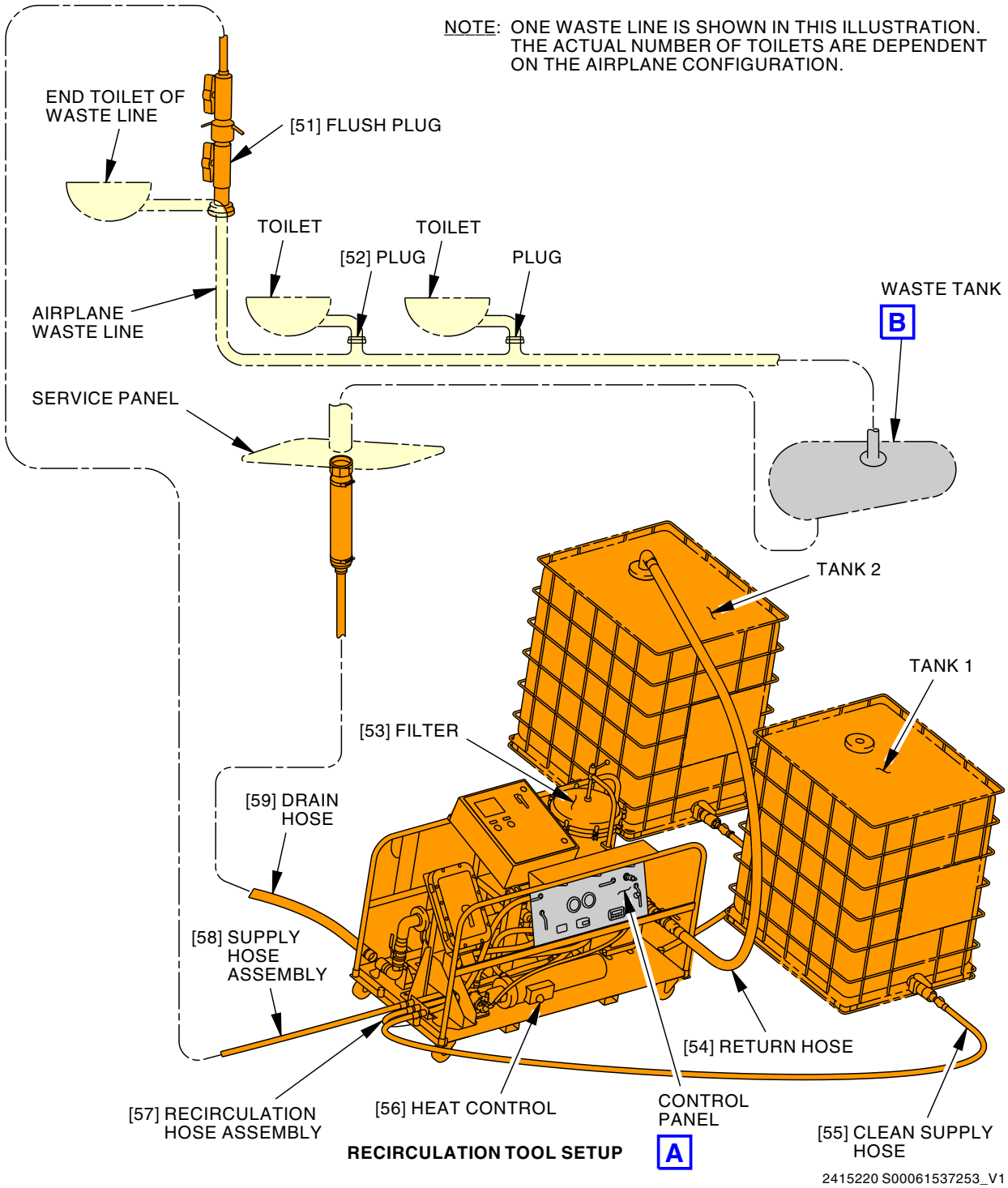
(11) Close the access to the passenger compartment.

SUBTASK 38-32-00-500-001

(12) Obey the local authorities to discard the chemical solution.

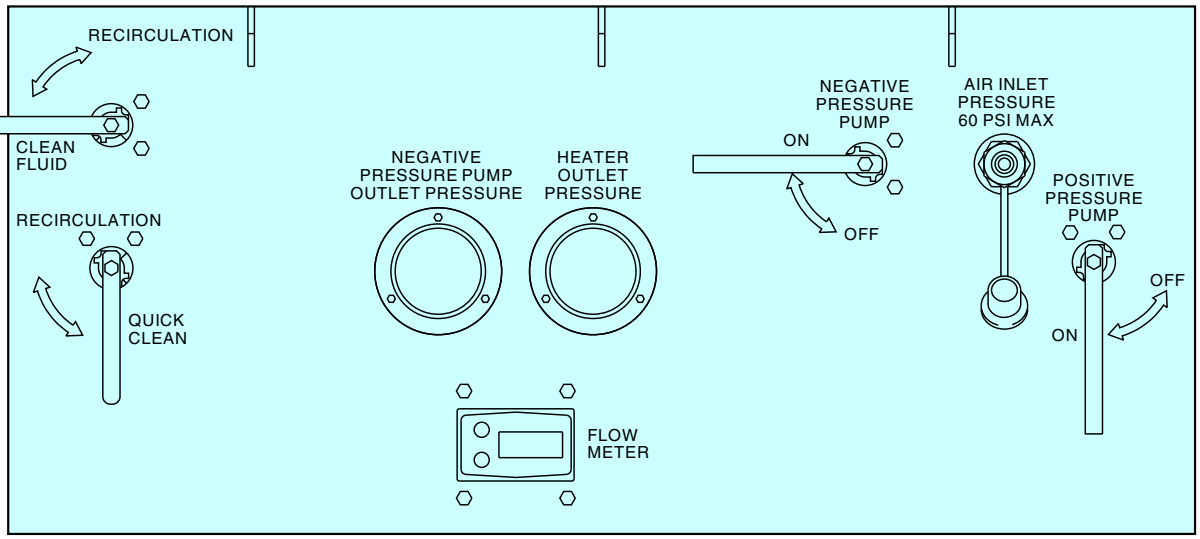
——— END OF TASK ———

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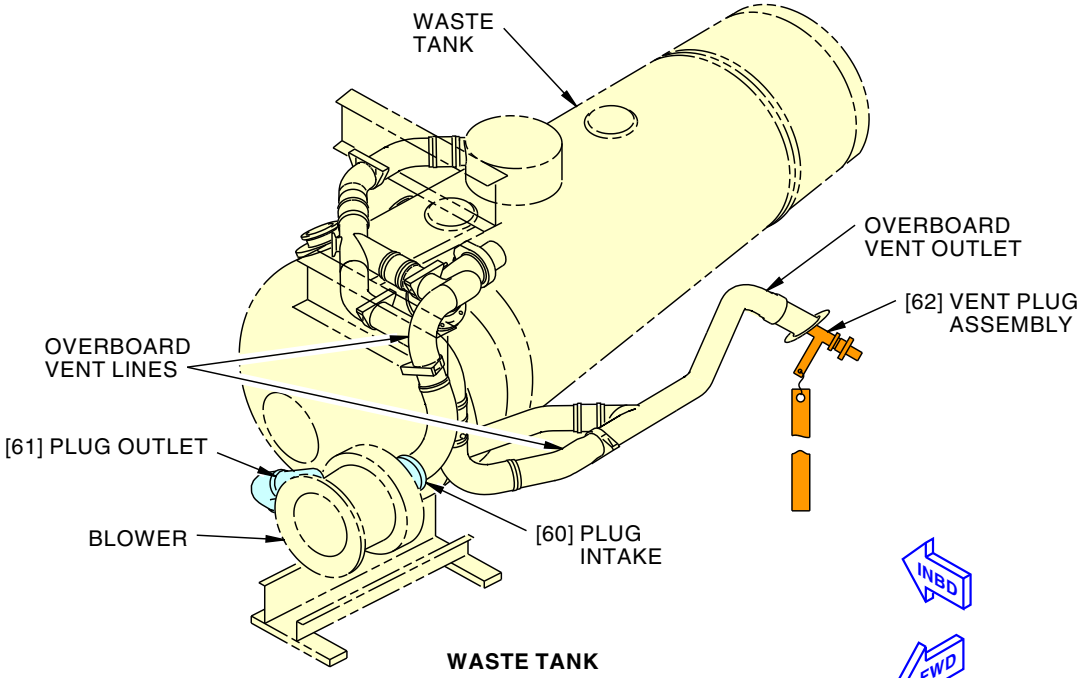
Recirculating Waste Line Cleaning
Figure 704/38-32-00-990-815 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CONTROL PANEL

A



WASTE TANK

B

2415221 S00061537254_V1

Recirculating Waste Line Cleaning
Figure 704/38-32-00-990-815 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-780-802

7. Pressure Test Procedure

(Figure 705)

A. General

(1) This procedure is used before the Soak or Recirculate procedures to make sure that the waste lines will not leak.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1953	Equipment - Pump, Static Soak, Vacuum Lav. System Part #: A38012-39 Supplier: 81205 Opt Part #: A38012-1 Supplier: 81205 Opt Part #: A38012-31 Supplier: 81205
SPL-4780	Pump Equipment - Vacuum Lav System Part #: A38012-40 Supplier: 81205 Opt Part #: A38012-18 Supplier: 81205 Opt Part #: A38012-32 Supplier: 81205
STD-77	Air Source, Regulated - Dry Filtered, 0-50 psig

D. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
200	Upper Half of Fuselage

E. Prepare for Pressurization

SUBTASK 38-32-00-010-028

(1) Get access to the passenger compartment.

SUBTASK 38-32-00-910-021

(2) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-00-610-025

(3) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: After 5 minutes, drain the precharge from the waste tank.

SUBTASK 38-32-00-780-004

(4) Do a pressure decay check for each waste line to be cleaned as follows:

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CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(a) In the farthest upstream lavatory of the waste line, install a Flush Plug and an Air Pressure Assembly from the static soak pump equipment, SPL-1953, or vacuum lav system pump equipment, SPL-4780.

(b) If you want to test more than one line at the same time, install a Flush Plug in the furthest upstream lavatory of another waste line and connect the lavatories with the Jumper Hose from the vacuum lav system pump equipment, SPL-4780. Additional Jumper hoses must be added for distances over ten feet.

NOTE: Make sure the valves on the Flush Plugs and Jumper Hose are open.

(c) At the end of the waste line in each lavatory between the most upstream toilet and the waste tank, install a Plug.

(d) Install a Drain Valve at the end of the waste tube inlet where the section of tubing has been removed.

NOTE: Make sure the valve is in the closed position.

(e) Connect the pressurization equipment (Figure 705) to the Air Pressure Assembly in the lavatory.

(f) Pressurize the waste line system to 5 psi (34 kPa) to 10 psi (69 kPa) and then turn off the regulated air source, STD-77.

(g) Stop for 5 minutes.

(h) If the pressure has degraded more than 0.5 psi (3.4 kPa) over the 5 minutes, the system cannot safely be cleaned.

NOTE: You must repair all leaks. When you repair the leak, then do this pressure test.

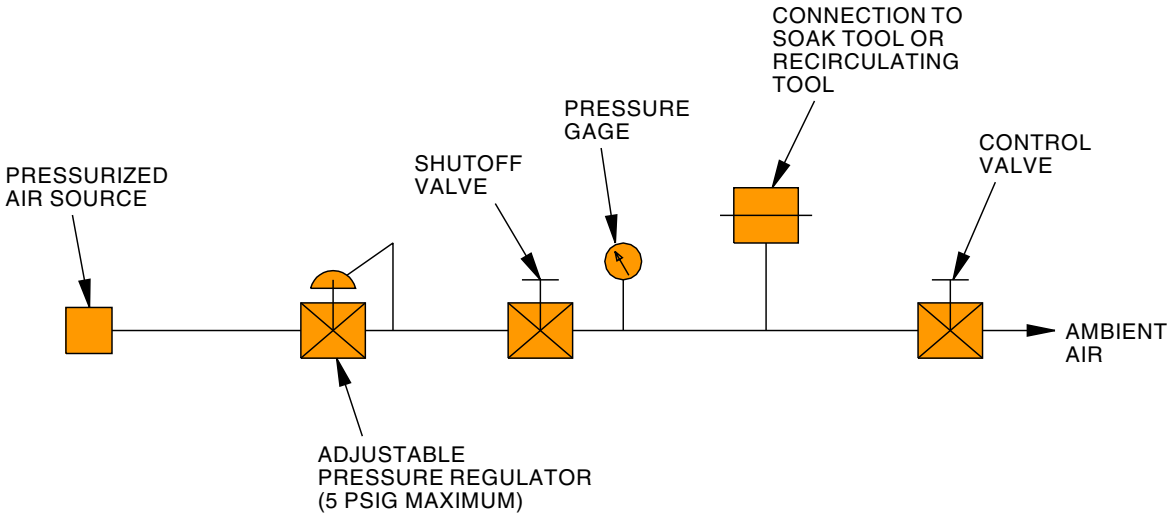
(i) If the pressure has not degraded by more than 0.5 psi (3.4 kPa), the system is fluid tight.

(j) Release the pressure from the waste system.

(k) Disconnect the pressurization equipment, but keep the Air Pressure Assembly, the Plugs and the Flush Plugs in their positions.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2415222 S00061537256_V1

Soak Pressure Test
Figure 705/38-32-00-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-00-100-804

8. Vacuum Waste Tube Off Airplane Soak

A. General

(1) This task gives the steps for an off-aircraft soak procedure for cleaning a vacuum waste tube that is nearly or completely blocked and can not be cleared by other means.

NOTE: You use this procedure to clean a vacuum waste tube that has a heavy waste build-up or blockage.

(2) If there is too much or a quickly reoccurring scale buildup, make sure that the toilet flush valves fully close. A valve that is not fully closed can cause a drying effect in the tubes. This drying effect can increase the collection of scale. A valve that is not fully closed will cause air noise at the toilet when a different toilet on the system operates.

(3) There are additional tasks that give instructions to clean the vacuum waste tubes. Use these tasks to decrease the scale that collects in the vacuum waste tubes. Use operational experience to find which task or mixture of tasks are necessary and the necessary time period between tasks.

These are the additional tasks that you can use:

(a) A cleaning by a periodic flush of a chemical through the vacuum waste system, Periodic Flush - Vacuum Waste System Cleaning, TASK 38-32-00-100-801.

NOTE: You use this procedure to keep the vacuum waste lines clean of the waste build-up. To get the maximum effect, you must frequently do this task. Frequency and amount is operator determined from experience.

(b) A pressure washer cleaning of the vacuum waste lines, Pressure Washer Cleaning of the Vacuum Waste Lines, TASK 38-32-00-100-802.

NOTE: You use this procedure to clean the vacuum waste lines of the waste build-up. To get the maximum effect, you must use this procedure before you have frequent problems with the vacuum waste lines.

(c) An in-situ soak method of cleaning the vacuum waste lines, Soak Method for Cleaning of the Vacuum Waste Lines, TASK 38-32-00-100-803.

NOTE: You use this procedure to clean the vacuum waste lines of the heavy waste build-up. To get the maximum effect, you must use this procedure before the waste lines are fully blocked.

(d) An in-situ recirculate method of cleaning the vacuum waste lines (TASK 38-32-00-110-801).

NOTE: You use this procedure to clean the vacuum waste lines of the heavy waste build-up. To get the maximum effect, you must use this procedure before the waste lines are fully blocked. This method will work quicker than the in-situ soak method.

(e) A pressure test procedure, Pressure Test Procedure, TASK 38-32-00-780-802.

NOTE: This procedure is used for the in-situ Soak or Recirculate procedures to make sure that the waste lines will not leak while you do the cleaning procedure.

B. References

Reference	Title
38-32-00-020-801	Vacuum Waste Tubes Removal (P/B 401)
38-32-00-420-802	Vacuum Waste Tubes Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-197	Container - Plastic, Polyethylene or Polypropylene

D. Consumable Materials

Reference	Description	Specification
B00046	Acid - Corrosion Removing, Metal Conditioning, Phosphoric	MIL-C-10578
B00636	Acid, Acetic (Vinegar) (5%-10% Acetic Acid)	Commercially available product
B00637	Acid, Citric	A-A-59147
B00638	Cleaner - Acidic Liquid - Honey Bee 60 (McGean-Rohco)	
B01021	Cleaner - Drain (Enzyme) - Dispoz-All	
B01022	Cleaner - enzyme septic	
B50091	Cleaner - Gly-Vak Toilet Line Scale Remover	AMS 1550B
B50110	Cleaner - Acidic Liquid - Cedarinse (Seiwa Systems Co., Ltd.)	
B50329	Cleaner - Acidic Liquid - Honey Bee 60 T	
B50330	Cleaner - Acidic Liquid - Honey Bee 60 TNP	
G00022	Compound - Chlorine Dioxide For Water Purification - Purogene or Oxine	

E. Location Zones

Zone	Area
100	Lower Half of Fuselage
200	Upper Half of Fuselage

F. Prepare for the Off-Aircraft Soak Procedure for Cleaning a Vacuum Waste Tube

SUBTASK 38-32-00-670-001

(1) Disinfect the waste lines. Flush the waste line with 1 gal (4 l) to 3 gal (11 l) of the disinfectant solution.

NOTE: The disinfectant is a solution of chlorine dioxide, citric acid, and water.

(a) Prepare the disinfectant solution as follows:

WARNING DO NOT BREATHE CHLORINE DIOXIDE GAS. PUT ON A RESPIRATOR, AND GOGGLES. WHEN THE TWO CHEMICALS MIX, THEY MAKE CHLORINE DIOXIDE GAS. CHLORINE DIOXIDE GAS CAN CAUSE INJURIES TO PERSONNEL IF THEY BREATHE IT.

1) For each 1 gal (4 l) of disinfectant solution desired, mix 20 fl-oz (0.6 l) of chlorine dioxide Purogene or Oxine compound, G00022 with 2 oz (57 g) of citric acid, B00637 in a plastic container, STD-197.

2) Stop for 5 minutes (activation period).

3) Use a clean instrument to mix the solution fully.

4) Add water to make the desired quantity of solution.

(b) Disinfect the waste line(s) you are cleaning.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Pour and flush 1 gal (4 l) to 3 gal (11 l) of the disinfectant solution through the toilet bowl, at the vacuum waste line you are cleaning.

NOTE: The quantity of solution necessary to disinfect the vacuum toilet line changes with the length and condition of the line. Operator experience shows that 1 gal (4 l) is sufficient for short vacuum lines, but more than 1 gal (4 l) is necessary for longer vacuum lines. The quantity of disinfectant necessary for a specific vacuum toilet line is established by the operator's experience.

SUBTASK 38-32-00-020-009

(2) To remove a specific waste tube for cleaning, do this task: Vacuum Waste Tubes Removal, TASK 38-32-00-020-801.

G. Off-Aircraft Soak Procedure for Cleaning a Vacuum Waste Tube

SUBTASK 38-32-00-110-013

(1) Prepare and place in a suitable hydraulic fluid resistant container, the chemical solution to soak the waste tube. Use one of the chemical cleaners that follow:

NOTE: The quantity of the cleaner necessary for the inside of a waste tube is 0.16 gallons for each linear foot (2.0 liters for each linear meter). Calculate the maximum quantity of the cleaner necessary for each tube, or to submerse each tube.

The acid cleaners are primarily effective on hard salt and alkaline deposits. The bacterial and enzymatic cleaners are primarily effective on organic deposits (soap scums, grease, fat, starch and organic debris).

(a) A 15% glycolic acid, Gly-Vak toilet line scale remover, B50091.

(b) An 8% to 12% sulfamic acid, Honey Bee 60 cleaner, B00638, Honey Bee 60 T cleaner, B50329, or Honey Bee 60 TNP cleaner, B50330.

(c) A 7% phosphoric acid, B00046, solution.

(d) A 5% to 10% acetic acid, B00636, solution.

(e) A bacterial cleaner, such as Dispoz-All cleaner, B01021.

(f) An enzymatic cleaner, such as enzyme septic cleaner, B01022.

A bio-organic acid, Cedarinse, B50110.

SUBTASK 38-32-00-110-014

(2) Place the waste tube to be cleaned into the container so it is completely submerged in the cleaning solution.

SUBTASK 38-32-00-110-015

(3) Let the waste tube soak in the cleaner solution for a minimum of 8 hours.

NOTE: To fully clean the waste tube, more than 24 hours can be necessary to remove the heavy scale.

SUBTASK 38-32-00-140-001

(4) Make sure all solid material is out of the waste tube before re-installation.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-420-008

(1) To install the waste tube after cleaning, do this task: Vacuum Waste Tubes Installation, TASK 38-32-00-420-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-680-002

(2) Obey the local authorities to discard the chemical solution.

——— END OF TASK ———

TASK 38-32-00-140-801

9. Overboard Vent Tube Cleaning - Cleaning/Painting

A. General

(1) This procedure uses a brush, operated manually or mechanically turned with a variable speed drill. The brush must have these specifications.

(a) 3” round, meduim stiff, nylon bristle head.

(b) Flexible 3' shaft, capable of 90 degree turns.

(2) You can do this task in parts. Do the part of the task for the area of the overboard vent system that is necessary to clean.

B. References

Reference	Title
38-32-00-020-801	Vacuum Waste Tubes Removal (P/B 401)
38-32-00-420-802	Vacuum Waste Tubes Installation (P/B 401)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-32-06-000-801	Vacuum Check Valve Removal (P/B 401)
38-32-06-400-801	Vacuum Check Valve Installation (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-32-00-010-029

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-00-010-030

(2) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-00-860-050

WARNING BE CAREFUL WHEN YOU OPEN OR CLOSE CIRCUIT BREAKERS IN THE P91 AND P92 PANELS WHILE THE PANELS HAVE POWER. ELECTRICAL SHOCK CAN CAUSE INJURIES TO PERSONNEL.

(3) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-910-022

(4) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

F. Overboard Vent Tube Cleaning

SUBTASK 38-32-00-140-002

(1) Clean the vent tube from the port at the skin.

(a) Put the brush into the vent port opening.

(b) Turn the brush while you push it into the vent tube.

(c) Push the brush until it stops at the vent tube orifice approximately 3 feet down the tube.

(d) Remove the brush from the tube.

(e) Remove contamination with the vacuum if necessary.

(f) If the tube is not clean, do steps (a) through (e) again.

(g) If it is necessary to clean more, do these tasks.

NOTE: These tasks are general. To get access it can be necessary to remove more equipment.

1) Vacuum Waste Tubes Removal, TASK 38-32-00-020-801

2) Vacuum Waste Tube Off Airplane Soak, TASK 38-32-00-100-804

SUBTASK 38-32-00-140-003

(2) Clean the vent tubes from the vacuum check valve location.

(a) Remove the vacuum check valve (Vacuum Check Valve Removal, TASK 38-32-06-000-801).

(b) Push the brush down (outboard) into the tube opening where the check valve was installed.

(c) Turn the brush while you push it into the vent tube.

(d) Push the brush until it stops at the vent tube orifice approximately 1 foot down the tube.

(e) Remove the brush from the tube.

(f) Remove contamination with the vacuum if necessary.

(g) If the tube is not clean, do steps (b) through (f) again.

(h) If it is necessary to clean more, do these tasks.

NOTE: These tasks are general. To get access it can be necessary to remove more equipment.

1) Vacuum Waste Tubes Removal, TASK 38-32-00-020-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Vacuum Waste Tube Off Airplane Soak, TASK 38-32-00-100-804

CAUTION

DO NOT PUSH THE BRUSH WITH TOO MUCH FORCE OR TOO QUICKLY. YOU CAN CAUSE DAMAGE TO THE FLEXIBLE COUPLING.

(i) Push the brush up (inboard) into the tube opening where the check valve was installed.

(j) Turn the brush while you push it into the vent tube.

(k) Push the brush until it stops at the liquid separator approximately 3 feet down the tube.

(l) Remove the brush from the tube.

(m) Remove contamination with the vacuum if necessary.

(n) If the tube is not clean, do steps (i) through (m) again.

(o) If it is necessary to clean more, do these tasks.

NOTE: These tasks are general. To get access it can be necessary to remove more equipment.

1) Vacuum Waste Tubes Removal, TASK 38-32-00-020-801

2) Vacuum Waste Tube Off Airplane Soak, TASK 38-32-00-100-804

SUBTASK 38-32-00-140-004

(3) Clean the vent tubes from the vacuum blower location.

(a) Loosen the clamps and remove the flexible couplings from the intake and outflow of the vacuum blower.

CAUTION

DO NOT PUSH THE BRUSH WITH TOO MUCH FORCE OR TOO QUICKLY. YOU CAN CAUSE DAMAGE TO THE FLEXIBLE COUPLING.

(b) Push the brush down (outboard) into the tube opening where the flexible coupling was installed.

(c) Turn the brush while you push it into the vent tube.

(d) Push the brush until it stops at the vent tube Y approximately 3 feet down the tube.

(e) Remove the brush from the tube.

(f) Remove contamination with the vacuum if necessary.

(g) If the tube is not clean, do steps (a) though (f) again.

(h) If it is necessary to clean more, do these tasks.

NOTE: These tasks are general. To get access it can be necessary to remove more equipment.

1) Vacuum Waste Tubes Removal, TASK 38-32-00-020-801

2) Vacuum Waste Tube Off Airplane Soak, TASK 38-32-00-100-804

CAUTION

DO NOT PUSH THE BRUSH WITH TOO MUCH FORCE OR TOO QUICKLY. YOU CAN CAUSE DAMAGE TO THE FLEXIBLE COUPLING.

(i) Push the brush up (inboard) into the tube opening where the flexible coupling was installed.

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(j) Turn the brush while you push it into the vent tube.

(k) Push the brush until it stops at the vent tube Y approximately 2 feet down the tube.

(l) Remove the brush from the tube.

(m) Remove contamination with the vacuum if necessary.

(n) If the tube is not clean, do steps (i) through (m) again.

(o) If it is necessary to clean more, do these tasks.

NOTE: These tasks are general. To get access it can be necessary to remove more equipment.

1) Vacuum Waste Tubes Removal, TASK 38-32-00-020-801

2) Vacuum Waste Tube Off Airplane Soak, TASK 38-32-00-100-804

(4) If a vent tube was removed, do this task (Vacuum Waste Tubes Installation, TASK 38-32-00-420-802).

(5) Install the flexible couplings to the intake and outflow of the vacuum blower.

(6) Install the clamps to the flexible couplings.

(7) Do this task: (Vacuum Check Valve Installation, TASK 38-32-06-400-801)

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-00-860-051

WARNING

BE CAREFUL WHEN YOU OPEN OR CLOSE CIRCUIT BREAKERS IN THE P91 AND P92 PANELS WHILE THE PANELS HAVE POWER. ELECTRICAL SHOCK CAN CAUSE INJURIES TO PERSONNEL.

(1) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-00-410-018

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GET INTO THE PASSENGER COMPARTMENT DURING A FIRE.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-00-410-019

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TOILET ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the toilet shroud

(2) An installation of the toilet shroud

(3) A removal of the vacuum toilet assembly

(4) An installation of the vacuum toilet assembly

(5) A cleaning of the rinse water supply

(6) A removal of the rinse valve

(7) An installation of the rinse valve

(8) A removal of the flush valve

(9) An installation of the flush valve

(10) A removal of the Integrated System Controller (ISC)

(11) An installation of the Integrated System Controller (ISC).

TASK 38-32-01-000-801

2. Toilet Shroud Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the toilet shroud.

B. References

Reference	Title
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14266	Screwdriver - Driver Assembly Part #: 29-90-114-10 Supplier: 94222 Part #: 29-90-214-10 Supplier: 94222

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Prepare for the Removal

SUBTASK 38-32-01-010-001

(1) Get access to the lavatory for the toilet or toilet component removal.

SUBTASK 38-32-01-910-001

(2) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Toilet Shroud Removal

SUBTASK 38-32-01-020-001

(1) Do these steps to remove the toilet shroud [21]:

(a) Remove the center shroud panel [23] above the toilet.

1) For lavatory A:

a) Pull on the center shroud panel [23] to disengage it from the magnetic mounts.

b) Remove the center shroud panel [23] from the lavatory.

2) For all lavatories except A:

a) Disengage the two quarter-turn fasteners [26].

b) Pull the center shroud panel [23] up and out to disengage it from the bracket mounts.

c) Disconnect the center shroud panel [23] tether from the bracket mount and remove the panel from the lavatory.

(b) Disengage the tamper proof quarter-turn fasteners [22] at the top of the toilet shroud [21].

1) Use the driver assembly screwdriver, SPL-14266, to turn and release each quarter-turn fastener [22].

2) Make sure that all quarter-turn fasteners [22] are disengaged.

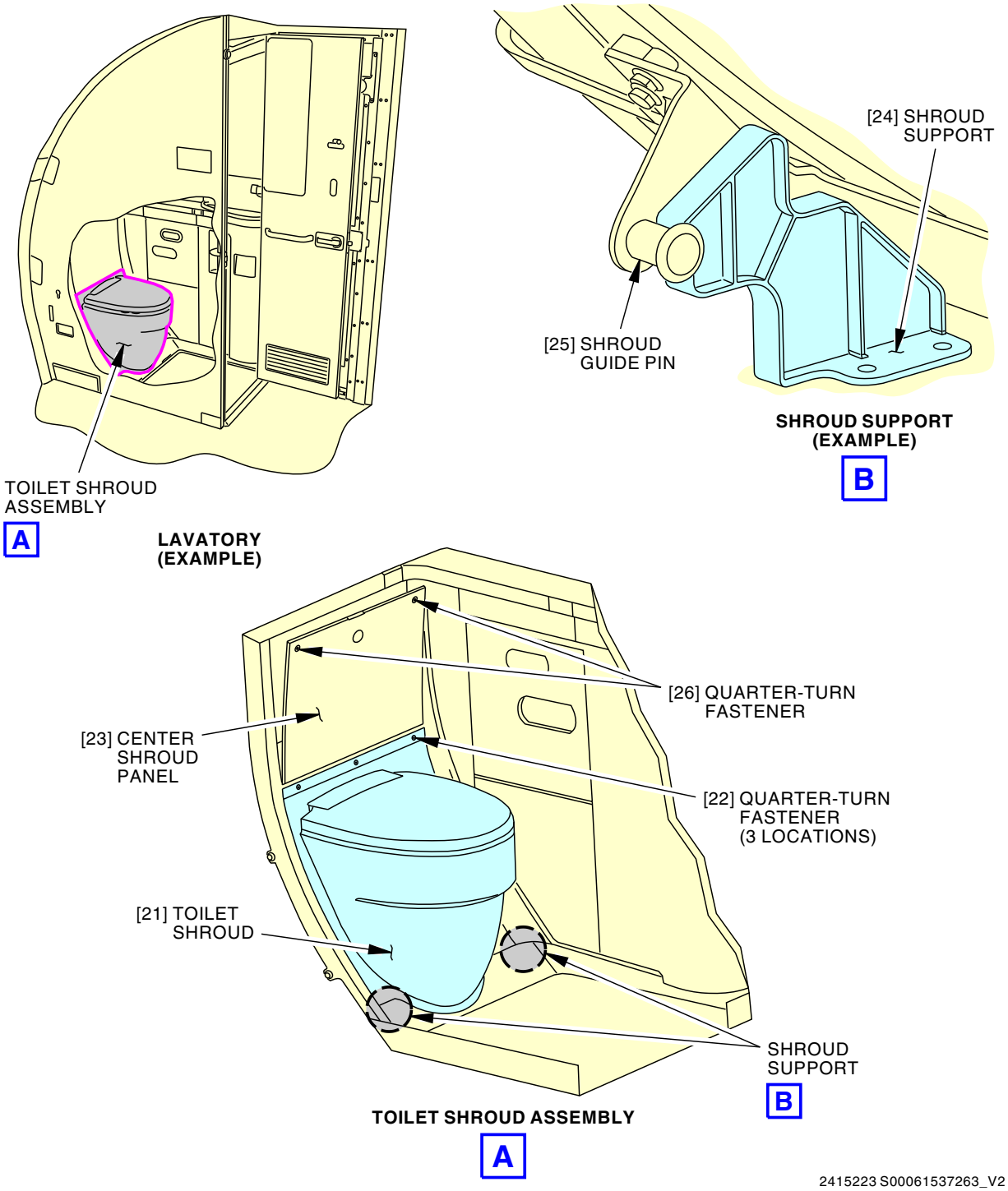
NOTE: The fastener head will move away from the shroud when disengaged.

(c) Remove the toilet shroud [21] from the lavatory.

1) Lift the toilet shroud [21] up and towards you to disengage the shroud guide pins [25] from the shroud supports [24] at the floor.

——— END OF TASK ———

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Toilet Shroud Installation
Figure 401/38-32-01-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-01-400-801

3. Toilet Shroud Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the toilet shroud.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14266	Screwdriver - Driver Assembly Part #: 29-90-114-10 Supplier: 94222 Part #: 29-90-214-10 Supplier: 94222

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Toilet shroud	25-40-11-24F-430	PTF ALL
		25-40-14-13X-420	PTF ALL
		25-40-15-11M-395	PTF ALL

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Toilet Shroud Installation

SUBTASK 38-32-01-410-001

(1) Do these steps to install the toilet shroud [21]:

(a) Put the bottom of the toilet shroud [21] into position with the shroud guide pins [25] against the shroud supports [24].

(b) Push the toilet shroud [21] into position against the lavatory surround structure.

(c) Engage the three tamper proof quarter-turn fasteners [22] at the top of the toilet shroud [21].

1) Use the driver assembly screwdriver, SPL-14266, to turn and tighten each quarter-turn fastener [22].

2) Make sure that the three quarter-turn fasteners [22] are engaged.

(d) Install the center shroud panel [23] above the toilet.

1) For lavatory A:

a) Put the center shroud panel [23] in its position to engage the magnetic mounts.

2) For all lavatories except A:

a) Move the center shroud panel [23] down on to its bracket mounts and attach the tether.

b) Push the center shroud panel [23] into position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Use the driver assembly screwdriver, SPL-14266, to turn and tighten each quarter-turn fastener [26].

——— END OF TASK ———

TASK 38-32-01-000-802

4. Vacuum Toilet Assembly Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the vacuum toilet assembly.

B. References

Reference	Title
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1946	Plug - Vacuum Waste System Part #: A38007-9 Supplier: 81205
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Prepare for the Removal

SUBTASK 38-32-01-010-002

(1) Get access to the lavatory for the toilet removal.

SUBTASK 38-32-01-010-016

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-01-040-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-01-860-001

(4) Set the water shutoff valve for the lavatory to the FAUCET ONLY or OFF position .

SUBTASK 38-32-01-910-002

(5) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-01-010-003

(6) To remove the toilet shroud, do this task: Toilet Shroud Removal, TASK 38-32-01-000-801.

H. Vacuum Toilet Assembly Removal

SUBTASK 38-32-01-020-002

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [2] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [2] from the power supply.

(b) Install the protective cover on the electrical connector [2].

SUBTASK 38-32-01-020-003

(2) Disconnect the nut [15], washer [20], washers [14], and bonding jumper [13] from the toilet ground stud [16].

SUBTASK 38-32-01-020-004

(3) Disconnect the coupling [4], O-ring [5], and rinse line [3] from the rinse valve assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-01-680-001

(4) Drain the rinse line [3] into a container.

SUBTASK 38-32-01-020-005

(5) Remove the coupling [8] that attaches the vacuum waste line [10] to the discharge drain [6].

SUBTASK 38-32-01-020-006

(6) Remove the sleeve [9] and packing [7] and disconnect the vacuum waste line [10].

SUBTASK 38-32-01-020-007

(7) Remove the screws [17] and disconnect the discharge drain [6] from the toilet assembly [1].

SUBTASK 38-32-01-020-008

(8) Remove the packing [18] from the discharge drain [6] and the toilet assembly [1].

SUBTASK 38-32-01-010-004

(9) Remove the hitch pins [19] and quick release pins [11] from the base supports [12].

SUBTASK 38-32-01-000-001

(10) Remove the vacuum toilet assembly [1].

SUBTASK 38-32-01-480-001

(11) If the vacuum waste system will be operated while the toilet is removed, then do these steps to install a plug in the vacuum waste line:

(a) Apply grease, D00463, or Molykote 111 Compound, D00189, to the packing [7] on the waste line vacuum waste plug, SPL-1946.

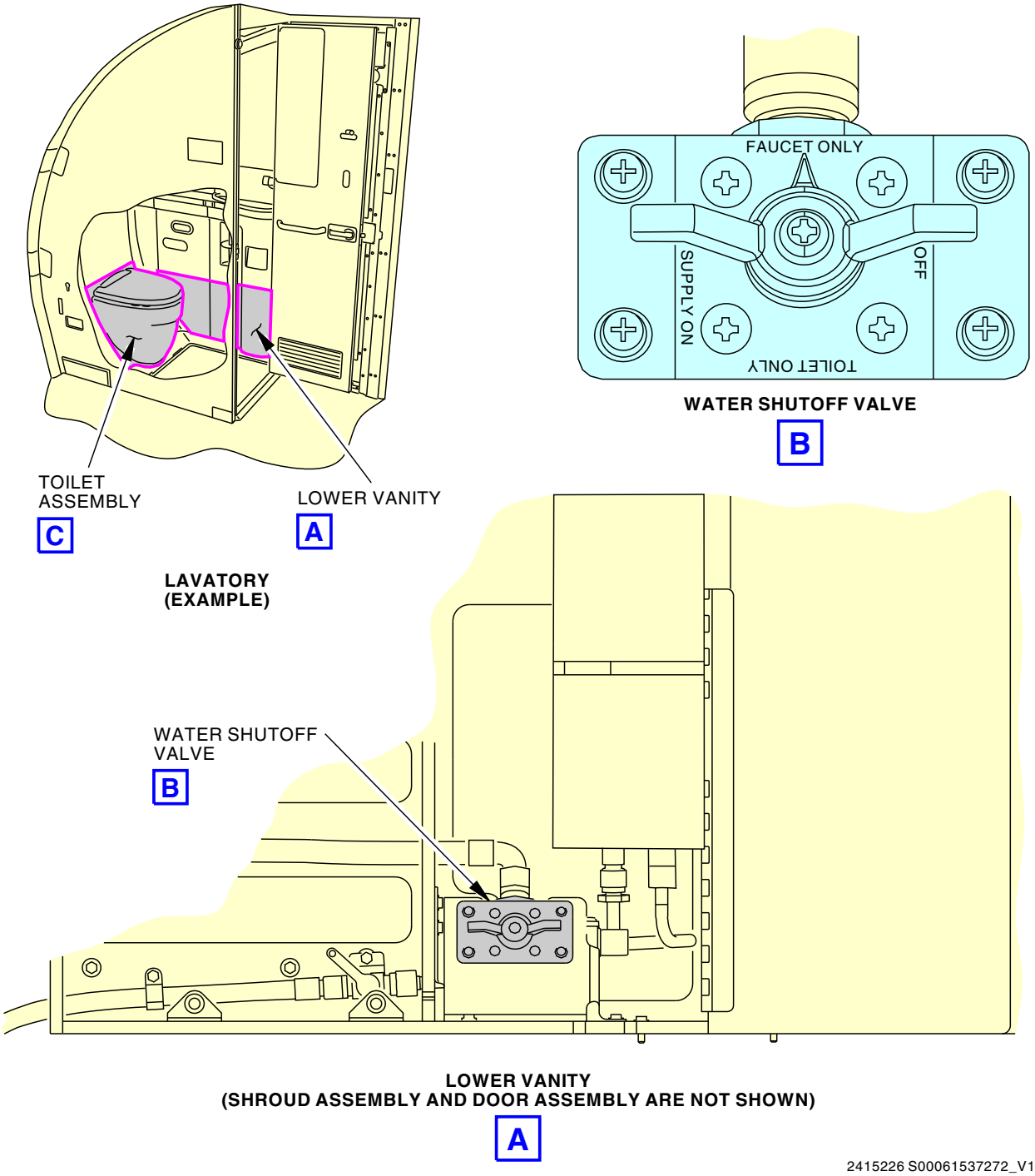
(b) Apply grease, D00463, or Molykote 111 Compound, D00189, to the packing [7] on the vacuum waste line [10].

(c) Put the waste line vacuum waste plug, SPL-1946, in the vacuum waste line [10].

(d) Attach the vacuum waste plug, SPL-1946, to the vacuum waste line [10] with the sleeve [9] and coupling [8].

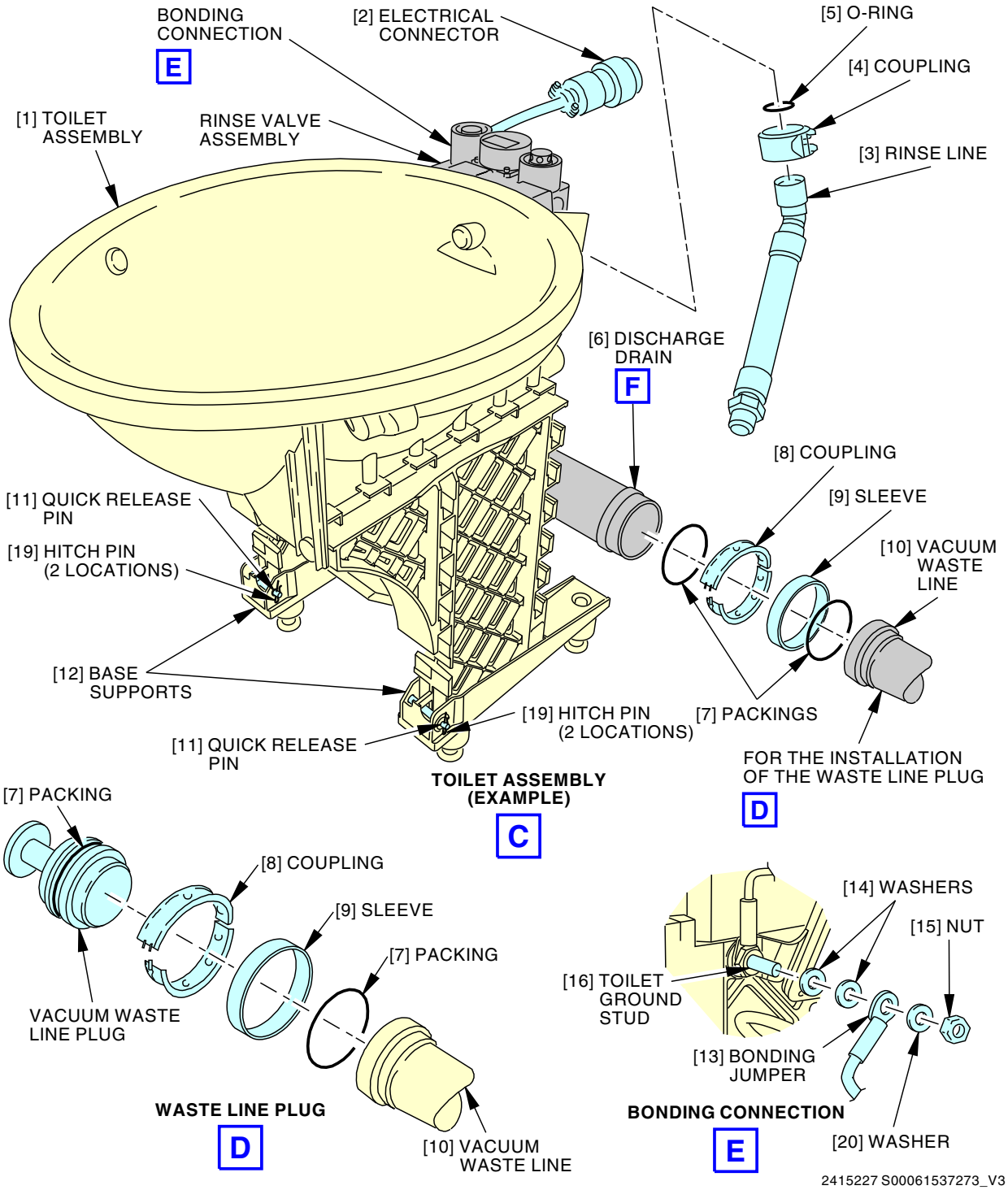
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



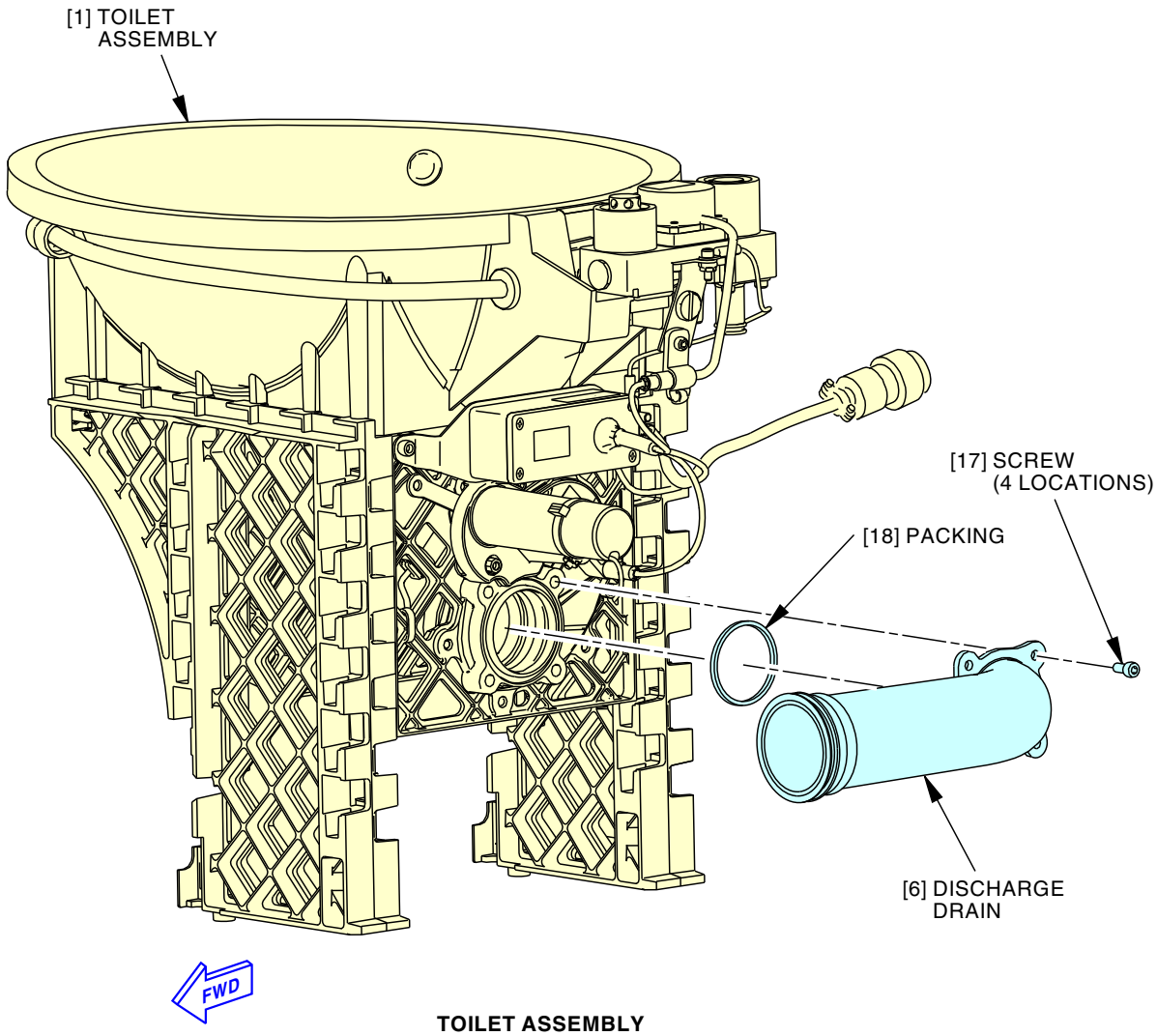
Toilet Assembly Installation
Figure 402/38-32-01-990-807 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Toilet Assembly Installation
Figure 402/38-32-01-990-807 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TOILET ASSEMBLY (EXAMPLE)

F

2809142 S0000645815_V1

Toilet Assembly Installation
Figure 402/38-32-01-990-807 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-01-400-802

5. Vacuum Toilet Assembly Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the vacuum toilet assembly.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1946	Plug - Vacuum Waste System Part #: A38007-9 Supplier: 81205
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Toilet assembly	25-41-21-01D-145	PTF ALL
		25-41-21-05Q-080	PTF ALL
5	O-ring	25-41-21-01D-175	PTF ALL
		25-41-21-05Q-075	PTF ALL
6	Discharge drain	25-41-21-01D-100	PTF ALL
		25-41-21-05Q-100	PTF ALL
7	Packing	38-31-51-70-035	PTF ALL
18	Packing	25-41-21-01D-110	PTF ALL
		25-41-21-05Q-110	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Vacuum Toilet Assembly Installation

SUBTASK 38-32-01-080-001

(1) If a plug is installed in the vacuum waste line, then do these steps to remove the waste line plug:

(a) Remove the coupling [8] and sleeve [9] that attaches the vacuum waste plug, SPL-1946, to the vacuum waste line [10].

(b) Pull the plug from the vacuum waste line [10].

SUBTASK 38-32-01-640-001

(2) Put grease, D00463, or Molykote 111 Compound, D00189, on the new packings [7].

SUBTASK 38-32-01-410-002

(3) Put the new packings [7] on the vacuum waste line [10] and the discharge drain [6].

SUBTASK 38-32-01-410-003

(4) Position the toilet assembly [1] in the base supports [12].

SUBTASK 38-32-01-410-004

(5) Put the quick release pins [11] into the base supports [12].

(a) Secure the quick release pins [11] with the hitch pins [19].

SUBTASK 38-32-01-410-007

(6) Put the new packing [18] in between the discharge drain [6] and the toilet assembly [1].

SUBTASK 38-32-01-410-008

(7) Connect the discharge drain [6] to the toilet assembly [1] and tighten the screws [17].

SUBTASK 38-32-01-410-005

(8) Install the sleeve [9] and the coupling [8] to connect the vacuum waste line [10] to the discharge drain [6].

(a) Turn the coupling [8] before you latch it, to put it in a position that will give the easiest access to the latch.

SUBTASK 38-32-01-670-001

(9) Do this task: Rinse Water Supply Cleaning, TASK 38-32-01-100-801.

SUBTASK 38-32-01-640-002

(10) Put grease, D00463, or Molykote 111 Compound, D00189, on the new O-ring [5].

SUBTASK 38-32-01-420-001

(11) Connect the rinse line [3] to the rinse valve assembly with the O-ring [5] and coupling [4].

SUBTASK 38-32-01-420-002

(12) Connect the nut [15], washer [20], washers [14], and bonding jumper [13] to the toilet ground stud [16].

SUBTASK 38-32-01-420-003

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(13) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [2] (TASK 70-00-01-910-803-G00).

(a) Remove the protective cover from the electrical connector [2].

(b) Connect the electrical connector [2] to the power supply.

1) Tighten the connector with the teflon-jawed pliers, STD-664.

I. Vacuum Toilet Assembly Installation Test

SUBTASK 38-32-01-040-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-01-410-016

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-01-610-001

(3) If it is empty, fill the potable water tank (TASK 12-14-00-600-802).

SUBTASK 38-32-01-860-003

(4) Set the water shutoff valve for the lavatory to the SUPPLY ON position (Figure 402).

SUBTASK 38-32-01-710-001

(5) Flush the toilet a minimum of two times to make sure that it operates correctly.

(a) If the toilet does not flush correctly, do the applicable task in Interactive Fault Isolation Manual (IFIM) sections 38-30 and 38-32 to correct the problem.

SUBTASK 38-32-01-790-006

(6) Make sure that the flush valve assembly and its connections do not leak.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If there are leaks, check for damaged parts or loose connections.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-01-410-006

(1) Do this task: Toilet Shroud Installation, TASK 38-32-01-400-801.

——— END OF TASK ———

TASK 38-32-01-100-801

6. Rinse Water Supply Cleaning

(Figure 402)

A. General

(1) Do this task to make sure the potable water system is not contaminated when the vacuum toilet assembly is connected.

B. Consumable Materials

Reference	Description	Specification
B00637	Acid, Citric	A-A-59147
G00022	Compound - Chlorine Dioxide For Water Purification - Purogene or Oxine	

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Rinse Water Supply Cleaning

SUBTASK 38-32-01-670-002

(1) Make a disinfectant as follows:

(a) Mix 0.3 fl-oz (0.01 l) of chlorine dioxide Purogene or Oxine compound, G00022, and 13.4 fl-oz (0.4 l) of citric acid, B00637, per 1 gal (4 l) of disinfectant.

(b) Stop for five minutes.

(c) Add 1 gal (4 l) of water to complete the disinfectant.

SUBTASK 38-32-01-670-003

(2) Clean the connections on the rinse line [3] and rinse valve assembly with disinfectant.

SUBTASK 38-32-01-670-004

(3) Put approximately one cup (8 fl-oz (0.24 l)) of the disinfectant in the rinse line [3].

SUBTASK 38-32-01-420-004

(4) Connect the rinse line [3] to the rinse valve assembly.

SUBTASK 38-32-01-170-001

(5) Flush the toilet a minimum of two times to make sure that the disinfectant is out of the supply line.

SUBTASK 38-32-01-790-007

(6) Make sure that the rinse valve assembly and its connections do not leak.

(a) If there are leaks, check for damaged parts or loose connections.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-01-000-804

7. Rinse Valve Removal

(Figure 403)

A. References

Reference	Title
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 38-32-01-010-008

(1) Get access to the lavatory for the toilet removal.

SUBTASK 38-32-01-010-015

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-01-040-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-01-860-006

(4) Set the water shutoff valve for the lavatory to the FAUCET ONLY or OFF position (Figure 402).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-01-910-004

(5) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-01-010-009

(6) To remove the toilet shroud, do this task: Toilet Shroud Removal, TASK 38-32-01-000-801.

SUBTASK 38-32-01-010-010

(7) If it is necessary, do this task: Vacuum Toilet Assembly Removal, TASK 38-32-01-000-802.

E. Rinse Valve Removal

SUBTASK 38-32-01-020-011

(1) Remove the rinse valve assembly [101], do these steps:

(a) Disconnect the coupling [4], O-ring [5], and rinse line [3] from the rinse valve assembly (Figure 402).

1) Discard the O-ring [5].

(b) Remove the ground cable [103] from the rinse valve assembly [101].

1) Remove the screw [102] and nut [105].

(c) Remove the electrical connector [104].

(d) Remove the rinse valve assembly [101] from the toilet bowl [109].

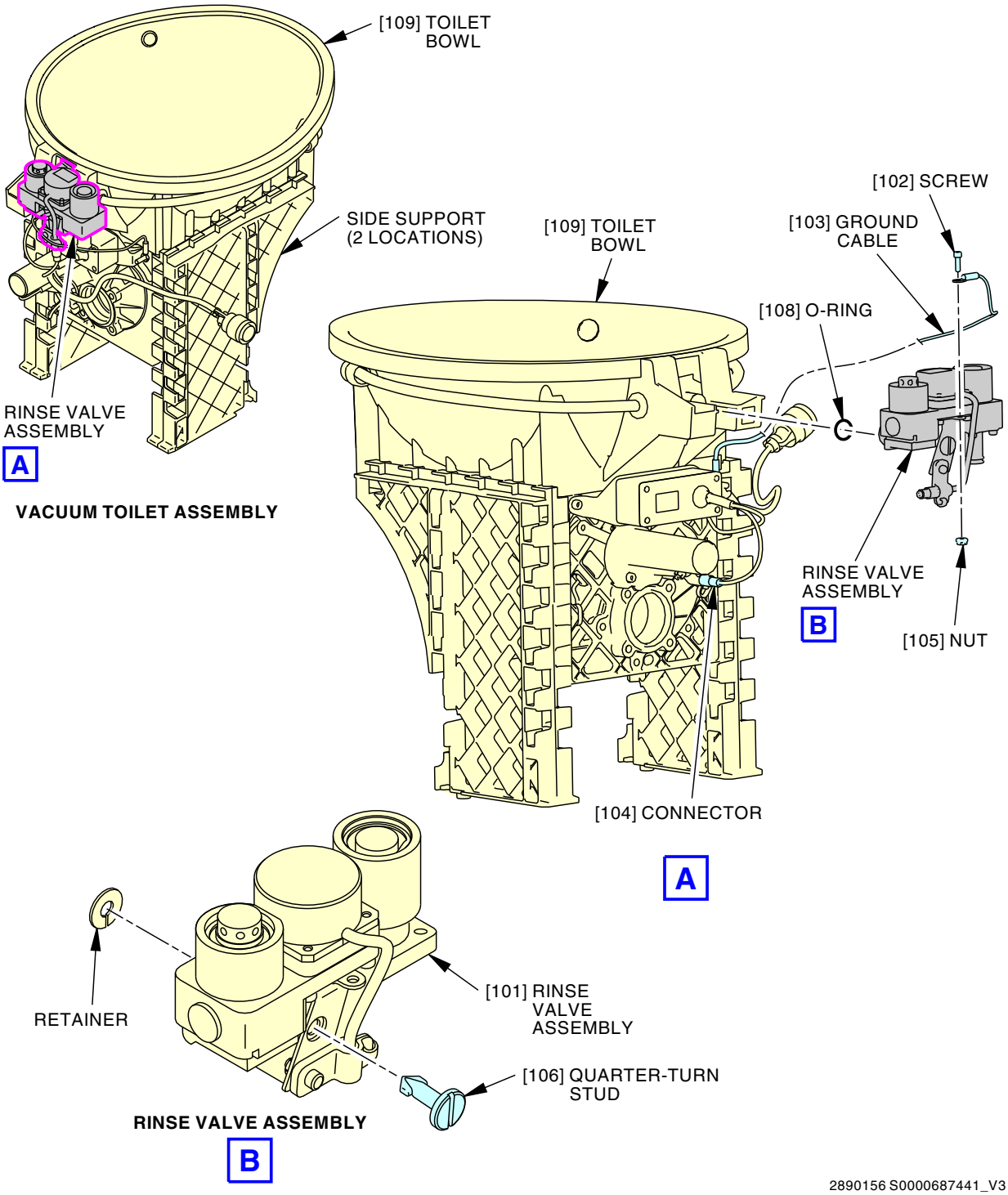
1) Rotate the quarter-turn stud [106] 1/4 turn counter-clockwise.

2) Slide the rinse valve assembly [101] off the rear of the toilet bowl [109].

3) Discard the O-ring [108].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Toilet Rinse Valve Installation
Figure 403/38-32-01-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-01-400-804

8. Rinse Valve Installation

(Figure 403)

A. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	O-ring	25-41-21-01D-175	PTF ALL
		25-41-21-05Q-075	PTF ALL
101	Rinse valve assembly	38-32-01-50-006	PTF ALL
108	O-ring	38-32-01-50-045	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Rinse Valve Installation

SUBTASK 38-32-01-420-007

(1) Position a new O-ring [108] on the toilet bowl [109].

(a) Lubricate the O-ring [108] with Molykote 111 Compound, D00189.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-01-420-008

(2) Slide the rinse valve assembly [101] into position on the rear of the toilet bowl [109].

SUBTASK 38-32-01-420-009

(3) Engage the quarter-turn stud [106] into the receptacle on the toilet bowl [109].

(a) Rotate the quarter-turn stud [106] 1/4 turn clockwise to lock the rinse valve assembly [101] into place.

SUBTASK 38-32-01-420-010

(4) Attach the electrical connector [104] to the rinse valve assembly [101].

SUBTASK 38-32-01-110-001

(5) Attach the ground cable [103] to the rinse valve assembly [101].

(a) Insert the screw [102] and nut [105].

(b) Tighten the nut [105] to 18 in-lb (2.0 N·m) - 20 in-lb (2.3 N·m).

SUBTASK 38-32-01-700-001

(6) Measure the electrical bonding resistance between the fasteners on both ends of the ground cable [103].

(a) Use a non-intrinsically safe bonding meter, COM-614.

(b) Make sure that the bonding resistance is 0.015 ohm (15 milliohms) or less.

(c) If the bonding resistance is more than 0.015 ohm (15 milliohms), do these steps:

1) Remove the ground cable [103].

2) Clean all connection points.

3) Install the ground cable [103].

4) Measure the electrical bonding resistance again.

SUBTASK 38-32-01-420-023

(7) Put grease, D00463, or Molykote 111 Compound, D00189, on the new O-ring [5].

SUBTASK 38-32-01-420-011

(8) Connect the rinse line [3] to the rinse valve assembly with the O-ring [5] and coupling [4] (Figure 402).

SUBTASK 38-32-01-420-012

(9) If it is necessary, connect the electrical connector [2] for the rinse valve assembly to the flush control assembly (Figure 402).

SUBTASK 38-32-01-410-012

(10) If removed, install the vacuum toilet assembly (TASK 38-32-01-400-802).

H. Rinse Valve Installation Test

SUBTASK 38-32-01-440-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-01-410-015

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-01-610-004

(3) If it is empty, fill the potable water tank (TASK 12-14-00-600-802).

SUBTASK 38-32-01-860-007

(4) Set the water shutoff valve for the lavatory to the SUPPLY ON position (Figure 402).

SUBTASK 38-32-01-710-003

(5) Flush the toilet a minimum of two times to make sure that it operates correctly.

(a) If the toilet does not flush correctly, do the applicable task in Interactive Fault Isolation Manual (IFIM) sections 38-30 and 38-32 to correct the problem.

SUBTASK 38-32-01-790-003

(6) Make sure that the rinse valve assembly and its connections do not leak.

(a) If there are leaks, check for damaged parts or loose connections.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-01-400-001

(1) Do this task: Toilet Shroud Installation, TASK 38-32-01-400-801.

——— END OF TASK ———

TASK 38-32-01-000-803

9. Flush Valve Removal

(Figure 404)

A. General

(1) This task gives the instructions to remove the flush valve.

B. References

Reference	Title
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Removal

SUBTASK 38-32-01-010-005

(1) Get access to the lavatory for the toilet removal.

SUBTASK 38-32-01-010-014

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-01-760-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-01-860-004

(4) Set the water shutoff valve for the lavatory to the FAUCET ONLY or OFF position (Figure 402).

SUBTASK 38-32-01-910-003

(5) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-01-010-006

(6) Do this task: Toilet Shroud Removal, TASK 38-32-01-000-801.

SUBTASK 38-32-01-010-007

(7) If it is necessary for better access, do this task: Vacuum Toilet Assembly Removal, TASK 38-32-01-000-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Flush Valve Removal

SUBTASK 38-32-01-020-009

(1) Remove the front support [147], do these steps:

(a) Lift and remove the front support [147] from the side supports [148].

(b) Attach identification tags on the side supports [148].

NOTE: Side supports are identical. Identification tags on slide supports will assure that receptacles at the base of each support are oriented correctly for toilet reinstallation into the aircraft.

SUBTASK 38-32-01-020-010

(2) Remove the flush valve assembly [146], do these steps:

NOTE: The flush valve and rear support structure are integrated as a single component.

(a) Slide the flush valve assembly [146] straight up to remove it from the side supports [148].

NOTE: Take a note of the vertical position of the flush valve on side supports to aid reassembly.

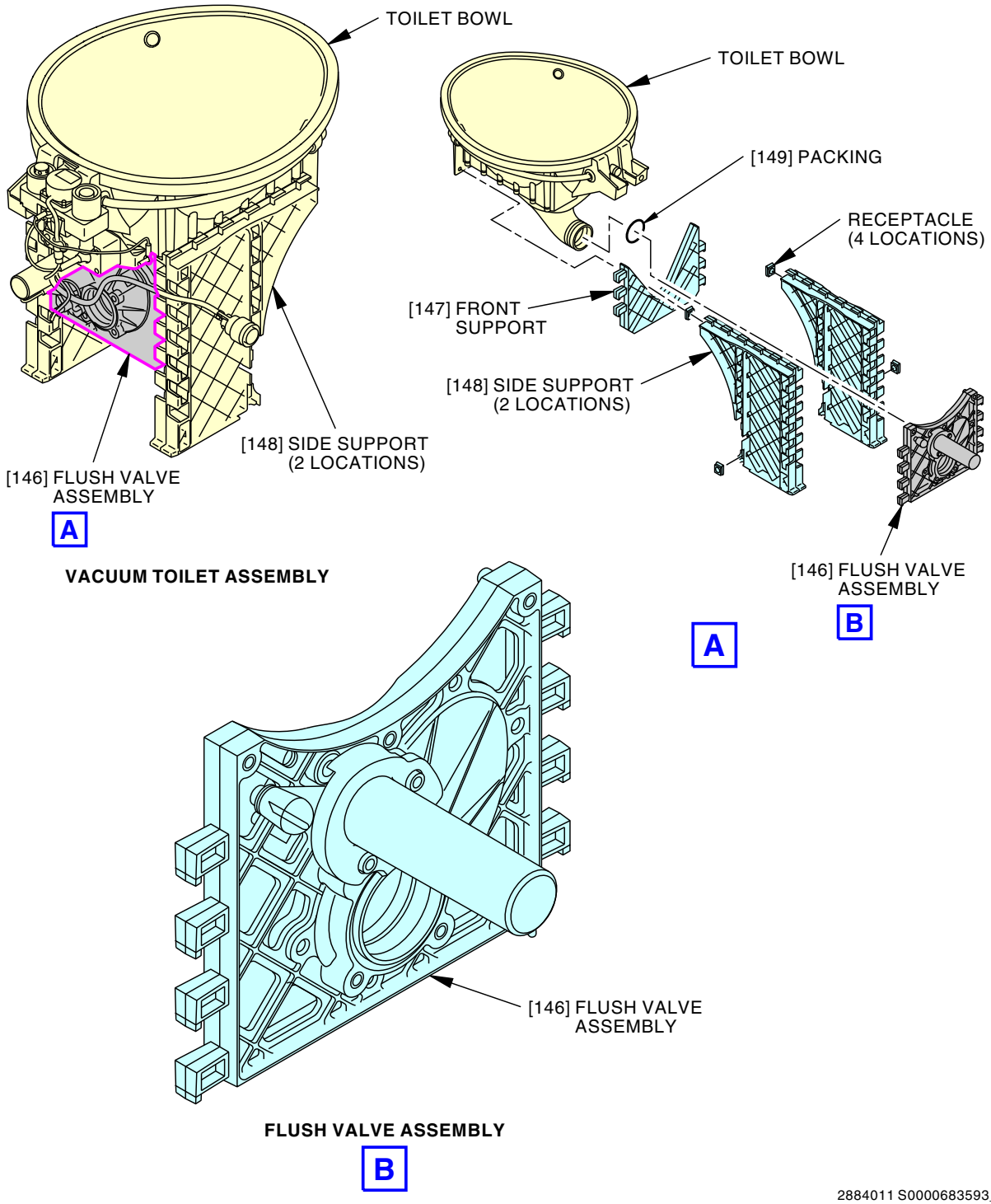
(b) Move rearward and lift the flush valve assembly [146] from the side supports [148].

SUBTASK 38-32-01-020-021

(3) Remove and discard the packing [149] from the toilet bowl outlet throat.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Toilet Flush Valve Installation
Figure 404/38-32-01-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-01-400-803

10. Flush Valve Installation

(Figure 404)

A. General

(1) This task gives instructions to install the flush valve.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
149	Packing	38-32-01-50-080	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

G. Flush Valve Installation

SUBTASK 38-32-01-640-003

(1) Apply Molykote 111 Compound, D00189, on the new packing [149].

SUBTASK 38-32-01-420-022

(2) Put new packing [149] to the toilet bowl outlet throat.

SUBTASK 38-32-01-420-005

CAUTION

MAKE SURE TO USE THE CORRECT VERTICAL POSITION OF THE FLUSH VALVE ASSEMBLY ON THE SUPPORT STRUCTURE. THIS CORRECTLY ALIGNS THE TOILET-BOWL OUTLET WITH THE FLUSH VALVE INLET.

(3) Face rear of the toilet assembly towards you, position the flush valve assembly [146] at rear of the side supports [148], such that the outlet is towards you.

SUBTASK 38-32-01-420-006

(4) Install the flush valve assembly [146], do these steps:

(a) Insert the tangs of the flush valve assembly [146] into the retainer slots of the side supports [148] such that top of the flush valve assembly [146] will be level with the side supports [148] when in the position.

(b) Slide the flush valve assembly [146] down to lock it into the position to the side supports [148].

(c) Position each side support [148] to the correct side of the front support [147] before assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Point the quarter-turn receptacles at the base of each side support [148] to engage the quarter-turn studs from the base supports during vacuum toilet installation (TASK 38-32-01-400-802).

(e) While holding the side supports [148] vertically, position the front support [147] into the highest slots and slide down into the position.

(f) Make sure that the front support [147] is fully seated in the retainer slots.

H. Flush Valve Installation Test

SUBTASK 38-32-01-410-011

(1) If removed, install the vacuum toilet assembly (TASK 38-32-01-400-802).

SUBTASK 38-32-01-760-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-01-410-014

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-01-610-002

(4) If it is empty, fill the potable water tank (TASK 12-14-00-600-802).

SUBTASK 38-32-01-860-005

(5) Set the water shutoff valve for the lavatory to the SUPPLY ON position (Figure 402).

SUBTASK 38-32-01-710-002

(6) Flush the toilet a minimum of two times to make sure that it operates correctly.

(a) If the toilet does not flush correctly, do the applicable task in Interactive Fault Isolation Manual (IFIM) sections 38-30 and 38-32 to correct the problem.

SUBTASK 38-32-01-790-004

(7) Make sure that the flush valve assembly and its connections do not leak.

(a) If there are leaks, check for damaged parts or loose connections.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. **Put the Airplane Back to Its Usual Condition**

SUBTASK 38-32-01-410-010

(1) Do this task: Toilet Shroud Installation, TASK 38-32-01-400-801.

——— END OF TASK ———

TASK 38-32-01-000-805

11. **Integrated System Controller (ISC) Removal**

(Figure 405)

A. **Location Zones**

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

B. **Access Panels**

Number	Name/Location
117A	Electronic Equipment Access Door

C. **Prepare for the Removal**

SUBTASK 38-32-01-010-011

(1) Get access to the lavatory for the toilet removal.

SUBTASK 38-32-01-010-013

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-01-040-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-01-010-012

(4) Set the water shutoff valve for the lavatory to the FAUCET ONLY or OFF position (Figure 402).

SUBTASK 38-32-01-020-012

(5) Remove the toilet shroud, do this task: Toilet Shroud Removal, TASK 38-32-01-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Integrated System Controller Removal

SUBTASK 38-32-01-020-013

(1) Cut and discard the cable ties [39].

SUBTASK 38-32-01-020-014

CAUTION DO NOT REMOVE THE POTENTIOMETER. REMOVAL OF THE POTENTIOMETER WILL CHANGE THE POSITIONAL CALIBRATION AND DISABLE THE TOILET. FOR THE REMOVAL OF THE INTEGRATED SYSTEM CONTROLLER SIMPLY DISCONNECT THE CONNECTOR.

(2) Disconnect the harness connector 1 from the flush valve assembly [146] (Figure 405).

SUBTASK 38-32-01-020-015

(3) Disconnect the harness connector 2 from the rinse valve assembly [101].

SUBTASK 38-32-01-020-016

(4) Disconnect the harness connector 3 from the potentiometer on the flush valve assembly [146].

SUBTASK 38-32-01-020-017

(5) Remove the nut [32], washer [33], and the grounding cable [34] from the attaching screw.

SUBTASK 38-32-01-020-018

(6) While facing the rear of the toilet, remove the left-most screw [35].

(a) Loosen the lower screw [35] enough so that the Integrated System Controller [31] with the controller bracket [37] can be removed as an assembly from the flush valve assembly [146].

(b) Leave the lower screw [35] in position.

SUBTASK 38-32-01-020-019

(7) Remove the screws [30] that attach the controller bracket [37] to the Integrated System Controller [31].

(a) Disconnect the Integrated System Controller [31] and controller bracket [37].

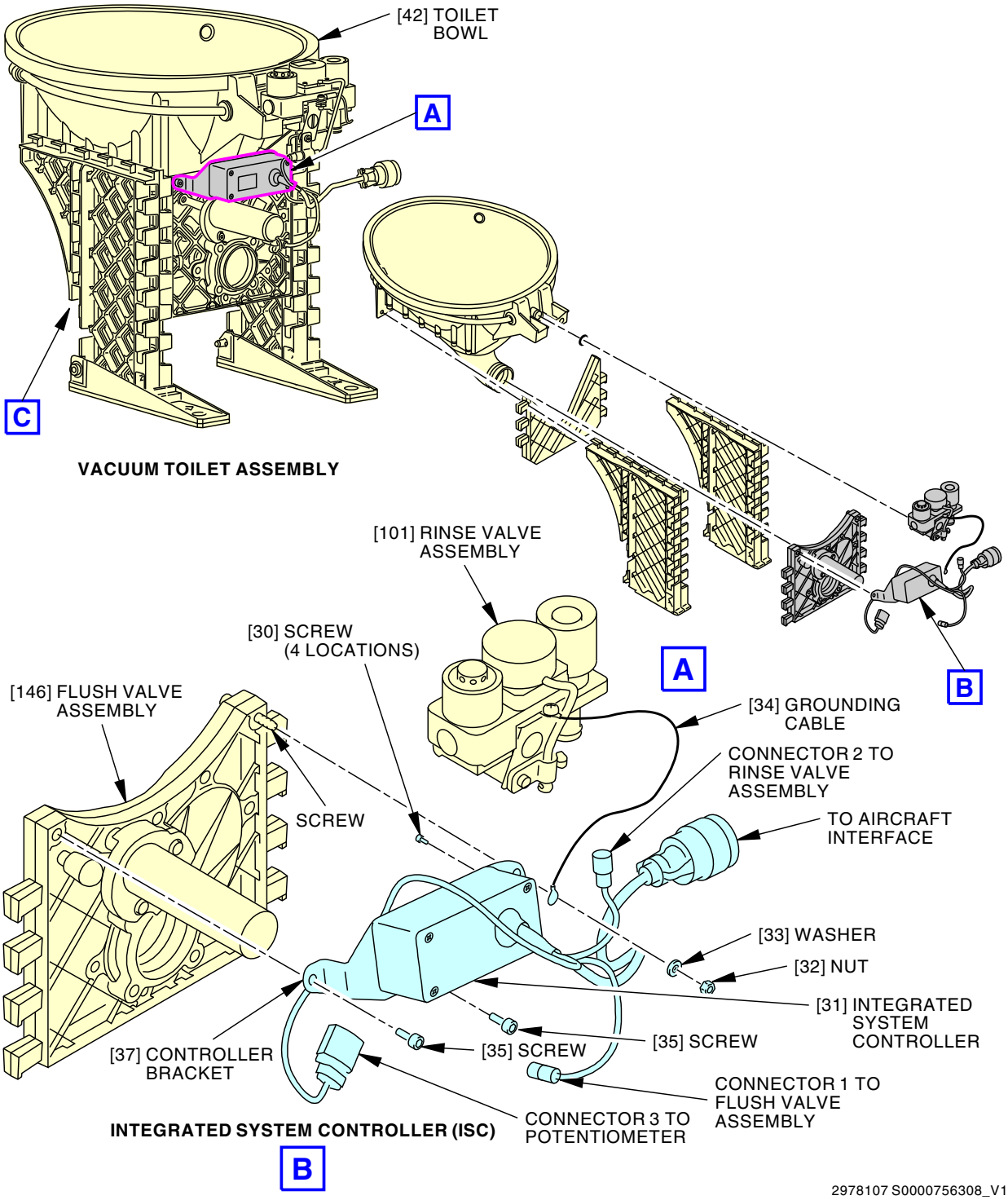
(b) Remove the screws as required to make the necessary repairs.

SUBTASK 38-32-01-020-020

(8) Remove the Integrated System Controller [31].

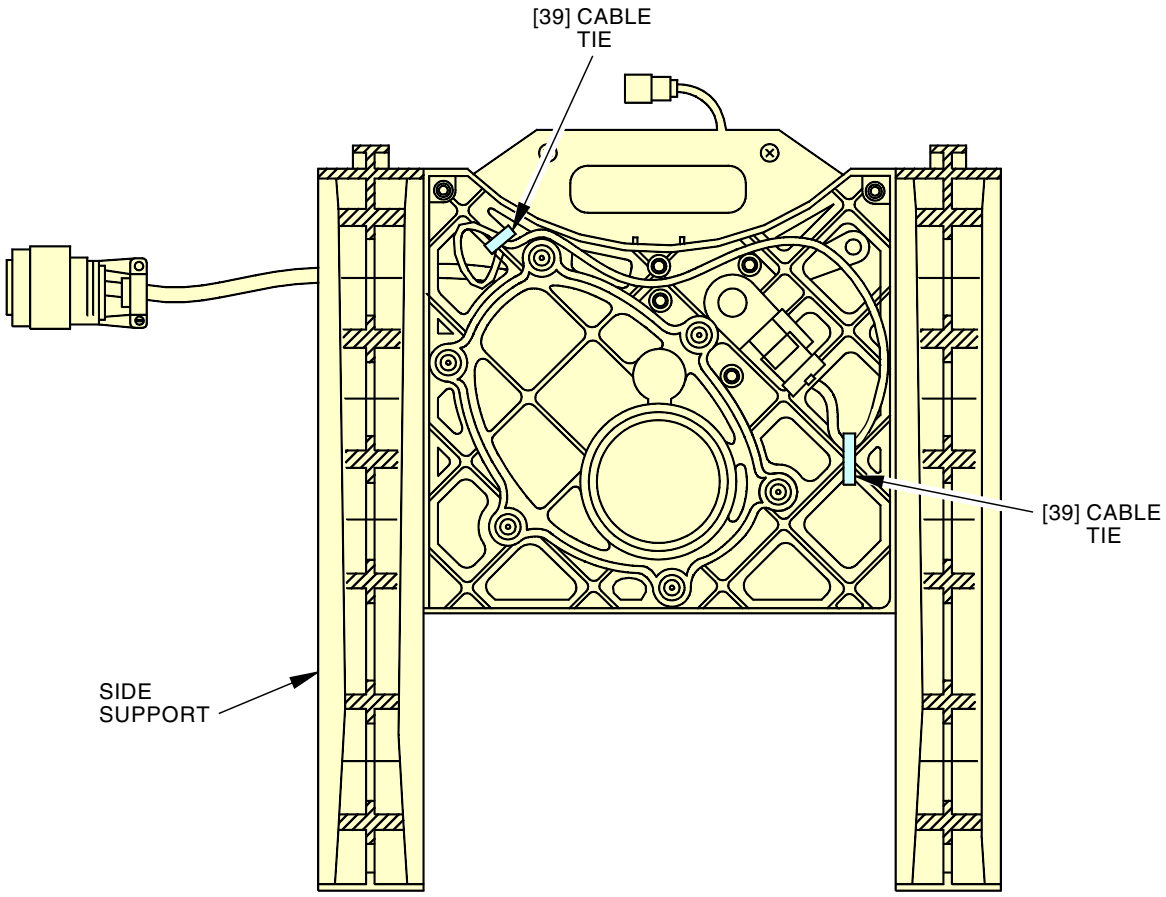
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Integrated System Controller Installation
Figure 405/38-32-01-990-810 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VACUUM TOILET ASSEMBLY (REAR VIEW)

C

NOTE:

HARNESS ATTACH POINTS, VIEWED INSIDE OF TOILET-STRUCTURE SECURING ISC WIRE HARNESS, TO PREVENT WIRE DAMAGE

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Integrated System Controller Installation
Figure 405/38-32-01-990-810 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-01-400-805

12. Integrated System Controller (ISC) Installation

(Figure 405)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Integrated System Controller Installation

SUBTASK 38-32-01-420-013

(1) Attach the controller bracket [37] to the Integrated System Controller [31] with the screws [30].

SUBTASK 38-32-01-420-014

(2) Install the controller bracket [37] with the attached Integrated System Controller [31] to the flush valve assemblies [146] with the screws [35].

(a) Do not tighten the screws [35] for now.

SUBTASK 38-32-01-420-015

(3) Place one end of the grounding cable [34] over the exposed threads of the attaching screw, followed by the washer [33] and nut [32].

(a) Do not tighten the nut [32] for now.

SUBTASK 38-32-01-420-016

(4) Tighten the attachment screws [35] and nut [32].

(a) Make sure that the screws [35] and nut [32] are tightened equally.

SUBTASK 38-32-01-420-017

(5) Connect the harness connector 1 to the flush valve assembly [146].

SUBTASK 38-32-01-420-018

(6) Connect the harness connector 2 to the rinse valve assembly [101].

SUBTASK 38-32-01-420-019

(7) Connect the harness connector 3 to the potentiometer on the flush valve assembly [146].

SUBTASK 38-32-01-420-020

(8) Secure the wiring harness from the Integrated System Controller [31] to the flush valve assembly [146] with new cable ties [39] (Figure 405).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Integrated System Controller Installation Test

SUBTASK 38-32-01-860-011

> **WARNING** WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

> **WARNING** DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-01-410-013

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-01-860-010

(3) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 38-32-01-860-008

(4) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-32-01-860-009

(5) Set the water shutoff valve for the lavatory to the ON position (Figure 402).

SUBTASK 38-32-01-710-004

(6) Flush the toilet a minimum of two times to make sure that it operates correctly.

(a) If the toilet does not flush correctly, do the applicable task in Interactive Fault Isolation Manual (IFIM) sections 38-30 and 38-32 to correct the problem.

SUBTASK 38-32-01-790-005

(7) Make sure that the flush and rinse valve assemblies and their connections do not leak.

(a) If there are leaks, check for damaged parts or loose connections.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-01-420-021

(1) Do this task: Toilet Shroud Installation, TASK 38-32-01-400-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LIQUID SEPARATOR - MAINTENANCE PRACTICES

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has these tasks:

(1) Liquid separator removal

(2) Liquid separator installation

(3) Liquid separator and liquid separator filter cleaning.

TASK 38-32-02-000-801

2. Liquid Separator Removal

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to remove the liquid separator.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-32-02-610-001

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-02-610-002

(2) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-32-02-010-004

(3) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-02-860-001

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-02-010-001

(5) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-02-010-002

(6) To remove the enclosure panels for the waste tanks, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

F. Liquid Separator Removal

SUBTASK 38-32-02-020-001

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Loosen the clamps [2] that attach the flexible coupling [7] to the upper cap [1].

SUBTASK 38-32-02-020-002

(2) Loosen the V band coupling [6].

SUBTASK 38-32-02-020-003

(3) Remove the upper cap [1] and packing [4].

(a) Discard the packing [4].

SUBTASK 38-32-02-020-004

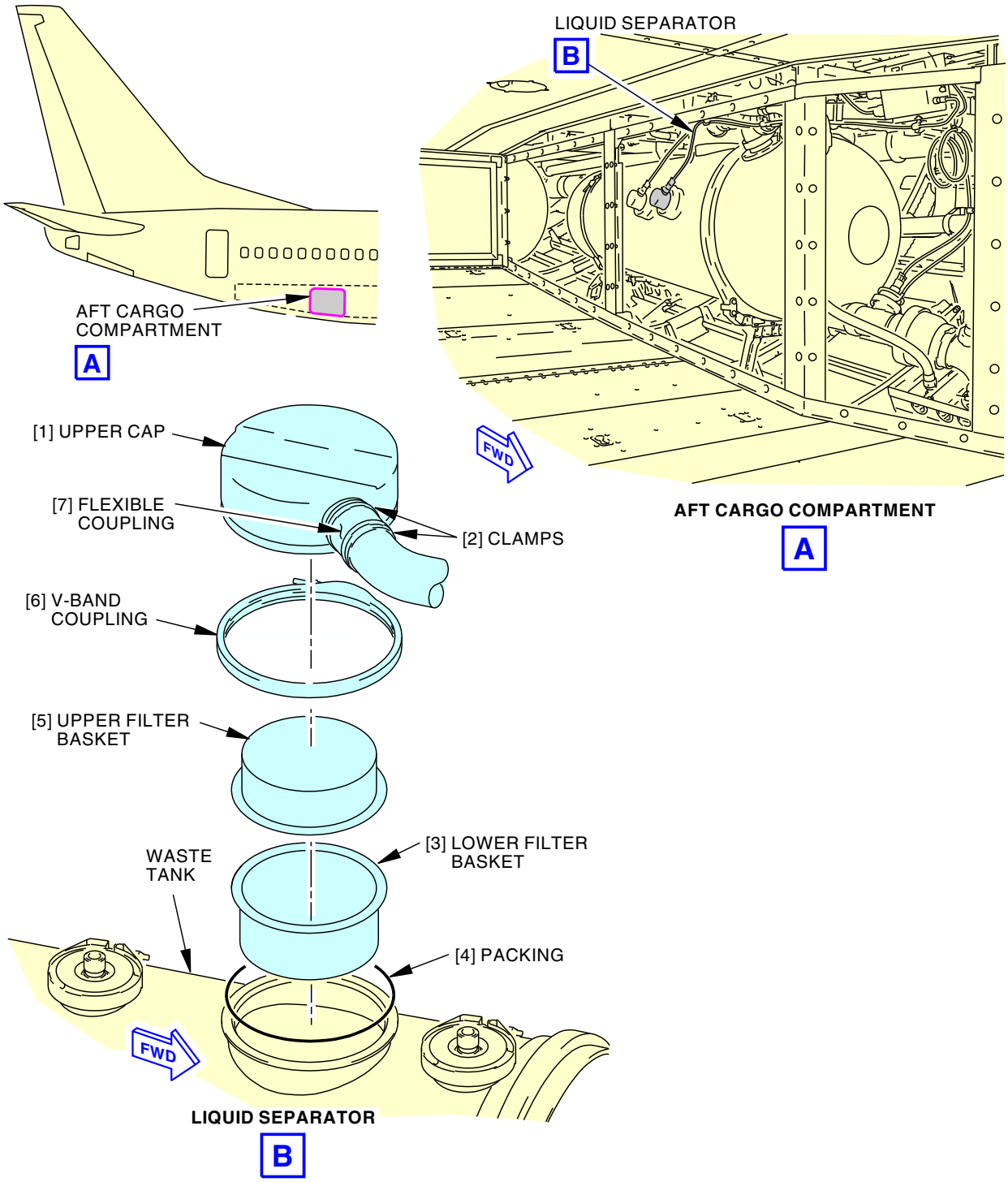
(4) Remove the lower filter basket [3] for the liquid separator from the waste tank.

SUBTASK 38-32-02-020-005

(5) Remove the upper filter basket [5] for the liquid separator from the upper cap [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Liquid Separator Maintenance Practices
Figure 201/38-32-02-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-02-400-801

3. Liquid Separator Installation

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to install the liquid separator.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650
D50007	Grease - Silicone-Based, Generic	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Lower filter basket	38-32-02-50-055	PTF ALL
4	Packing	38-32-02-50-060	PTF ALL
5	Upper filter basket	38-32-02-50-050	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

G. Liquid Separator Installation

SUBTASK 38-32-02-640-001

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Apply the grease, D00463 or silicone-based grease, D50007 or Molykote 111 Compound, D00189 to the new packing [4].

SUBTASK 38-32-02-420-001

(2) Install the lower filter basket [3] for the liquid separator in the waste tank.

SUBTASK 38-32-02-420-002

(3) Install the upper filter basket [5] for the liquid separator in the upper cap [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-02-420-003

(4) Install the upper cap [1] and packing [4].

SUBTASK 38-32-02-420-004

(5) Install the V band coupling [6].

(a) Tighten the V band coupling [6] to 90 in-lb (10 N·m) - 110 in-lb (12 N·m).

SUBTASK 38-32-02-420-005

(6) Tighten the clamps [2] that attach the flexible coupling [7] to the upper cap [1].

H. Liquid Separator Installation Test

SUBTASK 38-32-02-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-02-410-002

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-02-860-003

(3) Do this task: Supply Electrical Power, TASK 24-22-00-860-801

SUBTASK 38-32-02-710-001

(4) Flush a toilet connected to the waste tank.

SUBTASK 38-32-02-790-001

(5) Examine the upper cap and the connections for leakage.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-02-610-003

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-02-010-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-02-410-001

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

TASK 38-32-02-100-801

4. Liquid Separator and Liquid Separator Filter Cleaning

(Figure 201)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to clean the liquid separator filter canister.

B. Consumable Materials

Reference	Description	Specification
B00541	Cleaner - General Purpose Household Detergent	

C. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

D. Liquid Separator Cleaning

SUBTASK 38-32-02-020-006

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Do this task: Liquid Separator Removal, TASK 38-32-02-000-801.

SUBTASK 38-32-02-160-001

(2) Mix the general purpose household detergent cleaner, B00541 with water in a container.

SUBTASK 38-32-02-160-002

(3) Soak the two pieces of the liquid separator in the cleaner and water for a minimum of 15 minutes.

SUBTASK 38-32-02-170-001

(4) Flush the liquid separator with clean water.

SUBTASK 38-32-02-210-001

(5) Visually check the liquid separator and upper cap to see that they are clean.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Replace the upper filter basket [5] and lower filter basket [3] of the liquid separator if they are not clean.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-02-420-006

(1) Do this task: Liquid Separator Installation, TASK 38-32-02-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE DRAIN BALL VALVE AND LINKAGE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the waste drain ball valve

(2) An installation of the waste drain ball valve

(3) A removal of the waste drain ball valve linkage assembly

(4) An installation of the waste drain ball valve linkage assembly

(5) A removal of the waste drain ball valve limit switch

(6) An installation of the waste drain ball valve limit switch.

B. The waste drain ball valve is referred to as the ball valve in this procedure.

TASK 38-32-03-000-801

2. Waste Drain Ball Valve Removal

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to remove the waste drain ball valve.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
30-71-03-000-801	Waste Drain Line Heater Removal (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-32-03-610-001

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

(a) Do not add the chemical precharge after the task to service the waste tank is completed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-03-010-001

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-03-010-002

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-03-010-016

(4) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-03-860-001

(5) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	5	C00233	HEATERS DRAIN

SUBTASK 38-32-03-860-015

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

G. Waste Drain Ball Valve Removal

SUBTASK 38-32-03-010-004

(1) Do this task: Waste Drain Line Heater Removal, TASK 30-71-03-000-801.

SUBTASK 38-32-03-020-001

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(2) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector for the limit switch of the waste drain ball valve [11] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector for the limit switch of the waste drain ball valve [11].

(b) Install the protective cover on the electrical connector.

SUBTASK 38-32-03-020-002

(3) Remove the clamp [35], bolt [36], washer [37], and spacer [38] outboard of the ball valve attached to the limit switch cable.

SUBTASK 38-32-03-020-003

(4) Remove the bolt [14], bushing [12], washer [17], and nut [16] to disconnect the control rod [15].

SUBTASK 38-32-03-020-004

(5) Remove the clamshell [1], sleeve [2], and O-rings [3] that attach the waste drain ball valve [11] to the waste tube.

(a) Discard the O-rings [3].

SUBTASK 38-32-03-020-005

(6) Remove the V-band clamp [4] and V-band clamp [44] at the service panel and waste drain ball valve [11] connections for the drain elbow [18].

SUBTASK 38-32-03-020-006

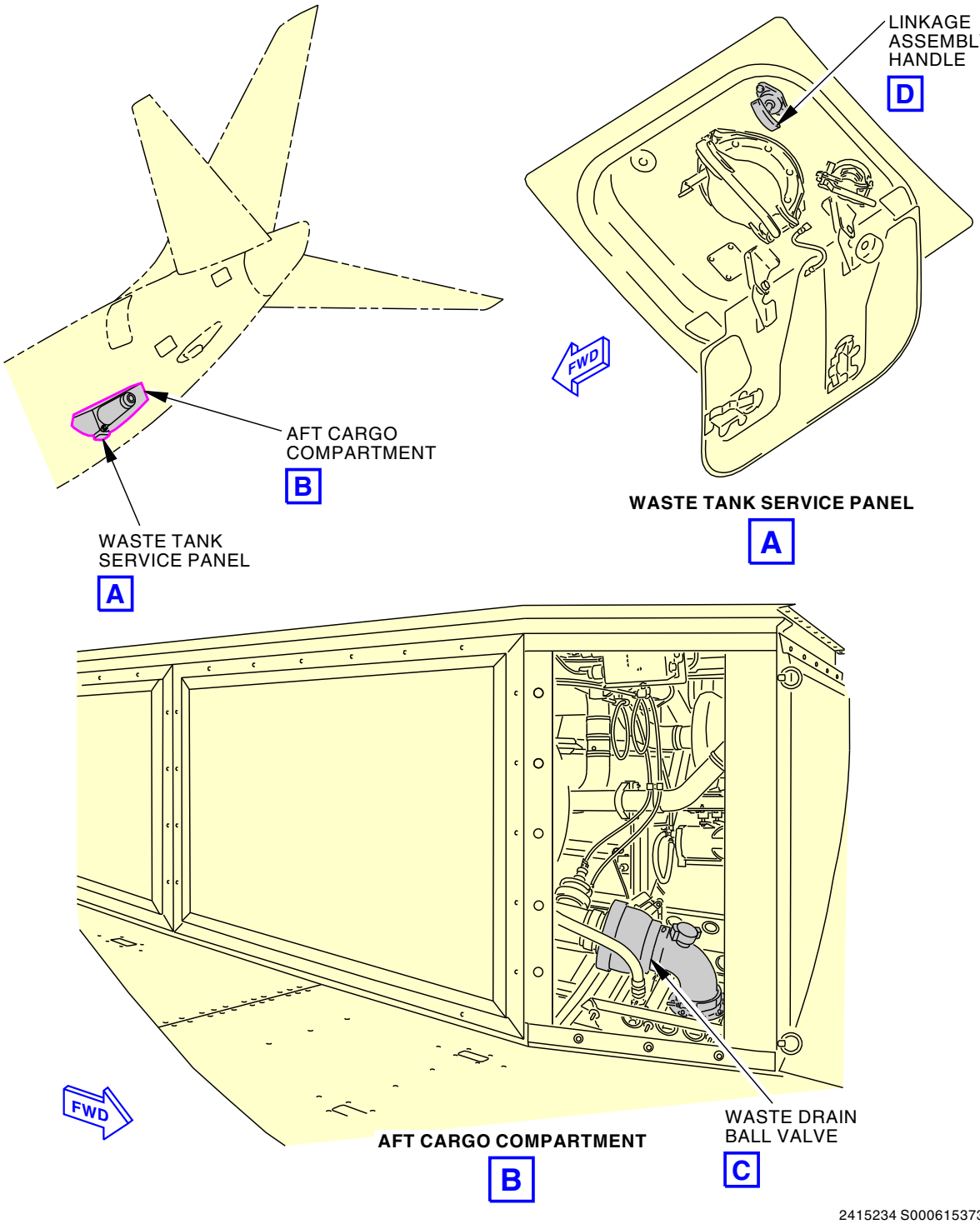
(7) Remove the waste drain ball valve [11] and drain elbow [18].

(a) Make sure that the pin [13] remains in the drain elbow [18] for the subsequent installation of the waste drain ball valve [11].

(b) Remove and discard the gask-O-seal [5] and gask-O-seal [45].

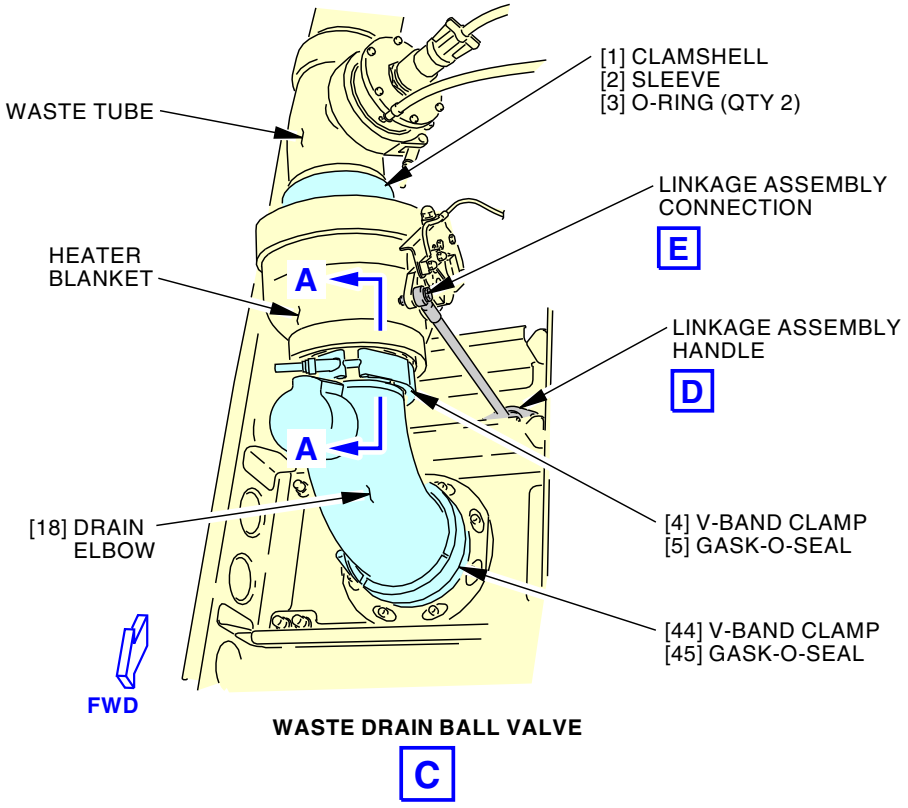
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Drain Ball Valve Installation
Figure 401/38-32-03-990-804 (Sheet 1 of 3)

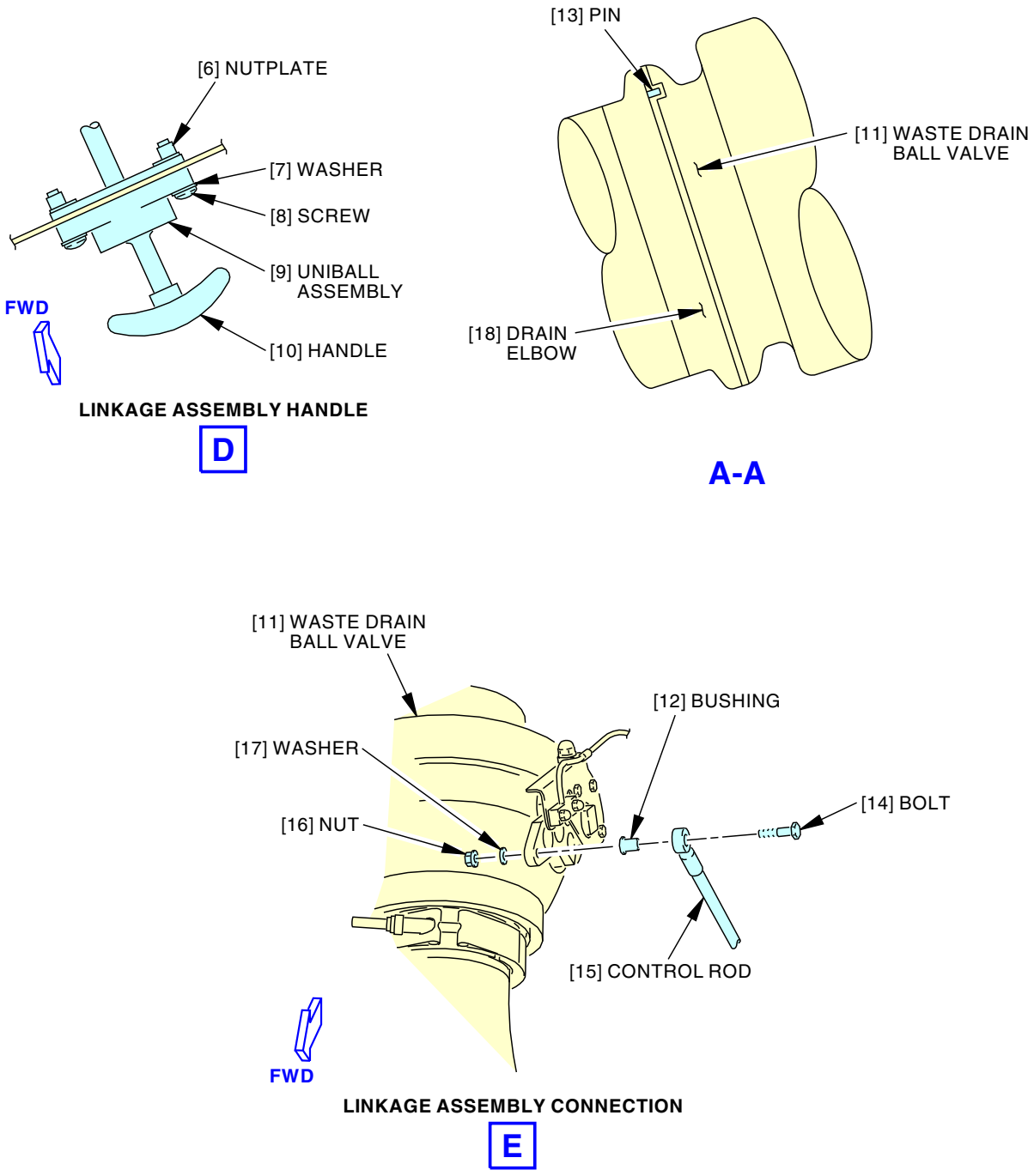
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Waste Drain Ball Valve Installation
Figure 401/38-32-03-990-804 (Sheet 2 of 3)

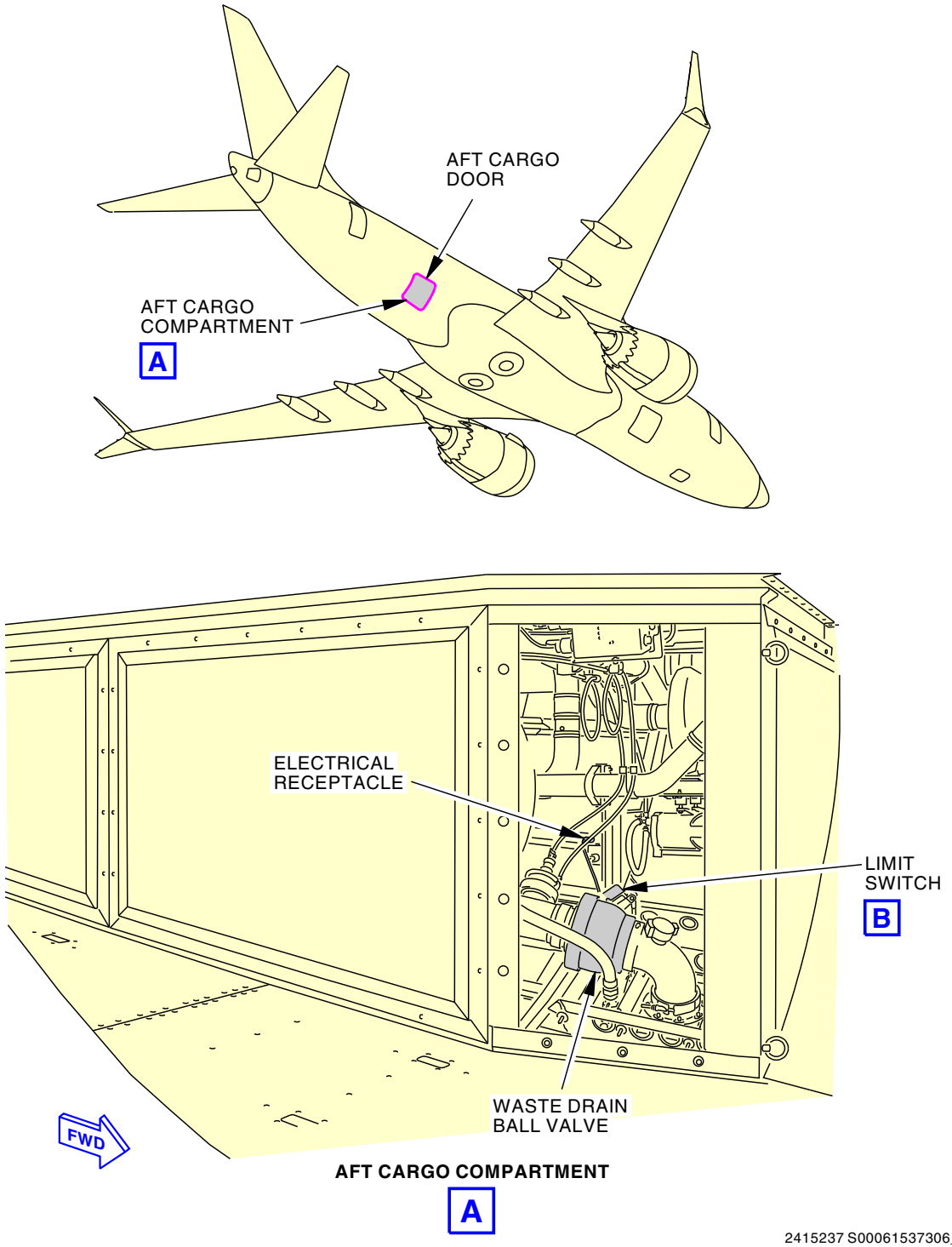
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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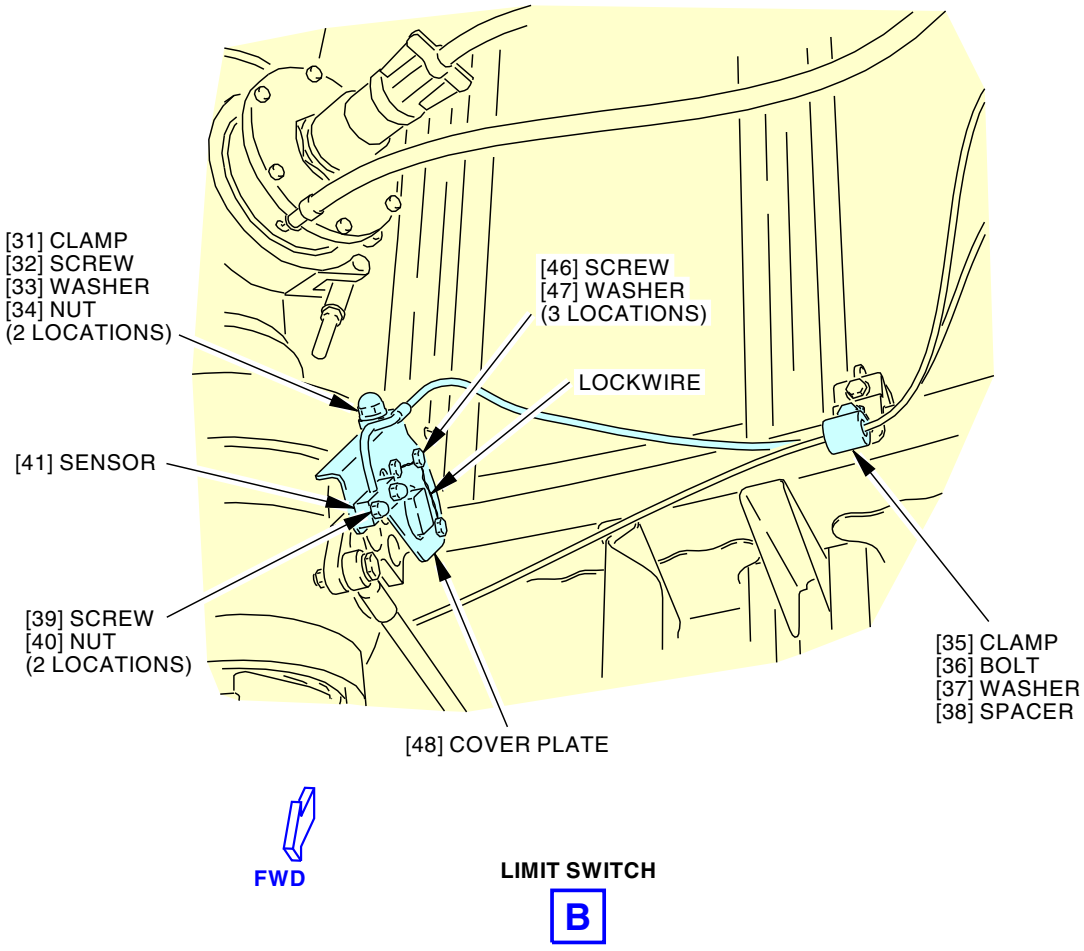
Waste Drain Ball Valve Installation
Figure 401/38-32-03-990-804 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Limit Switch Installation
Figure 402/38-32-03-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Limit Switch Installation
Figure 402/38-32-03-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-03-400-801

3. Waste Drain Ball Valve Installation

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to install the waste drain ball valve.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
30-71-03-400-801	Waste Drain Line Heater Installation (P/B 401)
38-32-03-710-801	Waste Drain Ball Valve and Control Rod - Operational Test (P/B 501)
38-32-03-820-801	Waste Drain Ball Valve and Control Rod - Adjustment (P/B 501)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-9123	Sleeve Installation, Waste Tank Part #: C38003-1 Supplier: 81205
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-1142	Equipment - Waste System Servicing

D. Consumable Materials

Reference	Description	Specification
D00016	Grease - Aircraft, General Purpose, Wide Temperature Range	MIL-PRF-81322

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	O-ring	38-32-03-02-255	PTF ALL
5	Gask-O-seal	38-32-03-02-230	PTF ALL
11	Waste drain ball valve	38-32-03-02-090	PTF ALL
45	Gask-O-seal	38-32-03-02-260	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
145AL	Waste Service Door
822	Aft Cargo Door

H. Waste Drain Ball Valve Installation

SUBTASK 38-32-03-640-004

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Apply grease, D00016, to the new O-rings [3], new gask-O-seals [5], and new gask-O-seal [45].

SUBTASK 38-32-03-420-022

(2) Install the gask-O-seals [5], gask-O-seal [45], and O-rings [3].

SUBTASK 38-32-03-420-023

(3) Put the sleeve [2] on the flange of the waste tube.

SUBTASK 38-32-03-420-024

(4) Put the waste drain ball valve [11] and drain elbow [18] into position.

(a) Align the hole in the waste drain ball valve [11] with the pin [13] on the flange.

1) If the waste drain ball valve [11] has a pin installed, remove the pin from the waste drain ball valve [11].

2) Use the pin [13] on the flange for the installation.

SUBTASK 38-32-03-420-025

(5) Install the V-band clamp [4] and V-band clamp [44] at the service panel and waste drain ball valve [11] connections for the drain elbow [18].

SUBTASK 38-32-03-420-026

(6) Tighten the V-band clamp [4] and V-band clamp [44] to 60 in-lb (6.8 N·m) - 65 in-lb (7.3 N·m).

SUBTASK 38-32-03-420-027

(7) Install the sleeve [2] and clamshell [1] to connect the waste tube.

(a) Make sure that the ends between the waste tube flange and waste drain ball valve [11] flange have a nominal clearance of 0.15 in. (3.81 mm) or minimum clearance of 0.10 in. (2.54 mm) apart.

NOTE: If a minimum clearance of 0.10 in. (2.54 mm) is not obtained damage to the waste tube flange or ball valve may occur. A maximum clearance is limited by the proper installation of the clamshell clamp. The clamp must not bind, or bend when installed. All of the clamp latches must close.

(b) To install the sleeve [2] with the sleeve, SPL-9123, do the following steps:

NOTE: The sleeve, SPL-9123, is a P/N 14C02–56 clamp with one of the lips machined off to allow the clamp to be slipped on the sleeve while the clamp is closed.

NOTE: Use of the modified clamp tool, sleeve, SPL-9123, is optional.

1) Attach the sleeve, SPL-9123, to the sleeve [2] and lock.

NOTE: The inner seat of the sleeve, SPL-9123, must sit against the edge of the sleeve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Carefully tap around the periphery of the tool until the sleeve [2] moves over the second O-ring.

NOTE: The sleeve and O-rings must not be damaged or deformed during this step. Damage or deformation shows that alignment or clearance are not correct.

3) Remove the sleeve, SPL-9123.

4) Install the clamshell [1].

(c) To install the sleeve [2] without the sleeve, SPL-9123, do the following steps:

1) Obtain a wood or a plastic block to allow tapping on the edge of the sleeve [2].

2) Carefully tap around the periphery of the sleeve [2] until it moves over the second O-ring.

NOTE: The sleeve and O-rings must not be damaged or deformed during this step. Damage or deformation shows that alignment or clearance are not correct.

3) Install the clamshell [1].

SUBTASK 38-32-03-420-028

(8) Install the bolt [14], bushing [12], washer [17], and nut [16] to connect the control rod [15].

SUBTASK 38-32-03-820-004

(9) Adjust the linkage assembly (TASK 38-32-03-820-801).

SUBTASK 38-32-03-420-029

(10) Install the clamp [35], bolt [36], washer [37], and spacer [38] outboard of the ball valve for the limit switch cable.

SUBTASK 38-32-03-420-030

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(11) Use the teflon-jawed pliers, STD-664, to connect the electrical connector for the limit switch of the waste drain ball valve [11] (TASK 70-00-01-910-803-G00).

(a) Remove the protective cover from the electrical connector.

(b) Connect the electrical connector for the limit switch of the waste drain ball valve [11].

1) Tighten the connector with the teflon-jawed pliers, STD-664.

I. Waste Drain Ball Valve Installation Test

SUBTASK 38-32-03-860-016

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-03-860-017

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 38-32-03-760-005

(3) Do this task: Waste Drain Ball Valve and Control Rod - Operational Test, TASK 38-32-03-710-801.

SUBTASK 38-32-03-790-002

(4) Do a leak check of the waste drain ball valve [11], do these steps:

(a) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

(b) At the service panel, make sure that the control handles for the drain ball valves are in the closed position.

(c) Make sure that the cap for the waste drain fitting outlet is closed.

(d) Connect the water hose from the waste system servicing equipment, STD-1142, to the rinse fitting of the service panel.

(e) Use the pump on the service cart to put about 50 gal (189 l) of water into the waste tank.

(f) Open the cap and inner door (use the PUSH-TO-OPEN lever) of the service panel drain valve assembly.

(g) Make sure that there is no leak within the next 30 minutes after installation of the waste drain ball valve [11].

(h) Close the drain cap.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-03-410-013

(1) Do this task: Waste Drain Line Heater Installation, TASK 30-71-03-400-801.

SUBTASK 38-32-03-860-018

(2) Remove safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
E	5	C00233	HEATERS DRAIN

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-03-710-004

(3) Make sure that the heater blanket is serviceable.

SUBTASK 38-32-03-610-008

(4) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-03-410-014

WARNING SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(5) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-03-410-015

(6) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door
822	Aft Cargo Door

END OF TASK

TASK 38-32-03-000-802

4. Waste Drain Ball Valve Linkage Assembly Removal

(Figure 403)

A. General

(1) This task gives the instructions to remove the linkage assembly for the waste drain ball valve.

(2) The linkage assembly has two sub-assemblies, the control rod [15] with the handle [10] and the uniball assembly [9].

B. References

Reference	Title
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 38-32-03-010-005

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-03-010-006

(2) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-03-010-007

(3) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

SUBTASK 38-32-03-010-014

(4) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-03-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

F. Waste Drain Ball Valve Linkage Assembly Removal

SUBTASK 38-32-03-020-007

(1) Remove the screw [8] and washer [7] at the service panel to disconnect the uniball assembly [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-03-020-008

(2) Remove the nut [16], washer [17], bushing [12], and bolt [14] to disconnect the control rod [15] from the waste drain ball valve [11].

SUBTASK 38-32-03-020-009

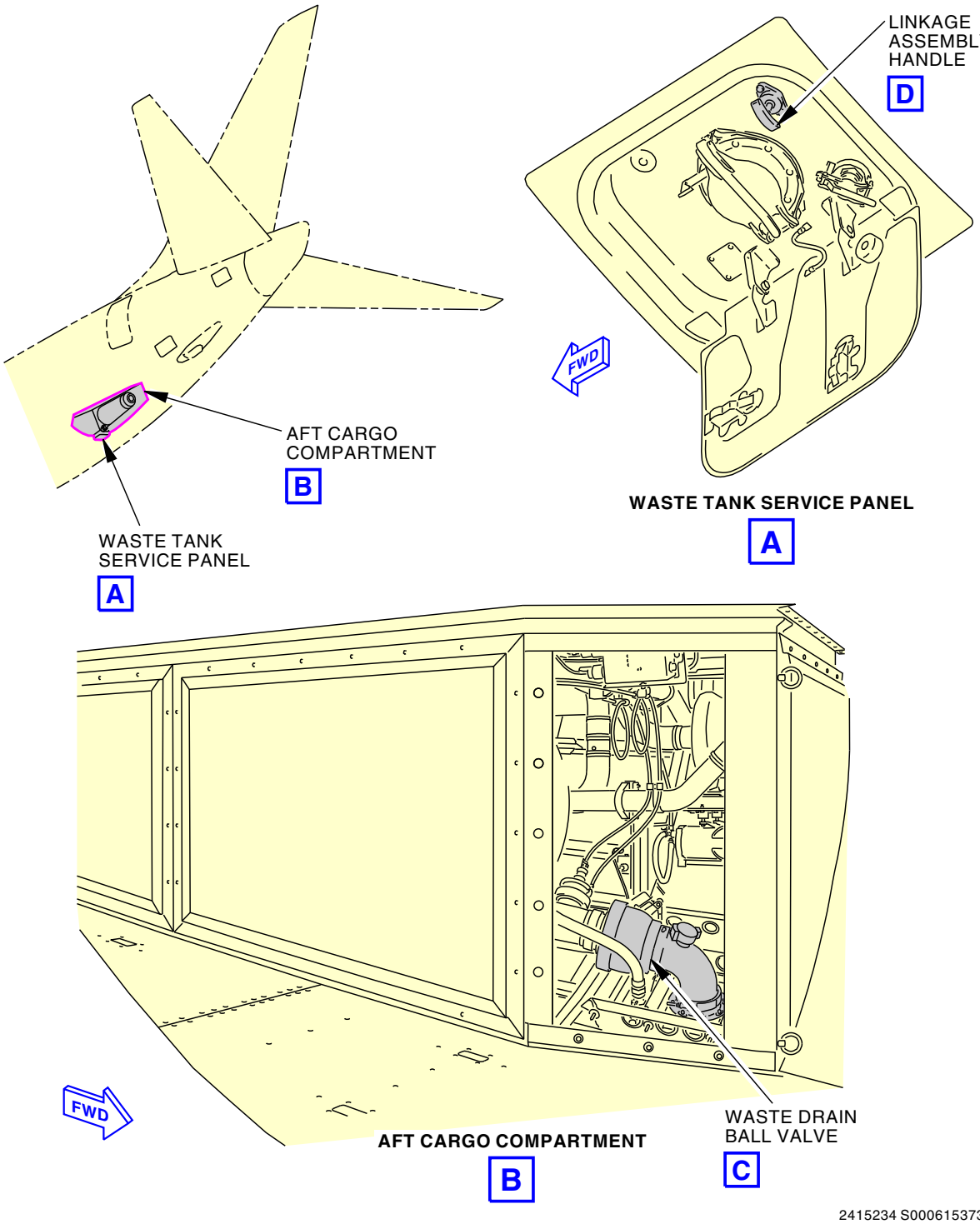
(3) Remove the linkage assembly and uniball assembly [9].

SUBTASK 38-32-03-020-010

(4) Disconnect the control rod [15] end from the handle [10] to remove the uniball assembly [9].

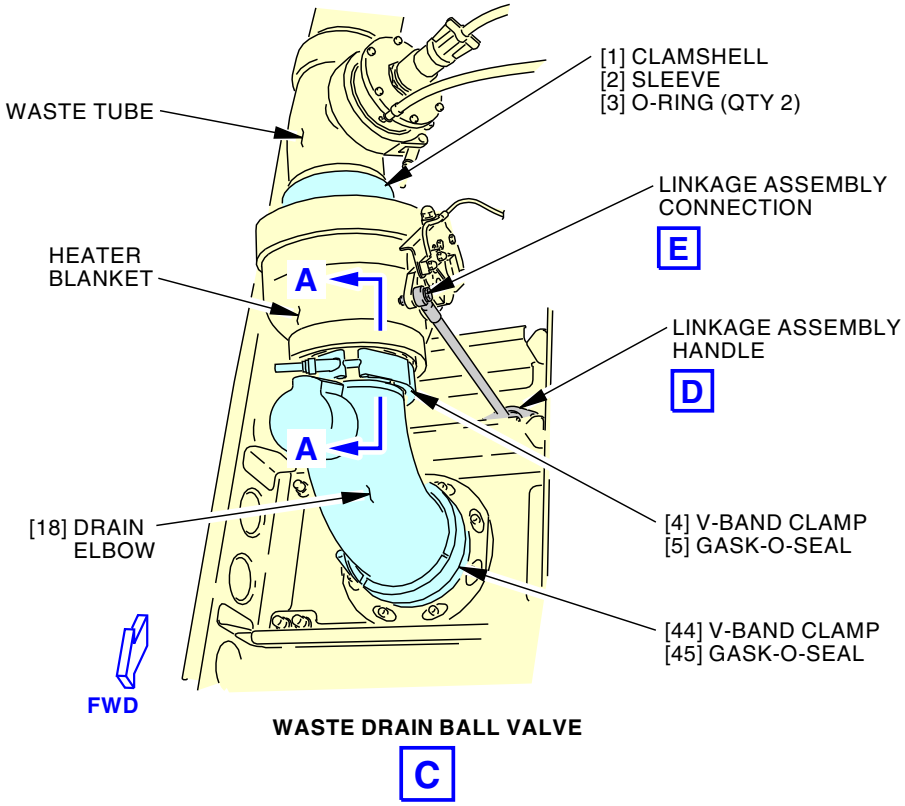
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Drain Ball Valve Installation
Figure 403/38-32-03-990-805 (Sheet 1 of 3)

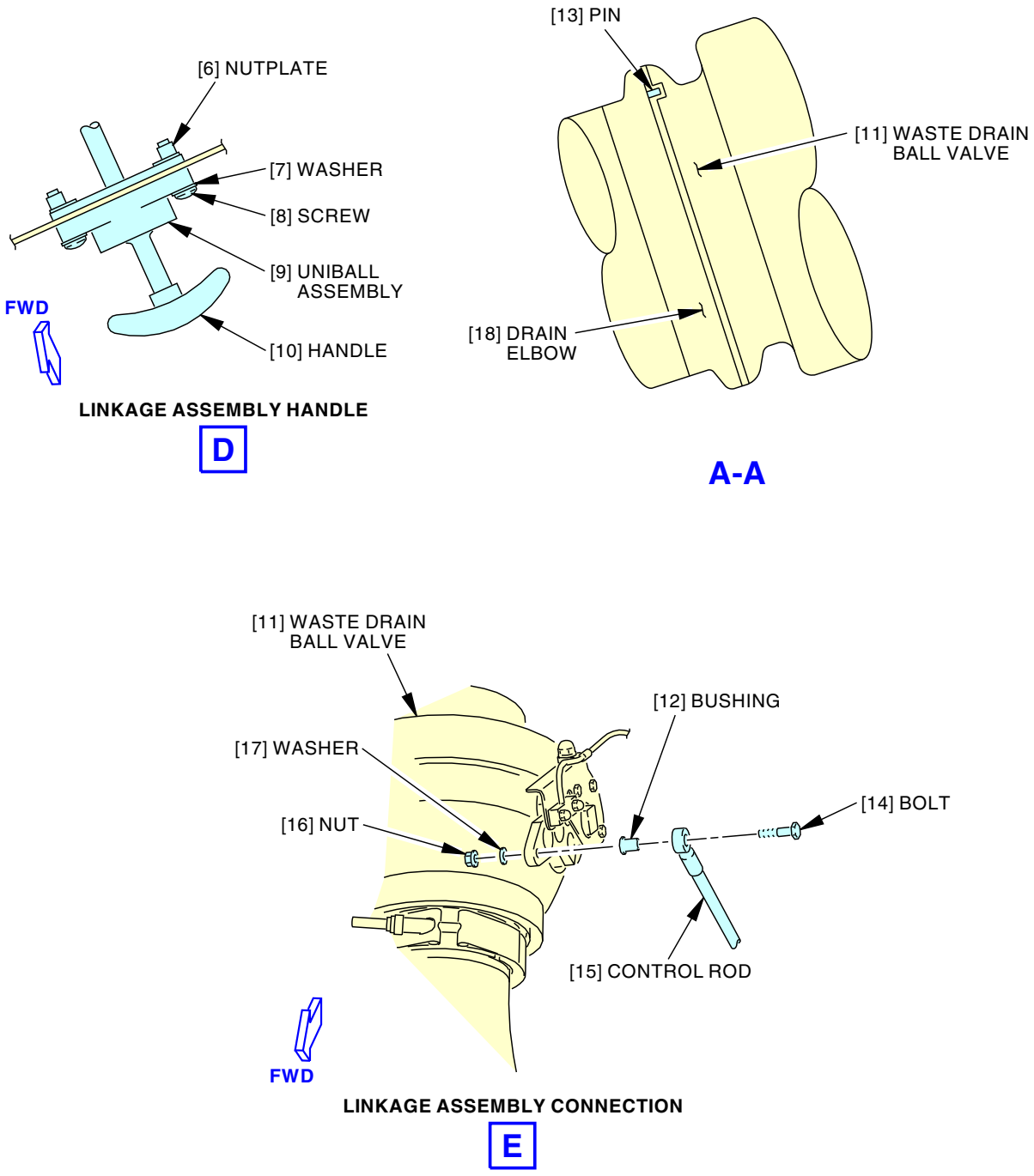
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2415235 S00061537303_V4

Waste Drain Ball Valve Installation
Figure 403/38-32-03-990-805 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2415236 S00061537304_V3

Waste Drain Ball Valve Installation
Figure 403/38-32-03-990-805 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-03-400-802

5. Waste Drain Ball Valve Linkage Assembly Installation

(Figure 403)

A. General

(1) This task gives the instructions to install the linkage assembly for the waste drain ball valve.

(2) The linkage assembly has two sub-assemblies, the control rod [15] with the handle [10] and the uniball assembly [9].

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
38-32-03-710-801	Waste Drain Ball Valve and Control Rod - Operational Test (P/B 501)
38-32-03-820-801	Waste Drain Ball Valve and Control Rod - Adjustment (P/B 501)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00062	Lubricant - Pneumatic System	SAE AMS-G-4343 (NATO G-392)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Uniball assembly	38-32-03-01-035	PTF ALL
10	Handle	38-32-03-02-280	PTF ALL
15	Control rod	38-32-03-02-265	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door
822	Aft Cargo Door

G. Waste Drain Ball Valve Linkage Assembly Installation

SUBTASK 38-32-03-640-002

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Apply the lubricant, D00062 to the internal portion of the uniball assembly [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-03-420-010

(2) Put the linkage assembly in its position in the uniball assembly [9].

SUBTASK 38-32-03-640-003

(3) Apply a fay surface seal with sealant, A00247 to the uniball assembly [9] and the service panel.

SUBTASK 38-32-03-420-011

(4) Put the uniball assembly [9] in its position.

SUBTASK 38-32-03-420-012

(5) Install the screws [8] and washers [7] to connect the uniball assembly [9] to the service panel.

SUBTASK 38-32-03-420-013

(6) Install the bolt [14], bushing [12], washer [17], and nut [16] to connect the control rod [15]to the ball waste drain ball valve [11].

SUBTASK 38-32-03-820-001

(7) Do this task: Waste Drain Ball Valve and Control Rod - Adjustment, TASK 38-32-03-820-801.

H. Waste Drain Ball Valve Linkage Assembly Installation Test

SUBTASK 38-32-03-860-006

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-03-860-007

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801

SUBTASK 38-32-03-760-002

(3) Do this task: Waste Drain Ball Valve and Control Rod - Operational Test, TASK 38-32-03-710-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-03-610-003

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-03-410-006

WARNING SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-03-410-007

(3) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door
822	Aft Cargo Door

——— END OF TASK ———

TASK 38-32-03-000-803

6. Waste Drain Ball Valve Limit Switch Removal

(Figure 404)

A. General

(1) This task gives the instructions to remove the waste drain ball valve limit switch.

B. References

Reference	Title
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-32-03-010-008

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-03-010-009

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(2) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-03-010-015

(3) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-03-860-008

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

G. Waste Drain Ball Valve Limit Switch Removal

SUBTASK 38-32-03-020-011

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector for the limit switch of the ball valve (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector for the limit switch of the ball valve.

(b) Install the protective cover on the electrical connector.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-03-020-015

(2) Remove the cover plate [48] as follows:

(a) Remove the lockwire from the screws [46].

(b) Remove the screws [46] and washers [47] that attach the cover plate [48] to the waste drain ball valve.

SUBTASK 38-32-03-020-012

(3) Remove the screw [32], washer [33], and nut [34] to disconnect the clamp [31] from the cover plate [48].

SUBTASK 38-32-03-020-013

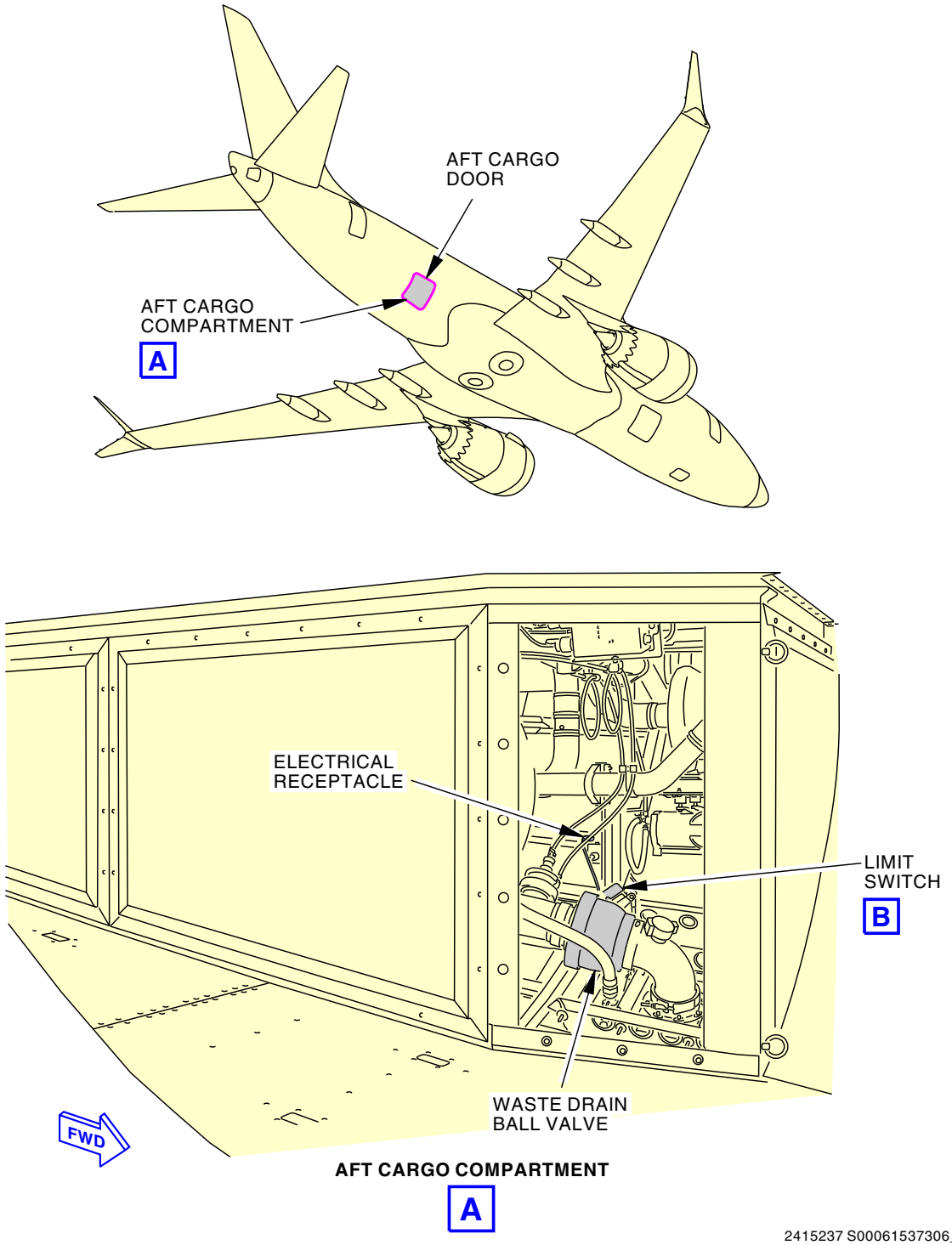
(4) Remove the clamp [35], bolt [36], washer [37], and spacer [38] from the cable for the limit switch.

SUBTASK 38-32-03-020-014

(5) Remove the screws [39] and nuts [40] to disconnect the sensor [41].

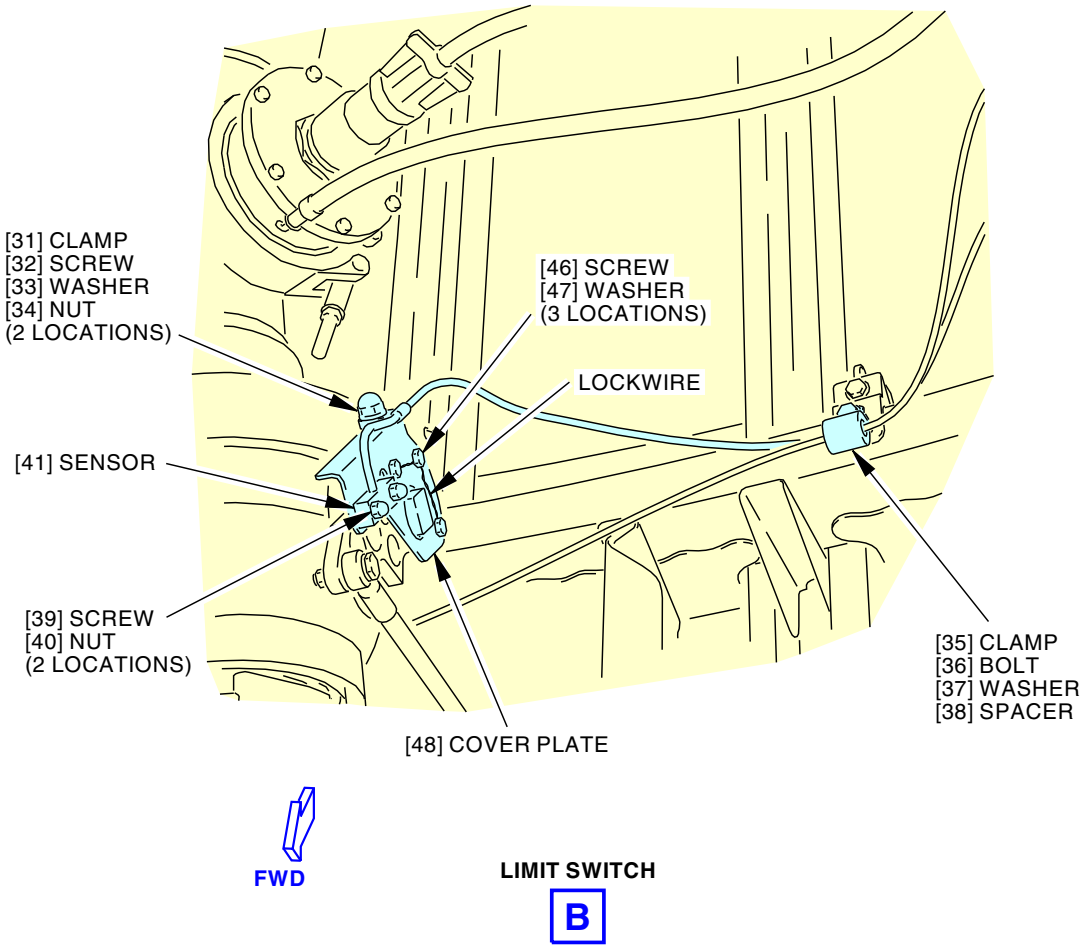
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Drain Ball Valve Limit Switch Installation
Figure 404/38-32-03-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Drain Ball Valve Limit Switch Installation
Figure 404/38-32-03-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-03-400-803

7. Waste Drain Ball Valve Limit Switch Installation

(Figure 404)

A. General

(1) This task gives the instructions to install the waste drain ball valve limit switch.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
38-32-03-710-801	Waste Drain Ball Valve and Control Rod - Operational Test (P/B 501)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
A00648	Compound - Retaining - Loctite 242	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
41	Sensor	38-32-03-02-110	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
143	Area Below Aft Cargo Compartment - Left
822	Aft Cargo Door

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

H. Waste Drain Ball Valve Limit Switch Installation

SUBTASK 38-32-03-420-019

(1) Apply Loctite 242 compound, A00648, to the threads of the screws [39] and screw [32].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-03-420-014

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(2) Put the sensor [41] in its position.

NOTE: See the waste drain ball valve manufactures instructions for further information on the sensor.

SUBTASK 38-32-03-420-015

(3) Install screws [39] and nuts [40], that attach the sensor [41] to the cover plate [48].

SUBTASK 38-32-03-420-016

(4) Install the clamp [31], screw [32], washer [33], and nut [34] to attach the sensor wire to the cover plate [48].

SUBTASK 38-32-03-420-017

(5) Install the clamp [35], bolt [36], washer [37], and spacer [38].

SUBTASK 38-32-03-420-020

(6) Install the screws [46] and washers [47] to attach the cover plate [48] to the waste drain ball valve.

SUBTASK 38-32-03-420-018

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(7) Use the teflon-jawed pliers, STD-664 to connect the electrical connector for the limit switch of the ball valve (TASK 70-00-01-910-803-G00).

(a) Remove the protective cover from the electrical connector.

(b) Connect the electrical connector for the limit switch of the ball valve.

1) Tighten the connector with the teflon-jawed pliers, STD-664.

I. Waste Drain Ball Valve Limit Switch Installation Test

SUBTASK 38-32-03-860-009

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-03-860-010

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801

SUBTASK 38-32-03-760-003

(3) Do this task: Waste Drain Ball Valve and Control Rod - Operational Test, TASK 38-32-03-710-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-03-420-021

(1) Install MS20995C32 lockwire, G01048, on the screws [46].

SUBTASK 38-32-03-610-004

(2) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-03-010-010

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(3) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-03-410-009

(4) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE DRAIN BALL-VALVE AND CONTROL ROD - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An operational test of the waste drain ball valve and control rod.

(2) An adjustment of the control rod for the waste drain ball valve.

B. The Waste Drain Ball Valve is referred to as the Ball Valve in this procedure.

TASK 38-32-03-710-801

2. Waste Drain Ball Valve and Control Rod - Operational Test

A. General

(1) This task does these tests at the same time:

(a) An operational test to measure the necessary force to use the control rod for the ball valve.

(b) An operational test to make sure that the switch on the ball valve operates correctly.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1142	Equipment - Waste System Servicing
STD-1184	Scale - Spring, 0-100 Lbs, Tension

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door

F. Prepare for the Test

SUBTASK 38-32-03-910-001

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-03-010-011

(2) At the waste service panel, do these steps:

(a) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

(b) Open the cap on the waste drain valve assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Connect the waste drain hose from the toilet service waste system servicing equipment, STD-1142, to the waste drain valve assembly.

(d) Push the PUSH-TO-OPEN lever on the waste drain valve assembly.

SUBTASK 38-32-03-010-017

(3) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-03-860-011

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

G. Operational Test

SUBTASK 38-32-03-480-001

(1) Connect the spring scale (0-100 Lbs), STD-1184, to the control handle for the ball valve.

SUBTASK 38-32-03-710-002

(2) Do a test of the force that is necessary to open the ball valve as follows:

(a) Pull down on the spring scale (0-100 Lbs), STD-1184, until the ball valve is fully open.

(b) At the same time, read the maximum force on the spring scale (0-100 Lbs), STD-1184.

(c) Make sure that the maximum force on the spring scale (0-100 Lbs), STD-1184, is less than 50 lbf (222 N).

(d) Disconnect the spring scale (0-100 Lbs), STD-1184, from the control handle for the ball valve.

(e) Push on the control handle for the ball valve until the ball valve is closed.

SUBTASK 38-32-03-710-003

(3) Make sure that the switch on the ball valve operates correctly as follows:

(a) Pull down on the control handle until the ball valve is fully open.

(b) Push the flush switch for a toilet.

(c) Make sure that the vacuum blower does not operate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Push on the control handle for the ball valve until the ball valve is closed.

(e) Make sure that this access panel closes freely and does not touch the control handle:

Number	Name/Location
145AL	Waste Service Door

1) If the access panel touches the control handle, do this task: Waste Drain Ball Valve and Control Rod - Adjustment, TASK 38-32-03-820-801.

(f) Push the flush switch for a toilet.

(g) Make sure that the vacuum blower does operate.

1) If the vacuum blower does not operate, do this task: Waste Drain Ball Valve and Control Rod - Adjustment, TASK 38-32-03-820-801.

(h) If the vacuum blower operates, then continue.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-03-610-005

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-03-410-010

(2) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door

——— END OF TASK ———

TASK 38-32-03-820-801

3. Waste Drain Ball Valve and Control Rod - Adjustment

(Figure 501)

A. General

(1) Use this procedure to do the adjustment of the control rod on the ball valve.

NOTE: When the ball valve is open, the limit switch stops the operation of the toilets.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A50086	Primer - Loctite 7471 Primer T Activator	
A50129	Primer - Loctite 7649	ASTM D5363
A50211	Compound - Threadlocking, High-Strength - Loctite 277	MIL-S-46163
A50319	Compound - Loctite 262 High Strength Red Threadlocker	
A50413	Primer - Activator	MIL-S-46163 Grade F

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door
822	Aft Cargo Door

F. Prepare for the Adjustment

SUBTASK 38-32-03-610-006

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

(a) Do not add the chemical precharge after the task to drain and flush the waste tanks is completed.

SUBTASK 38-32-03-010-012

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-03-010-013

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-03-010-018

(4) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-03-860-013

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Make sure that these circuit breakers are closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Ball Valve and Control Rod Adjustment

SUBTASK 38-32-03-820-002

(1) Do the control rod adjustment as follows:

(a) Make sure that the ball valve is fully closed.

(b) Make sure that this access panel is closed:

Number	Name/Location
145AL	Waste Service Door

(c) Adjust the length of the control rod.

1) Loosen the jam nut to turn the control rod in the rod end.

2) Remove the loose particle of solid Loctite 277 compound, A50211 or Loctite 262 compound, A50319, from the thread of the control rod end.

3) Rotate the control rod in an extended length direction until the control handle touches the waste service door.

4) Rotate the control rod back three full turns from the waste service door.

(d) Make sure that the threads of the control rod can be seen in the witness hole of the rod end.

(e) Tighten the jam nut for the rod end as follows:

1) Apply a Loctite 277 compound, A50211 or Loctite 262 compound, A50319, and primer, A50413, Loctite 7649 Primer, A50129, or Loctite 7471 Primer T Activator, A50086 to the thread of the control rod at the rod end.

2) Tighten the jam nut to 100 in-lb (11.3 N·m) - 130 in-lb (14.7 N·m).

SUBTASK 38-32-03-760-004

(2) Do this task: Waste Drain Ball Valve and Control Rod - Operational Test, TASK 38-32-03-710-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-03-610-007

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-03-410-011

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE FIRE EXTINGUISHING AGENT OR SMOKE CAN CAUSE INJURIES.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

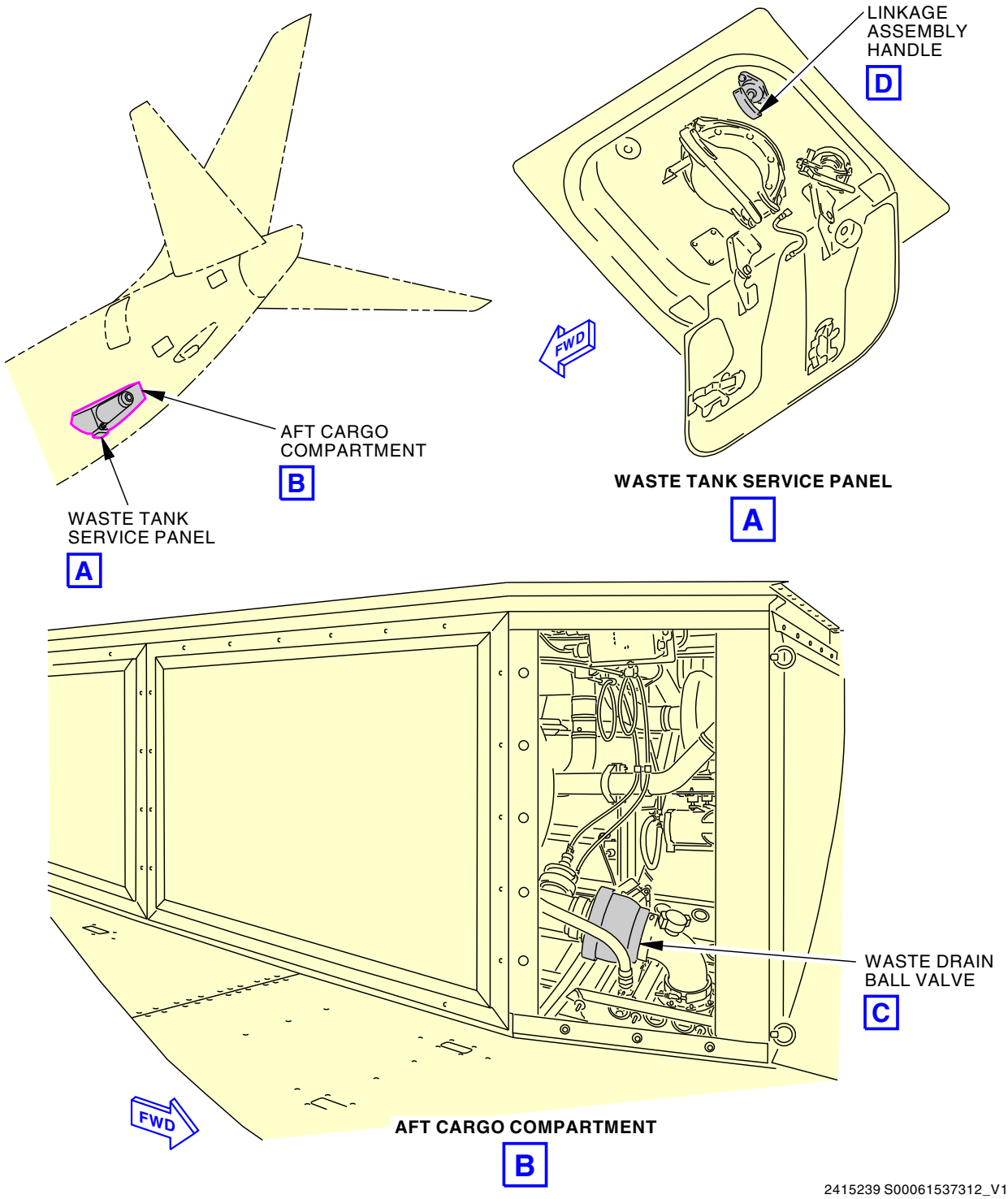
SUBTASK 38-32-03-410-012

(3) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

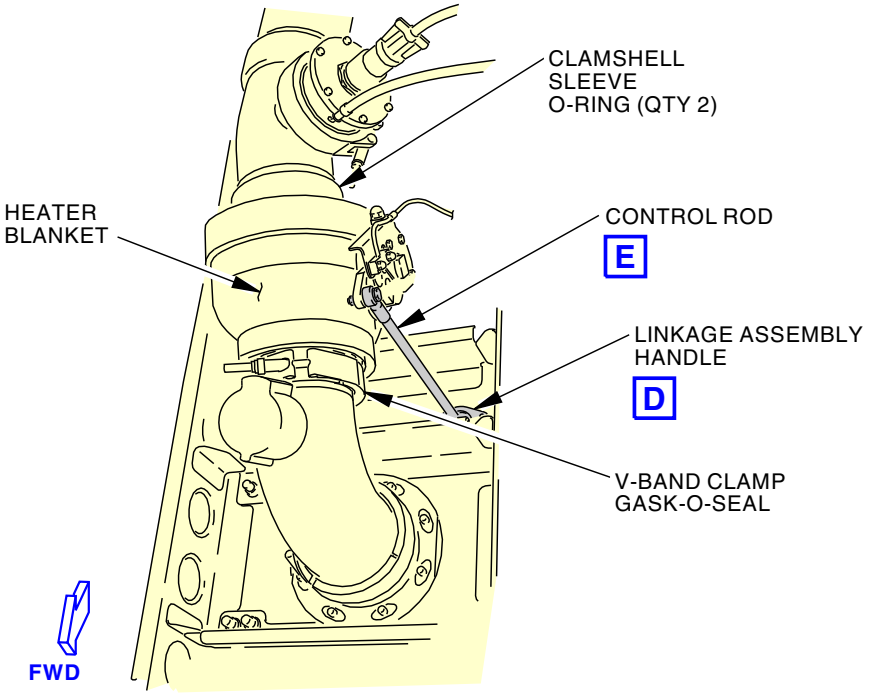
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



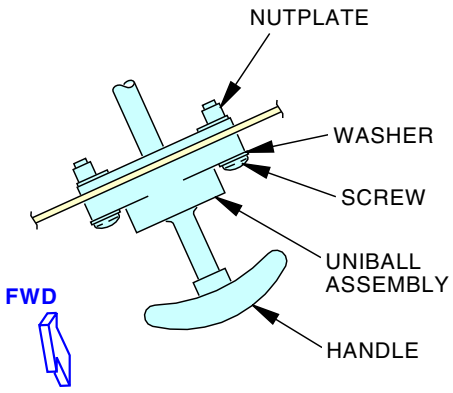
Waste Drain Ball Valve Adjustment
Figure 501/38-32-03-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



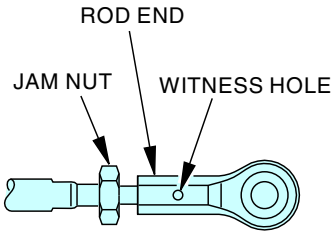
WASTE DRAIN BALL VALVE

C



LINKAGE ASSEMBLY HANDLE

D



CONTROL ROD

E

2415240 S00061537313_V3

Waste Drain Ball Valve Adjustment
Figure 501/38-32-03-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLUSH SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the flush switch

(2) An installation of the flush switch.

TASK 38-32-04-000-804

2. Flush Switch Removal

(Figure 401)

A. General

(1) This task is for the removal of the vacuum waste flush switch.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Removal

SUBTASK 38-32-04-010-002

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-04-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Flush Switch Removal

SUBTASK 38-32-04-020-007

(1) Do these steps to remove the standard flush switch [1].

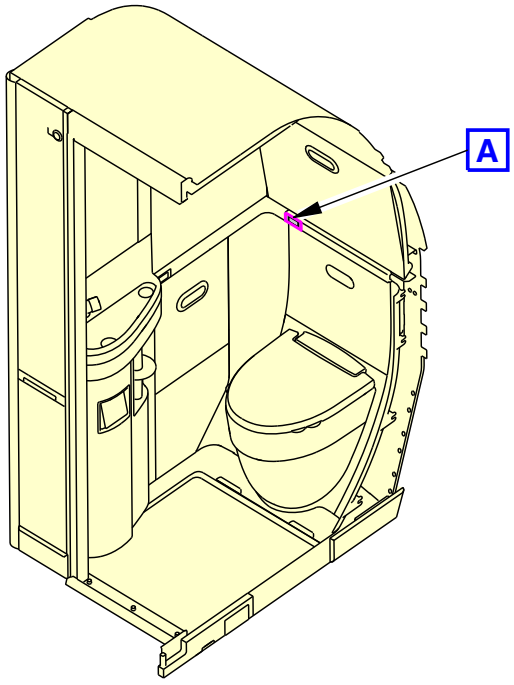
(a) Remove the flush switch [1] with a rounded edge flat tool.

1) Do not remove the mounting plate.

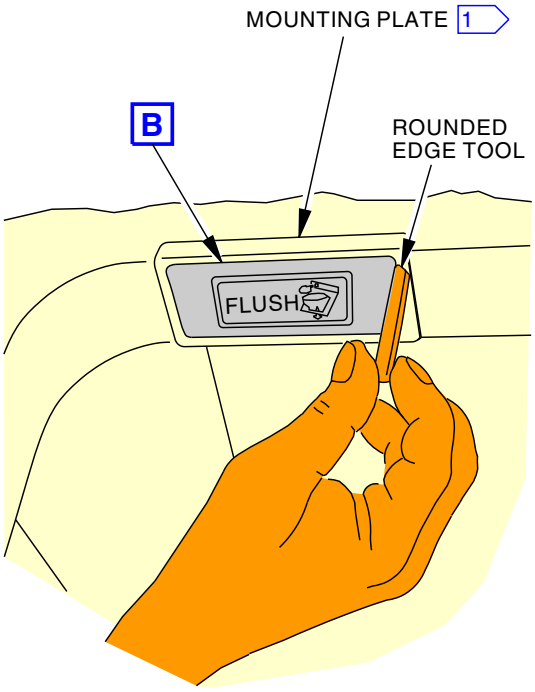
(b) Disconnect the electrical connector [2] from the flush switch [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

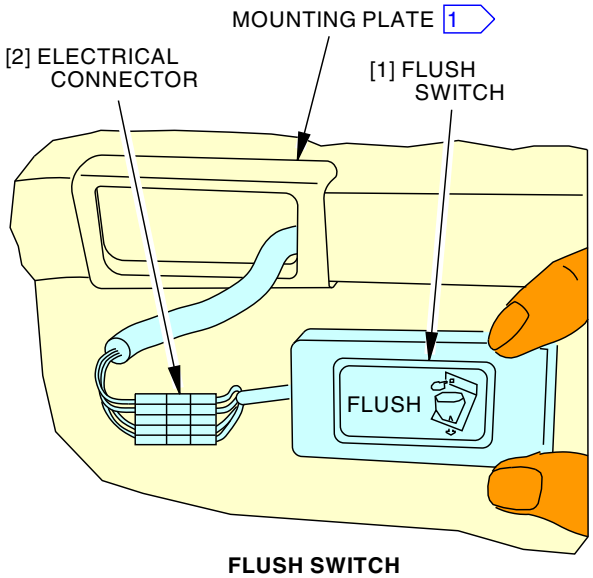


LAVATORY
(EXAMPLE)



FLUSH SWITCH

A



FLUSH SWITCH

B

1 IF INSTALLED

2827057 S0000655016_V2

Flush Switch Installation
Figure 401/38-32-04-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-04-400-804

3. Flush Switch Installation

(Figure 401)

A. General

(1) This task is for the installation of the vacuum waste flush switch.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
200	Upper Half of Fuselage

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Flush Switch Installation

SUBTASK 38-32-04-420-003

(1) Do these steps to install the standard flush switch [1]:

(a) Connect the flush switch [1] to the electrical connector [2].

(b) Push the flush switch [1] into its position on the switch mounting panel or mounting plate.

E. Flush Switch Installation Test

SUBTASK 38-32-04-860-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-04-410-004

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-04-710-002

(3) Activate the flush switch [1] to make sure that the toilet assembly is operational.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VACUUM BLOWER - REMOVAL/INSTALLATION

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has these tasks:

(1) Vacuum blower removal

(2) Vacuum blower installation

(3) Vacuum blower filter restore (clean) or replace.

NOTE: There are two possible vacuum blowers types. Vacuum blower, part number 606802–2, has a filter. Vacuum blower, part number 645172–2, does not have a filter. The two vacuum blower types are interchangeable.

TASK 38-32-05-000-801

2. Vacuum Blower Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the vacuum blower.

B. References

Reference	Title
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-32-05-010-001

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-05-010-002

(2) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-05-010-007

(3) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-05-040-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-05-910-001

(5) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

G. Vacuum Blower Removal

SUBTASK 38-32-05-020-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [8] from the vacuum blower [2] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [8] from the vacuum blower [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Install the protective covers on the electrical receptacle of the vacuum blower [2] and electrical connector [8].

SUBTASK 38-32-05-020-002

(2) To disconnect the vacuum blower [2], do these steps:

(a) For the outlet piping for the vacuum blower, disconnect the clamps [10] to disconnect the flexible coupling [9].

(b) For the inlet piping for the vacuum blower, disconnect the clamps [1] to disconnect the flexible coupling [11].

SUBTASK 38-32-05-020-003

(3) Remove the nut to disconnect the ground wire [7] to the vacuum blower [2].

(a) Install the nut on the vacuum blower [2].

SUBTASK 38-32-05-020-004

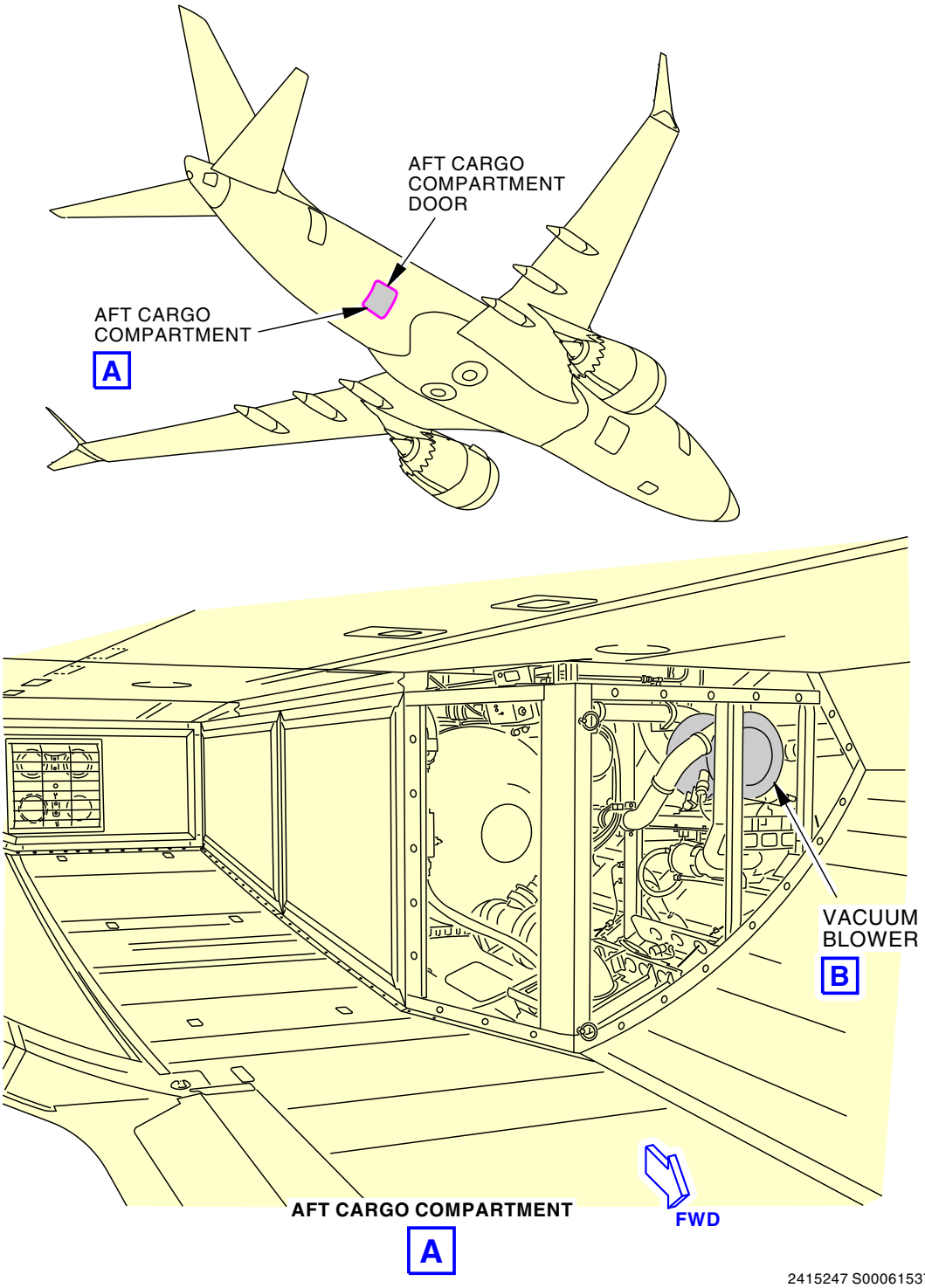
(4) Remove the bolts [3], countersunk washers [4], washers [5], and nuts [6].

SUBTASK 38-32-05-020-005

(5) Remove the vacuum blower [2].

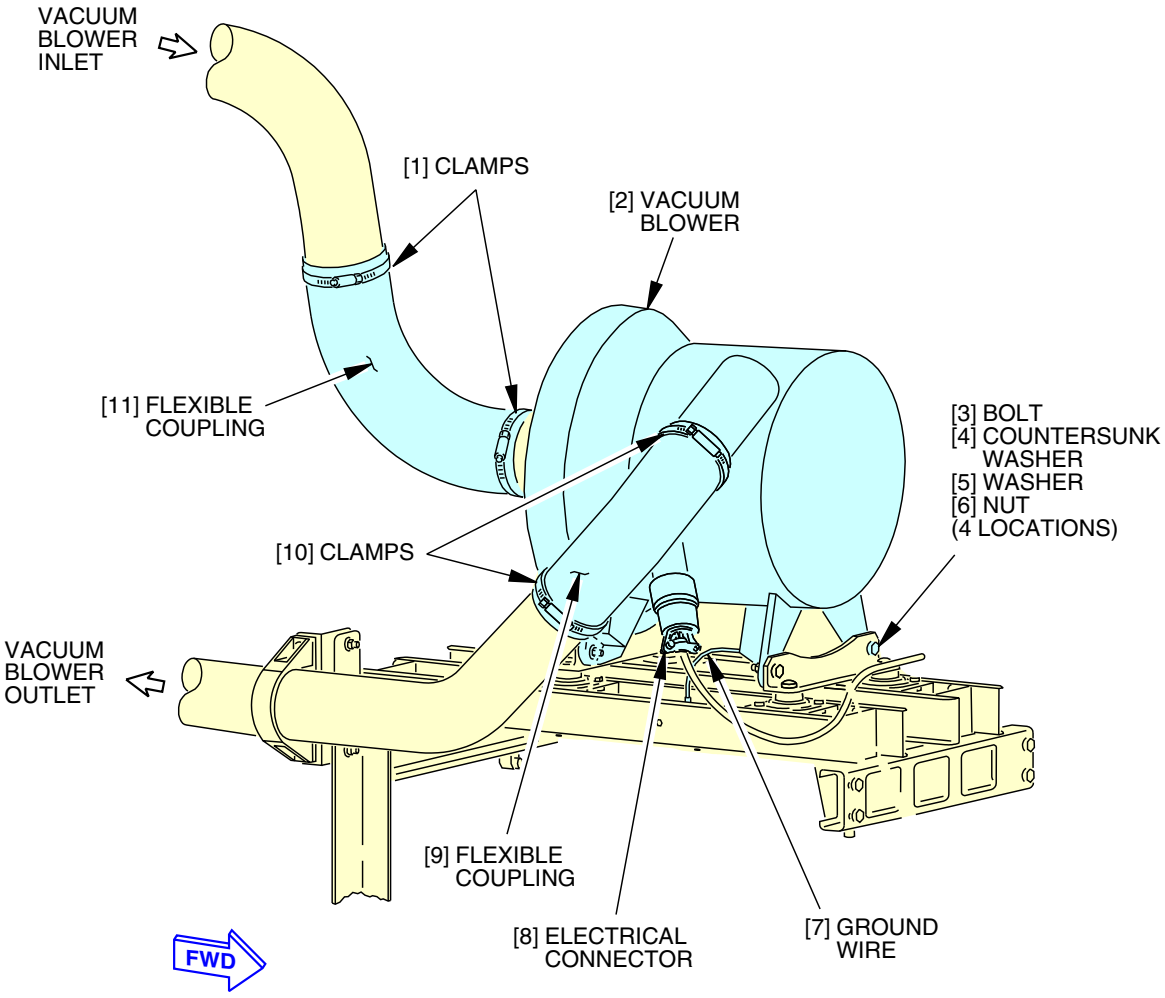
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Blower Installation
Figure 401/38-32-05-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Blower Installation
Figure 401/38-32-05-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-05-400-801

3. Vacuum Blower Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the vacuum blower.

B. References

Reference	Title
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00130	Alcohol - Isopropyl	TT-I-735
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Vacuum blower	38-32-05-01-030	PTF ALL
		38-32-05-01-205	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

H. Vacuum Blower Installation

SUBTASK 38-32-05-420-001

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Put the vacuum blower [2] in its position.

SUBTASK 38-32-05-420-002

(2) Install the bolts [3], countersunk washers [4], washers [5], and nuts [6].

NOTE: The installation of the aft bolts is with the head in the aft direction and the forward bolts with the head in the forward direction.

SUBTASK 38-32-05-420-003

(3) Connect the vacuum blower [2] as follows:

(a) For the inlet piping, connect the flexible coupling [11] with the clamps [1].

(b) For the outlet piping, connect the flexible coupling [9] with the clamps [10].

SUBTASK 38-32-05-020-006

(4) Remove the nut that attaches the ground wire [7] from the vacuum blower [2].

NOTE: The nut is on the vacuum blower.

SUBTASK 38-32-05-420-004

(5) Connect the ground wire [7] with the nut from the vacuum blower [2].

(a) Use cotton wiper, G00034, moist in alcohol, B00130, to clean the mating surfaces of the ground wire [7], vacuum blower [2], and nut.

(b) Install the nut to attach the ground wire [7] to the terminal.

(c) Tighten the nut to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

(d) Measure the electrical bonding resistance between the ground wire [7] and vacuum blower [2] structure (SWPM 20-20-00).

1) Use the non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

(e) Apply fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, to cover the nut and ground wire [7] terminal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-05-420-005

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(6) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [8] to the vacuum blower [2] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the vacuum blower [2] and electrical connector [8].

(b) Connect the electrical connector [8] to the vacuum blower [2].

1) Tighten the connector with the teflon-jawed pliers, STD-664.

I. Vacuum Blower Installation Test

SUBTASK 38-32-05-860-001

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-05-410-003

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-05-760-001

(3) Make sure that the vacuum blower operates as follows:

(a) Make sure that the vacuum blower starts when you push the flush switch for a toilet.

NOTE: If the vacuum blower operates before you push the flush switch, there is a problem in the control circuits.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the vacuum blower stops after approximately 15 seconds.

NOTE: If the vacuum blower continues to operate after approximately 15 seconds, there is a problem in the control circuits.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-05-010-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-05-410-001

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

TASK 38-32-05-960-801

4. Vacuum Blower Filter Restore (Clean) or Replace

(Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to restore (clean) or replace the vacuum blower filter.

B. References

Reference	Title
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1280	Source - Air, Regulated, Dry Filtered, 0-30 PSIG

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
31	Packing	38-32-05-01-150	PTF ALL
32	Vacuum blower filter	38-32-05-01-155	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Removal

NOTE: There are two possible vacuum blowers types. Vacuum blower, part number 606802–2, has a filter. Vacuum blower, part number 645172–2, does not have a filter. The two vacuum blower types are interchangeable.

SUBTASK 38-32-05-010-004

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-05-010-005

(2) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-05-010-008

(3) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-05-040-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

H. Vacuum Blower Filter Removal

SUBTASK 38-32-05-020-007

(1) Remove and discard the packings [31] that attach the vacuum blower filter [32] to the vacuum blower.

SUBTASK 38-32-05-020-008

(2) Remove the vacuum blower filter [32].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Vacuum Blower Filter Restore (Clean) or Replace

SUBTASK 38-32-05-420-006

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Clean the vacuum blower filter [32] with an 0-30 psig dry filtered regulated air source, STD-1280, in the opposite direction of the usual flow.

SUBTASK 38-32-05-420-007

(2) Do a check of the vacuum blower filter [32] for damage or signs of worn areas.

(a) If the vacuum blower filter [32] has damage or signs of worn areas, use a new vacuum blower filter [32].

(b) If the vacuum blower filter [32] is serviceable, then continue.

J. Vacuum Blower Filter Installation

SUBTASK 38-32-05-420-008

(1) Put the vacuum blower filter [32] in its position.

SUBTASK 38-32-05-420-009

(2) Install the new packings [31] that attach the vacuum blower filter [32] to the vacuum blower.

K. Vacuum Blower Filter Installation Test

SUBTASK 38-32-05-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-05-410-004

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-05-760-002

(3) Make sure that the vacuum blower operates as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the vacuum blower starts when you push the flush switch for a toilet.

NOTE: If the vacuum blower operates before you push the flush switch, there is a problem in the control circuits.

(b) Make sure that the vacuum blower stops after approximately 15 seconds.

NOTE: If the vacuum blower continues to operate after approximately 15 seconds, there is a problem in the control circuits.

L. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-05-010-006

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

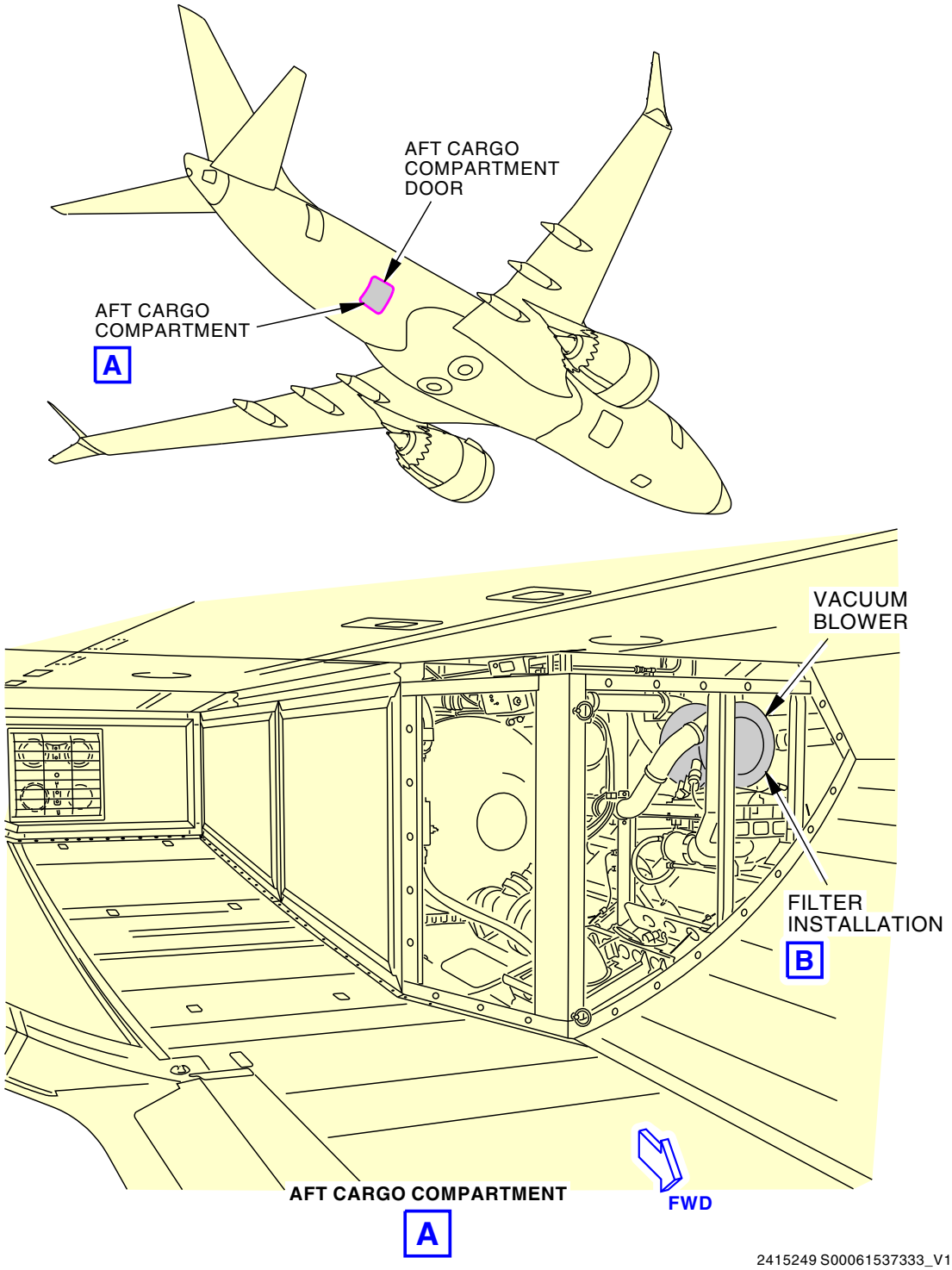
SUBTASK 38-32-05-410-002

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

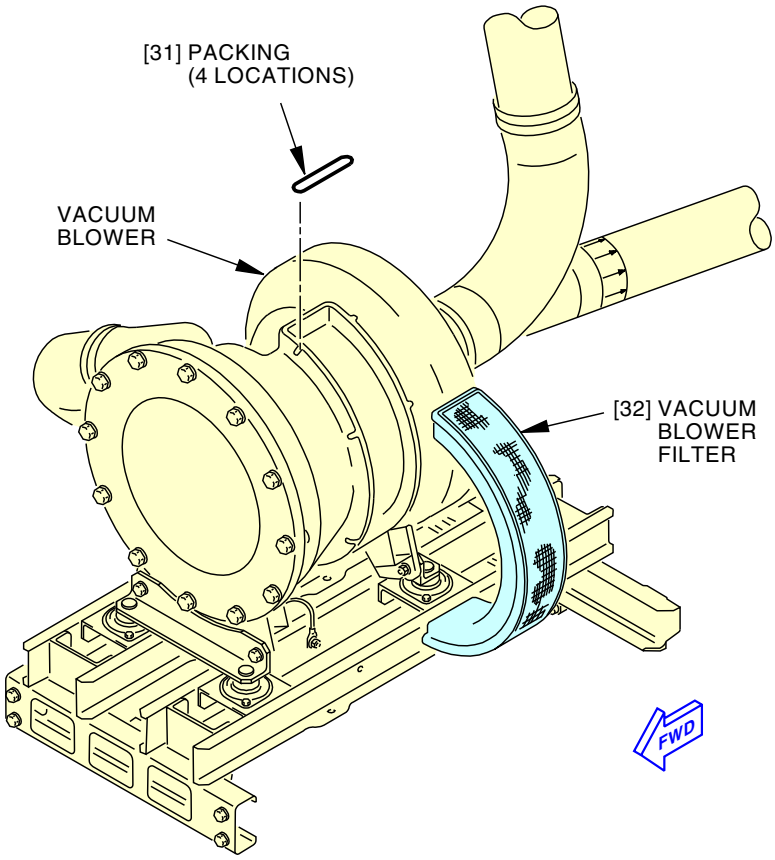
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Blower Filter Installation
Figure 402/38-32-05-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FILTER INSTALLATION

B

2415250 S00061537334_V1

Vacuum Blower Filter Installation
Figure 402/38-32-05-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VACUUM CHECK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Vacuum check valve removal

(2) Vacuum check valve installation.

B. The Vacuum Check Valve is referred to as the Check Valve in this procedure.

TASK 38-32-06-000-801

2. Vacuum Check Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the vacuum check valve.

B. References

Reference	Title
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-32-06-010-001

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-06-010-002

(2) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-06-010-003

(3) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-06-040-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-06-910-001

(5) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

F. Vacuum Check Valve Removal

SUBTASK 38-32-06-020-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

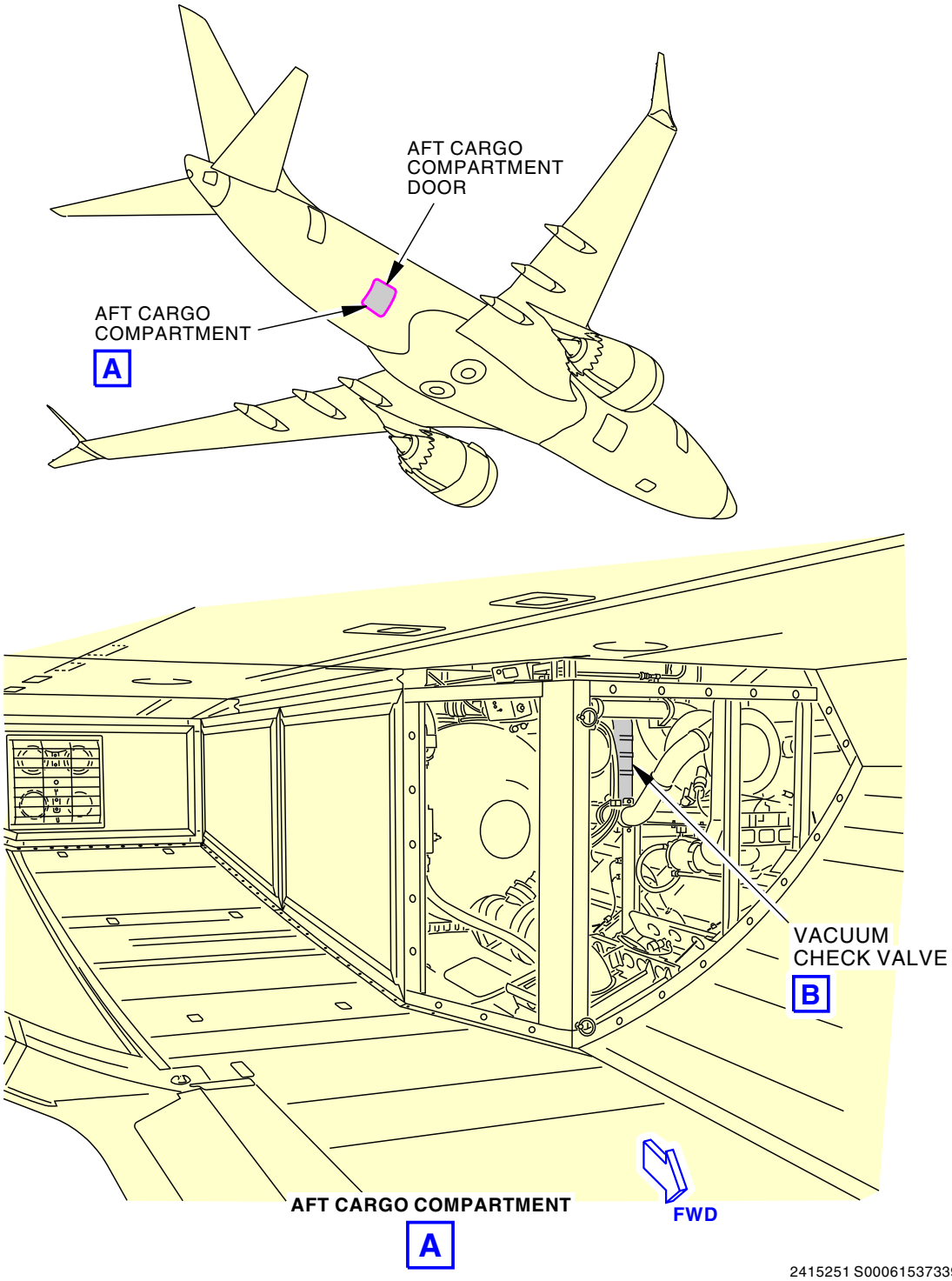
(1) Loosen the clamps [2] to disconnect the flexible coupling [1] from the check valve [3].

SUBTASK 38-32-06-020-002

(2) Remove the check valve [3].

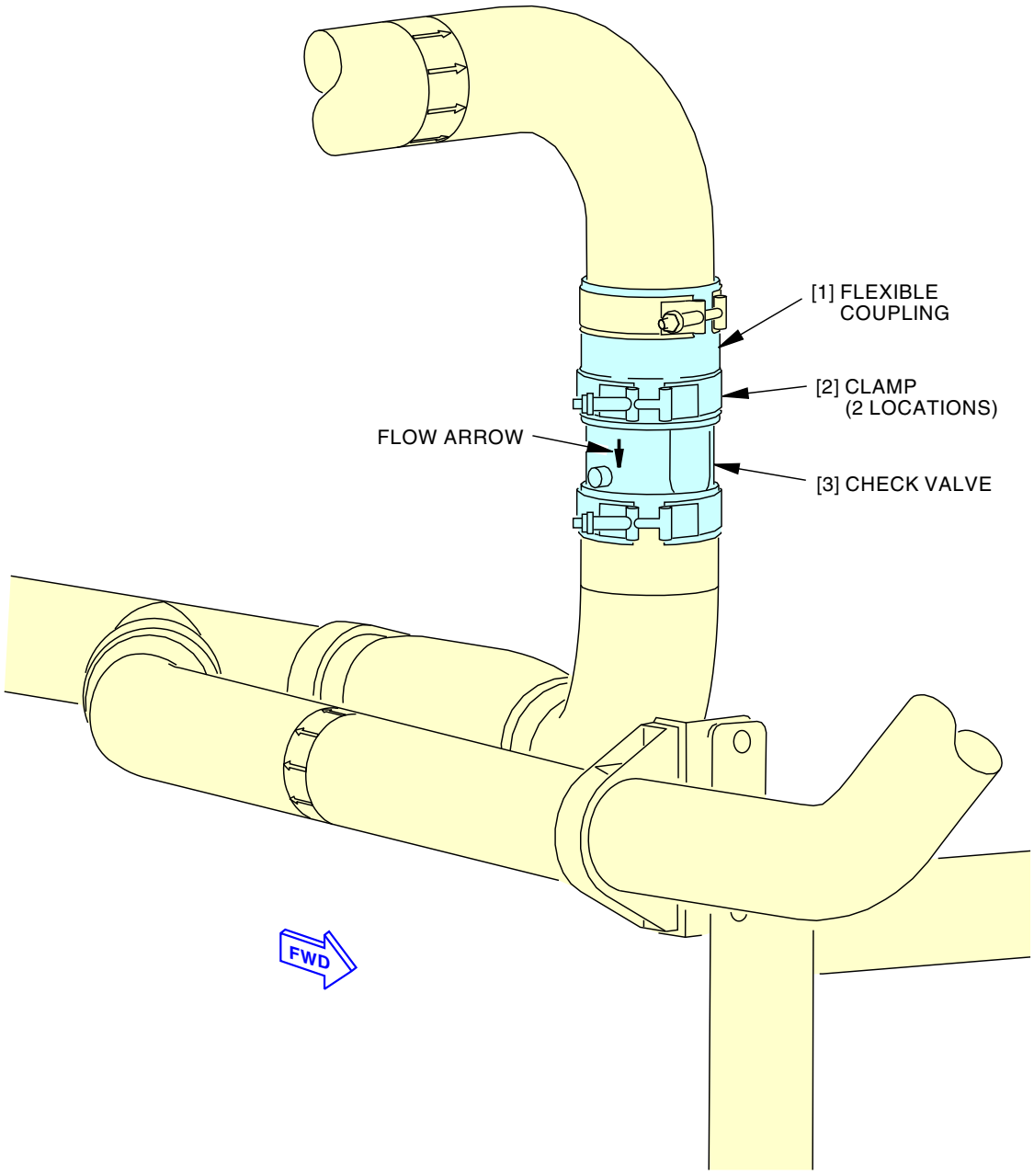
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Check Valve Installation
Figure 401/38-32-06-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



VACUUM CHECK VALVE

2415252 S00061537340_V1

Vacuum Check Valve Installation
Figure 401/38-32-06-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-06-400-801

3. Vacuum Check Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the vacuum check valve.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Check valve	38-32-06-01-440	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Vacuum Check Valve Installation

SUBTASK 38-32-06-420-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Put the check valve [3] in its position with the flow arrow in the down direction.

SUBTASK 38-32-06-420-002

(2) Connect the flexible coupling [1] with the clamps [2].

SUBTASK 38-32-06-420-003

(3) Tighten the clamps [2] to 17 ±1 in-lb (1.92 ±0.11 N·m).

G. Vacuum Check Valve Installation Test

SUBTASK 38-32-06-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-06-860-002

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801

SUBTASK 38-32-06-710-001

(3) Flush a toilet in the waste system.

SUBTASK 38-32-06-790-001

(4) Make sure that there is no leakage at the check valve connections.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-06-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-06-410-002

(2) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the waste tank

(2) An installation of the waste tank

(3) A cleaning of the waste tank.

TASK 38-32-07-000-801

2. Waste Tank Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove the waste tank.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-32-07-610-001

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-32-07-010-001

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-07-010-002

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-07-010-003

(4) Remove the pins [11] to disconnect the post [12] for removal access.

SUBTASK 38-32-07-040-001

(5) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-32-07-010-007

(6) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-07-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(7) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-07-910-001

(8) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

G. Waste Tank Removal

SUBTASK 38-32-07-020-001

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector from the point level sensor (TASK 70-00-01-910-803-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect the electrical connector from the point level sensor.

(b) Install the protective covers on the electrical receptacle of the point level sensor and the electrical connector.

SUBTASK 38-32-07-020-002

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(2) Remove the nuts [6] and washers [7] on the inboard side of the waste tank [28] to disconnect the bonding jumper [5].

SUBTASK 38-32-07-020-003

(3) Loosen the clamp [13] to disconnect the flexible coupling [14].

SUBTASK 38-32-07-020-004

(4) Disconnect the rinse hoses from the rinse nozzles.

SUBTASK 38-32-07-020-005

(5) Remove the clamshells [4], sleeve [3], and packings [29] to disconnect the waste elbows [2] from the waste line.

(a) Discard the packing [29].

SUBTASK 38-32-07-010-004

(6) Remove the vent tubes and the waste tubes above the waste tank [28] for removal access.

SUBTASK 38-32-07-020-006

(7) Remove the clamshell [8], sleeve [9], and packings [10] to disconnect the waste tube from the waste tank [28] outlet.

(a) To remove the sleeve [9] follow the procedure below:

1) Obtain a wood or plastic block to allow tapping on the edge of the sleeve [9] with a hammer.

2) Tap around the periphery, from forward or aft direction, whichever provides easiest access.

3) Once the sleeve [9] has slid off one of the o-rings, remove the sleeve [9] by hand.

(b) Discard the packing [10].

SUBTASK 38-32-07-020-007

(8) Loosen the strap [1] to disconnect the aft end of the waste tank [28].

SUBTASK 38-32-07-020-008

(9) Remove the bolt [20], washer [21], bushing [22], bushing [25], washer [23], and nut [24] to disconnect the tie rod assembly [26].

SUBTASK 38-32-07-020-009

(10) Remove the nuts [30], washers [31], and bolts [32] to disconnect the bracket [33] from the forward mount.

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SUBTASK 38-32-07-020-010

WARNING

BE CAREFUL WHEN YOU MOVE THE WASTE TANK. THE WASTE TANK IS HEAVY. INJURIES TO PERSONNEL CAN OCCUR.

(11) Remove the waste tank [28] from its position.

NOTE: The waste tank weighs more than 60 lb (27 kg).

SUBTASK 38-32-07-020-011

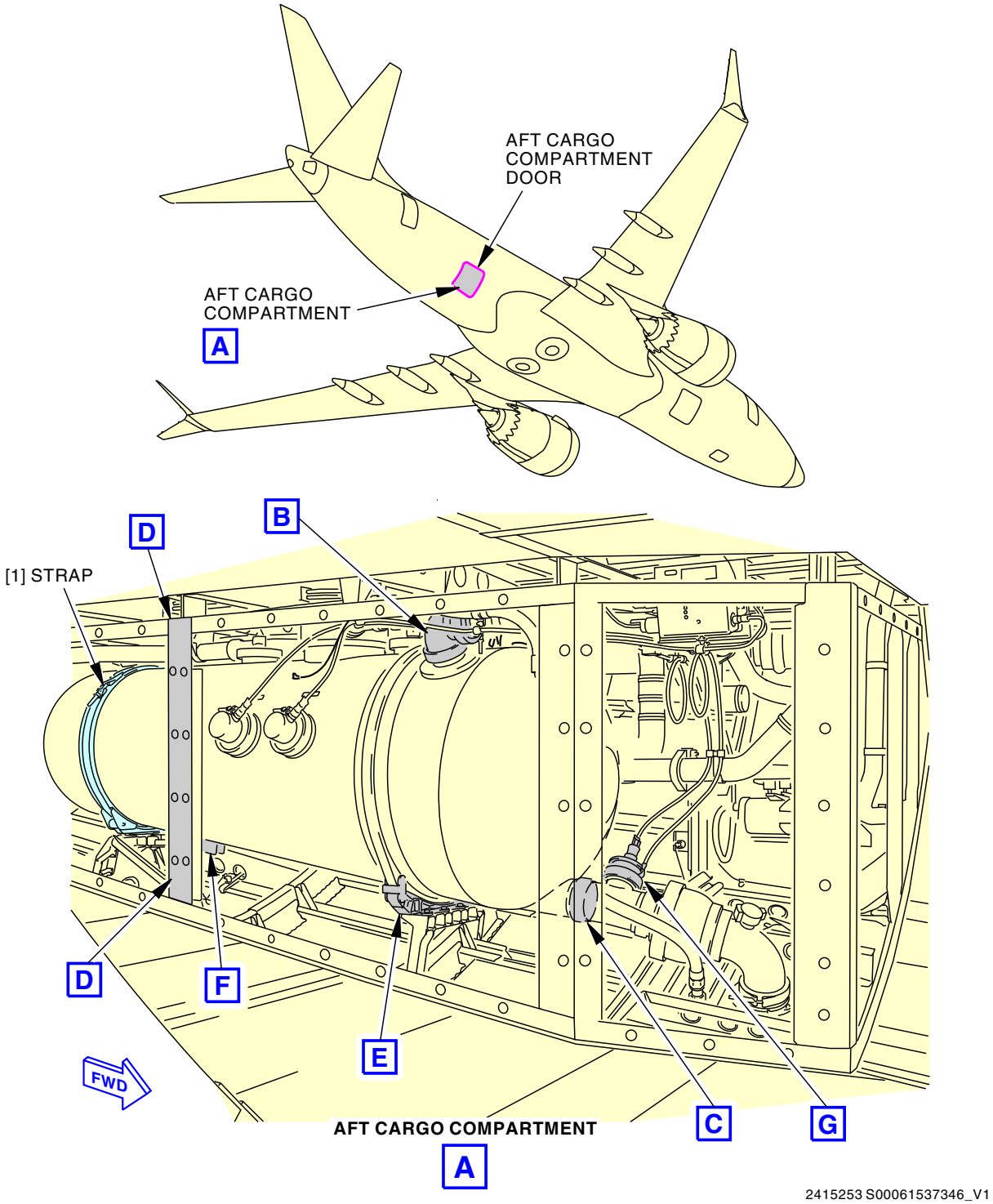
(12) To replace the waste tank [28] with a new waste tank [28], remove the bracket [33].

(a) Remove the bolts [16], nuts [18], washers [15], and washers [35] to disconnect the link [19] from the bracket [33] and waste tank [28].

(b) Remove the bolt [17], nut [18], washers [15], and bracket [33] from the waste tank [28].

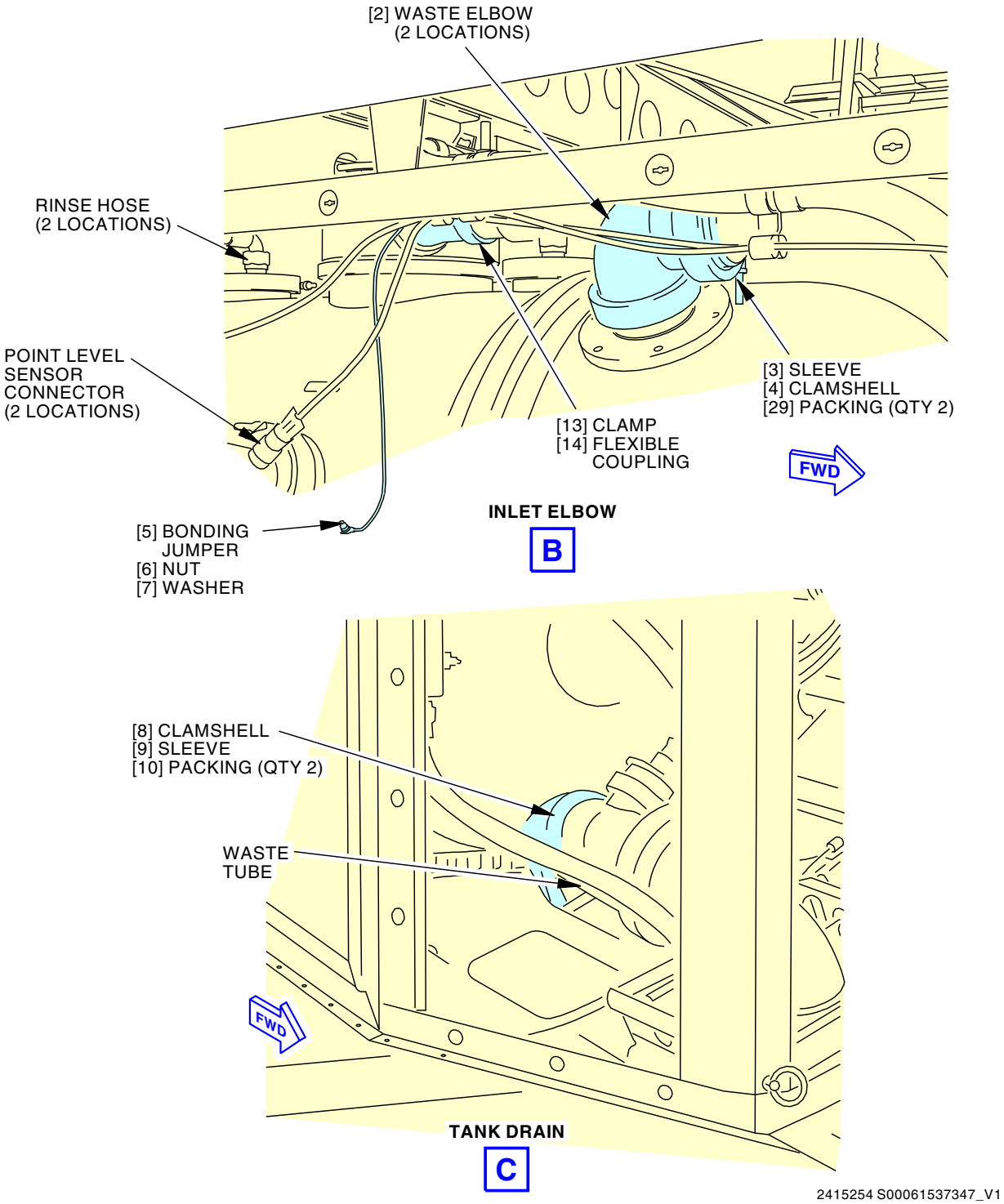
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



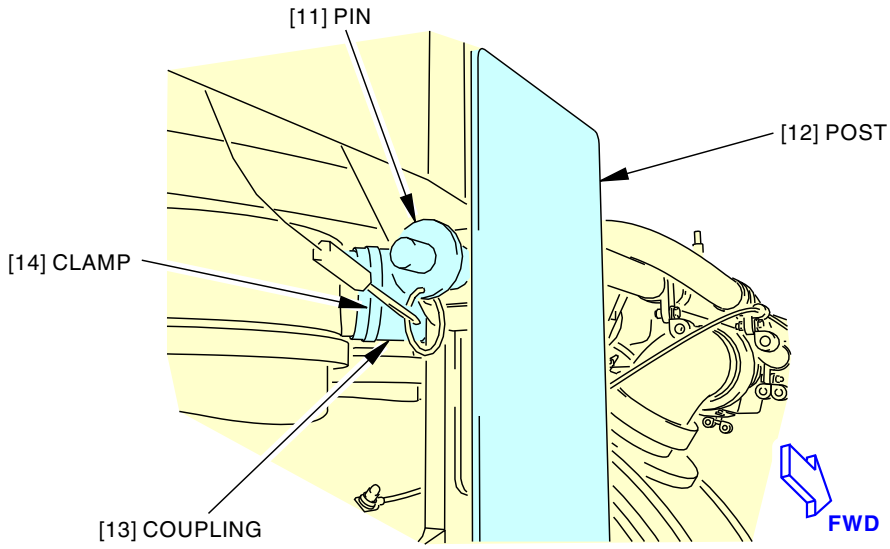
Waste Tank Installation
Figure 201/38-32-07-990-802 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



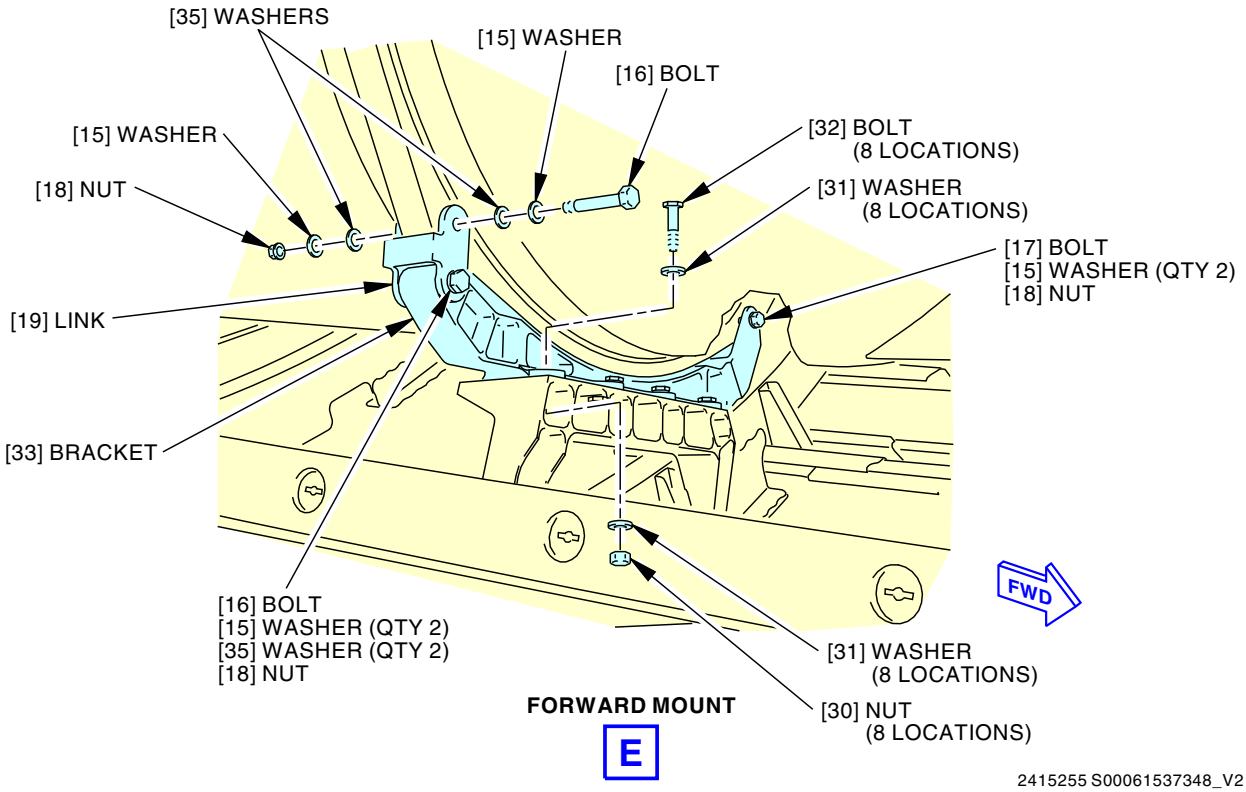
Waste Tank Installation
Figure 201/38-32-07-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POST RELEASE PIN (EXAMPLE)

D



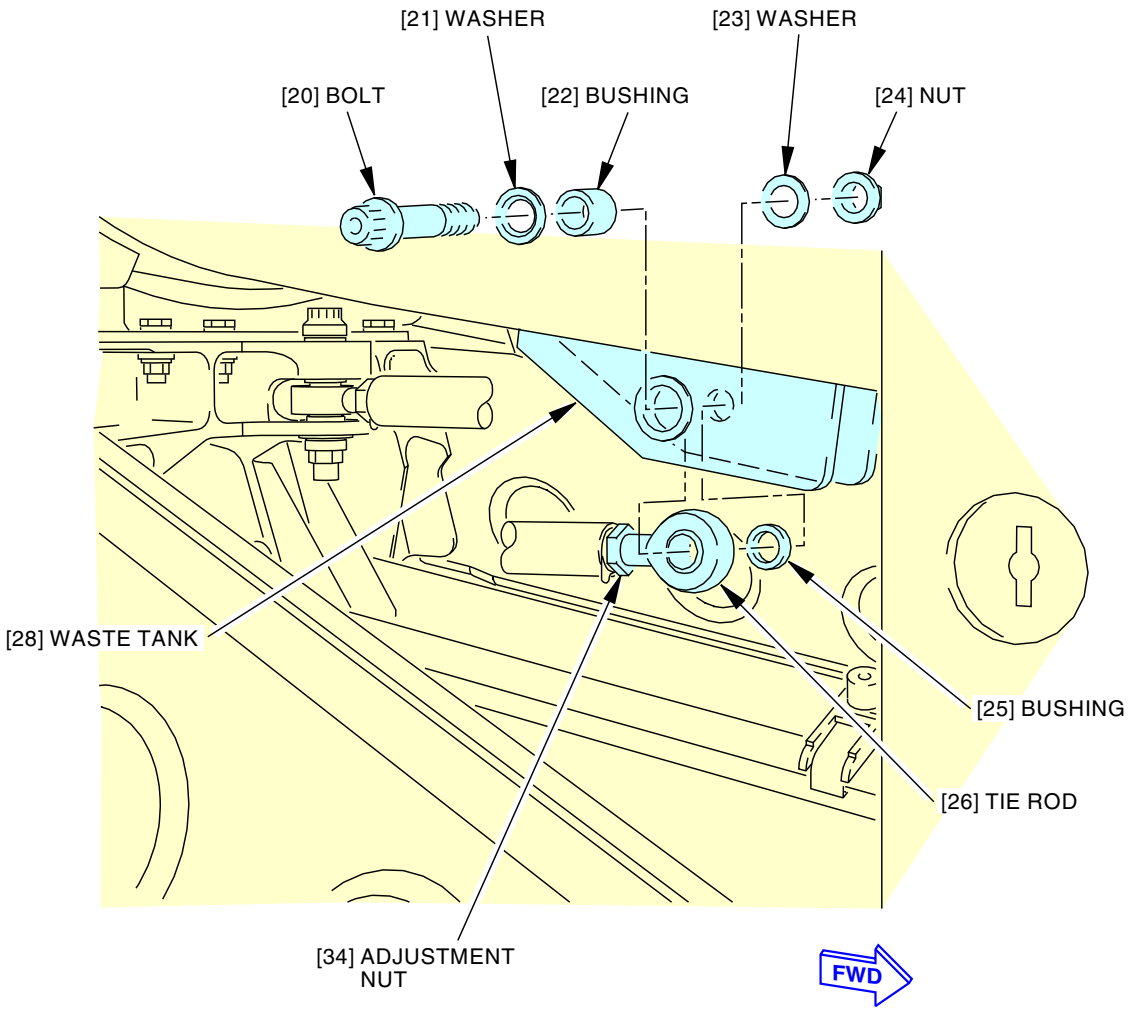
FORWARD MOUNT

E

2415255 S00061537348_V2

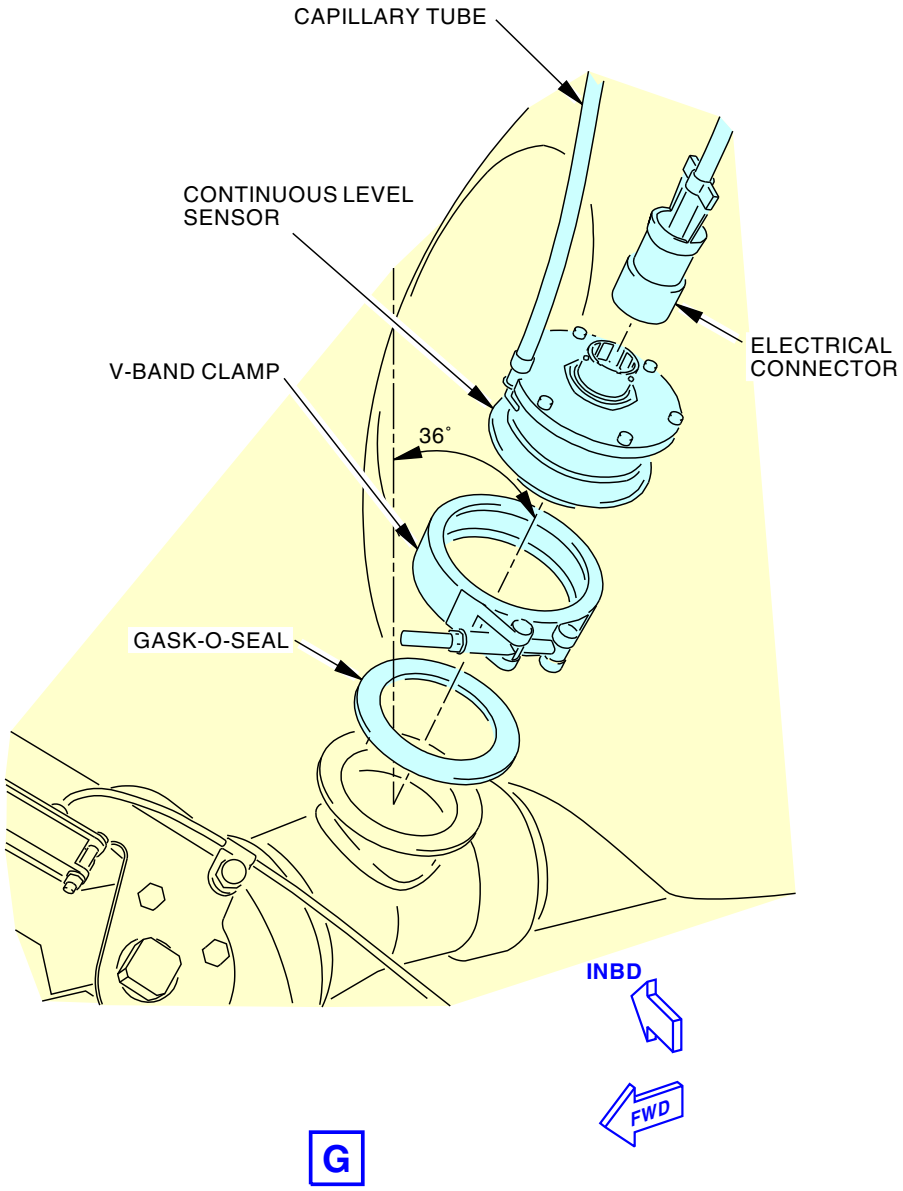
Waste Tank Installation
Figure 201/38-32-07-990-802 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Tank Installation
Figure 201/38-32-07-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2415257 S00061537350_V1

Waste Tank Installation
Figure 201/38-32-07-990-802 (Sheet 5 of 5)

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TASK 38-32-07-400-801

3. Waste Tank Installation

(Figure 201)

A. General

(1) This task gives the instructions to install the waste tank.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
38-33-00-710-801	Auto Zero Test (P/B 501)
38-33-00-740-802	LCM BITE Test (P/B 501)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-9123	Sleeve Installation, Waste Tank Part #: C38003-1 Supplier: 81205
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-1142	Equipment - Waste System Servicing

D. Consumable Materials

Reference	Description	Specification
D00504	Grease - Petrolatum	VV-P-236
D50007	Grease - Silicone-Based, Generic	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	Packing	38-32-03-02-050	PTF ALL
		38-32-03-02-255	PTF ALL
28	Waste tank	38-32-07-01-170	PTF ALL
29	Packing	38-32-07-01-015	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Waste Tank Installation

SUBTASK 38-32-07-420-001

(1) Do these steps before tank installation if the bracket [33] has been removed from the waste tank [28]:

(a) Install the bolt [17], nut [18], washers [15], and bracket [33] to the waste tank [28].

(b) Install the bolts [16], nuts [18], washers [15], and washers [35] to connect the link [19] to the bracket [33] and waste tank [28].

SUBTASK 38-32-07-640-001

(2) Apply grease, D00504, or silicone-based grease, D50007, to the new packings [10].

SUBTASK 38-32-07-420-002

(3) Put the packings [10] in their position to connect the waste tube to the waste tank [28] outlet.

SUBTASK 38-32-07-420-003

(4) Put the sleeve [9] on the flange of the waste tube.

SUBTASK 38-32-07-420-004

WARNING BE CAREFUL WHEN YOU MOVE THE WASTE TANK. THE WASTE TANK IS HEAVY. INJURIES TO PERSONNEL CAN OCCUR.

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(5) Put the waste tank [28] in its position and loosely install the nuts [30], washers [31], and bolts [32].

NOTE: The waste tank weighs more than 60 lb (27 kg).

SUBTASK 38-32-07-430-001

(6) Do the following steps to adjust the clearance for the waste tank [28] outlet connection.

(a) Make sure that the waste tube is positioned correctly on the waste tank [28] outlet flange.

NOTE: The center of the continuous sensor mount should be at 36 degrees outboard, + or - 2 degrees, from vertical of the waste tank outlet. Accuracy of the waste tube position affects the clearance dimensions between the waste tube and the ball valve.

(b) Adjust the tank in a forward or aft direction to obtain a nominal clearance of 0.15 in. (3.81 mm) or minimum clearance of 0.10 in. (2.54 mm) between the ends of the waste tube flange and the waste tank [28] outlet flange.

NOTE: If a minimum clearance of 0.10 in. (2.54 mm) is not obtained damage to the waste tank flange may occur. A maximum clearance is limited by the proper installation of the clamshell clamp. The clamp must not bind, or bend when installed. All of the clamp latches must close.

SUBTASK 38-32-07-430-002

(7) Tighten the nuts [30], washers [31], and bolts [32].

SUBTASK 38-32-07-420-005

(8) Install the clamshell [8] and sleeve [9] to connect the waste tank [28] outlet.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) To install the sleeve [9] with the sleeve, SPL-9123, follow the procedure below:

NOTE: The sleeve, SPL-9123, is a P/N 14C02–56 clamp with one of the lips machined off to allow the clamp to be slipped on the sleeve while the clamp is closed.

NOTE: Use of the modified clamp tool, sleeve, SPL-9123, is optional.

1) Attach the sleeve, SPL-9123, to the sleeve [9] and lock.

NOTE: The inner seat of the sleeve, SPL-9123, must sit against the edge of the sleeve.

2) Carefully tap around the periphery of the tool until the sleeve [9] moves over the second o-ring.

NOTE: The sleeve and o-rings must not be damaged or deformed during this step. Damage or deformation shows that alignment or clearances are not correct.

3) Remove the sleeve, SPL-9123.

4) Install the clamshell [8].

(b) To install the sleeve [9] without the sleeve, SPL-9123, follow the procedure below:

1) Obtain a wood or a plastic block to allow tapping on the edge of the sleeve [9].

2) Carefully tap around the periphery of the sleeve [9] until it moves over the second o-ring.

NOTE: The sleeve and o-rings must not be damaged or deformed during this step. Damage or deformation shows that alignment or clearances are not correct.

3) Install the clamshell [8]

SUBTASK 38-32-07-420-006

(9) Install the bolt [20], washer [21], bushing [22], bushing [25], washer [23], and nut [24] to connect the tie rod assembly [26].

SUBTASK 38-32-07-430-003

(10) Adjust the tie rod assembly [26] as required and lockwire the adjustment nut [34].

SUBTASK 38-32-07-420-007

(11) Install the strap [1] for the aft end of the waste tank [28].

SUBTASK 38-32-07-640-002

(12) Apply grease, D00504, or silicone-based grease, D50007, to the new packings [29].

SUBTASK 38-32-07-420-008

(13) Put the packings [29] in their position to connect the waste elbows [2] to the vacuum waste lines.

SUBTASK 38-32-07-420-009

(14) Install the sleeves [3] and clamshells [4] to connect the waste elbows [2].

SUBTASK 38-32-07-420-010

(15) Connect the rinse hoses to the rinse nozzles.

SUBTASK 38-32-07-420-011

(16) Tighten the clamp [13] to connect the flexible coupling [14].

SUBTASK 38-32-07-420-012

(17) Install the nuts [6] and washers [7] on the inboard side of the waste tank [28] to connect the bonding jumper [5].

(a) Apply sealant to both sides of the bonding jumper (SWPM 20-20-00).

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SUBTASK 38-32-07-420-013

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(18) Use the teflon-jawed pliers, STD-664, to connect the electrical connector to the point level sensor (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the point level sensor and the electrical connector.

(b) Connect the electrical connector to the point level sensor.

1) Tighten the connector with the teflon-jawed pliers, STD-664.

I. Waste Tank Installation Test

SUBTASK 38-32-07-860-004

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-07-410-007

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-07-860-001

(3) Remove safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-07-860-002

(4) Do this task: Supply Electrical Power, TASK 24-22-00-860-801

SUBTASK 38-32-07-790-001

(5) To do a leak check of the waste tank drain line connections, do these steps:

(a) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

(b) At the service panel, make sure that the control handle for the drain ball valve is in the closed position.

(c) Make sure that the cap for the waste drain fitting outlet is closed.

(d) Connect the water hose from the waste system servicing equipment, STD-1142, to the service panel drain valve assembly.

(e) Use the pump on the service cart to put about 50 gal (189 l) of water into the waste tank [28].

(f) Make sure that the waste tank drain line connections do not have a leak.

NOTE: You must wait for not less than 30 minutes after you put the water in the tank before you do this step.

SUBTASK 38-32-07-610-002

(6) Do this task: LCM BITE Test, TASK 38-33-00-740-802.

SUBTASK 38-32-07-610-003

(7) Do this task: Auto Zero Test, TASK 38-33-00-710-801.

SUBTASK 38-32-07-610-004

(8) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-07-610-005

(9) Flush a toilet attached to the waste tank [28].

SUBTASK 38-32-07-790-002

(10) Make sure that there are no leaks at the waste tank connections.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-07-410-001

(1) Install the pins [11] to connect the post [12].

SUBTASK 38-32-07-410-002

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-07-410-003

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-07-100-801

4. Waste Tank Cleaning

A. General

(1) This task gives the instructions to clean the waste tank.

(2) Use these methods to clean the waste tank:

(a) Clean the waste tank (in the airplane) with the cleaner tool

(b) Soak the waste tank with HOT water plus a cleaning solution.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-33-01-000-801	Waste Tank Point Level Sensor Removal (P/B 401)
38-33-01-400-801	Waste Tank Point Level Sensor Installation (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1944	Cleaner - Lavatory Waste Tank Part #: A38009-11 Supplier: 81205
STD-1088	Source - Cold Water, Regulated, 0-60 PSIG
STD-1142	Equipment - Waste System Servicing

D. Consumable Materials

Reference	Description	Specification
B00638	Cleaner - Acidic Liquid - Honey Bee 60 (McGean-Rohco)	
B50117	Detergent - General	
B50329	Cleaner - Acidic Liquid - Honey Bee 60 T	
B50330	Cleaner - Acidic Liquid - Honey Bee 60 TNP	

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Cleaning

SUBTASK 38-32-07-610-006

WARNING ALWAYS PUT ON RUBBER GLOVES BEFORE YOU DO MAINTENANCE ON THE TOILET SYSTEM, OR TOUCH PARTS THAT TOUCHED WASTE MATERIAL. FULLY CLEAN YOUR HANDS WITH SOAP AND WATER AFTER MAINTENANCE IS COMPLETED. THE TOILET WASTE CAN CAUSE ILLNESS AND INJURIES TO PERSONNEL.

WARNING DO NOT TOUCH THE POTABLE WATER SYSTEM AFTER YOU DO SERVICING OR MAINTENANCE TO THE WASTE SYSTEM. AT NO TIME LET THE SAME PERSONNEL DO WORK ON THE TWO SYSTEMS AT THE SAME TIME. CHANGE YOUR CLOTHES, AND CLEAN YOUR HANDS FULLY WITH SOAP AND WATER BEFORE YOU TOUCH THE POTABLE WATER SYSTEM. CONTAMINATION OF THE POTABLE WATER SYSTEM CAN CAUSE ILLNESS AND INJURIES TO PERSONS THAT DRINK THE WATER.

WARNING DO NOT TOUCH THE TOILET COMPONENTS OR MOVE OR SEND THEM TO A DIFFERENT LOCATION UNTIL THEY ARE CLEAN. DIRTY TOILET COMPONENTS CAN CAUSE ILLNESS AND INJURIES TO PERSONNEL.

(1) Do this task to drain and flush the waste tank, but do not add the chemical precharge fluid: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-07-010-005

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-07-010-006

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-07-040-002

(4) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-32-07-010-008

(5) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-07-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-07-910-002

(7) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

H. Clean the Waste Tank With the Cleaner Tool

SUBTASK 38-32-07-480-001

(1) Connect the drain hose as follows:

(a) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

(b) Open the cap on the waste drain valve assembly.

(c) Connect the waste drain hose from the service waste system servicing equipment, STD-1142, to the waste drain valve assembly.

(d) Open the waste drain valve assembly.

(e) Pull a handle to open the waste drain ball valve.

SUBTASK 38-32-07-480-002

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(2) To install the tank cleaner at the point level sensor, do these steps:

(a) Do this task: Waste Tank Point Level Sensor Removal, TASK 38-33-01-000-801.

(b) Install the tank lavatory waste tank cleaner, SPL-1944.

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SUBTASK 38-32-07-420-014

CAUTION

USE A CLEANING SOLUTION FROM THE WASTE-TANK TEST SPECIFICATION. OTHER CLEANING SOLUTIONS CAN CAUSE DAMAGE TO THE TANK.

(3) Connect a water source (0-60 PSIG), STD-1088, to the tank cleaner.

SUBTASK 38-32-07-140-001

(4) Make sure that the water pressure is a minimum of 45 psig (310 kPa) to the tank cleaner tool.

SUBTASK 38-32-07-140-002

CAUTION

KEEP THE WATER TEMPERATURE AND PRESSURE BELOW THE LIMITS. IF YOU USE WATER AT A HIGHER TEMPERATURE OR PRESSURE, YOU CAN CAUSE DAMAGE TO THE TANK.

(5) Make sure that water pressure is not more than 60 psig (414 kPa) and that water temperature is not more than 160°F (71°C).

NOTE: Hot water and higher water pressure will clean the waste tank faster.

SUBTASK 38-32-07-170-001

(6) Operate the tank cleaner as follows:

(a) Supply water pressure and flow to the tank cleaner tool.

NOTE: It is recommended that you operate the tank cleaner for approximately 60 minutes.

(b) Operate the tank cleaner for a minimum 15 minutes.

(c) Make sure that the waste system servicing equipment, STD-1142, has sufficient capacity to contain the waste water.

(d) Stop the operation of the tank cleaner.

SUBTASK 38-32-07-080-001

(7) Disconnect the water supply line from the tank cleaner.

SUBTASK 38-32-07-080-002

(8) To remove the tank cleaner at the point level sensor; do these steps:

(a) Remove the tank cleaner.

(b) Do a visual check of the inner side of the waste tank to make sure that the waste tank is clean.

1) If the inner side of the waste tank is not clean, do this procedure again.

(c) If the inner side of the waste tank is clean, do this task: Waste Tank Point Level Sensor Installation, TASK 38-33-01-400-801.

SUBTASK 38-32-07-080-003

(9) Disconnect the drain line as follows:

(a) Push the handle to close the waste drain ball valve.

(b) Disconnect the waste drain hose from the waste drain valve assembly.

(c) Close the waste drain valve assembly.

(d) Close the cap on the waste drain valve assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Close this access panel:

Number	Name/Location
145AL	Waste Service Door

(f) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

I. Soak the Waste Tank with Hot Water Plus a Cleaning Solution.

SUBTASK 38-32-07-680-001

(1) Drain the waste system servicing equipment, STD-1142, of all chemical precharge.

SUBTASK 38-32-07-480-003

(2) Connect the drain hose as follows:

(a) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

(b) Open the cap on the waste drain valve assembly.

(c) Connect the waste drain hose from the toilet service cart to the waste drain valve assembly.

(d) Open the waste drain valve assembly.

(e) Make sure that the waste drain ball valve is in the closed position.

SUBTASK 38-32-07-480-004

(3) Connect the rinse hose as follows:

(a) Open the cap on the rinse fitting assembly.

(b) Connect a rinse hose from the waste system servicing equipment, STD-1142, to the rinse fitting assembly.

SUBTASK 38-32-07-610-007

CAUTION

USE A CLEANING SOLUTION FROM THE WASTE-TANK TEST SPECIFICATION. OTHER CLEANING SOLUTIONS CAN CAUSE DAMAGE TO THE TANK.

(4) Fill the waste system servicing equipment, STD-1142, with a mixture of water and soap or cleaner solution. Make the cleaner solution using one of the following detergents/cleaners. Dilute the cleaner with water per the manufacturers instructions.

(a) Soap - general detergent, B50117 (comercially available).

NOTE: If you use a soap solution, it must not foam.

(b) Honey Bee 60 cleaner, B00638, Honey Bee 60 T cleaner, B50329, or Honey Bee 60 TNP cleaner, B50330.

SUBTASK 38-32-07-170-002

CAUTION

FILL THE WASTE TANK ONLY TO THE TOP OF THE WASTE TANK. TOO MUCH FLUID IN THE WASTE TANK CAN CAUSE DAMAGE TO THE VACUUM BLOWER.

(5) Fill the waste tank with 60 gallons of the disinfectant solution.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION KEEP THE WATER TEMPERATURE AND PRESSURE BELOW THE LIMITS. IF YOU USE WATER AT A HIGHER TEMPERATURE OR PRESSURE, YOU CAN CAUSE DAMAGE TO THE TANK.

(a) Make sure that water pressure is not more than 60 psig (414 kPa) and that water temperature is not more than 160°F (71°C).

(b) Make sure that the water pressure is a minimum of 30 psig (207 kPa).

NOTE: If the water pressure is less than the minimum pressure, the waste tank will not get clean.

SUBTASK 38-32-07-170-003

(6) Let the cleaning solution stay in the waste tank for 30 to 60 minutes.

SUBTASK 38-32-07-610-008

(7) Do this task to drain and flush the waste tank, but do not add the chemical precharge fluid: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-07-210-001

(8) Do a visual check of the inner side of the waste tank to make sure that the waste tank is clean.

(a) If the inner side of the waste tank is not clean, do this procedure again.

(b) If the inner side of the waste tank is clean, then continue.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-07-860-006

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-07-410-009

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-07-440-001

(3) Remove safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-32-07-610-009

(4) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-07-410-004

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(5) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-07-410-005

(6) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-07-410-006

(7) Close this access panel:

Number	Name/Location
145AL	Waste Service Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE DRAIN VALVE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Waste drain valve assembly removal

(2) Waste drain valve assembly installation.

B. The Waste Drain Valve Assembly is referred to as the Drain Valve in this procedure.

TASK 38-32-08-000-801

2. Waste Drain Valve Assembly Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the waste drain valve assembly.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
146AR	Water Service Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-32-08-610-001

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-32-08-010-001

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-08-010-002

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-08-010-003

(4) Open this access panel:

Number	Name/Location
146AR	Water Service Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Waste Drain Valve Assembly Removal

SUBTASK 38-32-08-020-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Remove the clamp [4] to disconnect the waste elbow tube [5] from the drain valve [1].

SUBTASK 38-32-08-020-002

(2) Remove the gask-O-seal [6] between the waste elbow tube [5] from the drain valve [1].

SUBTASK 38-32-08-020-003

(3) Remove the bolts [3] from the drain valve [1].

SUBTASK 38-32-08-020-004

(4) Remove the drain valve [1].

SUBTASK 38-32-08-020-005

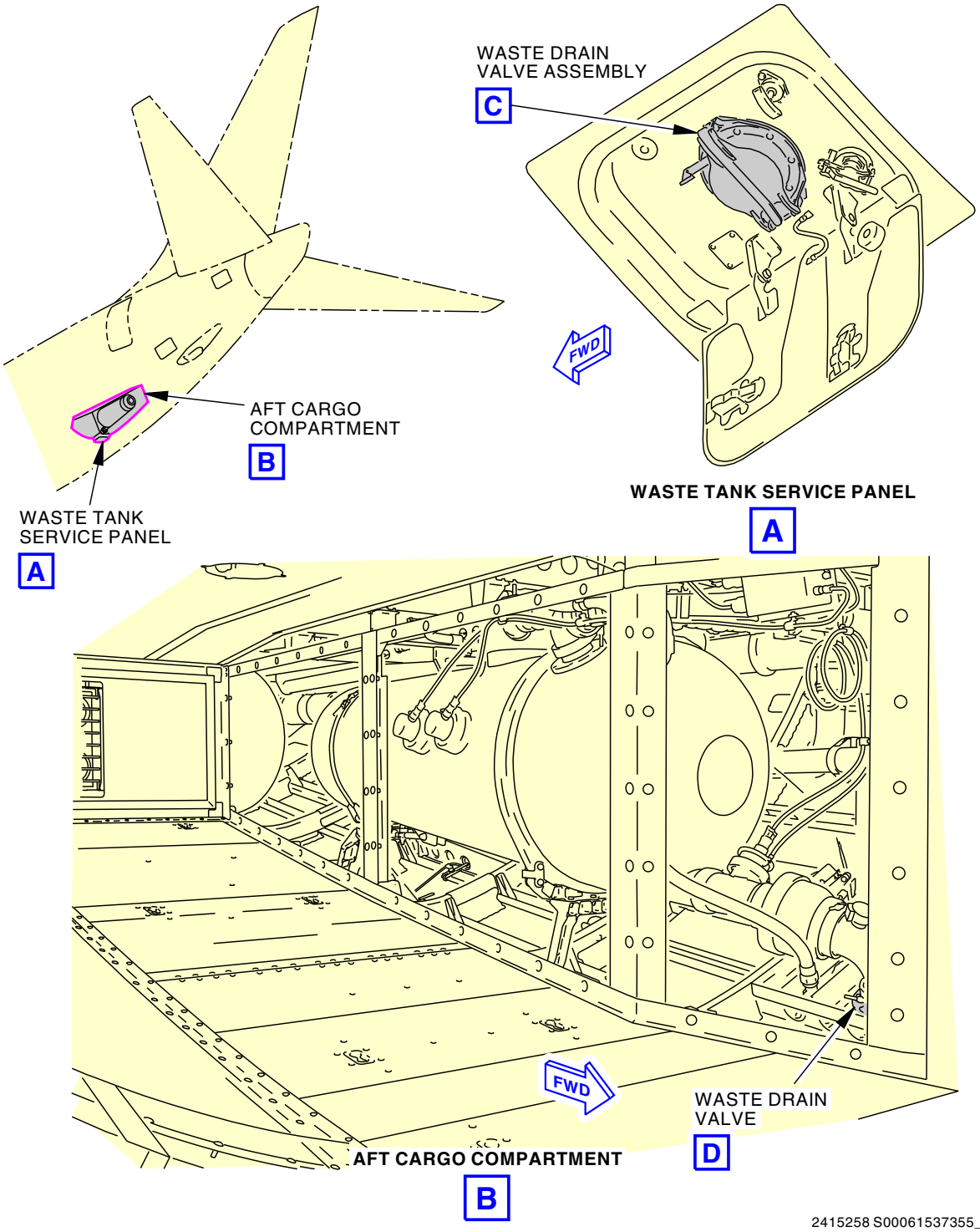
(5) Remove the toilet drain gasket [2] from the drain valve [1] or the waste tank service panel.

SUBTASK 38-32-08-020-006

(6) If it is damaged, remove the ring assembly [7].

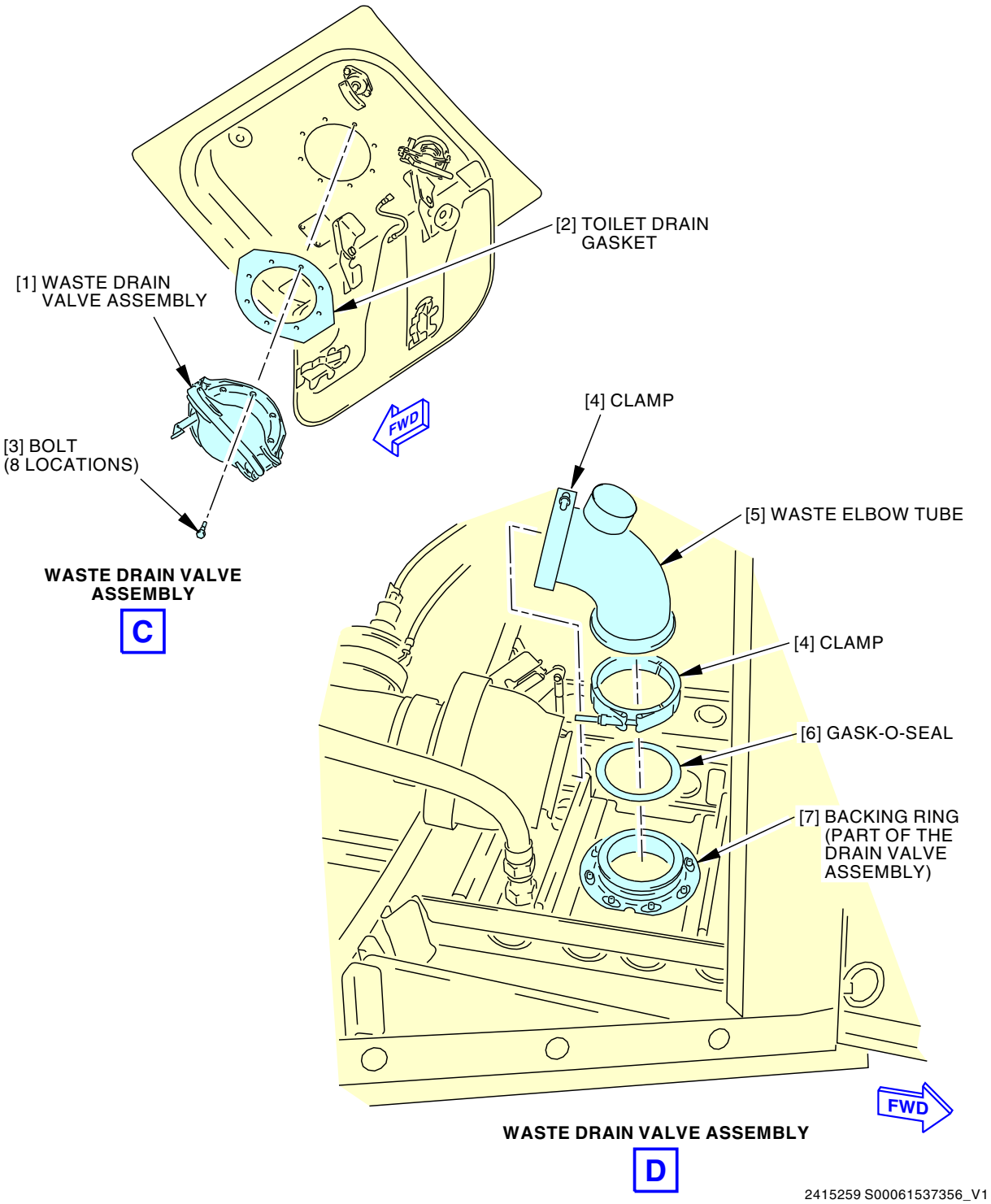
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Drain Valve Assembly Installation
Figure 401/38-32-08-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Drain Valve Assembly Installation
Figure 401/38-32-08-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-08-400-801

3. Waste Drain Valve Assembly Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the waste drain valve assembly.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1142	Equipment - Waste System Servicing

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B50073	Alcohol - Isopropyl	ASTM D 770
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Drain valve	38-32-08-01-020	PTF ALL
2	Toilet drain gasket	38-32-08-01-015	PTF ALL
6	Gask-O-seal	38-32-03-02-260	PTF ALL

F. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

G. Access Panels

Number	Name/Location
146AR	Water Service Door
822	Aft Cargo Door

H. Waste Drain Valve Assembly Installation

SUBTASK 38-32-08-420-001

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) If the ring assembly [7] was removed, apply a fay surface seal with sealant, A00247, to the waste tank service panel.

SUBTASK 38-32-08-420-002

(2) If it was removed, put the ring assembly [7] in its position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-08-110-001

CAUTION

USE ONLY ISOPROPYL ALCOHOL TO CLEAN GASKET SURFACES OF THE TOILET DRAIN. OTHER SOLVENTS TO CLEAN CAN CAUSE DAMAGE TO THE GASKET.

(3) Use cotton wiper, G00034, moist in alcohol, B50073, to clean the faying surfaces of the toilet drain gasket [2].

SUBTASK 38-32-08-390-001

(4) Apply a fay surface seal with sealant, A00247, to both sides of the toilet drain gasket [2].

SUBTASK 38-32-08-420-003

(5) Install the toilet drain gasket [2] to the waste tank service panel.

SUBTASK 38-32-08-390-002

(6) Apply a fay surface seal with sealant, A00247, to the drain valve [1].

SUBTASK 38-32-08-420-004

(7) Put the drain valve [1] in its position.

SUBTASK 38-32-08-420-005

(8) Install the bolts [3] for the drain valve [1].

SUBTASK 38-32-08-140-001

(9) Remove the unwanted sealant from the drain valve [1] and waste tank service panel.

SUBTASK 38-32-08-420-006

(10) Install the gask-O-seal [6] between the waste elbow tube [5] and drain valve [1].

SUBTASK 38-32-08-420-007

(11) Install the clamp [4] to connect the waste elbow tube [5] to the drain valve [1].

I. Waste Drain Valve Assembly Installation Test

SUBTASK 38-32-08-610-002

(1) Open the cap on the rinse fitting assembly.

SUBTASK 38-32-08-610-003

(2) Connect a rinse water hose from the service waste system servicing equipment, STD-1142, to the rinse fitting assembly.

SUBTASK 38-32-08-170-001

(3) Fill the waste tank with 30 gal (114 l) - 50 gal (189 l) of water.

SUBTASK 38-32-08-860-001

(4) Pull the handle to open the waste drain ball valve for the waste tank.

SUBTASK 38-32-08-860-002

(5) Open the cap for the drain valve assembly.

SUBTASK 38-32-08-790-001

(6) Let the water sit on the drain valve for 5 minutes.

SUBTASK 38-32-08-790-002

(7) Examine the drain valve and the connections for leakage.

SUBTASK 38-32-08-610-004

(8) Connect the drain hose from the service truck to the drain valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-08-610-005

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-08-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-08-410-002

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-08-410-003

(4) Close this access panel:

Number	Name/Location
146AR	Water Service Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK RINSE FITTING ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Waste tank rinse fitting assembly removal

(2) Waste tank rinse fitting assembly installation.

TASK 38-32-09-000-801

2. Waste Tank Rinse Fitting Assembly Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the waste tank rinse fitting assembly.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
30-71-04-000-801	Waste Tank Rinse Fitting Heater Removal (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
145AL	Waste Service Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-32-09-610-001

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-32-09-010-001

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-09-010-002

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-32-09-010-003

(4) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Waste Tank Rinse Fitting Assembly Removal

SUBTASK 38-32-09-020-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	18	C01463	WASTE/WTR LINE HEATERS

SUBTASK 38-32-09-020-002

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(2) Disconnect the rinse fitting heater [2] and rinse hose [1] from the rinse fitting assembly [3] (TASK 30-71-04-000-801).

SUBTASK 38-32-09-020-003

(3) Remove the bolts [5] that attaches the rinse fitting assembly [3] to the rinse fitting plate [4].

SUBTASK 38-32-09-020-004

(4) Remove the rinse fitting assembly [3].

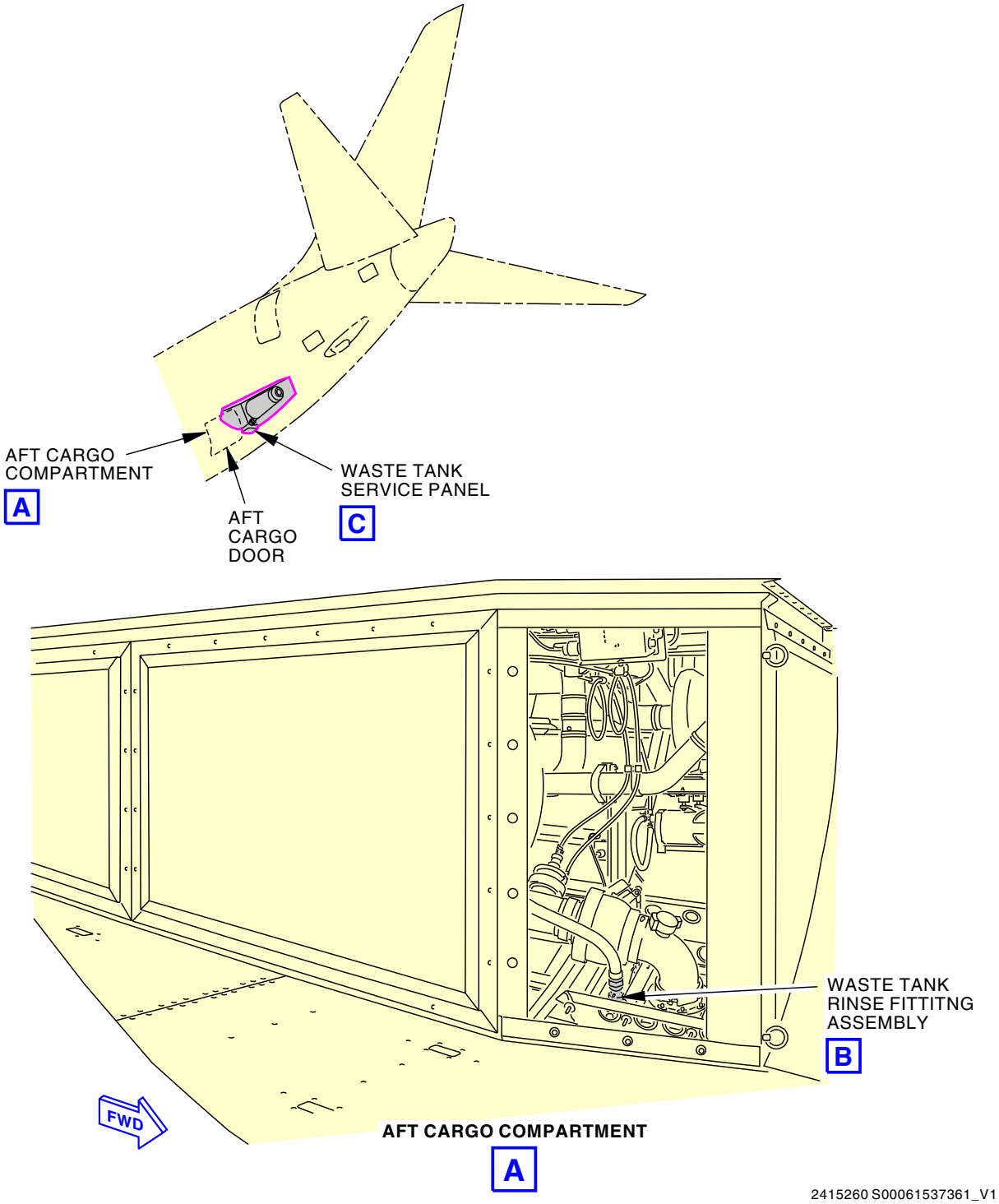
SUBTASK 38-32-09-210-001

(5) Do a check of the rinse fitting plate [4].

(a) If the rinse fitting plate [4] is not serviceable, then remove the rinse fitting plate [4].

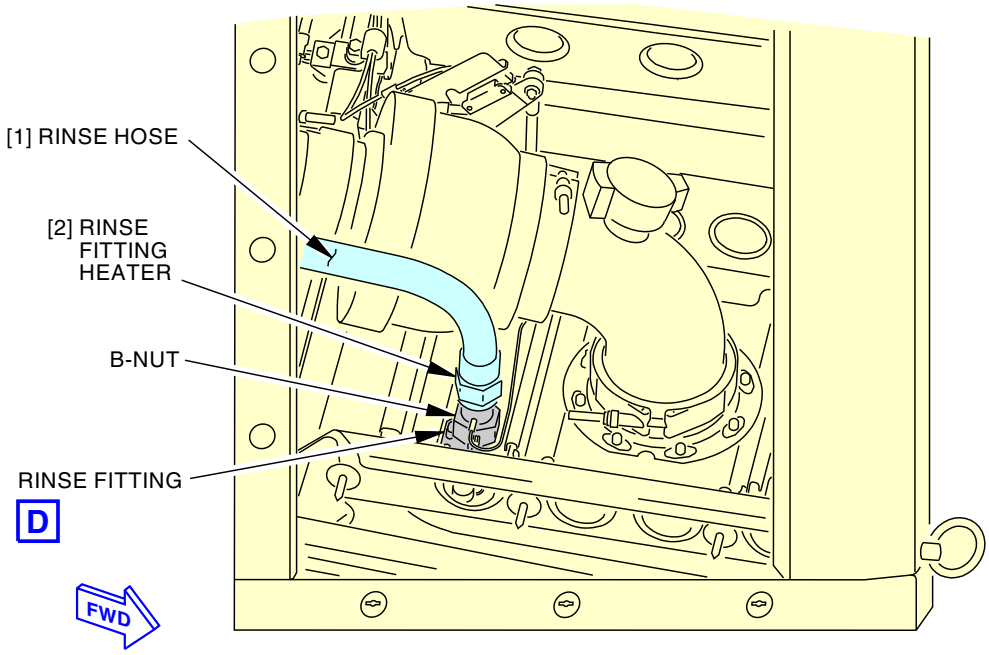
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



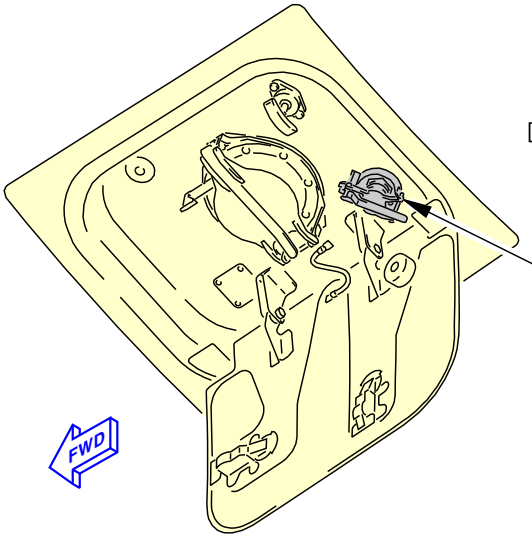
Waste Tank Rinse Fitting Assembly Installation
Figure 401/38-32-09-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



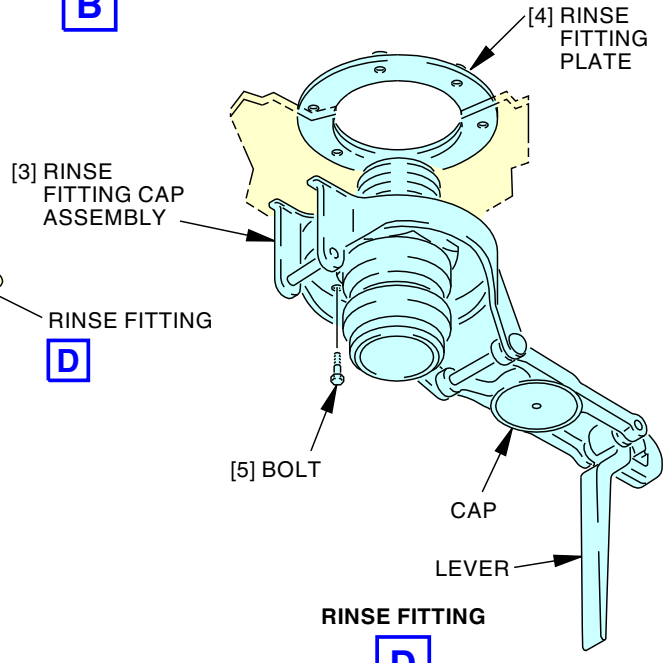
WASTE TANK RINSE FITTING ASSEMBLY (INTERNAL VIEW)

B



WASTE TANK SERVICE PANEL (EXTERNAL VIEW)

C



RINSE FITTING

D

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Waste Tank Rinse Fitting Assembly Installation
Figure 401/38-32-09-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-09-400-801

3. Waste Tank Rinse Fitting Assembly Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the waste tank rinse fitting assembly.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
30-71-04-400-801	Waste Tank Rinse Fitting Heater Installation (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1142	Equipment - Waste System Servicing

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Rinse fitting assembly	38-32-09-01-040	PTF ALL
4	Rinse fitting plate	38-32-09-01-010	PTF ALL

F. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

G. Access Panels

Number	Name/Location
145AL	Waste Service Door
822	Aft Cargo Door

H. Waste Tank Rinse Fitting Assembly Installation

SUBTASK 38-32-09-390-001

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) If the rinse fitting plate [4] is not in its position, apply a fay surface seal with sealant, A00247 to the rinse fitting plate [4].

SUBTASK 38-32-09-420-001

(2) Put the rinse fitting plate [4] in its position.

SUBTASK 38-32-09-390-002

(3) Apply a fay surface seal with sealant, A00247 to the rinse fitting assembly [3].

SUBTASK 38-32-09-420-002

(4) Put the rinse fitting assembly [3] in its position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-09-420-003

(5) Install the bolts [5] for the rinse fitting assembly [3].

SUBTASK 38-32-09-420-004

(6) Install the rinse fitting heater [2] and rinse hose [1] (TASK 30-71-04-400-801).

SUBTASK 38-32-09-860-001

(7) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	18	C01463	WASTE/WTR LINE HEATERS

I. Waste Tank Rinse Fitting Assembly Installation Test

SUBTASK 38-32-09-610-002

(1) Open the cap of the rinse fitting.

SUBTASK 38-32-09-610-003

(2) Connect a rinse water hose from the service waste system servicing equipment, STD-1142 to the rinse fitting.

SUBTASK 38-32-09-170-001

(3) Fill the waste tank with 30 gal (114 l) - 40 gal (151 l) of water.

SUBTASK 38-32-09-790-001

(4) Examine the rinse fitting assembly and the connections for leakage.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-09-610-004

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-09-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-09-410-002

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-09-410-003

(4) Close this access panel:

Number	Name/Location
145AL	Waste Service Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK RINSE NOZZLE - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the waste tank rinse nozzle

(2) An installation of the waste tank rinse nozzle

(3) A cleaning of the waste tank rinse nozzle.

B. The waste tank rinse nozzle is referred to as the rinse nozzle in this procedure.

TASK 38-32-10-000-801

2. Waste Tank Rinse Nozzle Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove the waste tank rinse nozzle.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-32-10-910-001

(1) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

SUBTASK 38-32-10-610-001

(2) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

(a) Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-32-10-010-001

(3) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-10-010-002

(4) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-10-010-004

(5) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-10-040-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

F. Waste Tank Rinse Nozzle Removal

SUBTASK 38-32-10-020-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Disconnect the rinse hose [3] from the rinse nozzle assembly [2].

SUBTASK 38-32-10-020-002

(2) Remove the V-band coupling [1].

SUBTASK 38-32-10-020-003

(3) Remove the rinse nozzle assembly [2] and packing [4] from the mounting flange of the waste tank.

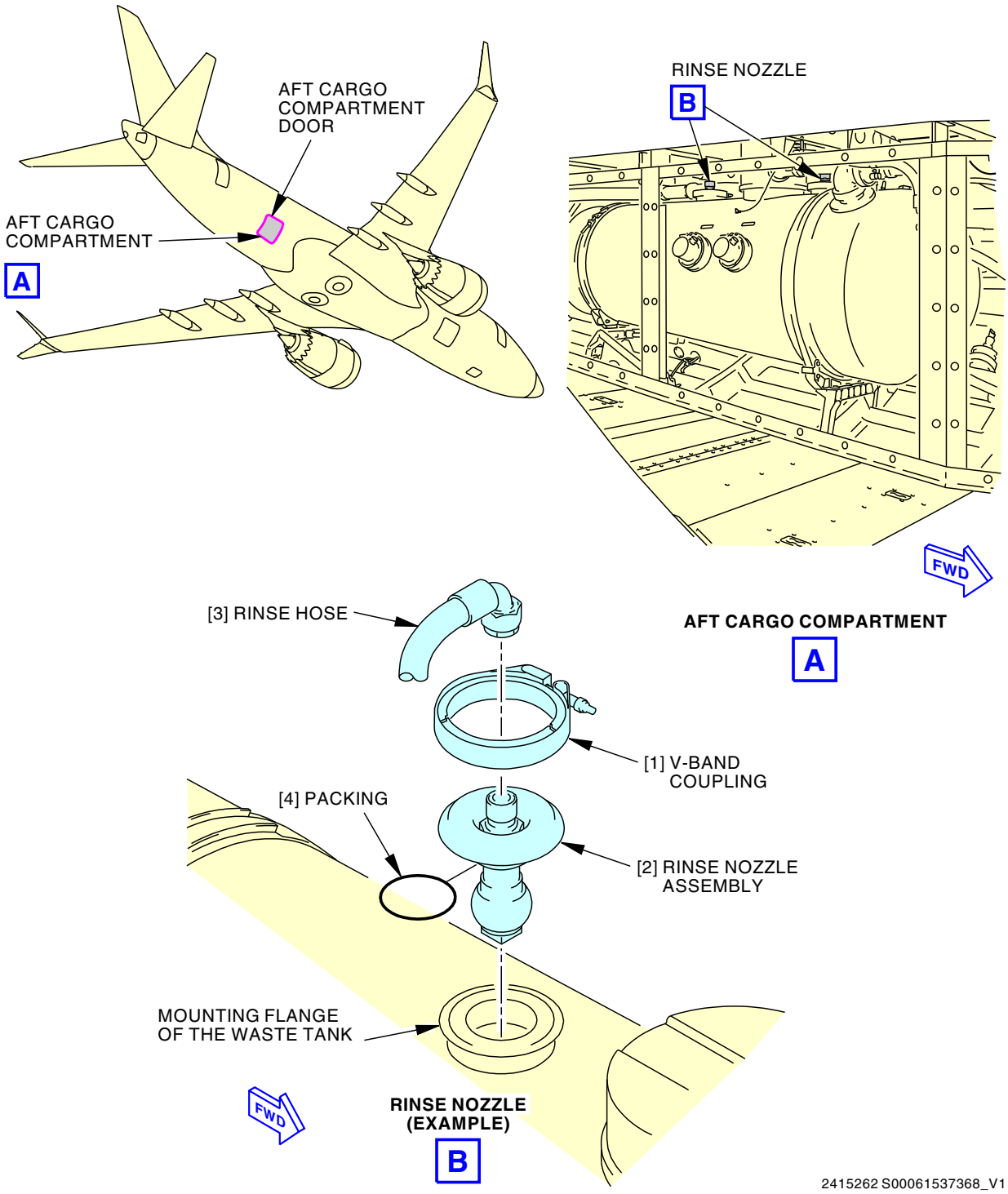
(a) Discard the packing [4].

SUBTASK 38-32-10-160-001

(4) Clean all the waste material from the hole in the waste tank for the rinse nozzle.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Tank Rinse Nozzle Installation
Figure 201/38-32-10-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-10-400-801

3. Waste Tank Rinse Nozzle Installation

(Figure 201)

A. General

(1) This task gives the instructions to install the waste tank rinse nozzle.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Rinse nozzle assembly	38-32-10-50-080	PTF ALL
4	Packing	38-32-10-50-085	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

G. Waste Tank Rinse Nozzle Installation

SUBTASK 38-32-10-640-001

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Apply grease, D00463, or Molykote 111 Compound, D00189, on the packing [4] and mounting flange of the waste tank.

SUBTASK 38-32-10-420-001

(2) Install the packing [4] on the rinse nozzle assembly [2].

SUBTASK 38-32-10-420-002

(3) Put the rinse nozzle assembly [2] in its position.

SUBTASK 38-32-10-420-003

(4) Install the V-band coupling [1] that attaches the rinse nozzle assembly [2].

SUBTASK 38-32-10-420-004

(5) Tighten the V-band coupling [1] to 40 in-lb (4.5 N·m) - 50 in-lb (5.6 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-10-420-005

(6) Connect the rinse hose [3] to the rinse nozzle assembly [2].

SUBTASK 38-32-10-420-008

(7) Tighten the rinse hose [3] to 300 in-lb (34 N·m) - 350 in-lb (40 N·m).

H. Waste Tank Rinse Nozzle Installation Test

SUBTASK 38-32-10-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tags and close these circuit breakers:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT
F	2	C01389	VACUUM WASTE BLOWER

SUBTASK 38-32-10-410-003

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-10-610-002

(3) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-10-790-001

(4) Check the rinse nozzle at the rinse line fitting connection and at the waste tank connection to make sure that no leaks occur during the servicing step.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-10-010-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-10-410-001

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

TASK 38-32-10-100-801

4. Waste Tank Rinse Nozzle Cleaning

(Figure 202)

A. General

(1) This task gives the instructions to clean the waste tank rinse nozzle.

B. Consumable Materials

Reference	Description	Specification
A00648	Compound - Retaining - Loctite 242	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
23	Gasket	38-32-10-50-070	PTF ALL

D. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

E. Procedure

SUBTASK 38-32-10-020-004

(1) Do this task: Waste Tank Rinse Nozzle Removal, TASK 38-32-10-000-801.

SUBTASK 38-32-10-140-001

(2) Clean the rinse nozzle assembly using liquid soap and a bristle brush.

SUBTASK 38-32-10-210-001

(3) Make sure the holes in the rotating rinse nozzle [22] are clean.

SUBTASK 38-32-10-710-001

(4) Make sure the rotating rinse nozzle [22] turns freely.

SUBTASK 38-32-10-020-005

(5) If there is a malfunction of the rinse nozzle, do the steps that follow to disassemble and clean the rinse nozzle:

NOTE: It is optional to disassemble the rinse nozzle.

(a) Remove the nut [25] and washer [24] from the housing [26].

(b) Remove the housing [26] and gasket [23] from the rinse mounting flange [21].

(c) Remove the rotating rinse nozzle [22] from the housing [26].

NOTE: These parts are assembled with a locking compound and will not disconnect easily.

(d) Clean the rotating rinse nozzle [22] with liquid soap and a bristle brush.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-10-420-006

(6) If the rinse nozzle was disassembled, do these steps to assemble the rinse nozzle:

(a) Apply Loctite 242 compound, A00648 to the threads of the housing [26].

(b) Install the rotating rinse nozzle [22] on the housing [26].

1) Tighten the rotating rinse nozzle [22] until it reaches the bottom of the housing [26].

2) Hold the square end of the rotating rinse nozzle [22] in a vise and tighten the assembly 1/4 turn maximum.

(c) Install the housing [26] and gasket [23] in the rinse mounting flange [21].

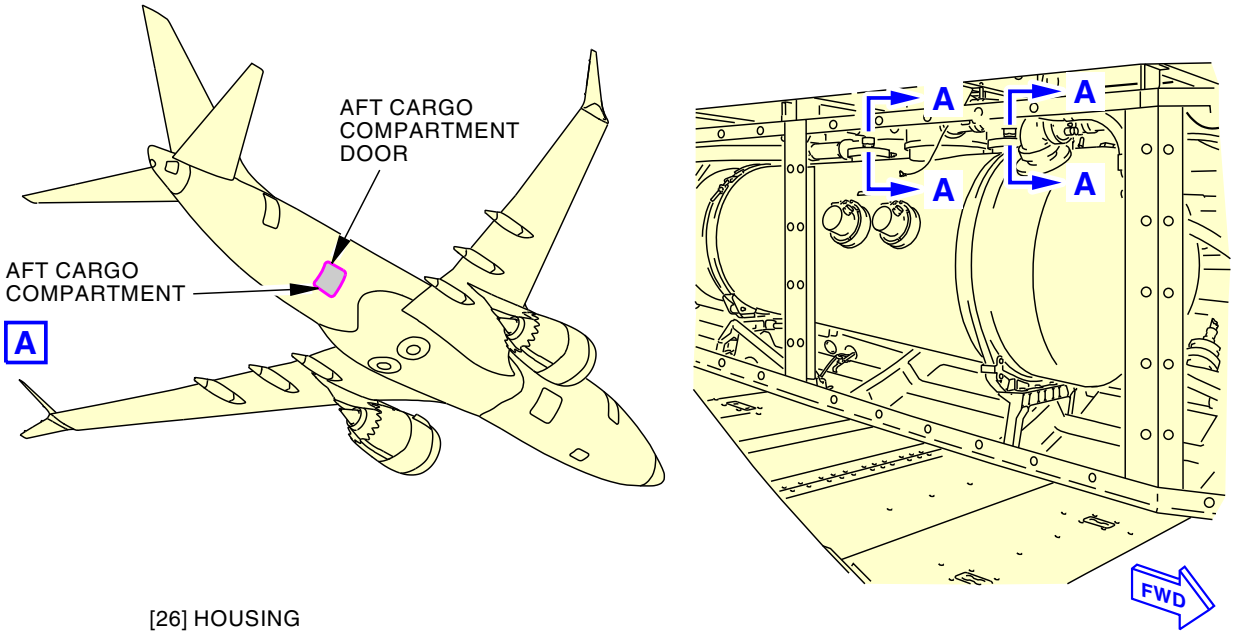
(d) Install the nut [25] and washer [24] to attach the rinse mounting flange [21] to the housing [26].

SUBTASK 38-32-10-420-007

(7) Do this task: Waste Tank Rinse Nozzle Installation, TASK 38-32-10-400-801.

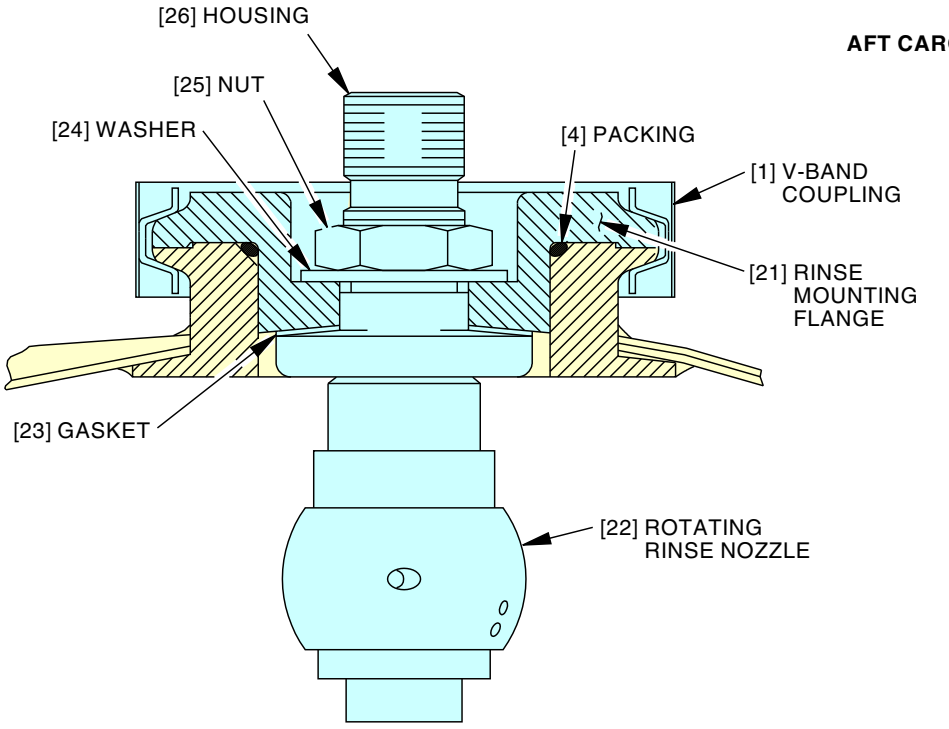
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AFT CARGO COMPARTMENT

A



RINSE NOZZLE ASSEMBLY
(EXAMPLE)

A-A

2415263 S00061537369_V1

Rinse Nozzle Cleaning
Figure 202/38-32-10-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK RINSE FILTER - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Waste tank rinse filter removal

(2) Waste tank rinse filter installation

(3) Waste tank rinse filter element cleaning.

TASK 38-32-11-000-801

2. Waste Tank Rinse Filter Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove the waste tank rinse filter.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-32-11-610-001

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-32-11-010-001

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-11-010-002

(3) To remove the enclosure panels for the waste tanks, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

F. Waste Tank Rinse Filter Removal

SUBTASK 38-32-11-020-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Loosen the B-nut of the rinse hose [5] to disconnect the rinse filter assembly [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-11-020-002

(2) Remove the straps [4].

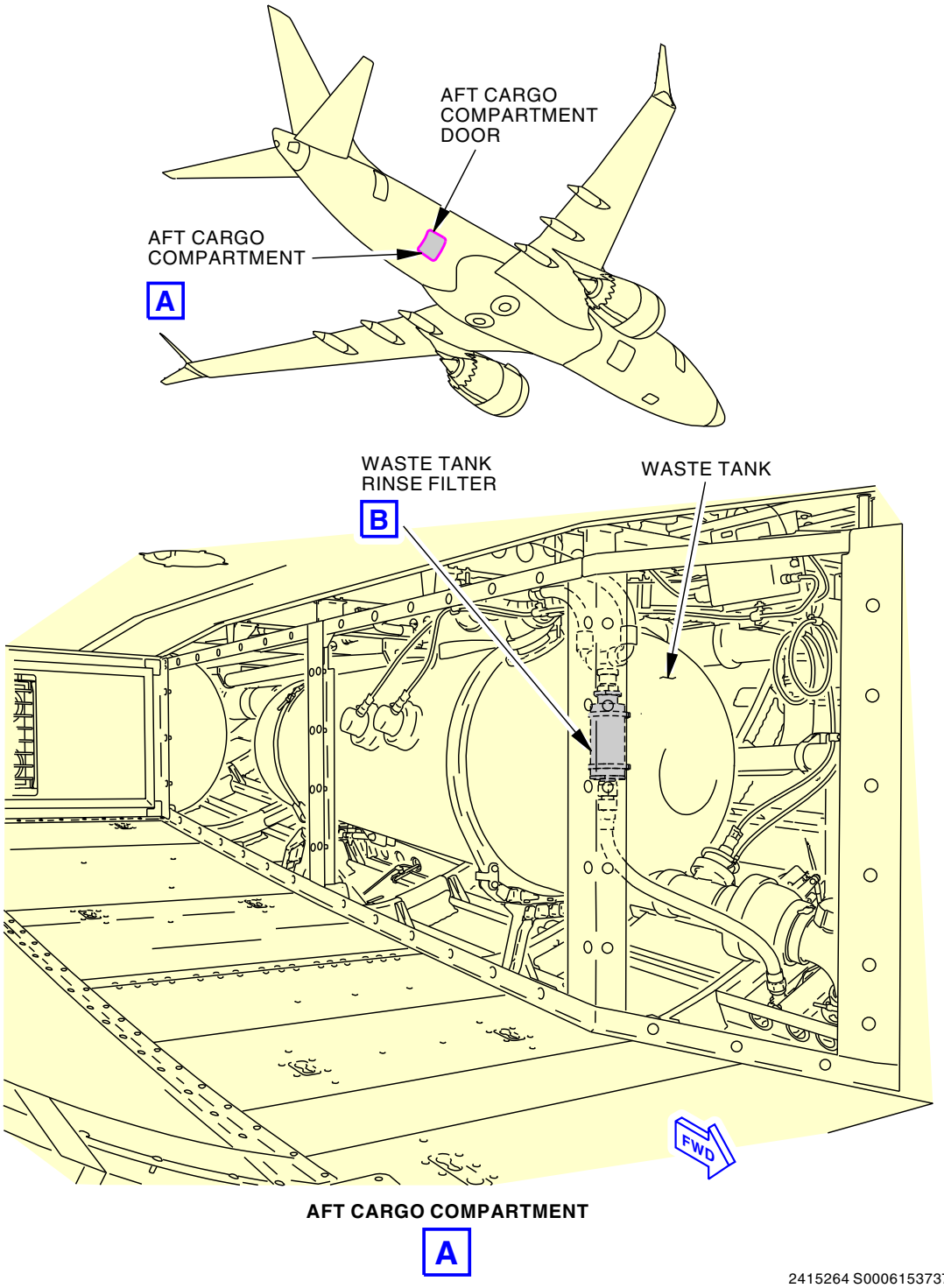
SUBTASK 38-32-11-020-003

(3) Remove the rinse filter assembly [3] and packing [2] from the union fitting [11]

(a) Discard the packing [2].

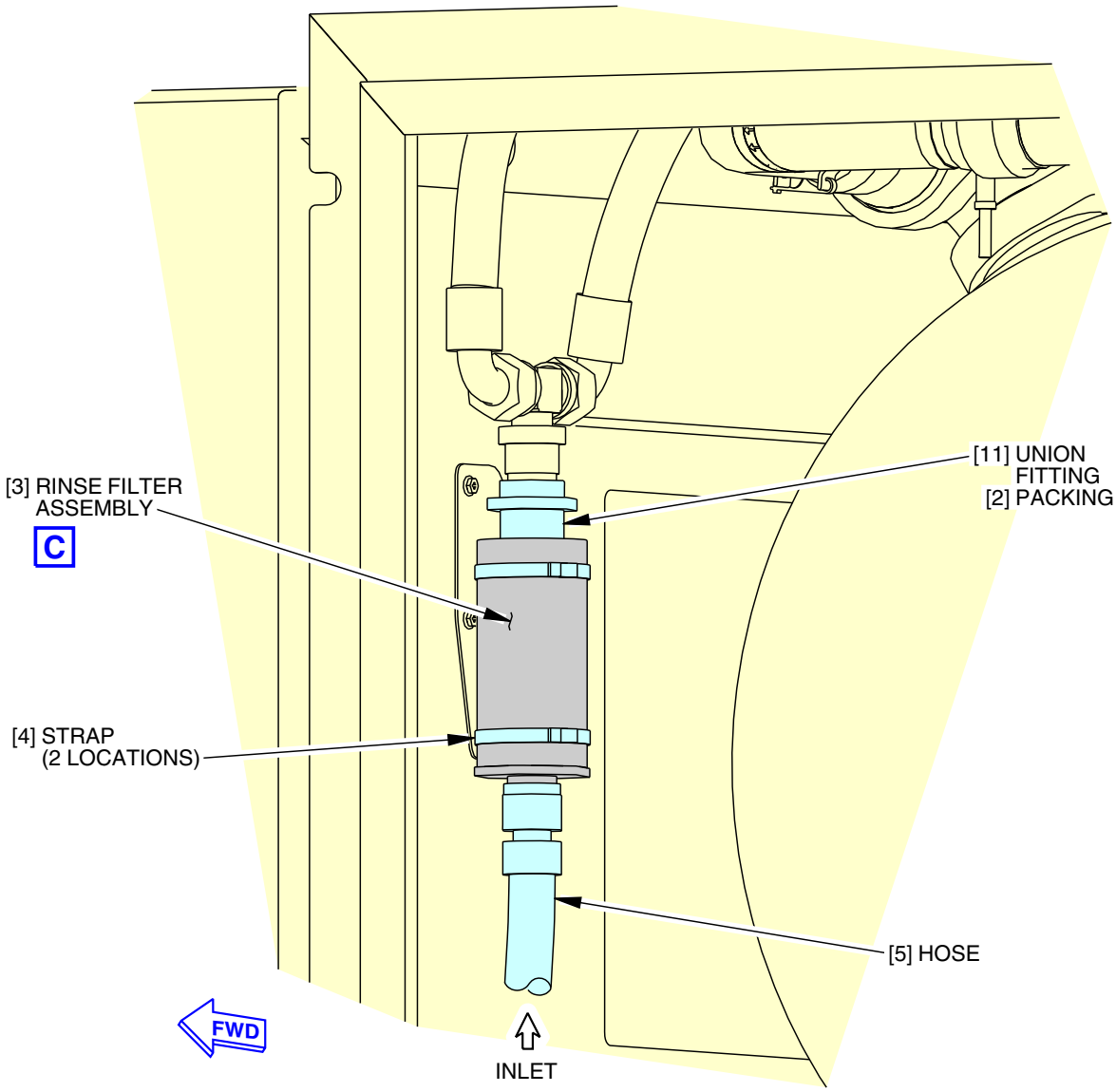
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Tank Rinse Filter Installation
Figure 201/38-32-11-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



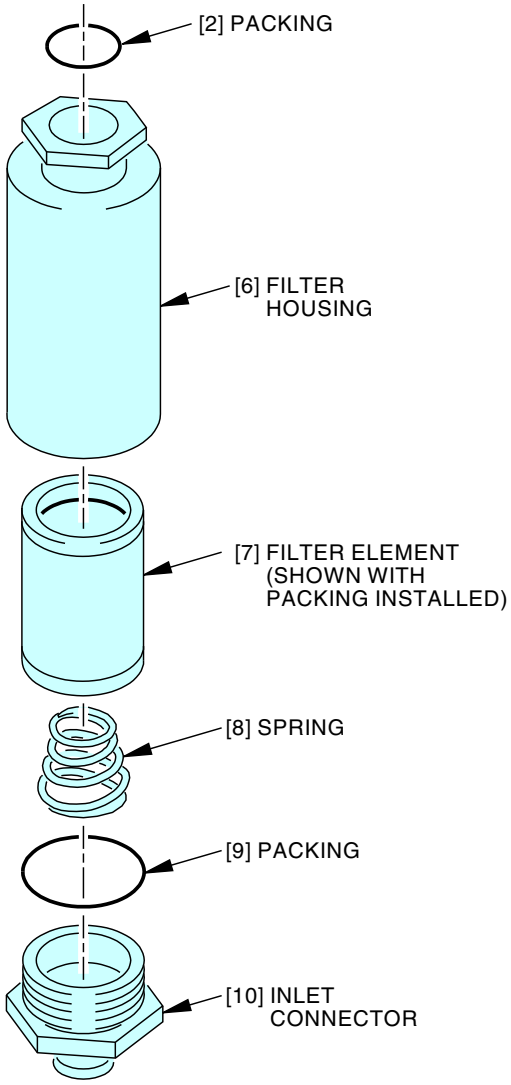
WASTE TANK RINSE FILTER

B

2415265 S00061537376_V2

Waste Tank Rinse Filter Installation
Figure 201/38-32-11-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RINSE FILTER ASSEMBLY

C

2415266 S00061537377_V1

Waste Tank Rinse Filter Installation
Figure 201/38-32-11-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-11-400-801

3. Waste Tank Rinse Filter Installation

(Figure 201)

A. General

(1) This task gives the instructions to install the waste tank rinse filter.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1142	Equipment - Waste System Servicing

D. Consumable Materials

Reference	Description	Specification
D00504	Grease - Petrolatum	VV-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Packing	38-32-11-01-130	PTF ALL
3	Rinse filter assembly	38-32-11-01-150	PTF ALL

F. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

G. Access Panels

Number	Name/Location
145AL	Waste Service Door
822	Aft Cargo Door

H. Waste Tank Rinse Filter Installation

SUBTASK 38-32-11-640-001

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Apply grease, D00504, to the new packing [2] and threads of the union fitting [11].

SUBTASK 38-32-11-420-001

(2) Install the packing [2] on the union fitting [11].

SUBTASK 38-32-11-420-002

(3) Install the rinse filter assembly [3] on the union fitting [11].

(a) Tighten the union fitting [11] to 80 in-lb (9.0 N·m) - 100 in-lb (11.3 N·m).

SUBTASK 38-32-11-420-003

(4) Install the straps [4] to attach the rinse filter assembly [3].

SUBTASK 38-32-11-420-004

(5) Install the rinse hose [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Tighten the hose [5] to 80 in-lb (9.0 N·m) - 100 in-lb (11.3 N·m).

I. Waste Tank Rinse Filter Installation Test

SUBTASK 38-32-11-860-001

(1) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

SUBTASK 38-32-11-610-002

(2) Open the cap on the rinse fitting assembly.

SUBTASK 38-32-11-610-003

(3) Connect a rinse water hose from the service waste system servicing equipment, STD-1142, to the rinse fitting assembly.

SUBTASK 38-32-11-170-001

(4) Fill the waste tank with 10 gal (38 l) to 20 gal (76 l) of water.

SUBTASK 38-32-11-790-001

(5) Examine the rinse filter and the connections for leakage.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-11-610-004

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-32-11-010-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-32-11-410-001

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-32-11-410-002

(4) Close this access panel:

Number	Name/Location
145AL	Waste Service Door

END OF TASK

TASK 38-32-11-100-801

4. Waste Tank Rinse Filter Element Cleaning

(Figure 201)

A. General

(1) This task gives the instructions to clean the waste tank rinse filter element.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)

C. Tools/Equipment

Reference	Description
STD-419	Gloves - Rubber, Elbow Length
STD-1136	Facepiece (Mask) - Particulate Filtering, Disposable (NIOSH N95 approved)
STD-1137	Glasses - Safety

D. Consumable Materials

Reference	Description	Specification
B00541	Cleaner - General Purpose Household Detergent	
B50099	Detergent - Concentrated, Anionic	
D00504	Grease - Petrolatum	VV-P-236
G02315	Clothing - Disposable Gown, Gloves For Sewage Handling	
G50321	Air - Clean, Dry	BAC5402 Table I

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Filter element	38-32-11-01-170	PTF ALL
9	Packing	38-32-11-01-160	PTF ALL

F. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left

G. Waste Tank Rinse Filter Element Cleaning

SUBTASK 38-32-11-160-001

(1) Make sure that you use the protective clothing (TASK 38-32-00-910-801).

NOTE: These are items that you can wear to give you protection when it is necessary.

(a) elbow length rubber gloves, STD-419

(b) filtering facepiece, STD-1136

(c) safety glasses, STD-1137

(d) Disposable clothing, G02315

SUBTASK 38-32-11-020-004

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(2) Do this task: Waste Tank Rinse Filter Removal, TASK 38-32-11-000-801.

SUBTASK 38-32-11-010-004

(3) Remove the inlet connector [10] and the packing [9] from the filter housing [6].

(a) Discard the packing [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-11-020-005

(4) Remove the filter element [7] and the spring [8].

SUBTASK 38-32-11-130-001

(5) Clean the filter element [7] using Cleaning Method I.

(a) Soak the filter element [7] in a weak detergent solution. Use the general purpose household detergent cleaner, B00541, or a liquid dishwashing detergent. Mix 4 fl-oz (118 cc)detergent per 1 gal (4 l) warm water.

(b) Brush each pleat of the filter element [7] with a soft bristle brush.

(c) Reverse flow flush with warm tap water.

(d) Blow dry with low pressure air.

(e) Make sure that the filter element [7] is in a serviceable condition.

SUBTASK 38-32-11-130-002

(6) Clean the filter element [7] using the alternate (ultrasonic) Cleaning Method II.

(a) Flush the filter element with a reverse flow of warm water (the opposite direction of normal fluid flow).

Remove as much contamination off the outside of the filter element as possible

(b) Ultrasonic clean the filter element for 15 minutes in the anionic detergent cleaning solution. Use a solution of 1 fl-oz (30 cc) concentrated, anionic detergent, B50099 per 1 gal (3.8 l) warm water.

(c) Flush the filter element with a reverse flow of hot water.

(d) Remove the entrapped water from the filter element with a reverse flow of clean dry air, G50321.

(e) Dry the filter element in an oven at 250°F (121°C) for 1/2 hour, or air dry the filter element until moisture is not detectable.

(f) Make sure that the filter element [7] is in a serviceable condition.

SUBTASK 38-32-11-960-001

(7) If the contamination is not removed, or if the filter element is damaged, use a new filter element [7].

SUBTASK 38-32-11-420-005

(8) Put the filter housing [6] in a position to assemble the rinse filter assembly [3].

SUBTASK 38-32-11-420-006

(9) Put the filter element [7] and the spring [8] in the filter housing [6].

SUBTASK 38-32-11-640-002

(10) Apply the grease, D00504 to the new packing [9] and the threads of the inlet connector [10].

SUBTASK 38-32-11-420-007

(11) Install the inlet connector [10] and the packing [9] in the filter housing [6].

SUBTASK 38-32-11-420-008

(12) Do this task: Waste Tank Rinse Filter Installation, TASK 38-32-11-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

VACUUM BLOWER BAROMETRIC SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the vacuum blower barometric switch

(2) An installation of the vacuum blower barometric switch.

B. The vacuum blower barometric switch is referred to as the altitude switch in this procedure.

(1) An altitude switch controls power to the vacuum blower. When the airplane is climbing, the switch cuts power to the vacuum blower at about 16,000 ft (4877 m). When the airplane is descending, the altitude sensing switch closes at about 12,000 ft (3658 m) providing power to the vacuum blower.

TASK 38-32-12-000-801

2. Vacuum Blower Barometric Switch Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the altitude switch.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
312	Area Aft of Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
311BL	Stabilizer Trim Access Door

F. Prepare for the Removal

SUBTASK 38-32-12-010-001

(1) On the stabilizer leading edge, open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 38-32-12-010-003

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-32-12-040-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

G. Altitude Switch Removal

SUBTASK 38-32-12-020-001

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [1] from the altitude switch [2] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [1] from the altitude switch [2].

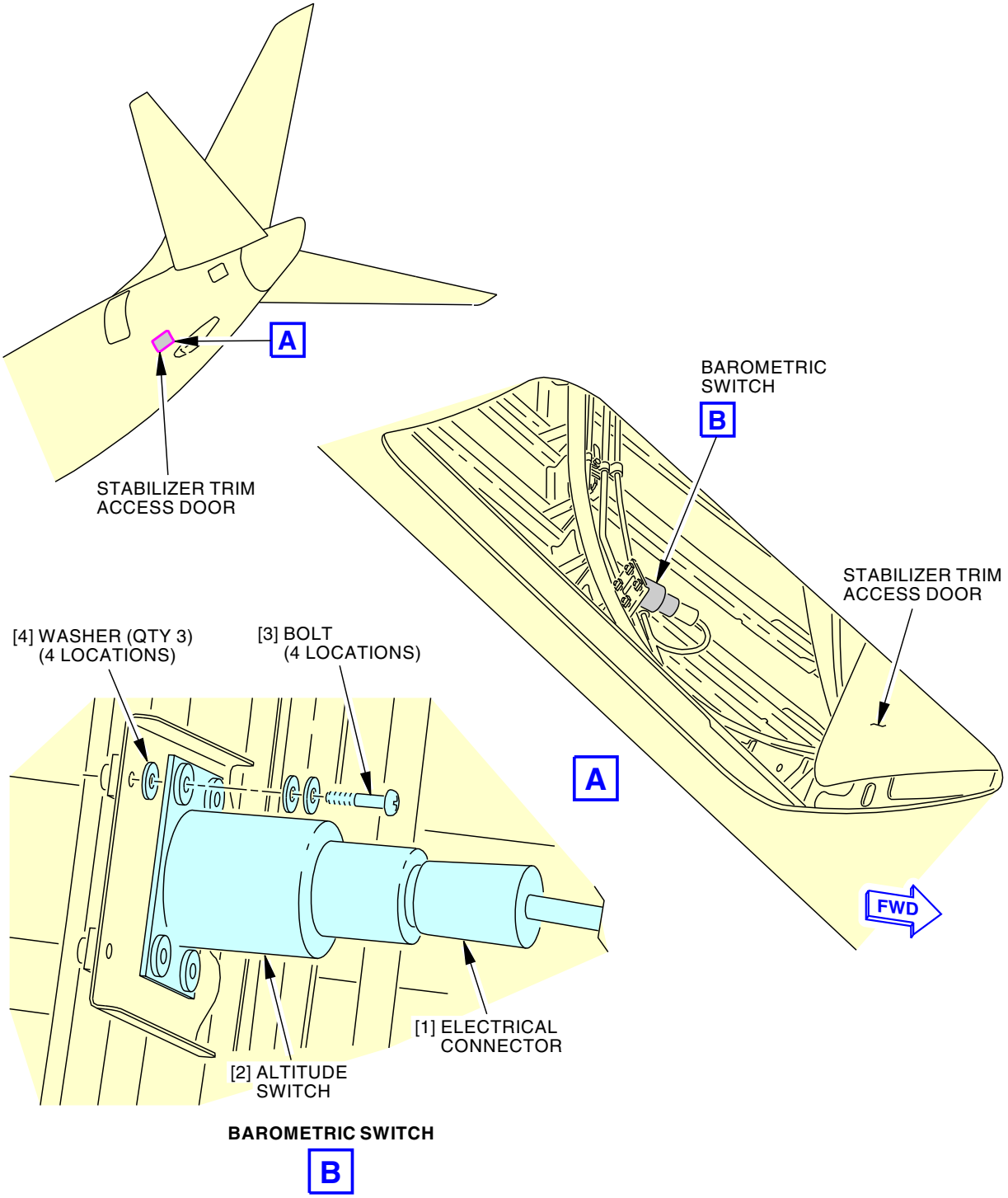
(b) Install the protective covers on the electrical receptacle of the altitude switch [2] and electrical connector [1].

SUBTASK 38-32-12-020-002

(2) Remove the bolts [3] and washers [4] to remove the altitude switch [2].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Vacuum Blower Barometric Switch Installation
Figure 401/38-32-12-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-32-12-400-801

3. Vacuum Blower Barometric Switch Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the altitude switch.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Altitude switch	38-32-12-01-005	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
312	Area Aft of Pressure Bulkhead - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
311BL	Stabilizer Trim Access Door

G. Altitude Switch Installation

SUBTASK 38-32-12-420-001

(1) Put the altitude switch [2] in its position.

SUBTASK 38-32-12-420-002

(2) Install the bolts [3] and washers [4].

SUBTASK 38-32-12-420-003

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(3) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [1] to the altitude switch [2] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the altitude switch [2] and electrical connector [1].

(b) Connect the electrical connector [1] to the altitude switch [2].

1) Tighten the connector with the teflon-jawed pliers, STD-664.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Altitude Switch Installation Test

SUBTASK 38-32-12-860-001

> **WARNING**
>
> WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

> **WARNING**
>
> DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-32-12-410-001

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-32-12-710-001

(3) Flush a toilet to make sure that the vacuum blower operates.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-32-12-010-002

(1) On the stabilizer leading edge, close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK QUANTITY INDICATION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) LAVS INOP BITE test at the attendant panel

(2) LCM BITE test

(3) Auto zero test

(4) Attendant panel waste tank quantity indication operational test.

TASK 38-33-00-740-801

2. LAVS INOP BITE Test at the Attendant Panel

(Figure 502)

A. General

(1) This task is for a test of the waste indication system at the attendant panel.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Prepare for the Test

SUBTASK 38-33-00-010-001

(1) Get access to the passenger compartment.

D. LAVS INOP BITE Test

SUBTASK 38-33-00-740-001

(1) Do the LAVS INOP BITE test:

(a) Select the Maintenance display on the attendant panel touch screen.

(b) Push and then release the VACUUM WASTE button.

(c) Push and then release the LAVS INOP Test button.

(d) Make sure that the LAVS INOP light is on for approximately 3 seconds and then is off.

(e) Make sure that the CLEAN/CHECK SENSOR light is off.

1) If the CLEAN/CHECK SENSOR light is on, do this task: LCM BITE Test, TASK 38-33-00-740-802.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-00-410-001

(1) Close the access to the passenger compartment.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-33-00-740-802

3. LCM BITE Test

(Figure 501)

A. General

(1) This task is for a full test of the LCM for the waste indication system at the LCM.

(2) There is a task for a test of the waste indication system from the attendant panel. The attendant panel test is equivalent but it is not necessary to remove the panel for the waste enclosure.

B. References

Reference	Title
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

D. Access Panels

Number	Name/Location
822	Aft Cargo Door

E. Prepare for the test

SUBTASK 38-33-00-860-003

(1) Make sure that this circuit breaker is closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-00-010-002

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-00-010-003

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

F. LCM BITE Test

SUBTASK 38-33-00-860-005

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Make sure that the POWER ON (green) light for the LCM is on.

(a) Make sure the J1, J2, J3, and TANK FULL (red) lights are off.

SUBTASK 38-33-00-760-001

(2) Move and then hold the switch to the TEST SENSOR position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-00-740-002

(3) Make sure the POWER ON (green), J1, J2, J3, and TANK FULL (red) lights are on for approximately 3 seconds.

NOTE: The lights on the LCM can flash irregularly. If you find a system error, the TANK FULL light will flash at 8 Hz, then go off. You will see this cycle again after 1.6 seconds until you correct the problem.

SUBTASK 38-33-00-760-002

(4) Release the switch from the TEST SENSOR position.

SUBTASK 38-33-00-740-003

(5) Make sure the J1, J2, J3, and TANK FULL (red) lights are off.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-00-410-002

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-33-00-410-003

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

TASK 38-33-00-710-801

4. Auto Zero Test

(Figure 501)

A. General

(1) This task is for the test of the auto zero of the waste tank indication system.

(2) You will use the auto zero procedure to calibrate the Continuous Level Sensor (CLS). The CLS measures the quantity of waste in the waste tank. When you replace the CLS, you must do the auto zero adjustment. You must also do the auto zero adjustment when you move the CLS diaphragm, the CLS sensor, or the waste tank. This procedure will calibrate the CLS with the LCM to give a more accurate value for the waste quantity indicator.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Access Panels

Number	Name/Location
822	Aft Cargo Door

E. Prepare for the test:

SUBTASK 38-33-00-610-001

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and then flush the waste tanks.

SUBTASK 38-33-00-860-006

(2) Make sure that this circuit breaker is closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-00-010-004

(3) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-00-010-005

(4) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

F. Auto Zero Adjustment

SUBTASK 38-33-00-860-008

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Make sure that the waste tanks are fully drained (TASK 12-17-01-610-801).

SUBTASK 38-33-00-860-009

(2) On the LCM, make sure that these lights are as follows:

(a) Make sure that the POWER ON (green) light is on.

(b) Make sure that the J1 and J2 (red) lights are off.

(c) Make sure that the J3 and TANK FULL (red) lights are off or flash.

SUBTASK 38-33-00-760-003

(3) Move and then hold the switch to the TEST LAMPS position.

(a) Make sure that the POWER ON (green), the J1 and J2, and the J3 and TANK FULL (red) lights are on.

SUBTASK 38-33-00-760-004

(4) Release the switch from the TEST LAMPS position.

(a) Make sure that the J1, J2, J3, and TANK FULL (red) lights are off.

NOTE: During this step the J3 or TANK FULL (red) lights can flash. This is permitted.

SUBTASK 38-33-00-020-001

(5) Disconnect the electrical connectors from the SENSOR J1 and SENSOR J2 positions on the LCM.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) After 30 seconds, make sure that the J1, J2, and TANK FULL (red) lights are on.

SUBTASK 38-33-00-710-001

(6) Move and then hold the switch to the TEST LAMPS position.

(a) After 10 seconds, make sure that the J1, J2, J3, and TANK FULL (red) lights are off for 3 seconds.

(b) Make sure that the POWER ON (green) light is on.

SUBTASK 38-33-00-760-005

(7) Release the switch from the TEST LAMPS position.

(a) Make sure that the POWER ON (green), J1, J2, and TANK FULL (red) lights are on.

SUBTASK 38-33-00-420-001

(8) Connect the electrical connector for the SENSOR J1 to the LCM.

(a) Make sure that the J1 and TANK FULL (red) lights are off.

(b) Make sure that the POWER ON (green) and J2 (red) lights are on.

SUBTASK 38-33-00-420-002

(9) Connect the electrical connector for the SENSOR J2 to the LCM.

(a) Make sure that the J1, J2, J3, and TANK FULL (red) lights are off.

(b) Make sure that the POWER ON (green) light is on.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-00-410-004

WARNING SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-33-00-410-005

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-00-610-002

(3) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

END OF TASK

TASK 38-33-00-710-802

5. Attendant Panel Waste Tank Quantity Indication Operational Test

(Figure 502)

A. General

(1) This task is for the operational test of the cabin attendant control panel waste tank quantity indication.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for the test:

SUBTASK 38-33-00-610-003

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks. Do not disconnect the rinse water connection from the rinse fitting assembly. Do not disconnect the waste connection from the waste drain valve assembly.

SUBTASK 38-33-00-860-010

(2) Make sure that this circuit breaker is closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

E. Attendant's Panel Waste Tank Quantity Indication Operational Test

SUBTASK 38-33-00-860-012

(1) Make sure that the waste drain ball valve is closed.

SUBTASK 38-33-00-860-014

(2) Touch environment on the attendant's panel touch screen.

(a) Make sure that the indicator for vacuum waste is near E.

SUBTASK 38-33-00-610-004

(3) Add 30 ±1 gallons (114 ±4 liters) of water at the rinse water connection for the waste tank.

SUBTASK 38-33-00-860-015

(4) Touch environment on the attendant panel touch screen.

(a) Make sure that the indicator for vacuum waste is near 1/2.

SUBTASK 38-33-00-860-017

(5) Pull the handle at the service panel to open the waste drain ball valve for the waste tank.

SUBTASK 38-33-00-860-016

(6) Touch environment on the attendant panel touch screen.

(a) Make sure that the indicator for vacuum waste is near E.

SUBTASK 38-33-00-860-013

(7) Push the handle at the service panel to close the waste drain ball valve for the waste tank.

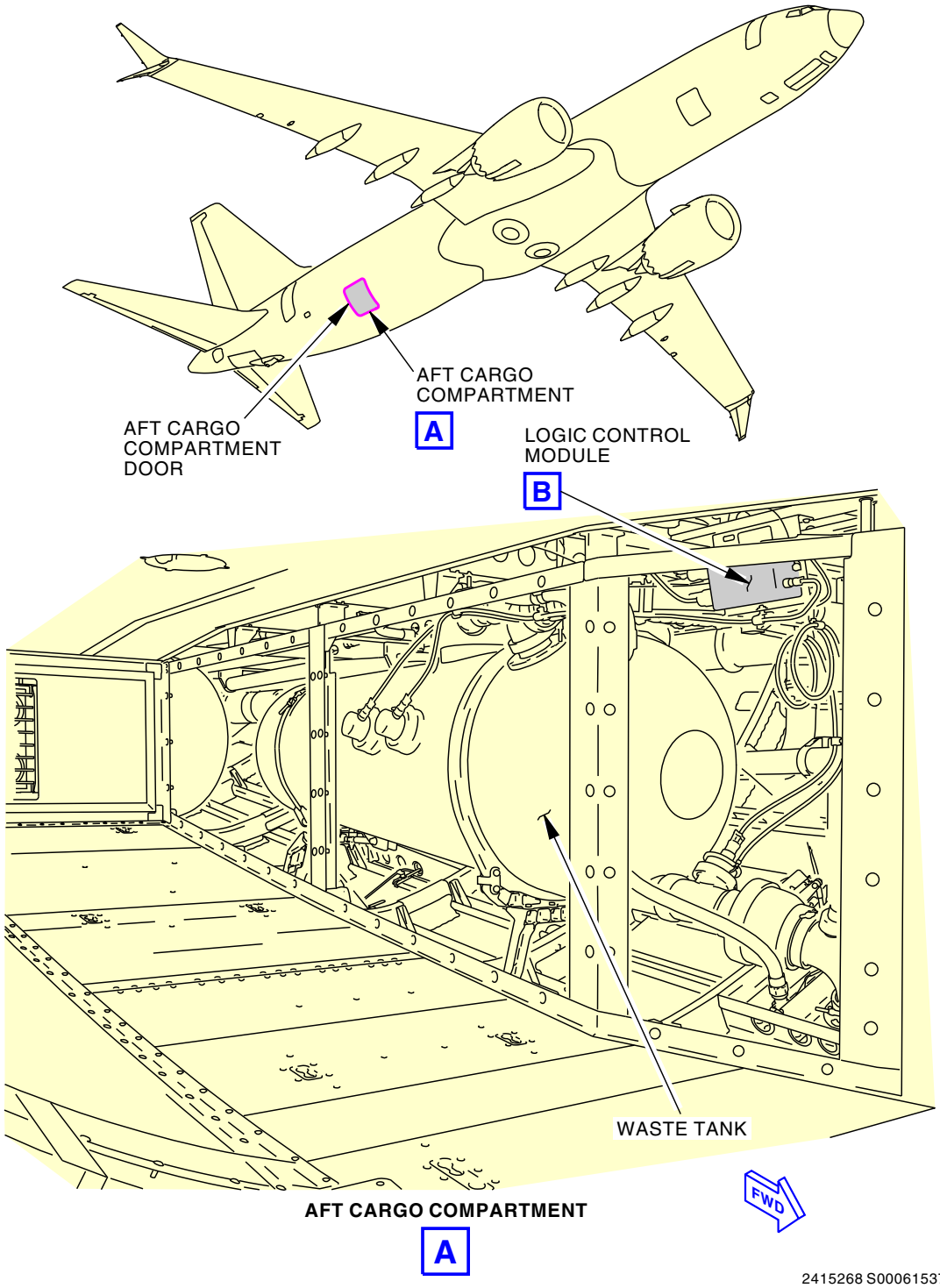
F. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-00-610-006

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

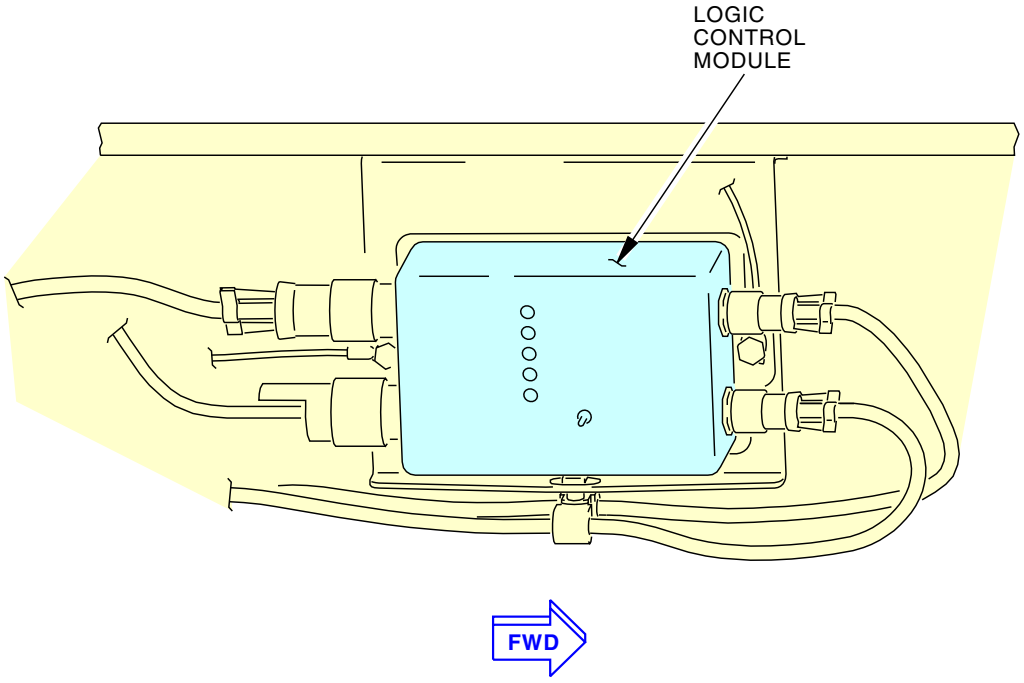
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Logic Control Module Test
Figure 501/38-33-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



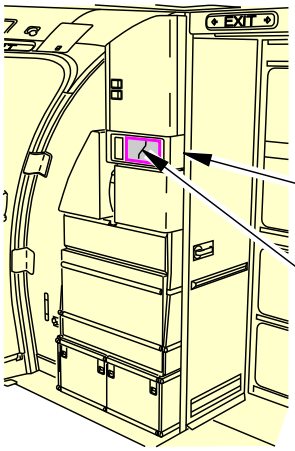
LOGIC CONTROL MODULE

B

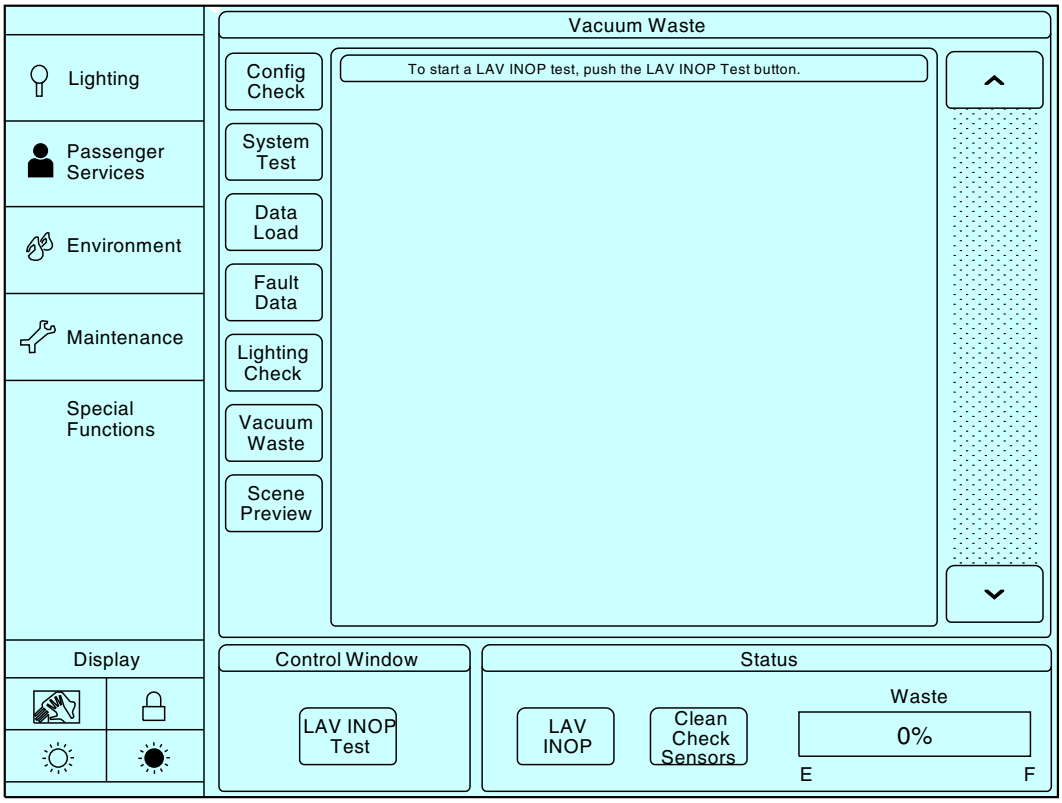
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Logic Control Module Test
Figure 501/38-33-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ENTRY DOOR (EXAMPLE)



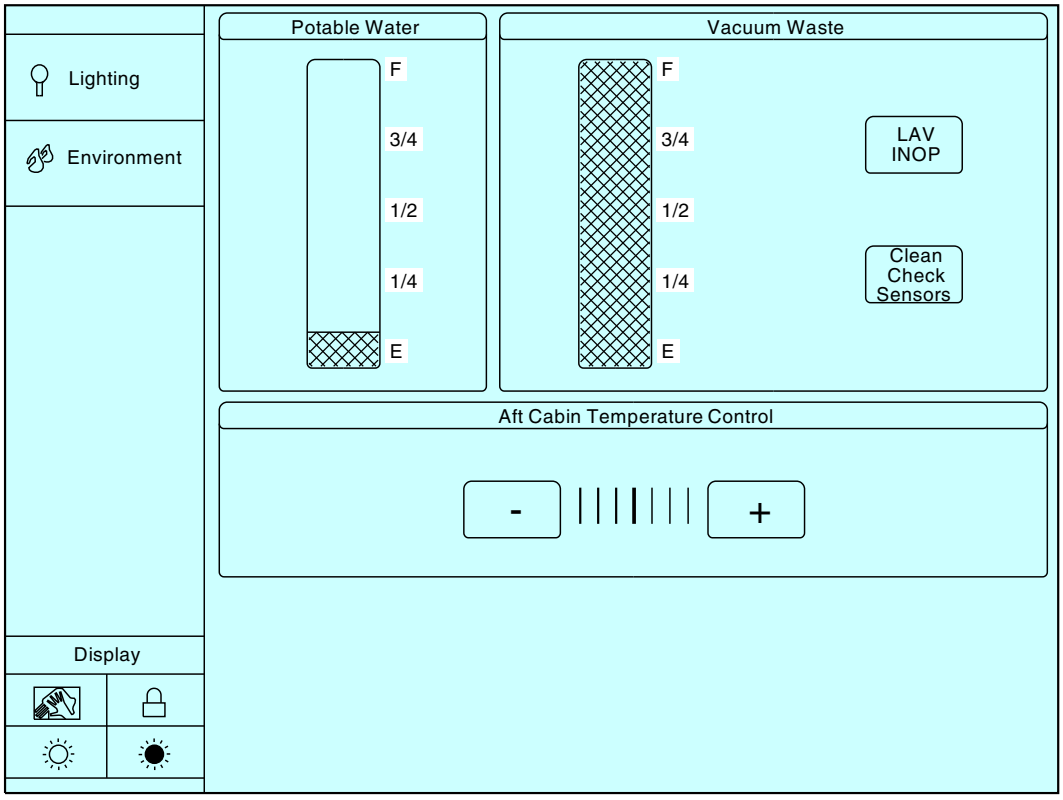
MAINTENANCE DISPLAY, FORWARD ACP

A

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Attendant's Panel Waste Quantity Indicator Operational Test
Figure 502/38-33-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ENVIRONMENT DISPLAY, AFT ACP

A

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Attendant's Panel Waste Quantity Indicator Operational Test
Figure 502/38-33-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK POINT LEVEL SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Waste tank point level sensor removal

(2) Waste tank point level sensor installation.

TASK 38-33-01-000-801

2. Waste Tank Point Level Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the waste tank point level sensor.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-33-01-610-001

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-33-01-010-001

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-01-010-002

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-01-040-001

(4) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-01-010-007

(5) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-33-01-040-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-01-910-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(7) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

G. Waste Tank Point Level Sensor Removal

SUBTASK 38-33-01-020-001

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the elecrical connector [2] from the point level sensor [5] (TASK 70-00-01-910-803-G00).

(a) Disconnect the elecrical connector [2] from the point level sensor [5].

(b) Install the protective covers on the electrical receptacle of the point level sensor [5] and the elecrical connector [2].

SUBTASK 38-33-01-020-002

(2) Remove the V-band clamp [1].

SUBTASK 38-33-01-020-003

(3) Remove the point level sensor [5] and the gask-O-seal [4] from the mounting flange of the waste tank.

(a) Discard the gask-O-seal [4].

SUBTASK 38-33-01-020-004

(4) Remove the packing [3] from the point level sensor [5].

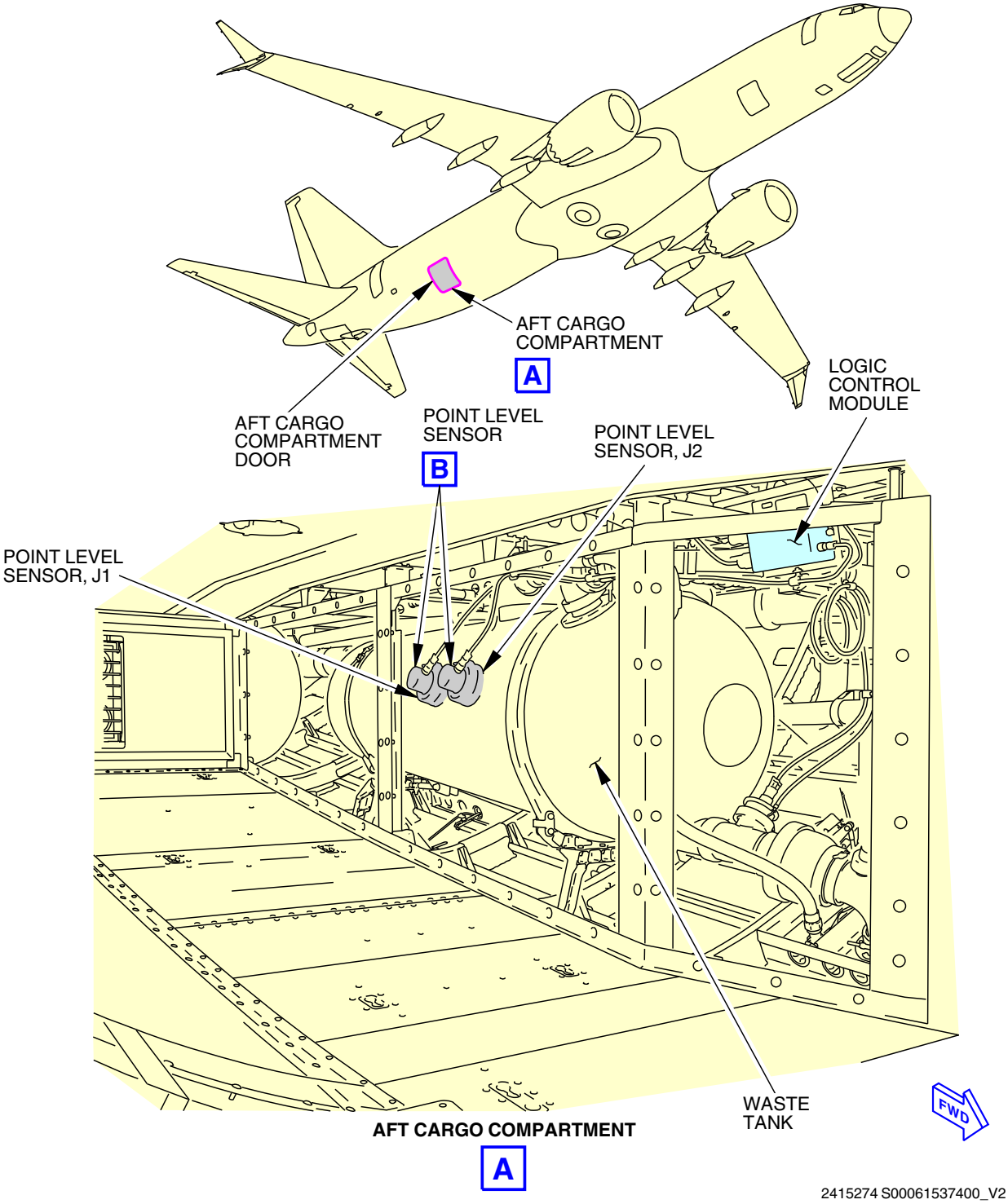
(a) Discard the packing [3].

SUBTASK 38-33-01-160-001

(5) Clean all the waste material from the hole in the waste tank for the point level sensor.

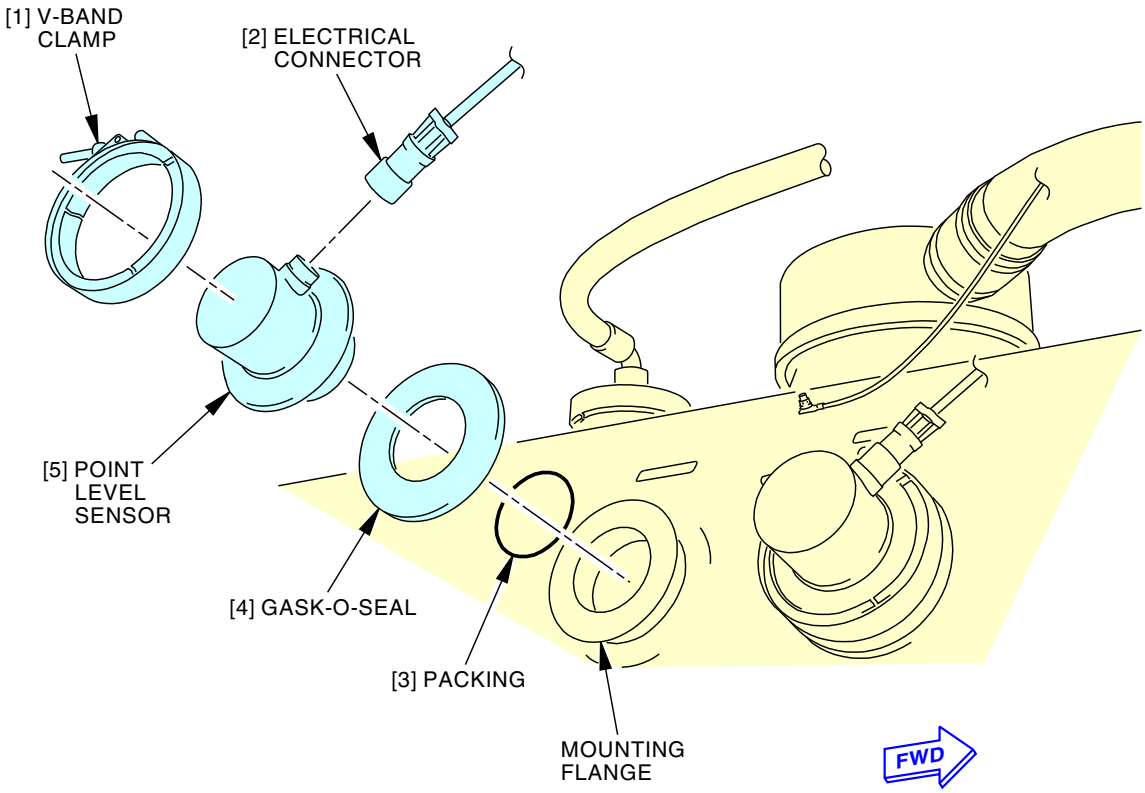
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Tank Point Level Sensor Installation
Figure 401/38-33-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POINT LEVEL SENSOR

B

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Waste Tank Point Level Sensor Installation
Figure 401/38-33-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-33-01-400-801

3. Waste Tank Point Level Sensor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the waste tank point level sensor.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-33-00-740-802	LCM BITE Test (P/B 501)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00128	Grease - Silicone	A-A-59173
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Packing	38-33-01-50-070	PTF ALL
4	Gask-O-seal	38-33-01-50-075	PTF ALL
5	Point level sensor	38-33-01-50-065	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

H. Waste Tank Point Level Sensor Installation

SUBTASK 38-33-01-640-001

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Apply grease, D00128, or Molykote 111 Compound, D00189, to the new gask-O-seal [4], new packing [3], and the mounting flange of the waste tank.

SUBTASK 38-33-01-420-001

(2) Install the packing [3] on the point level sensor [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-01-420-002

(3) Install the gask-O-seal [4] on the mounting flange of the waste tank.

SUBTASK 38-33-01-420-003

(4) Install the point level sensor [5] and V-band clamp [1].

SUBTASK 38-33-01-420-004

(5) Tighten the V-band clamp [1] to 40 in-lb (4.5 N·m) - 50 in-lb (5.6 N·m).

SUBTASK 38-33-01-420-005

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(6) Use the teflon-jawed pliers, STD-664, to connect the elecrical connector [2] to the point level sensor [5] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the point level sensor [5] and the elecrical connector [2].

(b) Connect the elecrical connector [2] to the point level sensor [5].

1) Tighten the connector with the teflon-jawed pliers, STD-664.

I. Waste Tank Point Level Sensor Installation Test

SUBTASK 38-33-01-630-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-01-630-002

(2) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-01-760-001

(3) Make sure that the Power On light on the Logic Control Module (LCM) is on.

SUBTASK 38-33-01-740-001

(4) Set and hold the switch on the LCM to the TEST SENSORS position.

SUBTASK 38-33-01-740-002

(5) Make sure that the J1 Sensor, J2 Sensor, J3 Sensor, Tank Full and Power On lights on the LCM come on.

SUBTASK 38-33-01-740-003

(6) Release the switch on the LCM.

SUBTASK 38-33-01-740-004

(7) Make sure that the J1 Sensor, J2 Sensor, J3 Sensor, and Tank Full lights on the LCM are off.

(a) If the J1 Sensor light or J2 Sensor light is on, do this task: LCM BITE Test, TASK 38-33-00-740-802.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-01-610-002

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-33-01-010-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-33-01-410-001

(3) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK POINT LEVEL SENSOR - CLEANING/PAINTING

1. General

A. This procedure has this task:

(1) A cleaning of the point level sensor for the waste tank.

TASK 38-33-01-100-801

2. Waste Tank Point Level Sensor Cleaning

A. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-33-01-000-801	Waste Tank Point Level Sensor Removal (P/B 401)
38-33-01-400-801	Waste Tank Point Level Sensor Installation (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle

C. Consumable Materials

Reference	Description	Specification
G00016	Acid, Hydrochloric	MIL-DTL-13528

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Cleaning

SUBTASK 38-33-01-610-003

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-33-01-010-004

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-01-010-005

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-01-040-002

(4) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-01-010-008

(5) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-33-01-040-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-01-910-002

(7) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

G. Point Level Sensor Cleaning

SUBTASK 38-33-01-020-005

(1) Do this task: Waste Tank Point Level Sensor Removal, TASK 38-33-01-000-801.

SUBTASK 38-33-01-140-001

(2) Use a rag to clean the loose material from the face of the point level sensor.

SUBTASK 38-33-01-140-002

CAUTION

BE VERY CAREFUL WITH THE FACE OF THE POINT LEVEL SENSOR. IF THERE ARE SCRATCHES OR OTHER DAMAGE, THE POINT LEVEL SENSOR WILL NOT OPERATE.

(3) Use a soft bristle soft bristle brush, STD-123, and soap to clean the face of the point level sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-01-140-003

(4) If necessary, use hydrochloric acid, G00016, to clean the face of the point level sensor.

NOTE: A neutral or alkali cleaning solution will not be effective in cleaning the material build-up on the sensor assembly after it has dried. Use an acidic-based solution to remove material deposit.

(a) Apply the hydrochloric acid, G00016, with a brush to the face of the point level sensor.

(b) Keep the hydrochloric acid, G00016, on the face of the point level sensor.

(c) Rinse with water.

SUBTASK 38-33-01-140-004

(5) Dry the point level sensor.

SUBTASK 38-33-01-160-002

(6) Clean all the waste material from the hole in the waste tank for the point level sensor.

SUBTASK 38-33-01-020-006

(7) Do this task: Waste Tank Point Level Sensor Installation, TASK 38-33-01-400-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-01-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Make sure that this circuit breaker is closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-01-860-004

(2) Make sure that this circuit breaker is closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-01-610-004

(3) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-01-010-006

WARNING SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GET INTO THE PASSENGER COMPARTMENT DURING A FIRE.

(4) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-33-01-410-002

(5) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK CONTINUOUS LEVEL SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Waste tank continuous level sensor removal

(2) Waste tank continuous level sensor installation.

TASK 38-33-02-000-801

2. Waste Tank Continuous Level Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove waste tank continuous level sensor.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-33-02-610-001

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-33-02-010-001

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-02-010-002

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-02-040-001

(4) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-02-010-007

(5) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-33-02-040-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-02-910-001

(7) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

G. Waste Tank Continuous Level Sensor Removal

SUBTASK 38-33-02-020-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector from the continuous level sensor [1] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector from the continuous level sensor [1].

(b) Install the protective covers on the electrical receptacle of the continuous level sensor [1] and electrical connector.

SUBTASK 38-33-02-020-002

(2) Remove the clamp [17] and clamp [18] adjacent to the waste tank, attached with screw [16], nut [20], and washer [19] to disconnect the capillary tube.

SUBTASK 38-33-02-020-003

(3) Remove the clamp [8] and clamp [9] outboard of the LCM for the waste tank, with screw [6], nut [10], and washer [7] to disconnect the capillary tube.

SUBTASK 38-33-02-020-004

(4) Remove the clamp [13] and clamp [14] aft of the LCM for the waste tank, with screw [11], nut [15], and washers [12] to disconnect the capillary tube.

SUBTASK 38-33-02-020-005

(5) Remove the V-band clamp [4].

SUBTASK 38-33-02-020-006

CAUTION

DO NOT BEND THE CAPILLARY TUBE OF THE CONTINUOUS LEVEL SENSOR TO LESS THAN A 3-INCH RADIUS. SHARP BENDS CAN CAUSE DAMAGE TO THE CAPILLARY TUBE.

(6) Remove the continuous level sensor [1] and gask-O-seal [5] from the waste line.

(a) Discard the gask-O-seal [5].

SUBTASK 38-33-02-020-007

(7) Remove the clamp [2].

SUBTASK 38-33-02-020-008

(8) Remove the diaphragm half of the continuous level sensor [1] and gask-O-seal [3].

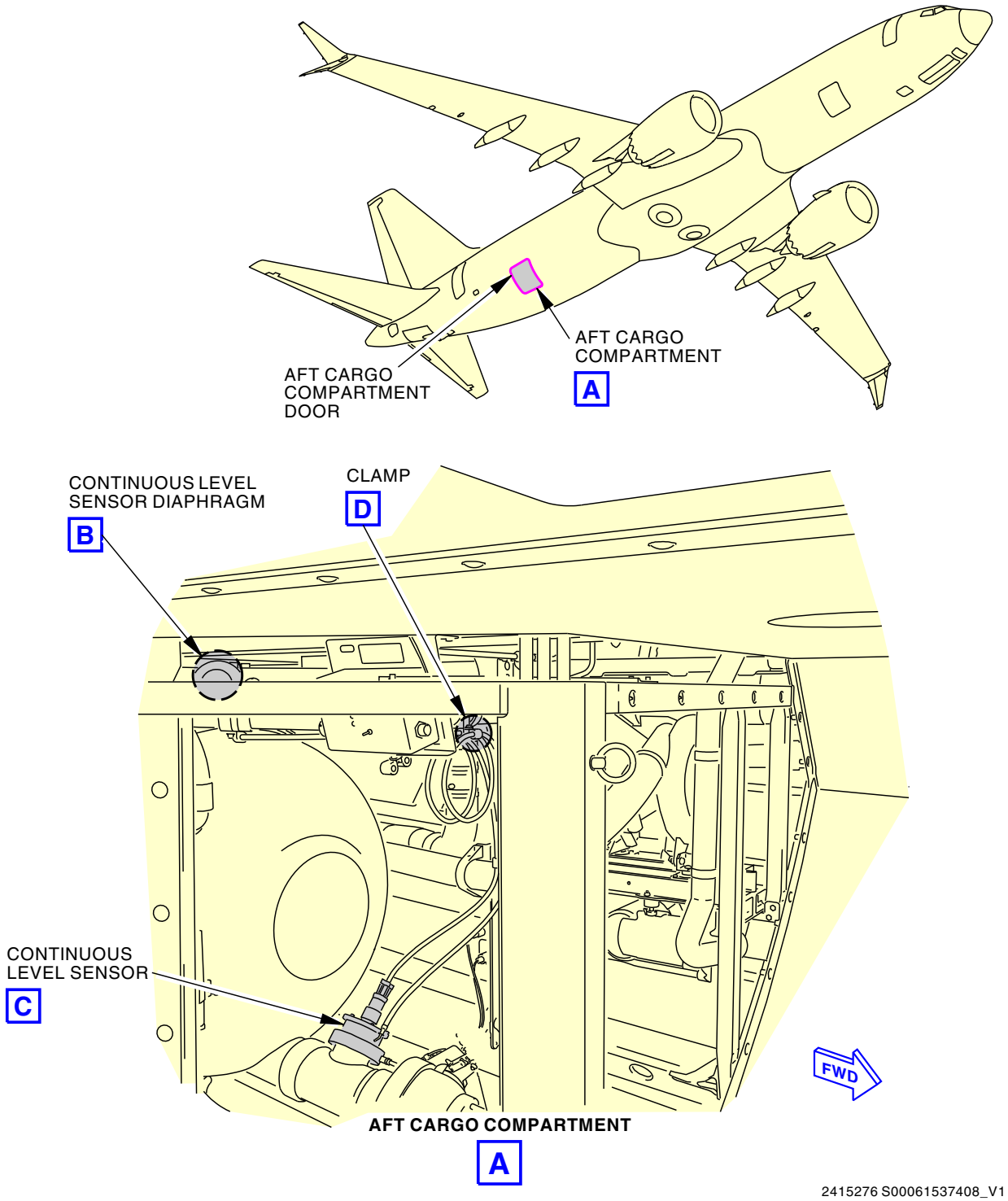
(a) Discard the gask-O-seal [3].

SUBTASK 38-33-02-160-001

(9) Clean all the waste material from the holes in the waste lines.

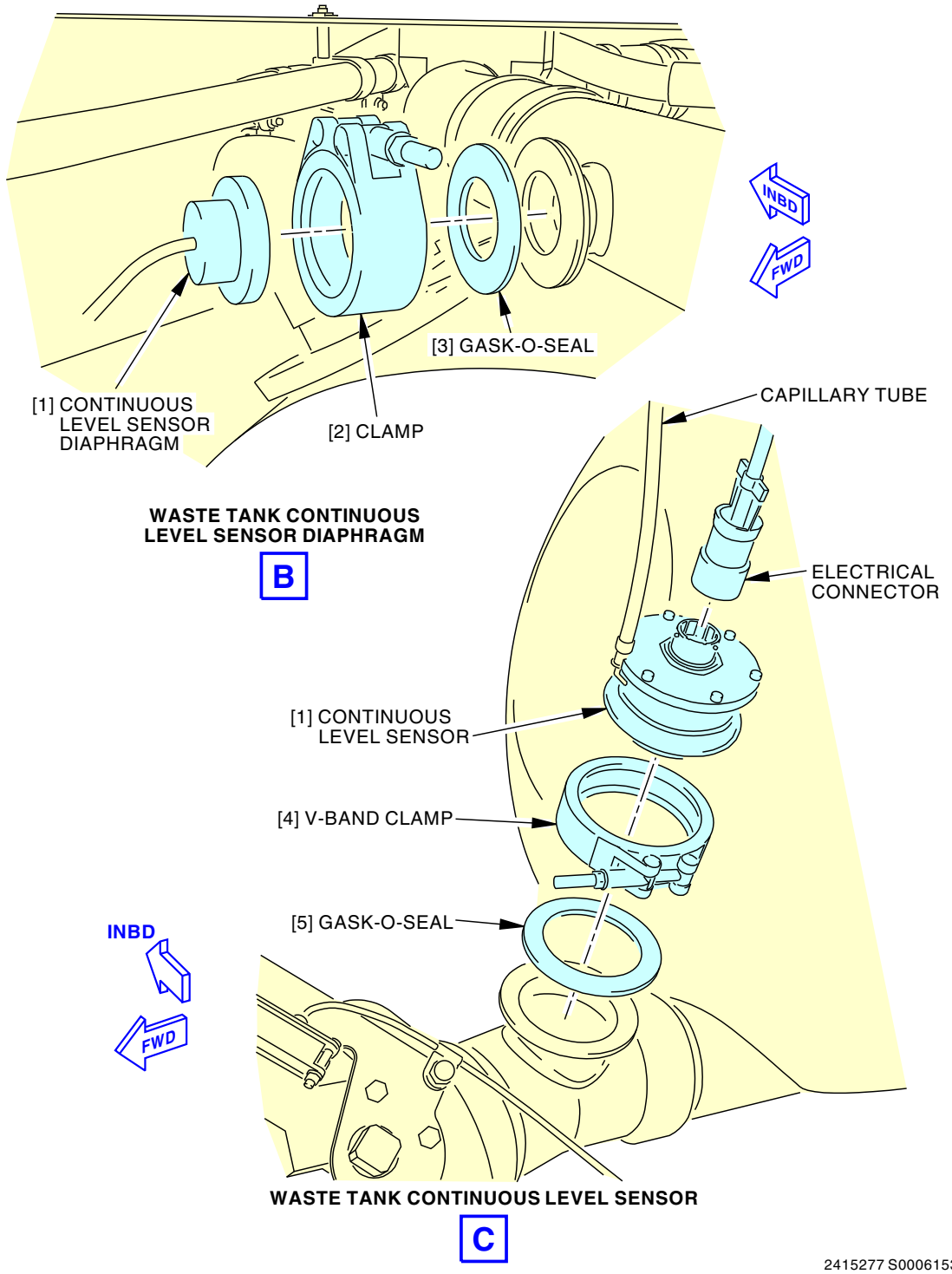
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



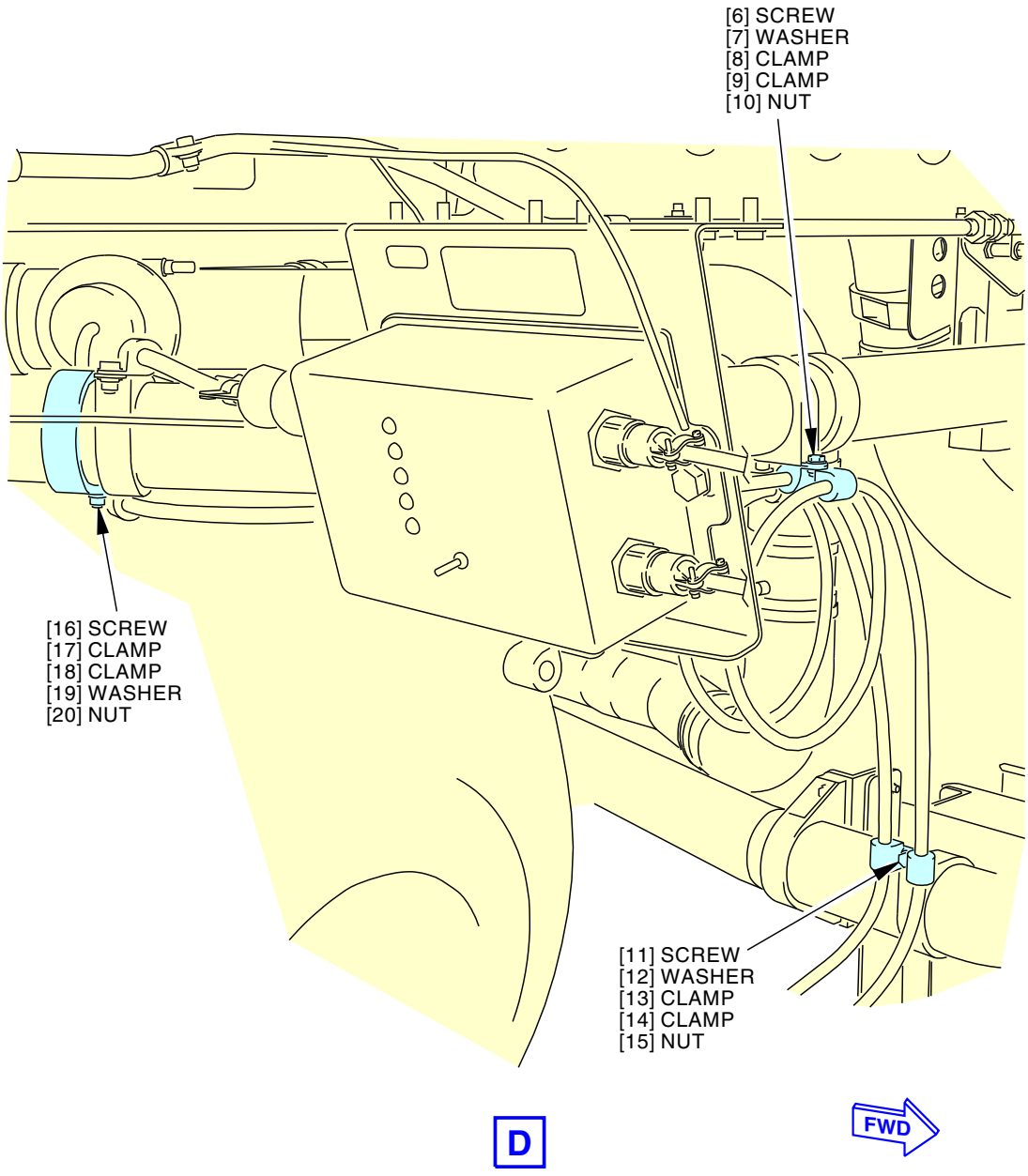
Waste Tank Continuous Level Sensor Installation
Figure 401/38-33-02-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Tank Continuous Level Sensor Installation
Figure 401/38-33-02-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Waste Tank Continuous Level Sensor Installation
Figure 401/38-33-02-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-33-02-400-801

3. Waste Tank Continuous Level Sensor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install waste tank continuous level sensor.

B. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-33-00-710-801	Auto Zero Test (P/B 501)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-1142	Equipment - Waste System Servicing

D. Consumable Materials

Reference	Description	Specification
B00490	Chemical, Toilet Flushing Deodorant	AMS 1476

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Continuous level sensor	38-33-02-01-090	PTF ALL
3	Gask-O-seal	38-33-02-01-080	PTF ALL
5	Gask-O-seal	38-33-02-01-085	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
145AL	Waste Service Door
822	Aft Cargo Door

H. Waste Tank Continuous Level Sensor Installation

SUBTASK 38-33-02-420-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

CAUTION

DO NOT BEND THE CAPILLARY TUBE OF THE CONTINUOUS LEVEL SENSOR TO LESS THAN A 3-INCH RADIUS. SHARP BENDS CAN CAUSE DAMAGE TO THE CAPILLARY TUBE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT BEND THE CAPILLARY TUBE OF THE CONTINUOUS LEVEL SENSOR TO LESS THAN A 3-INCH RADIUS. SHARP BENDS CAN CAUSE DAMAGE TO THE CAPILLARY TUBE.

(1) Install the diaphragm half of the continuous level sensor [1], clamp [2], and new gask-O-seal [3].

SUBTASK 38-33-02-420-002

(2) Tighten the clamp [2] to 40 in-lb (4.5 N·m).

SUBTASK 38-33-02-420-003

(3) Install the V-band clamp [4] and new gask-O-seal [5] to connect the continuous level sensor [1] in the waste line.

SUBTASK 38-33-02-420-004

(4) Tighten the V-band clamp [4] to 45 in-lb (5.1 N·m).

SUBTASK 38-33-02-420-005

(5) Install the clamp [17], clamp [18], screw [16], nut [20], and washers [19] for the capillary tube adjacent to the waste tank.

SUBTASK 38-33-02-420-006

(6) Install the clamp [8], clamp [9], screw [6], nut [10], and washers [7] to connect the capillary tube outboard of LCM for the waste tank.

SUBTASK 38-33-02-420-007

(7) Install the clamp [13], clamp [14], screw [11], nut [15], and washers [12] to connect the capillary tube aft of the LCM for the waste tank.

SUBTASK 38-33-02-420-008

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(8) Use the teflon-jawed pliers, STD-664, to connect the electrical connector to the continuous level sensor [1] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the continuous level sensor [1] and electrical connector.

(b) Connect the electrical connector to the continuous level sensor [1].

1) Tighten the connector with the teflon-jawed pliers, STD-664.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Waste Tank Continuous Level Sensor Installation Test

SUBTASK 38-33-02-630-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-02-630-002

(2) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-02-760-001

(3) Do this task: Auto Zero Test, TASK 38-33-00-710-801.

SUBTASK 38-33-02-790-001

(4) To do a leak check of the continuous level sensor connection, do these steps:

(a) Open this access panel:

Number	Name/Location
145AL	Waste Service Door

(b) At the service panel, make sure that the control handles for the drain ball valves are in the closed position.

(c) Connect the water servicing hose from the waste system servicing equipment, STD-1142, to the service panel rinse fitting.

WARNING

YOU MUST OBEY THE MANUFACTURERS INSTRUCTIONS WHEN YOU USE THE PRECHARGE CHEMICAL. THE PRECHARGE CHEMICAL IS POISONOUS AND CAN CAUSE CORROSION. THIS CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION

DO NOT ADD THE PRECHARGE CHEMICAL TO THE WASTE TANK IF THE AIRPLANE CAN FREEZE. DAMAGE TO THE WASTE SYSTEM CAN OCCUR IF THE PRECHARGE CHEMICAL BECOMES FROZEN.

(d) Use the pump on the service cart to put about 50 gal (189 l) of water or precharge chemical, B00490, into the waste tank (TASK 12-17-01-610-801).

NOTE: If chemical toilet flushing deodorant is used for leak check, then in case of leak check fail the cleaning process may be more difficult due to the chemicals present in the deodorant.

(e) Make sure that the continuous level sensor connection does not have a leak.

NOTE: You must wait at least 30 minutes after you put the water in the tank before you do this step.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-02-610-002

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-33-02-010-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-33-02-410-001

(3) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WASTE TANK CONTINUOUS LEVEL SENSOR - CLEANING/PAINTING

1. General

A. This procedure has this task:

(1) A cleaning of the continuous level sensor for the waste tank.

TASK 38-33-02-100-801

2. Waste Tank Continuous Level Sensor Cleaning

A. References

Reference	Title
12-17-01-610-801	Waste Tank Servicing (P/B 301)
38-32-00-910-801	Standard Practices for Work with the Toilet Waste and Equipment (P/B 201)
38-33-02-000-801	Waste Tank Continuous Level Sensor Removal (P/B 401)
38-33-02-400-801	Waste Tank Continuous Level Sensor Installation (P/B 401)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-123	Brush - Soft Bristle

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-33-02-610-003

CAUTION USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

NOTE: Do not add the chemical precharge after you do the task to drain and flush the waste tanks.

SUBTASK 38-33-02-010-004

(2) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-02-010-005

(3) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-02-040-002

(4) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-02-010-008

(5) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-33-02-040-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-02-910-002

(7) Do this task: Standard Practices for Work with the Toilet Waste and Equipment, TASK 38-32-00-910-801.

F. Continuous Level Sensor Cleaning

SUBTASK 38-33-02-020-009

(1) Do this task: Waste Tank Continuous Level Sensor Removal, TASK 38-33-02-000-801.

SUBTASK 38-33-02-140-001

(2) Use a rag to clean the loose material from the face of the continuous level sensor.

SUBTASK 38-33-02-140-002

CAUTION

BE CAREFUL TO PREVENT DAMAGE TO THE CONTINUOUS LEVEL-SENSOR DIAPHRAGMS (WETTED AREAS OF SENSOR LOCATED IN THE O-RINGS). BE CAREFUL NOT TO CAUSE DAMAGE TO THE DIAPHRAGMS BECAUSE THEY ARE VERY THIN (.001 INCH (.025 MM)). DAMAGE CAN CAUSE A FAILURE OF THE CONTINUOUS LEVEL-SENSOR.

(3) Use a soft bristle soft bristle brush, STD-123, and soap to clean the face of the continuous level sensor.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-02-140-003

(4) Dry the continuous level sensor.

SUBTASK 38-33-02-160-002

(5) Clean all the waste material from the hole in the waste tank for the continuous level sensor.

SUBTASK 38-33-02-020-010

(6) Do this task: Waste Tank Continuous Level Sensor Installation, TASK 38-33-02-400-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-02-610-004

(1) Do this task: Waste Tank Servicing, TASK 12-17-01-610-801.

SUBTASK 38-33-02-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-02-860-004

(3) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-02-010-006

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(4) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-33-02-410-002

(5) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOGIC CONTROL MODULE - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Logic control module removal

(2) Logic control module installation

(3) Logic control module operational test.

B. The Logic Control Module is referred to as the LCM in this procedure.

C. The LCM has four electrical connectors. J1 and J2 are connected to the point level sensors. J3 is connected to the continuous level sensor. J4 is connected to the power and control circuits for the airplane.

TASK 38-33-03-000-801

2. Logic Control Module Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove the logic control module.

B. References

Reference	Title
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-33-03-010-001

(1) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-03-010-002



CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(2) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-03-040-002

(3) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-03-010-006

(4) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-33-03-040-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

G. Logic Control Module Removal

SUBTASK 38-33-03-020-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

CAUTION

MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connectors from the logic control module [4] (TASK 70-00-01-910-803-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect the J3 electrical connector [1], J4 electrical connector [7], J1 electrical connector [5], and J2 electrical connector [6] from the logic control module [4].

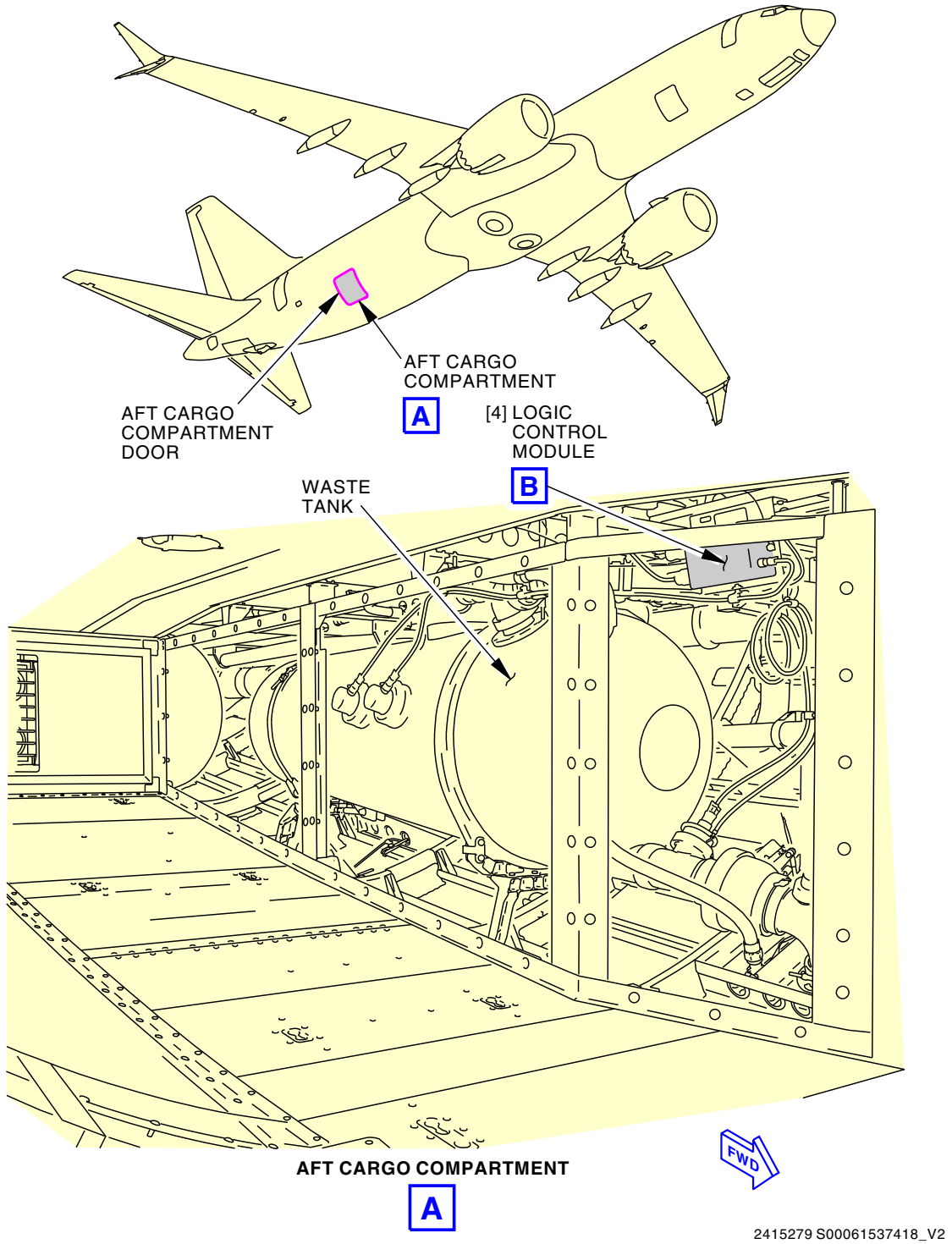
(b) Install the protective covers on the electrical receptacle of the logic control module [4] and the electrical connectors.

SUBTASK 38-33-03-020-004

(2) Remove the bolts [2] and washers [3] to disconnect the jumper assemblies [8] and remove the logic control module [4].

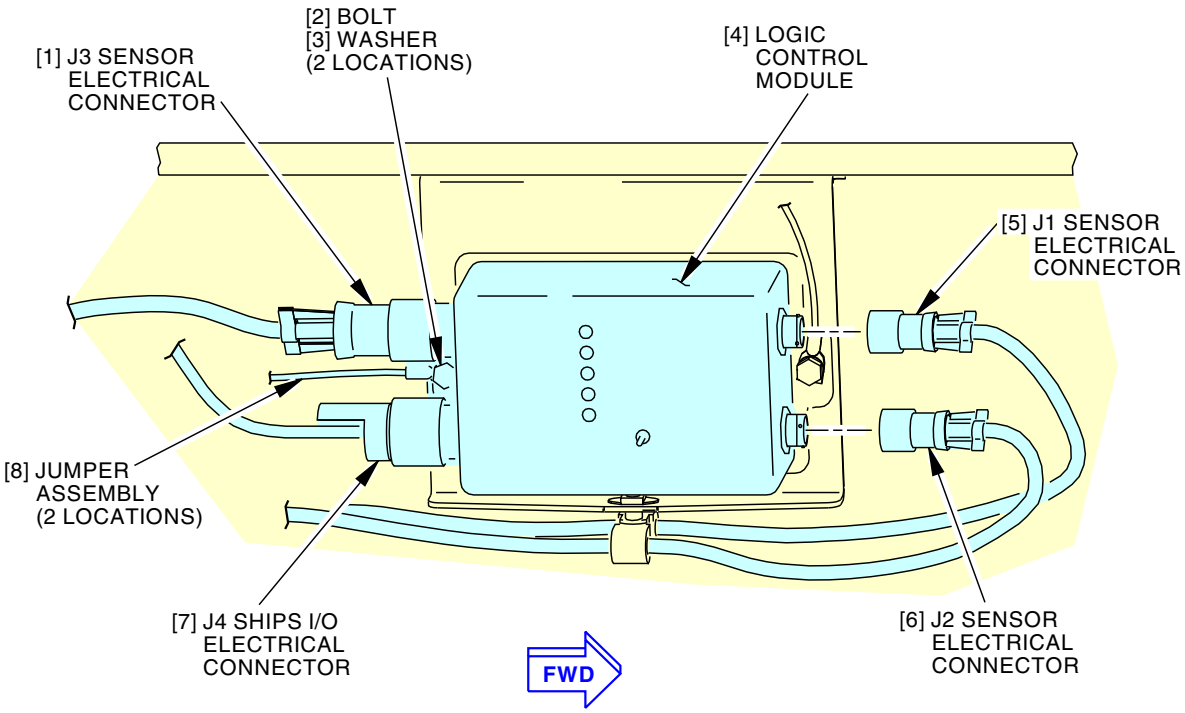
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Logic Control Module Installation
Figure 201/38-33-03-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LOGIC CONTROL MODULE

B

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Logic Control Module Installation
Figure 201/38-33-03-990-802 (Sheet 2 of 2)

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TASK 38-33-03-400-801

3. Logic Control Module Installation

(Figure 201)

A. General

(1) This task gives the instructions to install the logic control module.

B. References

Reference	Title
38-33-00-710-801	Auto Zero Test (P/B 501)
38-33-00-740-802	LCM BITE Test (P/B 501)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Logic control module	38-33-03-01-105	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

G. Logic Control Module Installation

SUBTASK 38-33-03-420-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Install the bolts [2] and washers [3] to connect the jumper assemblies [8] and install the logic control module [4].

SUBTASK 38-33-03-420-002

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Use the teflon-jawed pliers, STD-664 to connect the electrical connectors to the logic control module [4] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the logic control module [4] and the electrical connectors.

(b) Connect the J3 electrical connector [1], J4 electrical connector [7], J1 electrical connector [5], and J2 electrical connector [6] to the logic control module [4].

NOTE: You can connect the J1 connector to the J2 receptacle or the J2 connector to the J1 receptacle. Make sure that you install the J1 and J2 electrical connectors in the correct location on the LCM.

1) Tighten the connectors with the teflon-jawed pliers, STD-664.

H. Logic Control Module Installation Test

SUBTASK 38-33-03-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-03-860-008

(2) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-03-760-001

(3) Do this task: LCM BITE Test, TASK 38-33-00-740-802.

SUBTASK 38-33-03-760-002

(4) Do this task: Auto Zero Test, TASK 38-33-00-710-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-03-010-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-33-03-410-001

(2) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

——— END OF TASK ———

TASK 38-33-03-710-801

4. Logic Control Module Operational Test

A. General

(1) This task gives the instructions to test the logic control module.

B. References

Reference	Title
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Operational Test:

SUBTASK 38-33-03-010-004

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-03-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Make sure that this circuit breaker is closed:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	11	C01388	VACUUM WASTE CONT

SUBTASK 38-33-03-860-009

(3) Make sure that this circuit breaker is closed:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	19	C01423	VACUUM WASTE

SUBTASK 38-33-03-010-007

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-33-03-010-005

(5) Do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

F. Logic Control Module Operational Test

SUBTASK 38-33-03-710-001

CAUTION

USE TOOLS CAREFULLY BECAUSE THE WASTE TANKS HAVE A COMPOSITE SKIN. IT IS EASILY DAMAGED. IF YOU HIT THE TANK OR LET IT FALL, THIS CAN CAUSE DAMAGE THAT YOU CANNOT SEE.

(1) Make sure that the POWER ON light for the LCM is on.

SUBTASK 38-33-03-710-002

(2) Move and then release the switch to the TEST SENSOR position.

SUBTASK 38-33-03-710-003

(3) Make sure that the POWER ON, J1 SENSOR, J2 SENSOR, J3 SENSOR, and TANK FULL lights are on for approximately 3 seconds.

NOTE: The lights on the LCM can flash irregularly. If you find a system error, the TANK FULL light will flash at 8 Hz, then go off. You will see this cycle again after 1.6 seconds until you correct the defect.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-33-03-710-004

(4) Make sure that the J1 SENSOR, J2 SENSOR, J3 SENSOR, and TANK FULL lights are off.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-33-03-410-002

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-33-03-410-003

(2) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER TANK PRESSURIZATION - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A deactivation of the potable water system

(2) An activation of the potable water system.

TASK 38-42-00-800-801

2. Potable Water System - Deactivation

(Figure 201)

A. General

(1) This task will deactivate potable water air compressor in the potable water system.

B. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
144	Area Below Aft Cargo Compartment - Right
822	Aft Cargo Door

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
146AR	Water Service Door
822	Aft Cargo Door

E. Potable Water System Deactivation

SUBTASK 38-42-00-860-001

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 38-42-00-010-001

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-00-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-00-010-008

(4) Open this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-42-00-860-003

(5) Turn the handle of the fill/overflow valve to open.

SUBTASK 38-42-00-710-001

(6) When no more air comes out of the overflow port, close the fill/overflow valve.

F. Tryout

NOTE: This tryout is to make sure that the potable water system is in a zero energy state.

SUBTASK 38-42-00-410-001

(1) Close this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-42-00-010-002

(2) Open this door:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-00-010-003

(3) To remove the waste tank enclosure lining, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

SUBTASK 38-42-00-200-001

(4) Make sure that the potable water air compressor does not operate.

SUBTASK 38-42-00-410-011

(5) Close this door:

Number	Name/Location
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

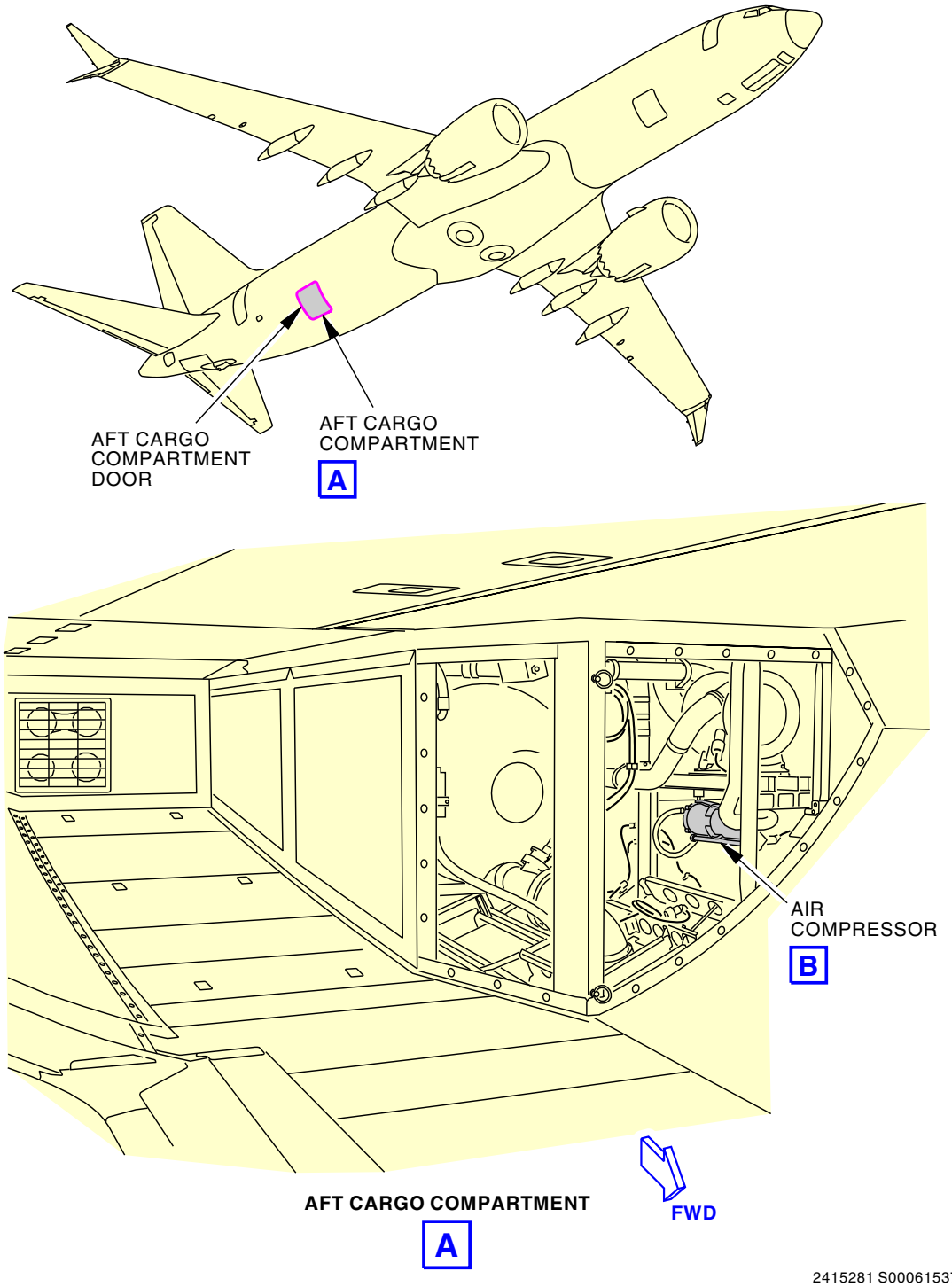
SUBTASK 38-42-00-410-009

(6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

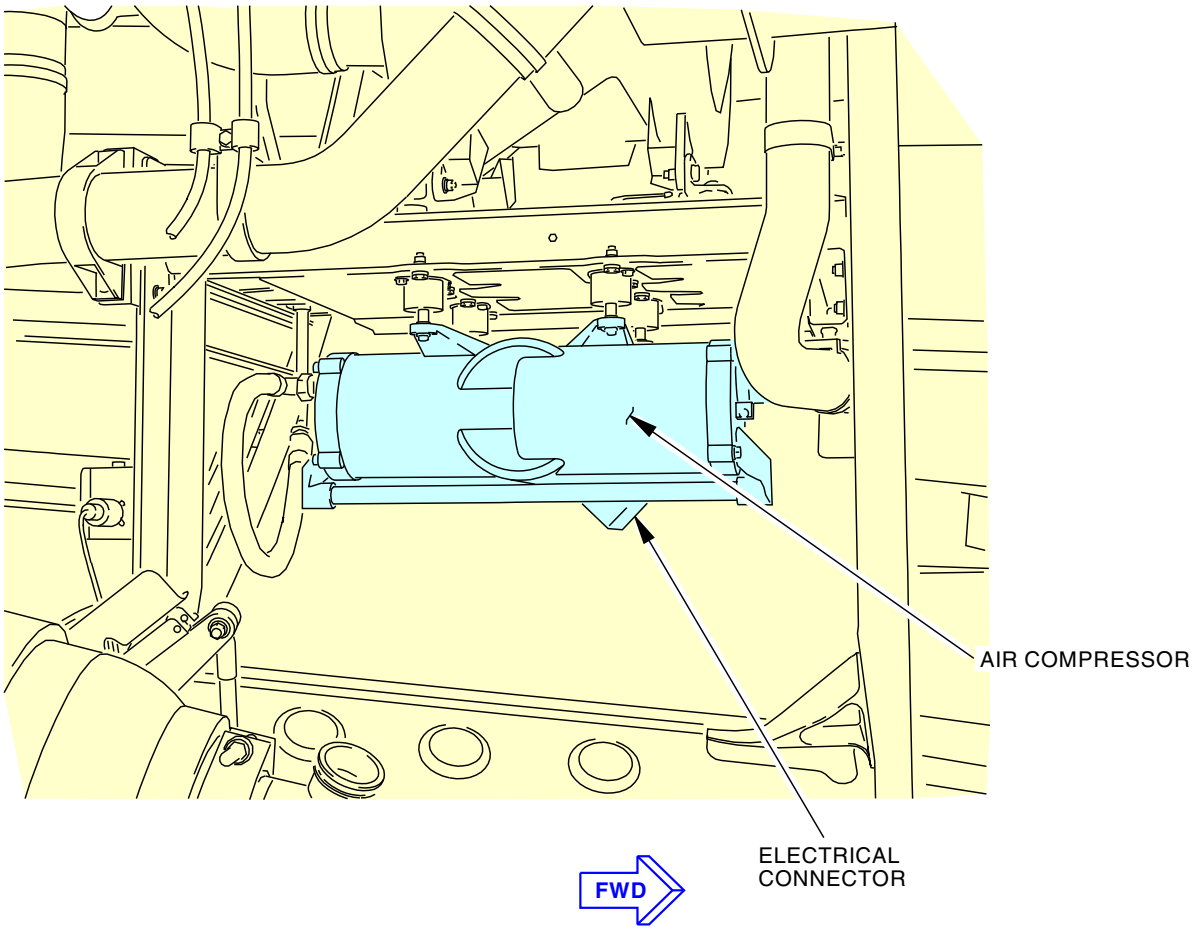
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Potable Water Air Compressor
Figure 201/38-42-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AIR COMPRESSOR

B

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Potable Water Air Compressor
Figure 201/38-42-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-42-00-800-802

3. Potable Water System - Activation

(Figure 201)

A. General

(1) This task will activate potable water air compressor in the potable water system.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
144	Area Below Aft Cargo Compartment - Right
822	Aft Cargo Door

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
146AR	Water Service Door
822	Aft Cargo Door

E. Potable Water System Activation

SUBTASK 38-42-00-010-010

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-00-610-001

(2) If the potable water tank does not have water in it, do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-42-00-010-009

(3) Open this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-42-00-860-004

(4) Make sure that the fill/overflow valve is closed.

SUBTASK 38-42-00-860-005

(5) Close this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-42-00-200-003

(6) Make sure that the CAB/UTIL switch, on the P5-13 panel, is in the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-00-200-004

(7) Make sure that the pressure from the pneumatic system is removed (TASK 36-00-00-860-806).

SUBTASK 38-42-00-860-006

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(8) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-00-410-012

(9) Open this door:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-00-200-002

(10) Make sure that the potable water air compressor operates.

SUBTASK 38-42-00-410-003

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GET INTO THE PASSENGER COMPARTMENT DURING A FIRE.

(11) To install the waste tank enclosure lining, do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-42-00-410-004

(12) Close this door:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-00-410-013

(13) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER TANK PRESSURIZATION - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An operational test of the water tank pressurization system.

(2) An operational test the pressure regulator.

TASK 38-42-00-800-803

2. Water Tank Pressurization System - Operational Test

A. References

Reference	Title
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
38-10-00-790-801	Potable Water System - Leak Test (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
141	Aft Cargo Compartment - Left
144	Area Below Aft Cargo Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
146AR	Water Service Door

D. Release Pressure from the Potable Water System

SUBTASK 38-42-00-860-009

(1) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 38-42-00-010-007

(2) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-00-860-010



WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-00-010-004

(4) Open this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-42-00-860-019

(5) Turn the handle of the fill/overflow valve to release the pressure.

SUBTASK 38-42-00-860-012

(6) When no more air comes out of the overflow port, close the fill/overflow valve.

SUBTASK 38-42-00-410-005

(7) Close this access panel:

Number	Name/Location
146AR	Water Service Door

SUBTASK 38-42-00-860-014

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(8) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-00-710-002

(9) Make sure the air compressor comes on and the air compressor shuts off.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-00-710-003

(10) If the air compressor shuts off after approximately 10 minutes, the system operation is satisfactory.

(a) If the air compressor does not shut off after approximately 10 minutes, do this task: Potable Water System - Leak Test, TASK 38-10-00-790-801.

SUBTASK 38-42-00-410-010

(11) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 38-42-00-800-804

3. Pressure Regulator - Operational Test

A. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
36-00-00-860-802	Supply Pressure to the Pneumatic System with an External Ground Air Source (P/B 201)
38-42-08-000-801	Pressure Regulator Removal (P/B 401)
38-42-08-400-801	Pressure Regulator Installation (P/B 401)
50-10-03-000-801	Cargo Compartment Ceiling Liner Removal (P/B 401)
50-10-03-400-801	Cargo Compartment Ceiling Liner Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-1091	Gauge - Air Pressure, 0-100 PSIG (0-690 KPa)

C. Pressure Regulator Operational Test

SUBTASK 38-42-00-020-001

(1) Do this task: Cargo Compartment Ceiling Liner Removal, TASK 50-10-03-000-801.

SUBTASK 38-42-00-020-002

(2) Remove the cap from the tee of the potable water line.

SUBTASK 38-42-00-480-001

(3) Connect the pressure gauge (0-100 PSIG) (0-690 KPa), STD-1091 to the tee fitting.

SUBTASK 38-42-00-610-002

(4) If the potable water tank is not full, do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

SUBTASK 38-42-00-610-003

(5) To pressurize the pneumatic system with at least 40 psig, do this task: Supply Pressure to the Pneumatic System with an External Ground Air Source, TASK 36-00-00-860-802.

SUBTASK 38-42-00-860-015

(6) Do a check of the pressure on the pneumatic manifold to make sure the pneumatic pressure is not less than 40 psig.

SUBTASK 38-42-00-860-016

(7) Read the pressure gauge (0-100 PSIG) (0-690 KPa), STD-1091 in the potable water system.

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(a) Make sure the pressure is 33 to 37 psig.

(b) If the pressure is not 33 to 37 psig, replace the regulator.
These are the tasks:
Pressure Regulator Removal, TASK 38-42-08-000-801,
Pressure Regulator Installation, TASK 38-42-08-400-801.

D. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-00-860-017

(1) Disconnect the pressure gauge (0-100 PSIG) (0-690 KPa), STD-1091 from the tee fitting.

SUBTASK 38-42-00-860-018

(2) Install the cap on the tee of the potable water line.

SUBTASK 38-42-00-020-003

(3) Do this task: Cargo Compartment Ceiling Liner Installation, TASK 50-10-03-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR COMPRESSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Air compressor removal

(2) Air compressor installation.

B. Leaking hose connections or a leaking check valve can cause frequent or continuous air compressor operation leading to compressor failure. The check valve can be replaced when replacing the compressor as a preventative step.

TASK 38-42-01-000-801

2. Air Compressor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the air compressor.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Removal

SUBTASK 38-42-01-010-003

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-01-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-01-860-002

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

G. Air Compressor Removal

SUBTASK 38-42-01-020-001

CAUTION MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [10] from the air compressor [7] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [10] from the air compressor [7].

(b) Install the protective covers on the electrical receptacle of the air compressor [7] and the electrical connector [10].

SUBTASK 38-42-01-020-002

(2) Disconnect the hose [8] from the union [1] on the outlet of the air compressor [7].

SUBTASK 38-42-01-020-003

(3) Disconnect the clamp [5], hose [6] and adapter [9] attached to the inlet of the air compressor [7].

SUBTASK 38-42-01-480-001

(4) Put a cap on the hose [6] to keep out contamination.

SUBTASK 38-42-01-020-004

(5) Remove the screws [3] and washers [4] to disconnect the air compressor [7] from the structure.

SUBTASK 38-42-01-020-005

(6) Remove the air compressor [7].

SUBTASK 38-42-01-020-006

(7) Put a cap on the hose [8] to keep out contamination.

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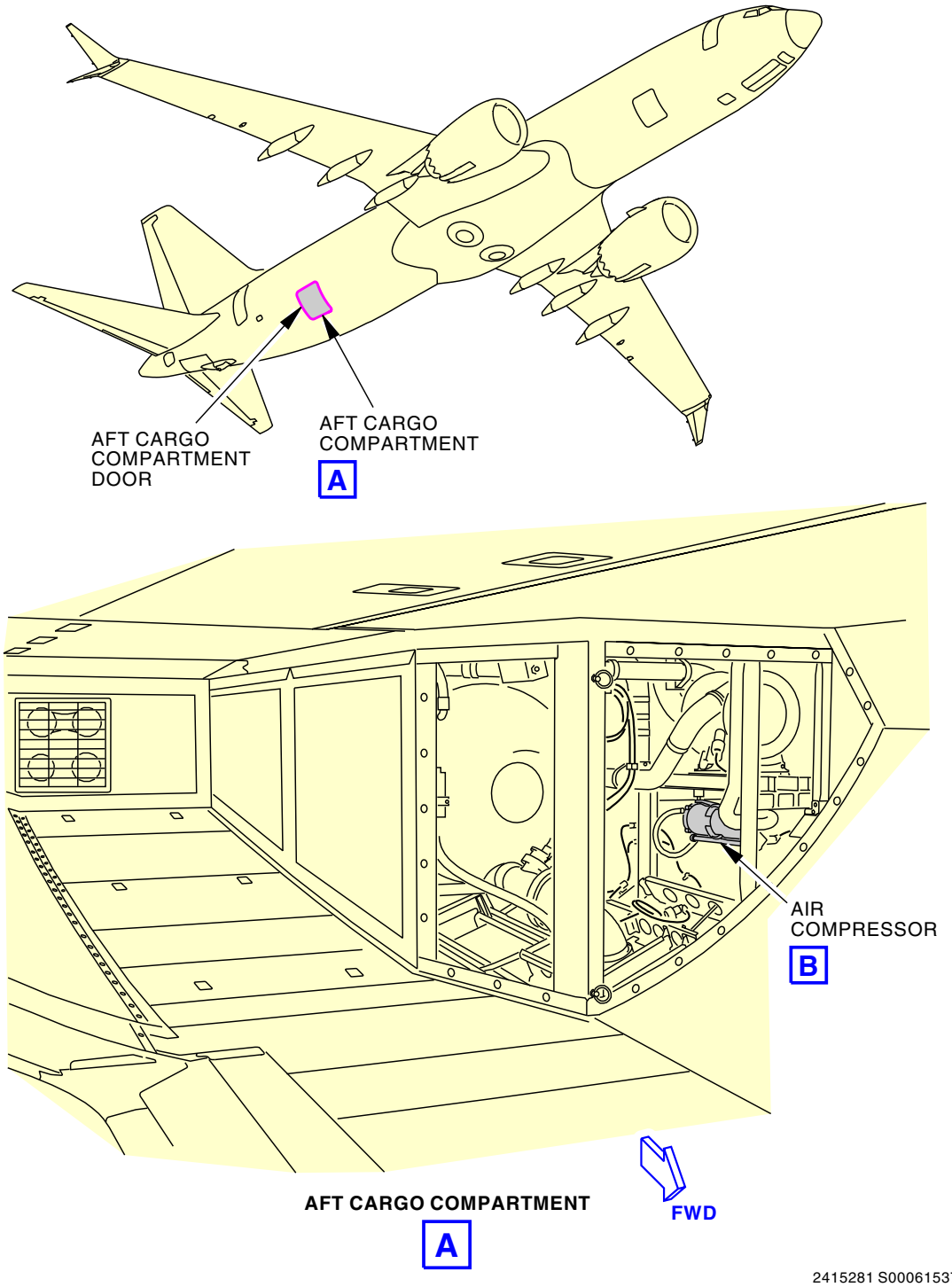
SUBTASK 38-42-01-020-007

(8) Remove the union [1] and packing [2] from the air compressor [7].

(a) Discard the packing [2].

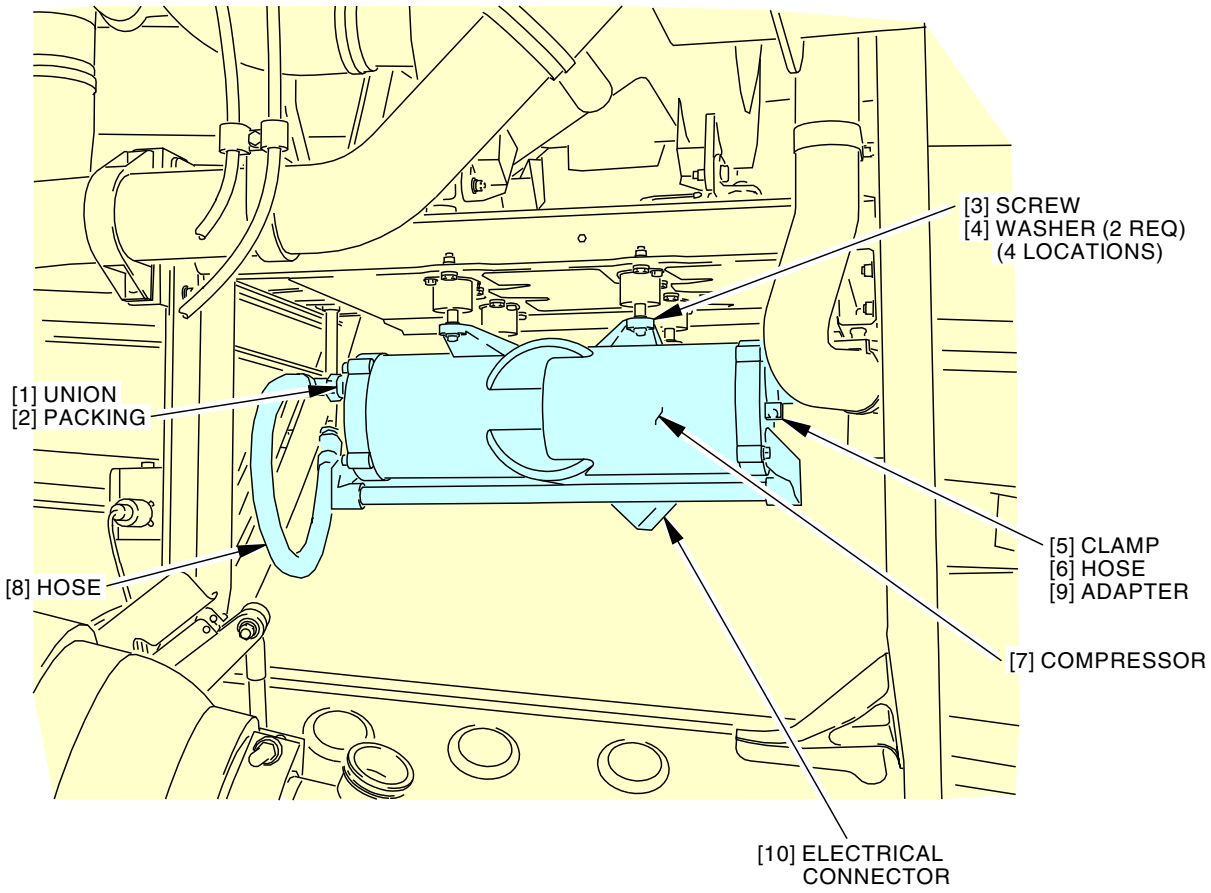
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Air Compressor Installation
Figure 401/38-42-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Air Compressor Installation
Figure 401/38-42-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-42-01-400-801

3. Air Compressor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the air compressor.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
38-42-09-020-801	Compressor Supply Check Valve Removal (P/B 401)
38-42-09-400-801	Compressor Check Valve Installation (P/B 401)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00091	Compound - Oxygen System Leak Detection - Snoop Leak Detector	MIL-PRF-25567
G50321	Air - Clean, Dry	BAC5402 Table I

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Packing	38-42-01-02-010	PTF ALL
7	Compressor	38-42-01-02-310	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Air Compressor Installation

SUBTASK 38-42-01-640-001

(1) Apply the grease, D00463, or Molykote 111 Compound, D00189, to the new packing [2] and threads of the union [1].

SUBTASK 38-42-01-420-001

(2) Install a packing [2] on the union [1].

SUBTASK 38-42-01-420-002

(3) Install the union [1] with the packing [2] in the air compressor [7].

SUBTASK 38-42-01-420-003

(4) Put the air compressor [7] in its position on the mounting plate.

SUBTASK 38-42-01-420-004

(5) Install the washers [4] and screws [3] for the air compressor [7] to the structure.

SUBTASK 38-42-01-020-008

(6) Remove the cap on the hose [8].

SUBTASK 38-42-01-420-005

(7) Connect the hose [8] to the union [1] in the outlet of the air compressor [7].

SUBTASK 38-42-01-420-006

(8) Put the adapter [9] in its position on the air compressors [7].

SUBTASK 38-42-01-020-009

(9) Do this task: Compressor Supply Check Valve Removal, TASK 38-42-09-020-801.

SUBTASK 38-42-01-080-001

(10) Remove the cap from the hose [6].

SUBTASK 38-42-01-617-001

(11) Clear the hose and line assembly, between the air compressor [7] and check valve, of any contamination.

NOTE: To clean the hose and line assembly, use 40 psig (3 kg/cm^2) to 100 psig (7 kg/cm^2) clean, dry shop air, or nitrogen, applied at the check valve end of the hose, for 1 to 2 minutes.

(a) Use clean dry air, G50321, or nitrogen, G00018.

SUBTASK 38-42-01-420-009

(12) Do this task: Compressor Check Valve Installation, TASK 38-42-09-400-801.

SUBTASK 38-42-01-420-007

(13) Connect the hose [6] and clamps [5] to the adapter [9].

SUBTASK 38-42-01-420-008

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(14) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [10] to the air compressor [7] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the air compressor [7] and the electrical connector [10].

(b) Connect the electrical connector [10] to the air compressor [7].

1) Tighten the connector with the teflon-jawed pliers, STD-664.

SUBTASK 38-42-01-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(15) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-01-860-004

(16) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 38-42-01-860-005

(17) Activate the portable water system (TASK 38-42-00-800-802).

SUBTASK 38-42-01-790-001

(18) Make sure there are no air leaks at the connections to the air compressor [7].

(a) Apply the Snoop Leak Detector compound, G00091, to all the fittings and connections.

1) Look for bubbles to find any leaks.

2) If you find leaks, tighten the fittings and connections.

(b) Remove the Snoop Leak Detector compound, G00091, with a clean cotton wiper, G00034, immediately after the check.

1) Make sure the fittings and connections are dry.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-01-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

SUBTASK 38-42-01-410-002

(2) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COMPRESSOR INLET AIR FILTER - REMOVAL/INSTALLATION

1. General

A. The air compressor has a built-in air filter assembly on its forward side.

NOTE: The air filter assembly has a filter element that you can remove.

B. This procedure has these tasks:

(1) A removal of the air filter element

(2) An installation of the air filter element.

NOTE: You can replace the used filter assembly with a new filter assembly.

TASK 38-42-02-000-801

2. Air Filter Element Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the air filter element.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
50-12-03-000-801	Waste Tank Enclosure Panel Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-42-02-010-003

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-02-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-02-860-002

(3) To release the pressure from the potable water system, do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-02-010-001

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-02-010-002

(5) To remove the waste tank enclosure panels, do this task: Waste Tank Enclosure Panel Removal, TASK 50-12-03-000-801.

F. Air Filter Element Removal

SUBTASK 38-42-02-020-001

(1) Loosen the clamp [1] to disconnect the hose [2] from the air compressor [4].

SUBTASK 38-42-02-020-002

(2) Remove the adapter [3].

SUBTASK 38-42-02-020-003

(3) Remove the air filter assembly [5].

SUBTASK 38-42-02-020-004

(4) Remove the retaining ring [8] from the air filter assembly [5].

NOTE: The retaining ring will be used again in the installation task.

SUBTASK 38-42-02-020-005

(5) Remove the screen [7] from the air filter assembly [5].

SUBTASK 38-42-02-020-006

(6) Remove and discard the packing [6].

SUBTASK 38-42-02-020-007

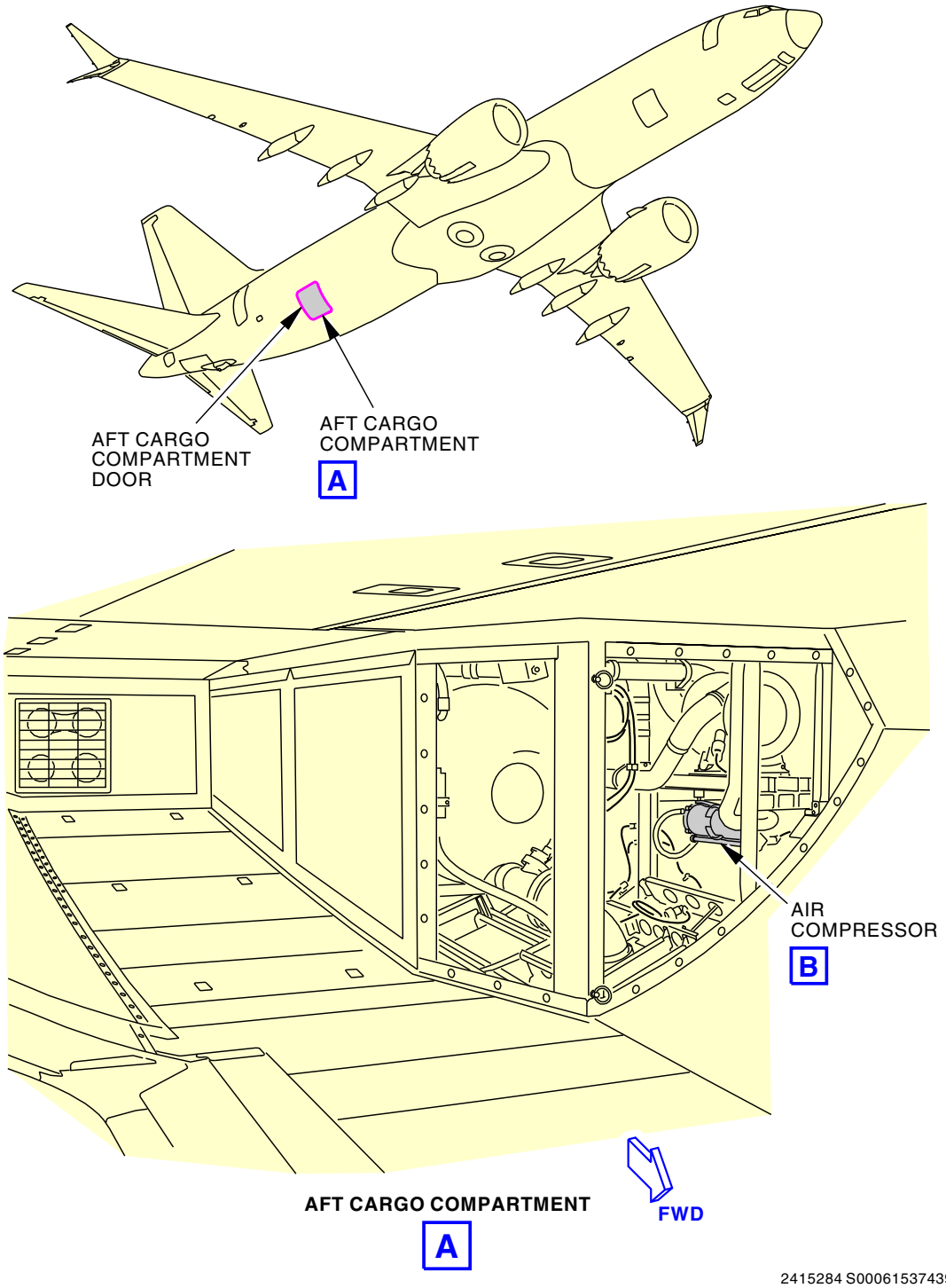
(7) Remove the filter [10].

SUBTASK 38-42-02-020-008

(8) Remove and discard the packing [11].

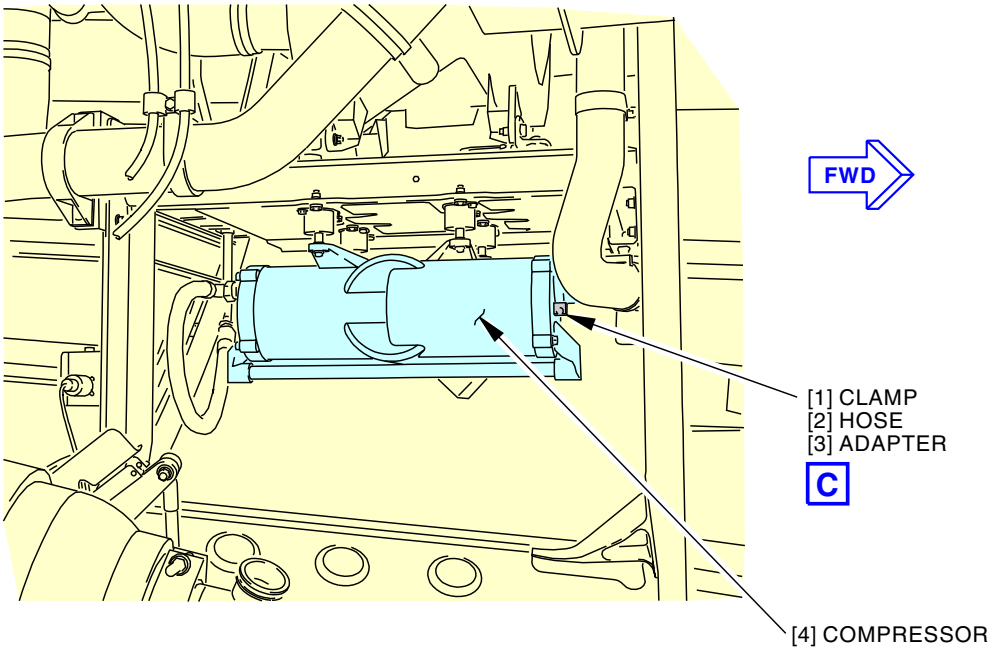
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



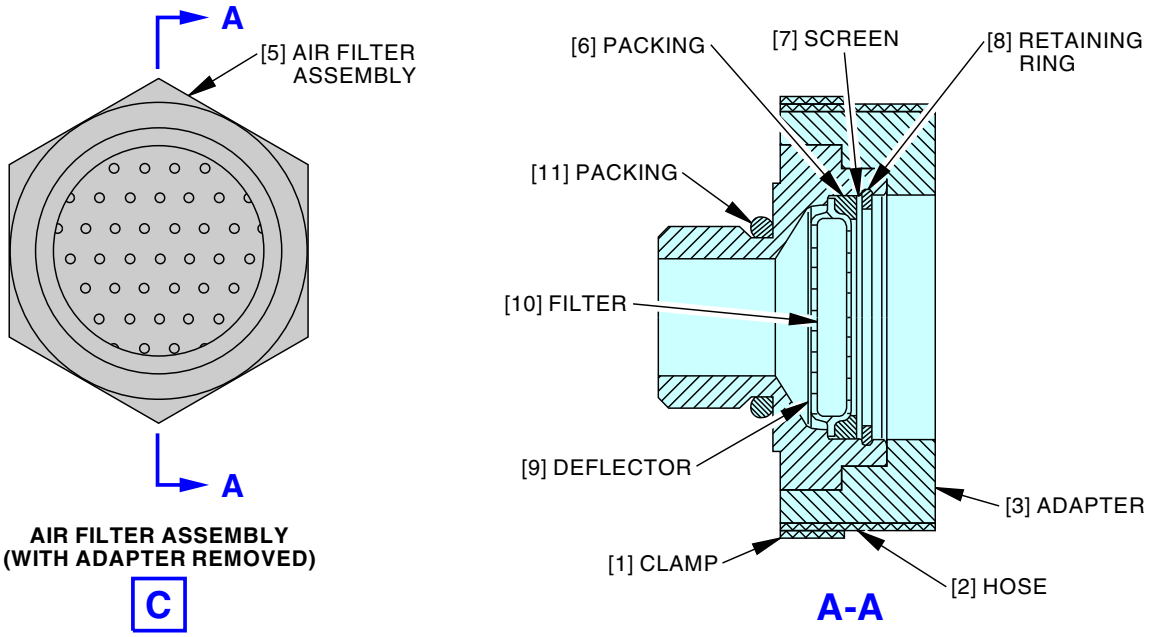
Compressor Inlet Air Filter Installation
Figure 401/38-42-02-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AIR COMPRESSOR

B



AIR FILTER ASSEMBLY (WITH ADAPTER REMOVED)

C

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Compressor Inlet Air Filter Installation
Figure 401/38-42-02-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-42-02-400-801

3. Air Filter Element Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the air filter element.

B. References

Reference	Title
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-12-03-400-801	Waste Tank Enclosure Panel Installation (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
6	Packing	38-42-02-01-030	PTF ALL
10	Filter	38-42-02-01-035	PTF ALL
11	Packing	38-42-02-01-015	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Air Filter Element Installation

SUBTASK 38-42-02-420-001

(1) Make sure that the deflector [9] is in its position in the air filter assembly [5].

SUBTASK 38-42-02-420-002

(2) Put a new or serviceable filter [10] into the air filter assembly [5].

(a) To make the inlet filter [10] serviceable, clean as follows:

1) Soak the filter element in a suitable solvent (such as aliphatic naphtha).

2) Reverse flow blow dry with low air pressure.

3) Make sure that the filter is in serviceable condition.

SUBTASK 38-42-02-420-003

(3) Put a new packing [6] into the air filter assembly [5].

SUBTASK 38-42-02-420-004

(4) Put the screen [7] into its position.

SUBTASK 38-42-02-420-005

(5) Install the retaining ring [8].

SUBTASK 38-42-02-420-006

(6) Install a new packing [11] on the air filter assembly [5].

SUBTASK 38-42-02-420-007

(7) Install the air filter assembly [5] into the air compressor [4].

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SUBTASK 38-42-02-420-008

(8) Torque the air filter assembly [5] to 77.5 ±2.5 in-lb (8.8 ±0.3 N·m).

SUBTASK 38-42-02-420-009

(9) Install the adapter [3] on the air filter assembly [5].

SUBTASK 38-42-02-420-010

(10) Put the hose [2] in its position and then install the clamp [1].

G. Air Filter Element Installation Test

SUBTASK 38-42-02-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-02-410-003

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-02-860-004

(3) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-42-02-710-001

(4) Make sure that the air compressor operates without leakage.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-02-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Waste Tank Enclosure Panel Installation, TASK 50-12-03-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-02-410-002

(2) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRESSURE RELIEF VALVE - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Pressure relief valve removal

(2) Pressure relief valve installation

(3) Pressure relief valve operational test.

TASK 38-42-06-000-801

2. Pressure Relief Valve Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove the pressure relief valve.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-42-06-010-003

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-06-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-06-860-002

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-06-010-001

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-06-010-004

(5) Remove the electrical equipment rack, E6, aft access panel.

F. Pressure Relief Valve Removal

SUBTASK 38-42-06-020-001

WARNING

DO NOT TURN THE DEFLECTOR CAP WHEN YOU REMOVE OR INSTALL THE PRESSURE RELIEF VALVE. THE PRESSURE AT WHICH THE RELIEF VALVE RELEASES PRESSURE CAN CHANGE IF THE DEFLECTOR CAP TURNS. IF THE PRESSURE RELIEF VALVE DOES NOT RELEASE AT THE CORRECT PRESSURE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Remove the pressure relief valve [2].

SUBTASK 38-42-06-020-002

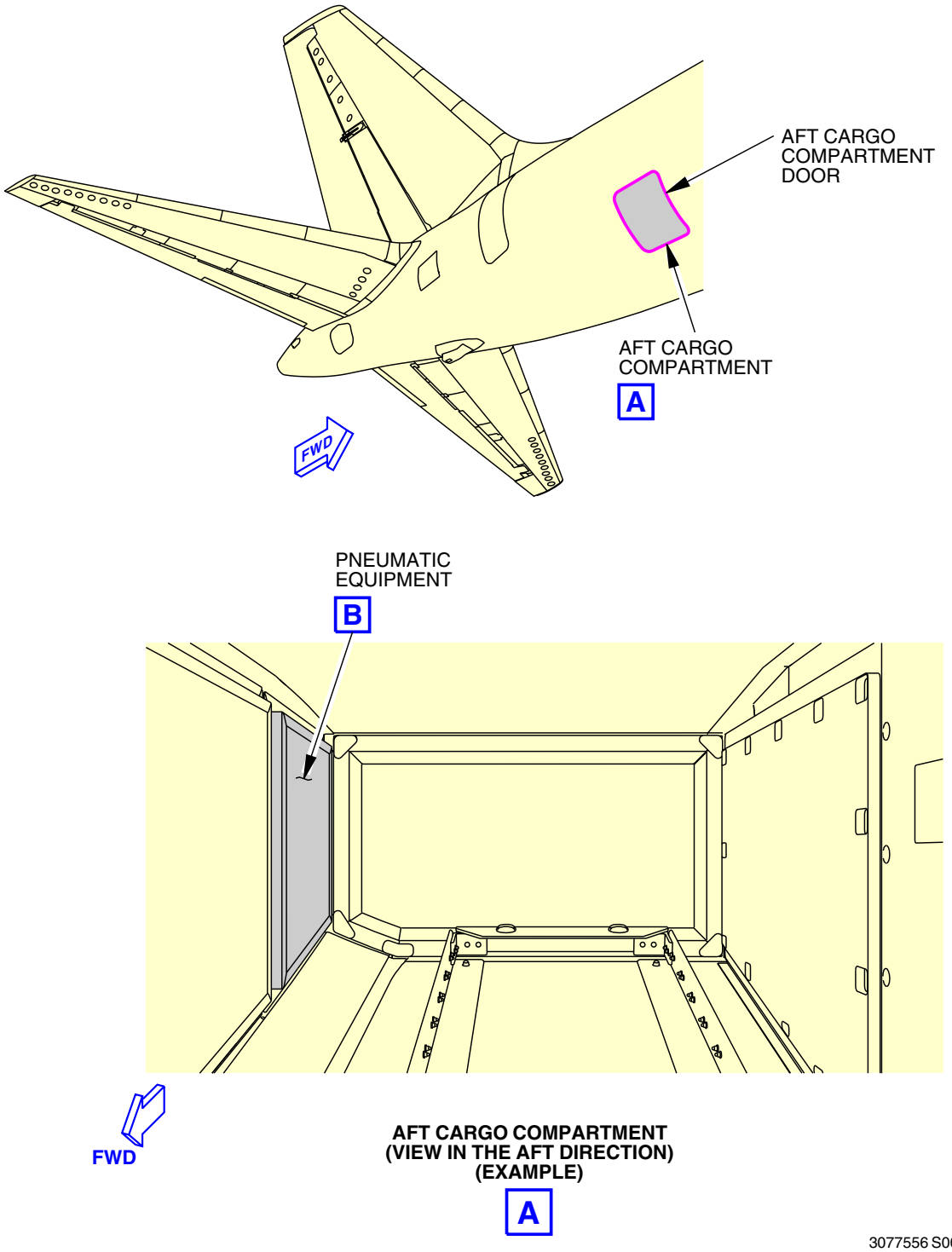
(2) Remove and discard the packing [1].

SUBTASK 38-42-06-910-001

(3) Install the protective covers on the pressure relief valve [2] and the opening on the potable water system.

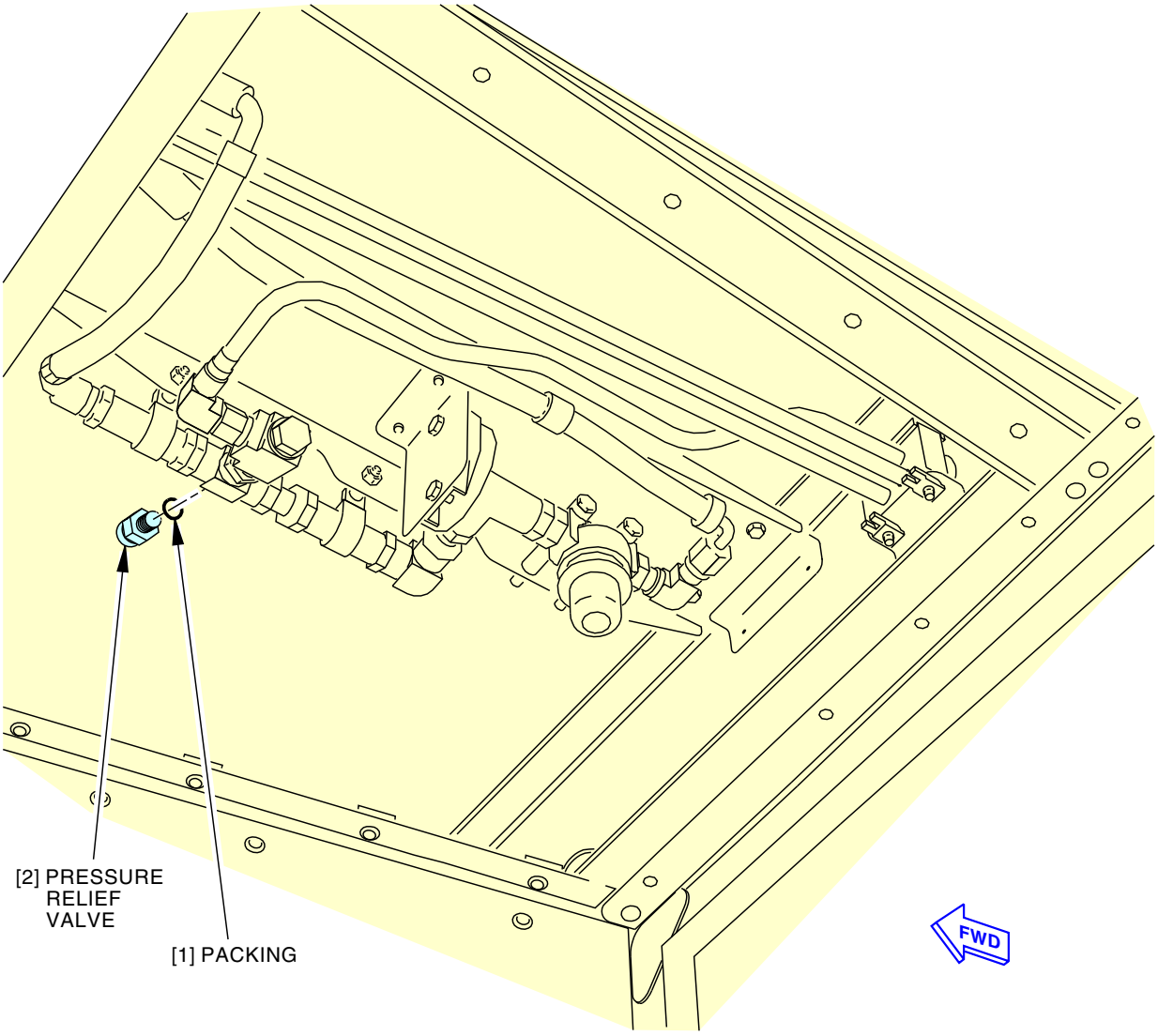
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pressure Relief Valve Installation
Figure 201/38-42-06-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Pressure Relief Valve Installation
Figure 201/38-42-06-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-42-06-400-801

3. Pressure Relief Valve Installation

(Figure 201)

A. General

(1) This task gives the instructions to install the pressure relief valve.

B. References

Reference	Title
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Packing	38-42-06-01-015	PTF ALL
2	Pressure relief valve	38-42-06-01-010	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

G. Pressure Relief Valve Installation

SUBTASK 38-42-06-420-001

(1) Remove the protective covers from the pressure relief valve [2] and the opening on the potable water system.

SUBTASK 38-42-06-640-001

(2) Apply the grease, D00463 or Molykote 111 Compound, D00189 to the new packing [1] and threads of the pressure relief valve [2].

SUBTASK 38-42-06-420-002

(3) Install the packing [1] on the pressure relief valve [2].

SUBTASK 38-42-06-420-003

WARNING

DO NOT TURN THE DEFLECTOR CAP WHEN YOU REMOVE OR INSTALL THE PRESSURE RELIEF VALVE. THE PRESSURE AT WHICH THE RELIEF VALVE RELEASES PRESSURE CAN CHANGE IF THE DEFLECTOR CAP TURNS. IF THE PRESSURE RELIEF VALVE DOES NOT RELEASE AT THE CORRECT PRESSURE, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Install the pressure relief valve [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-06-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-06-860-005

(6) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-42-06-710-001

(7) Make sure that there are no air leaks at the pressure relief valve.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-06-410-003

(1) Install the electrical equipment rack, E6, aft access panel.

SUBTASK 38-42-06-410-002

(2) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

END OF TASK

TASK 38-42-06-700-801

4. Pressure Relief Valve - Operational Test

(Figure 202)

A. General

(1) This task gives the instructions to test the pressure relief valve.

(2) The test of the pressure relief valve is an off airplane task.

B. Location Zones

Zone	Area
142	Aft Cargo Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Pressure Relief Valve - Operational Test

SUBTASK 38-42-06-020-003

(1) Do this task: Pressure Relief Valve Removal, TASK 38-42-06-000-801.

SUBTASK 38-42-06-780-001

(2) Do a test the relief valve as follows:

(a) With the initial pressure at zero psig, slowly increase the pressure to the relief valve.

(b) Keep a record of the initial pressure when you find the airflow through the pressure relief valve.

NOTE: The pressure must be less than 75 psig.

1) If the initial pressure is less than 53 psig or more than 75 psig, you must replace the pressure relief valve.

(c) If the initial pressure is satisfactory, then continue.

(d) With the pressure at relief pressure, slowly decrease the pressure to the relief valve.

(e) Keep a record of the reset pressure when you find that the airflow stops through the pressure relief valve.

NOTE: The pressure must be more than 53 psig.

1) If the reset pressure is less than 53 psig, you must replace the pressure relief valve.

(f) If the reset pressure is satisfactory, then continue.

(g) Decrease the pressure to zero psig.

(h) With the pressure at zero psig, slowly increase the pressure to the relief valve.

(i) Keep a record of the subsequent pressure when you find the airflow through the pressure relief valve.

NOTE: The pressure must be between 57 to 63 psig.

1) If the subsequent pressure is less than 57 psig or more than 63 psig, you must replace the pressure relief valve.

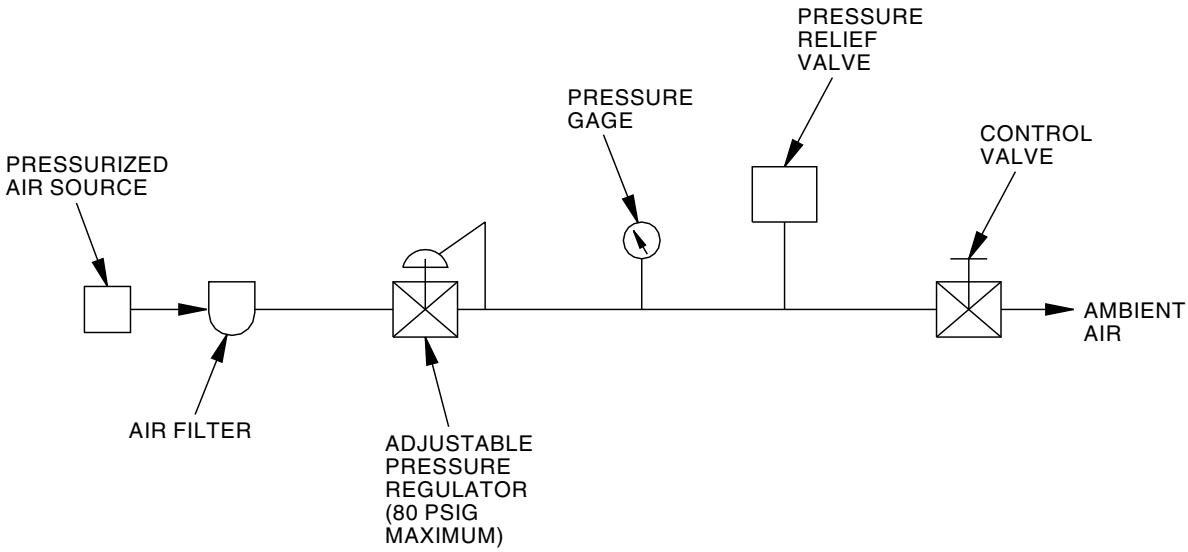
(j) If the subsequent pressure is satisfactory, then the test is completed.

SUBTASK 38-42-06-420-004

(3) Do this task: Pressure Relief Valve Installation, TASK 38-42-06-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Pressure Relief Valve Test
Figure 202/38-42-06-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WATER PRESSURE CUT-OUT SWITCH - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the water pressure cut-out switch

(2) An installation of the water pressure cut-out switch

(3) An operational test of the water pressure cut-out switch.

NOTE: This task is to make sure the air compressor starts and stops at the correct pressure.

TASK 38-42-07-000-801

2. Water Pressure Cut-Out Switch Removal

(Figure 201)

A. General

(1) This task gives the instructions to remove the water pressure cut-out switch.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
50-12-02-000-801	Aft Cargo Compartment Aft Bulkhead Liner Removal (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
146	Aft Cargo Compartment Equipment Bay - Right
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-42-07-010-007

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-07-860-001

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-07-860-002

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-07-010-001

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-07-010-002

(5) To remove the end wall of the aft cargo compartment, do this task: Aft Cargo Compartment Aft Bulkhead Liner Removal, TASK 50-12-02-000-801.

G. Water Pressure Cut-Out Switch Removal

SUBTASK 38-42-07-020-001

CAUTION MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector [2] from the water pressure cut-out switch [3] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [2] from the water pressure cut-out switch [3].

(b) Install the protective covers on the electrical receptacle of the water pressure cut-out switch [3] and electrical connector [2].

SUBTASK 38-42-07-020-002

(2) Remove the water pressure cut-out switch [3] from the water tank.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-07-020-003

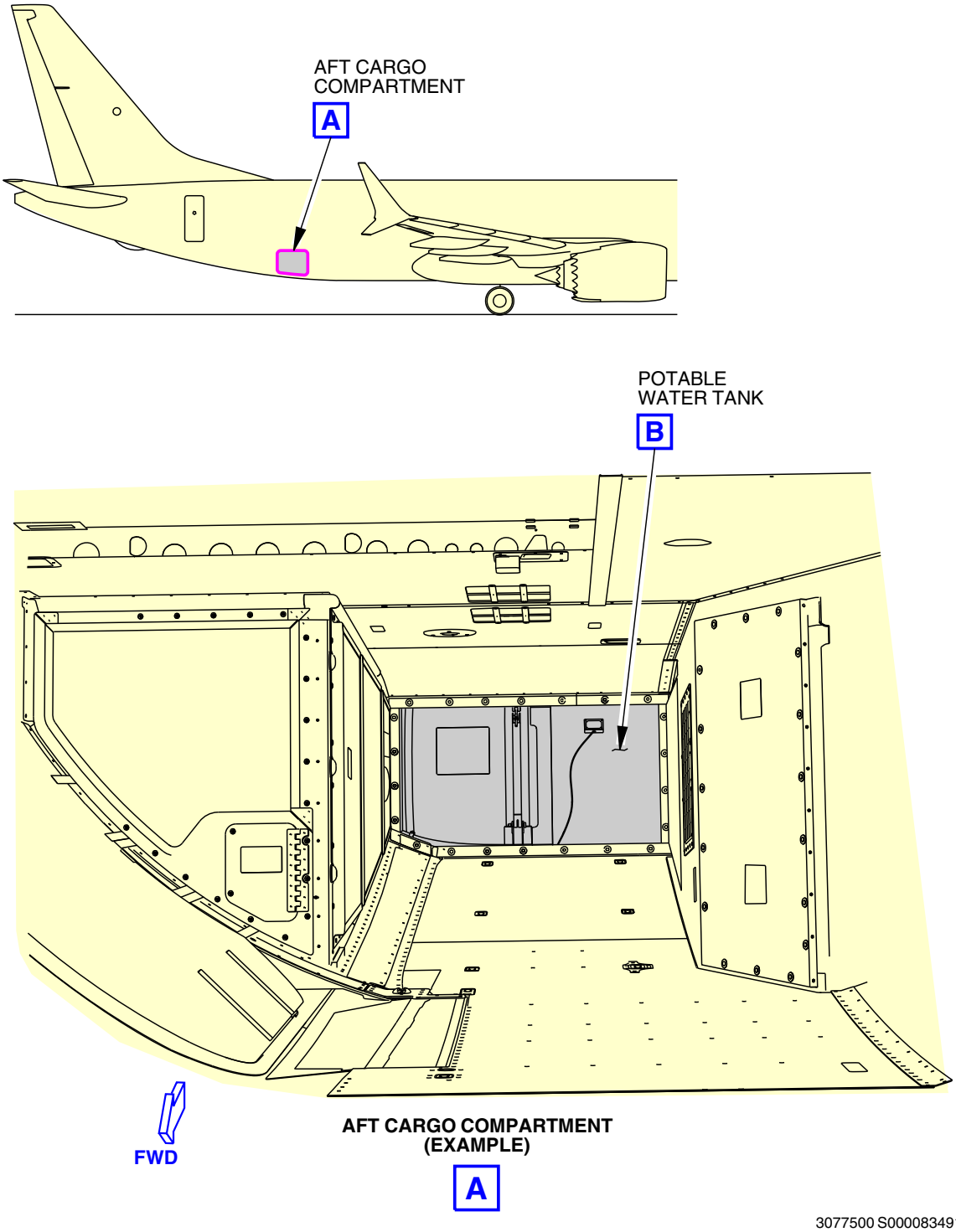
(3) Remove and discard the packing [1].

SUBTASK 38-42-07-020-004

(4) Install the protective covers on the water pressure cut-out switch [3] and opening on the water tank.

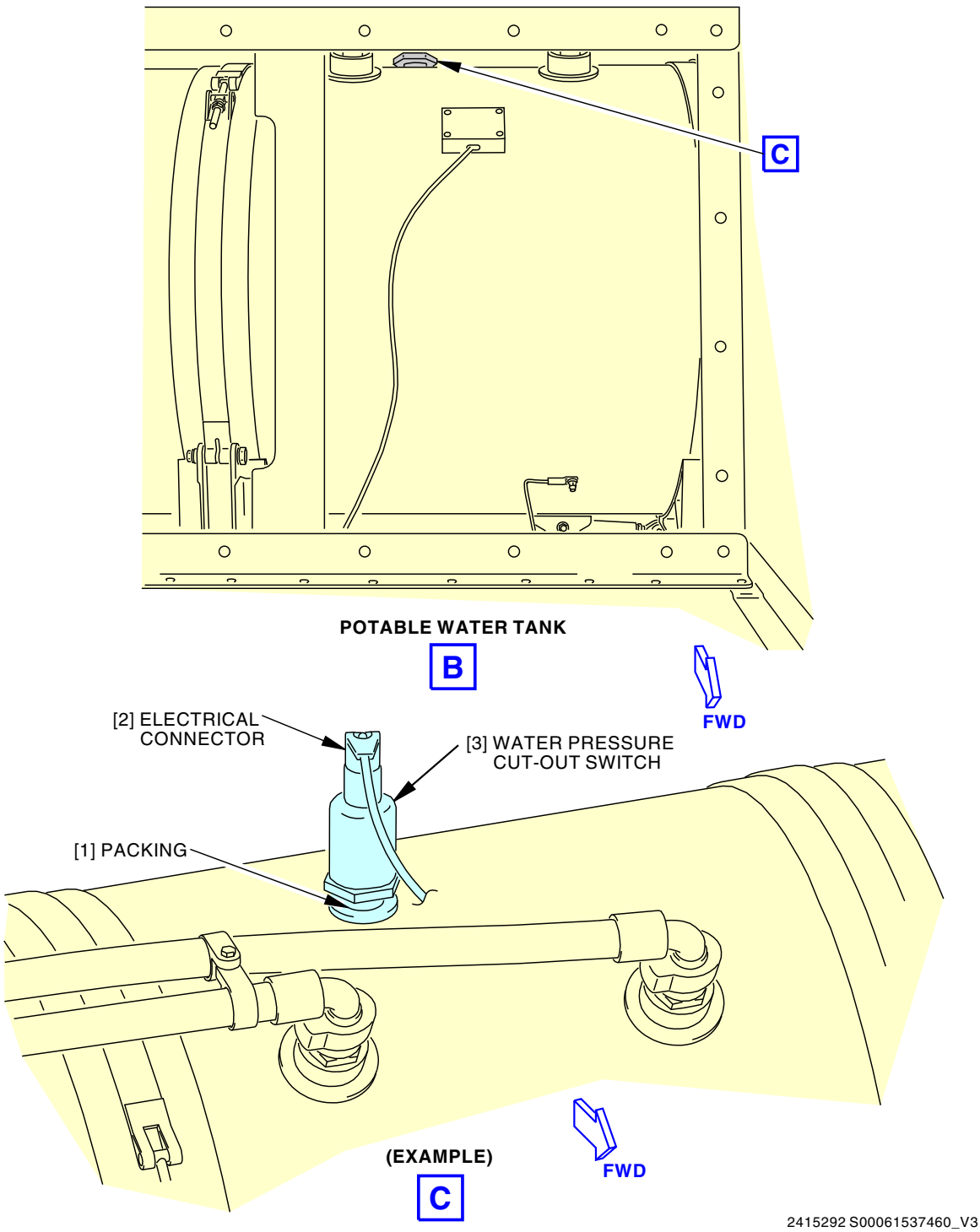
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Pressure Cut-Out Switch Installation
Figure 201/38-42-07-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Pressure Cut-Out Switch Installation
Figure 201/38-42-07-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-42-07-400-801

3. Water Pressure Cut-Out Switch Installation

(Figure 201)

A. General

(1) This task gives the instructions to install the water pressure cut-out switch.

B. References

Reference	Title
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-12-02-400-801	Aft Cargo Compartment Aft Bulkhead Liner Installation (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Packing	38-42-07-01-015	PTF ALL
3	Water pressure cut-out switch	38-42-07-01-010	PTF ALL

F. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
146	Aft Cargo Compartment Equipment Bay - Right
822	Aft Cargo Door

G. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

H. Water Pressure Cut-Out Switch Installation

SUBTASK 38-42-07-420-001

(1) Remove the protective covers from the water pressure cut-out switch [3] and opening on the water tank.

SUBTASK 38-42-07-640-001

(2) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packing [1] and threads of the water pressure cut-out switch [3].

SUBTASK 38-42-07-420-002

(3) Install the packing [1] on the water pressure cut-out switch [3].

SUBTASK 38-42-07-420-003

(4) Install the water pressure cut-out switch [3] on the water tank.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-07-420-004

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(5) Use the teflon-jawed pliers, STD-664, to connect the electrical connector [2] to the water pressure cut-out switch [3] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical receptacle of the water pressure cut-out switch [3] and electrical connector [2].

(b) Connect the electrical connector [2] to the water pressure cut-out switch [3].

1) Tighten the electrical connector [2] with the teflon-jawed pliers, STD-664.

SUBTASK 38-42-07-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-07-860-005

(7) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-42-07-710-001

(8) Make sure that the air compressor operates to supply the water tank with air pressure.

SUBTASK 38-42-07-710-002

(9) Make sure that there are no air leaks at the water pressure cut-out switch [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-07-410-001

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(1) Do this task: Aft Cargo Compartment Aft Bulkhead Liner Installation, TASK 50-12-02-400-801.

SUBTASK 38-42-07-410-002

(2) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

——— END OF TASK ———

TASK 38-42-07-700-801

4. Water Pressure Cut-Out Switch Operational Test

(Figure 202)

A. General

(1) This task gives the instructions to test the water pressure cut-out switch.

B. References

Reference	Title
12-14-00-600-802	Potable Water Tank - Fill (P/B 301)
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1201	Gauge - Pressure, 0-75 PSIG (0-518 KPa)

D. Location Zones

Zone	Area
146	Aft Cargo Compartment Equipment Bay - Right

E. Access Panels

Number	Name/Location
822	Aft Cargo Door

F. Water Pressure Cut-Out Switch Operational Test

SUBTASK 38-42-07-860-010

(1) On the P5 overhead panel, make sure that the CAB/UTIL switch is set to the ON position.

SUBTASK 38-42-07-860-007

(2) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-07-610-001

(3) Do this task: Potable Water Tank - Fill, TASK 12-14-00-600-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-07-010-003

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-07-010-005

(5) Remove the electrical equipment rack, E6, aft access panel.

SUBTASK 38-42-07-010-006

(6) Remove the cap [22] from the tee-fitting [21] in the pressurization line for the potable water system.

SUBTASK 38-42-07-480-001

(7) Connect the pressure gauge, STD-1201, to the tee-fitting [21] in the potable water line.

SUBTASK 38-42-07-780-001

(8) With the airplane compressor, do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

(a) Make sure that the air compressor stops when the pressure gets to 40 ±1 psi (276 ±7 kPa).

1) If the air compressor does not stop when the pressure gets to 40 ±1 psi (276 ±7 kPa), replace the water pressure cut-out switch.

(b) If the air compressor stops when the pressure gets to 40 ±1 psi (276 ±7 kPa), continue the test.

SUBTASK 38-42-07-780-002

(9) Turn the handle at the aft service panel to slowly bleed the air from the potable water system.

(a) Make sure that the air compressor starts when the pressure gets to 30 ±1 psi (207 ±7 kPa).

1) If the air compressor does not start when the pressure gets to 30 ±1 psi (207 ±7 kPa), replace the water pressure cut-out switch.

(b) If the air compressor starts when the pressure gets to 30 ±1 psi (207 ±7 kPa), the test is completed.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-07-860-008

(1) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-07-080-001

(2) Disconnect the pressure gauge, STD-1201, from the tee-fitting [21].

SUBTASK 38-42-07-420-005

(3) Install the cap [22] on the tee-fitting [21] in the pressurization line for the potable water system.

SUBTASK 38-42-07-860-009

(4) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-42-07-710-003

(5) Make sure that there are no air leaks at the cap.

SUBTASK 38-42-07-410-005

(6) Install the electrical equipment rack, E6, aft access panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

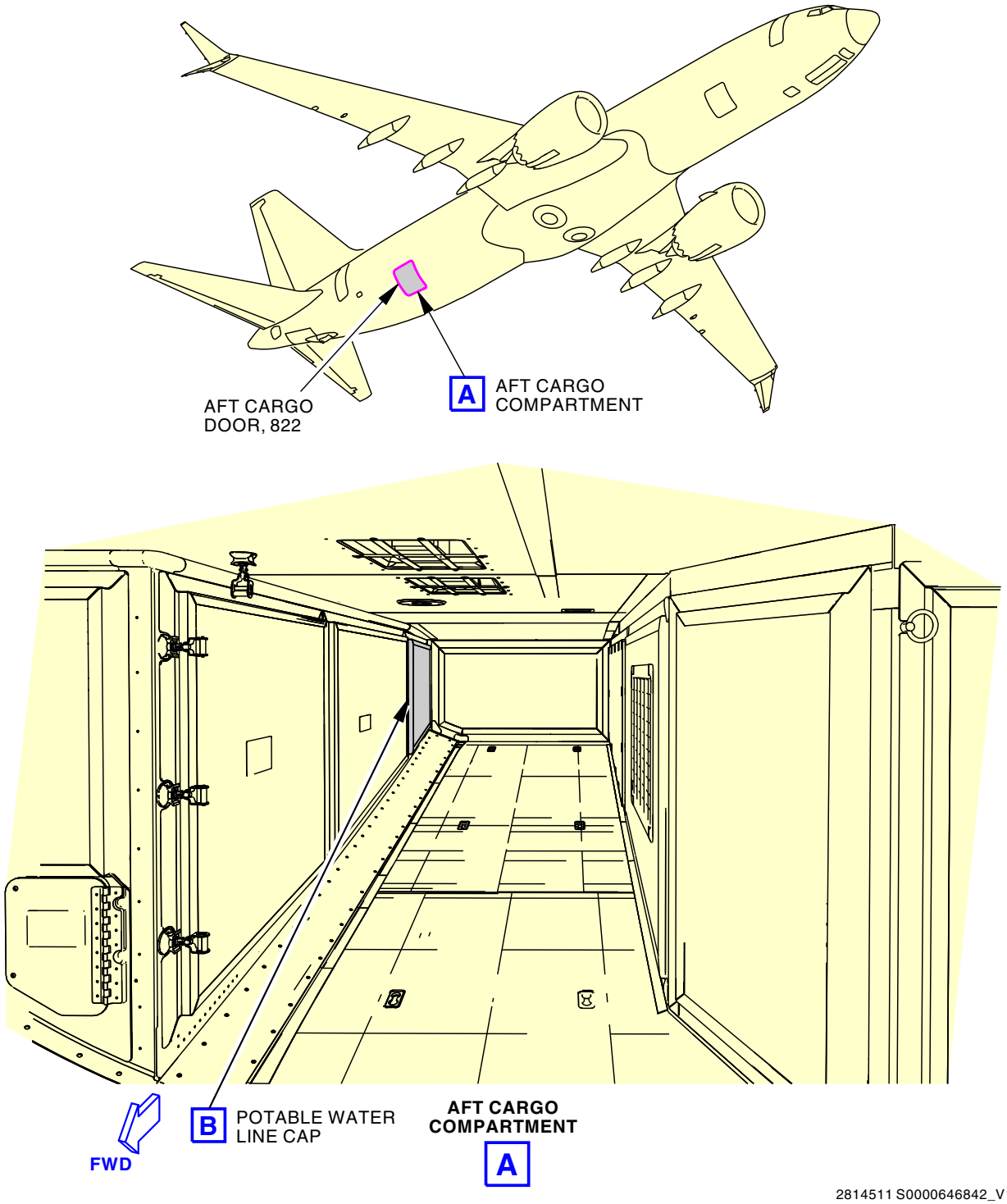
SUBTASK 38-42-07-410-004

(7) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

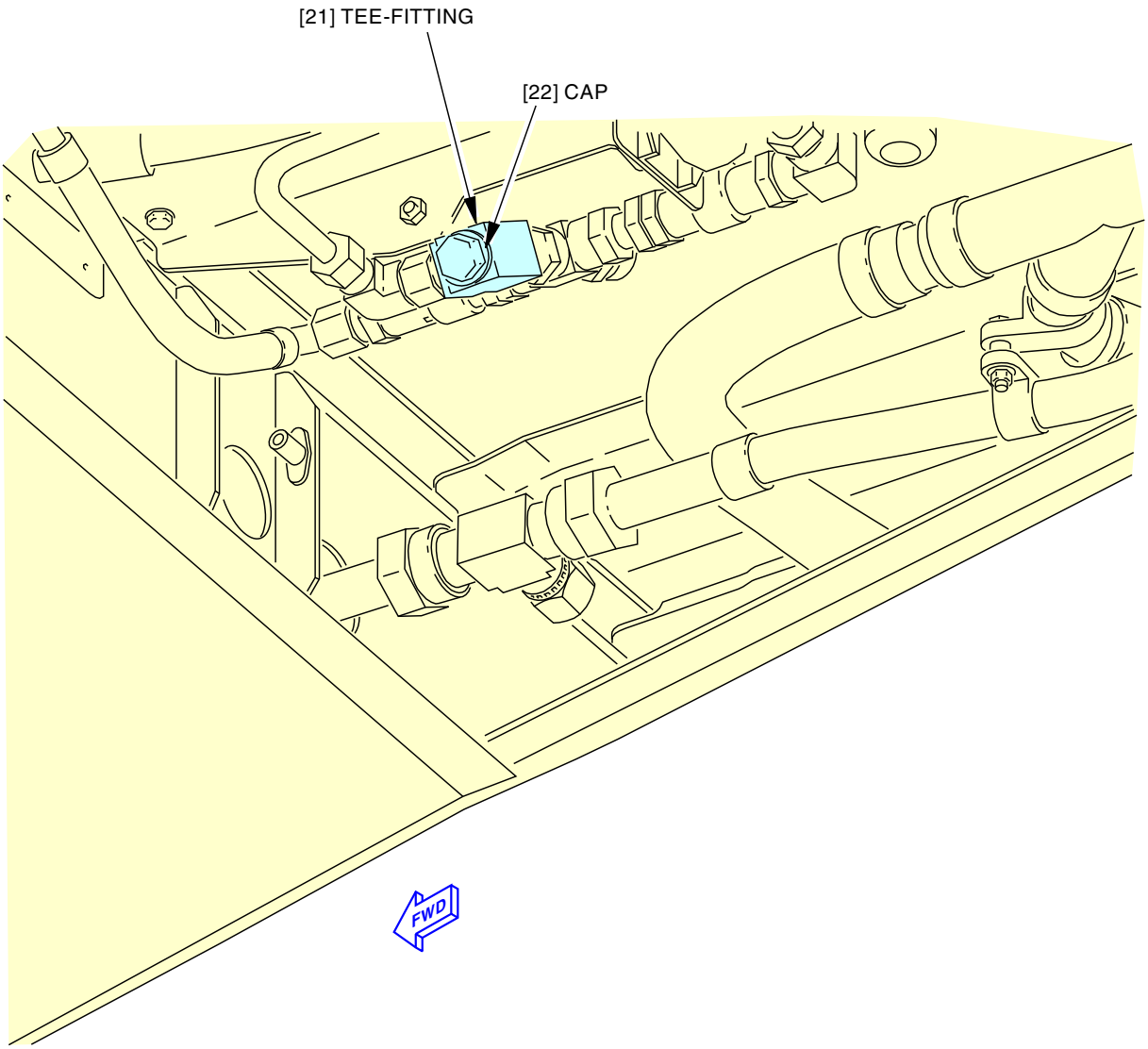
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Water Pressure Cut-Out Switch Operational Test
Figure 202/38-42-07-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POTABLE WATER LINE CAP

B

2415294 S00061537463_V2

Water Pressure Cut-Out Switch Operational Test
Figure 202/38-42-07-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRESSURE REGULATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Pressure regulator removal

(2) Pressure regulator installation.

TASK 38-42-08-000-801

2. Pressure Regulator Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the pressure regulator.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
38-42-10-000-801	Bleed Air Filter Assembly Removal (P/B 401)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Packing	38-42-08-02-035	PTF ALL

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

F. Prepare for the Removal

SUBTASK 38-42-08-010-003

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-08-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-08-860-002

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-08-010-001

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-08-010-004

(5) Remove the electrical equipment rack, E6, aft access panel.

G. Pressure Regulator Removal

SUBTASK 38-42-08-020-001

(1) Do this task: Bleed Air Filter Assembly Removal, TASK 38-42-10-000-801.

SUBTASK 38-42-08-020-002

(2) Loosen the nut [5] at the union [4].

(a) Remove the union [4].

(b) Discard the packings [3].

SUBTASK 38-42-08-020-003

(3) Remove the bolts [1] and washers [2] that attach the pressure regulator [6] to the mounting bracket.

SUBTASK 38-42-08-020-005

(4) Loosen the nut [10] at the union [8].

(a) Remove the union [8].

(b) Discard the packings [7].

SUBTASK 38-42-08-020-004

(5) Disconnect the elbow [9] from the pneumatic check valve.

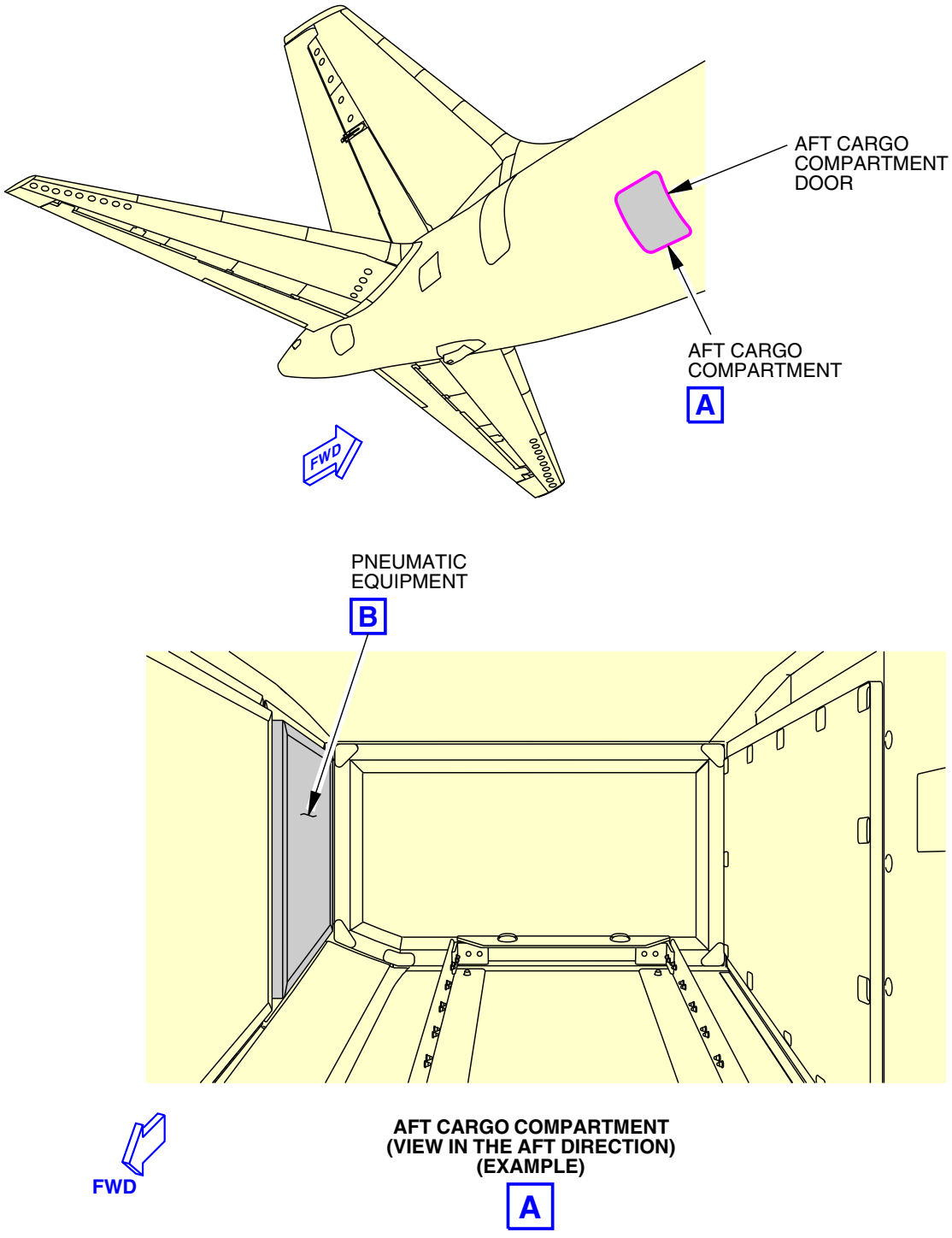
(a) Discard the packing [11].

SUBTASK 38-42-08-020-006

(6) Install the protective covers on the pressure regulator [6] and the air tubes.

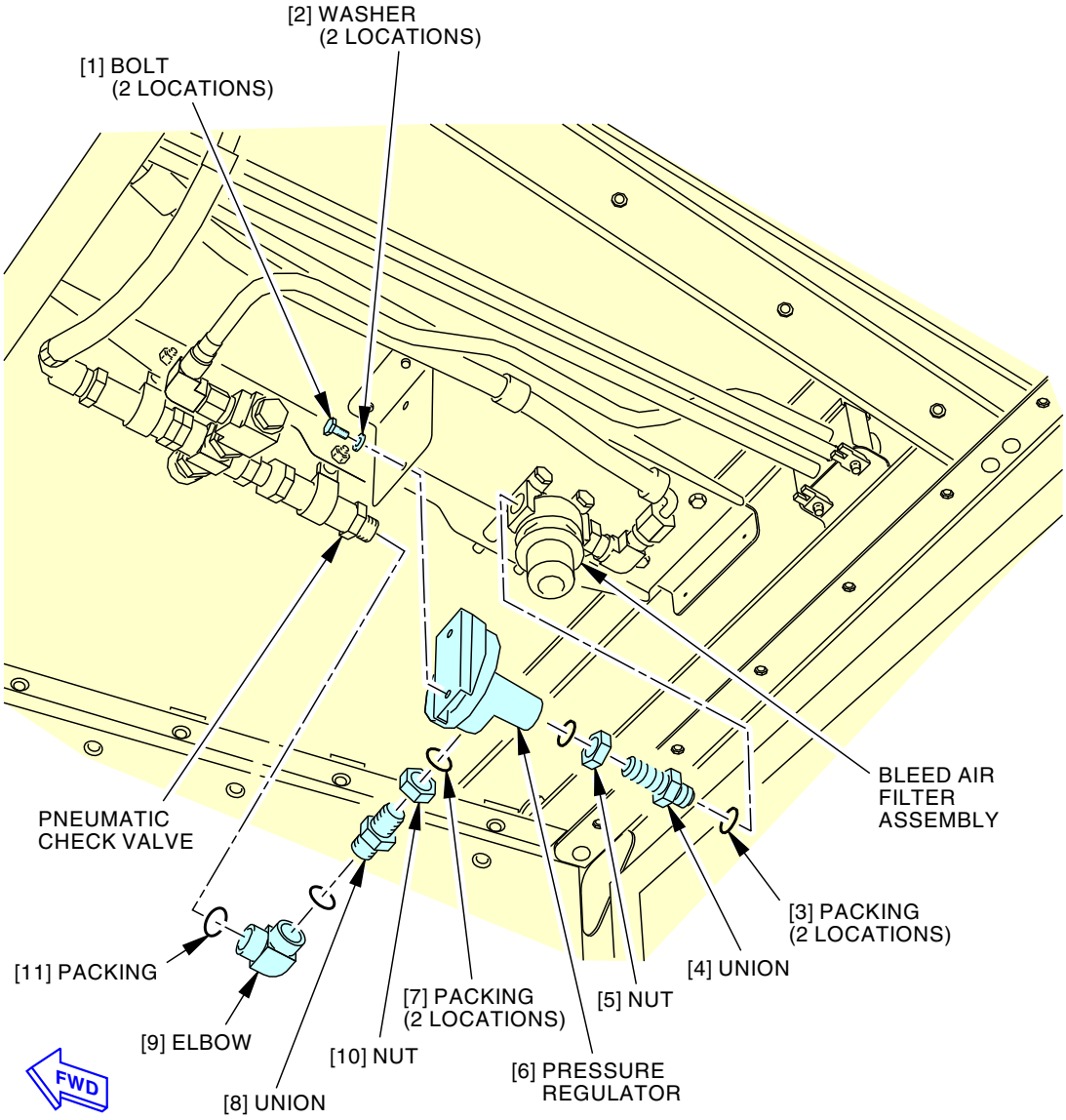
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pressure Regulator Installation
Figure 401/38-42-08-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PNEUMATIC EQUIPMENT

B

2415296 S00061537469_V3

Pressure Regulator Installation
Figure 401/38-42-08-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-42-08-400-801

3. Pressure Regulator Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the pressure regulator.

B. References

Reference	Title
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
38-42-00-800-803	Water Tank Pressurization System - Operational Test (P/B 501)
38-42-10-400-801	Bleed Air Filter Assembly Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Packing	38-42-08-02-035	PTF ALL
6	Pressure regulator	38-42-08-02-040	PTF ALL
7	Packing	38-42-08-02-035	PTF ALL
11	Packing	38-42-08-02-035	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right
822	Aft Cargo Door

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

G. Pressure Regulator Installation

SUBTASK 38-42-08-020-008

(1) Remove the protective covers from the pressure regulator [6] and air tubes.

SUBTASK 38-42-08-640-001

(2) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packings [3], new packings [7], new packing [11], and on the threads of the union [4] and union [8].

SUBTASK 38-42-08-420-001

(3) Install the packings [3] on the union [4].

(a) Make sure that the nut [5] is in its position on the union [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-08-420-002

(4) Install the packings [7] on the union [8].

(a) Make sure that the nut [10] is in its position on the union [8].

SUBTASK 38-42-08-420-003

(5) Install the pressure regulator [6] on the union [8] and union [4].

SUBTASK 38-42-08-420-004

(6) Connect the elbow [9] to the pneumatic check valve, as follows:

(a) Install the packing [11] on the pneumatic check valve.

(b) Connect the elbow [9] to the pneumatic check valve.

SUBTASK 38-42-08-420-009

(7) Connect the pressure regulator [6] and union [8] to the elbow [9].

SUBTASK 38-42-08-420-005

(8) Install the bolts [1] and washers [2] that attach the pressure regulator [6] to the mounting bracket.

SUBTASK 38-42-08-420-006

(9) Do this task: Bleed Air Filter Assembly Installation, TASK 38-42-10-400-801.

SUBTASK 38-42-08-420-007

(10) Tighten the nut [5] on the union [4].

SUBTASK 38-42-08-420-008

(11) Tighten the nut [10] on the union [8].

SUBTASK 38-42-08-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(12) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-08-860-005

(13) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-42-08-860-006

(14) To pressurize the pneumatic system, do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-08-710-001

(15) Make sure that the air compressor operates to pressurize the potable water tank.

SUBTASK 38-42-08-780-001

(16) Make sure that the pressure relief valve on the air line to the water tank line is off.

SUBTASK 38-42-08-780-002

(17) When the compressor is off, start the water at some lavatories or galleys and examine correct water pressure.

SUBTASK 38-42-08-710-002

(18) If the water pressure is not correct or the system does not operate correctly, do this task: Water Tank Pressurization System - Operational Test, TASK 38-42-00-800-803.

SUBTASK 38-42-08-780-003

(19) Make sure that there are no air leaks at the connections to the pressure regulator [6] or the piping connections.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-08-860-007

(1) To remove pressurization from the pneumatic system, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 38-42-08-410-003

(2) Install the electrical equipment rack, E6, aft access panel.

SUBTASK 38-42-08-410-002

(3) Close these access panels:

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHECK VALVES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the compressor supply check valve

(2) An installation of the compressor supply check valve

(3) A removal of the pneumatic supply check valve

(4) An installation of the pneumatic supply check valve.

B. The Air Compressor Supply Check Valve is between the air compressor and the common manifold assembly for the potable water tank pressurization air. The Air Compressor Supply Check Valve is referred to as the Compressor Check Valve in this procedure. The Compressor Check Valve connecting fittings must be removed before the Pneumatic Supply Check Valve can be removed.

C. The Pneumatic Supply Check Valve is between the pressure regulator and the common manifold assembly for the potable water tank pressurization air. The Pneumatic Supply Check Valve is referred to as the Pneumatic Check Valve in this procedure.

TASK 38-42-09-020-801

2. Compressor Supply Check Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the compressor supply check valve.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-42-09-010-005

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-09-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-09-860-002

WARNING

MAKE SURE THAT YOU REMOVE PRESSURE FROM THE PNEUMATIC MANIFOLD. IF YOU DO NOT REMOVE PRESSURE, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-09-010-001

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-09-010-007

(5) Remove the electrical equipment rack, E6, aft access panel.

F. Compressor Supply Check Valve Removal

SUBTASK 38-42-09-020-001

(1) Disconnect the hose [6] from the air compressor to the compressor check valve [10].

SUBTASK 38-42-09-020-002

(2) Loosen the nut [1], washers [2], spacer [3], screw [4], and clamp [5] on the compressor check valve [10].

SUBTASK 38-42-09-020-003

(3) Disconnect and remove the compressor check valve [10] from the cross fitting union [11].

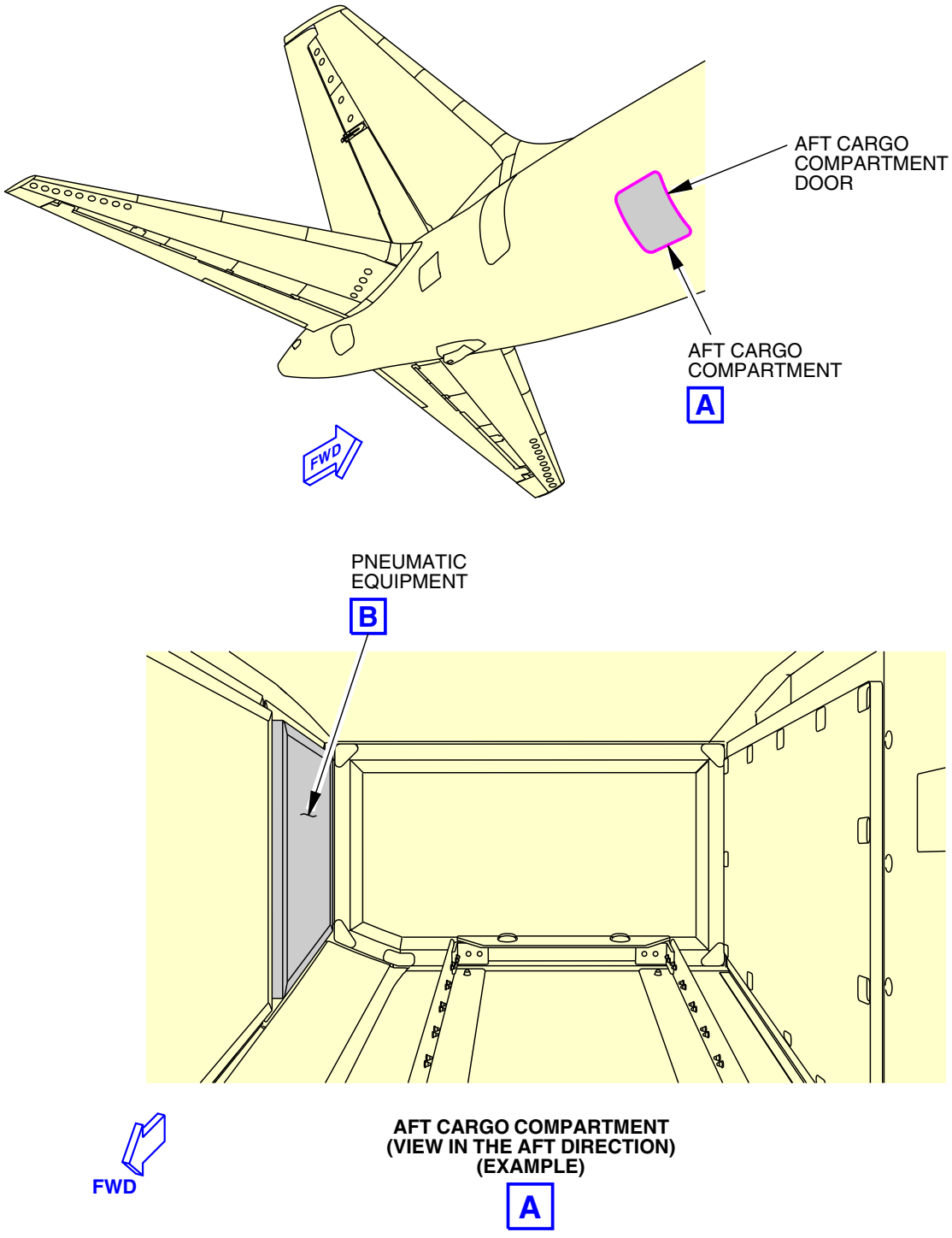
(a) Discard the packing [12].

SUBTASK 38-42-09-430-001

(4) Install the protective covers or caps on the compressor check valve [10] and all open hoses, tubes, and fittings.

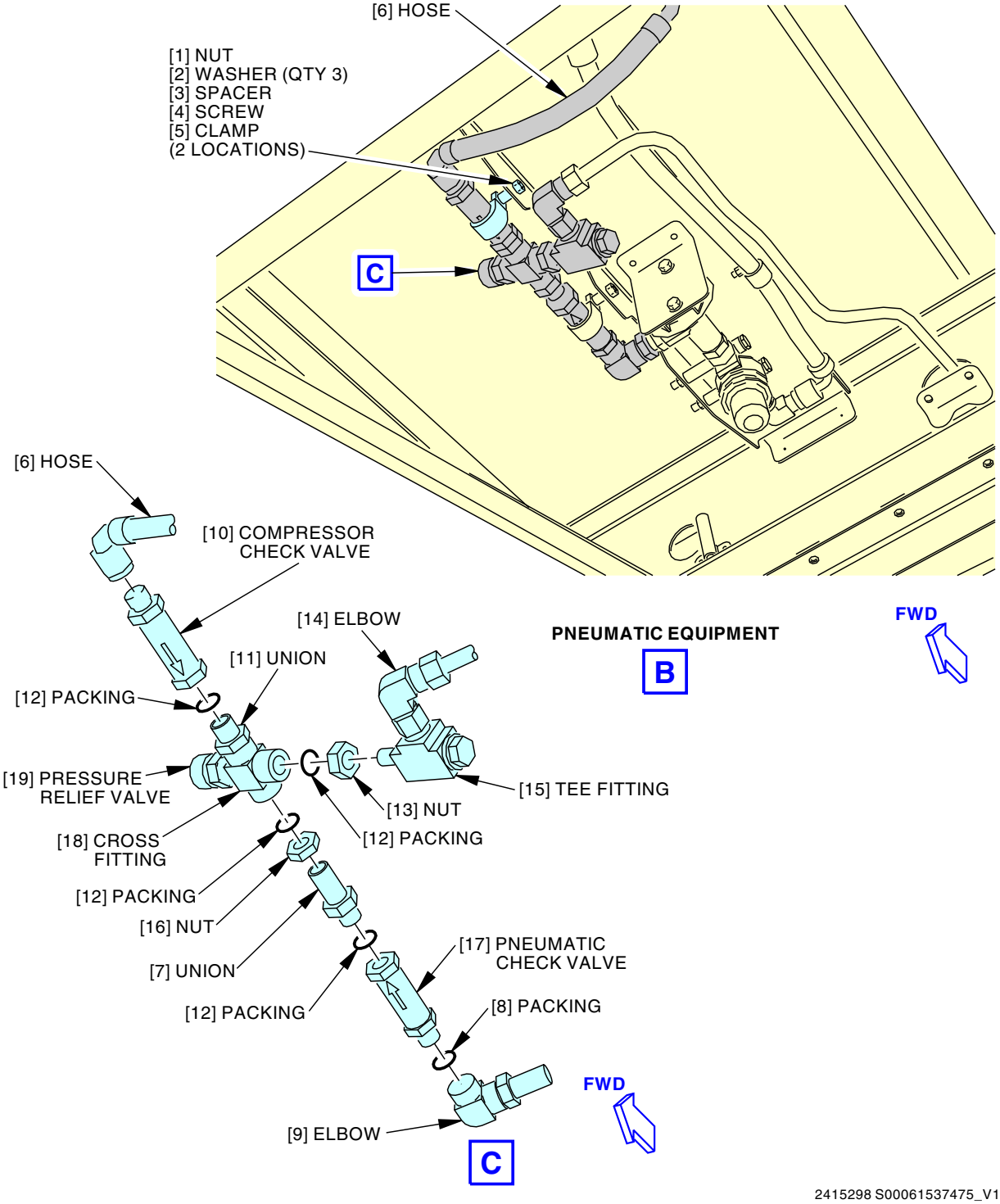
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Check Valve Installation
Figure 401/38-42-09-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Check Valve Installation
Figure 401/38-42-09-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-42-09-400-801

3. Compressor Check Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the compressor supply check valve.

B. References

Reference	Title
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00091	Compound - Oxygen System Leak Detection - Snoop Leak Detector	MIL-PRF-25567
G50321	Air - Clean, Dry	BAC5402 Table I

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	Compressor check valve	38-42-09-01-080	PTF ALL
12	Packing	38-42-09-01-075	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

G. Compressor Supply Check Valve Installation

SUBTASK 38-42-09-420-008

(1) Remove the protective covers or caps from the compressor check valve [10] and all hoses, tubes, and fittings.

SUBTASK 38-42-09-171-001

(2) Clear the hose [6] and tube assembly, between the air compressor and the check compressor check valve [10], of any contamination.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Use clean dry air, G50321, or nitrogen, G00018.

NOTE: To clean the hose and tube assembly, use 40 psig (3 kg/cm^2) to 100 psig (7 kg/cm^2) clean, dry shop air, or nitrogen, applied at the compressor check valve hose fitting , for 1 to 2 minutes.

SUBTASK 38-42-09-640-001

(3) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packing [12], threads of the compressor check valve [10], and threads of the union [11].

SUBTASK 38-42-09-420-001

(4) Install the packing [12] on the union [11].

SUBTASK 38-42-09-420-002

(5) Install the compressor check valve [10] on the union [11].

NOTE: The flow arrow on the check valve must point to the union and cross fitting, as shown in the figure.

SUBTASK 38-42-09-420-003

(6) Install the nut [1], washers [2], spacer [3], screw [4], and clamp [5] for the compressor check valve [10].

SUBTASK 38-42-09-420-004

(7) Install the air compressor hose [6] on the compressor check valve [10].

H. Compressor Supply Check Valve Installation Test

SUBTASK 38-42-09-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-09-410-005

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-09-860-005

(3) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-09-860-006

(4) To pressurize the Pneumatic System, do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 38-42-09-790-001

(5) Make sure that there are no air leaks at the check valves or air connections.

(a) Apply Snoop Leak Detector compound, G00091, to all the fittings and connections.

1) Look for bubbles to find any leaks.

2) If you find leaks, tighten the fittings and connections.

(b) Remove Snoop Leak Detector compound, G00091, with a clean cotton wiper, G00034, immediately after the check.

1) Make sure that the fittings and connections are dry.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-09-860-007

(1) To remove pressurization from the Pneumatic System, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 38-42-09-410-007

(2) Install the electrical equipment rack, E6, aft access panel.

SUBTASK 38-42-09-410-002

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

TASK 38-42-09-020-802

4. Pneumatic Supply Check Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the pneumatic supply check valve.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
50-10-02-000-801	Cargo Compartment Sidewall Liners Removal (P/B 401)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 38-42-09-010-006

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-09-862-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-09-864-001

WARNING

MAKE SURE THAT YOU REMOVE PRESSURE FROM THE PNEUMATIC MANIFOLD. IF YOU DO NOT REMOVE PRESSURE, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-09-010-003

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-09-010-008

(5) Remove the side panel adjacent to the potable water tank.

(a) Do this task: Cargo Compartment Sidewall Liners Removal, TASK 50-10-02-000-801.

F. Pneumatic Supply Check Valve Removal (Method I)

SUBTASK 38-42-09-030-001

(1) Disconnect the hose [6] from the air compressor to the compressor check valve [10].

SUBTASK 38-42-09-030-002

(2) Remove the nut [1], washers [2], spacer [3], screw [4], and clamp [5] on the compressor check valve [10].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-09-030-003

(3) Disconnect the elbow [14] from the tee fitting [15].

SUBTASK 38-42-09-030-004

(4) Loosen the nut [13] and remove the tee fitting [15], if necessary, for clearance to remove the cross fitting [18].

(a) Discard the tee fitting packing [12], if you removed the tee fitting [15].

SUBTASK 38-42-09-030-005

(5) Loosen the nut [16] and remove the compressor check valve [10], cross fitting [18], union [11], and pressure relief valve [19] as an assembly.

SUBTASK 38-42-09-030-006

(6) Loosen the nut [1], washers [2], spacer [3], screw [4], and clamp [5] on the pneumatic check valve [17].

SUBTASK 38-42-09-020-004

(7) Remove the pneumatic check valve [17] and the union [7] from the elbow [9].

(a) Discard the packing [8].

SUBTASK 38-42-09-030-007

(8) Hold the pneumatic check valve [17] with a wrench, and remove the union [7] from the pneumatic check valve [17].

(a) Discard the packing [12].

(b) Keep the union [7] and nut [16] for use on reinstallation of the pneumatic check valve [17].

SUBTASK 38-42-09-430-002

(9) Install the protective covers or caps on the pneumatic check valve [17] and all open hoses, tubes, and fittings.

G. Pneumatic Supply Check Valve Removal (Method II).

NOTE: You may find it faster to remove the check valves, pressure relief valve and fittings as one assembly. Then remove the pneumatic supply check valve from the assembly.

SUBTASK 38-42-09-030-008

(1) Disconnect the hose [6] from the compressor check valve [10].

SUBTASK 38-42-09-030-009

(2) Disconnect the elbow [14] from the tee fitting [15].

SUBTASK 38-42-09-030-010

(3) Disconnect the elbow [9] from the pneumatic check valve [17].

(a) Discard the packing [8].

SUBTASK 38-42-09-030-011

(4) Remove the nut [1], washers [2], spacer [3], screw [4], and clamp [5] on the pneumatic check valve [17] and the compressor check valve [10].

SUBTASK 38-42-09-020-005

(5) Remove the check valves, pressure relief valve, fittings, and unions as one assembly.

SUBTASK 38-42-09-020-006

(6) Hold the union [7] with a wrench, and remove the pneumatic check valve [17].

(a) Discard the packing [12].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-09-430-003

(7) Install the protective covers or caps on the pneumatic check valve [17] and all open hoses, tubes, and fittings.

——— END OF TASK ———

TASK 38-42-09-400-802

5. Pneumatic Supply Check Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the pneumatic supply check valve.

B. References

Reference	Title
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-10-02-400-801	Cargo Compartment Sidewall Liners Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G00091	Compound - Oxygen System Leak Detection - Snoop Leak Detector	MIL-PRF-25567
G50321	Air - Clean, Dry	BAC5402 Table I

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Packing	38-42-09-01-075	PTF ALL
12	Packing	38-42-09-01-075	PTF ALL
17	Pneumatic check valve	38-42-09-01-080	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Pneumatic Supply Check Valve Installation (Method I)

NOTE: Use Method I to install the pneumatic check valve, if you used Method I to remove the pneumatic check valve.

SUBTASK 38-42-09-030-012

(1) Remove the protective covers or caps from the pneumatic check valve [17] and all hoses, tubes, and fittings.

SUBTASK 38-42-09-170-001

(2) Clear the hose [6] and tube assembly, between the air compressor and the compressor check valve [10], of any contamination.

(a) Use clean dry air, G50321, or nitrogen, G00018.

NOTE: To clean the hose and tube assembly, use 40 psig (3 kg/cm^2) to 100 psig (7 kg/cm^2) clean, dry shop air, or nitrogen, applied at the check valve hose fitting for 1 to 2 minutes.

SUBTASK 38-42-09-640-002

(3) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packings [12], packing [8], threads of the pneumatic check valve [17], compressor check valve [10], unions, and fittings.

SUBTASK 38-42-09-430-004

(4) Install the packing [8] on the pneumatic check valve [17].

SUBTASK 38-42-09-420-005

(5) Install the pneumatic check valve [17] on the elbow [9].

NOTE: The flow arrow on the check valve must point to the union and cross fitting, as shown in the figure.

SUBTASK 38-42-09-430-005

(6) Install the nut [1], washers [2], spacer [3], screw [4], and clamp [5] for the pneumatic check valve [17].

SUBTASK 38-42-09-430-006

(7) Position the nut [16] on the union [7], and install packings [12] on the union [7].

SUBTASK 38-42-09-430-007

(8) Install the union [7] on the pneumatic check valve [17].

SUBTASK 38-42-09-420-006

(9) Install the assembled cross fitting [18], union [11], pressure relief valve [19], and compressor check valve [10] on the union [7].

NOTE: The flow arrow on the check valve must point to the union and cross fitting, as shown in the figure.

(a) Tighten the nut [16].

SUBTASK 38-42-09-430-008

(10) If you removed the tee fitting [15], do these steps:

(a) Install the packing [12] on the tee fitting [15].

(b) Put the nut [13] into its position.

(c) Install the tee fitting [15] on the cross fitting [18].

(d) Tighten the nut [13].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-09-430-009

(11) Install the elbow [14] on the tee fitting [15].

SUBTASK 38-42-09-430-010

(12) Install the nut [1], washers [2], spacer [3], screw [4], and clamp [5] for the compressor check valve [10].

SUBTASK 38-42-09-430-011

(13) Connect the hose [6] to the compressor check valve [10].

H. Pneumatic Supply Check Valve Installation (Method II)

NOTE: Use the Method II, if you are installing the check valves, pressure relief valve, fittings, and unions, as one assembly. Use Method II installation, if you used Method II removal.

SUBTASK 38-42-09-030-013

(1) Remove the protective covers or caps from the pneumatic check valve [17] and all hoses, tubes, and fittings.

SUBTASK 38-42-09-170-002

(2) Clear the hose [6] and tube assembly, between the air compressor and the compressor check valve [10], of any contamination.

(a) Use clean dry air, G50321, or nitrogen, G00018.

NOTE: To clean the hose and tube assembly, use 40 psig (3 kg/cm^2) to 100 psig (7 kg/cm^2) clean, dry shop air, or nitrogen, applied at the check valve hose fitting for 1 to 2 minutes.

SUBTASK 38-42-09-640-003

(3) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packings [12], threads of the pneumatic check valve [17], compressor check valve [10], unions, and fittings.

SUBTASK 38-42-09-430-012

(4) Install the packing [12] on the union [7] and install the packing [8] on the pneumatic check valve [17].

SUBTASK 38-42-09-430-013

(5) Hold the union [7] with a wrench, and install the pneumatic check valve [17] on the union [7].

NOTE: The flow arrow on the check valve must point to the union and cross fitting, as shown in the figure.

SUBTASK 38-42-09-420-007

(6) Put the check valves, pressure relief valve, fittings, and unions, as one assembly, in position.

SUBTASK 38-42-09-430-014

(7) Install the nut [1], washers [2], spacer [3], screw [4], and clamp [5] for the compressor check valve [10] and the pneumatic check valve [17].

SUBTASK 38-42-09-430-015

(8) Install the elbow [9] on the pneumatic check valve [17].

SUBTASK 38-42-09-430-016

(9) Install the hose [6] on the compressor check valve [10].

SUBTASK 38-42-09-430-017

(10) Install the elbow [14] on the tee fitting [15].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Pneumatic Check Valve Installation Test

SUBTASK 38-42-09-865-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-09-410-006

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-09-863-001

(3) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-42-09-863-002

(4) To pressurize the Pneumatic System, do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 38-42-09-790-002

(5) Make sure that there are no air leaks at the check valves or air connections.

(a) Apply Snoop Leak Detector compound, G00091, to all the fittings and connections.

1) Look for bubbles to find any leaks.

2) If you find leaks, tighten the fittings and connections.

(b) Remove Snoop Leak Detector compound, G00091, with a clean cotton wiper, G00034, immediately after the check.

1) Make sure that the fittings and connections are dry.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-09-860-008

(1) To remove pressurization from the Pneumatic System, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-09-410-008

WARNING

SEAL THE CARGO COMPARTMENT WITH THE LINING. OBEY THE INSTRUCTIONS IN THE SPECIFIED PROCEDURE WHEN YOU INSTALL THE LINING. IF YOU INSTALL THE LINING INCORRECTLY, THE SMOKE CAN GO INTO THE PASSENGER COMPARTMENT DURING A FIRE. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Do this task: Cargo Compartment Sidewall Liners Installation, TASK 50-10-02-400-801.

SUBTASK 38-42-09-410-004

(3) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BLEED AIR FILTER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of bleed air filter assembly

(2) An installation of bleed air filter assembly

(3) A replacement of bleed air filter assembly.

TASK 38-42-10-000-801

2. Bleed Air Filter Assembly Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the bleed air filter assembly.

B. References

Reference	Title
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

E. Prepare for the Removal

SUBTASK 38-42-10-010-006

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-10-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-10-860-002

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-10-010-001

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-10-010-007

(5) Remove the potable water line cap as follows:

(a) Release the quarter-turn fasteners that are around the edges of the potable water line cap.

(b) Remove the potable water line cap.

F. Bleed Air Filter Assembly Removal

SUBTASK 38-42-10-020-001

(1) Loosen the B-nut at the elbow [5].

SUBTASK 38-42-10-020-002

(2) Remove the elbow [5], nut [6], and packing [7] from the bleed air filter assembly [2].

(a) Discard the packing [7].

SUBTASK 38-42-10-020-003

(3) Remove the bolts [3], washers [4], spacers [8], washers [9], washers [1], and nuts [13].

SUBTASK 38-42-10-020-005

(4) Remove the bleed air filter assembly [2] from the pressure regulator as follows:

(a) Remove the bleed air filter assembly [2] with the union [11] and nut [12] from the pressure regulator.

(b) Disconnect the bleed air filter assembly [2] from the union [11].

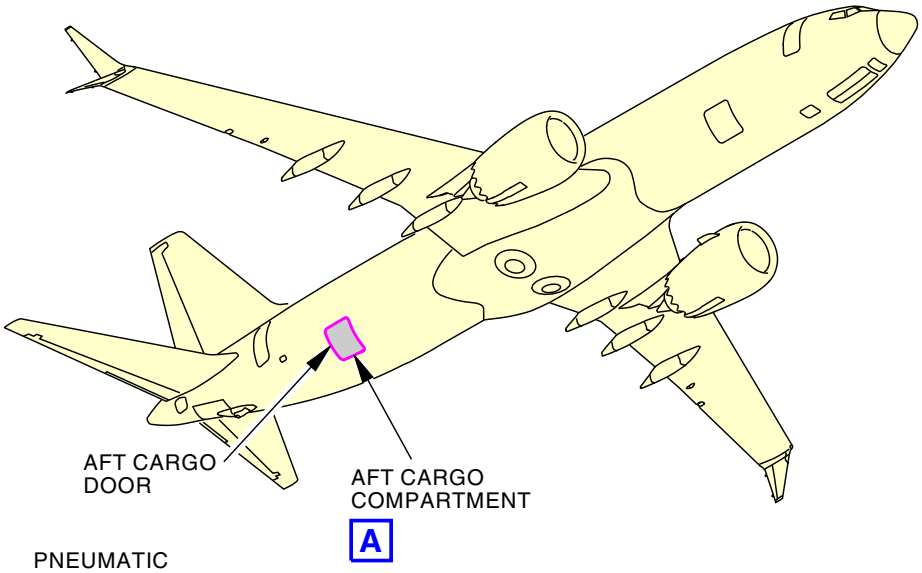
(c) Remove and discard the packings [10].

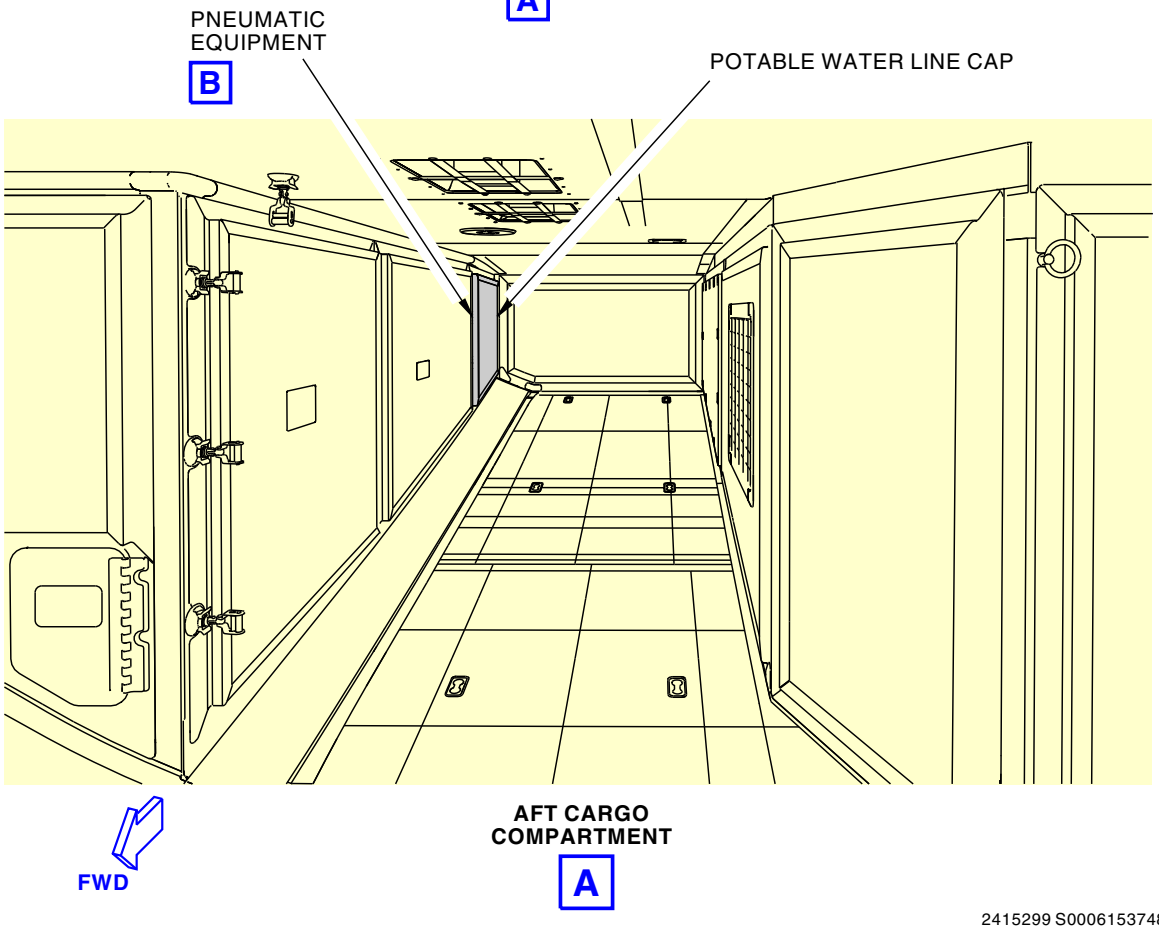
SUBTASK 38-42-10-020-006

(5) Install the protective covers or caps on the bleed air filter assembly [2] and open air tubes.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

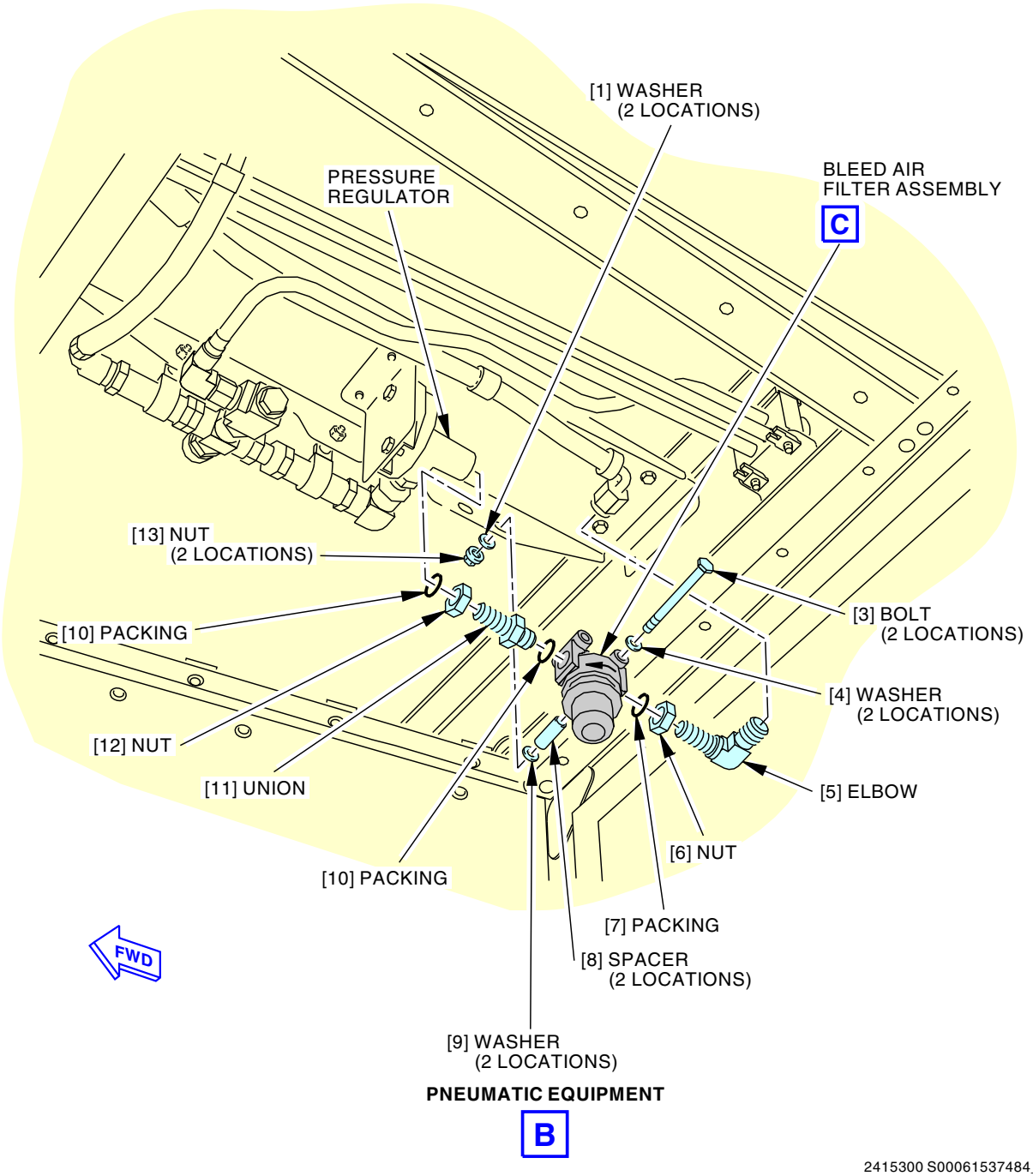




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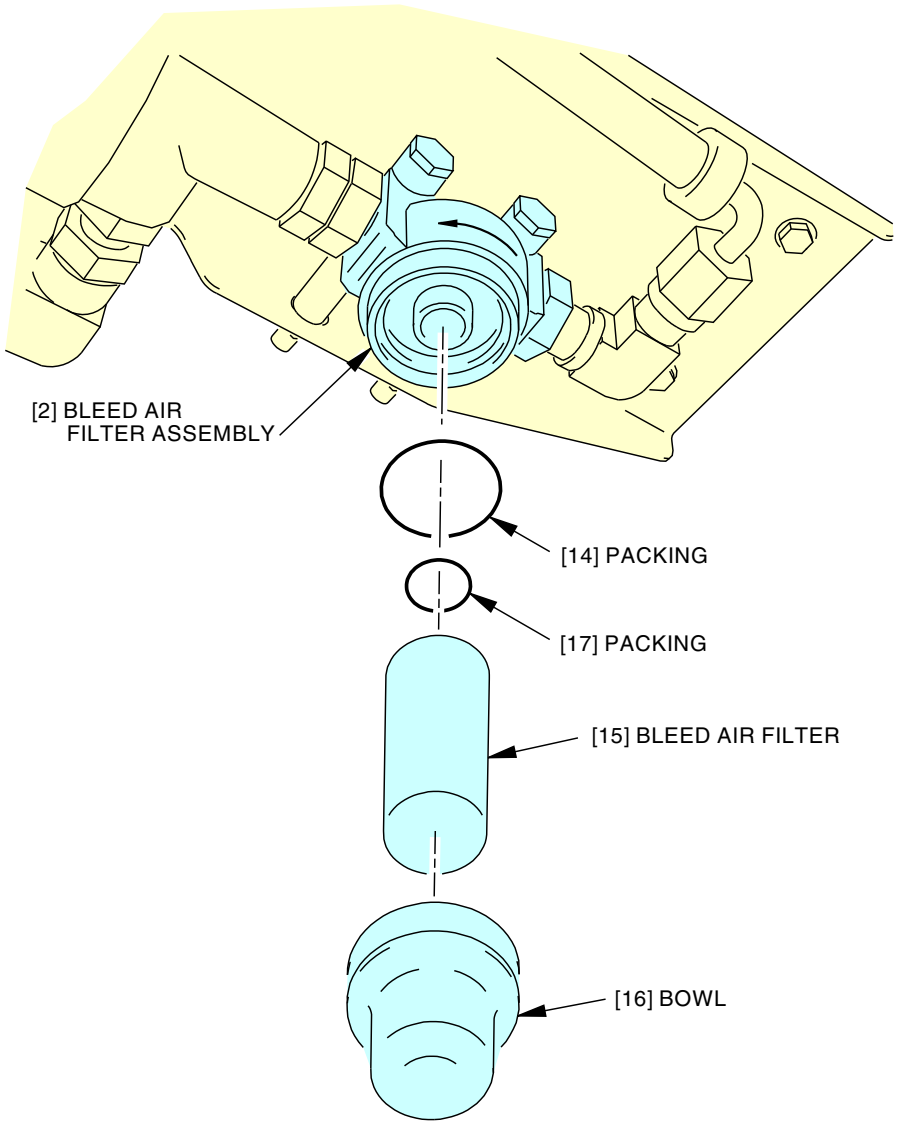
Bleed Air Filter Installation
Figure 401/38-42-10-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Bleed Air Filter Installation
Figure 401/38-42-10-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BLEED AIR FILTER ASSEMBLY

C

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Bleed Air Filter Installation
Figure 401/38-42-10-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 38-42-10-400-801

3. Bleed Air Filter Assembly Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the bleed air filter assembly.

B. References

Reference	Title
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00189	Compound - Molykote 111 (Formerly Dow Corning 111, Valve Sealant and Lubricant)	
D00463	Grease - Food Processing Equipment	DOD-G-24650
G50360	Seal - Closed Cell Silicone Foam Rubber, Hydraulic Fluid Resistant (with Adhesive)	BMS1-68 Grade A Form III

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Filter assembly	38-42-10-01-065	PTF ALL
7	Packing	38-42-10-01-055	PTF ALL
10	Packing	38-42-10-01-060	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

G. Bleed Air Filter Assembly Installation

SUBTASK 38-42-10-420-001

(1) Remove the protective covers or caps from the bleed air filter assembly [2] and air tubes.

SUBTASK 38-42-10-640-001

(2) Apply grease, D00463, or Molykote 111 Compound, D00189, to the new packing [7], new packings [10], and threads of the bleed air filter assembly [2].

SUBTASK 38-42-10-420-002

(3) Install the union [11], nut [12], and packing [10] on the bleed air filter assembly [2].

SUBTASK 38-42-10-420-003

(4) Install the bleed air filter assembly [2] and packing [10] on the pressure regulator.

SUBTASK 38-42-10-420-004

(5) Install the bolts [3], washers [4], spacers [8], washers [9], washers [1], and nuts [13].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 38-42-10-420-005

(6) Install the elbow [5], nut [6], and packing [7] on the bleed air filter assembly [2].

SUBTASK 38-42-10-420-006

(7) Connect the B-nut to the elbow [5].

H. Bleed Air Filter Assembly Installation Test

SUBTASK 38-42-10-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-10-410-006

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-10-860-005

(3) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-42-10-860-006

(4) To pressurize the pneumatic system, do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 38-42-10-790-001

(5) Make sure that there are no air leaks at the air filter.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-10-860-007

(1) To remove pressurization from the pneumatic system, do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 38-42-10-210-001

(2) Examine seal, G50360, on the potable water line cap frame.

(a) Replace seal, G50360, if it missed or damaged.

SUBTASK 38-42-10-410-009

(3) Install the potable water line cap as follows:

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(a) Install the potable water line cap.

(b) Lock the quarter-turn fasteners around the edges of the potable water line cap.

SUBTASK 38-42-10-410-002

(4) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

——— END OF TASK ———

TASK 38-42-10-960-801

4. Bleed Air Filter Element Replacement

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to replace the bleed air filter element.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
24-22-00-860-801	Supply Electrical Power (P/B 201)
36-00-00-860-801	Supply Pressure to the Pneumatic System (Selection) (P/B 201)
36-00-00-860-806	Remove Pressure from the Pneumatic System (P/B 201)
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D50232	Compound - Release, Molykote 7 (Formerly Dow Corning 7)	
G50360	Seal - Closed Cell Silicone Foam Rubber, Hydraulic Fluid Resistant (with Adhesive)	BMS1-68 Grade A Form III
G51934	Wire - Safety, Steel, Corrosion Resistant - 0.032 Inch (0.8128 mm) Diameter	SAE AS5685

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	Packing	38-42-10-01-080	PTF ALL
15	Bleed air filter	38-42-10-01-075	PTF ALL
17	Packing	38-42-10-01-085	PTF ALL

E. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
142	Aft Cargo Compartment - Right

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
822	Aft Cargo Door

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G. Prepare for the Removal

SUBTASK 38-42-10-010-005

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-10-860-008

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-10-860-009

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 38-42-10-010-003

(4) Open this access panel:

Number	Name/Location
822	Aft Cargo Door

SUBTASK 38-42-10-010-008

(5) Remove the potable water line cap as follows:

(a) Release the quarter-turn fasteners that are around the edges of the potable water line cap.

(b) Remove the potable water line cap.

H. Bleed Air Filter Element Replacement

SUBTASK 38-42-10-020-007

(1) Remove the lockwire and then loosen the filter bowl [16].

SUBTASK 38-42-10-020-009

(2) Remove and discard the packing [14].

SUBTASK 38-42-10-020-008

(3) Remove and discard the bleed air filter [15] and packing [17].

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SUBTASK 38-42-10-420-007

(4) Install the new bleed air filter [15] and packing [17].

NOTE: There is a packing in the bleed air filter that must be in its position before installation.

SUBTASK 38-42-10-420-010

(5) Apply a Molykote 7 release compound, D50232, on a new packing [14].

SUBTASK 38-42-10-420-008

(6) Put the new packing [14] on the filter bowl [16] and then put the filter bowl [16] in its position.

(a) Tighten the filter bowl [16] to 75 +10 / -0 in-lb (8.5 +1.1 / -0.0 N·m).

SUBTASK 38-42-10-420-009

(7) Install safety wire, G51934, do this task: Lockwire, Cotter Pins, and Lockrings - Installation, TASK 20-10-44-400-801.

I. Bleed Air Filter Element Installation Test

SUBTASK 38-42-10-860-010

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
D	11	C00873	POT WATER COMPRESSOR

SUBTASK 38-42-10-410-005

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 38-42-10-860-011

(3) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 38-42-10-860-012

(4) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 38-42-10-860-013

(5) Pressurize the Pneumatic System, do this task: Supply Pressure to the Pneumatic System (Selection), TASK 36-00-00-860-801.

SUBTASK 38-42-10-790-002

(6) Make sure that there are no air leaks at the air filter.

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J. Put the Airplane Back to Its Usual Condition

SUBTASK 38-42-10-860-014

(1) Remove pressure from the pneumatic system as follows:

(a) Do this task: Remove Pressure from the Pneumatic System, TASK 36-00-00-860-806.

SUBTASK 38-42-10-210-002

(2) Examine seal, G50360, on the potable water line cap frame.

(a) Replace seal, G50360, if it missed or damaged.

SUBTASK 38-42-10-410-008

(3) Install the potable water line cap as follows:

(a) Install the potable water line cap.

(b) Lock the quarter-turn fasteners around the edges of the potable water line cap.

SUBTASK 38-42-10-410-004

(4) Close this access panel:

Number	Name/Location
822	Aft Cargo Door

END OF TASK

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